

EXAMPLE 19.5 Thermodynamics of boiling water**WITH VARIATION PROBLEMS**

One gram of water (1 cm^3) becomes 1671 cm^3 of steam when boiled at a constant pressure of 1 atm ($1.013 \times 10^5 \text{ Pa}$). The heat of vaporization at this pressure is $L_v = 2.256 \times 10^6 \text{ J/kg}$. Compute (a) the work done by the water when it vaporizes and (b) its increase in internal energy.

IDENTIFY and SET UP The heat added causes the system (water) to change phase from liquid to vapor. We can analyze this process by using the first law of thermodynamics. The water is boiled at constant pressure, so we can use Eq. (19.3) to calculate the work W done by the vaporizing water as it expands. We are given the mass of water and the heat of vaporization, so we can use Eq. (17.20), $Q = mL_v$, to calculate the heat Q added to the water. We can then find the internal energy change from Eq. (19.4), $\Delta U = Q - W$.

EXECUTE (a) From Eq. (19.3), the water does work

$$W = p(V_2 - V_1) \\ = (1.013 \times 10^5 \text{ Pa})(1671 \times 10^{-6} \text{ m}^3 - 1 \times 10^{-6} \text{ m}^3) = 169 \text{ J}$$

(b) From Eq. (17.20), the heat added to the water is

$$Q = mL_v = (10^{-3} \text{ kg})(2.256 \times 10^6 \text{ J/kg}) = 2256 \text{ J}$$

Then from Eq. (19.4),

$$\Delta U = Q - W = 2256 \text{ J} - 169 \text{ J} = 2087 \text{ J}$$

EVALUATE To vaporize 1 g of water, we must add 2256 J of heat, most of which (2087 J) remains in the system as an increase in internal energy. The remaining 169 J leaves the system as the system expands from liquid to vapor and does work against the surroundings. (The increase in internal energy is associated mostly with the attractive intermolecular forces. The associated potential energies are greater after work has been done to pull apart the molecules in the liquid, forming the vapor state. It's like increasing gravitational potential energy by pulling an elevator farther from the center of the earth.)

KEYCONCEPT For a system that undergoes a thermodynamic process, if you can calculate any two of the quantities ΔU (internal energy change), Q (heat flow into the system), and W (work done by the system), you can calculate the third using the first law of thermodynamics, $\Delta U = Q - W$.

Infinitesimal Changes of State

In the preceding examples the initial and final states differ by a finite amount. Later we'll consider *infinitesimal* changes of state in which a small amount of heat dQ is added to the system, the system does a small amount of work dW , and its internal energy changes by an amount dU . For such a process,

$$\text{First law of thermodynamics, infinitesimal process:} \quad \text{Infinitesimal internal energy change} \quad dU = dQ - dW \quad (19.6)$$

↑ ↑
Infinitesimal heat added Infinitesimal work done

For the systems we'll discuss, the work dW is given by $dW = p dV$, so we can also state the first law as

$$dU = dQ - p dV \quad (19.7)$$

TEST YOUR UNDERSTANDING OF SECTION 19.4 Rank the following thermodynamic processes according to the change in internal energy in each process, from most positive to most negative. (i) As you do 250 J of work on a system, it transfers 250 J of heat to its surroundings; (ii) as you do 250 J of work on a system, it absorbs 250 J of heat from its surroundings; (iii) as a system does 250 J of work on you, it transfers 250 J of heat to its surroundings; (iv) as a system does 250 J of work on you, it absorbs 250 J of heat from its surroundings.

ANSWER

ANSWER (i) $\Delta U = 0$; (ii) $\Delta U = 500 \text{ J}$; (iii) $\Delta U = -500 \text{ J}$; (iv) $\Delta U = 0$. In the expression $\Delta U = Q - W$, Q is the heat added to the system and W is the work done by the system. If heat is transferred from the system to its surroundings, Q is negative; if work is done on the system, W is negative. Hence we have (i) $\Delta U = 250 \text{ J} - 250 \text{ J} = 0$; (ii) $\Delta U = 250 \text{ J} - (-250 \text{ J}) = 500 \text{ J}$; (iii) $\Delta U = (-250 \text{ J}) - 250 \text{ J} = -500 \text{ J}$; and (iv) $\Delta U = 250 \text{ J} - 250 \text{ J} = 0$.

19.5 KINDS OF THERMODYNAMIC PROCESSES

In this section we describe four specific kinds of thermodynamic processes that occur often in practical situations. We can summarize these briefly as “no heat transfer” or *adiabatic*, “constant volume” or *isochoric*, “constant pressure” or *isobaric*, and “constant temperature” or *isothermal*. For some of these processes we can use a simplified form of the first law of thermodynamics.

Figure 19.14 When the cork is popped on a bottle of champagne, the pressurized gases inside the bottle expand rapidly and do work on the outside air ($W > 0$). There is little time for the gases to exchange heat with their surroundings, so the expansion is nearly adiabatic ($Q = 0$). Hence the internal energy of the expanding gases decreases ($\Delta U = -W < 0$) and their temperature drops. This makes water vapor condense and form a miniature cloud.

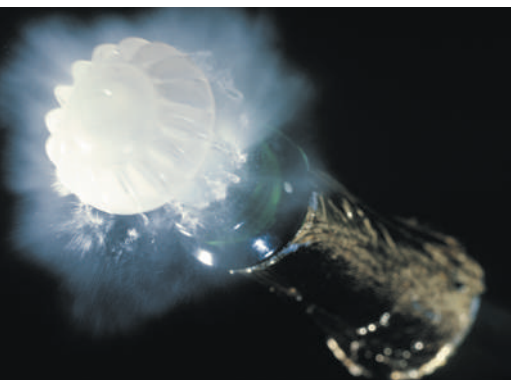


Figure 19.15 Most cooking involves isobaric processes. That's because the air pressure above a saucepan or frying pan, or inside a microwave oven, remains essentially constant while the food is being heated.



Adiabatic Process

An **adiabatic process** (pronounced “ay-dee-ah-bat-ic”) is defined as one with no heat transfer into or out of a system; $Q = 0$. We can prevent heat flow either by surrounding the system with thermally insulating material or by carrying out the process so quickly that there is not enough time for appreciable heat flow. From the first law we find that for every adiabatic process,

$$U_2 - U_1 = \Delta U = -W \quad (\text{adiabatic process}) \quad (19.8)$$

When a system expands adiabatically, W is positive (the system does work on its surroundings), so ΔU is negative and the internal energy decreases. When a system is *compressed* adiabatically, W is negative (work is done on the system by its surroundings) and U increases. In many (but not all) systems an increase of internal energy is accompanied by a rise in temperature, and a decrease in internal energy by a drop in temperature (Fig. 19.14).

The compression stroke in an internal-combustion engine is an approximately adiabatic process. The temperature rises as the air–fuel mixture in the cylinder is compressed. The expansion of the burned fuel during the power stroke is also an approximately adiabatic expansion with a drop in temperature. In Section 19.8 we’ll consider adiabatic processes in an ideal gas.

Isochoric Process

An **isochoric process** (pronounced “eye-so-kor-ic”) is a *constant-volume* process. When the volume of a thermodynamic system is constant, it does no work on its surroundings. Then $W = 0$ and

$$U_2 - U_1 = \Delta U = Q \quad (\text{isochoric process}) \quad (19.9)$$

In an isochoric process, all the energy added as heat remains in the system as an increase in internal energy. Heating a gas in a closed constant-volume container is an example of an isochoric process. The processes *ab* and *cd* in Example 19.4 are also examples of isochoric processes. (Note that there are types of work that do not involve a volume change. For example, we can do work on a fluid by stirring it. In some literature, “isochoric” is used to mean that no work of any kind is done.)

Isobaric Process

An **isobaric process** (pronounced “eye-so-bear-ic”) is a *constant-pressure* process. In general, none of the three quantities ΔU , Q , and W is zero in an isobaric process, but calculating W is easy nonetheless. From Eq. (19.3),

$$W = p(V_2 - V_1) \quad (\text{isobaric process}) \quad (19.10)$$

Boiling water at constant pressure is an isobaric process (Fig. 19.15).

Isothermal Process

An **isothermal process** is a *constant-temperature* process. For a process to be isothermal, any heat flow into or out of the system must occur slowly enough that thermal equilibrium is maintained. In general, none of the quantities ΔU , Q , or W is zero in an isothermal process.

In some special cases the internal energy of a system depends *only* on its temperature, not on its pressure or volume. The most familiar system having this special property is an ideal gas, as we’ll discuss in the next section. For such systems, if the temperature is constant, the internal energy is also constant; $\Delta U = 0$ and $Q = W$. That is, any energy entering the system as heat Q must leave it again as work W done by the system. Example 19.1, involving an ideal gas, is an example of an isothermal process in which U is also constant. For most systems other than ideal gases, the internal energy depends on pressure as well as temperature, so U may vary even when T is constant.

Figure 19.16 shows a pV -diagram for these four processes for a constant amount of an ideal gas. The path followed in an adiabatic process (a to 1) is called an **adiabat**. A vertical line (constant volume) is an **isochor**, a horizontal line (constant pressure) is an **isobar**, and a curve of constant temperature (shown as light blue lines in Fig. 19.16) is an **isotherm**.

TEST YOUR UNDERSTANDING OF SECTION 19.5 Which of the processes in Fig. 19.7 are isochoric? Which are isobaric? Is it possible to tell if any of the processes are isothermal or adiabatic?

ANSWER

I 1 $\leftarrow$ 4 and 3 $\rightarrow$ 2 are isochoric; 1 $\leftarrow$ 3 and 4 $\rightarrow$ 2 are isobaric; no 2 are isothermal; no In a pV -diagram like those shown in Fig. 19.7, isochoric processes are represented by vertical lines (lines of constant volume) and isobaric processes are represented by horizontal lines (lines of constant pressure). The process 1 $\rightarrow$ 2 in Fig. 19.7 is shown as a curved line, which superficially resembles the adiabatic and isothermal processes for an ideal gas in Fig. 19.16. Without more information we can't tell whether process 1 $\rightarrow$ 2 is isothermal, adiabatic, or neither.

19.6 INTERNAL ENERGY OF AN IDEAL GAS

We now show that for an ideal gas, the internal energy U depends only on temperature, not on pressure or volume. Let's think again about the free-expansion experiment described in Section 19.3. A thermally insulated container with rigid walls is divided into two compartments by a partition (**Fig. 19.17**). One compartment has a quantity of an ideal gas and the other is evacuated.

When the partition is removed or broken, the gas expands to fill both parts of the container. There is no heat flow through the insulation, and the gas does no work on its surroundings because the walls of the container don't move. So both Q and W are zero and the internal energy U is constant.

Does the *temperature* T change during a free expansion? Suppose it *does* change, while the internal energy stays the same. In that case we have to conclude that the internal energy depends on both T and the volume V or on both T and the pressure p , but certainly not on T alone. But if T is constant during a free expansion, for which we know that U is constant even though both p and V change, then we have to conclude that U depends only on T , not on p or V .

Many experiments have shown that when a low-density gas (essentially an ideal gas) undergoes a free expansion, its temperature *does not* change. The conclusion is:

The internal energy U of an ideal gas depends only on its temperature T , not on its pressure or volume.

This property, in addition to the ideal-gas equation of state, is part of the ideal-gas model. We'll make frequent use of this property.

For nonideal gases, some temperature change occurs during free expansions, even though the internal energy is constant. This shows that the internal energy cannot depend *only* on temperature; it must depend on pressure as well. From the microscopic viewpoint, in which internal energy U is the sum of the kinetic and potential energies for all the particles that make up the system, this is not surprising. Nonideal gases usually have attractive intermolecular forces, and when molecules move farther apart, the associated potential energies increase. If the total internal energy is constant, the kinetic energies must decrease. Temperature is directly related to molecular *kinetic* energy, and for such a gas a free expansion is usually accompanied by a *drop* in temperature.

TEST YOUR UNDERSTANDING OF SECTION 19.6 Is the internal energy of a solid likely to be independent of its volume, as is the case for an ideal gas? Explain your reasoning. (*Hint:* See Fig. 18.20.)

ANSWER **I** no Using the model of a solid in Fig. 18.20, we can see that the internal energy of a solid *does* depend on its volume. Compressing the solid means compressing the "springs" between the atoms, thereby increasing their stored potential energy and hence the internal energy of the solid.

Figure 19.16 Four different processes for a constant amount of an ideal gas, all starting at state a . For the adiabatic process, $Q = 0$; for the isochoric process, $W = 0$; and for the isothermal process, $\Delta U = 0$. The temperature increases only during the isobaric expansion.

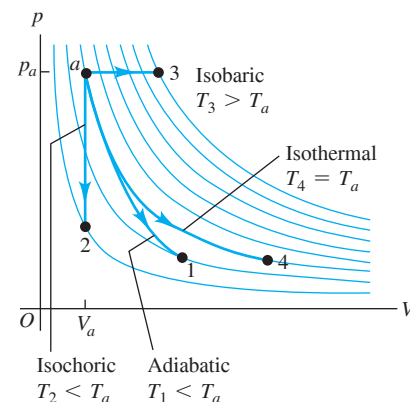


Figure 19.17 The partition is broken (or removed) to start the free expansion of gas into the vacuum region.

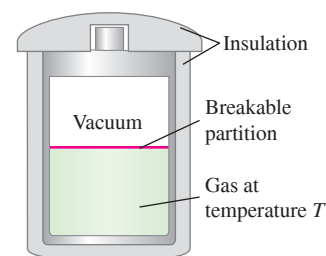
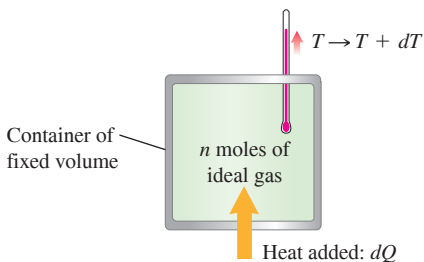


Figure 19.18 Measuring the molar heat capacity of an ideal gas (a) at constant volume and (b) at constant pressure.

(a) Constant volume: $dQ = nC_V dT$



(b) Constant pressure: $dQ = nC_p dT$

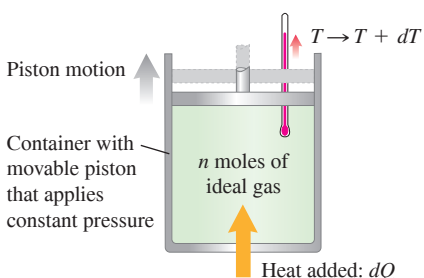
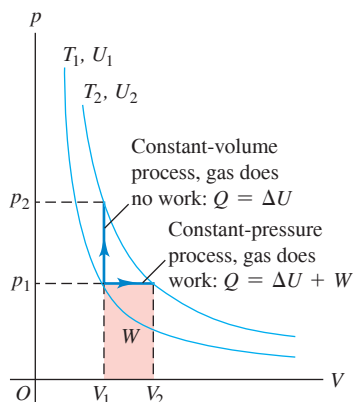


Figure 19.19 Raising the temperature of an ideal gas from T_1 to T_2 by a constant-volume or a constant-pressure process. For an ideal gas, U depends only on T , so ΔU is the same for both processes. But for the constant-pressure process, more heat Q must be added to both increase U and do work W . Hence $C_p > C_V$.



19.7 HEAT CAPACITIES OF AN IDEAL GAS

We defined specific heat and molar heat capacity in Section 17.5. We also remarked at the end of that section that the specific heat or molar heat capacity of a substance depends on the conditions under which the heat is added. The heat capacity of a gas is usually measured in a closed container under constant-volume conditions. The corresponding heat capacity is the **molar heat capacity at constant volume**, denoted by C_V . Heat capacity measurements for solids and liquids are usually carried out under constant atmospheric pressure, and we call the corresponding heat capacity the **molar heat capacity at constant pressure**, C_p .

Let's consider C_V and C_p for an ideal gas. To measure C_V , we raise the temperature of an ideal gas in a rigid container with constant volume, ignoring its thermal expansion (Fig. 19.18a). To measure C_p , we let the gas expand just enough to keep the pressure constant as the temperature rises (Fig. 19.18b).

Why should these two molar heat capacities be different? The answer lies in the first law of thermodynamics. In a constant-volume temperature increase, the system does no work, and the change in internal energy ΔU equals the heat added Q . In a constant-pressure temperature increase, on the other hand, the volume *must* increase; otherwise, the pressure (given by the ideal-gas equation of state, $p = nRT/V$) could not remain constant. As the material expands, it does an amount of work W . According to the first law,

$$Q = \Delta U + W \quad (19.11)$$

For a given temperature increase, the internal energy change ΔU of an ideal gas has the same value no matter what the process (remember that the internal energy of an ideal gas depends only on temperature, not on pressure or volume). Equation (19.11) then shows that the heat input for a constant-pressure process must be *greater* than that for a constant-volume process because additional energy must be supplied to account for the work W done during the expansion. So C_p is greater than C_V for an ideal gas. The pV -diagram in Fig. 19.19 shows this relationship. For air, C_p is 40% greater than C_V .

For a very few substances (one of which is water between 0°C and 4°C) the volume *decreases* during heating. In this case, W is negative and the internal energy change ΔU is greater than the heat input Q .

Relating C_p and C_V for an Ideal Gas

We can derive a simple relationship between C_p and C_V for an ideal gas. First consider the constant-volume process. We place n moles of an ideal gas at temperature T in a constant-volume container. We place it in thermal contact with a hotter object; an infinitesimal quantity of heat dQ flows into the gas, and its temperature increases by an infinitesimal amount dT . By the definition of C_V , the molar heat capacity at constant volume,

$$dQ = nC_V dT \quad (19.12)$$

The pressure increases during this process, but the gas does no work ($dW = 0$) because the volume is constant. The first law in differential form, Eq. (19.6), is $dQ = dU + dW$. Since $dW = 0$, $dQ = dU$ and Eq. (19.12) can also be written as

$$dU = nC_V dT \quad (19.13)$$

Now consider a constant-pressure process with the same temperature change dT . We place the same gas in a cylinder with a piston that we allow to move just enough to maintain constant pressure (Fig. 19.18b). Again we bring the system into contact with a hotter object. As heat flows into the gas, it expands at constant pressure and does work. By the definition of C_p , the molar heat capacity at constant pressure, the amount of heat entering the gas is

$$dQ = nC_p dT \quad (19.14)$$

The work dW done by the gas in this constant-pressure process is

$$dW = p dV$$

We can also express dW in terms of the temperature change dT by using the ideal-gas equation of state, $pV = nRT$. Because p is constant, the change in V is proportional to the change in T :

$$dW = p dV = nR dT \quad (19.15)$$

Now substitute Eqs. (19.14) and (19.15) into the first law, $dQ = dU + dW$:

$$nC_p dT = dU + nR dT \quad (19.16)$$

Here comes the crux of the calculation. The internal energy change dU for the constant-pressure process is again given by Eq. (19.13), $dU = nC_V dT$, *even though now the volume is not constant*. Why is this so? From Section 19.6, one of the special properties of an ideal gas is that its internal energy depends *only* on temperature. Thus the *change* in internal energy during any process must be determined only by the temperature change. If Eq. (19.13) is valid for an ideal gas for one particular kind of process, it must be valid for an ideal gas for *every* kind of process with the same dT . So we may replace dU in Eq. (19.16) by $nC_V dT$:

$$nC_p dT = nC_V dT + nR dT$$

When we divide each term by the common factor $n dT$, we get

$$\text{For an ideal gas:} \quad C_p = C_V + R \quad (19.17)$$

Molar heat capacity at constant pressure
Molar heat capacity at constant volume

As we predicted, the molar heat capacity of an ideal gas at constant pressure is *greater* than the molar heat capacity at constant volume; the difference is the gas constant R .

We have used the ideal-gas model to derive Eq. (19.17), but it turns out to be obeyed to within a few percent by many real gases at moderate pressures. **Table 19.1** gives measured values of C_p and C_V for several real gases at low pressures; the difference in most cases is approximately $R = 8.314 \text{ J/mol} \cdot \text{K}$.

The table also shows that the molar heat capacity of a gas is related to its molecular structure, as we discussed in Section 18.4. In fact, the first three columns of Table 19.1 are the same as Table 18.1.

The Ratio of Heat Capacities

The last column of Table 19.1 lists the values of the dimensionless **ratio of heat capacities**, C_p/C_V , denoted by γ (the Greek letter gamma):

$$\text{Ratio of heat capacities} \quad \gamma = \frac{C_p}{C_V} \quad (19.18)$$

Molar heat capacity at constant pressure
Molar heat capacity at constant volume

TABLE 19.1 Molar Heat Capacities of Gases at Low Pressure

Type of Gas	Gas	C_V (J/mol · K)	C_p (J/mol · K)	$C_p - C_V$ (J/mol · K)	$\gamma = C_p/C_V$
Monatomic	He	12.47	20.78	8.31	1.67
	Ar	12.47	20.78	8.31	1.67
Diatomic	H ₂	20.42	28.74	8.32	1.41
	N ₂	20.76	29.07	8.31	1.40
	O ₂	20.85	29.17	8.32	1.40
	CO	20.85	29.16	8.31	1.40
Polyatomic	CO ₂	28.46	36.94	8.48	1.30
	SO ₂	31.39	40.37	8.98	1.29
	H ₂ S	25.95	34.60	8.65	1.33

(This is sometimes called the “ratio of specific heats.”) For gases, C_p is always greater than C_V and γ is always greater than unity. We’ll see in the next section that γ plays an important role in adiabatic processes for an ideal gas.

We can use our kinetic-theory discussion of the molar heat capacity of an ideal gas (see Section 18.4) to predict values of γ . As an example, an ideal monatomic gas has $C_V = \frac{3}{2}R$. From Eq. (19.17),

$$C_p = C_V + R = \frac{3}{2}R + R = \frac{5}{2}R$$

so

$$\gamma = \frac{C_p}{C_V} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3} = 1.67$$

As Table 19.1 shows, this agrees well with values of γ computed from measured heat capacities. For most diatomic gases near room temperature, $C_V = \frac{5}{2}R$, $C_p = C_V + R = \frac{7}{2}R$, and

$$\gamma = \frac{C_p}{C_V} = \frac{\frac{7}{2}R}{\frac{5}{2}R} = \frac{7}{5} = 1.40$$

also in good agreement with measured values.

Here’s a final reminder: For an ideal gas, the internal energy change in *any* process is given by $\Delta U = nC_V \Delta T$, *whether the volume is constant or not*. This relationship holds for other substances *only* when the volume is constant.

CAUTION A few hints about ΔU , Q , and W for an ideal gas Remember that if the gas temperature T increases, the internal energy change ΔU of the gas is positive; if T decreases, ΔU is negative. Remember, too, that the work W done by the gas is positive if the gas expands and negative if the gas is compressed. Finally, you can calculate the heat flow Q into the gas directly only in certain special processes (isochoric, isobaric, or adiabatic processes); for other processes, you must first calculate ΔU and W and then find Q using the first law of thermodynamics. **|**

EXAMPLE 19.6 Cooling your room

WITH VARIATION PROBLEMS

A typical dorm room or bedroom contains about 2500 moles of air. Find the change in the internal energy of this much air when it is cooled from 35.0°C to 26.0°C at a constant pressure of 1.00 atm. Treat the air as an ideal gas with $\gamma = 1.400$.

IDENTIFY and SET UP Our target variable is the change in the internal energy ΔU of an ideal gas in a constant-pressure process. We are given the number of moles, the temperature change, and the value of γ for air. We use Eq. (19.13), $\Delta U = nC_V \Delta T$, which gives the internal energy change for an ideal gas in *any* process, *whether the volume is constant or not*. [See the discussion following Eq. (19.16).] We use Eqs. (19.17) and (19.18) to find C_V .

EXECUTE From Eqs. (19.17) and (19.18),

$$\gamma = \frac{C_p}{C_V} = \frac{C_V + R}{C_V} = 1 + \frac{R}{C_V}$$

$$C_V = \frac{R}{\gamma - 1} = \frac{8.314 \text{ J/mol} \cdot \text{K}}{1.400 - 1} = 20.79 \text{ J/mol} \cdot \text{K}$$

Then from Eq. (19.13),

$$\begin{aligned}\Delta U &= nC_V \Delta T \\ &= (2500 \text{ mol})(20.79 \text{ J/mol} \cdot \text{K})(26.0^\circ\text{C} - 35.0^\circ\text{C}) \\ &= -4.68 \times 10^5 \text{ J}\end{aligned}$$

EVALUATE To cool 2500 moles of air from 35.0°C to 26.0°C, a room air conditioner must extract this much internal energy from the air and transfer it to the air outside. In Chapter 20 we’ll discuss how this is done.

KEYCONCEPT When n moles of an ideal gas undergo a thermodynamic process in which the temperature changes by ΔT , the change in internal energy is equal to the product of n , ΔT , and the molar heat capacity at constant volume C_V : $\Delta U = nC_V \Delta T$. This is true whether or not the volume of the system remains constant.

TEST YOUR UNDERSTANDING OF SECTION 19.7 You want to cool a storage cylinder containing 10 moles of compressed gas from 30°C to 20°C. For which kind of gas would this be easiest? (i) A monatomic gas; (ii) a diatomic gas; (iii) a polyatomic gas; (iv) it would be equally easy for all of these.

ANSWER

(i) For a given number of moles n and a given temperature change ΔT , the amount of heat that must be transferred out of a fixed volume of air is $Q = nC_V \Delta T$. Hence the amount of heat transfer required is least for the gas with the smallest value of C_V . From Table 19.1, C_V is smallest for monatomic gases.

19.8 ADIABATIC PROCESSES FOR AN IDEAL GAS

An adiabatic process, defined in Section 19.5, is a process in which no heat transfer takes place between a system and its surroundings. Zero heat transfer is an idealization, but a process is approximately adiabatic if the system is well insulated or if the process takes place so quickly that there is not enough time for appreciable heat flow to occur.

In an adiabatic process, $Q = 0$, so from the first law, $\Delta U = -W$. An adiabatic process for an ideal gas is shown in the pV -diagram of **Fig. 19.20**. As the gas expands from volume V_a to V_b , it does positive work, so its internal energy decreases and its temperature drops. If point a , representing the initial state, lies on an isotherm at temperature $T + dT$, then point b for the final state is on a different isotherm at a lower temperature T . For an adiabatic *compression* from V_b to V_a the situation is reversed and the temperature rises.

The air in the output hoses of air compressors used to inflate tires and to fill scuba tanks is always warmer than the air entering the compressor; this is because the compression is rapid and hence approximately adiabatic. Adiabatic *cooling* occurs when you open a bottle of your favorite carbonated beverage. The gas just above the beverage surface expands rapidly in a nearly adiabatic process; the gas temperature drops so much that water vapor in the gas condenses, forming a miniature cloud (see Fig. 19.14).

CAUTION “Heating” and “cooling” without heat When we talk about “adiabatic heating” and “adiabatic cooling,” we really mean “raising the temperature” and “lowering the temperature,” respectively. In an adiabatic process, the temperature change is due to work done by or on the system; there is *no* heat flow at all. ■

Adiabatic Ideal Gas: Relating V , T , and p

We can derive a relationship between volume and temperature changes for an infinitesimal adiabatic process in an ideal gas. Equation (19.13) gives the internal energy change dU for *any* process for an ideal gas, adiabatic or not, so we have $dU = nC_V dT$. Also, the work done by the gas during the process is given by $dW = p dV$. Then, since $dU = -dW$ for an adiabatic process, we have

$$nC_V dT = -p dV \quad (19.19)$$

To obtain a relationship containing only the volume V and temperature T , we eliminate p by using the ideal-gas equation in the form $p = nRT/V$. Substituting this into Eq. (19.19) and rearranging, we get

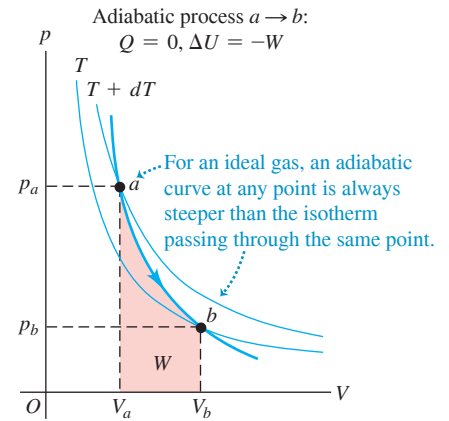
$$\begin{aligned} nC_V dT &= -\frac{nRT}{V} dV \\ \frac{dT}{T} + \frac{R}{C_V} \frac{dV}{V} &= 0 \end{aligned}$$

The coefficient R/C_V can be expressed in terms of $\gamma = C_p/C_V$. We have

$$\begin{aligned} \frac{R}{C_V} &= \frac{C_p - C_V}{C_V} = \frac{C_p}{C_V} - 1 = \gamma - 1 \\ \frac{dT}{T} + (\gamma - 1) \frac{dV}{V} &= 0 \end{aligned} \quad (19.20)$$

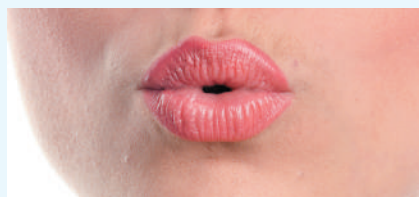
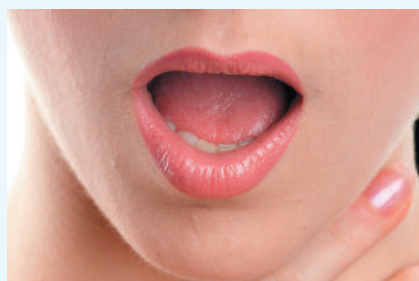
Because γ is always greater than unity for a gas, $(\gamma - 1)$ is always positive. This means that in Eq. (19.20), dV and dT always have opposite signs. An adiabatic *expansion* of an ideal gas ($dV > 0$) always occurs with a *drop* in temperature ($dT < 0$), and an adiabatic *compression* ($dV < 0$) always occurs with a *rise* in temperature ($dT > 0$); this confirms our earlier prediction.

Figure 19.20 A pV -diagram of an adiabatic ($Q = 0$) process for an ideal gas. As the gas expands from V_a to V_b , it does positive work W on its environment, its internal energy decreases ($\Delta U = -W < 0$), and its temperature drops from $T + dT$ to T . (An adiabatic process is also shown in Fig. 19.16.)



BIO APPLICATION Exhaling

Adiabatically Put your hand a few centimeters in front of your mouth, open your mouth wide, and exhale. Your breath will feel warm on your hand, because the exhaled gases emerge at roughly the temperature of your body's interior. Now purse your lips as though you were going to whistle, and again blow on your hand. The exhaled gases will feel much cooler. In this case the gases undergo a rapid, essentially adiabatic expansion as they emerge from between your lips, so the temperature of the exhaled gases decreases.



For finite changes in temperature and volume we integrate Eq. (19.20), obtaining

$$\ln T + (\gamma - 1) \ln V = \text{constant}$$

$$\ln T + \ln V^{\gamma-1} = \text{constant}$$

$$\ln(TV^{\gamma-1}) = \text{constant}$$

and finally,

$$TV^{\gamma-1} = \text{constant} \quad (19.21)$$

Thus for an initial state (T_1, V_1) and a final state (T_2, V_2) ,

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \quad (\text{adiabatic process, ideal gas}) \quad (19.22)$$

Because we have used the ideal-gas equation in our derivation of Eqs. (19.21) and (19.22), the T 's must always be *absolute* (Kelvin) temperatures.

We can also convert Eq. (19.21) into a relationship between pressure and volume by eliminating T , using the ideal-gas equation in the form $T = pV/nR$. Substituting this into Eq. (19.21), we find

$$\frac{pV}{nR} V^{\gamma-1} = \text{constant}$$

or, because n and R are constant,

$$pV^\gamma = \text{constant} \quad (19.23)$$

For an initial state (p_1, V_1) and a final state (p_2, V_2) , Eq. (19.23) becomes

$$p_1 V_1^\gamma = p_2 V_2^\gamma \quad (\text{adiabatic process, ideal gas}) \quad (19.24)$$

We can also calculate the *work* done by an ideal gas during an adiabatic process. We know that $Q = 0$ and $W = -\Delta U$ for *any* adiabatic process. For an ideal gas, $\Delta U = nC_V(T_2 - T_1)$. If the number of moles n and the initial and final temperatures T_1 and T_2 are known, we have simply

$$W = nC_V(T_1 - T_2) \quad (19.25)$$

Diagram labels for Eq. (19.25):
 - Work done by an ideal gas, adiabatic process (points to W)
 - Number of moles (points to n)
 - Initial temperature (points to T_1)
 - Final temperature (points to T_2)
 - Molar heat capacity at constant volume (points to C_V)

We may also use $pV = nRT$ in this equation to obtain

$$W = \frac{C_V}{R}(p_1 V_1 - p_2 V_2) = \frac{1}{\gamma - 1}(p_1 V_1 - p_2 V_2) \quad (19.26)$$

Diagram labels for Eq. (19.26):
 - Work done by an ideal gas, adiabatic process (points to W)
 - Gas constant (points to R)
 - Molar heat capacity at constant volume (points to C_V)
 - Ratio of heat capacities (points to γ)
 - Initial pressure, volume (points to $p_1 V_1$)
 - Final pressure, volume (points to $p_2 V_2$)

(We used the result $C_V = R/(\gamma - 1)$ from Example 19.6.) If the process is an expansion, the temperature drops, T_1 is greater than T_2 , $p_1 V_1$ is greater than $p_2 V_2$, and the work is *positive*. If the process is a compression, the work is *negative*.

Throughout this analysis of adiabatic processes we have used the ideal-gas equation of state, which is valid only for *equilibrium* states. Strictly speaking, our results are valid only for a process that is fast enough to prevent appreciable heat exchange with the surroundings (so that $Q = 0$ and the process is adiabatic), yet slow enough that the system does not depart very much from thermal and mechanical equilibrium. Even when these conditions are not strictly satisfied, though, Eqs. (19.22), (19.24), and (19.26) give useful approximate results.

EXAMPLE 19.7 Adiabatic compression in a diesel engine**WITH VARIATION PROBLEMS**

The compression ratio of a diesel engine is 15.0 to 1; that is, air in a cylinder is compressed to $\frac{1}{15.0}$ of its initial volume (**Fig. 19.21**). (a) If the initial pressure is 1.01×10^5 Pa and the initial temperature is 27°C (300 K), find the final pressure and the temperature after adiabatic compression. (b) How much work does the gas do during the compression if the initial volume of the cylinder is $1.00\text{ L} = 1.00 \times 10^{-3}\text{ m}^3$? Use the values $C_V = 20.8\text{ J/mol} \cdot \text{K}$ and $\gamma = 1.400$ for air.

IDENTIFY and SET UP This problem involves the adiabatic compression of an ideal gas, so we can use the ideas of this section. In part (a) we are given the initial pressure and temperature $p_1 = 1.01 \times 10^5$ Pa and $T_1 = 300$ K; the ratio of initial and final volumes is $V_1/V_2 = 15.0$. We use Eq. (19.22) to find the final temperature T_2 and Eq. (19.24) to find the final pressure p_2 . In part (b) our target variable is W , the work done by the gas during the adiabatic compression. We use Eq. (19.26) to calculate W .

EXECUTE (a) From Eqs. (19.22) and (19.24),

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = (300\text{ K})(15.0)^{0.40} = 886\text{ K} = 613^\circ\text{C}$$

$$\begin{aligned} p_2 &= p_1 \left(\frac{V_1}{V_2} \right)^\gamma = (1.01 \times 10^5\text{ Pa})(15.0)^{1.40} \\ &= 44.8 \times 10^5\text{ Pa} = 44\text{ atm} \end{aligned}$$

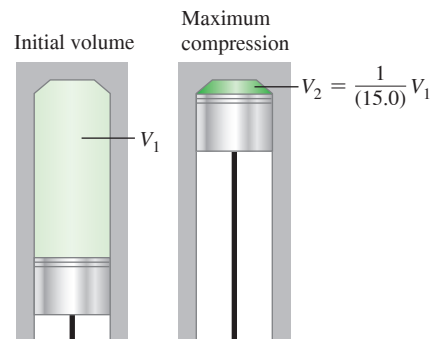
(b) From Eq. (19.26), the work done is

$$W = \frac{1}{\gamma - 1} (p_1 V_1 - p_2 V_2)$$

Using $V_1/V_2 = 15.0$, we have

$$\begin{aligned} W &= \frac{1}{1.400 - 1} \left[(1.01 \times 10^5\text{ Pa})(1.00 \times 10^{-3}\text{ m}^3) \right. \\ &\quad \left. - (44.8 \times 10^5\text{ Pa}) \left(\frac{1.00 \times 10^{-3}\text{ m}^3}{15.0} \right) \right] \\ &= -494\text{ J} \end{aligned}$$

Figure 19.21 Adiabatic compression of air in a cylinder of a diesel engine.



EVALUATE If the compression had been isothermal, the final pressure would have been 15.0 atm. Because the temperature also increases during an adiabatic compression, the final pressure is much greater. When fuel is injected into the cylinders near the end of the compression stroke, the high temperature of the air attained during compression causes the fuel to ignite spontaneously without the need for spark plugs.

We can check our result in part (b) by using Eq. (19.25). The number of moles of gas in the cylinder is

$$n = \frac{p_1 V_1}{RT_1} = \frac{(1.01 \times 10^5\text{ Pa})(1.00 \times 10^{-3}\text{ m}^3)}{(8.314\text{ J/mol} \cdot \text{K})(300\text{ K})} = 0.0405\text{ mol}$$

Then Eq. (19.25) gives

$$\begin{aligned} W &= nC_V(T_1 - T_2) \\ &= (0.0405\text{ mol})(20.8\text{ J/mol} \cdot \text{K})(300\text{ K} - 886\text{ K}) = -494\text{ J} \end{aligned}$$

The work is negative because the gas is compressed.

KEYCONCEPT In an adiabatic process there is no heat flow into or out of the system, so $Q = 0$ and the internal energy equals the negative of the work done by the system: $\Delta U = -W$. If the system is an ideal gas, the temperature T , volume V , and pressure p all change in the process, but the quantities $TV^{\gamma-1}$ and PV^γ remain constant (γ is the ratio of specific heats for the gas).

TEST YOUR UNDERSTANDING OF SECTION 19.8 You have four samples of ideal gas, each of which contains the same number of moles of gas and has the same initial temperature, volume, and pressure. You compress each sample to one-half of its initial volume. Rank the four samples in order from highest to lowest value of the final pressure. (i) A monatomic gas compressed isothermally; (ii) a monatomic gas compressed adiabatically; (iii) a diatomic gas compressed isothermally; (iv) a diatomic gas compressed adiabatically.

ANSWER

(i) and (iii) are compressed isothermally, so $pV = \text{constant}$. The volume of each sample decreases to one-half of its initial value, so the final pressure is twice the initial pressure. Samples (ii) and (iv) are compressed adiabatically, so $pV^\gamma = \text{constant}$ and the pressure increases by a factor of 2^γ . Sample (ii) is a monatomic gas for which $\gamma = \frac{5}{3}$, so its final pressure is $2^{5/3} = 3.17$ times greater than the initial pressure. Sample (iv) is a diatomic gas for which $\gamma = \frac{7}{5}$, so its final pressure is greater than the initial pressure by a factor of $2^{7/5} = 2.64$.

CHAPTER 19 SUMMARY

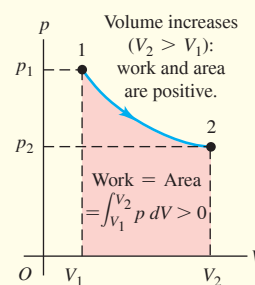
Heat and work in thermodynamic processes: A thermodynamic system has the potential to exchange energy with its surroundings by heat transfer or by mechanical work. When a system at pressure p changes volume from V_1 to V_2 , it does an amount of work W given by the integral of p with respect to volume. If the pressure is constant, the work done is equal to p times the change in volume. A negative value of W means that work is done on the system. (See Example 19.1.)

In any thermodynamic process, the heat added to the system and the work done by the system depend not only on the initial and final states, but also on the path (the series of intermediate states through which the system passes).

$$W = \int_{V_1}^{V_2} p \, dV \quad (19.2)$$

$$W = p(V_2 - V_1) \quad (19.3)$$

(constant pressure only)



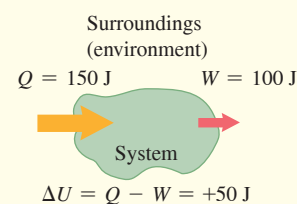
The first law of thermodynamics: The first law of thermodynamics states that when heat Q is added to a system while the system does work W , the internal energy U changes by an amount equal to $Q - W$. This law can also be expressed for an infinitesimal process. (See Examples 19.2, 19.3, and 19.5.)

The internal energy of any thermodynamic system depends only on its state. The change in internal energy in any process depends only on the initial and final states, not on the path. The internal energy of an isolated system is constant. (See Example 19.4.)

$$\Delta U = Q - W \quad (19.4)$$

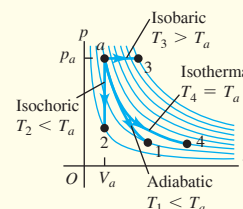
$$dU = dQ - dW \quad (19.6)$$

(infinitesimal process)



Important kinds of thermodynamic processes:

- Adiabatic process: No heat transfer into or out of a system; $Q = 0$.
- Isochoric process: Constant volume; $W = 0$.
- Isobaric process: Constant pressure; $W = p(V_2 - V_1)$.
- Isothermal process: Constant temperature.

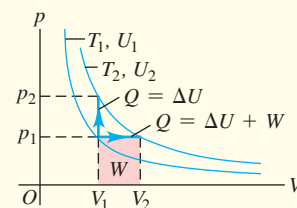


Thermodynamics of ideal gases: The internal energy of an ideal gas depends only on its temperature, not on its pressure or volume. For other substances the internal energy generally depends on both pressure and temperature.

The molar heat capacities C_V and C_p of an ideal gas differ by R , the ideal-gas constant. The dimensionless ratio of heat capacities, C_p/C_V , is denoted by γ . (See Example 19.6.)

$$C_p = C_V + R \quad (19.17)$$

$$\gamma = \frac{C_p}{C_V} \quad (19.18)$$

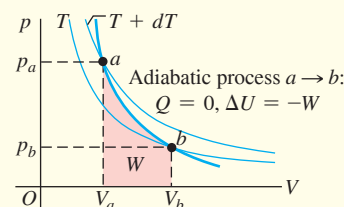


Adiabatic processes in ideal gases: For an adiabatic process for an ideal gas, the quantities $TV^{\gamma-1}$ and pV^γ are constant. The work done by an ideal gas during an adiabatic expansion can be expressed in terms of the initial and final values of temperature, or in terms of the initial and final values of pressure and volume. (See Example 19.7.)

$$W = nC_V(T_1 - T_2) \quad (19.25)$$

$$= \frac{C_V}{R}(p_1 V_1 - p_2 V_2) \quad (19.26)$$

$$= \frac{1}{\gamma - 1}(p_1 V_1 - p_2 V_2)$$





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 19.2, 19.3, 19.4, and 19.5** (Section 19.4) before attempting these problems.

VP19.5.1 Consider the following thermodynamic processes. (a) The internal energy of a quantity of gas increases by 2.50×10^3 J as the gas does 3.25×10^3 J of work on its surroundings. How much heat flows into the gas? (b) Ice enclosed in a container does 2.50×10^4 J of work on its container as 7.00×10^3 J of heat flows out of the ice. What is the internal energy change of the ice? (c) The internal energy of a metal block increases by 4.25×10^3 J as 2.40×10^3 J of heat flows into the block. How much work did the block do on its surroundings?

VP19.5.2 How much work does a quantity of gas (volume V , pressure p) do in each of these processes? (a) V increases from 2.00×10^{-3} m³ to 4.50×10^{-3} m³ while p is a constant 6.20×10^4 Pa. (b) V is a constant 2.00×10^{-3} m³ while p increases from 6.00×10^4 Pa to 9.00×10^4 Pa. (c) V decreases from 6.00×10^{-3} m³ to 3.00×10^{-3} m³ while p is a constant 1.25×10^5 Pa. (d) V increases from 2.00×10^{-3} m³ to 4.50×10^{-3} m³ while p increases from 1.50×10^5 Pa to 5.50×10^5 Pa. The graph of this process on a pV -diagram is a straight line.

VP19.5.3 A quantity of gas is enclosed in a cylinder that has a movable piston. The gas has volume $V = 3.60 \times 10^{-3}$ m³ and pressure $p = 2.00 \times 10^5$ Pa. (a) First V decreases to 2.40×10^{-3} m³ while p remains constant; then V remains constant while p increases to 6.00×10^5 Pa. During this process 1.56×10^3 J of heat flows into the gas. For this process, how much work does the gas do, and what is its internal energy change? (b) The gas returns to the same initial volume and pressure. Now p increases to 6.00×10^5 Pa while V remains constant; then V decreases to 2.40×10^{-3} m³ while p remains constant. For this process, how much work does the gas do, and how much heat flows into the gas?

VP19.5.4 At an atmospheric pressure of 1.01×10^5 Pa, carbon dioxide (CO₂) sublimates at a temperature of -78.5°C . The heat of sublimation is 5.71×10^5 J/kg. You let heat flow into 6.0 kg of solid CO₂ (volume 4.0×10^{-3} m³) at atmospheric pressure and a constant -78.5°C until it has completely sublimed to gas at -78.5°C . The CO₂ gas occupies a volume of 3.4 m³. For this process, calculate (a) how much heat flows into the CO₂, (b) how much work the CO₂ does, and (c) the internal energy change of the CO₂.

Be sure to review **EXAMPLE 19.6** (Section 19.7) before attempting these problems.

VP19.6.1 For neon gas, $C_V = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$. Calculate the change in internal energy of 4.00 mol of neon if its temperature (a) increases from 20.0°C to 40.0°C at constant volume; (b) increases from 15.0°C to

30.0°C at constant pressure; (c) increases from 12.0°C to 24.0°C while both the pressure and volume increase.

VP19.6.2 For nitrogen gas, $C_V = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$. You have 2.10 mol of nitrogen gas at temperature 27.0°C and pressure $p = 1.00 \times 10^5$ Pa. (a) Calculate the initial volume V of the gas. (b) Calculate the final temperature (in $^\circ\text{C}$) of the gas and its internal energy change if (i) p doubles while V remains constant, (ii) V doubles while p remains constant, and (iii) both p and V double.

VP19.6.3 A quantity of argon gas ($C_V = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$) is at pressure $p = 1.20 \times 10^5$ Pa and occupies volume $V = 0.250$ m³. Calculate the change in internal energy of the gas if the final pressure and volume of the gas are (a) $p = 2.40 \times 10^5$ Pa, $V = 0.250$ m³; (b) $p = 1.20 \times 10^5$ Pa, $V = 0.125$ m³; (c) $p = 1.80 \times 10^5$ Pa, $V = 0.600$ m³.

VP19.6.4 A quantity of n moles of oxygen gas ($C_V = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$) is at absolute temperature T . You increase the absolute temperature to $2T$. Find the change in internal energy of the gas, the heat flow into the gas, and the work done by the gas if the process you used to increase the temperature is (a) isochoric, (b) isobaric, or (c) adiabatic.

Be sure to review **EXAMPLE 19.7** (Section 19.8) before attempting these problems.

VP19.7.1 A quantity of argon gas ($C_V = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$) is at pressure 4.00×10^5 Pa and occupies volume 2.00×10^{-3} m³. Then the gas expands adiabatically to a new volume 6.00×10^{-3} m³. Calculate (a) the value of γ for argon, (b) the final pressure of the gas, and (c) the work done by the gas as it expands.

VP19.7.2 A quantity of oxygen gas ($C_V = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$) is at absolute temperature 325 K, is under pressure 1.00×10^5 Pa, and occupies volume 6.50×10^{-3} m³. After you compress this gas adiabatically, its absolute temperature is 855 K. Calculate (a) the value of γ for oxygen, (b) the final volume of the gas, (c) the final pressure of the gas, and (d) the work done by the gas as it is compressed.

VP19.7.3 Initially 5.00 mol of neon gas ($C_V = \frac{3}{2}R$ and $\gamma = \frac{5}{2}$) are at absolute temperature 305 K and occupy volume 4.00×10^{-2} m³. Then the gas expands adiabatically to a new volume of 9.00×10^{-2} m³. Calculate (a) the initial pressure of the gas, (b) the final pressure of the gas, (c) the final temperature of the gas, and (d) the work done by the gas as it expands.

VP19.7.4 You have 1.25 mol of hydrogen gas ($C_V = \frac{5}{2}R$ and $\gamma = \frac{7}{2}$) at absolute temperature 325 K. You allow the gas to expand adiabatically to a final temperature of 195 K. (a) How much work does the gas do while being compressed? (b) What is the ratio of its final volume to its initial volume? (c) What is the ratio of the final gas pressure to the initial gas pressure?

BRIDGING PROBLEM Work Done by a van der Waals Gas

The van der Waals equation of state, an approximate representation of the behavior of gases at high pressure, is given by Eq. (18.7): $[p + (an^2/V^2)](V - nb) = nRT$, where a and b are constants having different values for different gases. (In the special case of $a = b = 0$, this is the ideal-gas equation.) (a) Calculate the work done by a gas with this equation of state in an isothermal expansion from V_1 to V_2 . (b) For ethane gas (C₂H₆), $a = 0.554$ J·m³/mol² and $b = 6.38 \times 10^{-5}$ m³/mol. Calculate the work W done by 1.80 mol of ethane when it expands from 2.00×10^{-3} m³ to 4.00×10^{-3} m³ at

a constant temperature of 300 K. Do the calculation by using (i) the van der Waals equation of state and (ii) the ideal-gas equation of state. (c) For which equation of state is W larger? Why should this be so?

SOLUTION GUIDE

IDENTIFY and SET UP

- Review the discussion of the van der Waals equation of state in Section 18.1. What is the significance of the quantities a and b ?

Continued

- Decide how to find the work done by an expanding gas whose pressure p does not depend on V in the same way as for an ideal gas. (Hint: See Section 19.2.)
- How will you find the work done by an expanding ideal gas?

EXECUTE

- Find the general expression for the work done by a van der Waals gas as it expands from volume V_1 to volume V_2 (Fig. 19.22). (Hint: If you set $a = b = 0$ in your result, it should reduce to the expression for the work done by an expanding ideal gas.)
- Use your result from step 4 to solve part (b) for ethane treated as a van der Waals gas.
- Use the formula you chose in step 3 to solve part (b) for ethane treated as an ideal gas.

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q19.1 For the following processes, is the work done by the system (defined as the expanding or contracting gas) on the environment positive or negative? (a) expansion of the burned gasoline–air mixture in the cylinder of an automobile engine; (b) opening a bottle of champagne; (c) filling a scuba tank with compressed air; (d) partial crumpling of a sealed, empty water bottle as you drive from the mountains down to sea level.

Q19.2 It is not correct to say that an object contains a certain amount of heat, yet an object can transfer heat to another object. How can an object give away something it does not have in the first place?

Q19.3 In which situation must you do more work: inflating a balloon at sea level or inflating the same balloon to the same volume at the summit of Mt. McKinley? Explain in terms of pressure and volume change.

Q19.4 If you are told the initial and final states of a system and the associated change in internal energy, can you determine whether the internal energy change was due to work or to heat transfer? Explain.

Q19.5 Discuss the application of the first law of thermodynamics to a mountaineer who eats food, gets warm and perspires a lot during a climb, and does a lot of mechanical work in raising herself to the summit. The mountaineer also gets warm during the descent. Is the source of this energy the same as the source during the ascent?

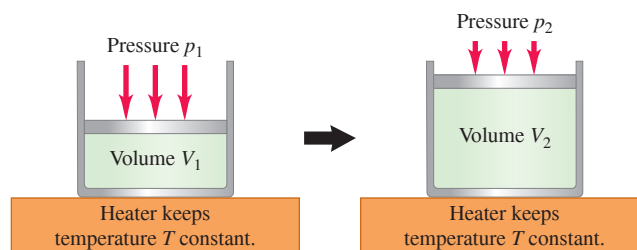
Q19.6 When ice melts at 0°C , its volume decreases. Is the internal energy change greater than, less than, or equal to the heat added? How can you tell?

Q19.7 You hold an inflated balloon over a hot-air vent in your house and watch it slowly expand. You then remove it and let it cool back to room temperature. During the expansion, which was larger: the heat added to the balloon or the work done by the air inside it? Explain. (Assume that air is an ideal gas.) Once the balloon has returned to room temperature, how does the net heat gained or lost by the air inside it compare to the net work done on or by the surrounding air?

Q19.8 You bake chocolate chip cookies and put them, still warm, in a container with a loose (not airtight) lid. What kind of process does the air inside the container undergo as the cookies gradually cool to room temperature (isothermal, isochoric, adiabatic, isobaric, or some combination)? Explain.

Q19.9 Imagine a gas made up entirely of negatively charged electrons. Like charges repel, so the electrons exert repulsive forces on each other. Would you expect that the temperature of such a gas would rise, fall, or stay the same in a free expansion? Why?

Figure 19.22 A gas undergoes an isothermal expansion.

**EVALUATE**

- Is the difference between W for the two equations of state large enough to be significant?
- Does the term with a in the van der Waals equation of state increase or decrease the amount of work done? What about the term with b ? Which one is more important for the ethane in this problem?

Q19.10 In an adiabatic process for an ideal gas, the pressure decreases. In this process does the internal energy of the gas increase or decrease? Explain.

Q19.11 When you blow on the back of your hand with your mouth wide open, your breath feels warm. But if you partially close your mouth to form an “o” and then blow on your hand, your breath feels cool. Why?

Q19.12 An ideal gas expands while the pressure is kept constant. During this process, does heat flow into the gas or out of the gas? Justify your answer.

Q19.13 A liquid is irregularly stirred in a well-insulated container and thereby undergoes a rise in temperature. Regard the liquid as the system. Has heat been transferred? How can you tell? Has work been done? How can you tell? Why is it important that the stirring is irregular? What is the sign of ΔU ? How can you tell?

Q19.14 When you use a hand pump to inflate the tires of your bicycle, the pump gets warm after a while. Why? What happens to the temperature of the air in the pump as you compress it? Why does this happen? When you raise the pump handle to draw outside air into the pump, what happens to the temperature of the air taken in? Again, why does this happen?

Q19.15 In the carburetor of an aircraft or automobile engine, air flows through a relatively small aperture and then expands. In cool, foggy weather, ice sometimes forms in this aperture even though the outside air temperature is above freezing. Why?

Q19.16 On a sunny day, large “bubbles” of air form on the sunwarmed earth, gradually expand, and finally break free to rise through the atmosphere. Soaring birds and glider pilots are fond of using these “thermals” to gain altitude easily. This expansion is essentially an adiabatic process. Why?

Q19.17 The prevailing winds on the Hawaiian island of Kauai blow from the northeast. The winds cool as they go up the slope of Mt. Waialeale (elevation 1523 m), causing water vapor to condense and rain to fall. There is much more precipitation at the summit than at the base of the mountain. In fact, Mt. Waialeale is the rainiest spot on earth, averaging 11.7 m of rainfall a year. But what makes the winds cool?

Q19.18 Applying the same considerations as in Question Q19.17, explain why the island of Niihau, a few kilometers to the southwest of Kauai, is almost a desert and farms there need to be irrigated.

Q19.19 In a constant-volume process, $dU = nC_V dT$. But in a constant-pressure process, it is *not* true that $dU = nC_p dT$. Why not?

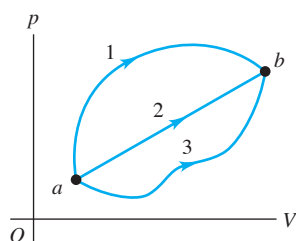
Q19.20 When a gas surrounded by air is compressed adiabatically, its temperature rises even though there is no heat input to the gas. Where does the energy come from to raise the temperature?

Q19.21 When a gas expands adiabatically, it does work on its surroundings. But if there is no heat input to the gas, where does the energy come from to do the work?

Q19.22 The gas used in separating the two uranium isotopes ^{235}U and ^{238}U has the formula UF_6 . If you added heat at equal rates to a mole of UF_6 gas and a mole of H_2 gas, which one's temperature would you expect to rise faster? Explain.

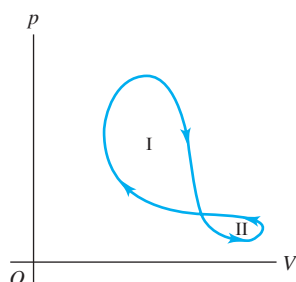
Q19.23 A system is taken from state a to state b along the three paths shown in **Fig. Q19.23**. (a) Along which path is the work done by the system the greatest? The least? (b) If $U_b > U_a$, along which path is the absolute value of the heat transfer, $|Q|$, the greatest? For this path, is heat absorbed or liberated by the system? Explain.

Figure Q19.23



Q19.24 A thermodynamic system undergoes a cyclic process as shown in **Fig. Q19.24**. The cycle consists of two closed loops: I and II. (a) Over one complete cycle, does the system do positive or negative work? (b) In each loop, is the net work done by the system positive or negative? (c) Over one complete cycle, does heat flow into or out of the system? (d) In each loop, does heat flow into or out of the system? Explain.

Figure Q19.24



EXERCISES

Section 19.2 Work Done During Volume Changes and

Section 19.3 Paths Between Thermodynamic States

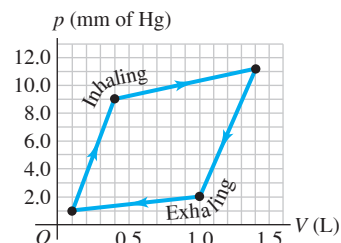
19.1 • Two moles of an ideal gas are heated at constant pressure from $T = 27^\circ\text{C}$ to $T = 107^\circ\text{C}$. (a) Draw a pV -diagram for this process. (b) Calculate the work done by the gas.

19.2 • Six moles of an ideal gas are in a cylinder fitted at one end with a movable piston. The initial temperature of the gas is 27.0°C and the pressure is constant. As part of a machine design project, calculate the final temperature of the gas after it has done $2.40 \times 10^3 \text{ J}$ of work.

19.3 • **CALC** Two moles of an ideal gas are compressed in a cylinder at a constant temperature of 65.0°C until the original pressure has tripled. (a) Sketch a pV -diagram for this process. (b) Calculate the amount of work done.

19.4 • **BIO Work Done by the Lungs.** The graph in **Fig. E19.4** shows a pV -diagram of the air in a human lung when a person is inhaling and then exhaling a deep breath. Such graphs, obtained in clinical practice, are normally somewhat curved, but we have modeled one as a set of straight lines of the same general shape. (*Important:* The pressure shown is the *gauge* pressure, *not* the absolute pressure.) (a) How many joules of *net* work does this person's lung do during one complete breath? (b) The process illustrated here is somewhat different from those we have been studying, because the pressure change is due to changes in the amount of gas in the lung, not to temperature changes. (Think of your own breathing. Your lungs do not expand because they've gotten hot.) If the temperature of the air in the lung remains a reasonable 20°C , what is the maximum number of moles in this person's lung during a breath?

Figure E19.4



19.5 • **CALC** During the time 0.305 mol of an ideal gas undergoes an isothermal compression at 22.0°C , 392 J of work is done on it by the surroundings. (a) If the final pressure is 1.76 atm, what was the initial pressure? (b) Sketch a pV -diagram for the process.

19.6 • A gas undergoes two processes. In the first, the volume remains constant at 0.200 m^3 and the pressure increases from $2.00 \times 10^5 \text{ Pa}$ to $5.00 \times 10^5 \text{ Pa}$. The second process is a compression to a volume of 0.120 m^3 at a constant pressure of $5.00 \times 10^5 \text{ Pa}$. (a) In a pV -diagram, show both processes. (b) Find the total work done by the gas during both processes.

19.7 • **Work Done in a Cyclic Process.** (a) In **Fig. 19.7a**, consider the closed loop $1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 1$. This is a *cyclic* process in which the initial and final states are the same. Find the total work done by the system in this cyclic process, and show that it is equal to the area enclosed by the loop. (b) How is the work done for the process in part (a) related to the work done if the loop is traversed in the opposite direction, $1 \rightarrow 4 \rightarrow 2 \rightarrow 3 \rightarrow 1$? Explain.

Section 19.4 Internal Energy and the First Law of Thermodynamics

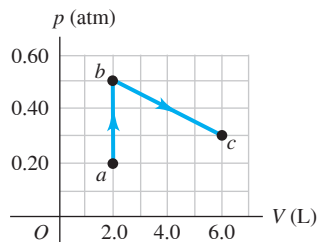
19.8 • **CALC** An ideal gas undergoes a process during which the pressure is kept directly proportional to the volume, so that $p = \alpha V$, where α is a positive constant. If the volume changes from V_1 to V_2 , how much work is done by the gas? Express your answer in terms of V_1 , V_2 , and α .

19.9 • A gas in a cylinder expands from a volume of 0.110 m^3 to 0.320 m^3 . Heat flows into the gas just rapidly enough to keep the pressure constant at $1.65 \times 10^5 \text{ Pa}$ during the expansion. The total heat added is $1.15 \times 10^5 \text{ J}$. (a) Find the work done by the gas. (b) Find the change in internal energy of the gas. (c) Does it matter whether the gas is ideal? Why or why not?

19.10 • Five moles of an ideal monatomic gas with an initial temperature of 127°C expand and, in the process, absorb 1500 J of heat and do 2100 J of work. What is the final temperature of the gas?

19.11 •• The process abc shown in the pV -diagram in **Fig. E19.11** involves 0.0175 mol of an ideal gas. (a) What was the lowest temperature the gas reached in this process? Where did it occur? (b) How much work was done by or on the gas from a to b ? From b to c ? (c) If 215 J of heat was put into the gas during abc , how many of those joules went into internal energy?

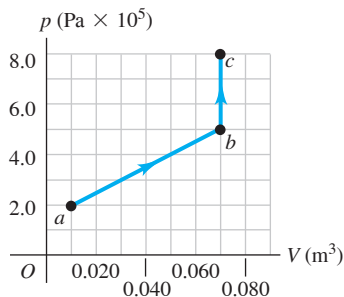
Figure E19.11



19.12 • A gas in a cylinder is held at a constant pressure of 1.80×10^5 Pa and is cooled and compressed from 1.70 m^3 to 1.20 m^3 . The internal energy of the gas decreases by 1.40×10^5 J. (a) Find the work done by the gas. (b) Find the absolute value of the heat flow, $|Q|$, into or out of the gas, and state the direction of the heat flow. (c) Does it matter whether the gas is ideal? Why or why not?

19.13 •• The pV -diagram in **Fig. E19.13** shows a process abc involving 0.450 mol of an ideal gas. (a) What was the temperature of this gas at points a , b , and c ? (b) How much work was done by or on the gas in this process? (c) How much heat had to be added during the process to increase the internal energy of the gas by 15,000 J?

Figure E19.13



19.14 • Boiling Water at High Pressure. When water is boiled at a pressure of 2.00 atm, the heat of vaporization is 2.20×10^6 J/kg and the boiling point is 120°C . At this pressure, 1.00 kg of water has a volume of $1.00 \times 10^{-3} \text{ m}^3$, and 1.00 kg of steam has a volume of 0.824 m^3 . (a) Compute the work done when 1.00 kg of steam is formed at this temperature. (b) Compute the increase in internal energy of the water.

19.15 • A thermodynamic system undergoes processes in which $|Q| = 100 \text{ J}$ and $|W| = 300 \text{ J}$. Find Q and W , including whether the quantity is positive or negative, if the change in internal energy is (a) +400 J, (b) +200 J, (c) -200 J, and (d) -400 J.

Section 19.5 Kinds of Thermodynamic Processes,

Section 19.6 Internal Energy of an Ideal Gas, and

Section 19.7 Heat Capacities of an Ideal Gas

19.16 • During an isothermal compression of an ideal gas, 410 J of heat must be removed from the gas to maintain constant temperature. How much work is done by the gas during the process?

19.17 • A cylinder contains 0.250 mol of carbon dioxide (CO_2) gas at a temperature of 27.0°C . The cylinder is provided with a frictionless piston, which maintains a constant pressure of 1.00 atm on the gas. The gas is heated until its temperature increases to 127.0°C . Assume that the CO_2 may be treated as an ideal gas. (a) Draw a pV -diagram for this process. (b) How much work is done by the gas in this process? (c) On what is this work done? (d) What is the change in internal energy of the gas? (e) How much heat was supplied to the gas? (f) How much work would have been done if the pressure had been 0.50 atm?

19.18 • A cylinder contains 0.0100 mol of helium at $T = 27.0^\circ\text{C}$. (a) How much heat is needed to raise the temperature to 67.0°C while keeping the volume constant? Draw a pV -diagram for this process. (b) If instead the pressure of the helium is kept constant, how much heat is needed to raise the temperature from 27.0°C to 67.0°C ? Draw a pV -diagram for this process. (c) What accounts for the difference between your answers to parts (a) and (b)? In which case is more heat required? What becomes of the additional heat? (d) If the gas is ideal, what is the change in its internal energy in part (a)? In part (b)? How do the two answers compare? Why?

19.19 • In an experiment to simulate conditions inside an automobile engine, 0.185 mol of air at 780 K and 3.00×10^6 Pa is contained in a cylinder of volume 40.0 cm^3 . Then 645 J of heat is transferred to the cylinder. (a) If the volume of the cylinder is constant while the heat is added, what is the final temperature of the air? Assume that the air is essentially nitrogen gas, and use the data in Table 19.1 even though the pressure is not low. Draw a pV -diagram for this process. (b) If instead the volume of the cylinder is allowed to increase while the pressure remains constant, repeat part (a).

19.20 •• A quantity of 5.00 mol of a monatomic ideal gas has an initial pressure of $p_a = 1.00 \times 10^5$ Pa and an initial volume of $V_a = 0.0200 \text{ m}^3$. Two paths are used to take the gas to a final state for which $p_b = 1.80 \times 10^5$ Pa and $V_b = 0.0280 \text{ m}^3$. (a) Along path 1 the pressure is increased to $p_a = 1.80 \times 10^5$ Pa while the volume is kept constant with a value of 0.0200 m^3 . Then the volume is increased to 0.0280 m^3 while the pressure is kept at a constant 1.80×10^5 Pa. Calculate the total Q and total W for this path. (b) Along path 2 the volume is increased to 0.0280 m^3 while the pressure remains 1.00×10^5 Pa and then the pressure is increased to 1.80×10^5 Pa while the volume remains constant at 0.0280 m^3 . Calculate the total Q and total W for this path. (c) Calculate the quantity $Q - W$ for each path. Your results should show that for these processes Q and W are path dependent but $Q - W$ is path independent.

19.21 • Heat Q flows into a monatomic ideal gas, and the volume increases while the pressure is kept constant. What fraction of the heat energy is used to do the expansion work of the gas?

19.22 • Three moles of an ideal monatomic gas expands at a constant pressure of 2.50 atm; the volume of the gas changes from $3.20 \times 10^{-2} \text{ m}^3$ to $4.50 \times 10^{-2} \text{ m}^3$. Calculate (a) the initial and final temperatures of the gas; (b) the amount of work the gas does in expanding; (c) the amount of heat added to the gas; (d) the change in internal energy of the gas.

19.23 • An experimenter adds 970 J of heat to 1.75 mol of an ideal gas to heat it from 10.0°C to 25.0°C at constant pressure. The gas does +223 J of work during the expansion. (a) Calculate the change in internal energy of the gas. (b) Calculate γ for the gas.

19.24 • A quantity of 2.00 mol of a monatomic ideal gas undergoes a compression during which the volume decreases from 0.0800 m^3 to 0.0500 m^3 while the pressure stays constant at a value of 1.80×10^4 Pa. (a) What is the work W ? Is work done by the gas or on the gas? (b) What is the heat flow Q ? Does heat enter or leave the gas? (c) What is the internal energy change for the gas? Does the internal energy of the gas increase or decrease?

19.25 • CALC The temperature of 0.150 mol of an ideal gas is held constant at 77.0°C while its volume is reduced to 25.0% of its initial volume. The initial pressure of the gas is 1.25 atm. (a) Determine the work done by the gas. (b) What is the change in its internal energy? (c) Does the gas exchange heat with its surroundings? If so, how much? Does the gas absorb or liberate heat?

19.26 •• For an ideal gas C_V and C_p are different because of the work W associated with the volume change for a constant-pressure process. To explore the difference between C_V and C_p for a liquid or a solid, consider the process in which 5.00 mol of ethanol is warmed from 10.0°C to 60.0°C while the applied pressure remains a constant 1.00 atm. The molar mass of ethanol is $M = 46.1\text{ g/mol}$. (a) What is the mass of 5.00 mol of ethanol? (b) How much heat Q enters the ethanol? (c) How much work W is done by the ethanol because of its thermal expansion? (d) What is the ratio W/Q ?

Section 19.8 Adiabatic Processes for an Ideal Gas

19.27 • A monatomic ideal gas that is initially at $1.50 \times 10^5\text{ Pa}$ and has a volume of 0.0800 m^3 is compressed adiabatically to a volume of 0.0400 m^3 . (a) What is the final pressure? (b) How much work is done by the gas? (c) What is the ratio of the final temperature of the gas to its initial temperature? Is the gas heated or cooled by this compression?

19.28 •• Five moles of monatomic ideal gas have initial pressure $2.50 \times 10^3\text{ Pa}$ and initial volume 2.10 m^3 . While undergoing an adiabatic expansion, the gas does 1480 J of work. What is the final pressure of the gas after the expansion?

19.29 • During an adiabatic expansion the temperature of 0.450 mol of argon (Ar) drops from 66.0°C to 10.0°C . The argon may be treated as an ideal gas. (a) Draw a pV -diagram for this process. (b) How much work does the gas do? (c) What is the change in internal energy of the gas?

19.30 •• A player bounces a basketball on the floor, compressing it to 80.0% of its original volume. The air (assume it is essentially N_2 gas) inside the ball is originally at 20.0°C and 2.00 atm. The ball's inside diameter is 23.9 cm. (a) What temperature does the air in the ball reach at its maximum compression? Assume the compression is adiabatic and treat the gas as ideal. (b) By how much does the internal energy of the air change between the ball's original state and its maximum compression?

19.31 •• On a warm summer day, a large mass of air (atmospheric pressure $1.01 \times 10^5\text{ Pa}$) is heated by the ground to 26.0°C and then begins to rise through the cooler surrounding air. (This can be treated approximately as an adiabatic process; why?) Calculate the temperature of the air mass when it has risen to a level at which atmospheric pressure is only $0.850 \times 10^5\text{ Pa}$. Assume that air is an ideal gas, with $\gamma = 1.40$. (This rate of cooling for dry, rising air, corresponding to roughly 1°C per 100 m of altitude, is called the *dry adiabatic lapse rate*.)

19.32 • A cylinder contains 0.100 mol of an ideal monatomic gas. Initially the gas is at $1.00 \times 10^5\text{ Pa}$ and occupies a volume of $2.50 \times 10^{-3}\text{ m}^3$. (a) Find the initial temperature of the gas in kelvins. (b) If the gas is allowed to expand to twice the initial volume, find the final temperature (in kelvins) and pressure of the gas if the expansion is (i) isothermal; (ii) isobaric; (iii) adiabatic.

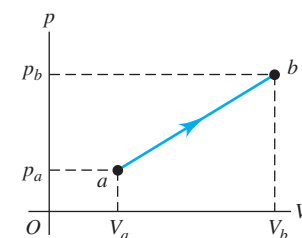
19.33 •• Helium gas undergoes an adiabatic process in which the Kelvin temperature doubles. By what factor does the pressure change?

19.34 • For a diatomic ideal gas, 3.01×10^{20} molecules undergo an adiabatic process in which the temperature increases from 25.0°C to 60.0°C . What is the change in the internal energy of the gas?

PROBLEMS

19.35 •• A quantity of air is taken from state a to state b along a path that is a straight line in the pV -diagram (Fig. P19.35). (a) In this process, does the temperature of the gas increase, decrease, or stay the same? Explain. (b) If $V_a = 0.0700\text{ m}^3$, $V_b = 0.1100\text{ m}^3$, $p_a = 1.00 \times 10^5\text{ Pa}$, and $p_b = 1.40 \times 10^5\text{ Pa}$, what is the work W done by the gas in this process? Assume that the gas may be treated as ideal.

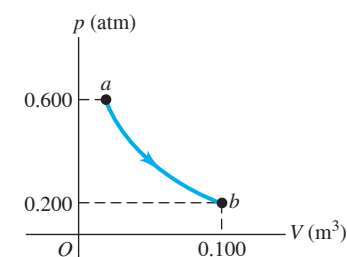
Figure P19.35



19.36 •• When water boils in a pot with a loose lid, the increased pressure forces the lid upward, releasing steam and reducing the pressure in the pot so that the lid drops back down. If the water continues to boil, the lid can rattle upward and downward. Consider a 1 L pot half filled with boiling water. In this case there is half a liter, or $\frac{1}{11.2}$ mol, of air and steam above the water. (a) Estimate the mass of the lid. (b) When the water is rapidly boiling, estimate the height to which the lid jumps during a steam-release event. (c) By multiplying the weight of the lid by the height of the jump, estimate the work done by the steam as the lid moves upward. (d) Treat the process as adiabatic and use the first law of thermodynamics to estimate the drop in the temperature of the steam as it pushes the lid upward.

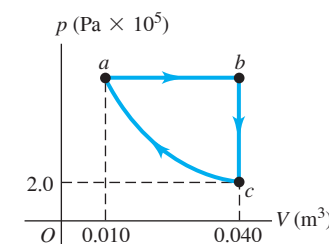
19.37 •• Figure P19.37 shows the pV -diagram for a process in which the temperature of the ideal gas remains constant at 85°C . (a) How many moles of gas are involved? (b) What volume does this gas occupy at a ? (c) How much work was done by or on the gas from a to b ? (d) By how much did the internal energy of the gas change during this process?

Figure P19.37



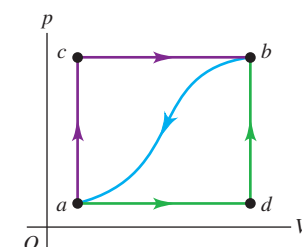
19.38 •• The graph in Fig. P19.38 shows a pV -diagram for 3.25 mol of ideal helium (He) gas. Part ca of this process is isothermal. (a) Find the pressure of the He at point a . (b) Find the temperature of the He at points a , b , and c . (c) How much heat entered or left the He during segments ab , bc , and ca ? In each segment, did the heat enter or leave? (d) By how much did the internal energy of the He change from a to b , from b to c , and from c to a ? Indicate whether this energy increased or decreased.

Figure P19.38



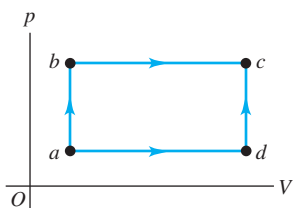
19.39 •• When a system is taken from state a to state b in Fig. P19.39 along path acb , 90.0 J of heat flows into the system and 60.0 J of work is done by the system. (a) How much heat flows into the system along path adb if the work done by the system is 15.0 J? (b) When the system is returned from b to a along the curved path, the absolute value of the work done by the system is 35.0 J. Does the system absorb or liberate heat? How much heat? (c) If $U_a = 0$ and $U_d = 8.0\text{ J}$, find the heat absorbed in the processes ad and db .

Figure P19.39



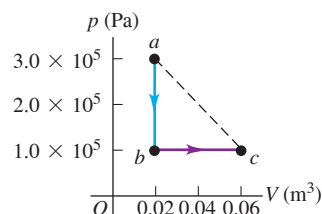
19.40 • A thermodynamic system is taken from state a to state c in **Fig. P19.40** along either path abc or path adc . Along path abc , the work W done by the system is 450 J. Along path adc , W is 120 J. The internal energies of each of the four states shown in the figure are $U_a = 150$ J, $U_b = 240$ J, $U_c = 680$ J, and $U_d = 330$ J. Calculate the heat flow Q for each of the four processes ab , bc , ad , and dc . In each process, does the system absorb or liberate heat?

Figure P19.40



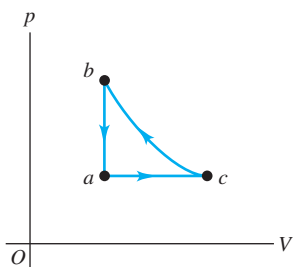
19.41 • A volume of air (assumed to be an ideal gas) is first cooled without changing its volume and then expanded without changing its pressure, as shown by path abc in **Fig. P19.41**. (a) How does the final temperature of the gas compare with its initial temperature? (b) How much heat does the air exchange with its surroundings during process abc ? Does the air absorb heat or release heat during this process? Explain. (c) If the air instead expands from state a to state c by the straight-line path shown, how much heat does it exchange with its surroundings?

Figure P19.41



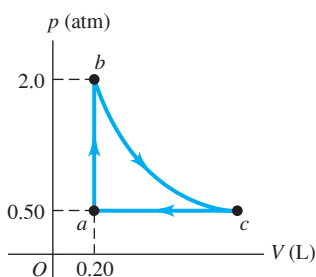
19.42 •• Three moles of an ideal gas are taken around cycle acb shown in **Fig. P19.42**. For this gas, $C_p = 29.1$ J/mol·K. Process ac is at constant pressure, process ba is at constant volume, and process cb is adiabatic. The temperatures of the gas in states a , c , and b are $T_a = 300$ K, $T_c = 492$ K, and $T_b = 600$ K. Calculate the total work W for the cycle.

Figure P19.42



19.43 •• **Figure P19.43** shows a pV -diagram for 0.0040 mol of ideal H_2 gas. The temperature of the gas does not change during segment bc . (a) What volume does this gas occupy at point c ? (b) Find the temperature of the gas at points a , b , and c . (c) How much heat went into or out of the gas during segments ab , ca , and bc ? Indicate whether the heat has gone into or out of the gas.

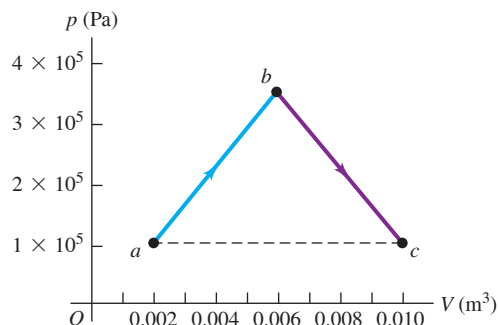
Figure P19.43



(d) Find the change in the internal energy of this hydrogen during segments ab , bc , and ca . Indicate whether the internal energy increased or decreased during each segment.

19.44 •• (a) One-third of a mole of He gas is taken along the path abc shown in **Fig. P19.44**. Assume that the gas may be treated as ideal. How much heat is transferred into or out of the gas? (b) If the gas instead went directly from state a to state c along the horizontal dashed line in **Fig. P19.44**, how much heat would be transferred into or out of the gas? (c) How does Q in part (b) compare with Q in part (a)? Explain.

Figure P19.44



19.45 •• DATA You place a quantity of gas into a metal cylinder that has a movable piston at one end. No gas leaks out of the cylinder as the piston moves. The external force applied to the piston can be varied to change the gas pressure as you move the piston to change the volume of the gas. A pressure gauge attached to the interior wall of the cylinder measures the gas pressure, and you can calculate the volume of the gas from a measurement of the piston's position in the cylinder.

You start with a pressure of 1.0 atm and a gas volume of 3.0 L. Holding the pressure constant, you increase the volume to 5.0 L. Then, keeping the volume constant at 5.0 L, you increase the pressure to 3.0 atm. Next you decrease the pressure linearly as a function of volume until the volume is 3.0 L and the pressure is 2.0 atm. Finally, you keep the volume constant at 3.0 L and decrease the pressure to 1.0 atm, returning the gas to its initial pressure and volume. The walls of the cylinder are good conductors of heat, and you provide the required heat sources and heat sinks so that the necessary heat flows can occur. At these relatively high pressures, you suspect that the ideal-gas equation will not apply with much accuracy. You don't know what gas is in the cylinder or whether it is monatomic, diatomic, or polyatomic. (a) Plot the cycle in the pV -plane. (b) What is the net heat flow for the gas during this cycle? Is there net heat flow into or out of the gas?

19.46 •• Nitrogen gas in an expandable container is cooled from 50.0°C to 10.0°C with the pressure held constant at 3.00×10^5 Pa. The total heat liberated by the gas is 2.50×10^4 J. Assume that the gas may be treated as ideal. Find (a) the number of moles of gas; (b) the change in internal energy of the gas; (c) the work done by the gas. (d) How much heat would be liberated by the gas for the same temperature change if the volume were constant?

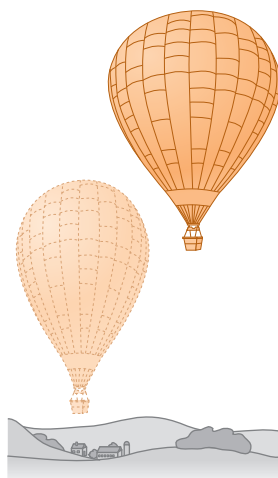
19.47 • CALC A cylinder with a frictionless, movable piston like that shown in **Fig. 19.5** contains a quantity of helium gas. Initially the gas is at 1.00×10^5 Pa and 300 K and occupies a volume of 1.50 L. The gas then undergoes two processes. In the first, the gas is heated and the piston is allowed to move to keep the temperature at 300 K. This continues until the pressure reaches 2.50×10^4 Pa. In the second process, the gas is compressed at constant pressure until it returns to its original volume of 1.50 L. Assume that the gas may be treated as ideal. (a) In a pV -diagram, show both processes. (b) Find the volume of the gas at the end of the first process, and the pressure and temperature at the end of the second process. (c) Find the total work done by the gas during both processes. (d) What would you have to do to the gas to return it to its original pressure and temperature?

19.48 ••• CP A steel cargo drum has a height of 880 mm and a diameter of 610 mm. With its top removed it has a mass of 17.3 kg. The drum is turned upside down at the surface of the North Atlantic and is pulled downward into the ocean by a robotic submarine. On this day the surface temperature is 23.0°C and the surface air pressure is $p_0 = 101 \text{ kPa}$. The water temperature decreases linearly with depth to 3.0°C at 1000 m below the surface. As the drum moves downward in the ocean, the air inside the drum is compressed, reducing the upward buoyant force. (a) At what depth y_{neutral} is the barrel neutrally buoyant? (*Hint:* The pressure in the drum is equal to the sea pressure, which at depth y is $p_0 + \rho gy$, where $\rho = 1025 \text{ kg/m}^3$ is the density of seawater. The temperature at depth can be determined using the information above. Together with the ideal-gas law, you can derive a formula for the volume of air at depth y , and therefore a formula for the upward buoyant force as a function of depth.) (b) What is the volume of the air in the drum at depth y_{neutral} ?

19.49 ••• Chinook. During certain seasons strong winds called chinooks blow from the west across the eastern slopes of the Rockies and downhill into Denver and nearby areas. Although the mountains are cool, the wind in Denver is very hot; within a few minutes after the chinook wind arrives, the temperature can climb 20°C (“chinook” refers to a Native American people of the Pacific Northwest). Similar winds occur in the Alps (called foehns) and in southern California (called Santa Anas). (a) Explain why the temperature of the chinook wind rises as it descends the slopes. Why is it important that the wind be fast moving? (b) Suppose a strong wind is blowing toward Denver (elevation 1630 m) from Grays Peak (80 km west of Denver, at an elevation of 4350 m), where the air pressure is $5.60 \times 10^4 \text{ Pa}$ and the air temperature is -15.0°C . The temperature and pressure in Denver before the wind arrives are 2.0°C and $8.12 \times 10^4 \text{ Pa}$. By how many Celsius degrees will the temperature in Denver rise when the chinook arrives?

19.50 ••• High-Altitude Research. Figure P19.50

A large research balloon containing $2.00 \times 10^3 \text{ m}^3$ of helium gas at 1.00 atm and a temperature of 15.0°C rises rapidly from ground level to an altitude at which the atmospheric pressure is only 0.900 atm (**Fig. P19.50**). Assume the helium behaves like an ideal gas and the balloon’s ascent is too rapid to permit much heat exchange with the surrounding air. (a) Calculate the volume of the gas at the higher altitude. (b) Calculate the temperature of the gas at the higher altitude. (c) What is the change in internal energy of the helium as the balloon rises to the higher altitude?



19.51 ••• An air pump has a cylinder 0.250 m long with a movable piston. The pump is used to compress air from the atmosphere (at absolute pressure $1.01 \times 10^5 \text{ Pa}$) into a very large tank at $3.80 \times 10^5 \text{ Pa}$ gauge pressure. (For air, $C_V = 20.8 \text{ J/mol} \cdot \text{K}$.) (a) The piston begins the compression stroke at the open end of the cylinder. How far down the length of the cylinder has the piston moved when air first begins to flow from the cylinder into the tank? Assume that the compression is adiabatic. (b) If the air is taken into the pump at 27.0°C , what is the temperature of the compressed air? (c) How much work does the pump do in putting 20.0 mol of air into the tank?

19.52 •• A certain ideal gas has molar heat capacity at constant volume C_V . A sample of this gas initially occupies a volume V_0 at pressure p_0 and absolute temperature T_0 . The gas expands isobarically to a volume $2V_0$ and then expands further adiabatically to a final volume $4V_0$. (a) Draw a pV -diagram for this sequence of processes. (b) Compute the total work done by the gas for this sequence of processes. (c) Find the

final temperature of the gas. (d) Find the absolute value of the total heat flow $|Q|$ into or out of the gas for this sequence of processes, and state the direction of heat flow.

19.53 • A monatomic ideal gas expands slowly to twice its original volume, doing 450 J of work in the process. Find the heat added to the gas and the change in internal energy of the gas if the process is (a) isothermal; (b) adiabatic; (c) isobaric.

19.54 •• CALC A cylinder with a piston contains 0.250 mol of oxygen at $2.40 \times 10^5 \text{ Pa}$ and 355 K. The oxygen may be treated as an ideal gas. The gas first expands isobarically to twice its original volume. It is then compressed isothermally back to its original volume, and finally it is cooled isochorically to its original pressure. (a) Show the series of processes on a pV -diagram. Compute (b) the temperature during the isothermal compression; (c) the maximum pressure; (d) the total work done by the piston on the gas during the series of processes.

19.55 • Use the conditions and processes of Problem 19.54 to compute (a) the work done by the gas, the heat added to it, and its internal energy change during the initial expansion; (b) the work done, the heat added, and the internal energy change during the final cooling; (c) the internal energy change during the isothermal compression.

19.56 •• CALC A cylinder with a piston contains 0.150 mol of nitrogen at $1.80 \times 10^5 \text{ Pa}$ and 300 K. The nitrogen may be treated as an ideal gas. The gas is first compressed isobarically to half its original volume. It then expands adiabatically back to its original volume, and finally it is heated isochorically to its original pressure. (a) Show the series of processes in a pV -diagram. (b) Compute the temperatures at the beginning and end of the adiabatic expansion. (c) Compute the minimum pressure.

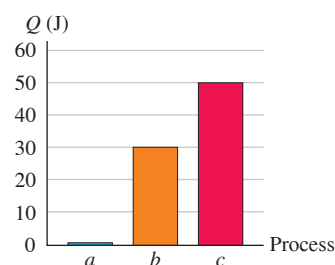
19.57 • Use the conditions and processes of Problem 19.56 to compute (a) the work done by the gas, the heat added to it, and its internal energy change during the initial compression; (b) the work done by the gas, the heat added to it, and its internal energy change during the adiabatic expansion; (c) the work done, the heat added, and the internal energy change during the final heating.

19.58 • Comparing Thermodynamic Processes. In a cylinder, 1.20 mol of an ideal monatomic gas, initially at $3.60 \times 10^5 \text{ Pa}$ and 300 K, expands until its volume triples. Compute the work done by the gas if the expansion is (a) isothermal; (b) adiabatic; (c) isobaric. (d) Show each process in a pV -diagram. In which case is the absolute value of the work done by the gas greatest? Least? (e) In which case is the absolute value of the heat transfer greatest? Least? (f) In which case is the absolute value of the change in internal energy of the gas greatest? Least?

19.59 •• DATA Figure P19.59

You have recorded measurements of the heat flow Q into 0.300 mol of a gas that starts at $T_1 = 20.0^{\circ}\text{C}$ and ends at a temperature T_2 . You measured Q for three processes: one isobaric, one isochoric, and one adiabatic. In each case, T_2 was the same. **Figure P19.59** summarizes your results. But you lost a page from your lab notebook and don’t have a record of the value

of T_2 ; you also don’t know which process was isobaric, isochoric, or adiabatic. Each process was done at a sufficiently low pressure for the gas to be treated as ideal. (a) Identify each process a , b , or c as isobaric, isochoric, or adiabatic. (b) What is the value of T_2 ? (c) How much work is done by the gas in each process? (d) For which process is the magnitude of the volume change the greatest? (e) For each process, does the volume of the gas increase, decrease, or stay the same?



19.60 •• DATA You compress a gas in an insulated cylinder—no heat flows into or out of the gas. The gas pressure is fairly low, so treating the gas as ideal is a good approximation. When you measure the pressure as a function of the volume of the gas, you obtain these results:

V (L)	2.50	2.02	1.48	1.01	0.50
p (atm)	0.101	0.139	0.202	0.361	0.952

(a) Graph $\log(p)$ versus $\log(V)$, with p in Pa and V in m^3 . Explain why the data points fall close to a straight line. (b) Use your graph to calculate γ for the gas. Is the gas monatomic, diatomic, or polyatomic? (c) When $p = 0.101$ atm and $V = 2.50$ L, the temperature is 22.0°C . Apply the ideal-gas equation and calculate the temperature for each of the other pairs of p and V values. In this compression, does the temperature of the gas increase, decrease, or stay constant?

19.61 ••• A pneumatic shock absorber consists of a cylinder with a radius of 1.50 cm and a length that varies from a maximum of 30.0 cm to a minimum of $30.0\text{ cm}/v$, where v is the compression ratio. For ease of manufacturing, v must be an integer. When the cylinder is fully extended and cooled, the air inside has an ambient pressure of 101 kPa and ambient temperature of 30.0°C . A quick adiabatic compression heats the air. The cylinder then expands isothermally until it reaches its maximum length. At that time the length remains fixed and the cylinder cools isochorically, returning to the ambient temperature. (a) How much work is done by the air during a full cycle, in terms of v ? (b) If the air temperature cannot exceed 400°C , what is the maximum integer value of v ? (c) If at least 25.0 J of work must be done by the gas in a full-compression cycle, what is the minimum allowable integer value of v ? (d) What is the only permissible value of v ? (e) How much heat leaves the air in the cylinder during the isochoric process if v has the value calculated in (d)?

CHALLENGE PROBLEM

19.62 •• Engine Turbochargers and Intercoolers. The power output of an automobile engine is directly proportional to the mass of air that can be forced into the volume of the engine's cylinders to react chemically with gasoline. Many cars have a *turbocharger*, which compresses the air before it enters the engine, giving a greater mass of air per volume. This rapid, essentially adiabatic compression also heats the air. To compress it further, the air then passes through an *intercooler* in which the air exchanges heat with its surroundings at essentially constant pressure. The air is then drawn into the cylinders. In a typical installation, air is taken into the turbocharger at atmospheric pressure (1.01×10^5 Pa), density $\rho = 1.23\text{ kg/m}^3$, and temperature 15.0°C . It is compressed adiabatically to 1.45×10^5 Pa. In the intercooler, the air is cooled to the original temperature of 15.0°C at a constant pressure of 1.45×10^5 Pa. (a) Draw a

pV -diagram for this sequence of processes. (b) If the volume of one of the engine's cylinders is 575 cm^3 , what mass of air exiting from the intercooler will fill the cylinder at 1.45×10^5 Pa? Compared to the power output of an engine that takes in air at 1.01×10^5 Pa at 15.0°C , what percentage increase in power is obtained by using the turbocharger and intercooler? (c) If the intercooler is not used, what mass of air exiting from the turbocharger will fill the cylinder at 1.45×10^5 Pa? Compared to the power output of an engine that takes in air at 1.01×10^5 Pa at 15.0°C , what percentage increase in power is obtained by using the turbocharger alone?

MCAT-STYLE PASSAGE PROBLEMS

BIO Anesthetic Gases. One type of gas mixture used in anesthesiology is a 50%/50% mixture (by volume) of nitrous oxide (N_2O) and oxygen (O_2), which can be premixed and kept in a cylinder for later use. Because these two gases don't react chemically at or below 2000 psi, at typical room temperatures they form a homogeneous single gas phase, which can be considered an ideal gas. If the temperature drops below -6°C , however, N_2O may begin to condense out of the gas phase. Then any gas removed from the cylinder will initially be nearly pure O_2 ; as the cylinder empties, the proportion of O_2 will decrease until the gas coming from the cylinder is nearly pure N_2O .

19.63 In a test of the effects of low temperatures on the gas mixture, a cylinder filled at 20.0°C to 2000 psi (gauge pressure) is cooled slowly and the pressure is monitored. What is the expected pressure at -5.00°C if the gas remains a homogeneous mixture? (a) 500 psi; (b) 1500 psi; (c) 1830 psi; (d) 1920 psi.

19.64 In another test, the valve of a 500 L cylinder full of the gas mixture at 2000 psi (gauge pressure) is opened wide so that the gas rushes out of the cylinder very rapidly. Why might some N_2O condense during this process? (a) This is an isochoric process in which the pressure decreases, so the temperature also decreases. (b) Because of the rapid expansion, heat is removed from the system, so the internal energy and temperature of the gas decrease. (c) This is an isobaric process, so as the volume increases, the temperature decreases proportionally. (d) With the rapid expansion, the expanding gas does work with no heat input, so the internal energy and temperature of the gas decrease.

19.65 You have a cylinder that contains 500 L of the gas mixture pressurized to 2000 psi (gauge pressure). A regulator sets the gas flow to deliver 8.2 L/min at atmospheric pressure. Assume that this flow is slow enough that the expansion is isothermal and the gases remain mixed. How much time will it take to empty the cylinder? (a) 1 h; (b) 33 h; (c) 57 h; (d) 140 h.

19.66 In a hospital, pure oxygen may be delivered at 50 psi (gauge pressure) and then mixed with N_2O . What volume of oxygen at 20°C and 50 psi (gauge pressure) should be mixed with 1.7 kg of N_2O to get a 50%/50% mixture by volume at 20°C ? (a) 0.21 m^3 ; (b) 0.27 m^3 ; (c) 1.9 m^3 ; (d) 100 m^3 .

ANSWERS

Chapter Opening Question ?

(ii) The work done by a gas as its volume changes from V_1 to V_2 is equal to the integral $\int p \, dV$ between those two volume limits. If the volume of the gas contracts, the final volume V_2 is less than the initial volume V_1 and the gas does negative work. Propelling the locomotive requires that the gas do positive work, so the gas doesn't contribute to propulsion while contracting.

Key Example VARIATION Problems

VP19.5.1 (a) $5.75 \times 10^3 \text{ J}$ (b) $-3.20 \times 10^4 \text{ J}$ (c) $-1.85 \times 10^3 \text{ J}$ (d) 875 J

VP19.5.2 (a) 155 J (b) zero (c) -375 J (d) 875 J

VP19.5.3 (a) $W = -2.40 \times 10^2 \text{ J}$, $\Delta U = 1.80 \times 10^3 \text{ J}$

(b) $W = -7.20 \times 10^2 \text{ J}$, $Q = 1.08 \times 10^3 \text{ J}$

VP19.5.4 (a) $3.4 \times 10^6 \text{ J}$ (b) $3.4 \times 10^5 \text{ J}$ (c) $3.1 \times 10^6 \text{ J}$

VP19.6.1 (a) 998 J (b) 748 J (c) 599 J

VP19.6.2 (a) $5.24 \times 10^{-2} \text{ m}^3$ (b) (i) 327°C , $1.31 \times 10^4 \text{ J}$ (ii) 327°C , $1.31 \times 10^4 \text{ J}$ (iii) 927°C , $3.93 \times 10^4 \text{ J}$

VP19.6.3 (a) $4.50 \times 10^4 \text{ J}$ (b) $-2.25 \times 10^4 \text{ J}$ (c) $1.17 \times 10^5 \text{ J}$

VP19.6.4 (a) $\Delta U = \frac{5}{2}nRT$, $Q = \frac{5}{2}nRT$, $W = 0$ (b) $\Delta U = \frac{5}{2}nRT$, $Q = \frac{7}{2}nRT$, $W = nRT$ (c) $\Delta U = \frac{5}{2}nRT$, $Q = 0$, $W = -\frac{5}{2}nRT$

VP19.7.1 (a) $\frac{5}{3}$ (b) $6.41 \times 10^4 \text{ Pa}$ (c) 623 J

VP19.7.2 (a) $\frac{7}{5}$ (b) $5.79 \times 10^{-4} \text{ m}^3$ (c) $2.95 \times 10^6 \text{ Pa}$ (d) $-2.65 \times 10^3 \text{ J}$

VP19.7.3 (a) $3.17 \times 10^5 \text{ Pa}$ (b) $8.20 \times 10^4 \text{ Pa}$ (c) 178 K

(d) $7.94 \times 10^3 \text{ J}$

VP19.7.4 (a) $-3.38 \times 10^3 \text{ J}$ (b) 3.59 (c) 0.167

Bridging Problem

$$(a) W = nRT \ln \left[\frac{V_2 - nb}{V_1 - nb} \right] + an^2 \left[\frac{1}{V_2} - \frac{1}{V_1} \right]$$

(b) (i) $W = 2.80 \times 10^3 \text{ J}$, (ii) $W = 3.11 \times 10^3 \text{ J}$

(c) ideal gas, for which there is no attraction between molecules

? The second law of thermodynamics tells us that heat naturally flows from a hot object (such as molten lava, shown here flowing into the ocean in Hawaii) to a cold one (such as ocean water, which is heated by the lava to make steam). Is it *ever* possible for heat to flow from a cold object to a hot one? (i) Yes, no matter what the temperature difference; (ii) yes, but for only certain temperature differences; (iii) no; (iv) answer depends on the compositions of the two objects.



20 The Second Law of Thermodynamics

LEARNING OUTCOMES

In this chapter, you'll learn...

- 20.1 The difference between reversible and irreversible processes.
- 20.2 What a heat engine is, and how to calculate its efficiency.
- 20.3 The physics of internal-combustion engines.
- 20.4 How refrigerators and heat engines are related, and how to analyze the performance of a refrigerator.
- 20.5 How the second law of thermodynamics sets limits on the efficiency of engines and the performance of refrigerators.
- 20.6 How to do calculations involving the idealized Carnot cycle for engines and refrigerators.
- 20.7 What is meant by entropy, and how to use this concept to analyze thermodynamic processes.
- 20.8 How to use the concept of microscopic states to understand entropy.

You'll need to review...

- 17.3 The Kelvin scale.
- 18.3 The Boltzmann constant.
- 19.1–19.8 Thermodynamic processes; first law of thermodynamics; free expansion of a gas.

Many thermodynamic processes proceed naturally in one direction but not the opposite. For example, heat by itself always flows from a hot object to a cooler object, never the reverse. Heat flow from a cool object to a hot object would not violate the first law of thermodynamics; energy would be conserved. But it doesn't happen in nature. Why not? As another example, note that it is easy to convert mechanical energy completely into heat; this happens every time we use a car's brakes to stop it. In the reverse direction, there are plenty of devices that convert heat *partially* into mechanical energy. (An automobile engine is an example.) But no one has ever managed to build a machine that converts heat *completely* into mechanical energy. Again, why not?

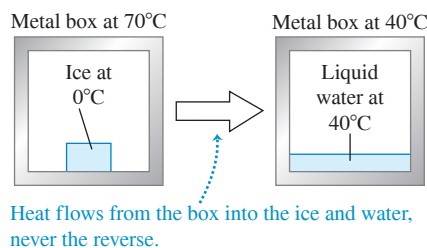
The answer to both of these questions has to do with the *directions* of thermodynamic processes and is called the *second law of thermodynamics*. This law places fundamental limitations on the efficiency of an engine or a power plant. It also places limitations on the minimum energy input needed to operate a refrigerator. So the second law is directly relevant for many important practical problems.

We can also state the second law in terms of the concept of *entropy*, a quantitative measure of the degree of randomness of a system. The idea of entropy helps explain why ink mixed with water never spontaneously unmixes and why we never observe a host of other seemingly possible processes.

20.1 DIRECTIONS OF THERMODYNAMIC PROCESSES

Thermodynamic processes that occur in nature are all **irreversible processes**. These are processes that proceed spontaneously in one direction but not the other (**Fig. 20.1a**). The flow of heat from a hot object to a cooler object is irreversible, as is the free expansion of a gas discussed in Sections 19.3 and 19.6. Sliding a book across a table converts mechanical energy into heat by friction; this process is irreversible, for no one has ever observed the reverse process (in which a book initially at rest on the table would spontaneously start moving and the table and book would cool down). Our main topic for this

(a) A block of ice melts *irreversibly* when we place it in a hot (70°C) metal box.



(b) A block of ice at 0°C can be melted *reversibly* if we put it in a 0°C metal box.

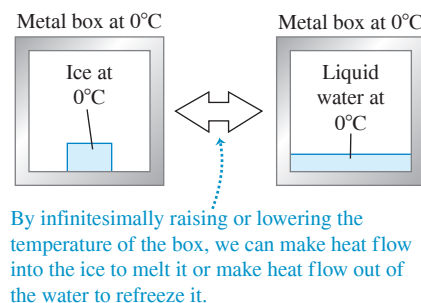


Figure 20.1 Reversible and irreversible processes.

chapter is the *second law of thermodynamics*, which determines the preferred direction for such processes.

Despite this preferred direction for every natural process, we can think of a class of idealized processes that *would* be reversible. A system that undergoes such an idealized **reversible process** is always very close to being in thermodynamic equilibrium within itself and with its surroundings. Any change of state that takes place can then be reversed by making only an infinitesimal change in the conditions of the system. For example, we can reverse heat flow between two objects whose temperatures differ only infinitesimally by making only a very small change in one temperature or the other (Fig. 20.1b).

Reversible processes are thus **equilibrium processes**, with the system always in thermodynamic equilibrium. Of course, if a system were *truly* in thermodynamic equilibrium, no change of state would take place. Heat would not flow into or out of a system with truly uniform temperature throughout, and a system that is truly in mechanical equilibrium would not expand and do work against its surroundings. A reversible process is an idealization that can never be precisely attained in the real world. But by making the temperature gradients and the pressure differences in the substance very small, we can keep the system very close to equilibrium states and make the process nearly reversible.

By contrast, heat flow with finite temperature difference, free expansion of a gas, and conversion of work to heat by friction are all *irreversible* processes; no small change in conditions could make any of them go the other way. They are also all *nonequilibrium* processes, in that the system is not in thermodynamic equilibrium at any point until the end of the process.

Disorder and Thermodynamic Processes

There is a relationship between the direction of a process and the *disorder* or *randomness* of the resulting state. For example, imagine a thousand names written on file cards and arranged in alphabetical order. Throw the alphabetized stack of cards into the air, and they will likely come down in a random, disordered state. In the free expansion of a gas discussed in Sections 19.3 and 19.6, the air is more disordered after it has expanded into the entire box than when it was confined in one side, just as your clothes are more disordered when scattered all over your floor than when confined to your closet.

Similarly, macroscopic kinetic energy is energy associated with organized, coordinated motions of many molecules, but heat transfer involves changes in energy of random, disordered molecular motion. Therefore conversion of mechanical energy into heat involves an increase of randomness or disorder.

In the following sections we'll introduce the second law of thermodynamics by considering two broad classes of devices: *heat engines*, which are partly successful in converting heat into work, and *refrigerators*, which are partly successful in transporting heat from cooler to hotter objects.

TEST YOUR UNDERSTANDING OF SECTION 20.1 Your left and right hands are normally at the same temperature, just like the metal box and ice in Fig. 20.1b. Is rubbing your hands together to warm them (i) a reversible process or (ii) an irreversible process?

ANSWER

(ii) Like sliding a book across a table, rubbing your hands together uses friction to convert mechanical energy into heat. The (impossible) reverse process would involve your hands spontaneously getting colder, with the released energy forcing your hands to move rhythmically back and forth!

20.2 HEAT ENGINES

The essence of our technological society is the ability to use sources of energy other than muscle power. Sometimes, mechanical energy is directly available; water power and wind power are examples. But most of our energy comes from the burning of fossil fuels (coal, oil, and gas) and from nuclear reactions. They supply energy that is transferred as *heat*. This is directly useful for heating buildings, for cooking, and for chemical processing, but to operate a machine or propel a vehicle, we need *mechanical energy*.

Thus it's important to know how to take heat from a source and convert as much of it as possible into mechanical energy or work. This is what happens in gasoline engines in automobiles, jet engines in airplanes, steam turbines in electric power plants, and many other systems. Closely related processes occur in the animal kingdom; food energy is “burned” (that is, carbohydrates combine with oxygen to yield water, carbon dioxide, and energy) and partly converted to mechanical energy as an animal’s muscles do work on its surroundings.

Any device that transforms heat partly into work or mechanical energy is called a **heat engine** (Fig. 20.2). Usually, a quantity of matter inside the engine undergoes inflow and outflow of heat, expansion and compression, and sometimes change of phase. We call this matter the **working substance** of the engine. In internal-combustion engines, such as those used in automobiles, the working substance is a mixture of air and fuel; in a steam turbine it is water.

The simplest kind of engine to analyze is one in which the working substance undergoes a **cyclic process**, a sequence of processes that eventually leaves the substance in the same state in which it started. In a steam turbine the water is recycled and used over and over. Internal-combustion engines do not use the same air over and over, but we can still analyze them in terms of cyclic processes that approximate their actual operation.

Hot and Cold Reservoirs

All heat engines *absorb* heat from a source at a relatively high temperature, perform some mechanical work, and *discard* or *reject* some heat at a lower temperature. As far as the engine is concerned, the discarded heat is wasted. In internal-combustion engines the waste heat is that discarded in the hot exhaust gases and the cooling system; in a steam turbine it is the heat that must flow out of the used steam to condense and recycle the water.

When a system is carried through a cyclic process, its initial and final internal energies are equal, so the first law of thermodynamics requires that

$$U_2 - U_1 = 0 = Q - W \quad \text{so} \quad Q = W$$

That is, the net heat flowing into the engine in a cyclic process equals the net work done by the engine.

When we analyze heat engines, it helps to think of two objects with which the working substance of the engine can interact. One of these, called the *hot reservoir*, represents the heat source; it can give the working substance large amounts of heat at a constant temperature T_H without appreciably changing its own temperature. The other object, called the *cold reservoir*, can absorb large amounts of discarded heat from the engine at a constant lower temperature T_C . In a steam-turbine system the flames and hot gases in the boiler are the hot reservoir, and the cold water and air used to condense and cool the used steam are the cold reservoir.

Figure 20.2 All motorized vehicles other than purely electric vehicles use heat engines for propulsion. (Hybrid vehicles use their internal-combustion engine to help charge the batteries for the electric motor.)



We denote the quantities of heat transferred from the hot and cold reservoirs as Q_H and Q_C , respectively. A quantity of heat Q is positive when heat is transferred *into* the working substance and is negative when heat leaves the working substance. Thus in a heat engine, Q_H is positive but Q_C is negative, representing heat *leaving* the working substance. This sign convention is consistent with the rules we stated in Section 19.1; we'll continue to use those rules here. For clarity, we'll often state the relationships in terms of the absolute values of the Q 's and W 's because absolute values are always positive.

Energy-Flow Diagrams and Efficiency

We can represent the energy transformations in a heat engine by the *energy-flow diagram* of **Fig. 20.3**. The engine itself is represented by the circle. The amount of heat Q_H supplied to the engine by the hot reservoir is proportional to the width of the incoming “pipeline” at the top of the diagram. The width of the outgoing pipeline at the bottom is proportional to the magnitude $|Q_C|$ of the heat rejected in the exhaust. The branch line to the right represents the portion of the heat supplied that the engine converts to mechanical work, W .

When an engine repeats the same cycle over and over, Q_H and Q_C represent the quantities of heat absorbed and rejected by the engine *during one cycle*; Q_H is positive, and Q_C is negative. The *net* heat Q absorbed per cycle is

$$Q = Q_H + Q_C = |Q_H| - |Q_C| \quad (20.1)$$

The useful output of the engine is the net work W done by the working substance. From the first law,

$$W = Q = Q_H + Q_C = |Q_H| - |Q_C| \quad (20.2)$$

Ideally, we would like to convert *all* the heat Q_H into work; in that case we would have $Q_H = W$ and $Q_C = 0$. Experience shows that this is impossible; there is always some heat wasted, and Q_C is *never* zero. We define the **thermal efficiency** of an engine, denoted by e , as the quotient

$$e = \frac{W}{Q_H} \quad (20.3)$$

The thermal efficiency e represents the fraction of Q_H that *is* converted to work. To put it another way, e is what you get divided by what you pay for. This is always less than unity, an all-too-familiar experience! In terms of the flow diagram of Fig. 20.3, the most efficient engine is one for which the branch pipeline representing the work output is as wide as possible and the exhaust pipeline representing the heat thrown away is as narrow as possible.

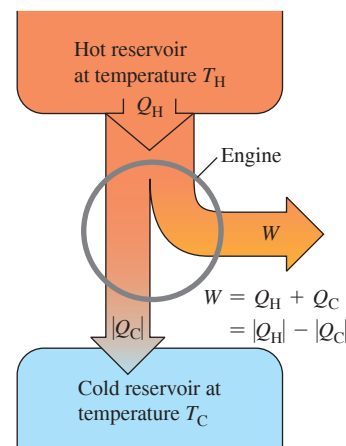
When we substitute the two expressions for W given by Eq. (20.2) into Eq. (20.3), we get the following equivalent expressions for e :

$$\text{Thermal efficiency of an engine } e = \frac{W}{Q_H} = 1 + \frac{Q_C}{Q_H} = 1 - \frac{|Q_C|}{|Q_H|} \quad (20.4)$$

Work done by engine Heat rejected by engine
Heat absorbed by engine

Note that e is a quotient of two energy quantities and thus is a pure number, without units. Of course, we must always express W , Q_H , and Q_C in the same units.

Figure 20.3 Schematic energy-flow diagram for a heat engine.



BIO APPLICATION Biological Efficiency

Although a biological organism is not a heat engine, the concept of efficiency still applies: Here e is the ratio of the work done to the energy that was used to do that work. To exercise on a stationary bike, your body must first convert the chemical-bond energy in glucose to chemical-bond energy in ATP (adenosine triphosphate), then convert energy from ATP into motion of your leg muscles, and finally convert muscular motion into motion of the pedals. The overall efficiency of this entire process is only about 25%. The remaining 75% of the energy liberated from glucose goes into heat.



PROBLEM-SOLVING STRATEGY 20.1 Heat Engines

Problems involving heat engines are, fundamentally, problems in the first law of thermodynamics. You should review Problem-Solving Strategy 19.1 (Section 19.4).

IDENTIFY *the relevant concepts:* A heat engine is any device that converts heat partially to work, as shown schematically in Fig. 20.3. We'll see in Section 20.4 that a refrigerator is essentially a heat engine running in reverse, so many of the same concepts apply.

SET UP *the problem* as suggested in Problem-Solving Strategy 19.1: Use Eq. (20.4) if the thermal efficiency of the engine is relevant. Sketch an energy-flow diagram like Fig. 20.3.

EXECUTE *the solution* as follows:

1. Be careful with the sign conventions for W and the various Q 's. W is positive when the system expands and does work; W is negative when the system is compressed and work is done on it.

Each Q is positive if it represents heat entering the system and is negative if it represents heat leaving the system. When you know that a quantity is negative, such as Q_C in the above discussion, it sometimes helps to write it as $Q_C = -|Q_C|$.

2. Power is work per unit time ($P = W/t$), and rate of heat transfer (heat current) H is heat transfer per unit time ($H = Q/t$). In problems involving these concepts it helps to ask, "What is W or Q in one second (or one hour)?"
3. Keeping steps 1 and 2 in mind, solve for the target variables.

EVALUATE *your answer:* Use the first law of thermodynamics to check your results. Pay particular attention to algebraic signs.

EXAMPLE 20.1 Analyzing a heat engine

WITH VARIATION PROBLEMS

A gasoline truck engine takes in 10,000 J of heat and delivers 2000 J of mechanical work per cycle. The heat is obtained by burning gasoline with heat of combustion $L_c = 5.0 \times 10^4$ J/g. (a) What is the thermal efficiency of this engine? (b) How much heat is discarded in each cycle? (c) If the engine goes through 25 cycles per second, what is its power output in watts? In horsepower? (d) How much gasoline is burned in each cycle? (e) How much gasoline is burned per second? Per hour?

IDENTIFY and SET UP This problem concerns a heat engine, so we can use the ideas of this section. **Figure 20.4** is our energy-flow diagram for one cycle. In each cycle the engine does $W = 2000$ J of work and takes in heat $Q_H = 10,000$ J. We use Eq. (20.4), in the form $e = W/Q_H$, to find the thermal efficiency. We use Eq. (20.2) to find the amount of heat Q_C rejected per cycle. The heat of combustion tells us how much gasoline must be burned per cycle and hence per unit time. The power output is the time rate at which the work W is done.

EXECUTE (a) From Eq. (20.4), the thermal efficiency is

$$e = \frac{W}{Q_H} = \frac{2000 \text{ J}}{10,000 \text{ J}} = 0.20 = 20\%$$

- (b) From Eq. (20.2), $W = Q_H + Q_C$, so

$$Q_C = W - Q_H = 2000 \text{ J} - 10,000 \text{ J} = -8000 \text{ J}$$

That is, 8000 J of heat leaves the engine during each cycle.

- (c) The power P equals the work per cycle multiplied by the number of cycles per second:

$$\begin{aligned} P &= (2000 \text{ J/cycle})(25 \text{ cycles/s}) = 50,000 \text{ W} = 50 \text{ kW} \\ &= (50,000 \text{ W}) \frac{1 \text{ hp}}{746 \text{ W}} = 67 \text{ hp} \end{aligned}$$

- (d) Let m be the mass of gasoline burned during each cycle. Then $Q_H = mL_c$ and

$$m = \frac{Q_H}{L_c} = \frac{10,000 \text{ J}}{5.0 \times 10^4 \text{ J/g}} = 0.20 \text{ g}$$

- (e) The mass of gasoline burned per second equals the mass per cycle multiplied by the number of cycles per second:

$$(0.20 \text{ g/cycle})(25 \text{ cycles/s}) = 5.0 \text{ g/s}$$

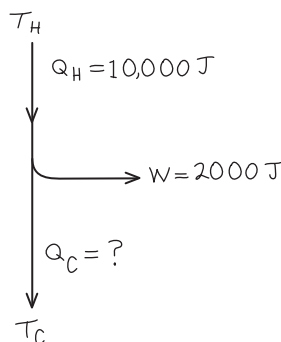
The mass burned per hour is

$$(5.0 \text{ g/s}) \frac{3600 \text{ s}}{1 \text{ h}} = 18,000 \text{ g/h} = 18 \text{ kg/h}$$

EVALUATE An efficiency of 20% is fairly typical for cars and trucks if W includes only the work delivered to the wheels. We can check the mass burned per hour by expressing it in miles per gallon ("mileage"). The density of gasoline is about 0.70 g/cm^3 , so this is about $25,700 \text{ cm}^3$, 25.7 L, or 6.8 gallons of gasoline per hour. If the truck is traveling at 55 mi/h (88 km/h), this represents fuel consumption of 8.1 miles/gallon (3.4 km/L). This is a fairly typical mileage for large trucks.

KEYCONCEPT A heat engine takes in energy in the form of heat from a hot reservoir, uses some of that energy to do work, and rejects the remaining energy as heat that goes into a cold reservoir. The fraction of the heat from the hot reservoir that is converted into work is called the thermal efficiency of the engine.

Figure 20.4 Our sketch for this problem.



TEST YOUR UNDERSTANDING OF SECTION 20.2 Rank the following heat engines in order from highest to lowest thermal efficiency. (i) An engine that in one cycle absorbs 5000 J of heat and rejects 4500 J of heat; (ii) an engine that in one cycle absorbs 25,000 J of heat and does 2000 J of work; (iii) an engine that in one cycle does 400 J of work and rejects 2800 J of heat.

ANSWER

(iii), (i), (ii) From Eq. (20.4) the efficiency is $e = W/Q_H$, and from Eq. (20.2)

$$W = Q_H + Q_C = |Q_H| - |Q_C|. \text{ For engine (i) } Q_H = 5000 \text{ J and } Q_C = -4500 \text{ J, so}$$

$$W = 5000 \text{ J} + (-4500 \text{ J}) = 500 \text{ J and } e = (500 \text{ J})/(5000 \text{ J}) = 0.100. \text{ For engine (ii)}$$

$$Q_H = 25,000 \text{ J and } W = 2000 \text{ J, so } e = (2000 \text{ J})/(25,000 \text{ J}) = 0.080. \text{ For engine (iii)}$$

$$W = 400 \text{ J and } Q_C = -2800 \text{ J, so } Q_H = W - Q_C = 400 \text{ J} - (-2800 \text{ J}) = 3200 \text{ J and}$$

$$e = (400 \text{ J})/(3200 \text{ J}) = 0.125.$$

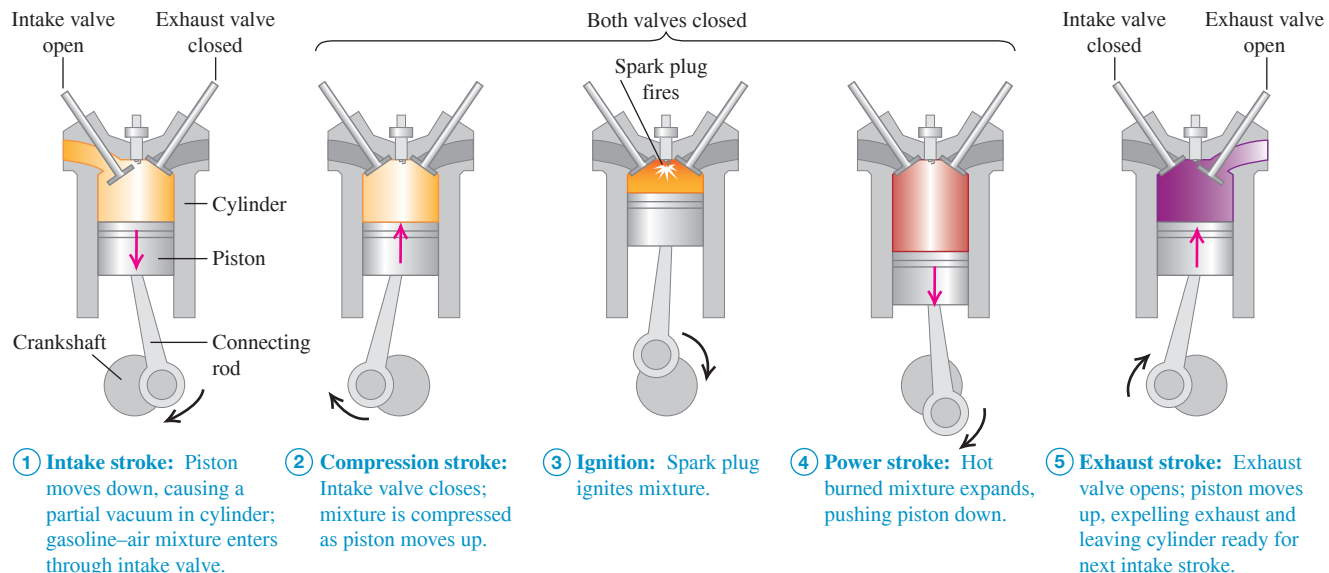
20.3 INTERNAL-COMBUSTION ENGINES

The gasoline engine, used in automobiles and many other types of machinery, is a familiar example of a heat engine. Let's look at its thermal efficiency. **Figure 20.5** shows the operation of one type of gasoline engine. First a mixture of air and gasoline vapor flows into a cylinder through an open intake valve while the piston descends, increasing the volume of the cylinder from a minimum of V (when the piston is all the way up) to a maximum of rV (when it is all the way down). The quantity r is called the **compression ratio**; for present-day automobile engines its value is typically 8 to 10. At the end of this *intake stroke*, the intake valve closes and the mixture is compressed, approximately adiabatically, to volume V during the *compression stroke*. The mixture is then ignited by the spark plug, and the heated gas expands, approximately adiabatically, back to volume rV , pushing on the piston and doing work; this is the *power stroke*. Finally, the exhaust valve opens, and the combustion products are pushed out (during the *exhaust stroke*), leaving the cylinder ready for the next intake stroke.

The Otto Cycle

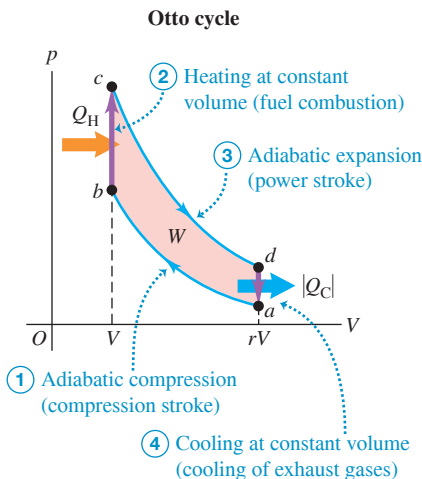
Figure 20.6 (next page) is a pV -diagram for an idealized model of the thermodynamic processes in a gasoline engine. This model is called the **Otto cycle**. At point a the gasoline–air mixture has entered the cylinder. The mixture is compressed adiabatically to point b and is then ignited. Heat Q_H is added to the system by the burning gasoline along line bc , and the power stroke is the adiabatic expansion to d . The gas is cooled to the temperature of the outside air along line da ; during this process, heat $|Q_C|$ is rejected. This gas leaves the engine as exhaust and does not enter the engine again. But since an equivalent amount of gasoline and air enters, we may consider the process to be cyclic.

Figure 20.5 Cycle of a four-stroke internal-combustion engine.



- ➊ **Intake stroke:** Piston moves down, causing a partial vacuum in cylinder; gasoline–air mixture enters through intake valve.
- ➋ **Compression stroke:** Intake valve closes; mixture is compressed as piston moves up.
- ➌ **Ignition:** Spark plug ignites mixture.
- ➍ **Power stroke:** Hot burned mixture expands, pushing piston down.
- ➎ **Exhaust stroke:** Exhaust valve opens; piston moves up, expelling exhaust and leaving cylinder ready for next intake stroke.

Figure 20.6 The pV -diagram for the Otto cycle, an idealized model of the thermodynamic processes in a gasoline engine.



We can calculate the efficiency of this idealized cycle. Processes bc and da are constant-volume, so the heats Q_H and Q_C are related simply to the temperatures at points a , b , c , and d :

$$Q_H = nC_V(T_c - T_b) > 0$$

$$Q_C = nC_V(T_a - T_d) < 0$$

The thermal efficiency is given by Eq. (20.4). Inserting the above expressions and canceling out the common factor nC_V , we find

$$e = \frac{Q_H + Q_C}{Q_H} = \frac{T_c - T_b + T_a - T_d}{T_c - T_b} \quad (20.5)$$

To simplify this further, we use the temperature–volume relationship for adiabatic processes for an ideal gas, Eq. (19.22). For the two adiabatic processes ab and cd ,

$$T_a(rV)^{\gamma-1} = T_b V^{\gamma-1} \quad \text{and} \quad T_d(rV)^{\gamma-1} = T_c V^{\gamma-1}$$

where γ is the ratio of heat capacities for the gas in the engine (see Section 19.7). We divide each of these equations by the common factor $V^{\gamma-1}$ and substitute the resulting expressions for T_b and T_c back into Eq. (20.5). The result is

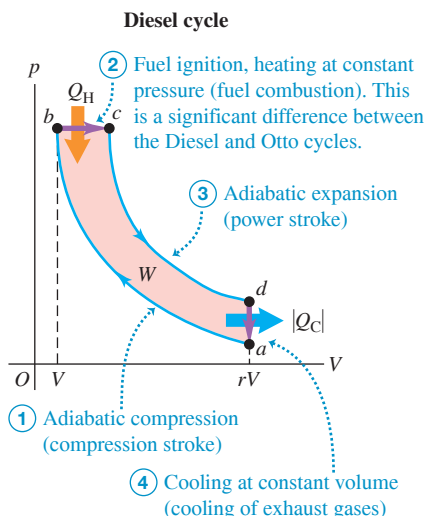
$$e = \frac{T_d r^{\gamma-1} - T_a r^{\gamma-1} + T_a - T_d}{T_d r^{\gamma-1} - T_a r^{\gamma-1}} = \frac{(T_d - T_a)(r^{\gamma-1} - 1)}{(T_d - T_a)r^{\gamma-1}}$$

Dividing out the common factor $(T_d - T_a)$, we get

$$\text{Thermal efficiency in Otto cycle} \quad e = 1 - \frac{1}{r^{\gamma-1}} \quad (20.6)$$

Compression ratio $\rightarrow r$ Ratio of heat capacities $\rightarrow \gamma$

Figure 20.7 The pV -diagram for the idealized Diesel cycle.



The thermal efficiency given by Eq. (20.6) is always less than unity, even for this idealized model. With $r = 8$ and $\gamma = 1.4$ (the value for air) the theoretical efficiency is $e = 0.56$, or 56%. The efficiency can be increased by increasing r . However, this also increases the temperature at the end of the adiabatic compression of the air–fuel mixture. If the temperature is too high, the mixture explodes spontaneously during compression instead of burning evenly after the spark plug ignites it. This is called *pre-ignition* or *detonation*; it causes a knocking sound and can damage the engine. The octane rating of a gasoline is a measure of its antiknock qualities. The maximum practical compression ratio for high-octane, or “premium,” gasoline is about 10 to 13.

The Otto cycle is a highly idealized model. It assumes that the mixture behaves as an ideal gas; it ignores friction, turbulence, loss of heat to cylinder walls, and many other effects that reduce the efficiency of an engine. Efficiencies of real gasoline engines are typically around 35%.

The Diesel Cycle

The Diesel engine is similar in operation to the gasoline engine. The most important difference is that there is no fuel in the cylinder at the beginning of the compression stroke. A little before the beginning of the power stroke, the injectors start to inject fuel directly into the cylinder, just fast enough to keep the pressure approximately constant during the first part of the power stroke. Because of the high temperature developed during the adiabatic compression, the fuel ignites spontaneously as it is injected; no spark plugs are needed.

Figure 20.7 shows the idealized **Diesel cycle**. Starting at point a , air is compressed adiabatically to point b , heated at constant pressure to point c , expanded adiabatically to point d , and cooled at constant volume to point a . Because there is no fuel in the cylinder during most of the compression stroke, pre-ignition cannot occur, and the compression ratio r can be much higher than for a gasoline engine. This improves efficiency

and ensures reliable ignition when the fuel is injected (because of the high temperature reached during the adiabatic compression). Values of r of 15 to 20 are typical; with these values and $\gamma = 1.4$, the theoretical efficiency of the idealized Diesel cycle is about 0.65 to 0.70. As with the Otto cycle, the efficiency of any actual engine is substantially less than this. While Diesel engines are very efficient, they must be built to much tighter tolerances than gasoline engines and the fuel-injection system requires careful maintenance.

TEST YOUR UNDERSTANDING OF SECTION 20.3 For an Otto-cycle engine with cylinders of a fixed size and a fixed compression ratio, which of the following aspects of the pV -diagram in Fig. 20.6 would change if you doubled the amount of fuel burned per cycle? (There may be more than one correct answer.) (i) The vertical distance between points b and c ; (ii) the vertical distance between points a and d ; (iii) the horizontal distance between points b and a .

ANSWER (i), (ii), (iii) Doubling the amount of fuel burned per cycle means that Q_H is doubled, so the resulting pressure increase from b to c in Fig. 20.6 is greater. The compression ratio and hence the efficiency remain the same, so $|Q_C|$ (the amount of heat rejected to the environment) must increase by the same factor as Q_H . Hence the pressure drop from d to a in Fig. 20.6 is also greater. The volume V and the compression ratio r don't change, so the horizontal dimensions of the pV -diagram don't change.

20.4 REFRIGERATORS

We can think of a **refrigerator** as a heat engine operating in reverse. A heat engine takes heat from a hot place and gives off heat to a colder place. A refrigerator does the opposite; it takes heat from a cold place (the inside of the refrigerator) and gives it off to a warmer place (usually the air in the room where the refrigerator is located). A heat engine has a net *output* of mechanical work; the refrigerator requires a net *input* of mechanical work. With the sign conventions from Section 20.2, for a refrigerator Q_C is positive but both W and Q_H are negative; hence $|W| = -W$ and $|Q_H| = -Q_H$.

Figure 20.8 shows an energy-flow diagram for a refrigerator. From the first law for a cyclic process,

$$Q_H + Q_C - W = 0 \quad \text{or} \quad -Q_H = Q_C - W$$

or, because both Q_H and W are negative,

$$|Q_H| = Q_C + |W| \quad (20.7)$$

Thus, as the diagram shows, the heat $|Q_H|$ leaving the working substance and given to the hot reservoir is always *greater* than the heat Q_C taken from the cold reservoir. Note that the absolute-value relationship

$$|Q_H| = |Q_C| + |W| \quad (20.8)$$

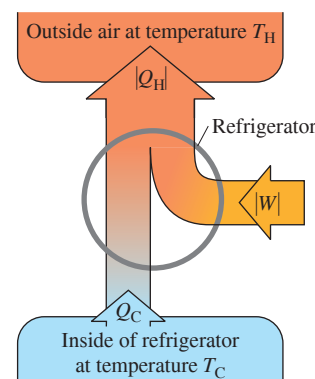
is valid for both heat engines and refrigerators.

From an economic point of view, the best refrigeration cycle is one that removes the greatest amount of heat $|Q_C|$ from the inside of the refrigerator for the least expenditure of mechanical work, $|W|$. The relevant ratio is therefore $|Q_C|/|W|$; the larger this ratio, the better the refrigerator. We call this ratio the **coefficient of performance**, K . From Eq. (20.8), $|W| = |Q_H| - |Q_C|$, so

$$K = \frac{|Q_C|}{|W|} = \frac{|Q_C|}{|Q_H| - |Q_C|} \quad (20.9)$$

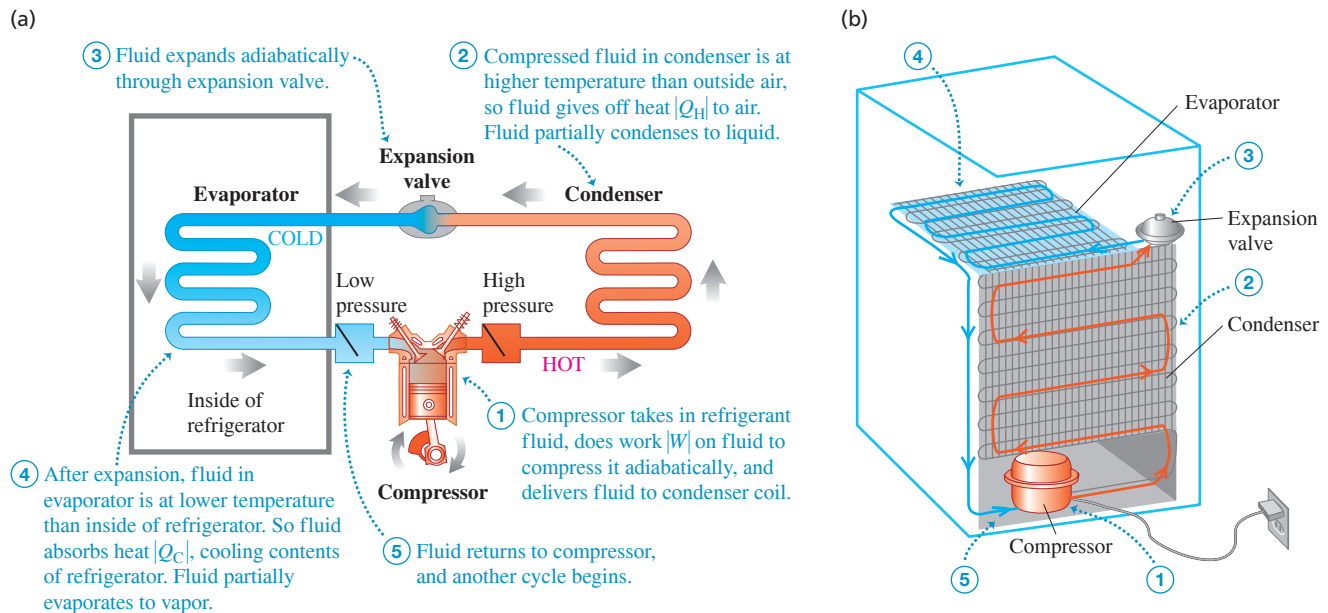
Coefficient of performance of a refrigerator Heat removed from inside of refrigerator
Work input of refrigerator Heat rejected to outside air

Figure 20.8 Schematic energy-flow diagram of a refrigerator.



As always, we measure Q_H , Q_C , and W all in the same energy units; K is then a dimensionless number.

Figure 20.9 (a) Principle of the mechanical refrigeration cycle. (b) How the key elements are arranged in a practical refrigerator.

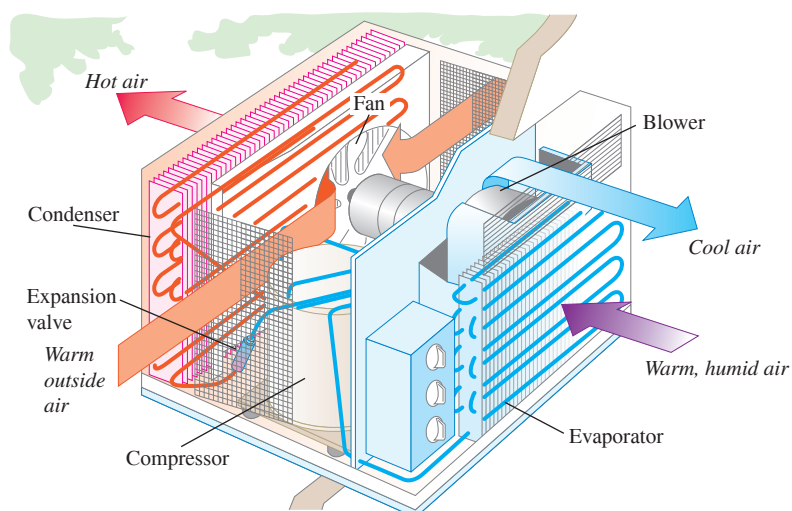


Practical Refrigerators

Figure 20.9a shows the principles of the cycle used in home refrigerators. The fluid “circuit” contains a refrigerant fluid (the working substance). The left side of the circuit (including the cooling coils inside the refrigerator) is at low temperature and low pressure; the right side (including the condenser coils outside the refrigerator) is at high temperature and high pressure. Ordinarily, both sides contain liquid and vapor in phase equilibrium. In each cycle the fluid absorbs heat $|Q_C|$ from the inside of the refrigerator on the left side and gives off heat $|Q_H|$ to the surrounding air on the right side. The compressor, usually driven by an electric motor (Fig. 20.9b), does work $|W|$ on the fluid during each cycle. So the compressor requires energy input, which is why refrigerators have to be plugged in.

An air conditioner operates on exactly the same principle. In this case the refrigerator box becomes a room or an entire building. The evaporator coils are inside, the condenser is outside, and fans circulate air through these (Fig. 20.10). In large installations the condenser coils are often cooled by water. For air conditioners the quantities of greatest practical importance are the *rate* of heat removal (the heat current H from the region being

Figure 20.10 An air conditioner works on the same principle as a refrigerator.



cooled) and the *power* input $P = W/t$ to the compressor. If heat $|Q_C|$ is removed in time t , then $H = |Q_C|/t$. Then we can express the coefficient of performance as

$$K = \frac{|Q_C|}{|W|} = \frac{Ht}{Pt} = \frac{H}{P}$$

Typical room air conditioners have heat removal rates H of about 1500–3000 W and require electric power input of about 600 to 1200 W. Typical coefficients of performance are about 3; the actual values depend on the inside and outside temperatures.

A variation on this theme is the **heat pump**, used to heat buildings by cooling the outside air. It functions like a refrigerator turned inside out. The evaporator coils are outside, where they take heat from cold air, and the condenser coils are inside, where they give off heat to the warmer air. With proper design, the heat $|Q_H|$ delivered to the inside per cycle can be considerably greater than the work $|W|$ required to get it there.

Work is *always* needed to transfer heat from a colder to a hotter object. Heat flows spontaneously from hotter to colder, and to reverse this flow requires the addition of work from the outside. Experience shows that it is impossible to make a refrigerator that transports heat from a colder object to a hotter object without the addition of work. If no work were needed, the coefficient of performance would be infinite. We call such a device a *workless refrigerator*; it is a mythical beast, like the unicorn and the free lunch.

TEST YOUR UNDERSTANDING OF SECTION 20.4 Can you cool your house by leaving the refrigerator door open?

ANSWER

A refrigerator uses an input of work to transfer heat from one system (the refrigerator's interior) to another system (its exterior, which includes the house in which the refrigerator is installed). If the door is open, these two systems are really the *same* system and will eventually come to the same temperature. By the first law of thermodynamics, all of the work input to the refrigerator motor will be converted into heat and the temperature in your house will actually *increase*. To cool the house you need a system that will transfer heat from it to the outside world, such as an air conditioner or heat pump.

20.5 THE SECOND LAW OF THERMODYNAMICS

Experimental evidence suggests strongly that it is *impossible* to build a heat engine that converts heat completely to work—that is, an engine with 100% thermal efficiency. This impossibility is the basis of one statement of the **second law of thermodynamics**, as follows:

SECOND LAW OF THERMODYNAMICS (“engine” statement) It is impossible for any system to undergo a process in which it absorbs heat from a reservoir at a single temperature and converts the heat completely into mechanical work, with the system ending in the same state in which it began.

This is also known to physicists as the *Kelvin–Planck statement* of the second law.

The basis of the second law of thermodynamics is the difference between the nature of internal energy and that of macroscopic mechanical energy. In a moving object the molecules have random motion, but superimposed on this is a coordinated motion of every molecule in the direction of the object's velocity. The kinetic energy associated with this *coordinated* macroscopic motion is what we call the kinetic energy of the moving object. The kinetic and potential energies associated with the *random* motion constitute the internal energy.

When an object sliding on a surface comes to rest as a result of friction, the organized motion of the object is converted to random motion of molecules in the object and in the surface. Since we cannot control the motions of individual molecules, we cannot convert this random motion completely back to organized motion. We can convert *part* of it, and this is what a heat engine does.

If the second law were *not* true, we could power an automobile or run a power plant by cooling the surrounding air. Neither of these impossibilities violates the *first* law of thermodynamics. The second law, therefore, is not a deduction from the first but stands by itself as a separate law of nature. The first law denies the possibility of creating or destroying energy; the second law limits the *availability* of energy and the ways in which it can be used and converted.

Restating the Second Law

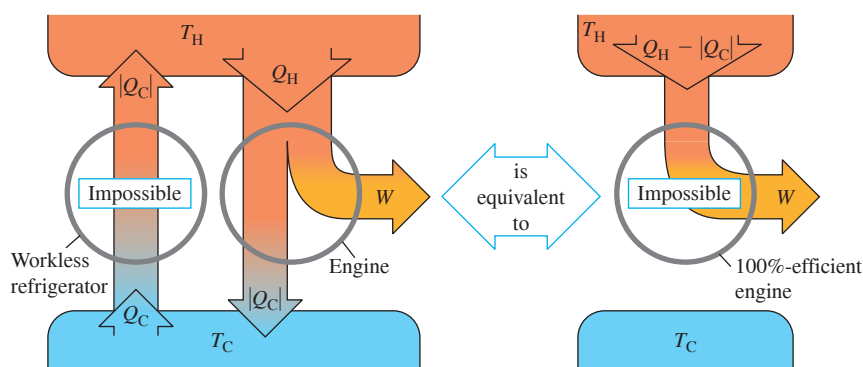
Our analysis of refrigerators in Section 20.4 forms the basis for an alternative statement of the second law of thermodynamics. Heat flows spontaneously from hotter to colder objects, never the reverse. A refrigerator does take heat from a colder to a hotter object, but its operation requires an input of mechanical energy or work. We can generalize this observation:

SECOND LAW OF THERMODYNAMICS (“refrigerator” statement) It is impossible for any process to have as its sole result the transfer of heat from a cooler to a hotter object.

This is also known as the *Clausius statement* of the second law. It may not seem to be very closely related to the “engine” statement. In fact, though, the two statements are completely equivalent. For example, if we could build a workless refrigerator, violating the second or “refrigerator” statement of the second law, we could use it in conjunction with a heat engine, pumping the heat rejected by the engine back to the hot reservoir to be reused. This composite machine (**Fig. 20.11a**) would violate the “engine” statement of the second

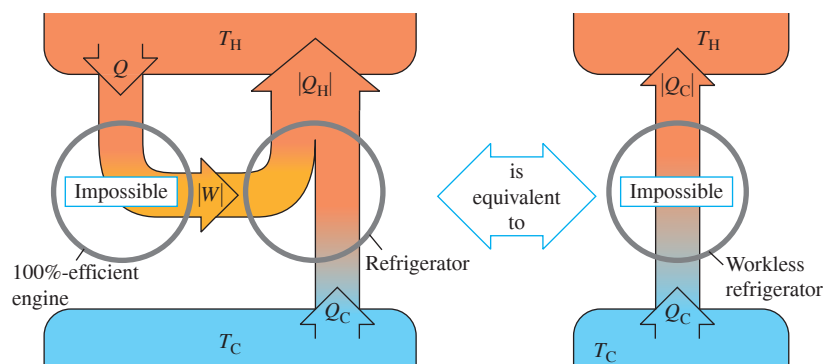
Figure 20.11 Energy-flow diagrams showing that the two forms of the second law are equivalent.

(a) The “engine” statement of the second law of thermodynamics



If a workless refrigerator were possible, it could be used in conjunction with an ordinary heat engine to form a 100%-efficient engine, converting heat $Q_H - |Q_C|$ completely to work.

(b) The “refrigerator” statement of the second law of thermodynamics



If a 100%-efficient engine were possible, it could be used in conjunction with an ordinary refrigerator to form a workless refrigerator, transferring heat Q_C from the cold to the hot reservoir with no input of work.

law because its net effect would be to take a net quantity of heat $Q_H - |Q_C|$ from the hot reservoir and convert it completely to work W .

Alternatively, if we could make an engine with 100% thermal efficiency, in violation of the first statement, we could run it using heat from the hot reservoir and use the work output to drive a refrigerator that pumps heat from the cold reservoir to the hot (Fig. 20.11b). This composite device would violate the “refrigerator” statement because its net effect would be to take heat Q_C from the cold reservoir and deliver it to the hot reservoir without requiring any input of work. Thus any device that violates one form of the second law can be used to make a device that violates the other form. If violations of the first form are impossible, so are violations of the second!

The conversion of work to heat and the heat flow from hot to cold across a finite temperature gradient are *irreversible* processes. The “engine” and “refrigerator” statements of the second law state that these processes can be only partially reversed. We could cite other examples. Gases naturally flow from a region of high pressure to a region of low pressure; gases and miscible liquids left by themselves always tend to mix, not to unmix. The second law of thermodynamics is an expression of the inherent one-way aspect of these and many other irreversible processes. Energy conversion is an essential aspect of all plant and animal life and of human technology, so the second law of thermodynamics is of fundamental importance.

TEST YOUR UNDERSTANDING OF SECTION 20.5 Would a 100%-efficient engine (Fig. 20.11a) violate the *first* law of thermodynamics? What about a workless refrigerator (Fig. 20.11b)?

ANSWER

no, no Both the 100%-efficient engine of Fig. 20.11a and the workless refrigerator of Fig. 20.11b return to the same state at the end of a cycle as at the beginning, so the net change in internal energy of each system is zero ($\Delta U = 0$). For the 100%-efficient engine, the net heat flow into the engine equals the net work done, so $Q = W$, and the first law ($\Delta U = 0$) is obeyed. For the workless refrigerator, no net work is done (so $W = 0$) and as much heat flows into it as out (so $Q = 0$), so again $Q = W = 0$ and $\Delta U = 0$ in accordance with the first law. It is the *second* law of thermodynamics that tells us that both the 100%-efficient engine and the workless refrigerator are impossible.

20.6 THE CARNOT CYCLE

According to the second law, no heat engine can have 100% efficiency. How great an efficiency *can* an engine have, given two heat reservoirs at temperatures T_H and T_C ? This question was answered in 1824 by the French engineer Sadi Carnot (1796–1832), who developed a hypothetical, idealized heat engine that has the maximum possible efficiency consistent with the second law. The cycle of this engine is called the **Carnot cycle**.

To understand the rationale of the Carnot cycle, we return to *reversibility* and its relationship to directions of thermodynamic processes. Conversion of work to heat is an irreversible process; the purpose of a heat engine is a *partial* reversal of this process, the conversion of heat to work with as great an efficiency as possible. For maximum heat-engine efficiency, therefore, *we must avoid all irreversible processes* (Fig. 20.12).

Heat flow through a finite temperature drop is an irreversible process. Therefore, during heat transfer in the Carnot cycle there must be *no* finite temperature difference. When the engine takes heat from the hot reservoir at temperature T_H , the working substance of the engine must also be at T_H ; otherwise, irreversible heat flow would occur. Similarly, when the engine discards heat to the cold reservoir at T_C , the engine itself must be at T_C . That is, every process that involves heat transfer must be *isothermal* at either T_H or T_C .

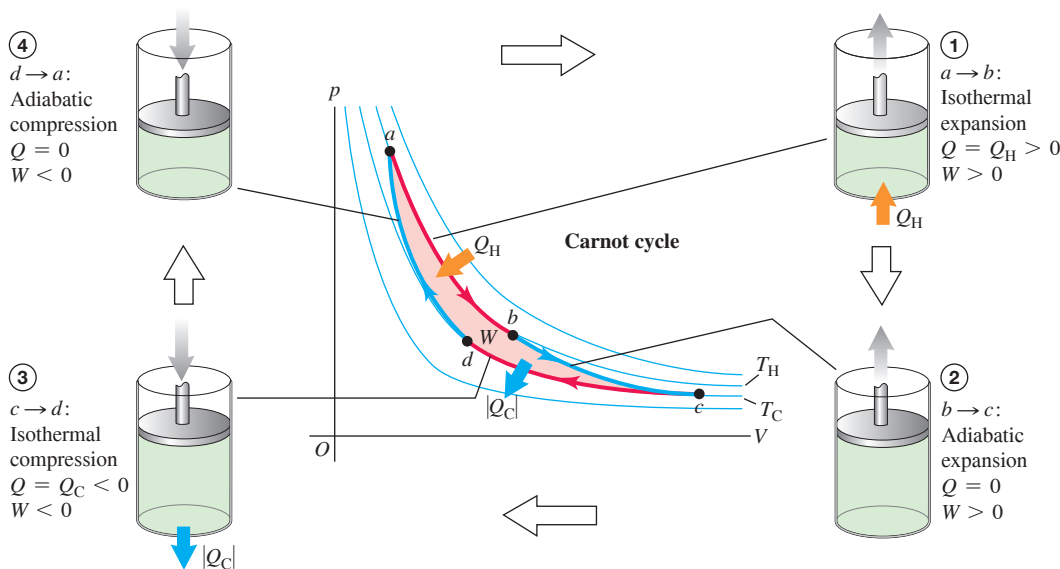
Conversely, in any process in which the temperature of the working substance of the engine is intermediate between T_H and T_C , there must be *no* heat transfer between the engine and either reservoir because such heat transfer could not be reversible. Therefore any process in which the temperature T of the working substance changes must be *adiabatic*.

The bottom line is that every process in our idealized cycle must be either isothermal or adiabatic. In addition, thermal and mechanical equilibrium must be maintained at all times so that each process is completely reversible.

Figure 20.12 The temperature of the firebox of a steam engine is much higher than the temperature of water in the boiler, so heat flows irreversibly from firebox to water. Carnot’s quest to understand the efficiency of steam engines led him to the idea that an ideal engine would involve only *reversible* processes.



Figure 20.13 The Carnot cycle for an ideal gas. The light blue lines in the pV -diagram are isotherms (curves of constant temperature) and the dark blue lines are adiabats (curves of zero heat flow).



Steps of the Carnot Cycle

The Carnot cycle consists of two reversible isothermal and two reversible adiabatic processes. **Figure 20.13** shows a Carnot cycle using as its working substance an ideal gas in a cylinder with a piston. It consists of the following steps:

1. The gas expands isothermally at temperature T_H , absorbing heat Q_H (ab).
2. It expands adiabatically until its temperature drops to T_C (bc).
3. It is compressed isothermally at T_C , rejecting heat $|Q_C|$ (cd).
4. It is compressed adiabatically back to its initial state at temperature T_H (da).

We can calculate the thermal efficiency e of a Carnot engine in the special case shown in Fig. 20.13 in which the working substance is an *ideal gas*. To do this, we'll first find the ratio Q_C/Q_H of the quantities of heat transferred in the two isothermal processes and then use Eq. (20.4) to find e .

For an ideal gas the internal energy U depends only on temperature and is thus constant in any isothermal process. For the isothermal expansion ab , $\Delta U_{ab} = 0$ and Q_H is equal to the work W_{ab} done by the gas during its isothermal expansion at temperature T_H . We calculated this work in Example 19.1 (Section 19.2); using that result, we have

$$Q_H = W_{ab} = nRT_H \ln \frac{V_b}{V_a} \quad (20.10)$$

Similarly,

$$Q_C = W_{cd} = nRT_C \ln \frac{V_d}{V_c} = -nRT_C \ln \frac{V_c}{V_d} \quad (20.11)$$

Because V_d is less than V_c , Q_C is negative ($Q_C = -|Q_C|$); heat flows out of the gas during the isothermal compression at temperature T_C .

The ratio of the two quantities of heat is thus

$$\frac{Q_C}{Q_H} = -\left(\frac{T_C}{T_H}\right) \frac{\ln(V_c/V_d)}{\ln(V_b/V_a)} \quad (20.12)$$

This can be simplified further by use of the temperature–volume relationship for an adiabatic process, Eq. (19.22). We find for the two adiabatic processes:

$$T_H V_b^{\gamma-1} = T_C V_c^{\gamma-1} \quad \text{and} \quad T_H V_a^{\gamma-1} = T_C V_d^{\gamma-1}$$

Dividing the first of these by the second, we find

$$\frac{V_b^{\gamma-1}}{V_a^{\gamma-1}} = \frac{V_c^{\gamma-1}}{V_d^{\gamma-1}} \quad \text{and} \quad \frac{V_b}{V_a} = \frac{V_c}{V_d}$$

Thus the two logarithms in Eq. (20.12) are equal, and that equation reduces to

$$\frac{Q_C}{Q_H} = -\frac{T_C}{T_H} \quad \text{or} \quad \frac{|Q_C|}{|Q_H|} = \frac{T_C}{T_H} \quad (\text{heat transfer in a Carnot engine}) \quad (20.13)$$

The ratio of the heat rejected at T_C to the heat absorbed at T_H is just equal to the ratio T_C/T_H . Then from Eq. (20.4) the efficiency of the Carnot engine is

$$\text{Efficiency of a Carnot engine } e_{\text{Carnot}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H} \quad (20.14)$$

Temperature of hot reservoir Temperature of cold reservoir

This simple result says that the efficiency of a Carnot engine depends only on the temperatures of the two heat reservoirs. The efficiency is large when the temperature *difference* is large, and it is very small when the temperatures are nearly equal. The efficiency can never be exactly unity unless $T_C = 0$; we'll see later that this, too, is impossible.

CAUTION Use Kelvin temperature in all Carnot calculations In Carnot-cycle calculations, you must use *absolute* (Kelvin) temperatures only. That's because Eqs. (20.10) through (20.14) come from the ideal-gas equation $pV = nRT$, in which T is absolute temperature. ■

EXAMPLE 20.2 Analyzing a Carnot engine I

WITH VARIATION PROBLEMS

A Carnot engine takes 2000 J of heat from a reservoir at 500 K, does some work, and discards some heat to a reservoir at 350 K. How much work does it do, how much heat is discarded, and what is its efficiency?

IDENTIFY and SET UP This problem involves a Carnot engine, so we can use the ideas of this section and those of Section 20.2 (which apply to heat engines of all kinds). **Figure 20.14** shows the energy-flow diagram. We have $Q_H = 2000$ J, $T_H = 500$ K, and $T_C = 350$ K. We use Eq. (20.13) to find Q_C , and then use the first law of thermodynamics as given by Eq. (20.2) to find W . We find the efficiency e from T_C and T_H from Eq. (20.14).

EXECUTE From Eq. (20.13),

$$Q_C = -Q_H \frac{T_C}{T_H} = -(2000 \text{ J}) \frac{350 \text{ K}}{500 \text{ K}} = -1400 \text{ J}$$

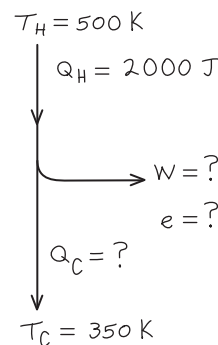
Then from Eq. (20.2), the work done is

$$W = Q_H + Q_C = 2000 \text{ J} + (-1400 \text{ J}) = 600 \text{ J}$$

From Eq. (20.14), the thermal efficiency is

$$e = 1 - \frac{T_C}{T_H} = 1 - \frac{350 \text{ K}}{500 \text{ K}} = 0.30 = 30\%$$

Figure 20.14 Our sketch for this problem.



EVALUATE The negative sign of Q_C is correct: It shows that 1400 J of heat flows *out* of the engine and into the cold reservoir. We can check our result for e by using the basic definition of thermal efficiency, Eq. (20.3):

$$e = \frac{W}{Q_H} = \frac{600 \text{ J}}{2000 \text{ J}} = 0.30 = 30\%$$

KEYCONCEPT An engine that utilizes the Carnot cycle, which consists of only reversible isothermal and reversible adiabatic processes, has the greatest efficiency possible consistent with the second law of thermodynamics. The smaller the ratio of the temperature of the cold reservoir to the temperature of the hot reservoir, the greater the Carnot efficiency.

EXAMPLE 20.3 Analyzing a Carnot engine II

WITH VARIATION PROBLEMS

Suppose 0.200 mol of an ideal diatomic gas ($\gamma = 1.40$) undergoes a Carnot cycle between 227°C and 27°C, starting at $p_a = 10.0 \times 10^5$ Pa at point a in the pV -diagram of Fig. 20.13. The volume doubles during the isothermal expansion step $a \rightarrow b$. (a) Find the pressure and volume at points a , b , c , and d . (b) Find Q , W , and ΔU for each step and for the entire cycle. (c) Find the efficiency directly from the results of part (b), and compare with the value calculated from Eq. (20.14).

IDENTIFY and SET UP This problem involves the properties of the Carnot cycle and those of an ideal gas. We are given the number of moles n and the pressure and temperature at point a (which is at the higher of the two reservoir temperatures); we can find the volume at a from the ideal-gas equation $pV = nRT$. We then find the pressure and volume at points b , c , and d from the known doubling of volume in step $a \rightarrow b$, from equations given in this section, and from $pV = nRT$. In each step we use Eqs. (20.10) and (20.11) to find the heat flow and work done and Eq. (19.13) to find the internal energy change.

EXECUTE (a) With $T_H = (227 + 273.15) \text{ K} = 500 \text{ K}$ and $T_C = (27 + 273.15) \text{ K} = 300 \text{ K}$, $pV = nRT$ yields

$$V_a = \frac{nRT_H}{p_a} = \frac{(0.200 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(500 \text{ K})}{10.0 \times 10^5 \text{ Pa}} \\ = 8.31 \times 10^{-4} \text{ m}^3$$

The volume doubles during the isothermal expansion $a \rightarrow b$:

$$V_b = 2V_a = 2(8.31 \times 10^{-4} \text{ m}^3) = 16.6 \times 10^{-4} \text{ m}^3$$

Because the expansion $a \rightarrow b$ is isothermal, $p_a V_a = p_b V_b$, so

$$p_b = \frac{p_a V_a}{V_b} = 5.00 \times 10^5 \text{ Pa}$$

For the adiabatic expansion $b \rightarrow c$, we use the equation $T_H V_b^{\gamma-1} = T_C V_c^{\gamma-1}$ that follows Eq. (20.12) as well as the ideal-gas equation:

$$V_c = V_b \left(\frac{T_H}{T_C} \right)^{1/(\gamma-1)} = (16.6 \times 10^{-4} \text{ m}^3) \left(\frac{500 \text{ K}}{300 \text{ K}} \right)^{2.5} \\ = 59.6 \times 10^{-4} \text{ m}^3 \\ p_c = \frac{nRT_C}{V_c} = \frac{(0.200 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(300 \text{ K})}{59.6 \times 10^{-4} \text{ m}^3} \\ = 0.837 \times 10^5 \text{ Pa}$$

For the adiabatic compression $d \rightarrow a$, we have $T_C V_d^{\gamma-1} = T_H V_a^{\gamma-1}$ and so

$$V_d = V_a \left(\frac{T_H}{T_C} \right)^{1/(\gamma-1)} = (8.31 \times 10^{-4} \text{ m}^3) \left(\frac{500 \text{ K}}{300 \text{ K}} \right)^{2.5} \\ = 29.8 \times 10^{-4} \text{ m}^3 \\ p_d = \frac{nRT_C}{V_d} = \frac{(0.200 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(300 \text{ K})}{29.8 \times 10^{-4} \text{ m}^3} \\ = 1.67 \times 10^5 \text{ Pa}$$

(b) For the isothermal expansion $a \rightarrow b$, $\Delta U_{ab} = 0$. From Eq. (20.10),

$$W_{ab} = Q_H = nRT_H \ln \frac{V_b}{V_a} \\ = (0.200 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(500 \text{ K})(\ln 2) = 576 \text{ J}$$

For the adiabatic expansion $b \rightarrow c$, $Q_{bc} = 0$. From the first law of thermodynamics, $\Delta U_{bc} = Q_{bc} - W_{bc} = -W_{bc}$; the work W_{bc} done by the gas in this adiabatic expansion equals the negative of the change in internal energy of the gas. From Eq. (19.13) we have $\Delta U = nC_V \Delta T$, where $\Delta T = T_C - T_H$. Using $C_V = 20.8 \text{ J/mol} \cdot \text{K}$ for an ideal diatomic gas, we find

$$W_{bc} = -\Delta U_{bc} = -nC_V(T_C - T_H) = nC_V(T_H - T_C) \\ = (0.200 \text{ mol})(20.8 \text{ J/mol} \cdot \text{K})(500 \text{ K} - 300 \text{ K}) = 832 \text{ J}$$

For the isothermal compression $c \rightarrow d$, $\Delta U_{cd} = 0$; Eq. (20.11) gives

$$W_{cd} = Q_C = nRT_C \ln \frac{V_d}{V_c} \\ = (0.200 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(300 \text{ K}) \left(\ln \frac{29.8 \times 10^{-4} \text{ m}^3}{59.6 \times 10^{-4} \text{ m}^3} \right) \\ = -346 \text{ J}$$

For the adiabatic compression $d \rightarrow a$, $Q_{da} = 0$ and

$$W_{da} = -\Delta U_{da} = -nC_V(T_H - T_C) = nC_V(T_C - T_H) \\ = (0.200 \text{ mol})(20.8 \text{ J/mol} \cdot \text{K})(300 \text{ K} - 500 \text{ K}) = -832 \text{ J}$$

We can tabulate these results as follows:

Process	Q	W	ΔU
$a \rightarrow b$	576 J	576 J	0
$b \rightarrow c$	0	832 J	-832 J
$c \rightarrow d$	-346 J	-346 J	0
$d \rightarrow a$	0	-832 J	832 J
Total	230 J	230 J	0

(c) From the above table, $Q_H = 576 \text{ J}$ and the total work is 230 J. Thus

$$e = \frac{W}{Q_H} = \frac{230 \text{ J}}{576 \text{ J}} = 0.40 = 40\%$$

We can compare this to the result from Eq. (20.14),

$$e = \frac{T_H - T_C}{T_H} = \frac{500 \text{ K} - 300 \text{ K}}{500 \text{ K}} = 0.40 = 40\%$$

EVALUATE The table in part (b) shows that for the entire cycle $Q = W$ and $\Delta U = 0$, just as we would expect: In a complete cycle, the *net* heat input is used to do work, and there is zero net change in the internal energy of the system. Note also that the quantities of work in the two adiabatic processes are negatives of each other. Can you show from the analysis leading to Eq. (20.13) that this must *always* be the case in a Carnot cycle?

KEYCONCEPT When analyzing any of the steps taken by a heat engine as it goes through a thermodynamic cycle, if you can calculate any two of the quantities ΔU , Q , and W , you can calculate the third using the first law of thermodynamics, $\Delta U = Q - W$. The efficiency of the engine equals the sum of the work done in all steps divided by the heat taken in from the hot reservoir.

The Carnot Refrigerator

Because each step in the Carnot cycle is reversible, the *entire cycle* may be reversed, converting the engine into a refrigerator. The coefficient of performance of the Carnot refrigerator is obtained by combining the general definition of K , Eq. (20.9), with Eq. (20.13) for the Carnot cycle. We first rewrite Eq. (20.9) as

$$K = \frac{|Q_C|}{|Q_H| - |Q_C|} \\ = \frac{|Q_C|/|Q_H|}{1 - |Q_C|/|Q_H|}$$

Then we substitute Eq. (20.13), $|Q_C|/|Q_H| = T_C/T_H$, into this expression:

$$\text{Coefficient of performance of a Carnot refrigerator} \rightarrow K_{\text{Carnot}} = \frac{T_C}{T_H - T_C} \quad (20.15)$$

Temperature of cold reservoir
Temperature of hot reservoir

When the temperature difference $T_H - T_C$ is small, K is much larger than unity; in this case a lot of heat can be “pumped” from the lower to the higher temperature with only a little expenditure of work. But the greater the temperature difference, the smaller the value of K and the more work is required to transfer a given quantity of heat.

EXAMPLE 20.4 Analyzing a Carnot refrigerator

WITH VARIATION PROBLEMS

If the cycle described in Example 20.3 is run backward as a refrigerator, what is its coefficient of performance?

IDENTIFY and SET UP This problem uses the ideas of Section 20.3 (for refrigerators in general) and the above discussion of Carnot refrigerators. Equation (20.9) gives the coefficient of performance K of *any* refrigerator in terms of the heat Q_C extracted from the cold reservoir per cycle and the work W that must be done per cycle.

EXECUTE In Example 20.3 we found that in one cycle the Carnot engine rejects heat $Q_C = -346$ J to the cold reservoir and does work $W = 230$ J. When run in reverse as a refrigerator, the system extracts heat $Q_C = +346$ J from the cold reservoir while requiring a work input of $W = -230$ J. From Eq. (20.9),

$$K = \frac{|Q_C|}{|W|} = \frac{346 \text{ J}}{230 \text{ J}} \\ = 1.50$$

Because this is a Carnot cycle, we can also use Eq. (20.15):

$$K = \frac{T_C}{T_H - T_C} = \frac{300 \text{ K}}{500 \text{ K} - 300 \text{ K}} \\ = 1.50$$

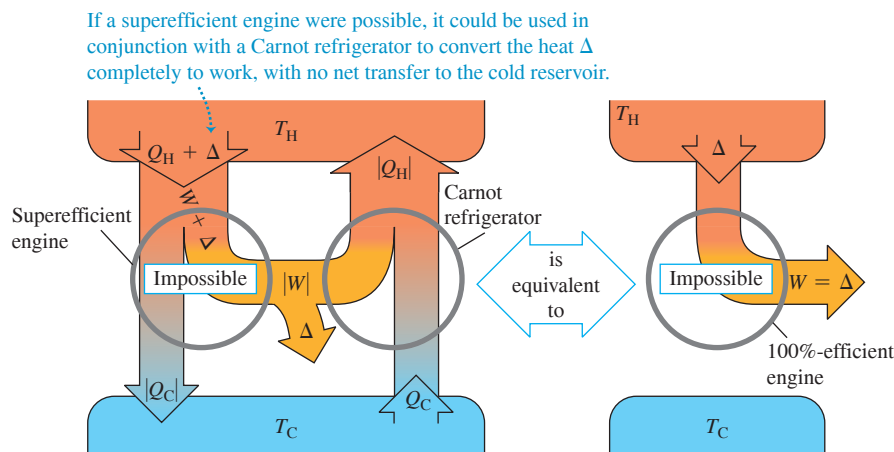
EVALUATE Equations (20.14) and (20.15) show that e and K for a Carnot cycle depend only on T_H and T_C , and we don’t need to calculate Q and W . For cycles containing irreversible processes, however, these two equations are not valid, and more detailed calculations are necessary.

KEYCONCEPT A refrigerator takes heat in from a cold reservoir and rejects heat to a hot reservoir. Work must be done on the refrigerator to make this happen. The coefficient of performance K of the refrigerator equals the amount of heat that is rejected to the hot reservoir divided by the amount of work that must be done on the refrigerator. The greater the value of K , the more heat can be removed from the cold reservoir for a given expenditure of work.

The Carnot Cycle and the Second Law

We can prove that **no engine can be more efficient than a Carnot engine operating between the same two temperatures**. The key to the proof is the above observation that since each step in the Carnot cycle is reversible, the *entire cycle* may be reversed. Run backward, the engine becomes a refrigerator. Suppose we have an engine that is more

Figure 20.15 Proving that the Carnot engine has the highest possible efficiency. A “superefficient” engine (more efficient than a Carnot engine) combined with a Carnot refrigerator could convert heat completely into work with no net heat transfer to the cold reservoir. This would violate the second law of thermodynamics.



efficient than a Carnot engine (Fig. 20.15). Let the Carnot engine, run backward as a refrigerator by negative work $-|W|$, take in heat Q_C from the cold reservoir and expel heat $|Q_H|$ to the hot reservoir. The superefficient engine expels heat $|Q_C|$, but to do this, it takes in a greater amount of heat $Q_H + \Delta$. Its work output is then $W + \Delta$, and the net effect of the two machines together is to take a quantity of heat Δ and convert it completely into work. This violates the “engine” statement of the second law. We could construct a similar argument that a superefficient engine could be used to violate the “refrigerator” statement of the second law. Note that we don’t have to assume that the superefficient engine is reversible. In a similar way we can show that *no refrigerator can have a greater coefficient of performance than a Carnot refrigerator operating between the same two temperatures*.

Thus the statement that no engine can be more efficient than a Carnot engine is yet another equivalent statement of the second law of thermodynamics. It also follows directly that **all Carnot engines operating between the same two temperatures have the same efficiency, irrespective of the nature of the working substance**. Although we derived Eq. (20.14) for a Carnot engine by using an ideal gas as its working substance, it is in fact valid for *any* Carnot engine, no matter what its working substance.

Equation (20.14), the expression for the efficiency of a Carnot engine, sets an upper limit to the efficiency of a real engine such as a steam turbine. To maximize this upper limit and the actual efficiency of the real engine, the designer must make the intake temperature T_H as high as possible and the exhaust temperature T_C as low as possible (Fig. 20.16).

The exhaust temperature cannot be lower than the lowest temperature available for cooling the exhaust. For a steam turbine at an electric power plant, T_C may be the temperature of river or lake water; then we want the boiler temperature T_H to be as high as possible. The vapor pressures of all liquids increase rapidly with temperature, so we are limited by the mechanical strength of the boiler. At 500°C the vapor pressure of water is about $240 \times 10^5 \text{ Pa}$ (235 atm); this is about the maximum practical pressure in large present-day steam boilers.

The Kelvin Temperature Scale

In Chapter 17 we expressed the need for a temperature scale that doesn’t depend on the properties of any particular material. We can now use the Carnot cycle to define such a scale. The thermal efficiency of a Carnot engine operating between two heat reservoirs at temperatures T_H and T_C is independent of the nature of the working substance and depends only on the temperatures. From Eq. (20.4), this thermal efficiency is

$$e = \frac{Q_H + Q_C}{Q_H} = 1 + \frac{Q_C}{Q_H}$$

Figure 20.16 To maximize efficiency, the temperatures inside a jet engine are made as high as possible. Exotic ceramic materials are used that can withstand temperatures in excess of 1000°C without melting or becoming soft.



Therefore the ratio Q_C/Q_H is the same for *all* Carnot engines operating between two given temperatures T_H and T_C .

Kelvin proposed that we *define* the ratio of the temperatures, T_C/T_H , to be equal to the magnitude of the ratio Q_C/Q_H of the quantities of heat absorbed and rejected:

$$\frac{T_C}{T_H} = \frac{|Q_C|}{|Q_H|} = -\frac{Q_C}{Q_H} \quad (\text{definition of Kelvin temperature}) \quad (20.16)$$

Equation (20.16) looks identical to Eq. (20.13), but there is a subtle and crucial difference. The temperatures in Eq. (20.13) are based on an ideal-gas thermometer, as defined in Section 17.3, while Eq. (20.16) *defines* a temperature scale based on the Carnot cycle and the second law of thermodynamics and is independent of the behavior of any particular substance. Thus the **Kelvin temperature scale** is truly *absolute*. To complete the definition of the Kelvin scale, we assign, as in Section 17.3, the arbitrary value of 273.16 K to the temperature of the triple point of water. When a substance is taken around a Carnot cycle, the ratio of the heats absorbed and rejected, $|Q_H|/|Q_C|$, is equal to the ratio of the temperatures of the reservoirs *as expressed on the gas-thermometer scale* defined in Section 17.3. Since the triple point of water is chosen to be 273.16 K in both scales, it follows that *the Kelvin and ideal-gas scales are identical*.

The zero point on the Kelvin scale is called **absolute zero**. At absolute zero a system has its *minimum* possible total internal energy (kinetic plus potential). Because of quantum effects, however, it is *not* true that at $T = 0$, all molecular motion ceases. There are theoretical reasons for believing that absolute zero cannot be attained experimentally, although temperatures below 10^{-8} K have been achieved. The more closely we approach absolute zero, the more difficult it is to get closer. One statement of the *third law of thermodynamics* is that it is impossible to reach absolute zero in a finite number of thermodynamic steps.

TEST YOUR UNDERSTANDING OF SECTION 20.6 An inventor looking for financial support comes to you with an idea for a gasoline engine that runs on a novel type of thermodynamic cycle. His design is made entirely of copper and is air-cooled. He claims that the engine will be 85% efficient. Should you invest in this marvelous new engine? (*Hint:* See Table 17.4.)

ANSWER 85% efficiency. You should invest your money elsewhere. The efficiency can be no better than that of a Carnot engine running between the same two temperature limits, $e_{\text{Carnot}} = 1 - (T_C/T_H)$ [Eq. (20.14)]. The temperature T_C of the cold reservoir for this air-cooled engine is about 300 K (ambient temperature), and the temperature T_H of the hot reservoir cannot exceed the melting point of copper, 1356 K (see Table 17.4). Hence the maximum possible Carnot efficiency is $e = 1 - (300 \text{ K})/(1356 \text{ K}) = 0.78$, or 78%. The temperature of any real engine would be less than this, so it would be impossible for the inventor's engine to attain 85% efficiency.

CAUTION Changes to the Kelvin scale

As we mentioned in Sections 17.3 and 18.3, in 2019 the Kelvin scale was redefined to be based not on the triple point of water but rather on the values of the joule ($1 \text{ J} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$) and the Boltzmann constant. Despite this change in definition, the Kelvin and ideal-gas scales remain identical. **|**

20.7 ENTROPY

The second law of thermodynamics, as we have stated it, is not an equation or a quantitative relationship but rather a statement of *impossibility*. However, the second law *can* be stated as a quantitative relationship with the concept of *entropy*, the subject of this section.

We have talked about several processes that proceed naturally in the direction of increasing randomness. Irreversible heat flow increases randomness: The molecules are initially sorted into hotter and cooler regions, but this sorting is lost when the system comes to thermal equilibrium. Adding heat to an object also increases average molecular speeds; therefore, molecular motion becomes more random. In the free expansion of a gas, the molecules have greater randomness of position after the expansion than before. **Figure 20.17** shows another process in which randomness increases.

Figure 20.17 When firecrackers explode, randomness increases: The neatly packaged chemicals within each firecracker are dispersed in all directions, and the stored chemical energy is converted to random kinetic energy of the fragments.



Entropy and Randomness

Entropy provides a *quantitative* measure of randomness. To introduce this concept, let's consider an infinitesimal isothermal expansion of an ideal gas. We add heat dQ and let the gas expand just enough to keep the temperature constant. Because the internal energy of an ideal gas depends on only its temperature, the internal energy is also constant; thus from the first law, the work dW done by the gas is equal to the heat dQ added. That is,

$$dQ = dW = p dV = \frac{nRT}{V} dV \quad \text{so} \quad \frac{dV}{V} = \frac{dQ}{nRT}$$

The gas is more disordered after the expansion than before: The molecules are moving in a larger volume and have more randomness of position. Thus the fractional volume change dV/V is a measure of the increase in randomness, and the above equation shows that it is proportional to the quantity dQ/T . We introduce the symbol S for the entropy of the system, and we define the infinitesimal entropy change dS during an infinitesimal reversible process at absolute temperature T as

$$dS = \frac{dQ}{T} \quad (\text{infinitesimal reversible process}) \quad (20.17)$$

If a total amount of heat Q is added during a reversible isothermal process at absolute temperature T , the total entropy change $\Delta S = S_2 - S_1$ is given by

$$\Delta S = S_2 - S_1 = \frac{Q}{T} \quad (\text{reversible isothermal process}) \quad (20.18)$$

Entropy has units of energy divided by temperature; the SI unit of entropy is 1 J/K.

We can see how the quotient Q/T is related to the increase in randomness. Higher temperature means greater randomness of motion. If the substance is initially cold, with little molecular motion, adding heat Q causes a substantial fractional increase in molecular motion and randomness. But if the substance is already hot, the same quantity of heat adds relatively little to the greater molecular motion already present. In the first case the quantity Q/T is large because T in the denominator is small; in the second case Q/T is small because T is large. So Q/T is an appropriate characterization of the increase in randomness when heat flows into a system.

CAUTION Be mindful of the signs of Q and ΔS When calculating the entropy change for a reversible process, remember that Q is positive if heat flows into a system and negative if heat flows out of a system. So the entropy of a system increases (ΔS is positive) when heat flows into it, and the entropy decreases (ΔS is negative) when heat flows out of it. ■

EXAMPLE 20.5 Entropy change in melting

What is the change of entropy of 1 kg of ice that is melted reversibly at 0°C and converted to water at 0°C ? The heat of fusion of water is $L_f = 3.34 \times 10^5 \text{ J/kg}$.

IDENTIFY and SET UP The melting occurs at a constant temperature $T = 0^\circ\text{C} = 273 \text{ K}$, so this is an *isothermal* reversible process. We can calculate the added heat Q required to melt the ice, then calculate the entropy change ΔS from Eq. (20.18).

EXECUTE The heat needed to melt the ice is $Q = mL_f = 3.34 \times 10^5 \text{ J}$. Then from Eq. (20.18),

$$\Delta S = S_2 - S_1 = \frac{Q}{T} = \frac{3.34 \times 10^5 \text{ J}}{273 \text{ K}} = 1.22 \times 10^3 \text{ J/K}$$

EVALUATE This entropy increase corresponds to the increase in disorder when the water molecules go from the state of a crystalline solid to the much more randomly arranged state of a liquid. In *any* isothermal reversible process, the entropy change equals the heat transferred divided by the absolute temperature. When we refreeze the water, Q has the opposite sign, and the entropy change is $\Delta S = -1.22 \times 10^3 \text{ J/K}$. The water molecules rearrange themselves into a crystal to form ice, so both randomness and entropy decrease.

KEYCONCEPT To find the entropy change of a system that goes through a reversible thermodynamic process at a constant temperature T , divide the heat Q that flows into the system during the process by T . If heat flows *in* (Q is positive), the entropy and disorder of the system *increase*; if heat flows *out* (Q is negative), the entropy and disorder of the system *decrease*.

Entropy in Reversible Processes

We can generalize the definition of entropy change to include *any* reversible process leading from one state to another, whether it is isothermal or not. We represent the process as a series of infinitesimal reversible steps. During a typical step, an infinitesimal quantity of heat dQ is added to the system at absolute temperature T . Then we sum (integrate) the quotients dQ/T for the entire process; that is,

$$\Delta S = \int_1^2 \frac{dQ}{T} \quad (20.19)$$

Entropy change in a reversible process ΔS Upper limit = final state
 Lower limit = initial state Infinitesimal heat flow into system
 Absolute temperature

Because entropy is a measure of the randomness of a system in any specific state, it must depend only on the current state of the system, not on its past history. (We'll verify this later.) When a system proceeds from an initial state with entropy S_1 to a final state with entropy S_2 , the change in entropy $\Delta S = S_2 - S_1$ defined by Eq. (20.19) does not depend on the path leading from the initial to the final state but is the same for *all possible* processes leading from state 1 to state 2. Thus the entropy of a system must also have a definite value for any given state of the system. *Internal energy*, introduced in Chapter 19, also has this property, although entropy and internal energy are very different quantities.

Since entropy is a function only of the state of a system, we can also compute entropy changes in *irreversible* (nonequilibrium) processes for which Eqs. (20.17) and (20.19) are not applicable. We simply invent a path connecting the given initial and final states that *does* consist entirely of reversible equilibrium processes and compute the total entropy change for that path. It is not the actual path, but the entropy change must be the same as for the actual path.

As with internal energy, the above discussion does not tell us how to calculate entropy itself, but only the change in entropy in any given process. Just as with internal energy, we may arbitrarily assign a value to the entropy of a system in a specified reference state and then calculate the entropy of any other state with reference to this.

EXAMPLE 20.6 Entropy change in a temperature change

WITH VARIATION PROBLEMS

One kilogram of water at 0°C is heated to 100°C . Compute its change in entropy. Assume that the specific heat of water is constant at $4190 \text{ J/kg} \cdot \text{K}$ over this temperature range.

IDENTIFY and SET UP The entropy change of the water depends only on the initial and final states of the system, no matter whether the process is reversible or irreversible. We can imagine a reversible process in which the water temperature is increased in a sequence of infinitesimal steps dT . We can use Eq. (20.19) to integrate over all these steps and calculate the entropy change for such a reversible process. (Heating the water on a stove whose cooking surface is maintained at 100°C would be an irreversible process. The entropy change would be the same, however.)

EXECUTE From Eq. (17.14) the heat required to carry out each infinitesimal step is $dQ = mc \, dT$. Substituting this into Eq. (20.19) and integrating, we find

$$\begin{aligned} \Delta S &= S_2 - S_1 = \int_1^2 \frac{dQ}{T} = \int_{T_1}^{T_2} mc \frac{dT}{T} = mc \ln \frac{T_2}{T_1} \\ &= (1.00 \text{ kg})(4190 \text{ J/kg} \cdot \text{K}) \left(\ln \frac{373 \text{ K}}{273 \text{ K}} \right) = 1.31 \times 10^3 \text{ J/K} \end{aligned}$$

EVALUATE The entropy change is positive, as it must be for a process in which the system absorbs heat. Our assumption about the specific heat is a pretty good one, since c for water varies by less than 1% between 0°C and 100°C (see Fig. 17.17).

CAUTION When $\Delta S = Q/T$ can (and cannot) be used In solving this problem you might be tempted to avoid doing an integral by using the simpler expression in Eq. (20.18), $\Delta S = Q/T$. This would be incorrect, however, because Eq. (20.18) is applicable only to *isothermal* processes, and the initial and final temperatures in our example are *not* the same. The *only* correct way to find the entropy change in a process with different initial and final temperatures is to use Eq. (20.19). I

KEYCONCEPT The entropy change ΔS of a thermodynamic system does not depend on the path taken between the initial and final states of the system. You can calculate ΔS for any initial and final states by analyzing a *reversible* process that takes the system between these states; in general this involves evaluating an integral.

CONCEPTUAL EXAMPLE 20.7 Entropy change in a reversible adiabatic process

WITH  VARIATION PROBLEMS

A gas expands adiabatically and reversibly. What is its change in entropy?

SOLUTION In an adiabatic process, no heat enters or leaves the system. Hence $dQ = 0$ and there is *no* change in entropy in this reversible process: $\Delta S = 0$. Every *reversible* adiabatic process is a constant-entropy process. (That's why such processes are also called *isentropic*

processes.) The increase in randomness resulting from the gas occupying a greater volume is exactly balanced by the decrease in randomness associated with the lowered temperature and reduced molecular speeds.

KEYCONCEPT In the special case where a system undergoes a reversible adiabatic process, the system undergoes *no* change in entropy.

EXAMPLE 20.8 Entropy change in a free expansion

WITH  VARIATION PROBLEMS

A partition divides a thermally insulated box into two compartments, each of volume V (Fig. 20.18). Initially, one compartment contains n moles of an ideal gas at temperature T , and the other compartment is evacuated. We break the partition and the gas expands, filling both compartments. What is the entropy change in this free-expansion process?

IDENTIFY and SET UP For this process, $Q = 0$, $W = 0$, $\Delta U = 0$, and therefore (because the system is an ideal gas) $\Delta T = 0$. We might think that the entropy change is zero because there is no heat exchange. But Eq. (20.19) can be used to calculate entropy changes for *reversible* processes only; this free expansion is *not* reversible, and there *is* an entropy change. As we mentioned at the beginning of this section, entropy increases in a free expansion because the positions of the molecules are more random than before the expansion. To calculate ΔS , we recall

that the entropy change depends only on the initial and final states. We can devise a *reversible* process having the same endpoints as this free expansion, and in general we can then use Eq. (20.19) to calculate its entropy change, which will be the same as for the free expansion. An appropriate reversible process is an *isothermal* expansion from V to $2V$ at temperature T , which allows us to use the simpler Eq. (20.18) to calculate ΔS . The gas does work W during this expansion, so an equal amount of heat Q must be supplied to keep the internal energy constant.

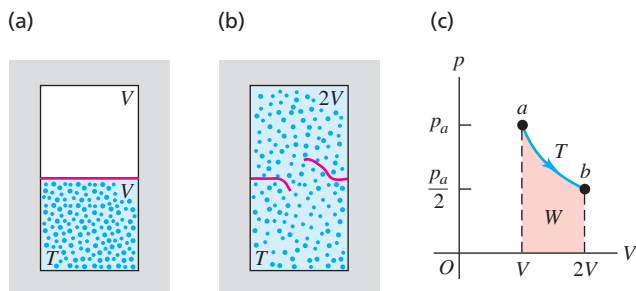
EXECUTE We saw in Example 19.1 that the work done by n moles of ideal gas in an isothermal expansion from V_1 to V_2 is $W = nRT \ln(V_2/V_1)$. With $V_1 = V$ and $V_2 = 2V$, we have

$$Q = W = nRT \ln \frac{2V}{V} = nRT \ln 2$$

From Eq. (20.18), the entropy change is

$$\Delta S = \frac{Q}{T} = nR \ln 2$$

Figure 20.18 (a, b) Free expansion of an insulated ideal gas. (c) The free-expansion process doesn't pass through equilibrium states from a to b . However, the entropy change $S_b - S_a$ can be calculated by using the isothermal path shown or *any* reversible path from a to b .



EVALUATE For 1 mole, $\Delta S = (1 \text{ mol})(8.314 \text{ J/mol} \cdot \text{K})(\ln 2) = 5.76 \text{ J/K}$. The entropy change is positive, as we predicted. The factor $(\ln 2)$ in our answer is a result of the volume having increased by a factor of 2, from V to $2V$. Can you show that if the volume increases in a free expansion from V to xV , where x is an arbitrary number, the entropy change is $\Delta S = nR \ln x$?

KEYCONCEPT To calculate the entropy change of a system in an *irreversible* process such as a free expansion, you must consider a *reversible* process that takes the system from the same initial state to the same final state.

EXAMPLE 20.9 Entropy and the Carnot cycle

WITH  VARIATION PROBLEMS

For the Carnot engine in Example 20.2 (Section 20.6), what is the total entropy change during one cycle?

IDENTIFY and SET UP All four steps in the Carnot cycle (see Fig. 20.13) are reversible, so we can use our expressions for the entropy change ΔS in a reversible process. We find ΔS for each step and add them to get ΔS for the complete cycle.

EXECUTE There is no entropy change during the adiabatic expansion $b \rightarrow c$ or the adiabatic compression $d \rightarrow a$. During the isothermal expansion $a \rightarrow b$ at $T_H = 500 \text{ K}$, the engine takes in 2000 J of heat, and from Eq. (20.18),

$$\Delta S_H = \frac{Q_H}{T_H} = \frac{2000 \text{ J}}{500 \text{ K}} = 4.0 \text{ J/K}$$

During the isothermal compression $c \rightarrow d$ at $T_C = 350 \text{ K}$, the engine gives off 1400 J of heat, and

$$\begin{aligned} \Delta S_C &= \frac{Q_C}{T_C} = \frac{-1400 \text{ J}}{350 \text{ K}} \\ &= -4.0 \text{ J/K} \end{aligned}$$

The total entropy change in the engine during one cycle is $\Delta S_{\text{tot}} = \Delta S_H + \Delta S_C = 4.0 \text{ J/K} + (-4.0 \text{ J/K}) = 0$.

EVALUATE The result $\Delta S_{\text{total}} = 0$ tells us that when the Carnot engine completes a cycle, it has the same entropy as it did at the beginning of the cycle. We'll explore this result in the next subsection.

What is the total entropy change of the engine's *environment* during this cycle? During the reversible isothermal expansion $a \rightarrow b$, the hot (500 K) reservoir gives off 2000 J of heat, so its entropy change is $(-2000 \text{ J})/(500 \text{ K}) = -4.0 \text{ J/K}$. During the reversible isothermal compression $c \rightarrow d$, the cold (350 K) reservoir absorbs 1400 J of heat, so its entropy change is $(+1400 \text{ J})/(350 \text{ K}) = +4.0 \text{ J/K}$. Thus the hot and cold reservoirs each have an entropy change, but the sum of these changes—that is, the total entropy change of the system's environment—is zero.

These results apply to the special case of the Carnot cycle, for which *all* of the processes are reversible. In this case the total entropy

change of the system and the environment together is zero. We'll see that if the cycle includes irreversible processes (as is the case for the Otto and Diesel cycles of Section 20.3), the total entropy change of the system and the environment *cannot* be zero, but rather must be positive.

KEYCONCEPT For the special case of a heat engine that uses the Carnot cycle, the net entropy change of the engine and its environment (the hot and cold reservoirs) in a complete cycle is zero. This is not the case for other, less ideal cycles.

Entropy in Cyclic Processes

Example 20.9 showed that the total entropy change for a cycle of a particular Carnot engine, which uses an ideal gas as its working substance, is zero. This result follows directly from Eq. (20.13), which we can rewrite as

$$\frac{Q_H}{T_H} + \frac{Q_C}{T_C} = 0 \quad (20.20)$$

The quotient Q_H/T_H equals ΔS_H , the entropy change of the engine that occurs at $T = T_H$. Likewise, Q_C/T_C equals ΔS_C , the (negative) entropy change of the engine that occurs at $T = T_C$. Hence Eq. (20.20) says that $\Delta S_H + \Delta S_C = 0$; that is, there is zero net entropy change in one cycle.

What about Carnot engines that use a different working substance? According to the second law, *any* Carnot engine operating between given temperatures T_H and T_C has the same efficiency $e = 1 - T_C/T_H$ [Eq. (20.14)]. Combining this expression for e with Eq. (20.4), $e = 1 + Q_C/Q_H$, just reproduces Eq. (20.20). So Eq. (20.20) is valid for any Carnot engine working between these temperatures, whether its working substance is an ideal gas or not. We conclude that *the total entropy change in one cycle of any Carnot engine is zero*.

This result can be generalized to show that the total entropy change during *any* reversible cyclic process is zero. A reversible cyclic process appears on a pV -diagram as a closed path (**Fig. 20.19a**). We can approximate such a path as closely as we like by a sequence of isothermal and adiabatic processes forming parts of many long, thin Carnot cycles (Fig. 20.19b). The total entropy change for the full cycle is the sum of the entropy changes for each small Carnot cycle, each of which is zero. So **the total entropy change during any reversible cycle is zero**:

$$\int \frac{dQ}{T} = 0 \quad (\text{reversible cyclic process}) \quad (20.21)$$

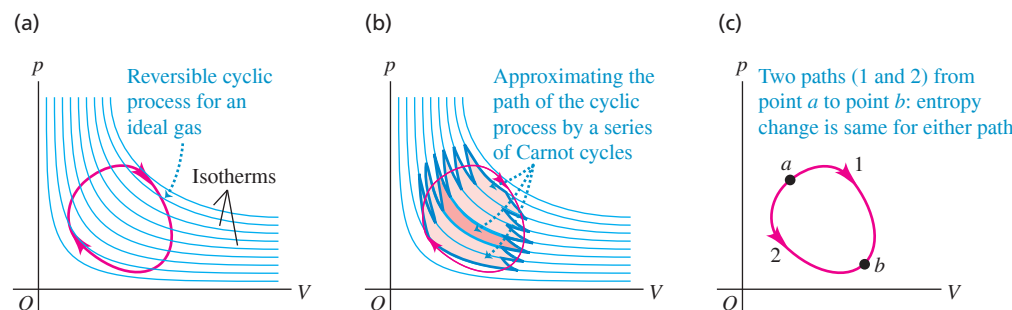


Figure 20.19 (a) A reversible cyclic process for an ideal gas is shown as a red closed path on a pV -diagram. Several ideal-gas isotherms are shown in blue. (b) We can approximate the path in (a) by a series of long, thin Carnot cycles; one of these is highlighted in color. The total entropy change is zero for each Carnot cycle and for the actual cyclic process. (c) The entropy change between points a and b is independent of the path.

BIO APPLICATION Entropy Changes in a Living Organism

When a puppy or other growing animal eats, it takes organized chemical energy from the food and uses it to make new cells that are even more highly organized. This process alone lowers entropy. But most of the energy in the food is either excreted in the animal's feces or used to generate heat, processes that lead to a large increase in entropy. So while the entropy of the animal alone decreases, the *total* entropy of animal plus food *increases*.



It follows that when a system undergoes a reversible process leading from any state *a* to any other state *b*, the *entropy change of the system is independent of the path* (Fig. 20.19c). If the entropy change for path 1 were different from the change for path 2, the system could be taken along path 1 and then backward along path 2 to the starting point, with a nonzero net change in entropy. This would violate the conclusion that the total entropy change in such a cyclic process must be zero. Because the entropy change in such processes is independent of path, we conclude that in any given state, the system has a definite value of entropy that depends only on the state, not on the processes that led to that state.

Entropy in Irreversible Processes

In an idealized, reversible process involving only equilibrium states, the total entropy change of the system and its surroundings is zero. But all *irreversible* processes involve an increase in entropy. Unlike energy, *entropy is not a conserved quantity*. The entropy of an isolated system *can* change, but as we shall see, it can never decrease. The free expansion of a gas (Example 20.8) is an irreversible process in an isolated system in which there is an entropy increase.

EXAMPLE 20.10 Entropy change in an irreversible process**WITH VARIATION PROBLEMS**

Suppose 1.00 kg of water at 100°C is placed in thermal contact with 1.00 kg of water at 0°C. What is the total change in entropy? Assume that the specific heat of water is constant at 4190 J/kg · K over this temperature range.

IDENTIFY and SET UP This process involves irreversible heat flow because of the temperature differences. There are equal masses of 0°C water and 100°C water, so the final temperature is the average of these two temperatures: 50°C = 323 K. Although the processes are irreversible, we can calculate the entropy changes for the (initially) hot water and the (initially) cold water by assuming that the process occurs reversibly. As in Example 20.6, we must use Eq. (20.19) to calculate ΔS for each substance because the temperatures are not constant.

EXECUTE The entropy changes of the hot water (subscript H) and the cold water (subscript C) are

$$\begin{aligned}\Delta S_H &= mc \int_{T_1}^{T_2} \frac{dT}{T} = (1.00 \text{ kg})(4190 \text{ J/kg} \cdot \text{K}) \int_{373 \text{ K}}^{323 \text{ K}} \frac{dT}{T} \\ &= (4190 \text{ J/K}) \left(\ln \frac{323 \text{ K}}{373 \text{ K}} \right) = -603 \text{ J/K} \\ \Delta S_C &= (4190 \text{ J/K}) \left(\ln \frac{323 \text{ K}}{273 \text{ K}} \right) = +705 \text{ J/K}\end{aligned}$$

The *total* entropy change of the system is

$$\Delta S_{\text{tot}} = \Delta S_H + \Delta S_C = (-603 \text{ J/K}) + 705 \text{ J/K} = +102 \text{ J/K}$$

EVALUATE An irreversible heat flow in an isolated system is accompanied by an increase in entropy. We could reach the same end state by mixing the hot and cold water, which is also an irreversible process; the total entropy change, which depends only on the initial and final states of the system, would again be 102 J/K.

Note that the entropy of the system increases *continuously* as the two quantities of water come to equilibrium. For example, the first 4190 J of heat transferred cools the hot water to 99°C and warms the cold water to 1°C. The net change in entropy for this step is approximately

$$\Delta S = \frac{-4190 \text{ J}}{373 \text{ K}} + \frac{4190 \text{ J}}{273 \text{ K}} = +4.1 \text{ J/K}$$

Can you show in a similar way that the net entropy change is positive for *any* one-degree temperature change leading to the equilibrium condition?

KEYCONCEPT When heat flows irreversibly from an object at higher temperature to one at lower temperature, the entropy of the high-temperature object decreases and the entropy of the low-temperature object increases. The *net* entropy of the two objects together always increases.

Entropy and the Second Law

The results of Example 20.10 about the flow of heat from a higher to a lower temperature are characteristic of *all* natural (that is, irreversible) processes. When we include the entropy changes of all the systems taking part in the process, the increases in entropy are always greater than the decreases. In the special case of a *reversible* process, the increases



Figure 20.20 The mixing of colored ink and water starts from a state of low entropy in which each fluid is separate and distinct from the other. In the final state, both the ink and water molecules are spread randomly throughout the volume of liquid, so the entropy is greater. Spontaneous unmixing of the ink and water, a process in which there would be a net decrease in entropy, is never observed.

and decreases are equal. Hence we can state the general principle: **When all systems taking part in a process are included, the entropy either remains constant or increases.** This gives us a third, alternative way of stating the second law of thermodynamics:

SECOND LAW OF THERMODYNAMICS (“entropy” statement) No process is possible in which the total entropy decreases, when all systems that take part in the process are included.

This is equivalent to the “engine” and “refrigerator” statements that we discussed in Section 20.5. **Figure 20.20** shows a specific example of this general principle.

The increase of entropy in every natural, irreversible process measures the increase of randomness in the universe associated with that process. Consider again the example of mixing hot and cold water (Example 20.10). We *might* have used the hot and cold water as the high- and low-temperature reservoirs of a heat engine. While removing heat from the hot water and giving heat to the cold water, we could have obtained some mechanical work. But once the hot and cold water have been mixed and have come to a uniform temperature, this opportunity to convert heat to mechanical work is lost irretrievably. The lukewarm water will never *unmix* itself and separate into hotter and colder portions. No decrease in *energy* occurs when the hot and cold water are mixed. What has been lost is the *opportunity* to convert part of the heat from the hot water into mechanical work. Hence when entropy increases, energy becomes less *available*, and the universe becomes more random or “run down.”

TEST YOUR UNDERSTANDING OF SECTION 20.7 Suppose 2.00 kg of water at 50°C spontaneously changes temperature, so that half of the water cools to 0°C while the other half spontaneously warms to 100°C. (All of the water remains liquid, so it doesn’t freeze or boil.) What would be the entropy change of the water? Is this process possible? (*Hint:* See Example 20.10.)

ANSWER

The result violates the second law of thermodynamics, which states that the entropy of an isolated system cannot decrease. **–102 J/K, no** The process described is exactly the opposite of the process used in Example 20.10.

20.8 MICROSCOPIC INTERPRETATION OF ENTROPY

We described in Section 19.4 how the internal energy of a system could be calculated, at least in principle, by adding up all the kinetic energies of its constituent particles and all the potential energies of interaction among the particles. This is called a *microscopic* calculation of the internal energy. We can also make a microscopic calculation of the entropy S of a system. Unlike energy, however, entropy is not something

Figure 20.21 All possible microscopic states of four coins. There can be several possible microscopic states for each macroscopic state.

Macroscopic state	Corresponding microscopic states
Four heads	
Three heads, one tails	
Two heads, two tails	
One heads, three tails	
Four tails	

that belongs to each individual particle or pair of particles in the system. Rather, entropy is a measure of the randomness of the system as a whole. To see how to calculate entropy microscopically, we first have to introduce the idea of *macroscopic* and *microscopic states*.

Suppose you toss N identical coins on the floor, and half of them show heads and half show tails. This is a description of the large-scale or **macroscopic state** of the system of N coins. A description of the **microscopic state** of the system includes information about each individual coin: Coin 1 was heads, coin 2 was tails, coin 3 was tails, and so on. There can be many microscopic states that correspond to the same macroscopic description. For instance, with $N = 4$ coins there are six possible states in which half are heads and half are tails (**Fig. 20.21**). The number of microscopic states grows rapidly with increasing N ; for $N = 100$ there are $2^{100} = 1.27 \times 10^{30}$ microscopic states, of which 1.01×10^{29} are half heads and half tails.

The least probable outcomes of the coin toss are the states that are either all heads or all tails. It is certainly possible that you could throw 100 heads in a row, but don't bet on it; the probability of doing this is only 1 in 1.27×10^{30} . The most probable outcome of tossing N coins is that half are heads and half are tails. The reason is that this *macroscopic* state has the greatest number of corresponding *microscopic* states, as **Fig. 20.21** shows.

To make the connection to the concept of entropy, note that the macroscopic description “all heads” completely specifies the state of each one of the N coins. The same is true if the coins are all tails. But the macroscopic description “half heads, half tails” by itself tells you very little about the state (heads or tails) of each individual coin. Compared to the state “all heads” or “all tails,” the state “half heads, half tails” has much greater *randomness* because the system could be in any of a much greater number of possible microscopic states. Hence the “half heads, half tails” state has much greater entropy (which is a quantitative measure of randomness).

Now instead of N coins, consider a mole of an ideal gas containing Avogadro's number of molecules. The macroscopic state of this gas is given by its pressure p , volume V , and temperature T ; a description of the microscopic state involves stating the position and velocity for each molecule in the gas. At a given pressure, volume, and temperature, the gas may be in any one of an astronomically large number of microscopic states, depending on the positions and velocities of its 6.02×10^{23} molecules. If the gas undergoes a free expansion into a greater volume, the range of possible positions increases, as does the number of possible microscopic states. The system becomes more random, and the entropy increases as calculated in Example 20.8 (Section 20.7).

We can draw the following general conclusion: **For any thermodynamic system, the most probable macroscopic state is the one with the greatest number of corresponding microscopic states, which is also the macroscopic state with the greatest randomness and the greatest entropy.**

Calculating Entropy: Microscopic States

Let w represent the number of possible microscopic states for a given macroscopic state. (For the four coins shown in **Fig. 20.21** the state of four heads has $w = 1$, the state of three heads and one tails has $w = 4$, and so on.) Then the entropy S of a macroscopic state can be shown to be given by

$$S = k \ln w \quad (20.22)$$

Expression for entropy in microscopic terms Boltzmann constant (gas constant per molecule) Number of microscopic states for the given macroscopic state

(We introduced the Boltzmann constant in Section 18.3.) As **Eq. (20.22)** shows, increasing the number of possible microscopic states w increases the entropy S .

What matters in a thermodynamic process is not the absolute entropy S but the *difference* in entropy between the initial and final states. Hence an equally valid and useful definition would be $S = k \ln w + C$, where C is a constant, since C cancels in any calculation of an entropy difference between two states. But it's convenient to set this constant equal to zero and use Eq. (20.22). With this choice, since the smallest possible value of w is unity, the smallest possible value of S for any system is $k \ln 1 = 0$. Entropy can *never* be negative.

In practice, calculating w is a difficult task, so Eq. (20.22) is typically used only to calculate the absolute entropy S of certain special systems. But we can use this relationship to calculate *differences* in entropy between one state and another. Consider a system that undergoes a thermodynamic process that takes it from macroscopic state 1, for which there are w_1 possible microscopic states, to macroscopic state 2, with w_2 associated microscopic states. The change in entropy in this process is

$$\Delta S = S_2 - S_1 = k \ln w_2 - k \ln w_1 = k \ln \frac{w_2}{w_1} \quad (20.23)$$

The *difference* in entropy between the two macroscopic states depends on the *ratio* of the numbers of possible microscopic states.

As the following example shows, using Eq. (20.23) to calculate a change in entropy from one macroscopic state to another gives the same results as considering a reversible process connecting those two states and using Eq. (20.19).

EXAMPLE 20.11 A microscopic calculation of entropy change

Use Eq. (20.23) to calculate the entropy change in the free expansion of n moles of gas at temperature T described in Example 20.8 (Fig. 20.22).

IDENTIFY and SET UP We are asked to calculate the entropy change by using the number of microscopic states in the initial and final macroscopic states (Figs. 20.22a and b). When the partition is broken, no work is done, so the velocities of the molecules are unaffected. But each molecule now has twice as much volume in which it can move and hence has twice the number of possible positions. This is all we need to calculate the entropy change using Eq. (20.23).

EXECUTE Let w_1 be the number of microscopic states of the system as a whole when the gas occupies volume V (Fig. 20.22a). The number of molecules is $N = nN_A$, and each of these N molecules has twice as many possible states after the partition is broken. Hence the number w_2 of microscopic states when the gas occupies volume $2V$ (Fig. 20.22b) is greater by a factor of 2^N ; that is, $w_2 = 2^N w_1$. The change in entropy in this process is

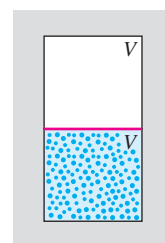
$$\begin{aligned} \Delta S &= k \ln \frac{w_2}{w_1} = k \ln \frac{2^N w_1}{w_1} \\ &= k \ln 2^N = Nk \ln 2 \end{aligned}$$

Since $N = nN_A$ and $k = R/N_A$, this becomes

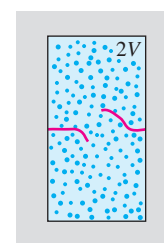
$$\begin{aligned} \Delta S &= (nN_A)(R/N_A) \ln 2 \\ &= nR \ln 2 \end{aligned}$$

Figure 20.22 In a free expansion of N molecules in which the volume doubles, the number of possible microscopic states increases by a factor of 2^N .

(a) Gas occupies volume V ; number of microscopic states $= w_1$.



(b) Gas occupies volume $2V$; number of microscopic states $= w_2 = 2^N w_1$.



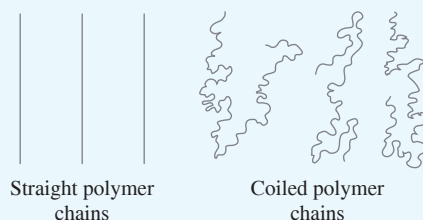
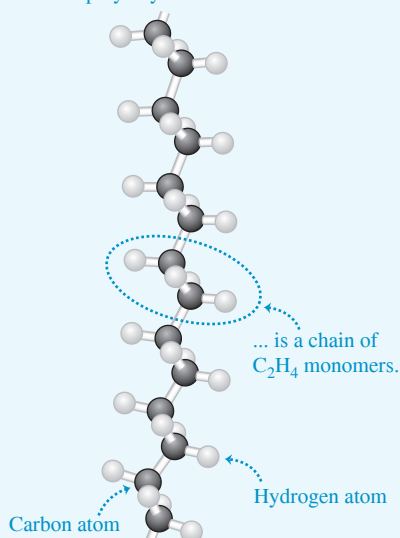
EVALUATE We found the same result as in Example 20.8, but without any reference to the thermodynamic path taken.

KEYCONCEPT An alternative way to calculate entropy changes is in terms of the number of microscopic states available to the system. If there are w possible microscopic states of the system for a given macroscopic state, the entropy for the macroscopic state equals the Boltzmann constant k multiplied by the natural logarithm of w .

APPLICATION Polymers Coil in

Solution A molecule of polyethylene, the most common plastic, is a polymer—a long chain of monomer units (C_2H_4). In solution these molecules coil on themselves, and the entropy concept explains why. The polymer can coil in many ways (microscopic states), but there is only one microscopic state in which the polymer is fully stretched out. Thus the entropy of the coiled polymer is much greater than that of a stretched-out polymer. The second law of thermodynamics says that isolated systems always move toward greater entropy, so we expect a polymer chain in solution to be in a coiled state.

A polyethylene molecule ...

**Microscopic States and the Second Law**

The relationship between entropy and the number of microscopic states gives us new insight into the entropy statement of the second law of thermodynamics: that the entropy of a closed system can never decrease. From Eq. (20.22) this means that a closed system can never spontaneously undergo a process that decreases the number of possible microscopic states.

An example of such a forbidden process would be if all of the air in your room spontaneously moved to one half of the room, leaving a vacuum in the other half. Such a “free compression” would be the reverse of the free expansion of Examples 20.8 and 20.11. This would decrease the number of possible microscopic states by a factor of 2^N . Strictly speaking, this process is not impossible! The probability of finding a given molecule in one half of the room is $\frac{1}{2}$, so the probability of finding all of the molecules in one half of the room at once is $(\frac{1}{2})^N$. (This is exactly the same as the probability of having a tossed coin come up heads N times in a row.) This probability is *not* zero. But lest you worry about suddenly finding yourself gasping for breath in the evacuated half of your room, consider that a typical room might hold 1000 moles of air, and so $N = 1000N_A = 6.02 \times 10^{26}$ molecules. The probability of all the molecules being in the same half of the room is therefore $(\frac{1}{2})^{6.02 \times 10^{26}}$. Expressed as a decimal, this number has more than 10^{26} zeros to the right of the decimal point!

Because the probability of such a “free compression” taking place is so vanishingly small, it has almost certainly never occurred anywhere in the universe since the beginning of time. We conclude that for all practical purposes the second law of thermodynamics is never violated.

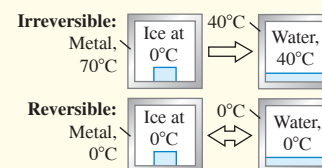
TEST YOUR UNDERSTANDING OF SECTION 20.8 A quantity of N molecules of an ideal gas initially occupies volume V . The gas then expands to volume $2V$. The number of microscopic states of the gas increases in this expansion. Under which of the following circumstances will this number increase the most? (i) If the expansion is reversible and isothermal; (ii) if the expansion is reversible and adiabatic; (iii) the number will change by the same amount for both circumstances.

ANSWER

(i) For case (i), we saw in Example 20.8 (Section 20.7) that for an ideal gas, the entropy change in a free expansion is the same as in an isothermal expansion. From Eq. (20.23), this implies that the ratio of the number of microscopic states after and before the expansion, w_2/w_1 , is also the same for these two cases. From Example 20.11, $w_2/w_1 = 2^N$, so the number of microscopic states increases by a factor 2^N . For case (ii), in a reversible expansion the entropy change is $\Delta S = \int dQ/T = 0$; if the expansion is adiabatic there is no heat flow, so $\Delta S = 0$. From Eq. (20.23), $w_2/w_1 = 1$ and there is *no* change in the number of microscopic states. The difference is that in an adiabatic expansion the temperature drops and the molecules move more slowly, so they have fewer microscopic states available to them than in an isothermal expansion.

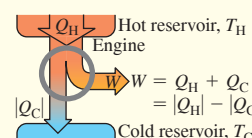
CHAPTER 20 SUMMARY

Reversible and irreversible processes: A reversible process is one whose direction can be reversed by an infinitesimal change in the conditions of the process, and in which the system is always in or very close to thermal equilibrium. All other thermodynamic processes are irreversible.



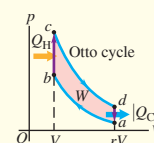
Heat engines: A heat engine takes heat Q_H from a source, converts part of it to work W , and discards the remainder $|Q_C|$ at a lower temperature. The thermal efficiency e of a heat engine measures how much of the absorbed heat is converted to work. (See Example 20.1.)

$$e = \frac{W}{Q_H} = 1 + \frac{Q_C}{Q_H} = 1 - \left| \frac{Q_C}{Q_H} \right| \quad (20.4)$$



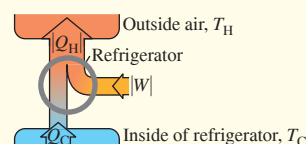
The Otto cycle: A gasoline engine operating on the Otto cycle has a theoretical maximum thermal efficiency e that depends on the compression ratio r and the ratio of heat capacities γ of the working substance.

$$e = 1 - \frac{1}{r^{\gamma-1}} \quad (20.6)$$

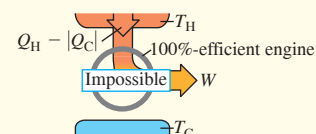


Refrigerators: A refrigerator takes heat Q_C from a colder place, has a work input $|W|$, and discards heat $|Q_H|$ at a warmer place. The effectiveness of the refrigerator is given by its coefficient of performance K .

$$K = \frac{|Q_C|}{|W|} = \frac{|Q_C|}{|Q_H| - |Q_C|} \quad (20.9)$$

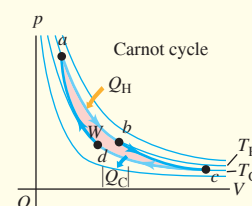


The second law of thermodynamics: The second law of thermodynamics describes the directionality of natural thermodynamic processes. It can be stated in two equivalent forms. The *engine* statement is that no cyclic process can convert heat completely into work. The *refrigerator* statement is that no cyclic process can transfer heat from a colder place to a hotter place with no input of mechanical work.



The Carnot cycle: The Carnot cycle operates between two heat reservoirs at temperatures T_H and T_C and uses only reversible processes. Its thermal efficiency depends only on T_H and T_C . An additional equivalent statement of the second law is that no engine operating between the same two temperatures can be more efficient than a Carnot engine. (See Examples 20.2 and 20.3.)

$$e_{\text{Carnot}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H} \quad (20.14)$$



A Carnot engine run backward is a Carnot refrigerator. Its coefficient of performance depends on only T_H and T_C . Another form of the second law states that no refrigerator operating between the same two temperatures can have a larger coefficient of performance than a Carnot refrigerator. (See Example 20.4.)

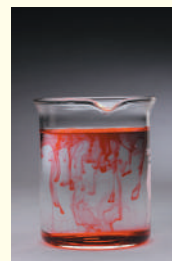
$$K_{\text{Carnot}} = \frac{T_C}{T_H - T_C} \quad (20.15)$$

Entropy: Entropy is a quantitative measure of the randomness of a system. The entropy change in any reversible process depends on the amount of heat flow and the absolute temperature T . Entropy depends only on the state of the system, and the change in entropy between given initial and final states is the same for all processes leading from one state to the other. This fact can be used to find the entropy change in an irreversible process. (See Examples 20.5–20.10.)

An important statement of the second law of thermodynamics is that the entropy of an isolated system may increase but can never decrease. When a system interacts with its surroundings, the total entropy change of system and surroundings can never decrease. When the interaction involves only reversible processes, the total entropy is constant and $\Delta S = 0$; when there is any irreversible process, the total entropy increases and $\Delta S > 0$.

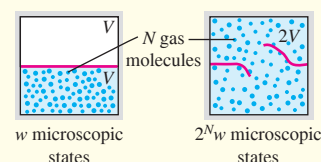
$$\Delta S = \int_1^2 \frac{dQ}{T} \quad (20.19)$$

(reversible process)



Entropy and microscopic states: When a system is in a particular macroscopic state, the particles that make up the system may be in any of w possible microscopic states. The greater the number w , the greater the entropy. (See Example 20.11.)

$$S = k \ln w \quad (20.22)$$



Chapter 20 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 20.1** (Section 20.2) before attempting these problems.

VP20.1.1 A diesel engine has efficiency 0.180. (a) In order for this engine to do 1.24×10^4 J of work, how many joules of heat must it take in? (b) How many joules of this heat is discarded?

VP20.1.2 In one cycle a heat engine absorbs 3.82×10^4 J of heat from the hot reservoir and rejects 3.16×10^4 J of heat to the cold reservoir. What is the efficiency of this engine?

VP20.1.3 Measurements of a gasoline engine show that it has an efficiency of 0.196 and that it exhausts 4.96×10^8 J of heat during 20 minutes of operation. During that time, (a) how much heat does the engine take in and (b) how much work does the engine do?

VP20.1.4 An aircraft piston engine that burns gasoline (heat of combustion 5.0×10^7 J/kg) has a power output of 1.10×10^5 W. (a) How much work does this engine do in 1.00 h? (b) This engine burns 34 kg of gasoline per hour. How much heat does the engine take in per hour? (c) What is the efficiency of the engine?

Be sure to review **EXAMPLES 20.2, 20.3 and 20.4** (Section 20.6) before attempting these problems.

VP20.4.1 In one cycle a Carnot engine takes in 8.00×10^4 J of heat and does 1.68×10^4 J of work. The temperature of the engine's cold reservoir is 25.0°C . (a) What is the efficiency of this engine? (b) How much heat does this engine exhaust per cycle? (c) What is the temperature (in $^\circ\text{C}$) of the hot reservoir?

VP20.4.2 For the Carnot cycle described in Example 20.3, you change the temperature of the cold reservoir from 27°C to -73°C . The initial pressure and volume at point a are unchanged, the volume still doubles

during the isothermal expansion $a \rightarrow b$, and the volume still decreases by one-half during the isothermal compression $c \rightarrow d$. For this modified cycle, calculate (a) the new efficiency of the cycle and (b) the amount of work done in each of the four steps of the cycle.

VP20.4.3 A Carnot refrigerator has a cold reservoir at -10.0°C and a hot reservoir at 25.0°C . (a) What is its coefficient of performance? (b) How much work input does this refrigerator require to remove 4.00×10^6 J of heat from the cold reservoir?

VP20.4.4 A Carnot engine uses the expansion and compression of n moles of argon gas, for which $C_V = \frac{3}{2}R$. This engine operates between temperatures T_C and T_H . During the isothermal expansion $a \rightarrow b$, the volume of the gas increases from V_a to $V_b = 2V_a$. (a) Calculate the work W_{ab} done during the isothermal expansion $a \rightarrow b$. Give your answer in terms of n , R , and T_H . (b) Calculate the work W_{bc} done during the adiabatic expansion $b \rightarrow c$. Give your answer in terms of n , R , T_C and T_H . (c) For this engine, $W_{ab} = W_{bc}$. Find the ratio T_C/T_H and the efficiency of the engine.

Be sure to review **EXAMPLES 20.6, 20.7, 20.8, 20.9, and 20.10** (Section 20.7) before attempting these problems.

VP20.10.1 Ethanol melts at 159 K (heat of fusion 1.042×10^5 J/kg) and boils at 351 K (heat of vaporization 8.54×10^5 J/kg). Liquid ethanol has a specific heat of 2428 J/kg $\cdot$ K (which we assume does not depend on temperature). If you have 1.00 kg of ethanol originally in the solid state at 159 K, calculate the change in entropy of the ethanol when it (a) melts at 159 K, (b) increases in temperature as a liquid from 159 K to 351 K, and (c) boils at 351 K.

VP20.10.2 Initially 5.00 mol of helium (which we can treat as an ideal gas) occupies volume 0.120 m³ and is at temperature 20.0°C . You allow

the helium to expand so that its final volume is 0.360 m^3 and its final temperature is also 20.0°C . Calculate the net change in entropy of the helium if (a) you make the helium expand isothermally from 0.120 m^3 to 0.360 m^3 and (b) you first increase the temperature of the helium so that it expands at constant pressure from 0.120 m^3 to 0.360 m^3 , then cool the helium at constant volume to 20.0°C .

VP20.10.3 In one cycle of operation, an engine that does *not* use the Carnot cycle takes in $8.00 \times 10^4 \text{ J}$ of heat from a reservoir at 260°C , does $1.60 \times 10^4 \text{ J}$ of work, and rejects $6.40 \times 10^4 \text{ J}$ of heat to a reservoir at 20°C . At the end of the cycle, the engine is in the same state as at the beginning of the cycle. Calculate the change in entropy in one cycle

for (a) the engine, (b) the reservoir at 260°C , (c) the reservoir at 20°C , and (d) the system of engine and two reservoirs. Does the net entropy of the system increase, decrease, or stay the same?

VP20.10.4 You combine 1.00 kg of ice (heat of fusion $3.34 \times 10^5 \text{ J/K}$) at 0.0°C and 0.839 kg of liquid water (specific heat $4.19 \times 10^3 \text{ J/K}$) at 95.0°C . When the system comes to equilibrium, all of the ice has melted and the temperature of the mixture is 0.0°C . Calculate the change in entropy for (a) the ice as it melts, (b) the water that was initially at 95.0°C , and (c) the combination of the two. Does the net entropy of the system increase, decrease, or stay the same?

BRIDGING PROBLEM Entropy Changes: Cold Ice in Hot Water

An insulated container of negligible mass holds 0.600 kg of water at 45.0°C . You put a 0.0500 kg ice cube at -15.0°C in the water (**Fig. 20.23**). (a) Calculate the final temperature of the water once the ice has melted. (b) Calculate the change in entropy of the system.

SOLUTION GUIDE

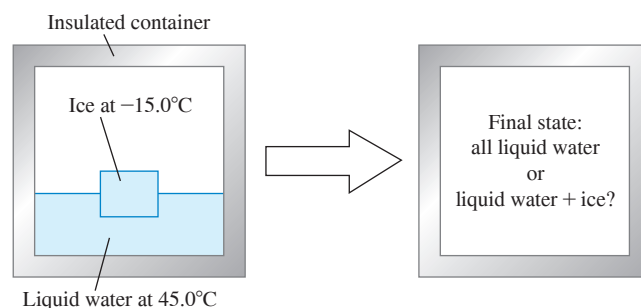
IDENTIFY and SET UP

1. Make a list of the known and unknown quantities, and identify the target variables.
2. How will you find the final temperature of the ice–water mixture? How will you decide whether or not all the ice melts?
3. Once you find the final temperature of the mixture, how will you determine the changes in entropy of (i) the ice initially at -15.0°C and (ii) the water initially at 45.0°C ?

EXECUTE

4. Use the methods of Chapter 17 to calculate the final temperature T . (*Hint*: First assume that all of the ice melts, then write an equation which says that the heat that flows into the ice equals the heat that flows out of the water. If your assumption is correct, the final temperature that you calculate will be greater than 0°C . If your assumption is incorrect, the final temperature will be 0°C or less, which means that some ice remains. You'll then need to redo the calculation to account for this.)

Figure 20.23 What becomes of this ice–water mixture?



5. Use your result from step 4 to calculate the entropy changes of the ice and the water. (*Hint*: You must include the heat flow associated with temperature changes, as in Example 20.6, as well as the heat flow associated with the change of phase.)
6. Find the total change in entropy of the system.

EVALUATE

7. Do the signs of the entropy changes make sense? Why or why not?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q20.1 A pot is half-filled with water, and a lid is placed on it, forming a tight seal so that no water vapor can escape. The pot is heated on a stove, forming water vapor inside the pot. The heat is then turned off and the water vapor condenses back to liquid. Is this cycle reversible or irreversible? Why?

Q20.2 Give two examples of reversible processes and two examples of irreversible processes in purely mechanical systems, such as blocks sliding on planes, springs, pulleys, and strings. Explain what makes each process reversible or irreversible.

Q20.3 Household refrigerators have arrays or coils of tubing on the outside, usually at the back or bottom. When the refrigerator is running, the tubing becomes quite hot. Where does the heat come from?

Q20.4 Suppose you try to cool the kitchen of your house by leaving the refrigerator door open. What happens? Why? Would the result be the same if you left open a picnic cooler full of ice? Explain the reason for any differences.

Q20.5 Why must a room air conditioner be placed in a window rather than just set on the floor and plugged in? Why can a refrigerator be set on the floor and plugged in?

Q20.6 Is it a violation of the second law of thermodynamics to convert mechanical energy completely into heat? To convert heat completely into work? Explain your answers.

Q20.7 Imagine a special air filter placed in a window of a house. The tiny holes in the filter allow only air molecules moving faster than a certain speed to exit the house, and allow only air molecules moving slower than that speed to enter the house from outside. Explain why such an air filter would cool the house, and why the second law of thermodynamics makes building such a filter an impossible task.

Q20.8 An electric motor has its shaft coupled to that of an electric generator. The motor drives the generator, and some current from the generator is used to run the motor. The excess current is used to light a home. What is wrong with this scheme?

Q20.9 When a wet cloth is hung up in a hot wind in the desert, it is cooled by evaporation to a temperature that may be 20°C or so below that of the air. Discuss this process in light of the second law of thermodynamics.

Q20.10 Compare the pV -diagram for the Otto cycle in Fig. 20.6 with the diagram for the Carnot heat engine in Fig. 20.13. Explain some of the important differences between the two cycles.

Q20.11 The efficiency of heat engines is high when the temperature difference between the hot and cold reservoirs is large. Refrigerators, on the other hand, work better when the temperature difference is small. Thinking of the mechanical refrigeration cycle shown in Fig. 20.9, explain in physical terms why it takes less work to remove heat from the working substance if the two reservoirs (the inside of the refrigerator and the outside air) are at nearly the same temperature, than if the outside air is much warmer than the interior of the refrigerator.

Q20.12 What would be the efficiency of a Carnot engine operating with $T_H = T_C$? What would be the efficiency if $T_C = 0\text{ K}$ and T_H were any temperature above 0 K ? Interpret your answers.

Q20.13 Real heat engines, like the gasoline engine in a car, always have some friction between their moving parts, although lubricants keep the friction to a minimum. Would a heat engine with completely frictionless parts be 100% efficient? Why or why not? Does the answer depend on whether or not the engine runs on the Carnot cycle? Again, why or why not?

Q20.14 Does a refrigerator full of food consume more power if the room temperature is 20°C than if it is 15°C ? Or is the power consumption the same? Explain your reasoning.

Q20.15 In Example 20.4, a Carnot refrigerator requires a work input of only 230 J to extract 346 J of heat from the cold reservoir. Doesn't this discrepancy imply a violation of the law of conservation of energy? Explain why or why not.

Q20.16 How can the thermal conduction of heat from a hot object to a cold object increase entropy when the same amount of heat that flows out of the hot object flows into the cold one?

Q20.17 Explain why each of the following processes is an example of increasing randomness: mixing hot and cold water; free expansion of a gas; irreversible heat flow; developing heat by mechanical friction. Are entropy increases involved in all of these? Why or why not?

Q20.18 The free expansion of an ideal gas is an adiabatic process and so no heat is transferred. No work is done, so the internal energy does not change. Thus, $Q/T = 0$, yet the randomness of the system and thus its entropy have increased after the expansion. Why does Eq. (20.19) not apply to this situation?

Q20.19 Are the earth and sun in thermal equilibrium? Are there entropy changes associated with the transmission of energy from the sun to the earth? Does radiation differ from other modes of heat transfer with respect to entropy changes? Explain your reasoning.

Q20.20 Suppose that you put a hot object in thermal contact with a cold object and observe (much to your surprise) that heat flows from the cold object to the hot object, making the cold one colder and the hot one hotter. Does this process necessarily violate the first law of thermodynamics? The second law of thermodynamics? Explain.

Q20.21 If you run a movie film backward, it is as if the direction of time were reversed. In the time-reversed movie, would you see processes that violate conservation of energy? Conservation of linear momentum? Would you see processes that violate the second law of thermodynamics? In each case, if law-breaking processes could occur, give some examples.

Q20.22 BIO Some critics of biological evolution claim that it violates the second law of thermodynamics, since evolution involves simple life forms developing into more complex and more highly ordered organisms. Explain why this is not a valid argument against evolution.

Q20.23 BIO A growing plant creates a highly complex and organized structure out of simple materials such as air, water, and trace minerals. Does this violate the second law of thermodynamics? Why or why not? What is the plant's ultimate source of energy? Explain.

EXERCISES

Section 20.2 Heat Engines

20.1 • A diesel engine performs 2200 J of mechanical work and discards 4300 J of heat each cycle. (a) How much heat must be supplied to the engine in each cycle? (b) What is the thermal efficiency of the engine?

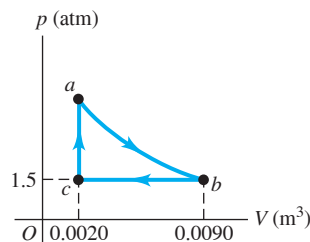
20.2 • An aircraft engine takes in 9000 J of heat and discards 6400 J each cycle. (a) What is the mechanical work output of the engine during one cycle? (b) What is the thermal efficiency of the engine?

20.3 • A Gasoline Engine. A gasoline engine takes in $1.61 \times 10^4\text{ J}$ of heat and delivers 3700 J of work per cycle. The heat is obtained by burning gasoline with a heat of combustion of $4.60 \times 10^4\text{ J/g}$. (a) What is the thermal efficiency? (b) How much heat is discarded in each cycle? (c) What mass of fuel is burned in each cycle? (d) If the engine goes through 60.0 cycles per second, what is its power output in kilowatts? In horsepower?

20.4 • A gasoline engine has a power output of 180 kW (about 241 hp). Its thermal efficiency is 28.0%. (a) How much heat must be supplied to the engine per second? (b) How much heat is discarded by the engine per second?

20.5 •• The pV -diagram in Fig. E20.5 shows a cycle of a heat engine that uses 0.250 mol of an ideal gas with $\gamma = 1.40$. Process ab is adiabatic. (a) Find the pressure of the gas at point a . (b) How much heat enters this gas per cycle, and where does it happen? (c) How much heat leaves this gas in a cycle, and where does it occur? (d) How much work does this engine do in a cycle? (e) What is the thermal efficiency of the engine?

Figure E20.5



20.6 •• CP A heat engine uses a large insulated tank of ice water as its cold reservoir. In 100 cycles the engine takes in 8000 J of heat energy from the hot reservoir and the rejected heat melts 0.0180 kg of ice in the tank. During these 100 cycles, how much work is performed by the engine?

Section 20.3 Internal-Combustion Engines

20.7 • The theoretical efficiency of an Otto cycle is 63%. The ratio of the C_p and C_V heat capacities for the gas in the engine is $\frac{7}{5}$. (a) What is the compression ratio r ? (b) If the net work done per cycle is 12.6 kJ, how much heat is rejected and how much is absorbed by the engine during each cycle?

20.8 • (a) Calculate the theoretical efficiency for an Otto-cycle engine with $\gamma = 1.40$ and $r = 9.50$. (b) If this engine takes in 10,000 J of heat from burning its fuel, how much heat does it discard to the outside air?

20.9 •• The Otto-cycle engine in a Mercedes-Benz SLK230 has a compression ratio of 8.8. (a) What is the ideal efficiency of the engine? Use $\gamma = 1.40$. (b) The engine in a Dodge Viper GT2 has a slightly higher compression ratio of 9.6. How much increase in the ideal efficiency results from this increase in the compression ratio?

Section 20.4 Refrigerators

20.10 •• A freezer has a coefficient of performance of 2.40. The freezer is to convert 1.80 kg of water at 25.0°C to 1.80 kg of ice at -5.0°C in one hour. (a) What amount of heat must be removed from the water at 25.0°C to convert it to ice at -5.0°C? (b) How much electrical energy is consumed by the freezer during this hour? (c) How much wasted heat is delivered to the room in which the freezer sits?

20.11 •• A refrigerator has a coefficient of performance of 2.25, runs on an input of 135 W of electrical power, and keeps its inside compartment at 5°C. If you put a dozen 1.0 L plastic bottles of water at 31°C into this refrigerator, how long will it take for them to be cooled down to 5°C? (Ignore any heat that leaves the plastic.)

20.12 • A refrigerator has a coefficient of performance of 2.10. In each cycle it absorbs 3.10×10^4 J of heat from the cold reservoir. (a) How much mechanical energy is required each cycle to operate the refrigerator? (b) During each cycle, how much heat is discarded to the high-temperature reservoir?

Section 20.6 The Carnot Cycle

20.13 •• In a Carnot engine the hot reservoir is 72.0°C warmer than the cold reservoir. The engine's efficiency is 12.5%. What are the Kelvin temperatures of the two reservoirs?

20.14 • A Carnot engine is operated between two heat reservoirs at temperatures of 520 K and 300 K. (a) If the engine receives 6.45 kJ of heat energy from the reservoir at 520 K in each cycle, how many joules per cycle does it discard to the reservoir at 300 K? (b) How much mechanical work is performed by the engine during each cycle? (c) What is the thermal efficiency of the engine?

20.15 • A Carnot engine whose high-temperature reservoir is at 620 K takes in 550 J of heat at this temperature in each cycle and gives up 335 J to the low-temperature reservoir. (a) How much mechanical work does the engine perform during each cycle? What is (b) the temperature of the low-temperature reservoir; (c) the thermal efficiency of the cycle?

20.16 •• An ice-making machine operates in a Carnot cycle. It takes heat from water at 0.0°C and rejects heat to a room at 24.0°C. Suppose that 85.0 kg of water at 0.0°C are converted to ice at 0.0°C. (a) How much heat is discharged into the room? (b) How much energy must be supplied to the device?

20.17 • A Carnot engine performs 2.5×10^4 J of work in each cycle and has an efficiency of 66%. (a) How much heat does the engine extract from its heat source in each cycle? (b) If the engine exhausts heat at room temperature (20.0°C), what is the temperature of its heat source?

20.18 •• A certain brand of freezer is advertised to use 730 kW·h of energy per year. (a) Assuming the freezer operates for 5 hours each day, how much power does it require while operating? (b) If the freezer keeps its interior at -5.0°C in a 20.0°C room, what is its theoretical maximum performance coefficient? (c) What is the theoretical maximum amount of ice this freezer could make in an hour, starting with water at 20.0°C?

20.19 • A Carnot refrigerator is operated between two heat reservoirs at temperatures of 320 K and 270 K. (a) If in each cycle the refrigerator receives 415 J of heat energy from the reservoir at 270 K, how many joules of heat energy does it deliver to the reservoir at 320 K? (b) If the refrigerator completes 165 cycles each minute, what power input is required to operate it? (c) What is the coefficient of performance of the refrigerator?

20.20 •• A Carnot heat engine uses a hot reservoir consisting of a large amount of boiling water and a cold reservoir consisting of a large tub of ice and water. In 5 minutes of operation, the heat rejected by the engine melts 0.0400 kg of ice. During this time, how much work W is performed by the engine?

20.21 •• Carnot refrigerator A has a 16% higher coefficient of performance than Carnot refrigerator B. The temperature difference between the hot and cold reservoirs is 30% greater for B than for A. If the cold-reservoir temperature for refrigerator B is 180 K, what is the cold-reservoir temperature for refrigerator A?

Section 20.7 Entropy

20.22 • CALC You decide to take a nice hot bath but discover that your thoughtless roommate has used up most of the hot water. You fill the tub with 195 kg of 30.0°C water and attempt to warm it further by pouring in 5.00 kg of boiling water from the stove. (a) Is this a reversible or an irreversible process? Use physical reasoning to explain. (b) Calculate the final temperature of the bath water. (c) Calculate the net change in entropy of the system (bath water + boiling water), assuming no heat exchange with the air or the tub itself.

20.23 • A sophomore with nothing better to do adds heat to 0.350 kg of ice at 0.0°C until it is all melted. (a) What is the change in entropy of the water? (b) The source of heat is a very massive object at 25.0°C. What is the change in entropy of this object? (c) What is the total change in entropy of the water and the heat source?

20.24 •• CALC You make tea with 0.250 kg of 85.0°C water and let it cool to room temperature (20.0°C). (a) Calculate the entropy change of the water while it cools. (b) The cooling process is essentially isothermal for the air in your kitchen. Calculate the change in entropy of the air while the tea cools, assuming that all of the heat lost by the water goes into the air. What is the total entropy change of the system tea + air?

20.25 • Three moles of an ideal gas undergo a reversible isothermal compression at 20.0°C. During this compression, 1850 J of work is done on the gas. What is the change of entropy of the gas?

20.26 •• What is the change in entropy of 0.130 kg of helium gas at the normal boiling point of helium when it all condenses isothermally to 1.00 L of liquid helium? (Hint: See Table 17.4 in Section 17.6.)

20.27 • (a) Calculate the change in entropy when 1.00 kg of water at 100°C is vaporized and converted to steam at 100°C (see Table 17.4). (b) Compare your answer to the change in entropy when 1.00 kg of ice is melted at 0°C, calculated in Example 20.5 (Section 20.7). Is the change in entropy greater for melting or for vaporization? Interpret your answer using the idea that entropy is a measure of the randomness of a system.

Section 20.8 Microscopic Interpretation of Entropy

20.28 • CALC Two moles of an ideal gas occupy a volume V . The gas expands isothermally and reversibly to a volume $3V$. (a) Is the velocity distribution changed by the isothermal expansion? Explain. (b) Use Eq. (20.23) to calculate the change in entropy of the gas. (c) Use Eq. (20.18) to calculate the change in entropy of the gas. Compare this result to that obtained in part (b).

20.29 • CALC A lonely party balloon with a volume of 2.40 L and containing 0.100 mol of air is left behind to drift in the temporarily uninhabited and depressurized International Space Station. Sunlight coming through a porthole heats and explodes the balloon, causing the air in it to undergo a free expansion into the empty station, whose total volume is 425 m³. Calculate the entropy change of the air during the expansion.

PROBLEMS

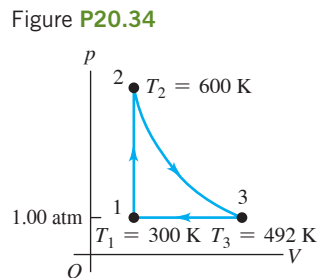
20.30 • CP As a budding mechanical engineer, you are called upon to design a Carnot engine that has 2.00 mol of a monatomic ideal gas as its working substance and operates from a high-temperature reservoir at 500°C. The engine is to lift a 15.0 kg weight 2.00 m per cycle, using 500 J of heat input. The gas in the engine chamber can have a minimum volume of 5.00 L during the cycle. (a) Draw a pV -diagram for this cycle. Show in your diagram where heat enters and leaves the gas. (b) What must be the temperature of the cold reservoir? (c) What is the thermal efficiency of the engine? (d) How much heat energy does this engine waste per cycle? (e) What is the maximum pressure that the gas chamber will have to withstand?

20.31 •• CP An ideal Carnot engine operates between 500°C and 100°C with a heat input of 250 J per cycle. (a) How much heat is delivered to the cold reservoir in each cycle? (b) What minimum number of cycles is necessary for the engine to lift a 500 kg rock through a height of 100 m?

20.32 •• BIO Entropy of Metabolism. An average sleeping person metabolizes at a rate of about 80 W by digesting food or burning fat. Typically, 20% of this energy goes into bodily functions, such as cell repair, pumping blood, and other uses of mechanical energy, while the rest goes to heat. Most people get rid of all this excess heat by transferring it (by conduction and the flow of blood) to the surface of the body, where it is radiated away. The normal internal temperature of the body (where the metabolism takes place) is 37°C, and the skin is typically 7°C cooler. By how much does the person's entropy change per second due to this heat transfer?

20.33 •• CP A certain heat engine operating on a Carnot cycle absorbs 410 J of heat per cycle at its hot reservoir at 135°C and has a thermal efficiency of 22.0%. (a) How much work does this engine do per cycle? (b) How much heat does the engine waste each cycle? (c) What is the temperature of the cold reservoir? (d) By how much does the engine change the entropy of the world each cycle? (e) What mass of water could this engine pump per cycle from a well 35.0 m deep?

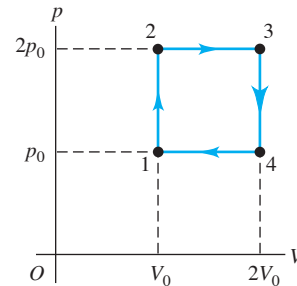
20.34 • A heat engine takes 0.350 mol of a diatomic ideal gas around the cycle shown in the pV -diagram of **Fig. P20.34**. Process 1→2 is at constant volume, process 2→3 is adiabatic, and process 3→1 is at a constant pressure of 1.00 atm. The value of γ for this gas is 1.40. (a) Find the pressure and volume at points 1, 2, and 3. (b) Calculate Q , W , and ΔU for each of the three processes. (c) Find the net work done by the gas in the cycle. (d) Find the net heat flow into the engine in one cycle. (e) What is the thermal efficiency of the engine? How does this compare to the efficiency of a Carnot-cycle engine operating between the same minimum and maximum temperatures T_1 and T_2 ?



20.35 •• BIO Entropy Change from Digesting Fat. Digesting fat produces 9.3 food calories per gram of fat, and typically 80% of this energy goes to heat when metabolized. (One food calorie is 1000 calories and therefore equals 4186 J.) The body then moves all this heat to the surface by a combination of thermal conductivity and motion of the blood. The internal temperature of the body (where digestion occurs) is normally 37°C, and the surface is usually about 30°C. By how much do the digestion and metabolism of a 2.50 g pat of butter change your body's entropy? Does it increase or decrease?

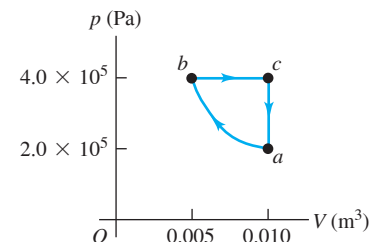
20.36 •• What is the thermal efficiency of an engine that takes n moles of diatomic ideal gas through the cycle 1→2→3→4→1 shown in **Fig. P20.36**?

Figure P20.36



20.37 •• CALC You build a heat engine that takes 1.00 mol of an ideal diatomic gas through the cycle shown in **Fig. P20.37**. (a) Show that process ab is an isothermal compression.

Figure P20.37



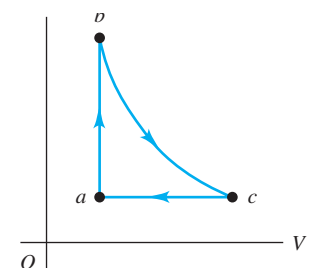
(b) During which process(es) of the cycle is heat absorbed by the gas? During which process(es) is heat rejected? How do you know? Calculate

(c) the temperature at points a , b , and c ; (d) the net heat exchanged with the surroundings and net work done by the engine in one cycle; (e) the thermal efficiency of the engine.

20.38 •• CALC One winter evening you fill a pint-sized plastic cup with ice and water to drink before going to bed. Estimate the volume of the ice and the volume of the water in the cup. Use the densities of ice and liquid water to find the mass of each. You set the cup on the kitchen counter, forget about it, and go to bed. The air in the kitchen is maintained at a constant 22.2°C by the central heating system of your apartment. When you wake up in the morning, the cup contains liquid water at 22.2°C. (a) What is the entropy change of the ice water during the process in which it went from 0°C to liquid water at 22.2°C as you slept? (b) What is the entropy change of the air in the room due to the heat that flows from it into the contents of the cup? (c) What is the total entropy change of the air-ice water system due to the heat exchange that changes the phase and temperature of the ice water? Is the total entropy change positive or negative?

20.39 • CALC A heat engine operates using the cycle shown in **Fig. P20.39**. The working substance is 2.00 mol of helium gas, which reaches a maximum temperature of 327°C. Assume the helium can be treated as an ideal gas. Process bc is isothermal. The pressure in states a and c is 1.00×10^5 Pa, and the pressure in state b is 3.00×10^5 Pa. (a) How much heat enters the gas and how much leaves the gas each cycle? (b)

Figure P20.39



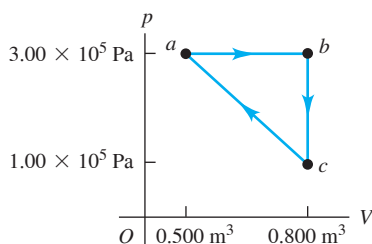
How much work does the engine do each cycle, and what is its efficiency? (c) Compare this engine's efficiency with the maximum possible efficiency attainable with the hot and cold reservoirs used by this cycle.

20.40 •• CP BIO A Human Engine. You decide to use your body as a Carnot heat engine. The operating gas is in a tube with one end in your mouth (where the temperature is 37.0°C) and the other end at the surface of your skin, at 30.0°C . (a) What is the maximum efficiency of such a heat engine? Would it be a very useful engine? (b) Suppose you want to use this human engine to lift a 2.50 kg box from the floor to a tabletop 1.20 m above the floor. How much must you increase the gravitational potential energy, and how much heat input is needed to accomplish this? (c) If your favorite candy bar has 350 food calories ($1\text{ food calorie} = 4186\text{ J}$) and 80% of the food energy goes into heat, how many of these candy bars must you eat to lift the box in this way?

20.41 ••• An experimental power plant at the Natural Energy Laboratory of Hawaii generates electricity from the temperature gradient of the ocean. The surface and deep-water temperatures are 27°C and 6°C , respectively. (a) What is the maximum theoretical efficiency of this power plant? (b) If the power plant is to produce 210 kW of power, at what rate must heat be extracted from the warm water? At what rate must heat be absorbed by the cold water? Assume the maximum theoretical efficiency. (c) The cold water that enters the plant leaves it at a temperature of 10°C . What must be the flow rate of cold water through the system? Give your answer in kg/h and in L/h .

20.42 • A monatomic ideal gas is taken around the cycle shown in **Fig. P20.42** in the direction shown in the figure. The path for process $c \rightarrow a$ is a straight line in the pV -diagram. (a) Calculate Q , W , and ΔU for each process $a \rightarrow b$, $b \rightarrow c$, and $c \rightarrow a$. (b) What are Q , W , and ΔU for one complete cycle? (c) What is the efficiency of the cycle?

Figure P20.42



20.43 • CALC A cylinder contains oxygen at a pressure of 2.00 atm . The volume is 4.00 L , and the temperature is 300 K . Assume that the oxygen may be treated as an ideal gas. The oxygen is carried through the following processes:

- Heated at constant pressure from the initial state (state 1) to state 2, which has $T = 450\text{ K}$.
- Cooled at constant volume to 250 K (state 3).
- Compressed at constant temperature to a volume of 4.00 L (state 4).
- Heated at constant volume to 300 K , which takes the system back to state 1.

(a) Show these four processes in a pV -diagram, giving the numerical values of p and V in each of the four states. (b) Calculate Q and W for each of the four processes. (c) Calculate the net work done by the oxygen in the complete cycle. (d) What is the efficiency of this device as a heat engine? How does this compare to the efficiency of a Carnot-cycle engine operating between the same minimum and maximum temperatures of 250 K and 450 K ?

20.44 ••• A typical coal-fired power plant generates 1000 MW of usable power at an overall thermal efficiency of 40% . (a) What is the rate of heat input to the plant? (b) The plant burns anthracite coal, which has a heat of combustion of $2.65 \times 10^7\text{ J/kg}$. How much coal does the plant use per day, if it operates continuously? (c) At what rate is heat ejected into the cool reservoir, which is the nearby river? (d) The river is at 18.0°C before it reaches the power plant and 18.5°C after it has received the plant's waste heat. Calculate the river's flow rate, in cubic meters per second. (e) By how much does the river's entropy increase each second?

20.45 •• A Carnot engine operates between two heat reservoirs at temperatures T_H and T_C . An inventor proposes to increase the efficiency by running one engine between T_H and an intermediate temperature T' and a second engine between T' and T_C , using as input the heat expelled by the first engine. Compute the efficiency of this composite system, and compare it to that of the original engine.

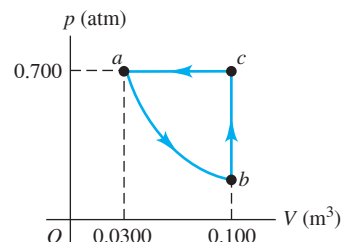
20.46 •• (a) Estimate the fuel efficiency, in miles per gallon, of a typical car. (b) The combustion of a gallon of gasoline releases 120 MJ of energy. If a typical car is driven at 40 mph and the engine is turning at 3000 revolutions per minute, then based on your estimate of fuel efficiency, how many joules of energy are released by combustion during each cycle of the engine? (c) A typical car engine runs with 20% efficiency. Based on this figure, how many watts of power does the car supply when moving at a steady 40 mph ? (d) What is that figure in horsepower?

20.47 • Automotive Thermodynamics. A Volkswagen Passat has a six-cylinder Otto-cycle engine with compression ratio $r = 10.6$. The diameter of each cylinder, called the *bore* of the engine, is 82.5 mm . The distance that the piston moves during the compression in **Fig. 20.5**, called the *stroke* of the engine, is 86.4 mm . The initial pressure of the air-fuel mixture (at point a in **Fig. 20.6**) is $8.50 \times 10^4\text{ Pa}$, and the initial temperature is 300 K (the same as the outside air). Assume that 200 J of heat is added to each cylinder in each cycle by the burning gasoline, and that the gas has $C_V = 20.5\text{ J/mol} \cdot \text{K}$ and $\gamma = 1.40$. (a) Calculate the total work done in one cycle in each cylinder of the engine, and the heat released when the gas is cooled to the temperature of the outside air. (b) Calculate the volume of the air-fuel mixture at point a in the cycle. (c) Calculate the pressure, volume, and temperature of the gas at points b , c , and d in the cycle. In a pV -diagram, show the numerical values of p , V , and T for each of the four states. (d) Compare the efficiency of this engine with the efficiency of a Carnot-cycle engine operating between the same maximum and minimum temperatures.

20.48 •• (a) Estimate the size of your room. (b) One mole of an ideal gas at STP occupies a volume of 22.4 L . Using this figure, estimate the number of moles of air in your room. (c) How many molecules is that? (d) There are N^N ways to distribute N things onto N sites. As a rough approximation, assume there are as many states for air molecules as there are air molecules in your room. Compute the entropy of the air in your room.

20.49 •• The pV -diagram in **Fig. P20.49** shows the cycle for a refrigerator operating on 0.850 mol of H_2 . Assume that the gas can be treated as ideal. Process ab is isothermal. Find the coefficient of performance of this refrigerator.

Figure P20.49



20.50 •• BIO Human Entropy. A person with skin of surface area 1.85 m^2 and temperature 30.0°C is resting in an insulated room where the ambient air temperature is 20.0°C . Assume that this person gets rid of excess heat by radiation only. By how much does the person change the entropy of the air in this room each second? (Recall that the room radiates back into the person and that the emissivity of the skin is 1.00 .)

20.51 •• CALC An object of mass m_1 , specific heat c_1 , and temperature T_1 is placed in contact with a second object of mass m_2 , specific heat c_2 , and temperature $T_2 > T_1$. As a result, the temperature of the first object increases to T and the temperature of the second object decreases to T' . (a) Show that the entropy increase of the system is

$$\Delta S = m_1 c_1 \ln \frac{T}{T_1} + m_2 c_2 \ln \frac{T'}{T_2}$$

and show that energy conservation requires that

$$m_1 c_1 (T - T_1) = m_2 c_2 (T_2 - T')$$

(b) Show that the entropy change ΔS , considered as a function of T , is a *maximum* if $T = T'$, which is just the condition of thermodynamic equilibrium. (c) Discuss the result of part (b) in terms of the idea of entropy as a measure of randomness.

20.52 •• CALC To heat 1 cup of water (250 cm^3) to make coffee, you place an electric heating element in the cup. As the water temperature increases from 20°C to 78°C , the temperature of the heating element remains at a constant 120°C . Calculate the change in entropy of (a) the water; (b) the heating element; (c) the system of water and heating element. (Make the same assumption about the specific heat of water as in Example 20.10 in Section 20.7, and ignore the heat that flows into the ceramic coffee cup itself.) (d) Is this process reversible or irreversible? Explain.

20.53 •• DATA In your summer job with a venture capital firm, you are given funding requests from four inventors of heat engines. The inventors claim the following data for their operating prototypes:

	Prototype			
	A	B	C	D
T_C ($^\circ\text{C}$), low-temperature reservoir	47	17	-33	37
T_H ($^\circ\text{C}$), high-temperature reservoir	177	197	247	137
Claimed efficiency e (%)	21	35	56	20

(a) Based on the T_C and T_H values for each prototype, find the maximum possible efficiency for each. (b) Are any of the claimed efficiencies impossible? Explain. (c) For all prototypes with an efficiency that is possible, rank the prototypes in decreasing order of the ratio of claimed efficiency to maximum possible efficiency.

20.54 •• DATA For a refrigerator or air conditioner, the coefficient of performance K (often denoted as COP) is, as in Eq. (20.9), the ratio of cooling output $|Q_C|$ to the required electrical energy input $|W|$, both in joules. The coefficient of performance is also expressed as a ratio of powers,

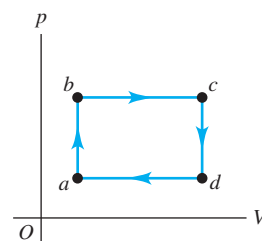
$$K = \frac{|Q_C|/t}{|W|/t}$$

where $|Q_C|/t$ is the cooling power and $|W|/t$ is the electrical power input to the device, both in watts. The energy efficiency ratio (EER) is the same quantity expressed in units of Btu for $|Q_C|$ and $\text{W} \cdot \text{h}$ for $|W|$. (a) Derive a general relationship that expresses EER in terms of K . (b) For a home air conditioner, EER is generally determined for a 95°F outside temperature and an 80°F return air temperature. Calculate EER

for a Carnot device that operates between 95°F and 80°F . (c) You have an air conditioner with an EER of 10.9. Your home on average requires a total cooling output of $|Q_C| = 1.9 \times 10^{10} \text{ J}$ per year. If electricity costs you 15.3 cents per $\text{kW} \cdot \text{h}$, how much do you spend per year, on average, to operate your air conditioner? (Assume that the unit's EER accurately represents the operation of your air conditioner. A *seasonal energy efficiency ratio* (SEER) is often used. The SEER is calculated over a range of outside temperatures to get a more accurate seasonal average.) (d) You are considering replacing your air conditioner with a more efficient one with an EER of 14.6. Based on the EER, how much would that save you on electricity costs in an average year?

20.55 ••• DATA You are conducting experiments to study prototype heat engines. In one test, 4.00 mol of argon gas are taken around the cycle shown in Fig. P20.55. The pressure is low enough for the gas to be treated as ideal. You measure the gas temperature in states a , b , c , and d and find $T_a = 250.0 \text{ K}$, $T_b = 300.0 \text{ K}$, $T_c = 380.0 \text{ K}$, and $T_d = 316.7 \text{ K}$. (a) Calculate the efficiency e of the cycle. (b) Disappointed

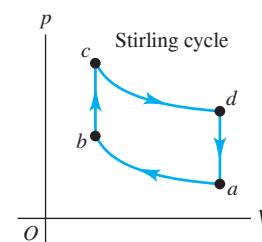
Figure P20.55



by the cycle's low efficiency, you consider doubling the number of moles of gas while keeping the pressure and volume the same. What would e be then? (c) You remember that the efficiency of a Carnot cycle increases if the temperature of the hot reservoir is increased. So, you return to using 4.00 mol of gas but double the volume in states c and d while keeping the pressures the same. The resulting temperatures in these states are $T_c = 760.0 \text{ K}$ and $T_d = 633.4 \text{ K}$. T_a and T_b remain the same as in part (a). Calculate e for this cycle with the new T_c and T_d values. (d) Encouraged by the increase in efficiency, you raise T_c and T_d still further. But e doesn't increase very much; it seems to be approaching a limiting value. If $T_a = 250.0 \text{ K}$ and $T_b = 300.0 \text{ K}$ and you keep volumes V_a and V_b the same as in part (a), then $T_c/T_d = T_b/T_a$ and $T_c = 1.20T_d$. Derive an expression for e as a function of T_d for this cycle. What value does e approach as T_d becomes very large?

20.56 •• CP In an ideal Stirling engine, n moles of an ideal gas are isothermally compressed, pressurized at constant volume, allowed to expand isothermally, and then cooled at constant volume, as shown in Fig. P20.56. The expansion takes place at temperature T_H and the compression at temperature T_C . (a) Show that $p_c/p_d = p_b/p_a$. This quantity is called the compression ratio of the engine, or the CR. Determine the work

Figure P20.56



done in terms of the CR (b) during the expansion phase and (c) during the compression phase. (d) An ideal Stirling engine uses 1 mole of helium as its working substance and has a CR of 10, temperature reservoirs at 100°C and 20°C , and a frequency of operation of 100 Hz. What is its power output?

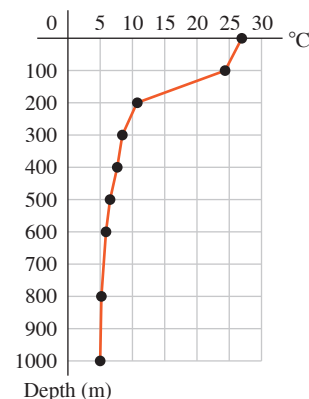
CHALLENGE PROBLEMS

20.57 ●●● On planet Tralfamadore the cool atmosphere is rich in froklom molecules. Spherical life forms called sporons feed on these molecules and obtain energy through a Carnot cycle. When a sporon that has injected froklom gas has isothermally contracted to a trigger volume, it violently squeezes itself, reducing its volume by half. At this point, its interior achieves the combustion temperature at which froklom fuses exothermically to produce klombactic acid. At the point of combustion, a sporon has radius 8.0 cm and internal pressure 20.3 kPa. The sporon swells during the reaction period, maintaining constant temperature until its injected froklom is depleted. The sporon continues to swell rapidly until its internal temperature matches the ambient temperature of 123 K and its gauge pressure is zero. Then it contracts slowly, emitting klombactic acid while simultaneously injecting froklom from the atmosphere. When its radius shrinks by a factor of 3 it reaches the trigger volume, which starts another cycle. Aside from the sporon's skin membrane, all substances behave like ideal gases. Froklom gas has a ratio of heat capacities of 1.30. (a) What is the combustion temperature at which froklom fuses? (b) What is the ambient pressure on Tralfamadore? (c) If a sporon generates 60 kJ of work each hour, how much heat does it draw in per minute? (d) How much heat is rejected back to the atmosphere per minute?

20.58 ●●● Consider a Diesel cycle that starts (at point *a* in Fig. 20.7) with air at temperature T_a . The air may be treated as an ideal gas. (a) If the temperature at point *c* is T_c , derive an expression for the efficiency of the cycle in terms of the compression ratio r . (b) What is the efficiency if $T_a = 300$ K, $T_c = 950$ K, $\gamma = 1.40$, and $r = 21.0$?

MCAT-STYLE PASSAGE PROBLEMS

Power from the Sea. *Ocean thermal energy conversion* is a process that uses the temperature difference between the warm surface water of tropical oceans and the cold deep-ocean water to run a heat engine. The graph shows a typical decrease of temperature with depth below the surface in tropical oceans. In the heat engine, the warmer surface water vaporizes a low-boiling-point fluid, such as ammonia. The heat of vaporization of ammonia is 260 cal/g at 27°C, the surface-water temperature. The vapor is used to turn a turbine and is then condensed back into a liquid by means of cold water



brought from deep below the surface through a large intake pipe. A power plant producing 10 MW of useful power would require a cold seawater flow rate of about 30,000 kg/s.

20.59 If the power plant uses a Carnot cycle and the desired theoretical efficiency is 6.5%, from what depth must cold water be brought? (a) 100 m; (b) 400 m; (c) 800 m; (d) deeper than 1000 m.

20.60 What is the change in entropy of the ammonia vaporized per second in the 10 MW power plant, assuming an ideal Carnot efficiency of 6.5%? (a) $+6 \times 10^6$ J/K per second; (b) $+5 \times 10^5$ J/K per second; (c) $+1 \times 10^5$ J/K per second; (d) 0.

20.61 Compare the entropy change of the warmer water to that of the colder water during one cycle of the heat engine, assuming an ideal Carnot cycle. (a) The entropy does not change during one cycle in either case. (b) The entropy of both increases, but the entropy of the colder water increases by more because its initial temperature is lower. (c) The entropy of the warmer water decreases by more than the entropy of the colder water increases, because some of the heat removed from the warmer water goes to the work done by the engine. (d) The entropy of the warmer water decreases by the same amount that the entropy of the colder water increases.

20.62 If the proposed plant is built and produces 10 MW but the rate at which waste heat is exhausted to the cold water is 165 MW, what is the plant's actual efficiency? (a) 5.7%; (b) 6.1%; (c) 6.5%; (d) 16.5%.

ANSWERS

Chapter Opening Question ?

(i) This is what a refrigerator does: It makes heat flow from the cold interior of the refrigerator to the warm outside. The second law of thermodynamics says that heat cannot *spontaneously* flow from a cold object to a hot one. A refrigerator has a motor that does work on the system to *force* the heat to flow in that way.

Key Example VARIATION Problems

VP20.1.1 (a) 6.89×10^4 J (b) 5.65×10^4 J

VP20.1.2 0.173

VP20.1.3 (a) 6.17×10^8 J (b) 1.21×10^8 J

VP20.1.4 (a) 3.96×10^8 J (b) 1.7×10^9 J (c) 0.23

VP20.4.1 (a) 0.210 (b) 6.32×10^4 J (c) 104°C

VP20.4.2 (a) 0.60

(b) $W_{ab} = 576$ J, $W_{bc} = 1250$ J, $W_{cd} = -231$ J, $W_{da} = -1250$ J

VP20.4.3 (a) 7.52 (b) 5.32×10^5 J

VP20.4.4 (a) $nRT_H \ln 2$ (b) $\frac{3}{2}nR(T_H - T_C)$

(c) $T_C/T_H = 1 - \frac{2}{3} \ln 2 = 0.538$, $e = \frac{2}{3} \ln 2 = 0.462$

VP20.10.1 (a) 6.55×10^2 J/K (b) 1.92×10^3 J/K (c) 2.43×10^3 J/K

VP20.10.2 (a) +45.7 J/K (b) +45.7 J/K

VP20.10.3 (a) zero (b) -150 J/K (c) +218 J/K (d) +68 J/K; increase

VP20.10.4 (a) $+1.22 \times 10^3$ J/K (b) -1.05×10^3 J/K

(c) $+1.7 \times 10^2$ J/K; increase

Bridging Problem

(a) 34.8°C

(b) +10 J/K

? Water makes life possible: The cells of your body could not function without water in which to dissolve essential biological molecules. Water is such a good solvent because its molecules (i) have zero net charge; (ii) have zero net charge, but the positive and negative charges are separated; (iii) have nonzero net charge; (iv) do not respond to electric forces; (v) exert repulsive electric forces on each other.



21 Electric Charge and Electric Field

LEARNING OUTCOMES

In this chapter, you'll learn...

- 21.1 The nature of electric charge, and how we know that electric charge is conserved.
- 21.2 How objects become electrically charged.
- 21.3 How to use Coulomb's law to calculate the electric force between charges.
- 21.4 The distinction between electric force and electric field.
- 21.5 How to calculate the electric field due to a collection of charges.
- 21.6 How to use the idea of electric field lines to visualize and interpret electric fields.
- 21.7 How to calculate the properties of electric dipoles.

You'll need to review...

- 1.7–1.10 Vector algebra, including the scalar (dot) product and the vector (cross) product.
- 4.3 Newton's second law.
- 7.5 Stable and unstable equilibria.
- 12.5 Streamlines in fluid flow.

In Chapter 5 we mentioned the four kinds of fundamental forces. To this point the only one of these forces that we have examined in any detail is gravity. Now we are ready to examine the force of *electromagnetism*, which encompasses both electricity and magnetism.

Electromagnetic interactions involve particles that have *electric charge*, an attribute that is as fundamental as mass. Just as objects with mass are accelerated by gravitational forces, so electrically charged objects are accelerated by electric forces. The shock you feel when you scuff your shoes across a carpet and then reach for a metal doorknob is due to charged particles leaping between your finger and the doorknob. Electric currents are simply streams of charged particles flowing within wires in response to electric forces. Even the forces that hold atoms together to form solid matter, and that keep the atoms of solid objects from passing through each other, are fundamentally due to electric interactions between the charged particles within atoms.

We begin our study of electromagnetism in this chapter by examining the nature of electric charge. We'll find that charge is quantized and obeys a conservation principle. When charges are at rest in our frame of reference, they exert *electrostatic* forces on each other. These forces are of tremendous importance in chemistry and biology and have many technological applications. Electrostatic forces are governed by a simple relationship known as *Coulomb's law* and are most conveniently described by using the concept of *electric field*. In later chapters we'll expand our discussion to include electric charges in motion. This will lead us to an understanding of magnetism and, remarkably, of the nature of light.

While the key ideas of electromagnetism are conceptually simple, applying them to practical problems will make use of many of your mathematical skills, especially your knowledge of geometry and integral calculus. For this reason you may find this chapter and those that follow to be more mathematically demanding than earlier chapters. The reward for your extra effort will be a deeper understanding of principles that are at the heart of modern physics and technology.

21.1 ELECTRIC CHARGE

The ancient Greeks discovered as early as 600 B.C. that after they rubbed amber with wool, the amber could attract other objects. Today we say that the amber has acquired a net **electric charge**, or has become *charged*. The word “electric” is derived from the Greek word *elektron*, meaning amber. When you scuff your shoes across a nylon carpet, you become electrically charged, and you can charge a comb by passing it through dry hair.

Plastic rods and fur (real or fake) are particularly good for demonstrating **electrostatics**, the interactions between electric charges that are at rest (or nearly so). After we charge both plastic rods in **Fig. 21.1a** by rubbing them with the piece of fur, we find that the rods repel each other.

When we rub glass rods with silk, the glass rods also become charged and repel each other (Fig. 21.1b). But a charged plastic rod *attracts* a charged glass rod; furthermore, the plastic rod and the fur attract each other, and the glass rod and the silk attract each other (Fig. 21.1c).

These experiments and many others like them have shown that there are exactly two kinds of electric charge: the kind on the plastic rod rubbed with fur and the kind on the glass rod rubbed with silk. Benjamin Franklin (1706–1790) suggested calling these two kinds of charge *negative* and *positive*, respectively, and these names are still used. The plastic rod and the silk have negative charge; the glass rod and the fur have positive charge.

Two positive charges or two negative charges repel each other. A positive charge and a negative charge attract each other.

CAUTION **Electric attraction and repulsion** The attraction and repulsion of two charged objects are sometimes summarized as “Like charges repel, and opposite charges attract.” But “like charges” does *not* mean that the two charges are exactly identical, only that both charges have the same algebraic *sign* (both positive or both negative). “Opposite charges” means that both objects have an electric charge, and those charges have different signs (one positive and the other negative). ▮

Figure 21.1 Experiments in electrostatics. (a) Negatively charged objects repel each other. (b) Positively charged objects repel each other. (c) Positively charged objects and negatively charged objects attract each other.

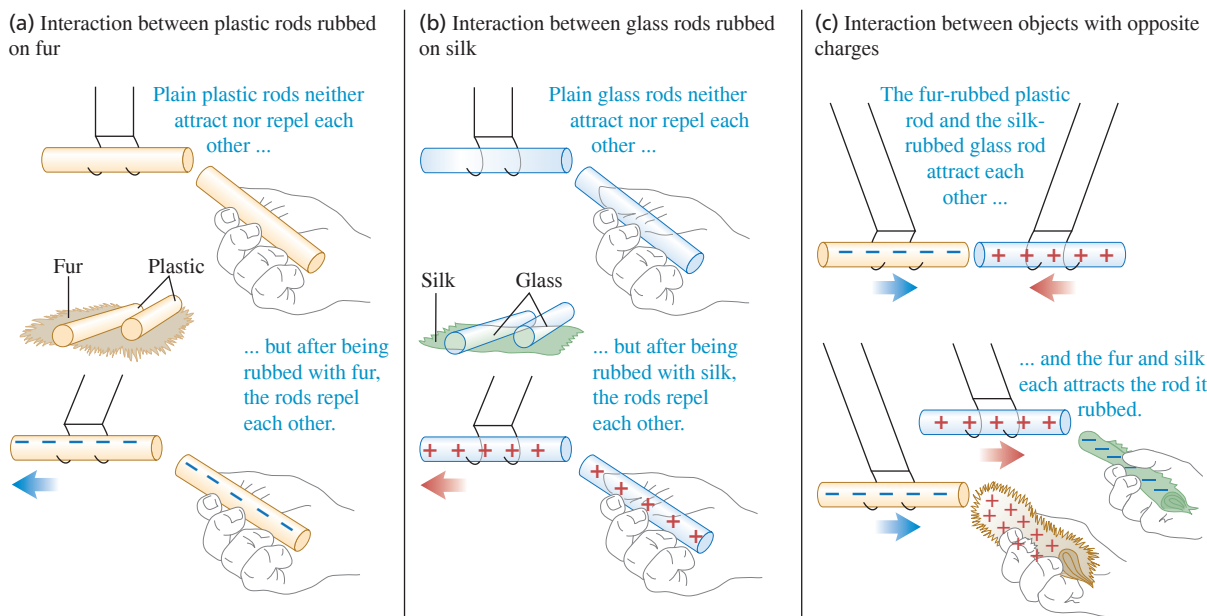
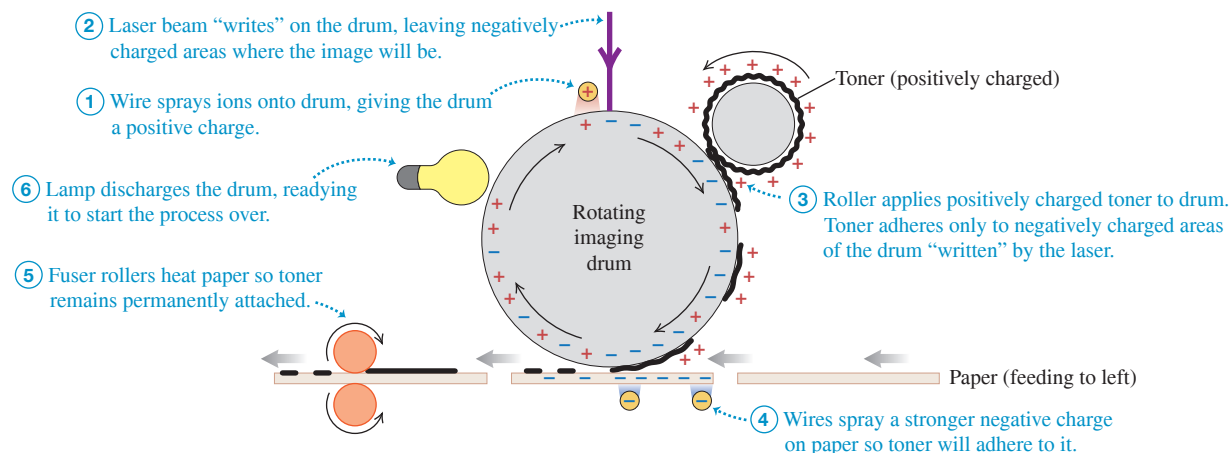


Figure 21.2 Schematic diagram of the operation of a laser printer.



A laser printer (**Fig. 21.2**) utilizes the forces between charged objects. The printer's light-sensitive imaging drum is given a positive charge. As the drum rotates, a laser beam shines on selected areas of the drum, leaving those areas with a *negative* charge. Positively charged particles of toner adhere only to the areas of the drum "written" by the laser. When a piece of paper is placed in contact with the drum, the toner particles stick to the paper and form an image.

Electric Charge and the Structure of Matter

When you charge a rod by rubbing it with fur or silk as in Fig. 21.1, there is no visible change in the appearance of the rod. What, then, actually happens to the rod when you charge it? To answer this question, we must look more closely at the structure of atoms, the building blocks of ordinary matter.

The structure of atoms can be described in terms of three particles: the negatively charged **electron**, the positively charged **proton**, and the uncharged **neutron** (**Fig. 21.3**). The proton and neutron are combinations of other entities called *quarks*, which have charges of $\pm\frac{1}{3}$ and $\pm\frac{2}{3}$ times the electron charge. Isolated quarks have not been observed, and there are theoretical reasons to believe that it is impossible in principle to observe a quark in isolation.

The protons and neutrons in an atom make up a small, very dense core called the **nucleus**, with dimensions of the order of 10^{-15} m. Surrounding the nucleus are the electrons, extending out to distances of the order of 10^{-10} m from the nucleus. If an atom were a few kilometers across, its nucleus would be the size of a tennis ball. The negatively charged electrons are held within the atom by the attractive electric forces exerted on them by the positively charged nucleus. (The protons and neutrons are held within stable atomic nuclei by an attractive interaction, called the *strong nuclear force*, that overcomes the electric repulsion of the protons. The strong nuclear force has a short range, and its effects do not extend far beyond the nucleus.)

The masses of the individual particles, to the precision that they are known (as of this writing), are

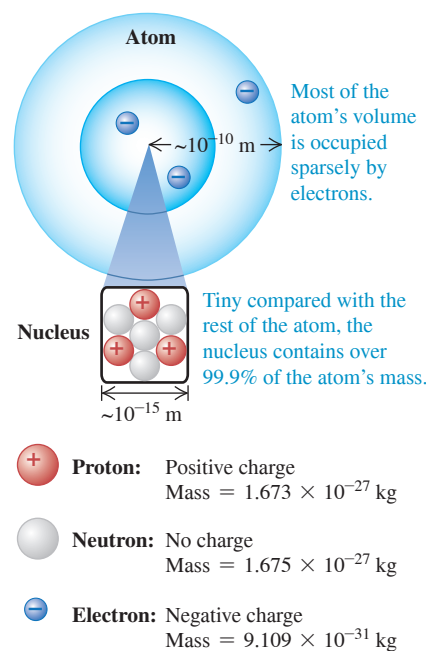
$$\text{Mass of electron} = m_e = 9.10938356(11) \times 10^{-31} \text{ kg}$$

$$\text{Mass of proton} = m_p = 1.672621898(21) \times 10^{-27} \text{ kg}$$

$$\text{Mass of neutron} = m_n = 1.674927471(21) \times 10^{-27} \text{ kg}$$

The numbers in parentheses are the uncertainties in the last two digits. Note that the masses of the proton and neutron are nearly equal and are roughly 2000 times the mass of the electron. Over 99.9% of the mass of any atom is concentrated in its nucleus.

Figure 21.3 The structure of an atom. The particular atom depicted here is lithium (see Fig. 21.4a).



The charges of the electron and proton are equal in magnitude.

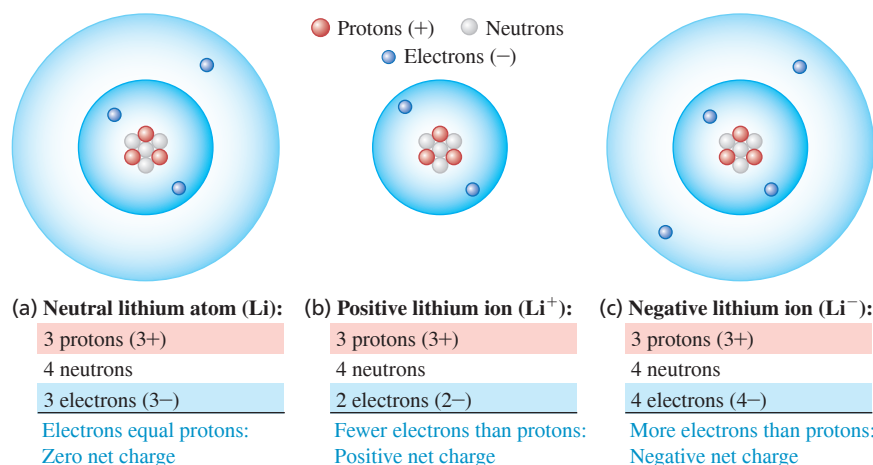


Figure 21.4 (a) A neutral atom has as many electrons as it does protons. (b) A positive ion has a deficit of electrons. (c) A negative ion has an excess of electrons. (The electron “shells” are a schematic representation of the actual electron distribution, a diffuse cloud many times larger than the nucleus.)

The negative charge of the electron has (within experimental error) *exactly* the same magnitude as the positive charge of the proton. In a neutral atom the number of electrons equals the number of protons in the nucleus, and the net electric charge (the algebraic sum of all the charges) is exactly zero (**Fig. 21.4a**). The number of protons or electrons in a neutral atom of an element is called the **atomic number** of the element. If one or more electrons are removed from an atom, what remains is called a **positive ion** (Fig. 21.4b). A **negative ion** is an atom that has *gained* one or more electrons (Fig. 21.4c). This gain or loss of electrons is called **ionization**.

When the total number of protons in a macroscopic object equals the total number of electrons, the total charge is zero and the object as a whole is electrically neutral. To give an object an excess negative charge, we may either *add negative* charges to a neutral object or *remove positive* charges from that object. Similarly, we can create an excess positive charge by either *adding positive* charge or *removing negative* charge. In most cases, negatively charged (and highly mobile) electrons are added or removed, and a “positively charged object” is one that has lost some of its normal complement of electrons. When we speak of the charge of an object, we always mean its *net* charge. The net charge is always a very small fraction (typically no more than 10^{-12}) of the total positive charge or negative charge in the object.

Electric Charge Is Conserved

Implicit in the foregoing discussion are two very important principles, the first of which is:

PRINCIPLE OF CONSERVATION OF CHARGE The algebraic sum of all the electric charges in any closed system is constant.

If we rub together a plastic rod and a piece of fur, both initially uncharged, the rod acquires a negative charge (since it takes electrons from the fur) and the fur acquires a positive charge of the *same* magnitude (since it has lost as many electrons as the rod has gained). Hence the total electric charge on the two objects together does not change. In any charging process, charge is not created or destroyed; it is merely *transferred* from one object to another.

Conservation of charge is thought to be a *universal* conservation law. No experimental evidence for any violation of this principle has ever been observed. Even in high-energy interactions in which particles are created and destroyed, such as the creation of electron–positron pairs, the total charge of any closed system is exactly constant.

Figure 21.5 Most of the forces on this water skier are electric. Electric interactions between adjacent molecules give rise to the force of the water on the ski, the tension in the tow rope, and the resistance of the air on the skier's body. Electric interactions also hold the atoms of the skier's body together. Only one wholly nonelectric force acts on the skier: the force of gravity.



The second important principle is:

The magnitude of charge of the electron or proton is a natural unit of charge.

Every observable amount of electric charge is always an integer multiple of this basic unit. We say that charge is *quantized*. A familiar example of quantization is money. When you pay cash for an item in a store, you have to do it in one-cent increments. Cash can't be divided into amounts smaller than one cent, and electric charge can't be divided into amounts smaller than the charge of one electron or proton. (The quark charges, $\pm\frac{1}{3}$ and $\pm\frac{2}{3}$ of the electron charge, are probably not observable as isolated charges.) Thus the charge on any macroscopic object is always zero or an integer multiple (negative or positive) of the electron charge.

Understanding the electric nature of matter gives us insight into many aspects of the physical world (**Fig. 21.5**). The chemical bonds that hold atoms together to form molecules are due to electric interactions between the atoms. They include the strong ionic bonds that hold sodium and chlorine atoms together to make table salt and the relatively weak bonds between the strands of DNA that record your body's genetic code. When you stand, the normal force exerted on you by the floor arises from electric forces between charged particles in the atoms of your shoes and the atoms of the floor. The tension force in a stretched string and the adhesive force of glue are likewise due to electric interactions of atoms.

TEST YOUR UNDERSTANDING OF SECTION 21.1 Two charged objects repel each other through the electric force. The charges on the objects are (i) one positive and one negative; (ii) both positive; (iii) both negative; (iv) either (ii) or (iii); (v) any of (i), (ii), or (iii).

ANSWER

(iv) Two charged objects repel if their charges are of the same sign (either both positive or both negative).

21.2 CONDUCTORS, INSULATORS, AND INDUCED CHARGES

Some materials permit electric charge to move easily from one region of the material to another, while others do not. For example, **Fig. 21.6a** shows a copper wire supported by a nylon thread. Suppose you touch one end of the wire to a charged plastic rod and attach the other end to a metal ball that is initially uncharged; you then remove the charged rod and the wire. When you bring another charged object up close to the ball (**Figs. 21.6b** and **21.6c**), the ball is attracted or repelled, showing that the ball has become electrically charged. Electric charge has been transferred through the copper wire between the ball and the surface of the plastic rod.

The copper wire is called a **conductor** of electricity. If you repeat the experiment using a rubber band or nylon thread in place of the wire, you find that *no* charge is transferred to the ball. These materials are called **insulators**. Conductors permit the easy movement of charge through them, while insulators do not. (The supporting nylon threads shown in **Fig. 21.6** are insulators, which prevents charge from leaving the metal ball and copper wire.)

As an example, carpet fibers on a dry day are good insulators. As you walk across a carpet, the rubbing of your shoes against the fibers causes charge to build up on you, and this charge remains on you because it can't flow through the insulating fibers. If you then touch a conducting object such as a doorknob, a rapid charge transfer takes place between your finger and the doorknob, and you feel a shock. One way to prevent this is to wind some of the carpet fibers around conducting cores so that any charge that builds up on you can be transferred harmlessly to the carpet. Another solution is to coat the carpet fibers with an antistatic layer that does not easily transfer electrons to or from your shoes; this prevents any charge from building up on you in the first place.

Most metals are good conductors, while most nonmetals are insulators. Within a solid metal such as copper, one or more outer electrons in each atom become detached and can move freely throughout the material, just as the molecules of a gas can move through the spaces between the grains in a bucket of sand. The other electrons remain bound to the positively charged nuclei, which themselves are bound in nearly fixed positions within the material. In an insulator there are no, or very few, free electrons, and electric charge cannot move freely through the material. Some materials called *semiconductors* are intermediate in their properties between good conductors and good insulators.

Charging by Induction

We can charge a metal ball by using a copper wire and an electrically charged plastic rod, as in Fig. 21.6a. In this process, some of the excess electrons on the rod are transferred from it to the ball, leaving the rod with a smaller negative charge. But there is a different technique in which the plastic rod can give another object a charge of *opposite* sign without losing any of its own charge. This process is called charging by **induction**.

Figure 21.7 shows an example of charging by induction. An uncharged metal ball is supported on an insulating stand (Fig. 21.7a). When you bring a negatively charged rod near it, without actually touching it (Fig. 21.7b), the free electrons in the metal ball are repelled by the excess electrons on the rod, and they shift toward the right, away from the rod. They cannot escape from the ball because the supporting stand and the surrounding air are insulators. So we get excess negative charge at the right surface of the ball and a deficiency of negative charge (that is, a net positive charge) at the left surface. These excess charges are called **induced charges**.

Not all of the free electrons move to the right surface of the ball. As soon as any induced charge develops, it exerts forces toward the *left* on the other free electrons. These electrons are repelled by the negative induced charge on the right and attracted toward the positive induced charge on the left. The system reaches an equilibrium state in which the force toward the right on an electron, due to the charged rod, is just balanced by the force toward the left due to the induced charge. If we remove the charged rod, the free electrons shift back to the left, and the original neutral condition is restored.

What happens if, while the plastic rod is nearby, you touch one end of a conducting wire to the right surface of the ball and the other end to the earth (Fig. 21.7c)? The earth is a conductor, and it is so large that it can act as a practically infinite source of extra electrons or sink of unwanted electrons. Some of the negative charge flows through the wire to the earth. Now suppose you disconnect the wire (Fig. 21.7d) and then remove the rod (Fig. 21.7e); a net positive charge is left on the ball. The charge on the negatively charged rod has not changed during this process. The earth acquires a negative charge that is equal in magnitude to the induced positive charge remaining on the ball.

Figure 21.6 Copper is a good conductor of electricity; nylon is a good insulator. (a) The copper wire conducts charge between the metal ball and the charged plastic rod to charge the ball negatively. Afterward, the metal ball is (b) repelled by a negatively charged plastic rod and (c) attracted to a positively charged glass rod.

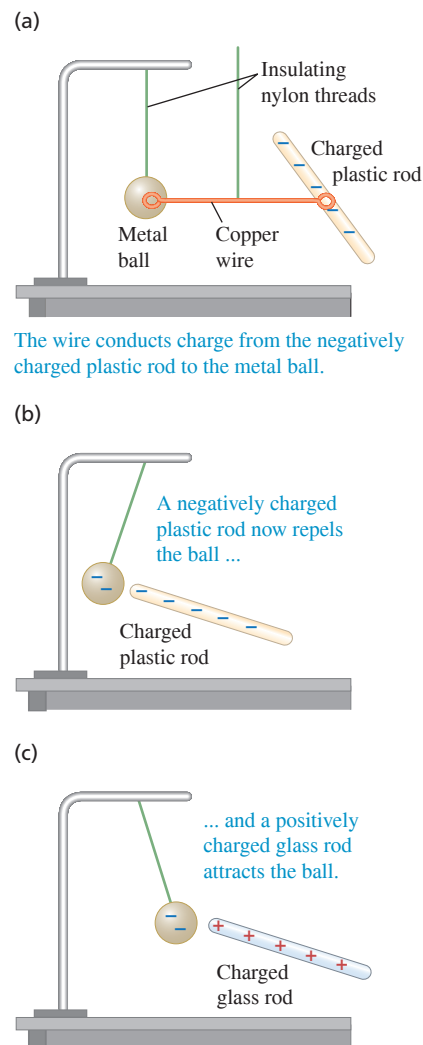


Figure 21.7 Charging a metal ball by induction.

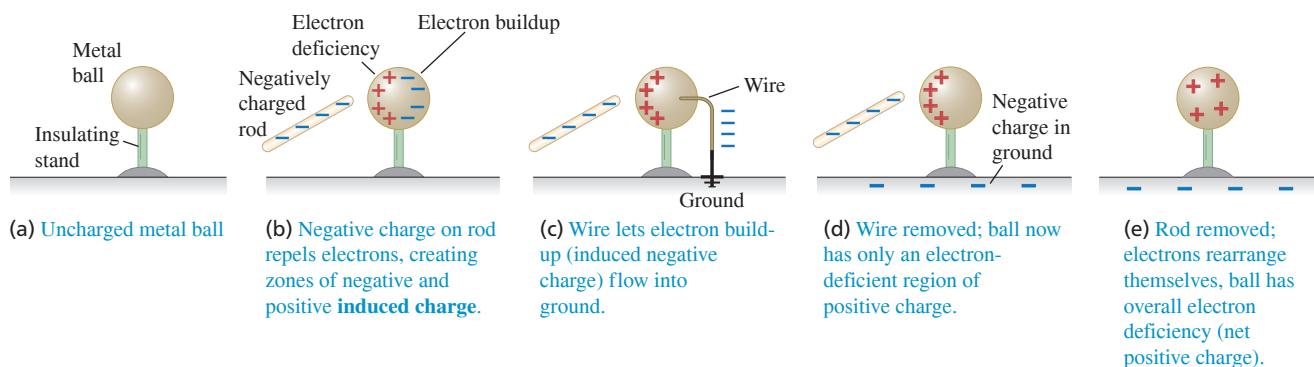
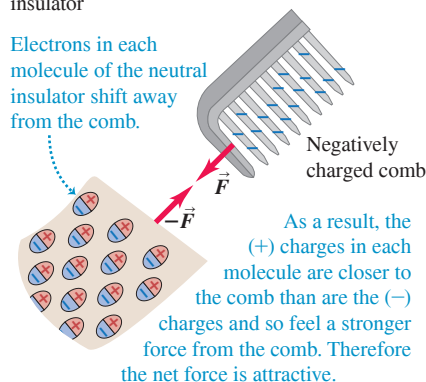


Figure 21.8 The charges within the molecules of an insulating material can shift slightly. As a result, a comb with either sign of charge attracts a neutral insulator. By Newton's third law the neutral insulator exerts an equal-magnitude attractive force on the comb.

(a) A charged comb picking up uncharged pieces of plastic



(b) How a negatively charged comb attracts an insulator



(c) How a positively charged comb attracts an insulator

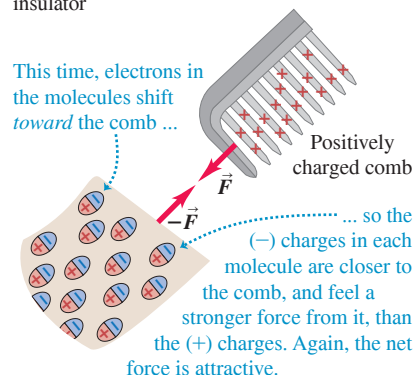
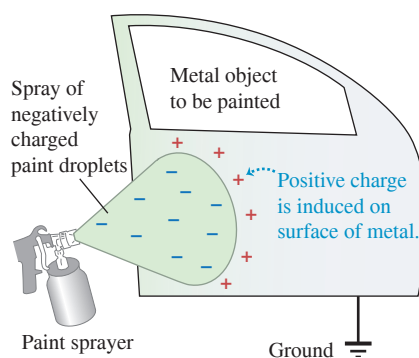


Figure 21.9 The electrostatic painting process (compare Figs. 21.7b and 21.7c). A metal object to be painted is connected to the earth ("ground"), and the paint droplets are given an electric charge as they exit the sprayer nozzle. Induced charges of the opposite sign appear in the object as the droplets approach, just as in Fig. 21.7b, and they attract the droplets to the surface. This process minimizes overspray from clouds of stray paint particles and gives a particularly smooth finish.



Electric Forces on Uncharged Objects

Finally, we note that a charged object can exert forces even on objects that are *not* charged themselves. If you rub a balloon on the rug and then hold the balloon against the ceiling, it sticks, even though the ceiling has no net electric charge. After you electrify a comb by running it through your hair, you can pick up uncharged bits of paper or plastic with it (Fig. 21.8a). How is this possible?

This interaction is an induced-charge effect. Even in an insulator, electric charges can shift back and forth a little when there is charge nearby. This is shown in Fig. 21.8b; the negatively charged plastic comb causes a slight shifting of charge within the molecules of the neutral insulator, an effect called *polarization*. The positive and negative charges in the material are present in equal amounts, but the positive charges are closer to the plastic comb and so feel an attraction that is stronger than the repulsion felt by the negative charges, giving a net attractive force. (In Section 21.3 we'll study how electric forces depend on distance.) Note that a neutral insulator is also attracted to a *positively* charged comb (Fig. 21.8c). Now the charges in the insulator shift in the opposite direction; the negative charges in the insulator are closer to the comb and feel an attractive force that is stronger than the repulsion felt by the positive charges in the insulator. Hence a charged object of *either* sign exerts an attractive force on an uncharged insulator. **Figure 21.9** shows an industrial application of this effect.

TEST YOUR UNDERSTANDING OF SECTION 21.2 You have two lightweight metal spheres, each hanging from an insulating nylon thread. One of the spheres has a net negative charge, while the other sphere has no net charge. (a) If the spheres are close together but do not touch, will they (i) attract each other, (ii) repel each other, or (iii) exert no force on each other? (b) You now allow the two spheres to touch. Once they have touched, will the two spheres (i) attract each other, (ii) repel each other, or (iii) exert no force on each other?

ANSWER

(a) (i) Before the two spheres touch, the negatively charged sphere exerts a repulsive force on the electrons in the other sphere, causing zones of positive and negative induced charge (see Fig. 21.7b). The positive zone is closer to the negatively charged sphere than the negative zone, so there is a net force of attraction that pulls the spheres together, like the comb and insulator in Fig. 21.8b. Once the two metal spheres touch, some of the excess electrons on the negatively charged sphere will flow onto the other sphere (because metals are conductors). Then both spheres will have a net negative charge and will repel each other.

21.3 COULOMB'S LAW

Charles Augustin de Coulomb (1736–1806) studied the interaction forces of charged particles in detail in 1784. He used a torsion balance (**Fig. 21.10a**) similar to the one used 13 years later by Cavendish to study the much weaker gravitational interaction, as we discussed in Section 13.1. For **point charges**, charged objects that are very small in comparison with the distance r between them, Coulomb found that the electric force is proportional to $1/r^2$. That is, when the distance r doubles, the force decreases to one-quarter of its initial value; when the distance is halved, the force increases to four times its initial value.

The electric force between two point charges also depends on the quantity of charge on each object, which we'll denote by q or Q . To explore this dependence, Coulomb divided a charge into two equal parts by placing a small charged spherical conductor into contact with an identical but uncharged sphere; by symmetry, the charge is shared equally between the two spheres. (Note the essential role of the principle of conservation of charge in this procedure.) Thus he could obtain one-half, one-quarter, and so on, of any initial charge. He found that the forces that two point charges q_1 and q_2 exert on each other are proportional to each charge and therefore are proportional to the *product* q_1q_2 of the two charges.

Thus Coulomb established the following law:

COULOMB'S LAW The magnitude of the electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.

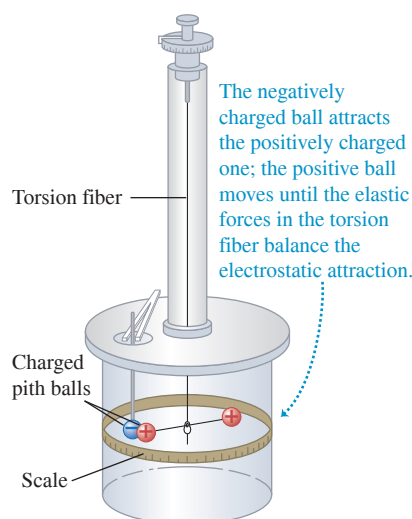
In mathematical terms, the magnitude F of the force that each of two point charges q_1 and q_2 a distance r apart exerts on the other can be expressed as

$$F = k \frac{|q_1 q_2|}{r^2} \quad (21.1)$$

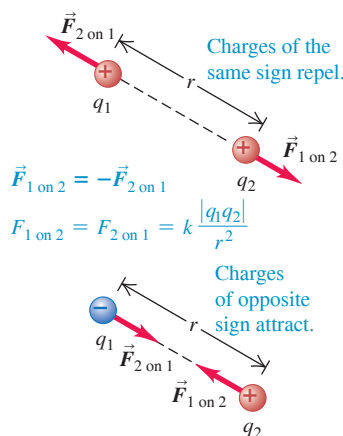
where k is a proportionality constant whose numerical value depends on the system of units used. The absolute value bars are used in Eq. (21.1) because the charges q_1 and q_2 can be either positive or negative, while the force magnitude F is always positive.

The directions of the forces the two charges exert on each other are always along the line joining them. When the charges q_1 and q_2 have the same sign, either both positive or both negative, the forces are repulsive; when the charges have opposite signs, the forces are attractive (Fig. 21.10b). The two forces obey Newton's third law; they are always equal in magnitude and opposite in direction, even when the charges are not equal in magnitude.

(a) A torsion balance of the type used by Coulomb to measure the electric force



(b) Interactions between point charges



BIO APPLICATION Electric Forces, Sweat, and Cystic Fibrosis

One way to test for the genetic disease cystic fibrosis (CF) is to measure the salt content of a person's sweat. Sweat is a mixture of water and ions, including the sodium (Na^+) and chloride (Cl^-) ions that make up ordinary salt (NaCl). When sweat is secreted by epithelial cells, some of the Cl^- ions flow from the sweat back into these cells (a process called reabsorption). The electric attraction between negative and positive charges pulls Na^+ ions along with the Cl^- . Water molecules cannot flow back into the epithelial cells, so sweat on the skin has a low salt content. However, in persons with CF the reabsorption of Cl^- ions is blocked. Hence the sweat of persons with CF is unusually salty, with up to four times the normal concentration of Cl^- and Na^+ .



Figure 21.10 (a) Measuring the electric force between point charges. (b) The electric forces between point charges obey Newton's third law: $\vec{F}_{1 \text{ on } 2} = -\vec{F}_{2 \text{ on } 1}$.

The proportionality of the electric force to $1/r^2$ has been verified with great precision. There is no reason to suspect that the exponent is different from precisely 2. Thus the form of Eq. (21.1) is the same as that of the law of gravitation. But electric and gravitational interactions are two distinct classes of phenomena. Electric interactions depend on electric charges and can be either attractive or repulsive, while gravitational interactions depend on mass and are always attractive (because there is no such thing as negative mass).

Fundamental Electric Constants

The value of the proportionality constant k in Coulomb's law depends on the system of units used. In our study of electricity and magnetism we'll use SI units exclusively. The SI electric units include most of the familiar units such as the volt, the ampere, the ohm, and the watt. (There is *no* British system of electric units.) The SI unit of electric charge is called one **coulomb** (1 C). As of this writing (2018) the value of the constant k in Eq. (21.1) is, to nine significant figures,

$$k = 8.987551787 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2 \cong 8.988 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

Prior to 2018 k was *defined* to have a precise value in terms of the speed of light in vacuum c , whose value is in turn related to the definitions of the second and the meter that we stated in Section 1.3. (We'll see the connection between k and c in Chapter 32 when we study electromagnetic radiation.) With the revisions to the definitions of units in the International System (SI) in 2018, the value of k is no longer exact, but has a very small fractional uncertainty of about 2×10^{-10} . Because this uncertainty is so small, these revisions to the definitions of the units have no effect on the calculations that we present in this book, and don't affect the calculations that you'll do.

In principle we can measure the electric force F between two equal charges q at a measured distance r and use Coulomb's law to determine the charge. Thus we could regard the value of k as an operational definition of the coulomb. For reasons of experimental precision it is better to define the coulomb instead in terms of the most fundamental unit of charge. This is the magnitude of the charge of an electron or a proton, which is denoted by e . The value of e is *defined* to be a certain fraction of a coulomb:

$$e = 1.602176634 \times 10^{-19} \text{ C}$$

So **one coulomb (C) is that amount of charge such that $1.602176634 \times 10^{-19} \text{ C}$ is equal to the magnitude of the charge on an electron or proton.** (An older definition of the coulomb involving the unit of electric *current*, or charge per unit time, was abandoned in 2018.)

One coulomb represents the negative of the total charge of about 6×10^8 electrons. For comparison, a copper cube 1 cm on a side contains about 2.4×10^{24} electrons. About 10^{19} electrons pass through the glowing filament of a flashlight bulb every second.

In SI units we usually write the constant k in Eq. (21.1) as $1/4\pi\epsilon_0$, where ϵ_0 ("epsilon-nought" or "epsilon-zero") is called the **electric constant**. This shorthand simplifies many formulas that we'll encounter in later chapters. From now on, we'll usually write Coulomb's law as

$$\text{Coulomb's law:} \quad \text{Magnitude of electric force between two point charges} \quad F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} \quad \text{Electric constant} \quad \text{Values of the two charges} \quad \text{Distance between the two charges} \quad (21.2)$$

The constants in Eq. (21.2) are approximately

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2 \quad \text{and} \quad \frac{1}{4\pi\epsilon_0} = k = 8.988 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

In examples and problems we'll often use the approximate value

$$\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

In electrostatics problems (problems that involve charges at rest), it's very unusual to encounter charges as large as 1 coulomb. Two 1 C charges separated by 1 m would exert forces on each other of magnitude $9 \times 10^9 \text{ N}$ (about 1 million tons)! The total charge of all the electrons in a copper one-cent coin is even greater, about $1.4 \times 10^5 \text{ C}$, which shows that we can't disturb electric neutrality very much without using enormous forces. More typical values of charge range from about a microcoulomb ($1 \mu\text{C} = 10^{-6} \text{ C}$) to about a nanocoulomb ($1 \text{ nC} = 10^{-9} \text{ C}$).

EXAMPLE 21.1 Electric force versus gravitational force

An α particle (the nucleus of a helium atom) has mass $m = 6.64 \times 10^{-27} \text{ kg}$ and charge $q = +2e = 3.2 \times 10^{-19} \text{ C}$. Compare the magnitude of the electric repulsion between two α ("alpha") particles with that of the gravitational attraction between them.

IDENTIFY and SET UP This problem involves Newton's law for the gravitational force F_g between particles (see Section 13.1) and Coulomb's law for the electric force F_e between point charges. To compare these forces, we make our target variable the ratio F_e/F_g . We use Eq. (21.2) for F_e and Eq. (13.1) for F_g .

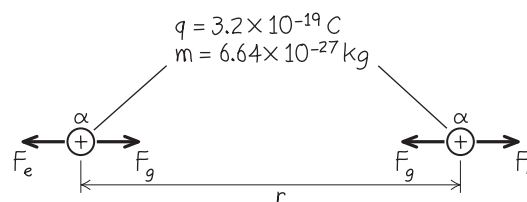
EXECUTE Figure 21.11 shows our sketch. From Eqs. (21.2) and (13.1),

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} \quad F_g = G \frac{m^2}{r^2}$$

These are both inverse-square forces, so the r^2 factors cancel when we take the ratio:

$$\begin{aligned} \frac{F_e}{F_g} &= \frac{1}{4\pi\epsilon_0 G} \frac{q^2}{m^2} \\ &= \frac{9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2}{6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2} \frac{(3.2 \times 10^{-19} \text{ C})^2}{(6.64 \times 10^{-27} \text{ kg})^2} = 3.1 \times 10^{35} \end{aligned}$$

Figure 21.11 Our sketch for this problem.



EVALUATE This astonishingly large number shows that the gravitational force in this situation is completely negligible in comparison to the electric force. This is always true for interactions of atomic and subnuclear particles. But within objects the size of a person or a planet, the positive and negative charges are nearly equal in magnitude, and the net electric force is usually much *smaller* than the gravitational force.

KEYCONCEPT The magnitude of the electric force that two point charges exert on each other is (a) proportional to the product of the magnitudes of their charges and (b) inversely proportional to the square of the distance between them.

Superposition of Forces

Coulomb's law as we have stated it describes only the interaction of two *point* charges. Experiments show that when two charges exert forces simultaneously on a third charge, the total force acting on that charge is the *vector sum* of the forces that the two charges would exert individually. This important property, called the **principle of superposition of forces**, holds for any number of charges. By using this principle, we can apply Coulomb's law to *any* collection of charges. Two of the examples at the end of this section use the superposition principle.

Strictly speaking, Coulomb's law as we have stated it should be used only for point charges *in a vacuum*. If matter is present in the space between the charges, the net force acting on each charge is altered because charges are induced in the molecules of the intervening material. We'll describe this effect later. As a practical matter, though, we can use Coulomb's law unaltered for point charges in air. At normal atmospheric pressure, the presence of air changes the electric force from its vacuum value by only about one part in 2000.

PROBLEM-SOLVING STRATEGY 21.1 Coulomb's Law

IDENTIFY the relevant concepts: Coulomb's law describes the electric force between charged particles.

SET UP the problem using the following steps:

1. Sketch the locations of the charged particles and label each particle with its charge.
2. If the charges do not all lie on a single line, set up an xy -coordinate system.
3. The problem will ask you to find the electric force on one or more particles. Identify which these are.

EXECUTE the solution as follows:

1. For each particle that exerts an electric force on a given particle of interest, use Eq. (21.2) to calculate the magnitude of that force.
2. Using those magnitudes, sketch a free-body diagram showing the electric-force vectors acting on each particle of interest. The force exerted by particle 1 on particle 2 points from particle 2 toward particle 1 if the charges have opposite signs, but points from particle 2 directly away from particle 1 if the charges have the same sign.
3. Use the principle of superposition to calculate the total electric force—a vector sum—on each particle of interest. (Review the vector algebra in Sections 1.7 through 1.9. The method of components is often helpful.)

4. Use consistent units; SI units are completely consistent. With $1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$, distances must be in meters, charges in coulombs, and forces in newtons.
5. Some examples and problems in this and later chapters involve *continuous* distributions of charge along a line, over a surface, or throughout a volume. In these cases the vector sum in step 3 becomes a vector *integral*. We divide the charge distribution into infinitesimal pieces, use Coulomb's law for each piece, and integrate to find the vector sum. Sometimes this can be done without actual integration.
6. Exploit any symmetries in the charge distribution to simplify your problem solving. For example, two identical charges q exert zero net electric force on a charge Q midway between them, because the forces on Q have equal magnitude and opposite direction.

EVALUATE your answer: Check whether your numerical results are reasonable. Confirm that the direction of the net electric force agrees with the principle that charges of the same sign repel and charges of opposite sign attract.

EXAMPLE 21.2 Force between two point charges

Two point charges, $q_1 = +25 \text{ nC}$ and $q_2 = -75 \text{ nC}$, are separated by a distance $r = 3.0 \text{ cm}$ (Fig. 21.12a). Find the magnitude and direction of the electric force (a) that q_1 exerts on q_2 and (b) that q_2 exerts on q_1 .

IDENTIFY and SET UP This problem asks for the electric forces that two charges exert on each other. We use Coulomb's law, Eq. (21.2), to calculate the magnitudes of the forces. The signs of the charges will determine the directions of the forces.

EXECUTE (a) After converting the units of r to meters and the units of q_1 and q_2 to coulombs, Eq. (21.2) gives us

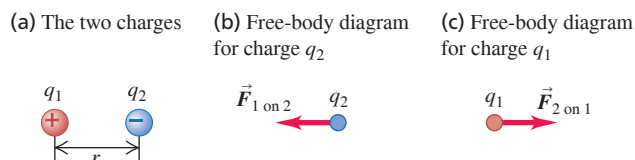
$$\begin{aligned} F_{1 \text{ on } 2} &= \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{|(+25 \times 10^{-9} \text{ C})(-75 \times 10^{-9} \text{ C})|}{(0.030 \text{ m})^2} \\ &= 0.019 \text{ N} \end{aligned}$$

The charges have opposite signs, so the force is attractive (to the left in Fig. 21.12b); that is, the force that acts on q_2 is directed toward q_1 along the line joining the two charges.

(b) Proceeding as in part (a), we have

$$F_{2 \text{ on } 1} = \frac{1}{4\pi\epsilon_0} \frac{|q_2 q_1|}{r^2} = F_{1 \text{ on } 2} = 0.019 \text{ N}$$

Figure 21.12 What force does q_1 exert on q_2 , and what force does q_2 exert on q_1 ? Gravitational forces are negligible.



The attractive force that acts on q_1 is to the right, toward q_2 (Fig. 21.12c).

EVALUATE Even though the charges have different magnitudes, the magnitude of the force that q_2 exerts on q_1 is the same as the magnitude of the force that q_1 exerts on q_2 .

KEYCONCEPT The electric forces that two point charges exert on each other obey Newton's third law: The force that the first charge exerts on the second charge has the same magnitude as the force that the second charge exerts on the first charge, but points in the opposite direction. The forces are repulsive if the two charges are both positive or both negative, and attractive if one charge is positive and the other negative.

EXAMPLE 21.3 Vector addition of electric forces on a line**WITH VARIATION PROBLEMS**

Two point charges are located on the x -axis of a coordinate system: $q_1 = 1.0 \text{ nC}$ is at $x = +2.0 \text{ cm}$, and $q_2 = -3.0 \text{ nC}$ is at $x = +4.0 \text{ cm}$. What is the total electric force exerted by q_1 and q_2 on a charge $q_3 = 5.0 \text{ nC}$ at $x = 0$?

IDENTIFY and SET UP Figure 21.13a shows the situation. To find the total force on q_3 , our target variable, we find the vector sum of the two electric forces on it.

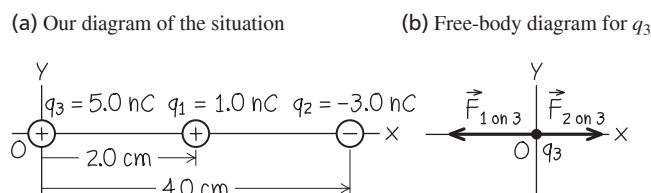
EXECUTE Figure 21.13b is a free-body diagram for q_3 , which is repelled by q_1 (which has the same sign) and attracted to q_2 (which has the opposite sign): $\vec{F}_{1 \text{ on } 3}$ is in the $-x$ -direction and $\vec{F}_{2 \text{ on } 3}$ is in the $+x$ -direction. After unit conversions, we have from Eq. (21.2)

$$\begin{aligned} F_{1 \text{ on } 3} &= \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_3|}{r_{13}^2} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(1.0 \times 10^{-9} \text{ C})(5.0 \times 10^{-9} \text{ C})}{(0.020 \text{ m})^2} \\ &= 1.12 \times 10^{-4} \text{ N} = 112 \mu\text{N} \end{aligned}$$

In the same way you can show that $F_{2 \text{ on } 3} = 84 \mu\text{N}$. We thus have $\vec{F}_{1 \text{ on } 3} = (-112 \mu\text{N})\hat{i}$ and $\vec{F}_{2 \text{ on } 3} = (84 \mu\text{N})\hat{i}$. The net force on q_3 is

$$\vec{F}_3 = \vec{F}_{1 \text{ on } 3} + \vec{F}_{2 \text{ on } 3} = (-112 \mu\text{N})\hat{i} + (84 \mu\text{N})\hat{i} = (-28 \mu\text{N})\hat{i}$$

Figure 21.13 Our sketches for this problem.



EVALUATE As a check, note that the magnitude of q_2 is three times that of q_1 , but q_2 is twice as far from q_3 as q_1 . Equation (21.2) then says that $F_{2 \text{ on } 3}$ must be $3/2^2 = 3/4 = 0.75$ as large as $F_{1 \text{ on } 3}$. This agrees with our calculated values: $F_{2 \text{ on } 3}/F_{1 \text{ on } 3} = (84 \mu\text{N})/(112 \mu\text{N}) = 0.75$. Because $F_{2 \text{ on } 3}$ is the weaker force, the direction of the net force is that of $\vec{F}_{1 \text{ on } 3}$ —that is, in the negative x -direction.

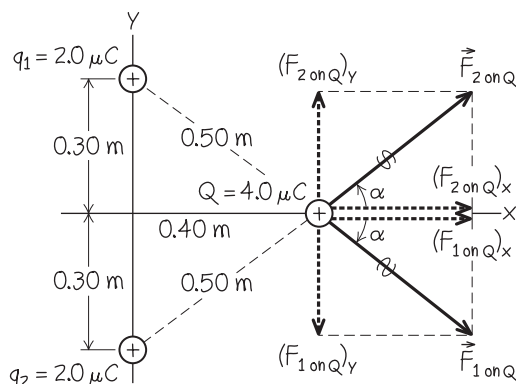
KEYCONCEPT Electric forces obey the principle of superposition of forces: The net electric force on a point charge due to two or more other charges is the vector sum of the individual electric forces that each of the other charges exert on it.

EXAMPLE 21.4 Vector addition of electric forces in a plane**WITH VARIATION PROBLEMS**

Two equal positive charges $q_1 = q_2 = 2.0 \mu\text{C}$ are located at $x = 0, y = 0.30 \text{ m}$ and $x = 0, y = -0.30 \text{ m}$, respectively. What are the magnitude and direction of the total electric force that q_1 and q_2 exert on a third charge $Q = 4.0 \mu\text{C}$ at $x = 0.40 \text{ m}, y = 0$?

IDENTIFY and SET UP As in Example 21.3, we must compute the force that each charge exerts on Q and then find the vector sum of those forces. Figure 21.14 shows the situation. Since the three charges do not all lie on a line, the best way to calculate the forces is to use components.

Figure 21.14 Our sketch for this problem.



EXECUTE Figure 21.14 shows the forces $\vec{F}_{1 \text{ on } Q}$ and $\vec{F}_{2 \text{ on } Q}$ due to the identical charges q_1 and q_2 , which are at equal distances from Q . From Coulomb's law, *both* forces have magnitude

$$\begin{aligned} F_{1 \text{ or } 2 \text{ on } Q} &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(4.0 \times 10^{-6} \text{ C})(2.0 \times 10^{-6} \text{ C})}{(0.50 \text{ m})^2} \\ &= 0.29 \text{ N} \end{aligned}$$

The x -components of the two forces are equal:

$$(F_{1 \text{ or } 2 \text{ on } Q})_x = (F_{1 \text{ or } 2 \text{ on } Q})\cos\alpha = (0.29 \text{ N})\frac{0.40 \text{ m}}{0.50 \text{ m}} = 0.23 \text{ N}$$

From symmetry we see that the y -components of the two forces are equal and opposite. Hence their sum is zero and the total force $\vec{F}$ on Q has only an x -component $F_x = 0.23 \text{ N} + 0.23 \text{ N} = 0.46 \text{ N}$. The total force on Q is in the $+x$ -direction, with magnitude 0.46 N .

EVALUATE The total force on Q points neither directly away from q_1 nor directly away from q_2 . Rather, this direction is a compromise that points away from the *system* of charges q_1 and q_2 . Can you see that the total force would *not* be in the $+x$ -direction if q_1 and q_2 were not equal or if the geometrical arrangement of the charges were not so symmetric?

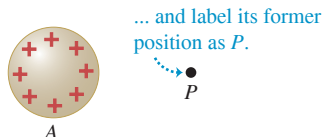
KEYCONCEPT When a point charge experiences electric forces due to two or more other charges, you must use vector addition to find the magnitude and direction of the net electric force on the first charge. This is most easily done by first finding the vector components of each individual electric force.

Figure 21.15 A charged object creates an electric field in the space around it.

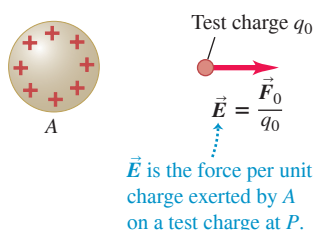
(a) *A* and *B* exert electric forces on each other.



(b) Remove object *B* ...



(c) Object *A* sets up an electric field $\vec{E}$ at point *P*.



BIO APPLICATION Sharks and the “Sixth Sense”

Sharks have the ability to locate prey (such as flounder and other bottom-dwelling fish) that are completely hidden beneath the sand at the bottom of the ocean. They do this by sensing the weak electric fields produced by muscle contractions in their prey. Sharks derive their sensitivity to electric fields (a “sixth sense”) from jelly-filled canals in their bodies. These canals end in pores on the shark’s skin (shown in this photograph). An electric field as weak as 5×10^{-7} N/C causes charge flow within the canals and triggers a signal in the shark’s nervous system. Because the shark has canals with different orientations, it can measure different components of the electric-field vector and hence determine the direction of the field.



TEST YOUR UNDERSTANDING OF SECTION 21.3 Suppose that charge q_2 in Example 21.4 were $-2.0 \mu\text{C}$. In this case, the total electric force on Q would be (i) in the positive x -direction; (ii) in the negative x -direction; (iii) in the positive y -direction; (iv) in the negative y -direction; (v) zero; (vi) none of these.

ANSWER

The force exerted by q_1 on Q is still as in Example 21.4. The magnitude of the force exerted by q_2 on Q is still equal to $F_{1 \text{ on } Q}$, but the direction of the force is now *toward* q_2 at an angle α below the x -axis. Hence the x -components of the two forces cancel while the (negative) y -components add together, and the total electric force is in the negative y -direction.

21.4 ELECTRIC FIELD AND ELECTRIC FORCES

When two electrically charged particles in empty space interact, how does each one know the other is there? We can begin to answer this question, and at the same time reformulate Coulomb’s law in a very useful way, by using the concept of *electric field*.

Electric Field

To introduce this concept, let’s look at the mutual repulsion of two positively charged objects *A* and *B* (Fig. 21.15a). Suppose *B* has charge q_0 , and let $\vec{F}_0$ be the electric force of *A* on *B*. One way to think about this force is as an “action-at-a-distance” force—that is, as a force that acts across empty space without needing physical contact between *A* and *B*. (Gravity can also be thought of as an “action-at-a-distance” force.) But a more fruitful way to visualize the repulsion between *A* and *B* is as a two-stage process. We first envision that object *A*, as a result of the charge that it carries, somehow *modifies the properties of the space around it*. Then object *B*, as a result of the charge that it carries, senses how space has been modified at its position. The response of object *B* is to experience the force $\vec{F}_0$.

To clarify how this two-stage process occurs, we first consider object *A* by itself: We remove object *B* and label its former position as point *P* (Fig. 21.15b). We say that the charged object *A* produces or causes an **electric field** at point *P* (and at all other points in the neighborhood). This electric field is present at *P* even if there is no charge at *P*; it is a consequence of the charge on object *A* only. If a point charge q_0 is then placed at point *P*, it experiences the force $\vec{F}_0$. We take the point of view that this force is exerted on q_0 by the field at *P* (Fig. 21.15c). Thus the electric field is the intermediary through which *A* communicates its presence to q_0 . Because the point charge q_0 would experience a force at *any* point in the neighborhood of *A*, the electric field that *A* produces exists at all points in the region around *A*.

We can likewise say that the point charge q_0 produces an electric field in the space around it and that this electric field exerts the force $-\vec{F}_0$ on object *A*. For each force (the force of *A* on q_0 and the force of q_0 on *A*), one charge sets up an electric field that exerts a force on the second charge. We emphasize that this is an *interaction* between *two* charged objects. A single charge produces an electric field in the surrounding space, but this electric field cannot exert a net force on the charge that created it; as we discussed in Section 4.3, an object cannot exert a net force on itself. (If this wasn’t true, you would be able to lift yourself to the ceiling by pulling up on your belt!)

The electric force on a charged object is exerted by the electric field created by other charged objects.

To find out experimentally whether there is an electric field at a particular point, we place a small charged object, which we call a **test charge**, at the point (Fig. 21.15c). If the test charge experiences an electric force, then there is an electric field at that point. This field is produced by charges other than q_0 .

Force is a vector quantity, so electric field is also a vector quantity. (Note the use of vector signs as well as boldface letters and plus, minus, and equals signs in the following discussion.) We define the *electric field* $\vec{E}$ at a point as the electric force $\vec{F}_0$

experienced by a test charge q_0 at the point, divided by the charge q_0 . That is, the electric field at a certain point is equal to the *electric force per unit charge* experienced by a charge at that point:

$$\text{Electric field} = \vec{E} = \frac{\vec{F}_0}{q_0} \quad (21.3)$$

Electric force on a test charge q_0 due to other charges
Value of test charge

In SI units, in which the unit of force is 1 N and the unit of charge is 1 C, the unit of electric-field magnitude is 1 newton per coulomb (1 N/C).

If the field $\vec{E}$ at a certain point is known, rearranging Eq. (21.3) gives the force $\vec{F}_0$ experienced by a point charge q_0 placed at that point. This force is just equal to the electric field $\vec{E}$ produced at that point by charges other than q_0 , multiplied by the charge q_0 :

$$\vec{F}_0 = q_0 \vec{E} \quad \begin{array}{l} \text{(force exerted on a point charge } q_0 \\ \text{by an electric field } \vec{E}) \end{array} \quad (21.4)$$

The charge q_0 can be either positive or negative. If q_0 is *positive*, the force $\vec{F}_0$ experienced by the charge is in the same direction as $\vec{E}$; if q_0 is *negative*, $\vec{F}_0$ and $\vec{E}$ are in opposite directions (**Fig. 21.16**).

While the electric field concept may be new to you, the basic idea—that one object sets up a field in the space around it and a second object responds to that field—is one that you’ve actually used before. Compare Eq. (21.4) to the familiar expression for the gravitational force $\vec{F}_g$ that the earth exerts on a mass m_0 :

$$\vec{F}_g = m_0 \vec{g} \quad (21.5)$$

In this expression, $\vec{g}$ is the acceleration due to gravity. If we divide both sides of Eq. (21.5) by the mass m_0 , we obtain

$$\vec{g} = \frac{\vec{F}_g}{m_0}$$

Thus $\vec{g}$ can be regarded as the gravitational force per unit mass. By analogy to Eq. (21.3), we can interpret $\vec{g}$ as the *gravitational field*. Thus we treat the gravitational interaction between the earth and the mass m_0 as a two-stage process: The earth sets up a gravitational field $\vec{g}$ in the space around it, and this gravitational field exerts a force given by Eq. (21.5) on the mass m_0 (which we can regard as a *test mass*). The gravitational field $\vec{g}$, or gravitational force per unit mass, is a useful concept because it does not depend on the mass of the object on which the gravitational force is exerted; likewise, the electric field $\vec{E}$, or electric force per unit charge, is useful because it does not depend on the charge of the object on which the electric force is exerted.

CAUTION $\vec{F}_0 = q_0 \vec{E}$ is for **point test charges only** The electric force experienced by a test charge q_0 can vary from point to point, so the electric field can also be different at different points. For this reason, use Eq. (21.4) to find the electric force on a *point* charge only. If a charged object is large enough in size, the electric field $\vec{E}$ may be noticeably different in magnitude and direction at different points on the object, and calculating the net electric force on it can be complicated. ■

Electric Field of a Point Charge

If the source distribution is a point charge q , it is easy to find the electric field that it produces. We call the location of the charge the **source point**, and we call the point P where we are determining the field the **field point**. It is also useful to introduce a *unit vector* $\hat{r}$ that points along the line from source point to field point (**Fig. 21.17a**, next page). This unit vector is equal to the displacement vector $\vec{r}$ from the source point to the field point, divided by the distance $r = |\vec{r}|$ between these two

Figure 21.16 The force $\vec{F}_0 = q_0 \vec{E}$ exerted on a point charge q_0 placed in an electric field $\vec{E}$.

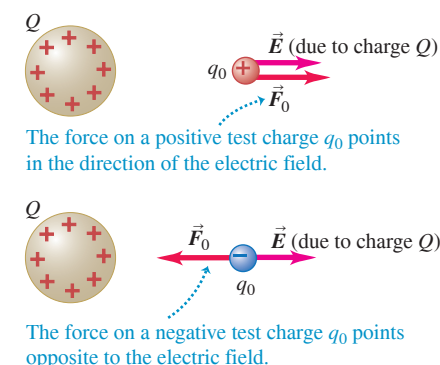
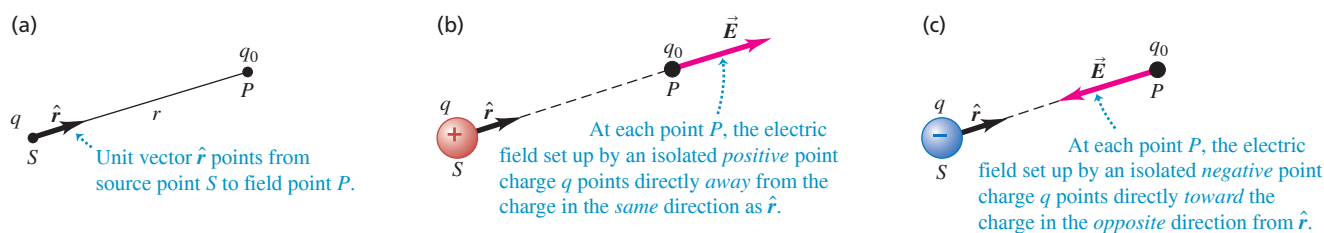


Figure 21.17 The electric field $\vec{E}$ produced at point P by an isolated point charge q at S . Note that in both (b) and (c), $\vec{E}$ is produced by q [see Eq. (21.7)] but acts on the charge q_0 at point P [see Eq. (21.4)].



points; that is, $\hat{r} = \vec{r}/r$. If we place a small test charge q_0 at the field point P , at a distance r from the source point, the magnitude F_0 of the force is given by Coulomb's law, Eq. (21.2):

$$F_0 = \frac{1}{4\pi\epsilon_0} \frac{|qq_0|}{r^2}$$

From Eq. (21.3) the magnitude E of the electric field at P is

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2} \quad (\text{magnitude of electric field due to a point charge}) \quad (21.6)$$

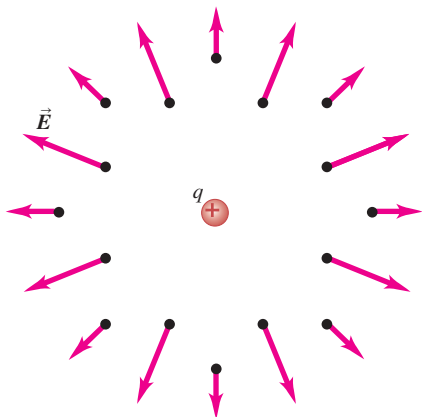
Using the unit vector $\hat{r}$, we can write a vector equation that gives both the magnitude and direction of the electric field $\vec{E}$:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (21.7)$$

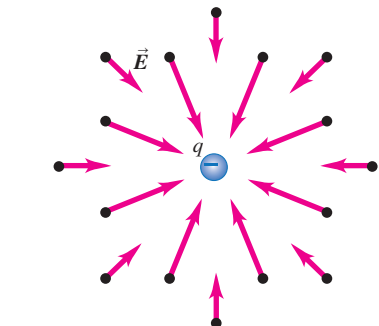
Labels for Eq. (21.7):
 - Electric field due to a point charge: $\vec{E}$
 - Value of point charge: q
 - Unit vector from point charge toward where field is measured: $\hat{r}$
 - Electric constant: $1/(4\pi\epsilon_0)$
 - Distance from point charge to where field is measured: r^2

Figure 21.18 A point charge q produces an electric field $\vec{E}$ at all points in space. The field strength decreases with increasing distance.

(a) The field produced by a positive point charge points away from the charge.



(b) The field produced by a negative point charge points toward the charge.



By definition, the electric field of a point charge always points *away from* a positive charge (that is, in the same direction as $\hat{r}$; see Fig. 21.17b) but *toward* a negative charge (that is, in the direction opposite $\hat{r}$; see Fig. 21.17c).

CAUTION The electric field at a point does not depend on the charge at that point. Equation (21.7) tells you the value of the electric field $\vec{E}$ at a certain point due to a point charge q at a *different* point. The electric field at a certain point is present whether or not there is a second point charge at that point to experience the field. **I**

We have emphasized calculating the electric field $\vec{E}$ at a certain point. But since $\vec{E}$ can vary from point to point, it is not a single vector quantity but rather an *infinite* set of vector quantities, one associated with each point in space. This is an example of a **vector field**. Figure 21.18 shows a number of the field vectors produced by a positive or negative point charge. If we use a rectangular (x, y, z) coordinate system, each component of $\vec{E}$ at any point is in general a function of the coordinates (x, y, z) of the point. We can represent the functions as $E_x(x, y, z)$, $E_y(x, y, z)$, and $E_z(x, y, z)$. Another example of a vector field is the velocity $\vec{v}$ of wind currents; the magnitude and direction of $\vec{v}$, and hence its vector components, vary from point to point in the atmosphere.

In some situations the magnitude and direction of the field (and hence its vector components) have the same values everywhere throughout a certain region; we then say that the field is *uniform* in this region. An important example of this is the electric field inside a *conductor*. If there is an electric field within a conductor, the field exerts a force on every charge in the conductor, giving the free charges a net motion. By definition an electrostatic situation is one in which the charges have *no* net motion. We conclude that *in electrostatics the electric field at every point within the material of a conductor must be zero*. (Note that we are not saying that the field is necessarily zero in a *hole* inside a conductor.)

In summary, our description of electric interactions has two parts. First, a given charge distribution acts as a source of electric field. Second, the electric field exerts a force on any

charge that is present in the field. Our analysis often has two corresponding steps: first, calculating the field caused by a source charge distribution; second, looking at the effect of the field in terms of force and motion. The second step often involves Newton's laws as well as the principles of electric interactions. In the next section we show how to calculate fields caused by various source distributions, but first here are three examples of calculating the field due to a point charge and of finding the force on a charge due to a given field $\vec{E}$.

EXAMPLE 21.5 Electric-field magnitude for a point charge

What is the magnitude of the electric field $\vec{E}$ at a field point 2.0 m from a point charge $q = 4.0$ nC?

IDENTIFY and SET UP This problem concerns the electric field due to a point charge. We are given the magnitude of the charge and the distance from the charge to the field point, so we use Eq. (21.6) to calculate the field magnitude E .

EXECUTE From Eq. (21.6),

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2} = (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{4.0 \times 10^{-9} \text{ C}}{(2.0 \text{ m})^2} = 9.0 \text{ N/C}$$

EVALUATE Our result $E = 9.0$ N/C means that if we placed a 1.0 C charge at a point 2.0 m from q , it would experience a 9.0 N force. The force on a 2.0 C charge at that point would be $(2.0 \text{ C})(9.0 \text{ N/C}) = 18 \text{ N}$, and so on.

KEYCONCEPT The magnitude of the electric field that a point charge produces at a given point is (a) proportional to the magnitude of the charge and (b) inversely proportional to the square of the distance from the point charge to the point where the field is measured.

EXAMPLE 21.6 Electric-field vector for a point charge

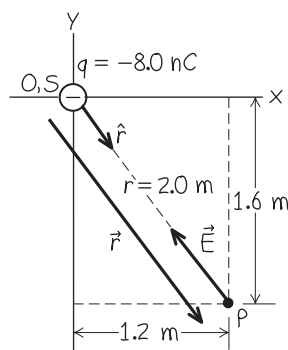
A point charge $q = -8.0$ nC is located at the origin. Find the electric-field vector at the field point $x = 1.2$ m, $y = -1.6$ m.

IDENTIFY and SET UP We must find the electric-field vector $\vec{E}$ due to a point charge. **Figure 21.19** shows the situation. We use Eq. (21.7); to do this, we must find the distance r from the source point S (the position of the charge q , which in this example is at the origin O) to the field point P , and we must obtain an expression for the unit vector $\hat{r} = \vec{r}/r$ that points from S to P .

EXECUTE The distance from S to P is

$$r = \sqrt{x^2 + y^2} = \sqrt{(1.2 \text{ m})^2 + (-1.6 \text{ m})^2} = 2.0 \text{ m}$$

Figure 21.19 Our sketch for this problem.



The unit vector $\hat{r}$ is then

$$\begin{aligned} \hat{r} &= \frac{\vec{r}}{r} = \frac{x\hat{i} + y\hat{j}}{r} \\ &= \frac{(1.2 \text{ m})\hat{i} + (-1.6 \text{ m})\hat{j}}{2.0 \text{ m}} = 0.60\hat{i} - 0.80\hat{j} \end{aligned}$$

Then, from Eq. (21.7),

$$\begin{aligned} \vec{E} &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(-8.0 \times 10^{-9} \text{ C})}{(2.0 \text{ m})^2} (0.60\hat{i} - 0.80\hat{j}) \\ &= (-11 \text{ N/C})\hat{i} + (14 \text{ N/C})\hat{j} \end{aligned}$$

EVALUATE Since q is negative, $\vec{E}$ points from the field point to the charge (the source point), in the direction opposite to $\hat{r}$ (compare Fig. 21.17c). We leave the calculation of the magnitude and direction of $\vec{E}$ to you.

KEYCONCEPT When you calculate the electric field due to a point charge q , the unit vector $\hat{r}$ always points from the charge to the point where the field is measured. If q is positive, the field is in the same direction as $\hat{r}$ (away from the charge); if q is negative, the field is in the direction opposite to $\hat{r}$ (toward the charge.)

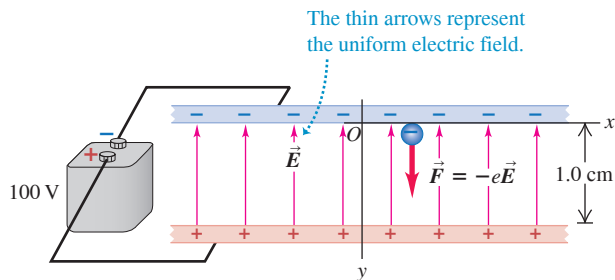
EXAMPLE 21.7 Electron in a uniform field

When the terminals of a battery are connected to two parallel conducting plates with a small gap between them, the resulting charges on the plates produce a nearly uniform electric field $\vec{E}$ between the plates. (In the next section we'll see why this is.) If the plates are 1.0 cm apart and are connected to a 100 volt battery as shown in **Fig. 21.20** (next page), the field

is vertically upward and has magnitude $E = 1.00 \times 10^4$ N/C. (a) If an electron (charge $-e = -1.60 \times 10^{-19}$ C, mass $m = 9.11 \times 10^{-31}$ kg) is released from rest at the upper plate, what is its acceleration? (b) What speed and kinetic energy does it acquire while traveling 1.0 cm to the lower plate? (c) How long does it take to travel this distance?

Continued

Figure 21.20 A uniform electric field between two parallel conducting plates connected to a 100 volt battery. (The separation of the plates is exaggerated in this figure relative to the dimensions of the plates.)



IDENTIFY and SET UP This example involves the relationship between electric field and electric force. It also involves the relationship between force and acceleration, the definition of kinetic energy, and the kinematic relationships among acceleration, distance, velocity, and time. Figure 21.20 shows our coordinate system. We are given the electric field, so we use Eq. (21.4) to find the force on the electron and Newton's second law to find its acceleration. Because the field is uniform, the force is constant and we can use the constant-acceleration formulas from Chapter 2 to find the electron's velocity and travel time. We find the kinetic energy from $K = \frac{1}{2}mv^2$.

EXECUTE (a) Although $\vec{E}$ is upward (in the $+y$ -direction), $\vec{F}$ is downward (because the electron's charge is negative) and so F_y is negative. Because F_y is constant, the electron's acceleration is constant:

$$a_y = \frac{F_y}{m} = \frac{-eE}{m} = \frac{(-1.60 \times 10^{-19} \text{ C})(1.00 \times 10^4 \text{ N/C})}{9.11 \times 10^{-31} \text{ kg}} = -1.76 \times 10^{15} \text{ m/s}^2$$

(b) The electron starts from rest, so its motion is in the y -direction only (the direction of the acceleration). We can find the electron's speed at any position y from the constant-acceleration Eq. (2.13), $v_y^2 = v_{0y}^2 + 2a_y(y - y_0)$. We have $v_{0y} = 0$ and $y_0 = 0$, so at $y = -1.0 \text{ cm} = -1.0 \times 10^{-2} \text{ m}$ we have

$$|v_y| = \sqrt{2a_y y} = \sqrt{2(-1.76 \times 10^{15} \text{ m/s}^2)(-1.0 \times 10^{-2} \text{ m})} = 5.9 \times 10^6 \text{ m/s}$$

The velocity is downward, so $v_y = -5.9 \times 10^6 \text{ m/s}$. The electron's kinetic energy is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}(9.11 \times 10^{-31} \text{ kg})(5.9 \times 10^6 \text{ m/s})^2 = 1.6 \times 10^{-17} \text{ J}$$

(c) From Eq. (2.8) for constant acceleration, $v_y = v_{0y} + a_y t$, so

$$t = \frac{v_y - v_{0y}}{a_y} = \frac{(-5.9 \times 10^6 \text{ m/s}) - (0)}{-1.76 \times 10^{15} \text{ m/s}^2} = 3.4 \times 10^{-9} \text{ s}$$

EVALUATE Our results show that in problems concerning subatomic particles such as electrons, many quantities—including acceleration, speed, kinetic energy, and time—will have *very* different values from those typical of everyday objects such as baseballs and automobiles.

KEYCONCEPT If you place a point charge q_0 at a location where the electric field is $\vec{E}$, the electric force on the point charge is $q_0 \vec{E}$. This force is in the same direction as $\vec{E}$ if q_0 is positive; the force is in the direction opposite to $\vec{E}$ if q_0 is negative.

TEST YOUR UNDERSTANDING OF SECTION 21.4 (a) A negative point charge moves along a straight-line path directly toward a stationary positive point charge. Which aspect(s) of the electric force on the negative point charge will remain constant as it moves? (i) Magnitude; (ii) direction; (iii) both magnitude and direction; (iv) neither magnitude nor direction. (b) A negative point charge moves along a circular orbit around a positive point charge. Which aspect(s) of the electric force on the negative point charge will remain constant as it moves? (i) Magnitude; (ii) direction; (iii) both magnitude and direction; (iv) neither magnitude nor direction.

ANSWER

(a) (i) The electric field $\vec{E}$ produced by a positive point charge points directly away from the charge (see Fig. 21.18a) and has a magnitude that depends on the distance r from the charge to the field point. Hence a second, negative point charge $q < 0$ will feel a force $\vec{F} = q\vec{E}$ that points directly toward the positive charge and has a magnitude that depends on the distance r between the two charges. If the negative charge moves directly toward the positive charge, the direction of the force remains the same but the force magnitude increases as the distance r decreases. If the negative charge moves in a circle around the positive charge, the force magnitude stays the same (because the distance r is constant) but the force direction changes.

21.5 ELECTRIC-FIELD CALCULATIONS

Equation (21.7) gives the electric field caused by a single point charge. But in most realistic situations that involve electric fields and forces, we encounter charge that is *distributed* over space. The charged plastic and glass rods in Fig. 21.1 have electric charge distributed over their surfaces, as does the imaging drum of a laser printer (Fig. 21.2). In this section we'll learn to calculate electric fields caused by various distributions of electric charge. Calculations of this kind are of tremendous importance for technological applications of electric forces. To determine the trajectories of atomic nuclei in an accelerator for cancer

radiotherapy or of charged particles in a semiconductor electronic device, you have to know the detailed nature of the electric field acting on the charges.

The Superposition of Electric Fields

To find the field caused by a charge distribution, we imagine the distribution to be made up of many point charges $q_1, q_2, q_3, \dots$ (This is actually quite a realistic description, since we have seen that charge is carried by electrons and protons that are so small as to be almost pointlike.) At any given point P , each point charge produces its own electric field $\vec{E}_1, \vec{E}_2, \vec{E}_3, \dots$, so a test charge q_0 placed at P experiences a force $\vec{F}_1 = q_0\vec{E}_1$ from charge q_1 , a force $\vec{F}_2 = q_0\vec{E}_2$ from charge q_2 , and so on. From the principle of superposition of forces discussed in Section 21.3, the *total* force $\vec{F}_0$ that the charge distribution exerts on q_0 is the vector sum of these individual forces:

$$\vec{F}_0 = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots = q_0\vec{E}_1 + q_0\vec{E}_2 + q_0\vec{E}_3 + \dots$$

The combined effect of all the charges in the distribution is described by the *total* electric field $\vec{E}$ at point P . From the definition of electric field, Eq. (21.3), this is

$$\vec{E} = \frac{\vec{F}_0}{q_0} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$$

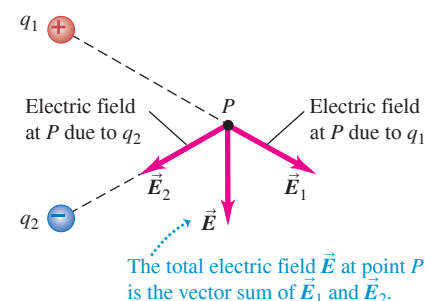
This equation is the mathematical expression of the *principle of superposition of electric fields*, illustrated in **Fig. 21.21**:

PRINCIPLE OF SUPERPOSITION OF ELECTRIC FIELDS The total electric field at a point P is the vector sum of the fields at P due to each point charge in the charge distribution.

When charge is distributed along a line, over a surface, or through a volume, a few additional terms are useful. For a line charge distribution (such as a long, thin, charged plastic rod), we use λ (the Greek letter lambda) to represent the **linear charge density** (charge per unit length, measured in C/m). When charge is distributed over a surface (such as the surface of the imaging drum of a laser printer), we use σ (sigma) to represent the **surface charge density** (charge per unit area, measured in C/m²). And when charge is distributed through a volume, we use ρ (rho) to represent the **volume charge density** (charge per unit volume, C/m³).

Some of the calculations in the following examples may look complex. After you've worked through the examples one step at a time, the process will seem less formidable. We'll use many of the calculational techniques in these examples in Chapter 28 to calculate the *magnetic* fields caused by charges in motion.

Figure 21.21 Illustrating the principle of superposition of electric fields.



PROBLEM-SOLVING STRATEGY 21.2 Electric-Field Calculations

IDENTIFY the relevant concepts: Use the principle of superposition to calculate the electric field due to a discrete or continuous charge distribution.

SET UP the problem using the following steps:

1. Make a drawing showing the locations of the charges and your choice of coordinate axes.
2. On your drawing, indicate the position of the *field point* P (the point at which you want to calculate the electric field $\vec{E}$).

EXECUTE the solution as follows:

1. Use consistent units. Distances must be in meters and charge must be in coulombs. If you are given centimeters or nanocoulombs, don't forget to convert.
2. Distinguish between the source point S and the field point P . The field produced by a point charge always points from S to P if the charge is positive, and from P to S if the charge is negative.

3. Use *vector* addition when applying the principle of superposition; review the treatment of vector addition in Chapter 1 if necessary.
4. Simplify your calculations by exploiting any symmetries in the charge distribution.
5. If the charge distribution is continuous, define a small element of charge that can be considered as a point, find its electric field at P , and find a way to add the fields of all the charge elements by doing an integral. Usually it is easiest to do this for each component of $\vec{E}$ separately, so you may need to evaluate more than one integral. Ensure that the limits on your integrals are correct; especially when the situation has symmetry, don't count a charge twice.

EVALUATE your answer: Check that the direction of $\vec{E}$ is reasonable. If your result for the electric-field magnitude E is a function of position (say, the coordinate x), check your result in any limits for which you know what the magnitude should be. When possible, check your answer by calculating it in a different way.

EXAMPLE 21.8 Field of an electric dipole

WITH VARIATION PROBLEMS

Point charges $q_1 = +12 \text{ nC}$ and $q_2 = -12 \text{ nC}$ are 0.100 m apart (**Fig. 21.22**). (Such pairs of point charges with equal magnitude and opposite sign are called *electric dipoles*.) Compute the electric field caused by q_1 , the field caused by q_2 , and the total field (a) at point a ; (b) at point b ; and (c) at point c .

IDENTIFY and SET UP We must find the total electric field at various points due to two point charges. We use the principle of superposition: $\vec{E} = \vec{E}_1 + \vec{E}_2$. Figure 21.22 shows the coordinate system and the locations of the field points a , b , and c .

EXECUTE At each field point, $\vec{E}$ depends on $\vec{E}_1$ and $\vec{E}_2$ there; we first calculate the magnitudes E_1 and E_2 at each field point. At a the magnitude of the field $\vec{E}_{1a}$ caused by q_1 is

$$\begin{aligned} E_{1a} &= \frac{1}{4\pi\epsilon_0} \frac{|q_1|}{r^2} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{12 \times 10^{-9} \text{ C}}{(0.060 \text{ m})^2} \\ &= 3.0 \times 10^4 \text{ N/C} \end{aligned}$$

We calculate the other field magnitudes in a similar way. The results are

$$\begin{aligned} E_{1a} &= 3.0 \times 10^4 \text{ N/C} \\ E_{1b} &= 6.8 \times 10^4 \text{ N/C} \\ E_{1c} &= 6.39 \times 10^3 \text{ N/C} \\ E_{2a} &= 6.8 \times 10^4 \text{ N/C} \\ E_{2b} &= 0.55 \times 10^4 \text{ N/C} \\ E_{2c} &= E_{1c} = 6.39 \times 10^3 \text{ N/C} \end{aligned}$$

The *directions* of the corresponding fields are in all cases *away* from the positive charge q_1 and *toward* the negative charge q_2 .

(a) At a , $\vec{E}_{1a}$ and $\vec{E}_{2a}$ are both directed to the right, so

$$\vec{E}_a = E_{1a}\hat{i} + E_{2a}\hat{i} = (9.8 \times 10^4 \text{ N/C})\hat{i}$$

(b) At b , $\vec{E}_{1b}$ is directed to the left and $\vec{E}_{2b}$ is directed to the right, so

$$\vec{E}_b = -E_{1b}\hat{i} + E_{2b}\hat{i} = (-6.2 \times 10^4 \text{ N/C})\hat{i}$$

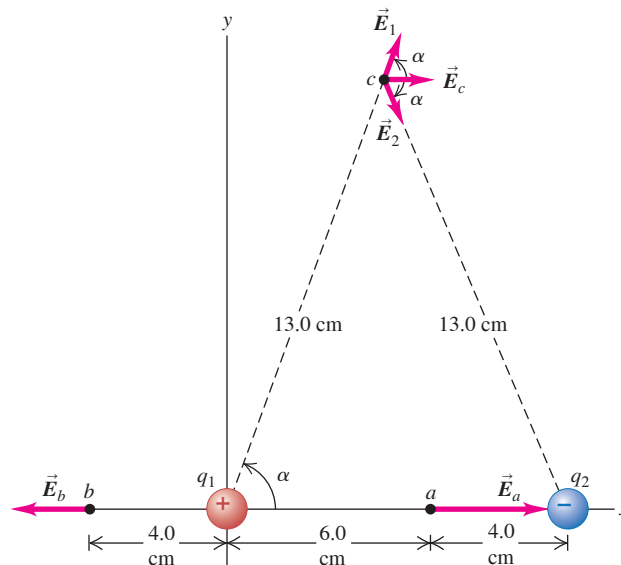
(c) Figure 21.22 shows the directions of $\vec{E}_1$ and $\vec{E}_2$ at c . Both vectors have the same x -component:

$$\begin{aligned} E_{1cx} &= E_{2cx} = E_{1c} \cos \alpha = (6.39 \times 10^3 \text{ N/C}) \left(\frac{5}{13} \right) \\ &= 2.46 \times 10^3 \text{ N/C} \end{aligned}$$

From symmetry, E_{1y} and E_{2y} are equal and opposite, so their sum is zero. Hence

$$\vec{E}_c = 2(2.46 \times 10^3 \text{ N/C})\hat{i} = (4.9 \times 10^3 \text{ N/C})\hat{i}$$

Figure 21.22 Electric field at three points, a , b , and c , set up by charges q_1 and q_2 , which form an electric dipole.



EVALUATE We can also find $\vec{E}_c$ by using Eq. (21.7) for the field of a point charge. The displacement vector $\vec{r}_1$ from q_1 to point c is $\vec{r}_1 = r \cos \alpha \hat{i} + r \sin \alpha \hat{j}$. Hence the unit vector that points from q_1 to point c is $\hat{r}_1 = \vec{r}_1/r = \cos \alpha \hat{i} + \sin \alpha \hat{j}$. By symmetry, the unit vector that points from q_2 to point c has the opposite x -component but the same y -component: $\hat{r}_2 = -\cos \alpha \hat{i} + \sin \alpha \hat{j}$. We can now use Eq. (21.7) to write the fields $\vec{E}_{1c}$ and $\vec{E}_{2c}$ at c in vector form, then find their sum. Since $q_2 = -q_1$ and the distance r to c is the same for both charges,

$$\begin{aligned} \vec{E}_c &= \vec{E}_{1c} + \vec{E}_{2c} = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2} \hat{r}_1 + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r^2} \hat{r}_2 \\ &= \frac{1}{4\pi\epsilon_0 r^2} (q_1 \hat{r}_1 + q_2 \hat{r}_2) \\ &= \frac{q_1}{4\pi\epsilon_0 r^2} (\hat{r}_1 - \hat{r}_2) \\ &= \frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2} (2 \cos \alpha \hat{i}) \\ &= 2(9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{12 \times 10^{-9} \text{ C}}{(0.13 \text{ m})^2} \left(\frac{5}{13} \right) \hat{i} \\ &= (4.9 \times 10^3 \text{ N/C})\hat{i} \end{aligned}$$

This is the same as we calculated in part (c).

KEYCONCEPT To find the net electric field at a point due to two or more point charges, first find the field at that point due to each individual charge. Then use vector addition to find the magnitude and direction of the net field at that point.

EXAMPLE 21.9 Field of a ring of charge**WITH VARIATION PROBLEMS**

Charge Q is uniformly distributed around a conducting ring of radius a (Fig. 21.23). Find the electric field at a point P on the ring axis at a distance x from its center.

IDENTIFY and SET UP This is a problem in the superposition of electric fields. Each bit of charge around the ring produces an electric field at an arbitrary point on the x -axis; our target variable is the total field at this point due to all such bits of charge.

EXECUTE We divide the ring into infinitesimal segments ds as shown in Fig. 21.23. In terms of the linear charge density $\lambda = Q/2\pi a$, the charge in a segment of length ds is $dQ = \lambda ds$. Consider two identical segments, one as shown in the figure at $y = a$ and another halfway around the ring at $y = -a$. From Example 21.4, we see that the net force $d\vec{F}$ they exert on a point test charge at P , and thus their net field $d\vec{E}$, are directed along the x -axis. The same is true for any such pair of segments around the ring, so the *net* field at P is along the x -axis: $\vec{E} = E_x \hat{i}$.

To calculate E_x , note that the square of the distance r from a single ring segment to the point P is $r^2 = x^2 + a^2$. Hence the magnitude of this segment's contribution $d\vec{E}$ to the electric field at P is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dQ}{x^2 + a^2}$$

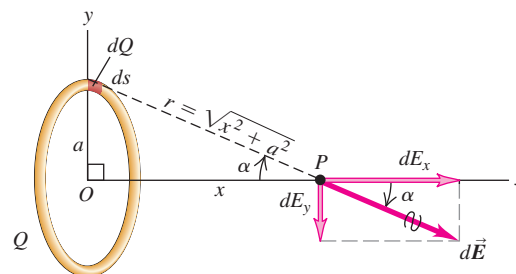
The x -component of this field is $dE_x = dE \cos \alpha$. We know $dQ = \lambda ds$ and Fig. 21.23 shows that $\cos \alpha = x/r = x/(x^2 + a^2)^{1/2}$, so

$$dE_x = dE \cos \alpha = \frac{1}{4\pi\epsilon_0} \frac{dQ}{x^2 + a^2} \frac{x}{\sqrt{x^2 + a^2}} = \frac{1}{4\pi\epsilon_0} \frac{\lambda x}{(x^2 + a^2)^{3/2}} ds$$

To find E_x we integrate this expression over the entire ring—that is, for s from 0 to $2\pi a$ (the circumference of the ring). The integrand has the same value for all points on the ring, so it can be taken outside the integral. Hence we get

$$\begin{aligned} E_x &= \int dE_x = \frac{1}{4\pi\epsilon_0} \frac{\lambda x}{(x^2 + a^2)^{3/2}} \int_0^{2\pi a} ds \\ &= \frac{1}{4\pi\epsilon_0} \frac{\lambda x}{(x^2 + a^2)^{3/2}} (2\pi a) \end{aligned}$$

Figure 21.23 Calculating the electric field on the axis of a ring of charge. In this figure, the charge is assumed to be positive.



$$\vec{E} = E_x \hat{i} = \frac{1}{4\pi\epsilon_0} \frac{Qx}{(x^2 + a^2)^{3/2}} \hat{i} \quad (21.8)$$

EVALUATE Equation (21.8) shows that $\vec{E} = \mathbf{0}$ at the center of the ring ($x = 0$). This makes sense; charges on opposite sides of the ring push in opposite directions on a test charge at the center, and the vector sum of each such pair of forces is zero. When the field point P is much farther from the ring than the ring's radius, we have $x \gg a$ and the denominator in Eq. (21.8) becomes approximately equal to x^3 . In this limit the electric field at P is

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{x^2} \hat{i}$$

That is, when the ring is so far away that its radius is negligible in comparison to the distance x , its field is the same as that of a point charge.

KEYCONCEPT To find the vector components of the net electric field at a point due to a continuous distribution of charge, first divide the distribution into infinitesimally small segments. Then find the components of the field at the point due to one such segment. Finally, integrate each component of the field due to a segment over all segments in the charge distribution.

EXAMPLE 21.10 Field of a charged line segment**WITH VARIATION PROBLEMS**

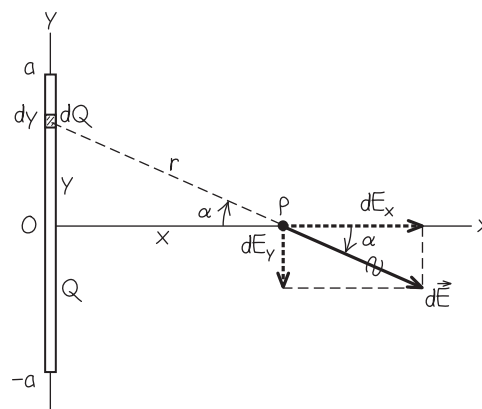
Positive charge Q is distributed uniformly along the y -axis between $y = -a$ and $y = +a$. Find the electric field at point P on the x -axis at a distance x from the origin.

IDENTIFY and SET UP Figure 21.24 shows the situation. As in Example 21.9, we must find the electric field due to a continuous distribution of charge. Our target variable is an expression for the electric field at P as a function of x . The x -axis is a perpendicular bisector of the segment, so we can use a symmetry argument.

EXECUTE We divide the line charge of length $2a$ into infinitesimal segments of length dy . The linear charge density is $\lambda = Q/2a$, and the charge in a segment is $dQ = \lambda dy = (Q/2a)dy$. The distance r from a segment at height y to the field point P is $r = (x^2 + y^2)^{1/2}$, so the magnitude of the field at P due to the segment at height y is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dQ}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{dy}{(x^2 + y^2)}$$

Figure 21.24 Our sketch for this problem.



Continued

Figure 21.24 shows that the x - and y -components of this field are $dE_x = dE \cos \alpha$ and $dE_y = -dE \sin \alpha$, where $\cos \alpha = x/r$ and $\sin \alpha = y/r$. Hence

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{xdy}{(x^2 + y^2)^{3/2}}$$

$$dE_y = -\frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{ydy}{(x^2 + y^2)^{3/2}}$$

To find the total field at P , we must sum the fields from all segments along the line—that is, we must integrate from $y = -a$ to $y = +a$. You should work out the details of the integration (a table of integrals will help). The results are

$$E_x = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \int_{-a}^{+a} \frac{xdy}{(x^2 + y^2)^{3/2}} = \frac{Q}{4\pi\epsilon_0} \frac{1}{x\sqrt{x^2 + a^2}}$$

$$E_y = -\frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \int_{-a}^{+a} \frac{ydy}{(x^2 + y^2)^{3/2}} = 0$$

or, in vector form,

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{x\sqrt{x^2 + a^2}} \hat{i} \quad (21.9)$$

$\vec{E}$ points away from the line of charge if λ is positive and toward the line of charge if λ is negative.

EVALUATE Using a symmetry argument as in Example 21.9, we could have guessed that E_y would be zero; if we place a positive test charge at P , the upper half of the line of charge pushes downward on it, and the lower half pushes up with equal magnitude. Symmetry also tells us that the upper and lower halves of the segment contribute equally to the total field at P .

If the segment is very *short* (or the field point is very far from the segment) so that $x \gg a$, we can ignore a in the denominator of Eq. (21.9). Then the field becomes that of a point charge, just as in Example 21.9:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{x^2} \hat{i}$$

To see what happens if the segment is very *long* (or the field point is very close to it) so that $a \gg x$, we first rewrite Eq. (21.9) slightly:

$$\vec{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{x\sqrt{(x^2/a^2) + 1}} \hat{i} \quad (21.10)$$

In the limit $a \gg x$ we can ignore x^2/a^2 in the denominator of Eq. (21.10), so

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 x} \hat{i}$$

This is the field of an *infinitely long* line of charge. At any point P at a perpendicular distance r from the line in *any* direction, $\vec{E}$ has magnitude

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{infinite line of charge})$$

Note that this field is proportional to $1/r$ rather than to $1/r^2$ as for a point charge.

There's really no such thing in nature as an infinite line of charge. But when the field point is close enough to the line, there's very little difference between the result for an infinite line and the real-life finite case. For example, if the distance r of the field point from the center of the line is 1% of the length of the line, the value of E differs from the infinite-length value by less than 0.02%.

KEYCONCEPT The electric field due to a symmetrical distribution of charge is most easily calculated at a point of symmetry. Whenever possible, take advantage of the symmetry of the situation to check your results.

EXAMPLE 21.11 Field of a uniformly charged disk

A nonconducting disk of radius R has a uniform positive surface charge density σ . Find the electric field at a point along the axis of the disk a distance x from its center. Assume that x is positive.

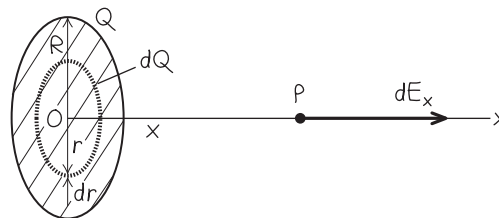
IDENTIFY and SET UP Figure 21.25 shows the situation. We represent the charge distribution as a collection of concentric rings of charge dQ . In Example 21.9 we obtained Eq. (21.8) for the field on the axis of a single uniformly charged ring, so all we need do here is integrate the contributions of our rings.

EXECUTE A typical ring has charge dQ , inner radius r , and outer radius $r + dr$. Its area is approximately equal to its width dr times its circumference $2\pi r$, or $dA = 2\pi r dr$. The charge per unit area is $\sigma = dQ/dA$, so the charge of the ring is $dQ = \sigma dA = 2\pi\sigma r dr$. We use dQ in place of Q in Eq. (21.8), the expression for the field due to a ring that we found in Example 21.9, and replace the ring radius a with r . Then the field component dE_x at point P due to this ring is

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{2\pi\sigma r x dr}{(x^2 + r^2)^{3/2}}$$

To find the total field due to all the rings, we integrate dE_x over r from $r = 0$ to $r = R$ (not from $-R$ to R):

Figure 21.25 Our sketch for this problem.



$$E_x = \int_0^R \frac{1}{4\pi\epsilon_0} \frac{(2\pi\sigma r dr)x}{(x^2 + r^2)^{3/2}} = \frac{\sigma x}{4\epsilon_0} \int_0^R \frac{2r dr}{(x^2 + r^2)^{3/2}}$$

You can evaluate this integral by making the substitution $t = x^2 + r^2$ (which yields $dt = 2r dr$); you can work out the details. The result is

$$E_x = \frac{\sigma x}{2\epsilon_0} \left[-\frac{1}{\sqrt{x^2 + R^2}} + \frac{1}{x} \right]$$

$$= \frac{\sigma}{2\epsilon_0} \left[1 - \frac{1}{\sqrt{(R^2/x^2) + 1}} \right] \quad (21.11)$$

EVALUATE If the disk is very large (or if we are very close to it), so that $R \gg x$, the term $1/\sqrt{(R^2/x^2) + 1}$ in Eq. (21.11) is very much less than 1. Then Eq. (21.11) becomes

$$E = \frac{\sigma}{2\epsilon_0} \quad (21.12)$$

Our final result does not contain the distance x from the plane. Hence the electric field produced by an *infinite* plane sheet of charge is *independent of the distance from the sheet*. The field direction is everywhere perpendicular to the sheet, away from it. There is no such thing as an infinite sheet of charge, but if the dimensions of the sheet are

much larger than the distance x of the field point P from the sheet, the field is very nearly given by Eq. (21.12).

If P is to the *left* of the plane ($x < 0$), the result is the same except that the direction of $\vec{E}$ is to the left instead of the right. If the surface charge density is negative, the directions of the fields on both sides of the plane are toward it rather than away from it.

KEYCONCEPT To find the electric field due to a two-dimensional charge distribution such as a disk, divide the distribution into infinitesimal segments such as rings for which you know the components of the electric field. Then integrate over these segments to find the net field.

EXAMPLE 21.12 Field of two oppositely charged infinite sheets

Two infinite plane sheets with uniform surface charge densities $+\sigma$ and $-\sigma$ are placed parallel to each other with separation d (Fig. 21.26). Find the electric field between the sheets, above the upper sheet, and below the lower sheet.

IDENTIFY and SET UP Equation (21.12) gives the electric field due to a single infinite plane sheet of charge. To find the field due to *two* such sheets, we combine the fields by using the principle of superposition (Fig. 21.26).

EXECUTE From Eq. (21.12), both $\vec{E}_1$ and $\vec{E}_2$ have the same magnitude at all points, independent of distance from either sheet:

$$E_1 = E_2 = \frac{\sigma}{2\epsilon_0}$$

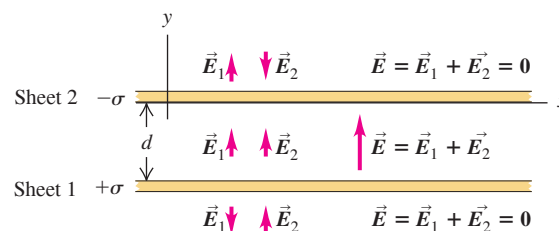
From Example 21.11, $\vec{E}_1$ is everywhere directed away from sheet 1, and $\vec{E}_2$ is everywhere directed toward sheet 2.

Between the sheets, $\vec{E}_1$ and $\vec{E}_2$ reinforce each other; above the upper sheet and below the lower sheet, they cancel each other. Thus the total field is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = \begin{cases} 0 & \text{above the upper sheet} \\ \frac{\sigma}{\epsilon_0} \hat{j} & \text{between the sheets} \\ 0 & \text{below the lower sheet} \end{cases}$$

EVALUATE Because we considered the sheets to be infinite, our result does not depend on the separation d . Our result shows that the field between oppositely charged plates is essentially uniform if the plate separation is much smaller than the dimensions of the plates. We actually used this result in Example 21.7 (Section 21.4).

Figure 21.26 Finding the electric field due to two oppositely charged infinite sheets. The sheets are seen edge-on; only a portion of the infinite sheets can be shown!



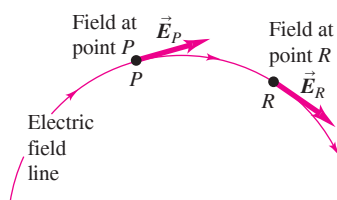
CAUTION Electric fields are not “flows” You may have thought that the field $\vec{E}_1$ of sheet 1 would be unable to “penetrate” sheet 2, and that field $\vec{E}_2$ caused by sheet 2 would be unable to “penetrate” sheet 1. You might conclude this if you think of the electric field as some kind of physical substance that “flows” into or out of charges. But there is no such substance, and the electric fields $\vec{E}_1$ and $\vec{E}_2$ depend on only the individual charge distributions that create them. The *total* field at every point is just the vector sum of $\vec{E}_1$ and $\vec{E}_2$.

KEYCONCEPT An infinite, uniform sheet of charge produces a uniform electric field at all points. If the charge is positive, the field points away from the sheet; if the charge is negative, the field points toward the sheet.

TEST YOUR UNDERSTANDING OF SECTION 21.5 Suppose that the line of charge in Fig. 21.24 (Example 21.10) had charge $+Q$ distributed uniformly between $y = 0$ and $y = +a$ and had charge $-Q$ distributed uniformly between $y = 0$ and $y = -a$. In this situation, the electric field at P would be (i) in the positive x -direction; (ii) in the negative x -direction; (iii) in the positive y -direction; (iv) in the negative y -direction; (v) zero; (vi) none of these.

ANSWER (iv) Think of a pair of segments of length dy , one at coordinate $y > 0$ and the other at coordinate $-y > 0$. The upper segment has a positive charge and produces an electric field $d\vec{E}$ at P that points away from the segment, so this $d\vec{E}$ has a positive x -component and a negative y -component, like the vector $d\vec{E}$ in Fig. 21.24. The lower segment has the same amount of negative charge. It produces a $d\vec{E}$ that has the same magnitude but points *toward* the lower segment, so it has a negative x -component and a negative y -component. By symmetry, the two x -components are equal but opposite, so they cancel. Thus the total electric field has only a negative y -component.

Figure 21.27 The direction of the electric field at any point is tangent to the field line through that point.



21.6 ELECTRIC FIELD LINES

The concept of an electric field can be a little elusive because you can't see an electric field directly. Electric field *lines* can be a big help for visualizing electric fields and making them seem more real. An **electric field line** is an imaginary line or curve drawn through a region of space so that its tangent at any point is in the direction of the electric-field vector at that point. **Figure 21.27** shows the basic idea. (We used a similar concept in our discussion of fluid flow in Section 12.5. A *streamline* is a line or curve whose tangent at any point is in the direction of the velocity of the fluid at that point. However, the similarity between electric field lines and fluid streamlines is a mathematical one only; there is nothing “flowing” in an electric field.) The English scientist Michael Faraday (1791–1867) first introduced the concept of field lines. He called them “lines of force,” but the term “field lines” is preferable.

Electric field lines show the direction of $\vec{E}$ at each point, and their spacing gives a general idea of the *magnitude* of $\vec{E}$ at each point. Where $\vec{E}$ is strong, we draw lines close together; where $\vec{E}$ is weaker, they are farther apart. At any particular point, the electric field has a unique direction, so only one field line can pass through each point of the field. In other words, *field lines never intersect*.

Figure 21.28 shows some of the electric field lines in a plane containing (a) a single positive charge; (b) two equal-magnitude charges, one positive and one negative (a dipole); and (c) two equal positive charges. Such diagrams are called *field maps*; they are cross sections of the actual three-dimensional patterns. The direction of the total electric field at every point in each diagram is along the tangent to the electric field line passing through the point. Arrowheads indicate the direction of the $\vec{E}$ -field vector along each field line. The actual field vectors have been drawn at several points in each pattern. Notice that in general, the magnitude of the electric field is different at different points on a given field line; a field line is *not* a curve of constant electric-field magnitude!

Figure 21.28 shows that field lines are directed *away* from positive charges (since close to a positive point charge, $\vec{E}$ points away from the charge) and *toward* negative charges (since close to a negative point charge, $\vec{E}$ points toward the charge). In regions where the field magnitude is large, such as between the positive and negative charges in Fig. 21.28b, the field lines are drawn close together. In regions where the field magnitude is small, such as between the two positive charges in Fig. 21.28c, the lines are widely separated. In a *uniform* field, the field lines are straight, parallel, and uniformly spaced, as in Fig. 21.20.

Figure 21.28 Electric field lines for three different charge distributions. In general, the magnitude of $\vec{E}$ is different at different points along a given field line.

(a) A single positive charge

(b) Two equal and opposite charges (a dipole)

(c) Two equal positive charges

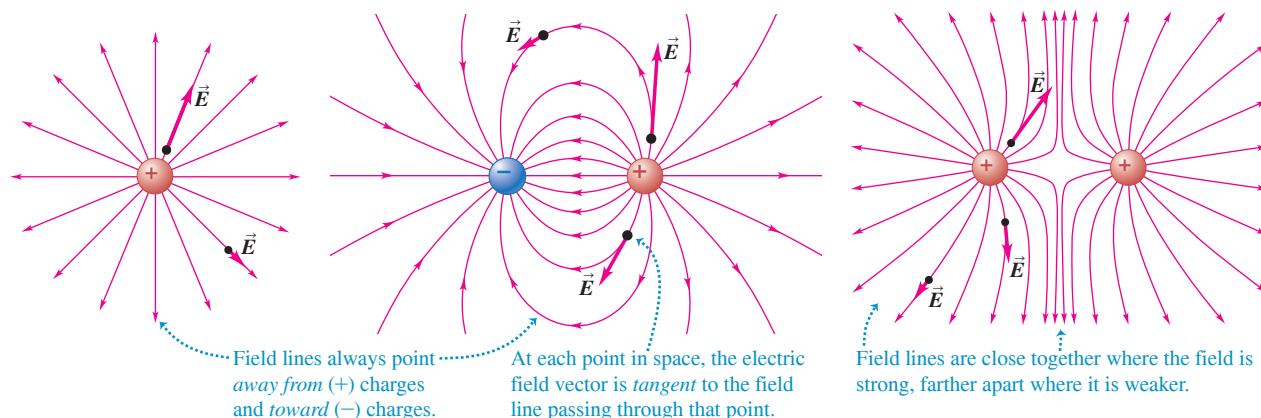


Figure 21.29a is a view from above of a demonstration setup for visualizing electric field lines. In the arrangement shown here, the tips of two positively charged wires are inserted in a container of insulating liquid, and some grass seeds are floated on the liquid. The grass seeds are electrically neutral insulators, but the electric field of the two charged wires causes *polarization* of the grass seeds; there is a slight shifting of the positive and negative charges within the molecules of each seed, like that shown in Fig. 21.8. The positively charged end of each grass seed is pulled in the direction of $\vec{E}$ and the negatively charged end is pulled opposite $\vec{E}$. Hence the long axis of each grass seed tends to orient parallel to the electric field, in the direction of the field line that passes through the position of the seed (Fig. 21.29b).

CAUTION Electric field lines are not trajectories It's a common misconception that if a particle of charge q is in motion where there is an electric field, the particle must move along an electric field line. Because $\vec{E}$ at any point is tangent to the field line that passes through that point, it is true that the force $\vec{F} = q\vec{E}$ on the particle, and hence the particle's acceleration, are tangent to the field line. But we learned in Chapter 3 that when a particle moves on a curved path, its acceleration *cannot* be tangent to the path. In general, the trajectory of a charged particle is *not* the same as a field line. ■

TEST YOUR UNDERSTANDING OF SECTION 21.6 Suppose the electric field lines in a region of space are straight lines. If a charged particle is released from rest in that region, will the trajectory of the particle be along a field line?

ANSWER

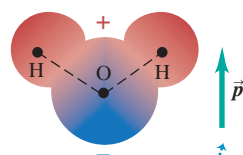
■ **yes** If the field lines are straight, $\vec{E}$ must point in the same direction throughout the region. Hence the force $\vec{F} = q\vec{E}$ on a particle of charge q is always in the same direction. A particle released from rest accelerates in a straight line in the direction of $\vec{F}$, and so its trajectory is a straight line along a field line.

21.7 ELECTRIC DIPOLES

An **electric dipole** is a pair of point charges with equal magnitude and opposite sign (a positive charge q and a negative charge $-q$) separated by a distance d . We introduced electric dipoles in Example 21.8 (Section 21.5); the concept is worth exploring further because many physical systems, from molecules to TV antennas, can be described as electric dipoles. We'll also use this concept extensively in our discussion of dielectrics in Chapter 24.

Figure 21.30a shows a molecule of water (H_2O), which in many ways behaves **?** like an electric dipole. The water molecule as a whole is electrically neutral, but the chemical bonds within the molecule cause a displacement of charge; the result is a net negative charge on the oxygen end of the molecule and a net positive charge on the hydrogen end, forming an electric dipole. The effect is equivalent to shifting one electron only about 4×10^{-11} m (about the radius of a hydrogen atom), but the consequences of this shift are profound. Water is an excellent solvent for ionic substances such as table salt (sodium chloride, NaCl) precisely because the water molecule is an electric dipole (Fig. 21.30b). When dissolved in water, salt dissociates into a positive sodium ion (Na^+)

(a) A water molecule, showing positive charge as red and negative charge as blue



The electric dipole moment $\vec{p}$ is directed from the negative end to the positive end of the molecule.

(b) Various substances dissolved in water



Figure 21.29 (a) Electric field lines produced by two opposite charges. The pattern is formed by grass seeds floating in mineral oil; the charges are on two wires whose tips are inserted into the oil. Compare this pattern with Fig. 21.28b. (b) The electric field causes polarization of the grass seeds, which in turn causes the seeds to align with the field.

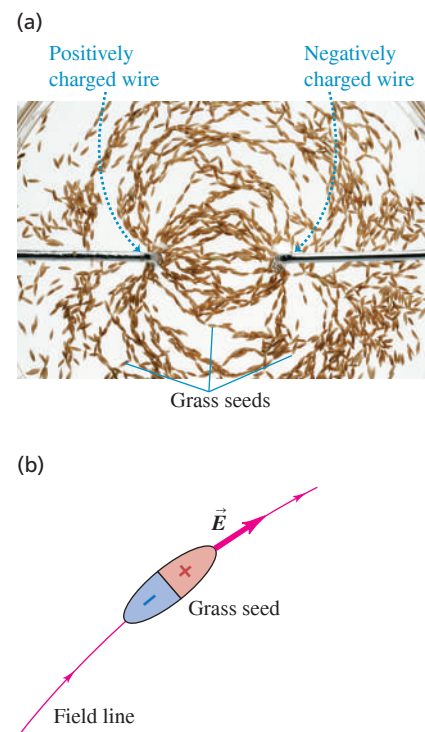
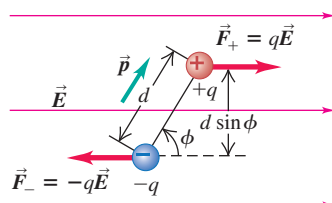


Figure 21.30 (a) A water molecule is an example of an electric dipole. (b) Each test tube contains a solution of a different substance in water. The large electric dipole moment of water makes it an excellent solvent.

and a negative chlorine ion (Cl^-), which tend to be attracted to the negative and positive ends, respectively, of water molecules; this holds the ions in solution. If water molecules were not electric dipoles, water would be a poor solvent, and almost all of the chemistry that occurs in aqueous solutions would be impossible. This includes all of the biochemical reactions that occur in all of the life on earth. In a very real sense, your existence as a living being depends on electric dipoles!

We examine two questions about electric dipoles. First, what forces and torques does an electric dipole experience when placed in an external electric field (that is, a field set up by charges outside the dipole)? Second, what electric field does an electric dipole itself produce?

Figure 21.31 The net force on this electric dipole is zero, but there is a torque directed into the page that tends to rotate the dipole clockwise.



Force and Torque on an Electric Dipole

To start with the first question, let's place an electric dipole in a *uniform* external electric field $\vec{E}$, as shown in Fig. 21.31. Both forces $\vec{F}_+$ and $\vec{F}_-$ on the two charges have magnitude qE , but their directions are opposite, and they add to zero. *The net force on an electric dipole in a uniform external electric field is zero.*

However, the two forces don't act along the same line, so their *torques* don't add to zero. We calculate torques with respect to the center of the dipole. Let the angle between the electric field $\vec{E}$ and the dipole axis be ϕ ; then the lever arm for both $\vec{F}_+$ and $\vec{F}_-$ is $(d/2) \sin \phi$. The torque of $\vec{F}_+$ and the torque of $\vec{F}_-$ both have the same magnitude of $(qE)(d/2) \sin \phi$, and both torques tend to rotate the dipole clockwise (that is, $\vec{\tau}$ is directed into the page in Fig. 21.31). Hence the magnitude of the net torque is twice the magnitude of either individual torque:

$$\tau = (qE)(d \sin \phi) \quad (21.13)$$

where $d \sin \phi$ is the perpendicular distance between the lines of action of the two forces.

The product of the charge q and the separation d is the magnitude of a quantity called the **electric dipole moment**, denoted by p :

$$p = qd \quad (\text{magnitude of electric dipole moment}) \quad (21.14)$$

The units of p are charge times distance ($\text{C} \cdot \text{m}$). For example, the magnitude of the electric dipole moment of a water molecule is $p = 6.13 \times 10^{-30} \text{ C} \cdot \text{m}$.

CAUTION The symbol p has multiple meanings Do not confuse dipole moment with momentum or pressure. There aren't as many letters in the alphabet as there are physical quantities, so some letters are used several times. The context usually makes it clear what we mean, but be careful. ■

We further define the electric dipole moment to be a *vector* quantity $\vec{p}$. The magnitude of $\vec{p}$ is given by Eq. (21.14), and its direction is along the dipole axis from the negative charge to the positive charge as shown in Fig. 21.31.

In terms of p , Eq. (21.13) for the magnitude τ of the torque exerted by the field becomes

$$\text{Magnitude of torque on an electric dipole} \quad \tau = pE \sin \phi \quad \begin{array}{l} \text{Magnitude of electric field } \vec{E} \\ \text{Angle between } \vec{p} \text{ and } \vec{E} \\ \text{Magnitude of electric dipole moment } \vec{p} \end{array} \quad (21.15)$$

Since the angle ϕ in Fig. 21.31 is the angle between the directions of the vectors $\vec{p}$ and $\vec{E}$, this is reminiscent of the expression for the magnitude of the *vector product* discussed in Section 1.10. (You may want to review that discussion.) Hence we can write the torque on the dipole in vector form as

$$\vec{\tau} = \vec{p} \times \vec{E} \quad (21.16)$$

Vector torque on an electric dipole Electric dipole moment Electric field

You can use the right-hand rule for the vector product to verify that in the situation shown in Fig. 21.31, $\vec{\tau}$ is directed into the page. The torque is greatest when $\vec{p}$ and $\vec{E}$ are perpendicular and is zero when they are parallel or antiparallel. The torque always tends to turn $\vec{p}$ to line it up with $\vec{E}$. The position $\phi = 0$, with $\vec{p}$ parallel to $\vec{E}$, is a position of stable equilibrium, and the position $\phi = \pi$, with $\vec{p}$ and $\vec{E}$ antiparallel, is a position of unstable equilibrium. The polarization of a grass seed in the apparatus of Fig. 21.29b gives it an electric dipole moment; the torque exerted by $\vec{E}$ then causes the seed to align with $\vec{E}$ and hence with the field lines.

Potential Energy of an Electric Dipole

When a dipole changes direction in an electric field, the electric-field torque does *work* on it, with a corresponding change in potential energy. The work dW done by a torque τ during an infinitesimal displacement $d\phi$ is given by Eq. (10.19): $dW = \tau d\phi$. Because the torque is in the direction of decreasing ϕ , we must write the torque as $\tau = -pE \sin \phi$, and

$$dW = \tau d\phi = -pE \sin \phi d\phi$$

In a finite displacement from ϕ_1 to ϕ_2 the total work done on the dipole is

$$\begin{aligned} W &= \int_{\phi_1}^{\phi_2} (-pE \sin \phi) d\phi \\ &= pE \cos \phi_2 - pE \cos \phi_1 \end{aligned}$$

The work is the negative of the change of potential energy, just as in Chapter 7: $W = U_1 - U_2$. So a suitable definition of potential energy U for this system is

$$U(\phi) = -pE \cos \phi \quad (21.17)$$

In this expression we recognize the *scalar product* $\vec{p} \cdot \vec{E} = pE \cos \phi$, so we can also write

$$U = -\vec{p} \cdot \vec{E} \quad (21.18)$$

Potential energy for an electric dipole in an electric field Electric field Electric dipole moment

The potential energy has its minimum (most negative) value $U = -pE$ at the stable equilibrium position, where $\phi = 0$ and $\vec{p}$ is parallel to $\vec{E}$. The potential energy is maximum when $\phi = \pi$ and $\vec{p}$ is antiparallel to $\vec{E}$; then $U = +pE$. At $\phi = \pi/2$, where $\vec{p}$ is perpendicular to $\vec{E}$, U is zero. We could define U differently so that it is zero at some other orientation of $\vec{p}$, but our definition is simplest.

Equation (21.18) gives us another way to look at the effect illustrated in Fig. 21.29. The electric field $\vec{E}$ gives each grass seed an electric dipole moment, and the grass seed then aligns itself with $\vec{E}$ to minimize the potential energy.

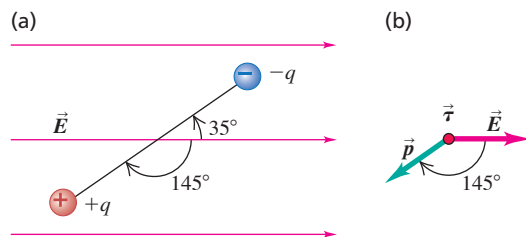
EXAMPLE 21.13 Force and torque on an electric dipole**WITH VARIATION PROBLEMS**

Figure 21.32a (next page) shows an electric dipole in a uniform electric field of magnitude $5.0 \times 10^5 \text{ N/C}$ that is directed parallel to the plane of the figure. The charges are $\pm 1.6 \times 10^{-19} \text{ C}$; both lie in the plane and are separated by $0.125 \text{ nm} = 0.125 \times 10^{-9} \text{ m}$. Find (a) the net force exerted by the field on the dipole; (b) the magnitude and direction of the electric dipole moment; (c) the magnitude and direction of the torque; (d) the potential energy of the system in the position shown.

IDENTIFY and SET UP This problem uses the ideas of this section about an electric dipole placed in an electric field. We use the relationship $\vec{F} = q\vec{E}$ for each point charge to find the force on the dipole as a whole. Equation (21.14) gives the dipole moment, Eq. (21.16) gives the torque on the dipole, and Eq. (21.18) gives the potential energy of the system.

EXECUTE (a) The field is uniform, so the forces on the two charges are equal and opposite. Hence the total force on the dipole is zero.

Figure 21.32 (a) An electric dipole. (b) Directions of the electric dipole moment, electric field, and torque ($\vec{\tau}$ points out of the page).



(b) The magnitude p of the electric dipole moment $\vec{p}$ is

$$p = qd = (1.6 \times 10^{-19} \text{ C})(0.125 \times 10^{-9} \text{ m}) \\ = 2.0 \times 10^{-29} \text{ C} \cdot \text{m}$$

The direction of $\vec{p}$ is from the negative to the positive charge, 145° clockwise from the electric-field direction (Fig. 21.32b).

(c) The magnitude of the torque is

$$\tau = pE \sin \phi = (2.0 \times 10^{-29} \text{ C} \cdot \text{m})(5.0 \times 10^5 \text{ N/C})(\sin 145^\circ) \\ = 5.7 \times 10^{-24} \text{ N} \cdot \text{m}$$

From the right-hand rule for vector products (see Section 1.10), the direction of the torque $\vec{\tau} = \vec{p} \times \vec{E}$ is out of the page. This corresponds to a counterclockwise torque that tends to align $\vec{p}$ with $\vec{E}$.

(d) The potential energy

$$U = -pE \cos \phi \\ = -(2.0 \times 10^{-29} \text{ C} \cdot \text{m})(5.0 \times 10^5 \text{ N/C})(\cos 145^\circ) \\ = 8.2 \times 10^{-24} \text{ J}$$

EVALUATE The charge magnitude, the distance between the charges, the dipole moment, and the potential energy are all very small, but are all typical of molecules.

KEYCONCEPT An electric dipole placed in an electric field tends to orient itself with its dipole moment in the same direction as the field.

BIO APPLICATION A Fish with an Electric Dipole Moment

Unlike the tiger shark (Section 21.4), which senses the electric fields produced by its prey, the African knifefish *Gymnarchus niloticus*—which is nocturnal and has poor vision—hunts other fish by generating its own electric field. It can make its tail negatively charged relative to its head, producing a dipole-like field similar to that in Fig. 21.28b. When a smaller fish ventures into the field, its body alters the field pattern and alerts *G. niloticus* that a meal is present.



In this discussion we have assumed that $\vec{E}$ is uniform, so there is no net force on the dipole. If $\vec{E}$ is not uniform, the forces at the ends may not cancel completely, and the net force may not be zero. Thus an object with zero net charge but an electric dipole moment can experience a net force in a nonuniform electric field. As we mentioned in Section 21.1, an uncharged object can be polarized by an electric field, giving rise to a separation of charge and an electric dipole moment. This is how uncharged objects can experience electrostatic forces (see Fig. 21.8).

Field of an Electric Dipole

Now let's think of an electric dipole as a *source* of electric field. Figure 21.28b shows the general shape of the field due to a dipole. At each point in the pattern the total $\vec{E}$ field is the vector sum of the fields from the two individual charges, as in Example 21.8 (Section 21.5). Try drawing diagrams showing this vector sum for several points.

To get quantitative information about the field of an electric dipole, we have to do some calculating, as illustrated in the next example. Notice the use of the principle of superposition of electric fields to add up the contributions to the field of the individual charges. Also notice that we need to use approximation techniques even for the relatively simple case of a field due to two charges. Field calculations often become very complicated, and computer analysis is typically used to determine the field due to an arbitrary charge distribution.

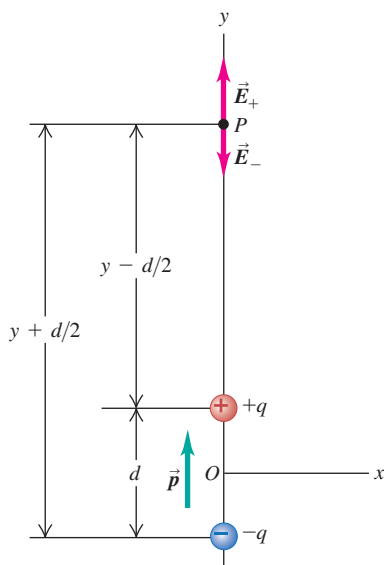
EXAMPLE 21.14 Field of an electric dipole, revisited

WITH VARIATION PROBLEMS

An electric dipole is centered at the origin, with $\vec{p}$ in the direction of the $+y$ -axis (**Fig. 21.33**). Derive an approximate expression for the electric field at a point P on the y -axis for which y is much larger than d . To do this, use the binomial expansion $(1 + x)^n \cong 1 + nx + n(n-1)x^2/2 + \cdots$ (valid for the case $|x| < 1$).

IDENTIFY and SET UP We use the principle of superposition: The total electric field is the vector sum of the field produced by the positive charge and the field produced by the negative charge. At the field point P shown in Fig. 21.33, the field $\vec{E}_+$ of the positive charge has a positive (upward) y -component and the field $\vec{E}_-$ of the negative charge has a negative (downward) y -component. We add these components to find the total field and then apply the approximation that y is much greater than d .

Figure 21.33 Finding the electric field of an electric dipole at a point on its axis.



EXECUTE The total y -component E_y of electric field from the two charges is

$$E_y = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(y - d/2)^2} - \frac{1}{(y + d/2)^2} \right] \\ = \frac{q}{4\pi\epsilon_0 y^2} \left[\left(1 - \frac{d}{2y}\right)^{-2} - \left(1 + \frac{d}{2y}\right)^{-2} \right]$$

We used this same approach in Example 21.8 (Section 21.5). Now the approximation: When we are far from the dipole compared to its size, so $y \gg d$, we have $d/2y \ll 1$. With $n = -2$ and with $d/2y$ replacing x in the binomial expansion, we keep only the first two terms (the terms we discard are much smaller). We then have

$$\left(1 - \frac{d}{2y}\right)^{-2} \cong 1 + \frac{d}{y} \quad \text{and} \quad \left(1 + \frac{d}{2y}\right)^{-2} \cong 1 - \frac{d}{y}$$

Hence E_y is given approximately by

$$E_y \cong \frac{q}{4\pi\epsilon_0 y^2} \left[1 + \frac{d}{y} - \left(1 - \frac{d}{y}\right) \right] = \frac{qd}{2\pi\epsilon_0 y^3} = \frac{p}{2\pi\epsilon_0 y^3}$$

EVALUATE An alternative route to this result is to put the fractions in the first expression for E_y over a common denominator, add, and then approximate the denominator $(y - d/2)^2(y + d/2)^2$ as y^4 . We leave the details to you.

For points P off the coordinate axes, the expressions are more complicated, but at *all* points far away from the dipole (in any direction) the field drops off as $1/r^3$. We can compare this with the $1/r^2$ behavior of a point charge, the $1/r$ behavior of a long line charge, and the independence of r for a large sheet of charge. There are charge distributions for which the field drops off even more quickly. At large distances, the field of an *electric quadrupole*, which consists of two equal dipoles with opposite orientation, separated by a small distance, drops off as $1/r^4$.

KEYCONCEPT The electric field produced by an electric dipole is the vector sum of the fields produced by the positive and negative charges that make up the dipole. At distances r from the dipole that are large compared to the dipole size, the field magnitude is inversely proportional to the cube of r .

TEST YOUR UNDERSTANDING OF SECTION 21.7 An electric dipole is placed in a region of uniform electric field $\vec{E}$, with the electric dipole moment $\vec{p}$ pointing in the direction opposite to $\vec{E}$. Is the dipole (i) in stable equilibrium, (ii) in unstable equilibrium, or (iii) neither? (*Hint:* You may want to review Section 7.5.)

ANSWER

that this is a situation of unstable equilibrium. maximum value that U can have. From our discussion of energy diagrams in Section 7.5, it follows is the point in opposite directions, so that $\phi = 180^\circ$, we have $\cos \phi = -1$ and $U = +pE$. This is the is $U = -p \cdot \vec{E} = -pE \cos \phi$, where ϕ is the angle between the directions of $\vec{p}$ and $\vec{E}$. If $\vec{p}$ and $\vec{E}$ are in the same direction, $\phi = 0^\circ$, we have $\cos \phi = 1$ and $U = -pE$. This is the minimum value that U can have. From our discussion of energy diagrams in Section 7.5, it follows that this is a situation of stable equilibrium.

CHAPTER 21 SUMMARY

Electric charge, conductors, and insulators: The fundamental quantity in electrostatics is electric charge. There are two kinds of charge, positive and negative. Charges of the same sign repel each other; charges of opposite sign attract. Charge is conserved; the total charge in an isolated system is constant.

All ordinary matter is made of protons, neutrons, and electrons. The positive protons and electrically neutral neutrons in the nucleus of an atom are bound together by the nuclear force; the negative electrons surround the nucleus at distances much greater than the nuclear size. Electric interactions are chiefly responsible for the structure of atoms, molecules, and solids.

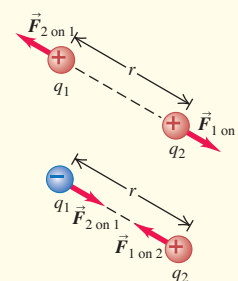
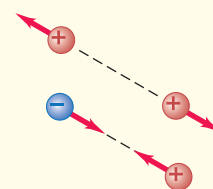
Conductors are materials in which charge moves easily; in insulators, charge does not move easily. Most metals are good conductors; most nonmetals are insulators.

Coulomb's law: For charges q_1 and q_2 separated by a distance r , the magnitude of the electric force on either charge is proportional to the product $q_1 q_2$ and inversely proportional to r^2 . The force on each charge is along the line joining the two charges—repulsive if q_1 and q_2 have the same sign, attractive if they have opposite signs. In SI units the unit of electric charge is the coulomb, abbreviated C. (See Examples 21.1 and 21.2.)

When two or more charges each exert a force on a charge, the total force on that charge is the vector sum of the forces exerted by the individual charges. (See Examples 21.3 and 21.4.)

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} \quad (21.2)$$

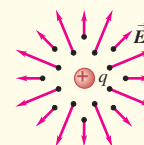
$$\frac{1}{4\pi\epsilon_0} = 8.988 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$



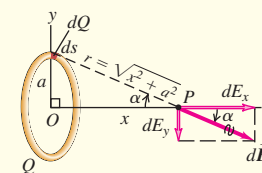
Electric field: Electric field $\vec{E}$, a vector quantity, is the force per unit charge exerted on a test charge at any point. The electric field produced by a point charge is directed radially away from a positive charge and radially toward a negative charge. (See Examples 21.5–21.7.)

$$\vec{E} = \frac{\vec{F}_0}{q_0} \quad (21.3)$$

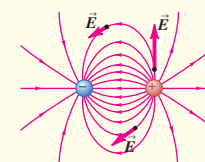
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (21.7)$$



Superposition of electric fields: The electric field $\vec{E}$ of any combination of charges is the vector sum of the fields caused by the individual charges. To calculate the electric field caused by a continuous distribution of charge, divide the distribution into small elements, calculate the field caused by each element, and then carry out the vector sum, usually by integrating. Charge distributions are described by linear charge density λ , surface charge density σ , and volume charge density ρ . (See Examples 21.8–21.12.)



Electric field lines: Field lines provide a graphical representation of electric fields. At any point on a field line, the tangent to the line is in the direction of $\vec{E}$ at that point. The number of lines per unit area (perpendicular to their direction) is proportional to the magnitude of $\vec{E}$ at the point.

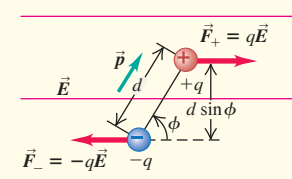


Electric dipoles: An electric dipole is a pair of electric charges of equal magnitude q but opposite sign, separated by a distance d . The electric dipole moment $\vec{p}$ has magnitude $p = qd$. The direction of $\vec{p}$ is from negative toward positive charge. An electric dipole in an electric field $\vec{E}$ experiences a torque $\vec{\tau}$ equal to the vector product of $\vec{p}$ and $\vec{E}$. The magnitude of the torque depends on the angle ϕ between $\vec{p}$ and $\vec{E}$. The potential energy U for an electric dipole in an electric field also depends on the relative orientation of $\vec{p}$ and $\vec{E}$. (See Examples 21.13 and 21.14.)

$$\tau = pE \sin \phi \quad (21.15)$$

$$\vec{\tau} = \vec{p} \times \vec{E} \quad (21.16)$$

$$U = -\vec{p} \cdot \vec{E} \quad (21.18)$$





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 21.3 and 21.4** (Section 21.3) before attempting these problems.

VP21.4.1 Charge $q_1 = +4.00$ nC is on the x -axis at $x = 0$ and charge $q_2 = -1.20$ nC is on the x -axis at $x = 6.00$ cm. Find the total electric force exerted by q_1 and q_2 on a third charge $q_3 = -2.00$ nC on the x -axis at $x = 4.00$ cm.

VP21.4.2 Charge $q_1 = +3.60$ nC is on the x -axis at $x = 0$ and an unknown charge q_2 is on the x -axis at $x = -4.00$ cm. The total electric force that q_1 and q_2 exert on a third charge $q_3 = +2.00$ nC on the x -axis at $x = 4.00$ cm is in the $+x$ -direction and has magnitude 54.0 μ N. Find (a) the magnitude and direction of the force that q_1 exerts on q_3 , (b) the magnitude and direction of the force that q_2 exerts on q_3 , and (c) the charge q_2 .

VP21.4.3 Charge $q_1 = +6.00$ nC is at $x = 0$, $y = 0.200$ m and charge $q_2 = -6.00$ nC is at $x = 0$, $y = -0.200$ m. Find the magnitude and direction of the total electric force exerted by q_1 and q_2 on a third charge $q_3 = -2.50$ nC at $x = 0.150$ m, $y = 0$.

VP21.4.4 A small piece of plastic with charge $q_1 = +4.00$ nC is at $x = 0$, $y = 0.200$ m and a second small piece of plastic with charge $q_2 = -4.00$ nC is at $x = 0$, $y = 0$. Find the x - and y -components of the total electric force exerted by q_1 and q_2 on a third small piece of plastic with charge $q_3 = -1.50$ nC at $x = 0.250$ m, $y = 0$.

Be sure to review **EXAMPLES 21.8, 21.9, and 21.10** (Section 21.5) before attempting these problems.

VP21.10.1 Charge $q_1 = +4.00$ nC is at $x = 0$, $y = 0.200$ m and charge $q_2 = -5.00$ nC is at $x = 0$, $y = 0$. Find the x - and y -components of the total electric field caused by q_1 and q_2 at (a) $x = 0$, $y = 0.100$ m; (b) $x = 0$, $y = 0.400$ m; (c) $x = 0.200$ m, $y = 0$.

VP21.10.2 Charge $q_1 = +1.80$ nC is on the x -axis at $x = 0$ and an unknown charge q_2 is on the x -axis at $x = +4.00$ cm. The total electric field caused by q_1 and q_2 at a point P on the x -axis at $x = +2.00$ cm is in the $+x$ -direction and has magnitude 6.75×10^4 N/C. Find (a) the magnitude and direction of the electric field that q_1 causes at P , (b) the magnitude and direction of the electric field that q_2 causes at P , and (c) the charge q_2 .

VP21.10.3 A simple model of a hydrogen atom is a positive point charge $+e$ (representing the proton) at the center of a ring of radius a

with negative charge $-e$ distributed uniformly around the ring (representing the electron in orbit around the proton). (a) Find the magnitude of the total electric field due to this charge distribution at a point a distance a from the proton and perpendicular to the plane of the ring. (b) Does the field point toward or away from the proton?

VP21.10.4 Positive charge Q is distributed uniformly along a rod of length L that lies along the x -axis from $x = L$ to $x = 2L$. Find (a) how much charge is contained within a segment of the rod of length dx , (b) the x - and y -components of the infinitesimal electric field caused at the origin ($x = 0$) by one such segment at coordinate x , and (c) the x - and y -components of the total electric field caused at the origin by the rod.

Be sure to review **EXAMPLES 21.13 and 21.14** (Section 21.7) before attempting these problems.

VP21.14.1 The electric dipole moment of a water molecule (H_2O) is 6.13×10^{-30} C $\cdot$ m. You place a water molecule in a uniform electric field of magnitude 3.00×10^5 N/C, with the dipole moment vector at an angle of 50.0° from the direction of the field. Calculate (a) the magnitude of the torque on the water molecule and (b) the potential energy for the water molecule.

VP21.14.2 An electric dipole is in a uniform electric field of magnitude 8.50×10^4 N/C. The charges in the dipole are separated by 1.10×10^{-10} m, and the torque on the dipole when its dipole moment is perpendicular to the electric field is 6.60×10^{-26} N $\cdot$ m. Calculate the magnitude of the charges that make up the dipole.

VP21.14.3 You place a molecule in a uniform electric field of magnitude 1.20×10^5 N/C. It requires 4.60×10^{-25} J of work to change the orientation of the molecule's dipole moment from parallel to the field to opposite to the field. Calculate the magnitude of the dipole moment.

VP21.14.4 The ionic molecule potassium bromide (KBr), made up of a positive potassium ion (K^+) of charge $+e = 1.60 \times 10^{-19}$ C and a negative bromine ion (Br^-) of charge $-e = -1.60 \times 10^{-19}$ C, has an electric dipole moment of 3.50×10^{-29} C $\cdot$ m. (a) Calculate the distance between the two ions. (b) At a certain point along the axis that connects the ions, the electric field due to the KBr molecule has magnitude 8.00×10^4 N/C. How far from the center of the molecule is this point?

BRIDGING PROBLEM Calculating Electric Field: Half a Ring of Charge

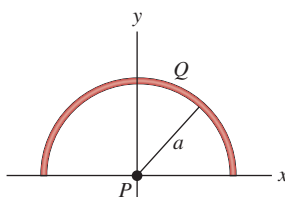
Positive charge Q is uniformly distributed around a semicircle of radius a as shown in **Fig. 21.34**. Find the magnitude and direction of the resulting electric field at point P , the center of curvature of the semicircle.

SOLUTION GUIDE

IDENTIFY and SET UP

1. The target variables are the components of the electric field at P .
2. Divide the semicircle into infinitesimal segments, each of which is a short circular arc of radius a and angle $d\theta$. What is the length of such a segment? How much charge is on a segment?
3. Consider an infinitesimal segment located at an angular position θ on the semicircle, measured from the lower right corner of the semicircle at $x = a$, $y = 0$. (Thus $\theta = \pi/2$ at

Figure 21.34 Charge uniformly distributed around a semicircle.



$x = 0$, $y = a$ and $\theta = \pi$ at $x = -a$, $y = 0$.) What are the x - and y -components of the electric field at P (dE_x and dE_y) produced by just this segment?

EXECUTE

4. Integrate your expressions for dE_x and dE_y from $\theta = 0$ to $\theta = \pi$. The results will be the x -component and y -component of the electric field at P .
5. Use your results from step 4 to find the magnitude and direction of the field at P .

EVALUATE

6. Does your result for the electric-field magnitude have the correct units?
7. Explain how you could have found the x -component of the electric field by using a symmetry argument.
8. What would be the electric field at P if the semicircle were extended to a full circle centered at P ?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q21.1 If you peel two strips of transparent tape off the same roll and immediately let them hang near each other, they will repel each other. If you then stick the sticky side of one to the shiny side of the other and rip them apart, they will attract each other. Give a plausible explanation, involving transfer of electrons between the strips of tape, for this sequence of events.

Q21.2 Two metal spheres are hanging from nylon threads. When you bring the spheres close to each other, they tend to attract. Based on this information alone, discuss all the possible ways that the spheres could be charged. Is it possible that after the spheres touch, they will cling together? Explain.

Q21.3 The electric force between two charged particles becomes weaker with increasing distance. Suppose instead that the electric force were *independent* of distance. In this case, would a charged comb still cause a neutral insulator to become polarized as in Fig. 21.8? Why or why not? Would the neutral insulator still be attracted to the comb? Again, why or why not?

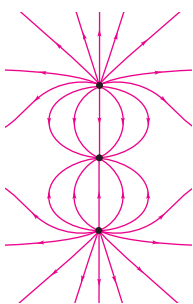
Q21.4 Your clothing tends to cling together after going through the dryer. Why? Would you expect more or less clinging if all your clothing were made of the same material (say, cotton) than if you dried different kinds of clothing together? Again, why? (You may want to experiment with your next load of laundry.)

Q21.5 An uncharged metal sphere hangs from a nylon thread. When a positively charged glass rod is brought close to the metal sphere, the sphere is drawn toward the rod. But if the sphere touches the rod, it suddenly flies away from the rod. Explain why the sphere is first attracted and then repelled.

Q21.6 **BIO** Estimate how many electrons there are in your body. Make any assumptions you feel are necessary, but clearly state what they are. (*Hint*: Most of the atoms in your body have equal numbers of electrons, protons, and neutrons.) What is the combined charge of all these electrons?

Q21.7 **Figure Q21.7** shows some of the electric field lines due to three point charges arranged along the vertical axis. All three charges have the same magnitude. (a) What are the signs of the three charges? Explain your reasoning. (b) At what point(s) is the magnitude of the electric field the smallest? Explain your reasoning. Explain how the fields produced by each individual point charge combine to give a small net field at this point or points.

Figure Q21.7



Q21.8 Good conductors of electricity, such as metals, are typically good conductors of heat; insulators, such as wood, are typically poor conductors of heat. Explain why there is a relationship between conduction of electricity and conduction of heat in these materials.

Q21.9 Suppose that the charge shown in Fig. 21.28a is fixed in position. A small, positively charged particle is then placed at some location and released from rest. Will the trajectory of the particle follow an electric field line? Why or why not? Suppose instead that the particle is placed at some point in Fig. 21.28b and released from rest (the positive and negative charges shown are fixed in position). Will its trajectory follow an electric field line? Again, why or why not? Explain any differences between your answers for the two situations.

Q21.10 Two identical metal objects are mounted on insulating stands. Describe how you could place charges of opposite sign but exactly equal magnitude on the two objects.

Q21.11 Because the charges on the electron and proton have the same absolute value, atoms are electrically neutral. Suppose that this is not precisely true, and the absolute value of the charge of the electron is less than the charge of the proton by 0.00100%. Estimate what the net charge of this textbook would be under these circumstances. Make any assumptions you feel are justified, but state clearly what they are. (*Hint*: Most of the atoms in this textbook have equal numbers of electrons, protons, and neutrons.) What would be the magnitude of the electric force between two textbooks placed 5.0 m apart? Would this force be attractive or repulsive? Discuss how the fact that ordinary matter is stable shows that the absolute values of the charges on the electron and proton must be identical to a *very* high level of accuracy.

Q21.12 If you walk across a nylon rug and then touch a large metal object such as a doorknob, you may get a spark and a shock. Why does this tend to happen more on dry days than on humid days? (*Hint*: See Fig. 21.30.) Why are you less likely to get a shock if you touch a *small* metal object, such as a paper clip?

Q21.13 You have a negatively charged object. How can you use it to place a net negative charge on an insulated metal sphere? To place a net positive charge on the sphere?

Q21.14 When two point charges of equal mass and charge are released on a frictionless table, each has an initial acceleration (magnitude) a_0 . If instead you keep one fixed and release the other one, what will be its initial acceleration: a_0 , $2a_0$, or $a_0/2$? Explain.

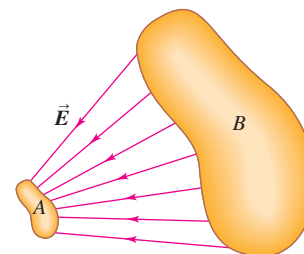
Q21.15 A point charge of mass m and charge Q and another point charge of mass m but charge $2Q$ are released on a frictionless table. If the charge Q has an initial acceleration a_0 , what will be the acceleration of $2Q$: a_0 , $2a_0$, $4a_0$, $a_0/2$, or $a_0/4$? Explain.

Q21.16 A proton is placed in a uniform electric field and then released. Then an electron is placed at this same point and released. Do these two particles experience the same force? The same acceleration? Do they move in the same direction when released?

Q21.17 In Example 21.1 (Section 21.3) we saw that the electric force between two α particles is of the order of 10^{35} times as strong as the gravitational force. So why do we readily feel the gravity of the earth but no electric force from it?

Q21.18 What similarities do electric forces have with gravitational forces? What are the most significant differences?

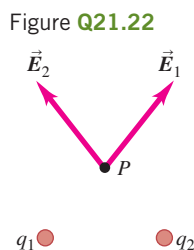
Q21.19 Two irregular objects A and B carry charges of opposite sign. **Figure Q21.19** shows the electric field lines near each of these objects. (a) Which object is positive, A or B ? How do you know? (b) Where is the electric field stronger, close to A or close to B ? How do you know?



Q21.20 Atomic nuclei are made of protons and neutrons. This shows that there must be another kind of interaction in addition to gravitational and electric forces. Explain.

Q21.21 Sufficiently strong electric fields can cause atoms to become positively ionized—that is, to lose one or more electrons. Explain how this can happen. What determines how strong the field must be to make this happen?

Q21.22 The electric fields at point P due to the positive charges q_1 and q_2 are shown in **Fig. Q21.22**. Does the fact that they cross each other violate the statement in Section 21.6 that electric field lines never cross? Explain.



Q21.23 The air temperature and the velocity of the air have different values at different places in the earth's atmosphere. Is the air velocity a vector field? Why or why not? Is the air temperature a vector field? Again, why or why not?

EXERCISES

Section 21.3 Coulomb's Law

21.1 • Excess electrons are placed on a small lead sphere with mass 8.00 g so that its net charge is $-3.20 \times 10^{-9}\text{ C}$. (a) Find the number of excess electrons on the sphere. (b) How many excess electrons are there per lead atom? The atomic number of lead is 82, and its atomic mass is 207 g/mol .

21.2 • Lightning occurs when there is a flow of electric charge (principally electrons) between the ground and a thundercloud. The maximum rate of charge flow in a lightning bolt is about $20,000\text{ C/s}$; this lasts for $100\text{ }\mu\text{s}$ or less. How much charge flows between the ground and the cloud in this time? How many electrons flow during this time?

21.3 • If a proton and an electron are released when they are $2.0 \times 10^{-10}\text{ m}$ apart (a typical atomic distance), find the initial acceleration of each particle.

21.4 • In Example 21.4, what is the net force (magnitude and direction) on charge q_1 exerted by the other two charges?

21.5 • **BIO Signal Propagation in Neurons.** Neurons are components of the nervous system of the body that transmit signals as electric impulses travel along their length. These impulses propagate when charge suddenly rushes into and then out of a part of the neuron called an *axon*. Measurements have shown that, during the inflow part of this cycle, approximately $5.6 \times 10^{11}\text{ Na}^+$ (sodium ions) per meter, each with charge $+e$, enter the axon. How many coulombs of charge enter a 1.5 cm length of the axon during this process?

21.6 • Two small spheres spaced 20.0 cm apart have equal charge. How many excess electrons must be present on each sphere if the magnitude of the force of repulsion between them is $3.33 \times 10^{-21}\text{ N}$?

21.7 • An average human weighs about 650 N . If each of two average humans could carry 1.0 C of excess charge, one positive and one negative, how far apart would they have to be for the electric attraction between them to equal their 650 N weight?

21.8 • Two small aluminum spheres, each having mass 0.0250 kg , are separated by 80.0 cm . (a) How many electrons does each sphere contain? (The atomic mass of aluminum is 26.982 g/mol , and its atomic number is 13.) (b) How many electrons would have to be removed from one sphere and added to the other to cause an attractive force between the spheres of magnitude $1.00 \times 10^4\text{ N}$ (roughly 1 ton)? Assume that the spheres may be treated as point charges. (c) What fraction of all the electrons in each sphere does this represent?

21.9 • Two small plastic spheres are given positive electric charges. When they are 15.0 cm apart, the repulsive force between them has magnitude 0.220 N . What is the charge on each sphere (a) if the two charges are equal and (b) if one sphere has four times the charge of the other?

21.10 • Two point charges are placed on the x -axis as follows: Charge $q_1 = +4.00\text{ nC}$ is located at $x = 0.200\text{ m}$, and charge $q_2 = +5.00\text{ nC}$ is at $x = -0.300\text{ m}$. What are the magnitude and direction of the total force exerted by these two charges on a negative point charge $q_3 = -6.00\text{ nC}$ that is placed at the origin?

21.11 • In an experiment in space, one proton is held fixed and another proton is released from rest a distance of 2.50 mm away. (a) What is the initial acceleration of the proton after it is released? (b) Sketch qualitative (no numbers!) acceleration–time and velocity–time graphs of the released proton's motion.

21.12 • A negative charge of $-0.550\text{ }\mu\text{C}$ exerts an upward 0.600 N force on an unknown charge that is located 0.300 m directly below the first charge. What are (a) the value of the unknown charge (magnitude and sign); (b) the magnitude and direction of the force that the unknown charge exerts on the $-0.550\text{ }\mu\text{C}$ charge?

21.13 • Three point charges are arranged on a line. Charge $q_3 = +5.00\text{ nC}$ and is at the origin. Charge $q_2 = -3.00\text{ nC}$ and is at $x = +4.00\text{ cm}$. Charge q_1 is at $x = +2.00\text{ cm}$. What is q_1 (magnitude and sign) if the net force on q_3 is zero?

21.14 • In Example 21.4, suppose the point charge on the y -axis at $y = -0.30\text{ m}$ has negative charge $-2.0\text{ }\mu\text{C}$, and the other charges remain the same. Find the magnitude and direction of the net force on Q . How does your answer differ from that in Example 21.4? Explain the differences.

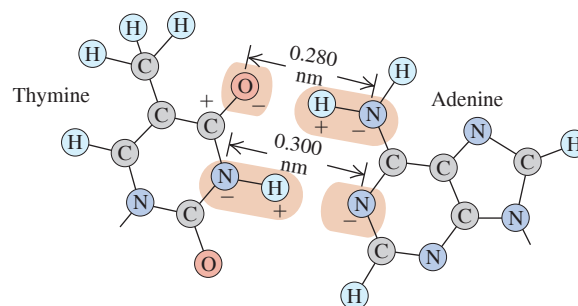
21.15 • In Example 21.3, calculate the net force on charge q_1 .

21.16 • Three point charges are arranged along the x -axis. Charge $q_1 = +3.00\text{ }\mu\text{C}$ is at the origin, and charge $q_2 = -5.00\text{ }\mu\text{C}$ is at $x = 0.200\text{ m}$. Charge $q_3 = -8.00\text{ }\mu\text{C}$. Where is q_3 located if the net force on q_1 is 7.00 N in the $-x$ -direction?

21.17 • Two point charges are located on the y -axis as follows: charge $q_1 = -1.50\text{ nC}$ at $y = -0.600\text{ m}$, and charge $q_2 = +3.20\text{ nC}$ at the origin ($y = 0$). What is the total force (magnitude and direction) exerted by these two charges on a third charge $q_3 = +5.00\text{ nC}$ located at $y = -0.400\text{ m}$?

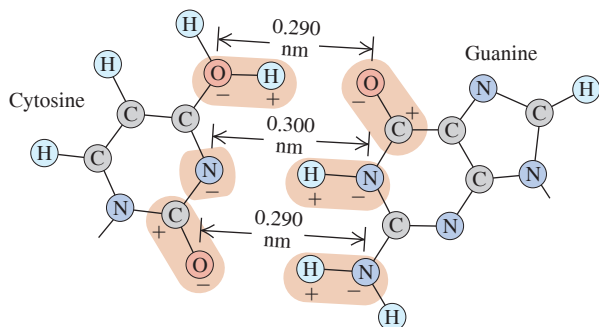
21.18 • **BIO Base Pairing in DNA, I.** The two sides of the DNA double helix are connected by pairs of bases (adenine, thymine, cytosine, and guanine). Because of the geometric shape of these molecules, adenine bonds with thymine and cytosine bonds with guanine. **Figure E21.18** shows the bonding of thymine and adenine. Each charge shown is $\pm e$, and the H–N distance is 0.110 nm . (a) Calculate the *net* force that thymine exerts on adenine. Is it attractive or repulsive? To keep the calculations fairly simple, yet reasonable, consider only the forces due to the O–H–N and the N–H–N combinations, assuming that these two combinations are parallel to each other. Remember, however, that in the O–H–N set, the O^- exerts a force on both the H^+ and the N^+ , and likewise along the N–H–N set. (b) Calculate the force on the electron in the hydrogen atom, which is 0.0529 nm from the proton. Then compare the strength of the bonding force of the electron in hydrogen with the bonding force of the adenine–thymine molecules.

Figure E21.18



21.19 •• BIO Base Pairing in DNA, II. Refer to Exercise 21.18. **Figure E21.19** shows the bonding of cytosine and guanine. The O—H and H—N distances are each 0.110 nm. In this case, assume that the bonding is due only to the forces along the O—H—O, N—H—N, and O—H—N combinations, and assume also that these three combinations are parallel to each other. Calculate the *net* force that cytosine exerts on guanine due to the preceding three combinations. Is this force attractive or repulsive?

Figure E21.19



Section 21.4 Electric Field and Electric Forces

21.20 •• CP A uniform electric field exists in the region between two oppositely charged plane parallel plates. A proton is released from rest at the surface of the positively charged plate and strikes the surface of the opposite plate, 1.60 cm distant from the first, in a time interval of 3.20×10^{-6} s. (a) Find the magnitude of the electric field. (b) Find the speed of the proton when it strikes the negatively charged plate.

21.21 • CP A proton is placed in a uniform electric field of 2.75×10^3 N/C. Calculate (a) the magnitude of the electric force felt by the proton; (b) the proton's acceleration; (c) the proton's speed after $1.00 \mu\text{s}$ in the field, assuming it starts from rest.

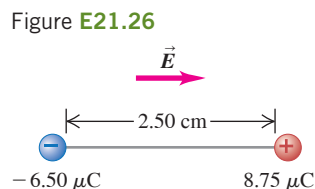
21.22 • A particle has charge -5.00 nC. (a) Find the magnitude and direction of the electric field due to this particle at a point 0.250 m directly above it. (b) At what distance from this particle does its electric field have a magnitude of 12.0 N/C?

21.23 • CP A proton is traveling horizontally to the right at 4.50×10^6 m/s. (a) Find the magnitude and direction of the weakest electric field that can bring the proton uniformly to rest over a distance of 3.20 cm. (b) How much time does it take the proton to stop after entering the field? (c) What minimum field (magnitude and direction) would be needed to stop an electron under the conditions of part (a)?

21.24 • CP An electron is released from rest in a uniform electric field. The electron accelerates vertically upward, traveling 4.50 m in the first $3.00 \mu\text{s}$ after it is released. (a) What are the magnitude and direction of the electric field? (b) Are we justified in ignoring the effects of gravity? Justify your answer quantitatively.

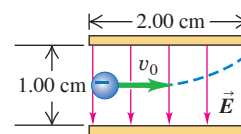
21.25 •• (a) What must the charge (sign and magnitude) of a 1.45 g particle be for it to remain stationary when placed in a downward-directed electric field of magnitude 650 N/C? (b) What is the magnitude of an electric field in which the electric force on a proton is equal in magnitude to its weight?

21.26 •• A $+8.75 \mu\text{C}$ point charge is glued down on a horizontal frictionless table. It is tied to a $-6.50 \mu\text{C}$ point charge by a light, nonconducting 2.50 cm wire. A uniform electric field of magnitude 1.85×10^8 N/C is directed parallel to the wire, as shown in **Fig. E21.26**. (a) Find the tension in the wire. (b) What would the tension be if both charges were negative?



21.27 •• CP An electron is projected with an initial speed $v_0 = 1.60 \times 10^6$ m/s into the uniform field between two parallel plates (**Fig. E21.27**). Assume that the field between the plates is uniform and directed vertically downward and that the field outside the plates is zero. The electron enters the field at a point midway between the plates. (a) If the electron just misses the upper plate as it emerges from the field, find the magnitude of the electric field. (b) Suppose that the electron in Fig. E21.27 is replaced by a proton with the same initial speed v_0 . Would the proton hit one of the plates? If not, what would be the magnitude and direction of its vertical displacement as it exits the region between the plates? (c) Compare the paths traveled by the electron and the proton, and explain the differences. (d) Discuss whether it is reasonable to ignore the effects of gravity for each particle.

Figure E21.27



21.28 •• CP In Exercise 21.27, what is the speed of the electron as it emerges from the field?

21.29 • A point charge is at the origin. With this point charge as the source point, what is the unit vector $\hat{r}$ in the direction of the field point (a) at $x = 0$, $y = -1.35$ m; (b) at $x = 12.0$ cm, $y = 12.0$ cm; (c) at $x = -1.10$ m, $y = 2.60$ m? Express your results in terms of the unit vectors $\hat{i}$ and $\hat{j}$.

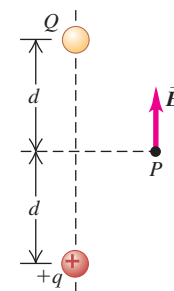
21.30 •• (a) An electron is moving east in a uniform electric field of 1.50 N/C directed to the west. At point A, the velocity of the electron is 4.50×10^5 m/s toward the east. What is the speed of the electron when it reaches point B, 0.375 m east of point A? (b) A proton is moving in the uniform electric field of part (a). At point A, the velocity of the proton is 1.90×10^4 m/s, east. What is the speed of the proton at point B?

Section 21.5 Electric-Field Calculations

21.31 •• A uniform line of charge with length 20.0 cm is along the x -axis, with its midpoint at $x = 0$. Its charge per length is $+4.80$ nC/m. A small sphere with charge $-2.00 \mu\text{C}$ is located at $x = 0$, $y = 5.00$ cm. What are the magnitude and direction of the force that the charged sphere exerts on the line of charge?

21.32 • Two point charges Q and $+q$ (where q is positive) produce the net electric field shown at point P in **Fig. E21.32**. The field points parallel to the line connecting the two charges. (a) What can you conclude about the sign and magnitude of Q ? Explain your reasoning. (b) If the lower charge were negative instead, would it be possible for the field to have the direction shown in the figure? Explain your reasoning.

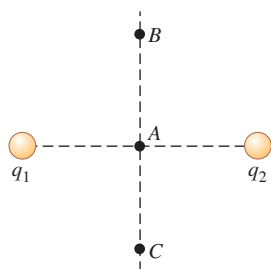
Figure E21.32



21.33 •• A very long line of charge with charge per unit length $+8.00 \mu\text{C}/\text{m}$ is on the x -axis and its midpoint is at $x = 0$. A second very long line of charge with charge per unit length $-4.00 \mu\text{C}/\text{m}$ is parallel to the x -axis at $y = 10.0$ cm and its midpoint is also at $x = 0$. At what point on the y -axis is the resultant electric field of the two lines of charge equal to zero?

21.34 • The two charges q_1 and q_2 shown in **Fig. E21.34** have equal magnitudes. What is the direction of the net electric field due to these two charges at points A (midway between the charges), B, and C if (a) both charges are negative, (b) both charges are positive, (c) q_1 is positive and q_2 is negative.

Figure E21.34

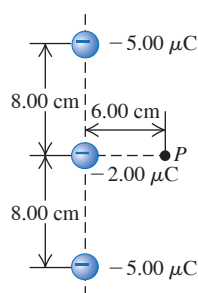


21.35 • A $+2.00$ nC point charge is at the origin, and a second -5.00 nC point charge is on the x -axis at $x = 0.800$ m. (a) Find the electric field (magnitude and direction) at each of the following points on the x -axis: (i) $x = 0.200$ m; (ii) $x = 1.20$ m; (iii) $x = -0.200$ m. (b) Find the net electric force that the two charges would exert on an electron placed at each point in part (a).

21.36 • Repeat Exercise 21.35, but now let the charge at the origin be -4.00 nC.

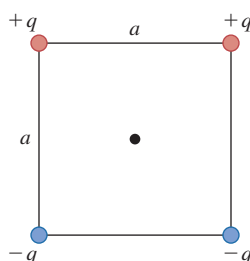
21.37 • Three negative point charges lie along a line as shown in Fig. E21.37. Find the magnitude and direction of the electric field this combination of charges produces at point P , which lies 6.00 cm from the -2.00 μC charge measured perpendicular to the line connecting the three charges.

Figure E21.37



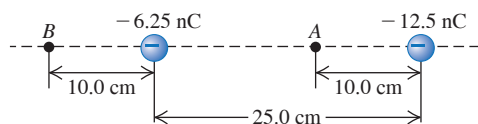
21.38 • A point charge is placed at each corner of a square with side length a . All charges have magnitude q . Two of the charges are positive and two are negative (Fig. E21.38). What is the direction of the net electric field at the center of the square due to the four charges, and what is its magnitude in terms of q and a ?

Figure E21.38



21.39 • Two point charges are separated by 25.0 cm (Fig. E21.39). Find the net electric field these charges produce at (a) point A and (b) point B . (c) What would be the magnitude and direction of the electric force this combination of charges would produce on a proton at A ?

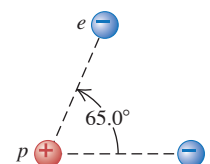
Figure E21.39



21.40 • Point charge $q_1 = -5.00$ nC is at the origin and point charge $q_2 = +3.00$ nC is on the x -axis at $x = 3.00$ cm. Point P is on the y -axis at $y = 4.00$ cm. (a) Calculate the electric fields $\vec{E}_1$ and $\vec{E}_2$ at point P due to the charges q_1 and q_2 . Express your results in terms of unit vectors (see Example 21.6). (b) Use the results of part (a) to obtain the resultant field at P , expressed in unit vector form.

21.41 • If two electrons are each 1.50×10^{-10} m from a proton (Fig. E21.41), find the magnitude and direction of the net electric force they will exert on the proton.

Figure E21.41



21.42 • **BIO Electric Field of Axons.** A nerve signal is transmitted through a neuron when an excess of Na^+ ions suddenly enters the axon, a long cylindrical part of the neuron. Axons are approximately 10.0 μm in diameter, and measurements show that about 5.6×10^{11} Na^+ ions per meter (each of charge $+e$) enter during this process. Although the axon is a long cylinder, the charge does not all enter everywhere at the same time. A plausible model would be a series of point charges moving along the axon. Consider a 0.10 mm length of the axon and model it as a point charge. (a) If the charge that enters each meter of the axon gets distributed uniformly along it, how many coulombs of charge enter a 0.10 mm length of the axon? (b) What electric field (magnitude and direction) does the sudden influx of charge produce at the surface of the body if the axon is 5.00 cm below the skin? (c) Certain sharks can respond to electric fields as weak as 1.0 $\mu\text{N/C}$. How far from this segment of axon could a shark be and still detect its electric field?

21.43 • In a rectangular coordinate system a positive point charge $q = 6.00 \times 10^{-9}$ C is placed at the point $x = +0.150$ m, $y = 0$, and an identical point charge is placed at $x = -0.150$ m, $y = 0$. Find the x - and y -components, the magnitude, and the direction of the electric field at the following points: (a) the origin; (b) $x = 0.300$ m, $y = 0$; (c) $x = 0.150$ m, $y = -0.400$ m; (d) $x = 0$, $y = 0.200$ m.

21.44 • A point charge $q_1 = -4.00$ nC is at the point $x = 0.600$ m, $y = 0.800$ m, and a second point charge $q_2 = +6.00$ nC is at the point $x = 0.600$ m, $y = 0$. Calculate the magnitude and direction of the net electric field at the origin due to these two point charges.

21.45 • Three parallel sheets of charge, large enough to be treated as infinite sheets, are perpendicular to the x -axis. Sheet A has surface charge density $\sigma_A = +8.00$ nC/m². Sheet B is 4.00 cm to the right of sheet A and has surface charge density $\sigma_B = -4.00$ nC/m². Sheet C is 4.00 cm to the right of sheet B , so is 8.00 cm to the right of sheet A , and has surface charge density $\sigma_C = +6.00$ nC/m². What are the magnitude and direction of the resultant electric field at a point that is midway between sheets B and C , or 2.00 cm from each of these two sheets?

21.46 • Point charge $q_1 = -5.00$ nC is on the x -axis at $x = -0.400$ m. Point P is on the x -axis at $x = +0.200$ m. Point charge q_2 is at the origin. What are the sign and magnitude of q_2 if the resultant electric field at point P is zero?

21.47 • Point charge A is on the x -axis at $x = -3.00$ cm. At $x = 1.00$ cm on the x -axis its electric field is 2700 N/C. Point charge B is also on the x -axis, at $x = 5.00$ cm. The absolute magnitude of charge B is twice that of A . Find the magnitude and direction of the total electric field at the origin if (a) both A and B are positive; (b) both are negative; (c) A is positive and B is negative; (d) A is negative and B is positive.

21.48 • A very long, straight wire has charge per unit length 3.20×10^{-10} C/m. At what distance from the wire is the electric-field magnitude equal to 2.50 N/C?

21.49 • A ring-shaped conductor with radius $a = 2.50$ cm has a total positive charge $Q = +0.125$ nC uniformly distributed around it (see Fig. 21.23). The center of the ring is at the origin of coordinates O . (a) What is the electric field (magnitude and direction) at point P , which is on the x -axis at $x = 40.0$ cm? (b) A point charge $q = -2.50$ μC is placed at P . What are the magnitude and direction of the force exerted by the charge q on the ring?

21.50 •• A straight, nonconducting plastic wire 8.50 cm long carries a charge density of $+175 \text{ nC/m}$ distributed uniformly along its length. It is lying on a horizontal tabletop. (a) Find the magnitude and direction of the electric field this wire produces at a point 6.00 cm directly above its midpoint. (b) If the wire is now bent into a circle lying flat on the table, find the magnitude and direction of the electric field it produces at a point 6.00 cm directly above its center.

Section 21.7 Electric Dipoles

21.51 • Point charges $q_1 = -4.5 \text{ nC}$ and $q_2 = +4.5 \text{ nC}$ are separated by 3.1 mm, forming an electric dipole. (a) Find the electric dipole moment (magnitude and direction). (b) The charges are in a uniform electric field whose direction makes an angle of 36.9° with the line connecting the charges. What is the magnitude of this field if the torque exerted on the dipole has magnitude $7.2 \times 10^{-9} \text{ N} \cdot \text{m}$?

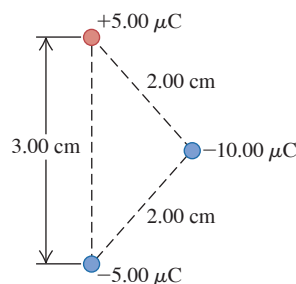
21.52 • The ammonia molecule (NH_3) has a dipole moment of $5.0 \times 10^{-30} \text{ C} \cdot \text{m}$. Ammonia molecules in the gas phase are placed in a uniform electric field $\vec{E}$ with magnitude $1.6 \times 10^6 \text{ N/C}$. (a) What is the change in electric potential energy when the dipole moment of a molecule changes its orientation with respect to $\vec{E}$ from parallel to perpendicular? (b) At what absolute temperature T is the average translational kinetic energy $\frac{3}{2}kT$ of a molecule equal to the change in potential energy calculated in part (a)? (Note: Above this temperature, thermal agitation prevents the dipoles from aligning with the electric field.)

21.53 • Torque on a Dipole. An electric dipole with dipole moment $\vec{p}$ is in a uniform external electric field $\vec{E}$. (a) Find the orientations of the dipole for which the torque on the dipole is zero. (b) Which of the orientations in part (a) is stable, and which is unstable? (Hint: Consider a small rotation away from the equilibrium position and see what happens.) (c) Show that for the stable orientation in part (b), the dipole's own electric field tends to oppose the external field.

21.54 • The dipole moment of the water molecule (H_2O) is $6.17 \times 10^{-30} \text{ C} \cdot \text{m}$. Consider a water molecule located at the origin whose dipole moment $\vec{p}$ points in the $+x$ -direction. A chlorine ion (Cl^-), of charge $-1.60 \times 10^{-19} \text{ C}$, is located at $x = 3.00 \times 10^{-9} \text{ m}$. Find the magnitude and direction of the electric force that the water molecule exerts on the chlorine ion. Is this force attractive or repulsive? Assume that x is much larger than the separation d between the charges in the dipole, so that the approximate expression for the electric field along the dipole axis derived in Example 21.14 can be used.

21.55 • Three charges are at the corners of an isosceles triangle as shown in Fig. E21.55. The $\pm 5.00 \mu\text{C}$ charges form a dipole. (a) Find the force (magnitude and direction) the $-10.00 \mu\text{C}$ charge exerts on the dipole. (b) For an axis perpendicular to the line connecting the $\pm 5.00 \mu\text{C}$ charges at the midpoint of this line, find the torque (magnitude and direction) exerted on the dipole by the $-10.00 \mu\text{C}$ charge.

Figure E21.55



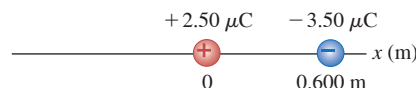
21.56 • An electric dipole with a dipole moment of magnitude p is placed at various orientations in an electric field $\vec{E}$ that is directed to the left. (a) What orientation of the dipole will result in maximum torque directed into the page? What then is the electric potential energy? (b) What orientation of the dipole will give zero torque and maximum electric potential energy? What type of equilibrium is this: stable, unstable, or neutral?

PROBLEMS

21.57 ••• Four identical charges Q are placed at the corners of a square of side L . (a) In a free-body diagram, show all of the forces that act on one of the charges. (b) Find the magnitude and direction of the total force exerted on one charge by the other three charges.

21.58 ••• Two charges are placed on the x -axis: one, of $2.50 \mu\text{C}$, at the origin and the other, of $-3.50 \mu\text{C}$, at $x = 0.600 \text{ m}$ (Fig. P21.58). Find the position on the x -axis where the net force on a small charge $+q$ would be zero.

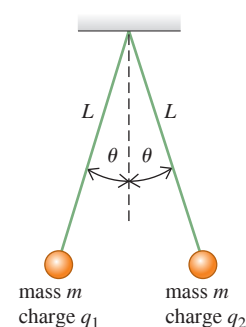
Figure P21.58



21.59 •• A charge $q_1 = +5.00 \text{ nC}$ is placed at the origin of an xy -coordinate system, and a charge $q_2 = -2.00 \text{ nC}$ is placed on the positive x -axis at $x = 4.00 \text{ cm}$. (a) If a third charge $q_3 = +6.00 \text{ nC}$ is now placed at the point $x = 4.00 \text{ cm}$, $y = 3.00 \text{ cm}$, find the x - and y -components of the total force exerted on this charge by the other two. (b) Find the magnitude and direction of this force.

21.60 •• CP Two identical spheres with mass m are hung from silk threads of length L (Fig. P21.60). The spheres have the same charge, so $q_1 = q_2 = q$. The radius of each sphere is very small compared to the distance between the spheres, so they may be treated as point charges. Show that if the angle θ is small, the equilibrium separation d between the spheres is $d = (q^2 L / 2\pi\epsilon_0 mg)^{1/3}$. (Hint: If θ is small, then $\tan \theta \approx \sin \theta$.)

Figure P21.60

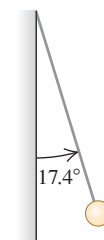


21.61 ••• CP Two small spheres with mass $m = 15.0 \text{ g}$ are hung by silk threads of length $L = 1.20 \text{ m}$ from a common point (Fig. P21.60). When the spheres are given equal quantities of negative charge, so that $q_1 = q_2 = q$, each thread hangs at $\theta = 25.0^\circ$ from the vertical. (a) Draw a diagram showing the forces on each sphere. Treat the spheres as point charges. (b) Find the magnitude of q . (c) Both threads are now shortened to length $L = 0.600 \text{ m}$, while the charges q_1 and q_2 remain unchanged. What new angle will each thread make with the vertical? (Hint: This part of the problem can be solved numerically by using trial values for θ and adjusting the values of θ until a self-consistent answer is obtained.)

21.62 •• CP A small sphere with charge $q = 5.00 \mu\text{C}$ and mass 0.500 g is traveling horizontally toward the east at a height of 60.0 cm above the ground. The sphere has a speed of 2.00 m/s as it enters a region of uniform electric field with magnitude E . What is E if the sphere has a speed of 5.00 m/s just before it strikes the ground?

21.63 •• CP A small 12.3 g plastic ball is tied to a very light 28.6 cm string that is attached to the vertical wall of a room (Fig. P21.63). A uniform horizontal electric field exists in this room. When the ball has been given an excess charge of $-1.11 \mu\text{C}$, you observe that it remains suspended, with the string making an angle of 17.4° with the wall. Find the magnitude and direction of the electric field in the room.

Figure P21.63



21.64 •• CP A small sphere with positive charge q and mass m is released from rest in a uniform electric field $\vec{E}$ that is directed vertically upward. The magnitude of the field is large enough for the sphere to travel upward when it is released. How long does it take the sphere to travel upward a distance d after being released from rest? Give your answer in terms of q , m , d , E , and the acceleration due to gravity, g .

21.65 •• If we rub a balloon on our hair, the balloon sticks to a wall or ceiling. This is because the rubbing transfers electrons from our hair to the balloon, giving it a net negative charge. When the balloon is placed near the ceiling, the extra electrons in it repel nearby electrons in the ceiling, creating a separation of charge on the ceiling, with positive charge closer to the balloon. Model the interaction as two point-like charges of equal magnitude and opposite signs, separated by a distance of $500\ \mu\text{m}$. Neglect the more distant negative charges on the ceiling. (a) A typical balloon has a mass of $4\ \text{g}$. Estimate the minimum magnitude of charge the balloon requires to stay attached to the ceiling. (b) Since a balloon sticks handily to the ceiling after being rubbed, assume that it has attained 10 times the estimated minimum charge. Estimate the number of electrons that were transferred to the balloon by the process of rubbing.

21.66 •• An American penny is 97.5% zinc and 2.5% copper and has a mass of $2.5\ \text{g}$. (a) Use the approximation that a penny is pure zinc, which has an atomic mass of $65.38\ \text{g/mol}$, to estimate the number of electrons in a penny. (Each zinc atom has 30 electrons.) (b) Estimate the net charge on all of the electrons in one penny. (c) The net positive charge on all of the protons in a penny has the same magnitude as the charge on the electrons. Estimate the force on either of two objects with this net magnitude of charge if the objects are separated by $2\ \text{cm}$. (d) Estimate the number of leaves on an oak tree that is 60 feet tall. (e) Imagine a forest filled with such trees, arranged in a square lattice, each $10\ \text{m}$ distant from its neighbors. Estimate how large such a forest would need to be to include as many leaves as there are electrons in one penny. (f) How does that area compare to the surface area of the earth?

21.67 •• Two particles having charges $q_1 = 0.500\ \text{nC}$ and $q_2 = 8.00\ \text{nC}$ are separated by a distance of $1.20\ \text{m}$. At what point along the line connecting the two charges is the total electric field due to the two charges equal to zero?

21.68 •• A $-3.00\ \text{nC}$ point charge is on the x -axis at $x = 1.20\ \text{m}$. A second point charge, Q , is on the x -axis at $-0.600\ \text{m}$. What must be the sign and magnitude of Q for the resultant electric field at the origin to be (a) $45.0\ \text{N/C}$ in the $+x$ -direction, (b) $45.0\ \text{N/C}$ in the $-x$ -direction?

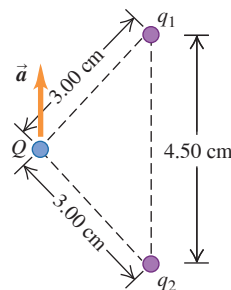
21.69 ••• A charge $+Q$ is located at the origin, and a charge $+4Q$ is at distance d away on the x -axis. Where should a third charge, q , be placed, and what should be its sign and magnitude, so that all three charges will be in equilibrium?

21.70 •• A charge of $-3.00\ \text{nC}$ is placed at the origin of an xy -coordinate system, and a charge of $2.00\ \text{nC}$ is placed on the y -axis at $y = 4.00\ \text{cm}$. (a) If a third charge, of $5.00\ \text{nC}$, is now placed at the point $x = 3.00\ \text{cm}$, $y = 4.00\ \text{cm}$, find the x - and y -components of the total force exerted on this charge by the other two charges. (b) Find the magnitude and direction of this force.

21.71 • Three identical point charges q are placed at each of three corners of a square of side L . Find the magnitude and direction of the net force on a point charge $-3q$ placed (a) at the center of the square and (b) at the vacant corner of the square. In each case, draw a free-body diagram showing the forces exerted on the $-3q$ charge by each of the other three charges.

21.72 ••• Two point charges q_1 and q_2 are held in place $4.50\ \text{cm}$ apart. Another point charge $Q = -1.75\ \mu\text{C}$, of mass $5.00\ \text{g}$, is initially located $3.00\ \text{cm}$ from both of these charges (Fig. P21.72) and released from rest. You observe that the initial acceleration of Q is $324\ \text{m/s}^2$ upward, parallel to the line connecting the two point charges. Find q_1 and q_2 .

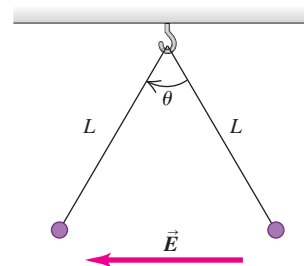
Figure P21.72



21.73 •• CP Strength of the Electric Force. Imagine two $1.0\ \text{g}$ bags of protons, one at the earth's north pole and the other at the south pole. (a) How many protons are in each bag? (b) Calculate the gravitational attraction and the electric repulsion that each bag exerts on the other. (c) Are the forces in part (b) large enough for you to feel if you were holding one of the bags?

21.74 ••• CP Two tiny spheres of mass $6.80\ \text{mg}$ carry charges of equal magnitude, $72.0\ \text{nC}$, but opposite sign. They are tied to the same ceiling hook by light strings of length $0.530\ \text{m}$. When a horizontal uniform electric field E that is directed to the left is turned on, the spheres hang at rest with the angle θ between the strings equal to 58.0° (Fig. P21.74). (a) Which ball (the one on the right or the one on the left) has positive charge? (b) What is the magnitude E of the field?

Figure P21.74



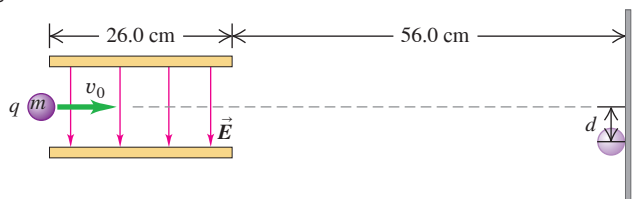
21.75 •• CP Consider a model of a hydrogen atom in which an electron is in a circular orbit of radius $r = 5.29 \times 10^{-11}\ \text{m}$ around a stationary proton. What is the speed of the electron in its orbit?

21.76 ••• CP A disk with radius R and uniform positive charge density σ lies horizontally on a tabletop. A small plastic sphere with mass M and positive charge Q hovers motionless above the center of the disk, suspended by the Coulomb repulsion due to the charged disk. (a) What is the magnitude of the net upward force on the sphere as a function of the height z above the disk? (b) At what height h does the sphere hover? Express your answer in terms of the dimensionless constant $v \equiv 2\epsilon_0 Mg / (Q\sigma)$. (c) If $M = 100\ \text{g}$, $Q = 1\ \mu\text{C}$, $R = 5\ \text{cm}$, and $\sigma = 10\ \text{nC/cm}^2$, what is h ?

21.77 •• CP A proton is projected into a uniform electric field that points vertically upward and has magnitude E . The initial velocity of the proton has a magnitude v_0 and is directed at an angle α below the horizontal. (a) Find the maximum distance h_{max} that the proton descends vertically below its initial elevation. Ignore gravitational forces. (b) After what horizontal distance d does the proton return to its original elevation? (c) Sketch the trajectory of the proton. (d) Find the numerical values of h_{max} and d if $E = 500\ \text{N/C}$, $v_0 = 4.00 \times 10^5\ \text{m/s}$, and $\alpha = 30.0^\circ$.

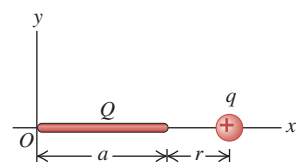
21.78 ••• A small object with mass m , charge q , and initial speed $v_0 = 5.00 \times 10^3$ m/s is projected into a uniform electric field between two parallel metal plates of length 26.0 cm (**Fig. P21.78**). The electric field between the plates is directed downward and has magnitude $E = 800$ N/C. Assume that the field is zero outside the region between the plates. The separation between the plates is large enough for the object to pass between the plates without hitting the lower plate. After passing through the field region, the object is deflected downward a vertical distance $d = 1.25$ cm from its original direction of motion and reaches a collecting plate that is 56.0 cm from the edge of the parallel plates. Ignore gravity and air resistance. Calculate the object's charge-to-mass ratio, q/m .

Figure P21.78



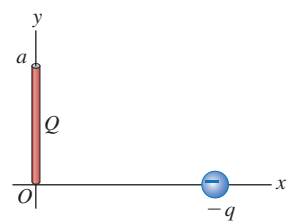
21.79 •• CALC Positive charge Q is distributed uniformly along the x -axis from $x = 0$ to $x = a$. A positive point charge q is located on the positive x -axis at $x = a + r$, a distance r to the right of the end of Q (**Fig. P21.79**). (a) Calculate the x - and y -components of the electric field produced by the charge distribution Q at points on the positive x -axis where $x > a$. (b) Calculate the force (magnitude and direction) that the charge distribution Q exerts on q . (c) Show that if $r \gg a$, the magnitude of the force in part (b) is approximately $Qq/4\pi\epsilon_0 r^2$. Explain why this result is obtained.

Figure P21.79



21.80 •• CALC Positive charge Q is distributed uniformly along the positive y -axis between $y = 0$ and $y = a$. A negative point charge $-q$ lies on the positive x -axis, a distance x from the origin (**Fig. P21.80**). (a) Calculate the x - and y -components of the electric field produced by the charge distribution Q at points on the positive x -axis. (b) Calculate the x - and y -components of the force that the charge distribution Q exerts on q . (c) Show that if $x \gg a$, $F_x \cong -Qq/4\pi\epsilon_0 x^2$ and $F_y \cong +Qqa/8\pi\epsilon_0 x^3$. Explain why this result is obtained.

Figure P21.80



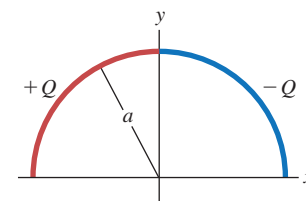
21.81 ••• A uniformly charged disk like the disk in Fig. 21.25 has radius 2.50 cm and carries a total charge of 7.0×10^{-12} C. (a) Find the electric field (magnitude and direction) on the x -axis at $x = 20.0$ cm. (b) Show that for $x \gg R$, Eq. (21.11) becomes $E = Q/4\pi\epsilon_0 x^2$, where Q is the total charge on the disk. (c) Is the magnitude of the electric field you calculated in part (a) larger or smaller than the electric field 20.0 cm from a point charge that has the same total charge as this disk? In terms of the approximation used in part (b) to derive $E = Q/4\pi\epsilon_0 x^2$ for a point charge from Eq. (21.11), explain why this is so. (d) What is the percent difference between the electric fields produced by the finite disk and by a point charge with the same charge at $x = 20.0$ cm and at $x = 10.0$ cm?

21.82 •• CP A small sphere with mass m carries a positive charge q and is attached to one end of a silk fiber of length L . The other end of the fiber is attached to a large vertical insulating sheet that has a positive surface charge density σ . Show that when the sphere is in equilibrium, the fiber makes an angle equal to $\arctan(q\sigma/2mg\epsilon_0)$ with the vertical sheet.

21.83 •• CALC Negative charge $-Q$ is distributed uniformly around a quarter-circle of radius a that lies in the first quadrant, with the center of curvature at the origin. Find the x - and y -components of the net electric field at the origin.

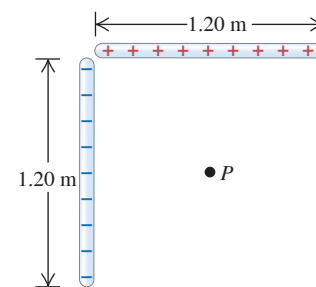
21.84 •• CALC A semicircle of radius a is in the first and second quadrants, with the center of curvature at the origin. Positive charge $+Q$ is distributed uniformly around the left half of the semicircle, and negative charge $-Q$ is distributed uniformly around the right half of the semicircle (**Fig. P21.84**). What are the magnitude and direction of the net electric field at the origin produced by this distribution of charge?

Figure P21.84



21.85 •• Two 1.20 m non-conducting rods meet at a right angle. One rod carries $+2.50 \mu\text{C}$ of charge distributed uniformly along its length, and the other carries $-2.50 \mu\text{C}$ distributed uniformly along it (**Fig. P21.85**). (a) Find the magnitude and direction of the electric field these rods produce at point P , which is 60.0 cm from each rod. (b) If an electron is released at P , what are the magnitude and direction of the net force that these rods exert on it?

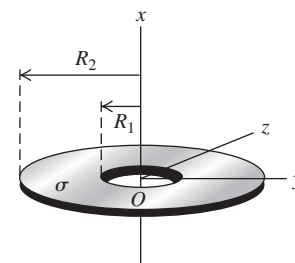
Figure P21.85



21.86 • Two very large parallel sheets are 5.00 cm apart. Sheet A carries a uniform surface charge density of $-8.80 \mu\text{C}/\text{m}^2$, and sheet B, which is to the right of A, carries a uniform charge density of $-11.6 \mu\text{C}/\text{m}^2$. Assume that the sheets are large enough to be treated as infinite. Find the magnitude and direction of the net electric field these sheets produce at a point (a) 4.00 cm to the right of sheet A; (b) 4.00 cm to the left of sheet A; (c) 4.00 cm to the right of sheet B.

21.87 •• CP A thin disk with a circular hole at its center, called an *annulus*, has inner radius R_1 and outer radius R_2 (**Fig. P21.87**). The disk has a uniform positive surface charge density σ on its surface. (a) Determine the total electric charge on the annulus. (b) The annulus lies in the yz -plane, with its center at the origin. For an arbitrary point on the x -axis (the axis of the annulus), find the magnitude and direction of the electric field $\vec{E}$. Consider points both above and below the annulus. (c) Show that at points on the x -axis that are sufficiently close to the origin, the magnitude of the electric field is approximately proportional to the distance between the center of the annulus and the point. How close is "sufficiently close"? (d) A point particle with mass m and negative charge $-q$ is free to move along the x -axis (but cannot move off the axis). The particle is originally placed at rest at $x = 0.01 R_1$ and released. Find the frequency of oscillation of the particle. (*Hint:* Review Section 14.2. The annulus is held stationary.)

Figure P21.87

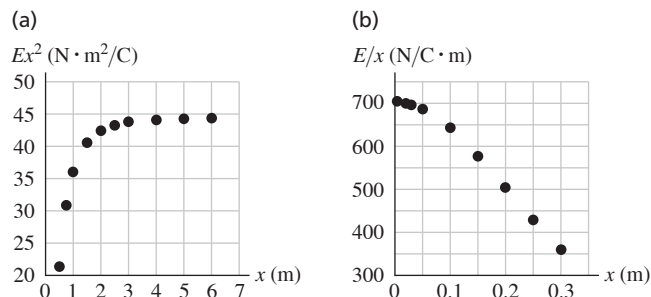


21.88 ••• DATA CP Design of an Inkjet Printer. Inkjet printers can be described as either continuous or drop-on-demand. In a continuous inkjet printer, letters are built up by squirting drops of ink at the paper from a rapidly moving nozzle. You are part of an engineering group working on the design of such a printer. Each ink drop will have a mass of 1.4×10^{-8} g. The drops will leave the nozzle and travel toward the paper at 50 m/s in a horizontal direction, passing through a charging unit that gives each drop a positive charge q by removing some electrons from it. The drops will then pass between parallel deflecting plates, 2.0 cm long, where there is a uniform vertical electric field with magnitude 8.0×10^4 N/C. Your team is working on the design of the charging unit that places the charge on the drops. (a) If a drop is to be deflected 0.30 mm by the time it reaches the end of the deflection plates, what magnitude of charge must be given to the drop? How many electrons must be removed from the drop to give it this charge? (b) If the unit that produces the stream of drops is redesigned so that it produces drops with a speed of 25 m/s, what q value is needed to achieve the same 0.30 mm deflection?

21.89 •• DATA Two small spheres, each carrying a net positive charge, are separated by 0.400 m. You have been asked to perform measurements that will allow you to determine the charge on each sphere. You set up a coordinate system with one sphere (charge q_1) at the origin and the other sphere (charge q_2) at $x = +0.400$ m. Available to you are a third sphere with net charge $q_3 = 4.00 \times 10^{-6}$ C and an apparatus that can accurately measure the location of this sphere and the net force on it. First you place the third sphere on the x -axis at $x = 0.200$ m; you measure the net force on it to be 4.50 N in the $+x$ -direction. Then you move the third sphere to $x = +0.600$ m and measure the net force on it now to be 3.50 N in the $+x$ -direction. (a) Calculate q_1 and q_2 . (b) What is the net force (magnitude and direction) on q_3 if it is placed on the x -axis at $x = -0.200$ m? (c) At what value of x (other than $x = \pm \infty$) could q_3 be placed so that the net force on it is zero?

21.90 ••• DATA Positive charge Q is distributed uniformly around a very thin conducting ring of radius a , as in Fig. 21.23. You measure the electric field E at points on the ring axis, at a distance x from the center of the ring, over a wide range of values of x . (a) Your results for the larger values of x are plotted in Fig. P21.90a as Ex^2 versus x . Explain why the quantity Ex^2 approaches a constant value as x increases. Use Fig. P21.90a to calculate the net charge Q on the ring. (b) Your results for smaller values of x are plotted in Fig. P21.90b as E/x versus x . Explain why E/x approaches a constant value as x approaches zero. Use Fig. P21.90b to calculate a .

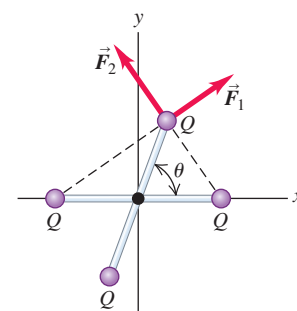
Figure P21.90



21.91 ••• CP Consider an infinite flat sheet with positive charge density σ in which a circular hole of radius R has been cut out. The sheet lies in the xy -plane with the origin at the center of the hole. The sheet is parallel to the ground, so that the positive z -axis describes the “upward” direction. If a particle of mass m and negative charge $-q$ sits at rest at the center of the hole and is released, the particle, constrained to the z -axis, begins to fall. As it drops farther beneath the sheet, the upward electric force increases. For a sufficiently low value of m , the upward electrical attraction eventually exceeds the particle’s weight and the particle will slow, come to a stop, and then rise back to its original position. This sequence of events will repeat indefinitely. (a) What is the electric field at a depth Δ beneath the origin along the negative z -axis? (b) What is the maximum mass $m_{\max}$ that would prevent the particle from falling indefinitely? (c) If $m < m_{\max}$, how much work is done by the electric field as the particle drops from $z = 0$ to $z = -\Delta$? (d) In this same interval, how much work is done by gravity? (e) To what ultimate depth $\Delta_{\max}$ will the particle drop? Express your answer in terms of the dimensionless parameter $\alpha \equiv m/m_{\max}$. (Use the work–energy theorem to find the speed as a function of depth, and then solve for the depth at which the speed is zero.) (f) What is the particle’s speed at depth $\Delta \leq \Delta_{\max}$? (g) If the sheet has charge density 1.00 nC/cm², the radius of the hole is $R = 10.0$ cm, and the particle has mass 25.0 g and charge 1.00 μ C, what are $m_{\max}$ and $\Delta_{\max}$?

21.92 ••• CP Two rigid insulating rods, each of length L and negligible mass, are connected at their centers by a frictionless hinge. One rod is fixed horizontally atop a vertical shaft, while the other rod is free to pivot about the hinge horizontally. A small ball with charge Q and mass M is affixed to each end of each rod. When the rods are perpendicular, there is no net torque on the second rod. We may describe the configuration of this system by the angle θ between the two rods. The first rod lies on the x -axis and the hinge sits at the origin. The second rod is rotated clockwise as seen from above in Fig. 21.92. (a) Determine the force $\vec{F}_1$ exerted by the charge on the left side of the fixed rod on the charge at the upper right side of the movable rod, as a function of θ . (b) Determine the force $\vec{F}_2$ exerted by the charge on the right side of the fixed rod on the charge at the upper right side of the movable rod, as a function of θ . (c) Determine the torque about the pivot point associated with the force $\vec{F}_1$. (d) Determine the torque about the pivot point associated with the force $\vec{F}_2$. (e) Given that the net torque exerted on the movable rod has contributions from the charges at both ends, write a formula for the net torque on the movable rod. (f) The equilibrium configuration at which the torque vanishes is $\theta = \pi/2$. Deviations from equilibrium may be parameterized as $\theta = \pi/2 - \epsilon$. Using power series expansions (see Appendix B), derive the torque for small ϵ , keeping only the lowest order term. (g) For small amplitudes, this system describes a torsional oscillator. Using the previous result, write an expression for the frequency of the small oscillations.

Figure P21.92



CHALLENGE PROBLEMS

21.93 ••• CP CALC Two thin rods, each with length L and total charge $+Q$, are parallel and separated by a distance a . The first rod has one end at the origin and its other end on the positive y -axis. The second rod has its lower end on the positive x -axis. (a) Explain why the y -component of the net force on the second rod vanishes. (b) Determine the x -component of the differential force dF_2 exerted on a small portion of the second rod, with length dy_2 and position y_2 , by the first rod. (This requires integrating over differential portions of the first rod, parameterized by dy_1 .) (c) Determine the net force $\vec{F}_2$ on the second rod by integrating dF_{2x} over the second rod. (d) Show that in

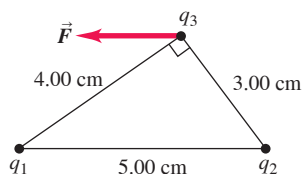
the limit $a \gg L$ the force determined in part (c) becomes $\frac{1}{4\pi\epsilon_0} \frac{Q^2}{a^2} \hat{i}$.

(e) Determine the external work required to move the second rod from very far away to the position $x = a$, provided the first rod is held fixed at $x = 0$. This describes the potential energy of the original configuration. (f) Suppose $L = 50.0$ cm, $a = 10.0$ cm, $Q = 10.0$ μC , and $m = 500$ g. If the two rods are released from the original configuration, they will fly apart and ultimately achieve a particular *relative speed*. What is that relative speed?

21.94 ••• CP CALC An insulating rigid rod of length $2a$ and negligible mass is attached at its center to a pivot at the origin and is free to rotate in the xy -plane. A small ball with mass M and charge Q is attached to one end of the rod. A second small ball with mass M and no charge is attached to the other end. A constant electric field $\vec{E} = -E\hat{i}$ is present in the region $y > 0$, while the region $y < 0$ has a vanishing electric field. Define $\vec{r}$ as the vector that points from the center of the rod to the charged end of the rod, and θ as the angle between $\vec{r}$ and the positive x -axis. The rod is oriented so that $\theta = 0$ and is given an infinitesimal nudge in the direction of increasing θ . (a) Write an expression for the vector $\vec{r}$. (b) Determine the torque $\vec{\tau}$ about the center of the rod when $0 \leq \theta \leq \pi$. (c) Determine the torque on the rod about its center when $\pi \leq \theta \leq 2\pi$. (d) What is the moment of inertia I of the system about the z -axis? (e) The potential energy $U(\theta)$ is determined by $\tau = -dU/d\theta$. Use this equation to write an expression for $U(\theta)$ over the range $0 \leq \theta \leq 4\pi$, using the convention that $U(0) = 0$. Make sure that $U(\theta)$ is continuous. (f) The angular velocity of the rod is $\omega = \omega(\theta)$. Using $\tau = I d^2\theta/dt^2$, show that the energy $\frac{1}{2}I\omega^2 + U(\theta)$ is conserved. (g) Using energy conservation, determine an expression for the angular velocity at the n th time the positive charge crosses the negative y -axis.

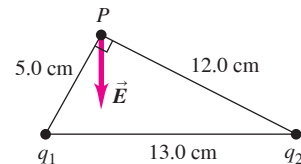
21.95 ••• Three charges are placed as shown in **Fig. P21.95**. The magnitude of q_1 is 2.00 μC , but its sign and the value of the charge q_2 are not known. Charge q_3 is $+4.00$ μC , and the net force $\vec{F}$ on q_3 is entirely in the negative x -direction. (a) Considering the different possible signs of q_1 , there are four possible force diagrams representing the forces $\vec{F}_1$ and $\vec{F}_2$ that q_1 and q_2 exert on q_3 . Sketch these four possible force configurations. (b) Using the sketches from part (a) and the direction of $\vec{F}$, deduce the signs of the charges q_1 and q_2 . (c) Calculate the magnitude of q_2 . (d) Determine F , the magnitude of the net force on q_3 .

Figure P21.95



21.96 ••• Two charges are placed as shown in **Fig. P21.96**. The magnitude of q_1 is 3.00 μC , but its sign and the value of the charge q_2 are not known. The direction of the net electric field $\vec{E}$ at point P is entirely in the negative y -direction. (a) Considering the different possible signs of q_1 and q_2 , four possible diagrams could represent the electric fields $\vec{E}_1$ and $\vec{E}_2$ produced by q_1 and q_2 . Sketch the four possible electric-field configurations. (b) Using the sketches from part (a) and the direction of $\vec{E}$, deduce the signs of q_1 and q_2 . (c) Determine the magnitude of $\vec{E}$.

Figure P21.96



21.97 ••• CALC Two thin rods of length L lie along the x -axis, one between $x = \frac{1}{2}a$ and $x = \frac{1}{2}a + L$ and the other between $x = -\frac{1}{2}a$ and $x = -\frac{1}{2}a - L$. Each rod has positive charge Q distributed uniformly along its length. (a) Calculate the electric field produced by the second rod at points along the positive x -axis. (b) Show that the magnitude of the force that one rod exerts on the other is

$$F = \frac{Q^2}{4\pi\epsilon_0 L^2} \ln \left[\frac{(a+L)^2}{a(a+2L)} \right]$$

(c) Show that if $a \gg L$, the magnitude of this force reduces to $F = Q^2/4\pi\epsilon_0 a^2$. (Hint: Use the expansion $\ln(1+z) = z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \dots$, valid for $|z| \ll 1$. Carry all expansions to at least order L^2/a^2 .) Interpret this result.

MCAT-STYLE PASSAGE PROBLEMS

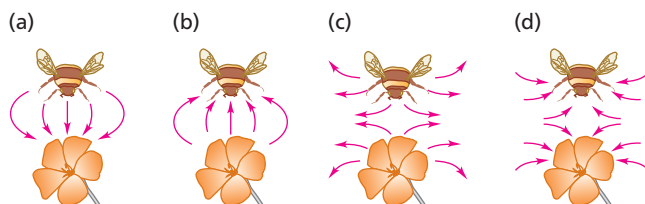
BIO Electric Bees. Flying insects such as bees may accumulate a small positive electric charge as they fly. In one experiment, the mean electric charge of 50 bees was measured to be $+(30 \pm 5)$ pC per bee. Researchers also observed the electrical properties of a plant consisting of a flower atop a long stem. The charge on the stem was measured as a positively charged bee approached, landed, and flew away. Plants are normally electrically neutral, so the measured net electric charge on the stem was zero when the bee was very far away. As the bee approached the flower, a small net positive charge was detected in the stem, even before the bee landed. Once the bee landed, the whole plant became positively charged, and this positive charge remained on the plant after the bee flew away. By creating artificial flowers with various charge values, experimenters found that bees can distinguish between charged and uncharged flowers and may use the positive electric charge left by a previous bee as a cue indicating whether a plant has already been visited (in which case, little pollen may remain).

21.98 Consider a bee with the mean electric charge found in the experiment. This charge represents roughly how many missing electrons? (a) 1.9×10^8 ; (b) 3.0×10^8 ; (c) 1.9×10^{18} ; (d) 3.0×10^{18} .

21.99 What is the best explanation for the observation that the electric charge on the stem became positive as the charged bee approached (before it landed)? (a) Because air is a good conductor, the positive charge on the bee's surface flowed through the air from bee to plant. (b) Because the earth is a reservoir of large amounts of charge, positive ions were drawn up the stem from the ground toward the charged bee. (c) The plant became electrically polarized as the charged bee approached. (d) Bees that had visited the plant earlier deposited a positive charge on the stem.

21.100 After one bee left a flower with a positive charge, that bee flew away and another bee with the same amount of positive charge flew close to the plant. Which diagram in **Fig. P21.100** (next page) best represents the electric field lines between the bee and the flower?

Figure P21.100



21.101 In a follow-up experiment, a charge of $+40 \text{ pC}$ was placed at the center of an artificial flower at the end of a 30-cm-long stem. Bees were observed to approach no closer than 15 cm from the center of this flower before they flew away. This observation suggests that the smallest external electric field to which bees may be sensitive is closest to which of these values? (a) 2.4 N/C ; (b) 16 N/C ; (c) $2.7 \times 10^{-10} \text{ N/C}$; (d) $4.8 \times 10^{-10} \text{ N/C}$.

ANSWERS

Chapter Opening Question ?

(ii) Water molecules have a permanent electric dipole moment: One end of the molecule has a positive charge and the other end has a negative charge. These ends attract negative and positive ions, respectively, holding the ions apart in solution. Water is less effective as a solvent for materials whose molecules do not ionize (called *nonionic* substances), such as oils.

Key Example VARIATION Problems

VP21.4.1 $(-98.9 \mu\text{N})\hat{i}$

VP21.4.2 (a) $40.5 \mu\text{N}$ in the $+x$ -direction (b) $13.6 \mu\text{N}$ in the $+x$ -direction (c) $+4.82 \text{ nC}$

VP21.4.3 $3.45 \mu\text{N}$ in the $+y$ -direction

VP21.4.4 $F_x = +0.452 \mu\text{N}$, $F_y = +0.329 \mu\text{N}$

VP21.10.1 (a) $E_x = 0$, $E_y = -8.09 \times 10^3 \text{ N/C}$ (b) $E_x = 0$, $E_y = +6.18 \times 10^2 \text{ N/C}$ (c) $E_x = -8.06 \times 10^2 \text{ N/C}$, $E_y = -3.18 \times 10^2 \text{ N/C}$

VP21.10.2 (a) $4.05 \times 10^4 \text{ N/C}$ in the $+x$ -direction (b) $2.70 \times 10^4 \text{ N/C}$ in the $+x$ -direction (c) -1.20 nC

VP21.10.3 (a) $E = \frac{e}{4\pi\epsilon_0 a^2} \left(1 - \frac{1}{2\sqrt{2}}\right)$ (b) away from the proton

VP21.10.4 (a) $dq = (Q/L)dx$ (b) $dE_x = -\frac{dq}{4\pi\epsilon_0 x^2} = -\frac{Qdx}{4\pi\epsilon_0 Lx^2}$, $dE_y = 0$ (c) $E_x = -\frac{Q}{8\pi\epsilon_0 L^2}$, $E_y = 0$

VP21.14.1 (a) $1.41 \times 10^{-24} \text{ N}\cdot\text{m}$ (b) $-1.18 \times 10^{-24} \text{ J}$

VP21.14.2 $7.06 \times 10^{-21} \text{ C}$

VP21.14.3 $1.92 \times 10^{-30} \text{ C}\cdot\text{m}$

VP21.14.4 (a) $2.19 \times 10^{-10} \text{ m}$ (b) $1.99 \times 10^{-8} \text{ m}$

Bridging Problem

$E = 2kQ/\pi a^2$ in the $-y$ -direction

? This child acquires an electric charge by touching the charged metal shell. The charged hairs on the child's head repel and stand out. What would happen if the child stood *inside* a large, charged metal shell? She would acquire (i) the same sign of charge as on the shell, and her hairs would stand out; (ii) the opposite sign of charge as on the shell, and her hairs would stand out; (iii) no charge, and her hairs would be relaxed; (iv) any of these, depending on the amount of charge on the shell.



22 Gauss's Law

LEARNING OUTCOMES

In this chapter, you'll learn...

- 22.1** How you can determine the amount of charge within a closed surface by examining the electric field on the surface.
- 22.2** What is meant by electric flux, and how to calculate it.
- 22.3** How Gauss's law relates the electric flux through a closed surface to the charge enclosed by the surface.
- 22.4** How to use Gauss's law to calculate the electric field due to a symmetric charge distribution.
- 22.5** Where the charge is located on a charged conductor.

You'll need to review...

- 21.4–21.6** Electric fields and their properties.

The discussion of Gauss's law in this section is based on and inspired by the innovative ideas of Ruth W. Chabay and Bruce A. Sherwood in *Electric and Magnetic Interactions* (John Wiley & Sons, 1994).

In physics, an important tool for simplifying problems is the *symmetry properties* of systems. Many physical systems have symmetry; for example, a cylindrical body doesn't look any different after you've rotated it around its axis, and a charged metal sphere looks just the same after you've turned it about any axis through its center.

In this chapter we'll use symmetry ideas along with a new principle, called *Gauss's law*, to simplify electric-field calculations. For example, the field of a straight-line or plane-sheet charge distribution, which we derived in Section 21.5 by using some fairly strenuous integrations, can be obtained in a few steps with the help of Gauss's law. But Gauss's law is more than just a way to make certain calculations easier. Indeed, it is a fundamental statement about the relationship between electric charges and electric fields. Among other things, Gauss's law can help us understand how electric charge distributes itself over conducting bodies.

Here's what Gauss's law is all about. Given any general distribution of charge, we surround it with an imaginary surface that encloses the charge. Then we look at the electric field at various points on this imaginary surface. Gauss's law is a relationship between the field at *all* the points on the surface and the total charge enclosed within the surface. This may sound like a rather indirect way of expressing things, but it turns out to be a tremendously useful relationship. In the next several chapters, we'll make frequent use of the insights that Gauss's law provides into the character of electric fields.

22.1 CHARGE AND ELECTRIC FLUX

In Chapter 21 we asked the question, "Given a charge distribution, what is the electric field produced by that distribution at a point P ?" We saw that the answer could be found by representing the distribution as an assembly of point charges, each of which produces an electric field $\vec{E}$ given by Eq. (21.7). The total field at P is then the vector sum of the fields due to all the point charges.

But there is an alternative relationship between charge distributions and electric fields. To discover this relationship, let's stand the question of Chapter 21 on its head and ask,

“If the electric-field pattern is known in a given region, what can we determine about the charge distribution in that region?”

Here’s an example. Consider the box shown in **Fig. 22.1a**, which may or may not contain electric charge. We’ll imagine that the box is made of a material that has no effect on any electric fields; it’s of the same breed as the massless rope and the frictionless incline. Better still, let the box represent an *imaginary* surface that may or may not enclose some charge. We’ll refer to the box as a **closed surface** because it completely encloses a volume. How can you determine how much (if any) electric charge lies within the box?

Knowing that a charge distribution produces an electric field and that an electric field exerts a force on a test charge, you move a test charge q_0 around the vicinity of the box. By measuring the force $\vec{F}$ experienced by the test charge at different positions, you make a three-dimensional map of the electric field $\vec{E} = \vec{F}/q_0$ outside the box. In the case shown in Fig. 22.1b, the map turns out to be the same as that of the electric field produced by a positive point charge (Fig. 21.28a). From the details of the map, you can find the exact value of the point charge inside the box.

To determine the contents of the box, we actually need to measure $\vec{E}$ on only the *surface* of the box. In **Fig. 22.2a** there is a single *positive* point charge inside the box, and in Fig. 22.2b there are two such charges. The field patterns on the surfaces of the boxes are different in detail, but in each case the electric field points *out* of the box. Figures 22.2c and 22.2d show cases with one and two *negative* point charges, respectively, inside the box. Again, the details of $\vec{E}$ are different for the two cases, but the electric field points *into* each box.

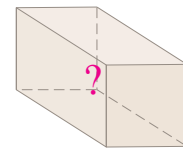
Electric Flux and Enclosed Charge

In Section 21.4 we mentioned the analogy between electric-field vectors and the velocity vectors of a fluid in motion. This analogy can be helpful, even though an electric field does not actually “flow.” Using this analogy, in Figs. 22.2a and 22.2b, in which the electric-field vectors point out of the surface, we say that there is an **outward electric flux**. (The word “flux” comes from a Latin word meaning “flow.”) In Figs. 22.2c and 22.2d the $\vec{E}$ vectors point into the surface, and the electric flux is **inward**.

Figure 22.2 suggests a simple relationship: Positive charge inside the box goes with an outward electric flux through the box’s surface, and negative charge inside goes with an inward electric flux. What happens if there is *zero* charge inside the box? In **Fig. 22.3a** (next page) the box is empty and $\vec{E} = \mathbf{0}$ everywhere, so there is no electric flux into or out of the box. In Fig. 22.3b, one positive and one negative point charge of equal magnitude are enclosed within the box, so the *net* charge inside the box is zero. There is an electric field, but it “flows into” the box on half of its surface and “flows out of” the box on the other half. Hence there is *no net* electric flux into or out of the box.

Figure 22.1 How can you measure the charge inside a box without opening it?

(a) A box containing an unknown amount of charge



(b) Using a test charge outside the box to probe the amount of charge inside the box

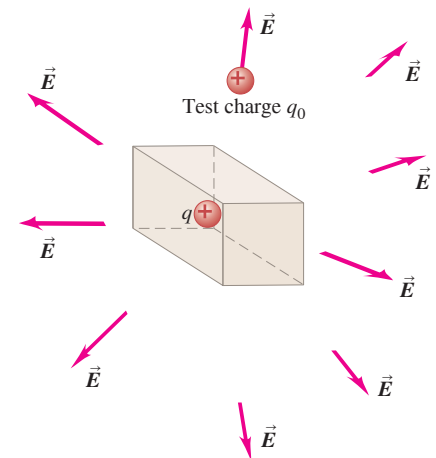
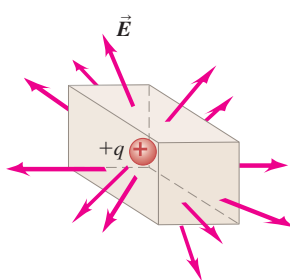
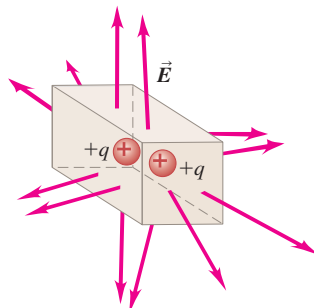


Figure 22.2 The electric field on the surface of boxes containing (a) a single positive point charge, (b) two positive point charges, (c) a single negative point charge, or (d) two negative point charges.

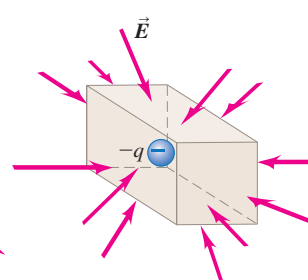
(a) Positive charge inside box, outward flux



(b) Positive charges inside box, outward flux



(c) Negative charge inside box, inward flux



(d) Negative charges inside box, inward flux

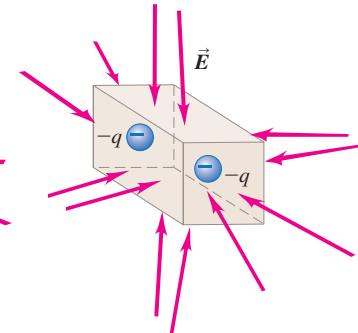


Figure 22.3 Three cases in which there is zero *net* charge inside a box and no net electric flux through the surface of the box. (a) An empty box with $\vec{E} = 0$. (b) A box containing one positive and one equal-magnitude negative point charge. (c) An empty box immersed in a uniform electric field.

(a) No charge inside box, zero flux

(b) Zero *net* charge inside box, inward flux cancels outward flux.

(c) No charge inside box, inward flux cancels outward flux.

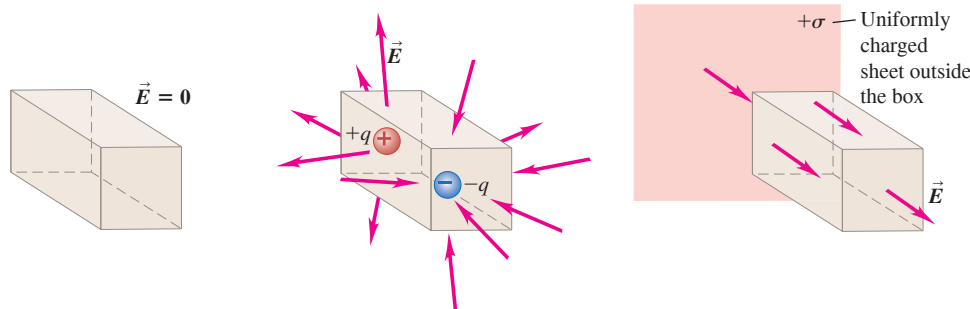
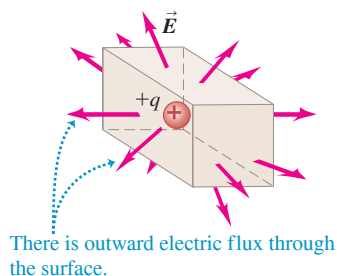
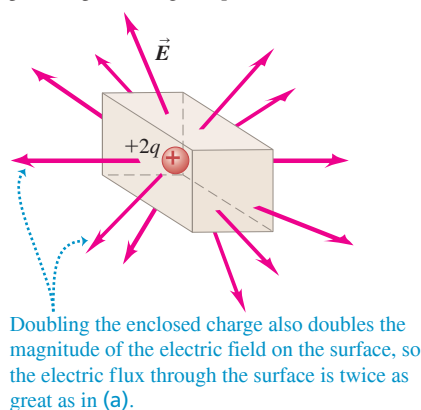


Figure 22.4 Three boxes, each of which encloses a positive point charge.

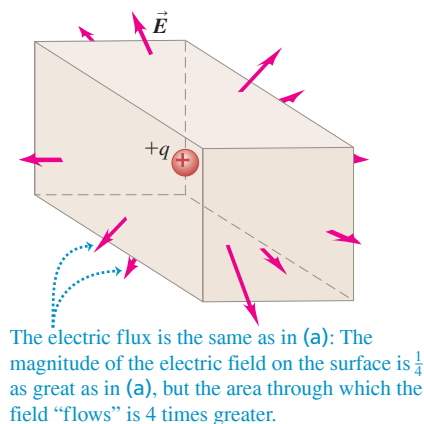
(a) A box containing a positive point charge $+q$



(b) The same box as in (a), but containing a positive point charge $+2q$



(c) The same positive point charge $+q$, but enclosed by a box with twice the dimensions



The box is again empty in Fig. 22.3c. However, there is charge present *outside* the box; the box has been placed with one end parallel to a uniformly charged infinite sheet, which produces a uniform electric field perpendicular to the sheet (see Example 21.11 of Section 21.5). On one end of the box, $\vec{E}$ points into the box; on the opposite end, $\vec{E}$ points out of the box; and on the sides, $\vec{E}$ is parallel to the surface and so points neither into nor out of the box. As in Fig. 22.3b, the inward electric flux on one part of the box exactly compensates for the outward electric flux on the other part. So in all of the cases shown in Fig. 22.3, there is no *net* electric flux through the surface of the box, and no *net* charge is enclosed in the box.

Figures 22.2 and 22.3 demonstrate a connection between the *sign* (positive, negative, or zero) of the *net* charge enclosed by a closed surface and the *sense* (outward, inward, or none) of the net electric flux through the surface. There is also a connection between the *magnitude* of the net charge inside the closed surface and the *strength* of the net “flow” of $\vec{E}$ over the surface. In both **Figs. 22.4a** and **22.4b** there is a single point charge inside the box, but in Fig. 22.4b the magnitude of the charge is twice as great, and so $\vec{E}$ is everywhere twice as great in magnitude as in Fig. 22.4a. If we keep in mind the fluid-flow analogy, this means that the net outward electric flux is also twice as great in Fig. 22.4b as in Fig. 22.4a. This suggests that the net electric flux through the surface of the box is *directly proportional* to the magnitude of the net charge enclosed by the box.

This conclusion is independent of the size of the box. In Fig. 22.4c the point charge $+q$ is enclosed by a box with twice the linear dimensions of the box in Fig. 22.4a. The magnitude of the electric field of a point charge decreases with distance according to $1/r^2$, so the average magnitude of $\vec{E}$ on each face of the large box in Fig. 22.4c is just $\frac{1}{4}$ of the average magnitude on the corresponding face in Fig. 22.4a. But each face of the large box has exactly four times the area of the corresponding face of the small box. Hence the outward electric flux is the *same* for the two boxes if we *define* electric flux as follows: For each face of the box, take the product of the average perpendicular component of $\vec{E}$ and the area of that face; then add up the results from all faces of the box. With this definition the net electric flux due to a single point charge inside the box is independent of the size of the box and depends only on the net charge inside the box.

To summarize, for the special cases of a closed surface in the shape of a rectangular box and charge distributions made up of point charges or infinite charged sheets, we have found:

1. Whether there is a net outward or inward electric flux through a closed surface depends on the sign of the enclosed charge.
2. Charges *outside* the surface do not give a net electric flux through the surface.
3. The net electric flux is directly proportional to the net amount of charge enclosed within the surface but is otherwise independent of the size of the closed surface.

These observations are a qualitative statement of *Gauss's law*.

Do these observations hold true for other kinds of charge distributions and for closed surfaces of arbitrary shape? The answer to these questions will prove to be yes. But to explain why this is so, we need a precise mathematical statement of what we mean by electric flux. We develop this in the next section.

TEST YOUR UNDERSTANDING OF SECTION 22.1 If all of the dimensions of the box in Fig. 22.2a are increased by a factor of 3, how will the electric flux through the box change?

(i) The flux will be $3^2 = 9$ times greater; (ii) the flux will be 3 times greater; (iii) the flux will be unchanged; (iv) the flux will be $\frac{1}{3}$ as great; (v) the flux will be $(\frac{1}{3})^2 = \frac{1}{9}$ as great; (vi) not enough information is given to decide.

ANSWER

(iii) Each part of the surface of the box will be three times farther from the charge $+q$, so the electric field will be $(\frac{1}{3})^2 = \frac{1}{9}$ as strong. But the area of the box will increase by a factor of $3^2 = 9$. Hence the electric flux will be multiplied by a factor of $(\frac{1}{9})(9) = 1$. In other words, the flux will be unchanged.

22.2 CALCULATING ELECTRIC FLUX

In the preceding section we introduced the concept of *electric flux*. We used this to give a rough qualitative statement of Gauss's law: The net electric flux through a closed surface is directly proportional to the net charge inside that surface. To be able to make full use of this law, we need to know how to *calculate* electric flux. To do this, let's again make use of the analogy between an electric field $\vec{E}$ and the field of velocity vectors $\vec{v}$ in a flowing fluid. (Again, keep in mind that this is only an analogy; an electric field is *not* a flow.)

Flux: Fluid-Flow Analogy

Figure 22.5 shows a fluid flowing steadily from left to right. Let's examine the volume flow rate dV/dt (in, say, cubic meters per second) through the wire rectangle with area A . When the area is perpendicular to the flow velocity $\vec{v}$ (Fig. 22.5a) and the flow velocity is the same at all points in the fluid, the volume flow rate dV/dt is the area A multiplied by the flow speed v :

$$\frac{dV}{dt} = vA$$

When the rectangle is tilted at an angle ϕ (Fig. 22.5b) so that its face is not perpendicular to $\vec{v}$, the area that counts is the silhouette area that we see when we look in the direction of $\vec{v}$. This area, which is outlined in red and labeled $A_{\perp}$ in Fig. 22.5b, is the *projection* of the area A onto a surface perpendicular to $\vec{v}$. Two sides of the projected rectangle have the same length as the original one, but the other two are foreshortened by a factor of $\cos \phi$, so the projected area $A_{\perp}$ is equal to $A \cos \phi$. Then the volume flow rate through A is

$$\frac{dV}{dt} = vA \cos \phi$$

If $\phi = 90^\circ$, $dV/dt = 0$; the wire rectangle is edge-on to the flow, and no fluid passes through the rectangle.

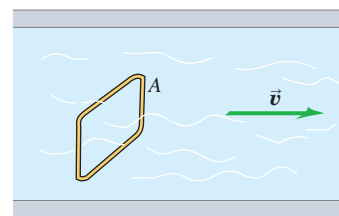
Also, $v \cos \phi$ is the component of the vector $\vec{v}$ perpendicular to the plane of the area A . Calling this component $v_{\perp}$, we can rewrite the volume flow rate as

$$\frac{dV}{dt} = v_{\perp} A$$

We can express the volume flow rate more compactly by using the concept of *vector area* $\vec{A}$, a vector quantity with magnitude A and a direction perpendicular to the plane of the area we are describing. The vector area $\vec{A}$ describes both the size of an area and its

Figure 22.5 The volume flow rate of fluid through the wire rectangle (a) is vA when the area of the rectangle is perpendicular to $\vec{v}$ and (b) is $vA \cos \phi$ when the rectangle is tilted at an angle ϕ .

(a) A wire rectangle in a fluid



(b) The wire rectangle tilted by an angle ϕ

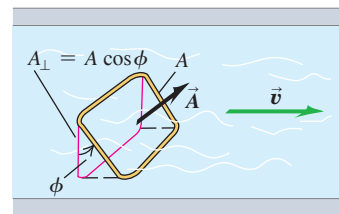
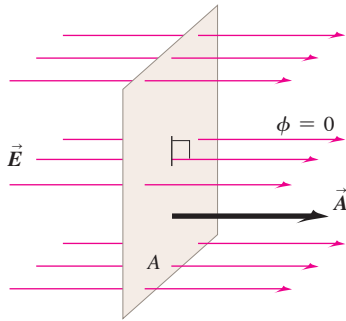
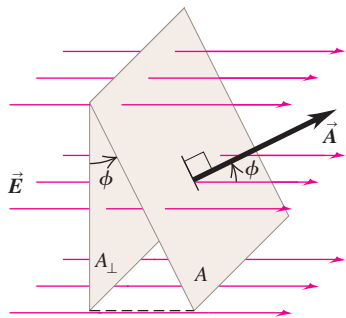


Figure 22.6 A flat surface in a uniform electric field. The electric flux Φ_E through the surface equals the scalar product of the electric field $\vec{E}$ and the area vector $\vec{A}$.

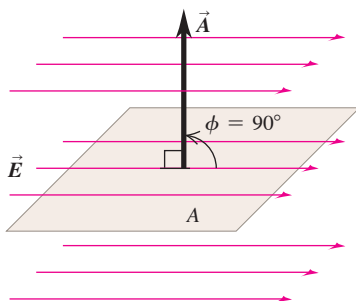
- (a) Surface is face-on to electric field:
- $\vec{E}$ and $\vec{A}$ are parallel (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 0$).
 - The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA$.



- (b) Surface is tilted from a face-on orientation by an angle ϕ :
- The angle between $\vec{E}$ and $\vec{A}$ is ϕ .
 - The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \phi$.



- (c) Surface is edge-on to electric field:
- $\vec{E}$ and $\vec{A}$ are perpendicular (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 90^\circ$).
 - The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos 90^\circ = 0$.



orientation in space. In terms of $\vec{A}$, we can write the volume flow rate of fluid through the rectangle in Fig. 22.5b as a scalar (dot) product:

$$\frac{dV}{dt} = \vec{v} \cdot \vec{A}$$

Flux of a Uniform Electric Field

Using the analogy between electric field and fluid flow, we now define electric flux in the same way as we have just defined the volume flow rate of a fluid; we simply replace the fluid velocity $\vec{v}$ by the electric field $\vec{E}$. The symbol that we use for electric flux is Φ_E (the capital Greek letter phi; the subscript E is a reminder that this is *electric* flux). Consider first a flat area A perpendicular to a uniform electric field $\vec{E}$ (Fig. 22.6a). We define the electric flux through this area to be the product of the field magnitude E and the area A :

$$\Phi_E = EA$$

Roughly speaking, we can picture Φ_E in terms of the field lines passing through A . Increasing the area means that more lines of $\vec{E}$ pass through the area, increasing the flux; a stronger field means more closely spaced lines of $\vec{E}$ and therefore more lines per unit area, so again the flux increases.

If the area A is flat but not perpendicular to the field $\vec{E}$, then fewer field lines pass through it. In this case the area that counts is the silhouette area that we see when looking in the direction of $\vec{E}$. This is the area $A_\perp$ in Fig. 22.6b and is equal to $A \cos \phi$ (compare to Fig. 22.5b). We generalize our definition of electric flux for a uniform electric field to

$$\Phi_E = EA \cos \phi \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface}) \quad (22.1)$$

Since $E \cos \phi$ is the component of $\vec{E}$ perpendicular to the area, we can rewrite Eq. (22.1) as

$$\Phi_E = E_\perp A \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface}) \quad (22.2)$$

In terms of the vector area $\vec{A}$ perpendicular to the area, we can write the electric flux as the scalar product of $\vec{E}$ and $\vec{A}$:

$$\Phi_E = \vec{E} \cdot \vec{A} \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface}) \quad (22.3)$$

Equations (22.1), (22.2), and (22.3) express the electric flux for a *flat* surface and a *uniform* electric field in different but equivalent ways. The SI unit for electric flux is $1 \text{ N} \cdot \text{m}^2/\text{C}$. Note that if the area is edge-on to the field, $\vec{E}$ and $\vec{A}$ are perpendicular and the flux is zero (Fig. 22.6c).

We can represent the direction of a vector area $\vec{A}$ by using a *unit vector* $\hat{n}$ perpendicular to the area; $\hat{n}$ stands for “normal.” Then

$$\vec{A} = A\hat{n} \quad (22.4)$$

A surface has two sides, so there are two possible directions for $\hat{n}$ and $\vec{A}$. We must always specify which direction we choose. In Section 22.1 we related the charge inside a *closed* surface to the electric flux through the surface. With a closed surface we’ll always choose the direction of $\hat{n}$ to be *outward*, and we’ll speak of the flux *out of* a closed surface. Thus what we called “outward electric flux” in Section 22.1 corresponds to a *positive* value of Φ_E , and what we called “inward electric flux” corresponds to a *negative* value of Φ_E .

Flux of a Nonuniform Electric Field

What happens if the electric field $\vec{E}$ isn't uniform but varies from point to point over the area A ? Or what if A is part of a curved surface? Then we divide A into many small elements dA , each of which has a unit vector $\hat{n}$ perpendicular to it and a vector area $d\vec{A} = \hat{n} dA$. We calculate the electric flux through each element and integrate the results to obtain the total flux:

$$\Phi_E = \int E \cos \phi \, dA = \int E_{\perp} dA = \int \vec{E} \cdot d\vec{A} \quad (22.5)$$

Labels in the diagram:
 - Electric flux through a surface: Φ_E
 - Angle between $\vec{E}$ and normal to surface: ϕ
 - Magnitude of electric field $\vec{E}$: E
 - Component of $\vec{E}$ perpendicular to surface: $E_{\perp}$
 - Element of surface area: dA
 - Vector element of surface area: $d\vec{A}$

We call this integral the **surface integral** of the component $E_{\perp}$ over the area, or the surface integral of $\vec{E} \cdot d\vec{A}$. In specific problems, one form of the integral is sometimes more convenient than another. Example 22.3 at the end of this section illustrates the use of Eq. (22.5).

In Eq. (22.5) the electric flux $\int E_{\perp} dA$ is equal to the *average* value of the perpendicular component of the electric field, multiplied by the area of the surface. This is the same definition of electric flux that we were led to in Section 22.1, now expressed more mathematically. In the next section we'll see the connection between the total electric flux through *any* closed surface, no matter what its shape, and the amount of charge enclosed within that surface.

BIO APPLICATION Flux Through a Basking Shark's Mouth

Unlike aggressive carnivorous sharks such as great whites, a basking shark feeds passively on plankton in the water that passes through the shark's gills as it swims. To survive on these tiny organisms requires a huge flux of water through a basking shark's immense mouth, which can be up to a meter across. The water flux—the product of the shark's speed through the water and the area of its mouth—can be up to $0.5 \text{ m}^3/\text{s}$ (500 liters per second, or almost 5×10^5 gallons per hour). In a similar way, the flux of electric field through a surface depends on the magnitude of the field and the area of the surface (as well as the relative orientation of the field and surface).

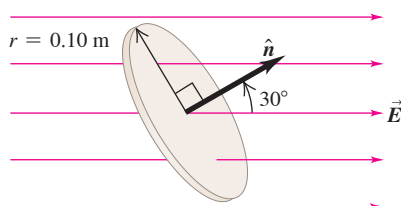


EXAMPLE 22.1 Electric flux through a disk

A disk of radius 0.10 m is oriented with its normal unit vector $\hat{n}$ at 30° to a uniform electric field $\vec{E}$ of magnitude $2.0 \times 10^3 \text{ N/C}$ (Fig. 22.7). (Since this isn't a closed surface, it has no "inside" or "outside." That's why we have to specify the direction of $\hat{n}$ in the figure.) (a) What is the electric flux through the disk? (b) What is the flux through the disk if it is turned so that $\hat{n}$ is perpendicular to $\vec{E}$? (c) What is the flux through the disk if $\hat{n}$ is parallel to $\vec{E}$?

IDENTIFY and SET UP This problem is about a flat surface in a uniform electric field, so we can apply the ideas of this section. We calculate the electric flux from Eq. (22.1).

Figure 22.7 The electric flux Φ_E through a disk depends on the angle between its normal $\hat{n}$ and the electric field $\vec{E}$.



EXECUTE (a) The area is $A = \pi(0.10 \text{ m})^2 = 0.0314 \text{ m}^2$ and the angle between $\vec{E}$ and $\vec{A} = A\hat{n}$ is $\phi = 30^\circ$, so from Eq. (22.1),

$$\Phi_E = EA \cos \phi = (2.0 \times 10^3 \text{ N/C})(0.0314 \text{ m}^2)(\cos 30^\circ) = 54 \text{ N} \cdot \text{m}^2/\text{C}$$

(b) The normal to the disk is now perpendicular to $\vec{E}$, so $\phi = 90^\circ$, $\cos \phi = 0$, and $\Phi_E = 0$.

(c) The normal to the disk is parallel to $\vec{E}$, so $\phi = 0$ and $\cos \phi = 1$:

$$\Phi_E = EA \cos \phi = (2.0 \times 10^3 \text{ N/C})(0.0314 \text{ m}^2)(1) = 63 \text{ N} \cdot \text{m}^2/\text{C}$$

EVALUATE Our answer to part (b) is smaller than that to part (a), which is in turn smaller than that to part (c). Is all this as it should be?

KEYCONCEPT To find the electric flux of a uniform electric field $\vec{E}$ through a flat surface, multiply the area of the surface by the component of $\vec{E}$ normal (perpendicular) to the surface.

EXAMPLE 22.2 Electric flux through a cube

WITH VARIATION PROBLEMS

An imaginary cubical surface of side L is in a region of uniform electric field $\vec{E}$. Find the electric flux through each face of the cube and the total flux through the cube when (a) it is oriented with two of its faces perpendicular to $\vec{E}$ (Fig. 22.8a) and (b) the cube is turned by an angle θ about a vertical axis (Fig. 22.8b).

IDENTIFY and SET UP Since $\vec{E}$ is uniform and each of the six faces of the cube is flat, we find the flux Φ_{Ei} through each face from Eqs. (22.3) and (22.4). The total flux through the cube is the sum of the six individual fluxes.

EXECUTE (a) Figure 22.8a shows the unit vectors $\hat{n}_1$ through $\hat{n}_6$ for each face; each unit vector points *outward* from the cube's closed surface. The angle between $\vec{E}$ and $\hat{n}_1$ is 180° , the angle between $\vec{E}$ and $\hat{n}_2$ is 0° , and the angle between $\vec{E}$ and each of the other four unit vectors is 90° . Each face of the cube has area L^2 , so the fluxes through the faces are

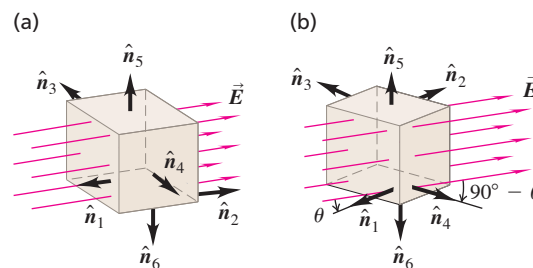
$$\begin{aligned}\Phi_{E1} &= \vec{E} \cdot \hat{n}_1 A = EL^2 \cos 180^\circ = -EL^2 \\ \Phi_{E2} &= \vec{E} \cdot \hat{n}_2 A = EL^2 \cos 0^\circ = +EL^2 \\ \Phi_{E3} &= \Phi_{E4} = \Phi_{E5} = \Phi_{E6} = EL^2 \cos 90^\circ = 0\end{aligned}$$

The flux is negative on face 1, where $\vec{E}$ is directed into the cube, and positive on face 2, where $\vec{E}$ is directed out of the cube. The total flux through the cube is

$$\begin{aligned}\Phi_E &= \Phi_{E1} + \Phi_{E2} + \Phi_{E3} + \Phi_{E4} + \Phi_{E5} + \Phi_{E6} \\ &= -EL^2 + EL^2 + 0 + 0 + 0 + 0 = 0\end{aligned}$$

(b) The field $\vec{E}$ is directed into faces 1 and 3, so the fluxes through them are negative; $\vec{E}$ is directed out of faces 2 and 4, so the fluxes through them are positive. We find

Figure 22.8 Electric flux of a uniform field $\vec{E}$ through a cubical box of side L in two orientations.



$$\begin{aligned}\Phi_{E1} &= \vec{E} \cdot \hat{n}_1 A = EL^2 \cos (180^\circ - \theta) = -EL^2 \cos \theta \\ \Phi_{E2} &= \vec{E} \cdot \hat{n}_2 A = +EL^2 \cos \theta \\ \Phi_{E3} &= \vec{E} \cdot \hat{n}_3 A = EL^2 \cos (90^\circ + \theta) = -EL^2 \sin \theta \\ \Phi_{E4} &= \vec{E} \cdot \hat{n}_4 A = EL^2 \cos (90^\circ - \theta) = +EL^2 \sin \theta \\ \Phi_{E5} &= \Phi_{E6} = EL^2 \cos 90^\circ = 0\end{aligned}$$

The total flux $\Phi_E = \Phi_{E1} + \Phi_{E2} + \Phi_{E3} + \Phi_{E4} + \Phi_{E5} + \Phi_{E6}$ through the surface of the cube is again zero.

EVALUATE We came to the same conclusion in our discussion of Fig. 22.3c: There is zero net flux of a uniform electric field through a closed surface that contains no electric charge.

KEYCONCEPT When calculating the net electric flux through a closed three-dimensional surface, always choose the normal at every location on the surface to point outward. The net flux equals the sum of the fluxes through each part of the surface.

EXAMPLE 22.3 Electric flux through a sphere

WITH VARIATION PROBLEMS

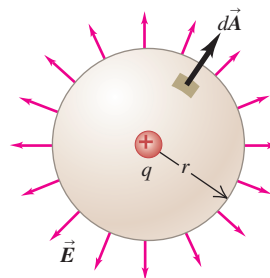
A point charge $q = +3.0 \mu\text{C}$ is surrounded by an imaginary sphere of radius $r = 0.20 \text{ m}$ centered on the charge (Fig. 22.9). Find the resulting electric flux through the sphere.

IDENTIFY and SET UP The surface is not flat and the electric field is not uniform, so to calculate the electric flux (our target variable) we must use the general definition, Eq. (22.5). Because the sphere is centered on the point charge, at any point on the spherical surface, $\vec{E}$ is directed out of the sphere perpendicular to the surface. The positive direction for both $\hat{n}$ and $E_\perp$ is outward, so $E_\perp = E$ and the flux through a surface element dA is $\vec{E} \cdot d\vec{A} = E dA$. This greatly simplifies the integral in Eq. (22.5).

EXECUTE We must evaluate the integral of Eq. (22.5), $\Phi_E = \int E dA$. At any point on the sphere of radius r the electric field has the same magnitude $E = q/4\pi\epsilon_0 r^2$. Hence E can be taken outside the integral, which becomes $\Phi_E = E \int dA = EA$, where A is the area of the spherical surface: $A = 4\pi r^2$. Hence the total flux through the sphere is

$$\begin{aligned}\Phi_E &= EA = \frac{q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{q}{\epsilon_0} \\ &= \frac{3.0 \times 10^{-6} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 3.4 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

Figure 22.9 Electric flux through a sphere centered on a point charge.



EVALUATE The radius r of the sphere cancels out of the result for Φ_E . We would have obtained the same flux with a sphere of radius 2.0 m or 200 m . We came to essentially the same conclusion in our discussion of Fig. 22.4 in Section 22.1, where we considered rectangular closed surfaces of two different sizes enclosing a point charge. There we found that the flux of $\vec{E}$ was independent of the size of the surface; the same result holds true for a spherical surface. Indeed, the flux through *any* surface enclosing a single point charge is independent of the shape or size of the surface, as we'll soon see.

KEYCONCEPT If a closed surface is curved, calculating the net electric flux through the surface requires integrating the normal component of the electric field over the entire surface. It greatly simplifies the integration if the field is symmetrical and the surface used has the same symmetry.

TEST YOUR UNDERSTANDING OF SECTION 22.2 Rank the following surfaces in order from most positive to most negative electric flux: (i) a flat rectangular surface with vector area $\vec{A} = (6.0 \text{ m}^2)\hat{i}$ in a uniform electric field $\vec{E} = (4.0 \text{ N/C})\hat{j}$; (ii) a flat circular surface with vector area $\vec{A} = (3.0 \text{ m}^2)\hat{j}$ in a uniform electric field $\vec{E} = (4.0 \text{ N/C})\hat{i} + (2.0 \text{ N/C})\hat{j}$; (iii) a flat square surface with vector area $\vec{A} = (3.0 \text{ m}^2)\hat{i} + (7.0 \text{ m}^2)\hat{j}$ in a uniform electric field $\vec{E} = (4.0 \text{ N/C})\hat{i} - (2.0 \text{ N/C})\hat{j}$; (iv) a flat oval surface with vector area $\vec{A} = (3.0 \text{ m}^2)\hat{i} - (7.0 \text{ m}^2)\hat{j}$ in a uniform electric field $\vec{E} = (4.0 \text{ N/C})\hat{i} - (2.0 \text{ N/C})\hat{j}$.

ANSWER

In each case the electric field is uniform, so the flux is $\Phi_E = \vec{E} \cdot \vec{A}$. We use the relationships for the scalar products of unit vectors: $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = 1$, $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{i} = 0$. In case (i) we have $\Phi_E = (4.0 \text{ N/C})(6.0 \text{ m}^2)\hat{j} \cdot \hat{i} = 0$ (the electric field and vector area are perpendicular, so there is zero flux). In case (ii) we have $\Phi_E = [(4.0 \text{ N/C})\hat{i} + (2.0 \text{ N/C})\hat{j}] \cdot [(3.0 \text{ m}^2)\hat{j}] = (2.0 \text{ N/C})(3.0 \text{ m}^2) = 6.0 \text{ N} \cdot \text{m}^2/\text{C}$. Similarly, in case (iii) we have $\Phi_E = [(4.0 \text{ N/C})\hat{i} + (2.0 \text{ N/C})\hat{j}] \cdot [(3.0 \text{ m}^2)\hat{i} + (7.0 \text{ m}^2)\hat{j}] = (4.0 \text{ N/C})(3.0 \text{ m}^2) + (2.0 \text{ N/C})(7.0 \text{ m}^2) = 12.0 \text{ N} \cdot \text{m}^2/\text{C} + 14.0 \text{ N} \cdot \text{m}^2/\text{C} = 26.0 \text{ N} \cdot \text{m}^2/\text{C}$. In case (iv) we have $\Phi_E = [(4.0 \text{ N/C})\hat{i} - (2.0 \text{ N/C})\hat{j}] \cdot [(3.0 \text{ m}^2)\hat{i} - (7.0 \text{ m}^2)\hat{j}] = (4.0 \text{ N/C})(3.0 \text{ m}^2) - (2.0 \text{ N/C})(-7.0 \text{ m}^2) = 12.0 \text{ N} \cdot \text{m}^2/\text{C} + 14.0 \text{ N} \cdot \text{m}^2/\text{C} = 26.0 \text{ N} \cdot \text{m}^2/\text{C}$. We have $\Phi_E = 0$ for (i), $\Phi_E = 6.0 \text{ N} \cdot \text{m}^2/\text{C}$ for (ii), $\Phi_E = 26.0 \text{ N} \cdot \text{m}^2/\text{C}$ for (iii), and $\Phi_E = 26.0 \text{ N} \cdot \text{m}^2/\text{C}$ for (iv). The ranking from most positive to most negative electric flux is (i), (ii), (iii), (iv).

22.3 GAUSS'S LAW

Gauss's law is an alternative to Coulomb's law. While completely equivalent to Coulomb's law, Gauss's law provides a different way to express the relationship between electric charge and electric field. It was formulated by Carl Friedrich Gauss (1777–1855), one of the greatest mathematicians of all time (**Fig. 22.10**).

Point Charge Inside a Spherical Surface

Gauss's law states that the total electric flux through any closed surface (a surface enclosing a definite volume) is proportional to the total (net) electric charge inside the surface. In Section 22.1 we observed this relationship qualitatively; now we'll develop it more rigorously. We'll start with the field of a single positive point charge q . The field lines radiate out equally in all directions. We place this charge at the center of an imaginary spherical surface with radius R . The magnitude E of the electric field at every point on the surface is given by

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

At each point on the surface, $\vec{E}$ is perpendicular to the surface, and its magnitude is the same at every point, as in Example 22.3 (Section 22.2). The total electric flux is the product of the field magnitude E and the total area $A = 4\pi R^2$ of the sphere:

$$\Phi_E = EA = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} (4\pi R^2) = \frac{q}{\epsilon_0} \quad (22.6)$$

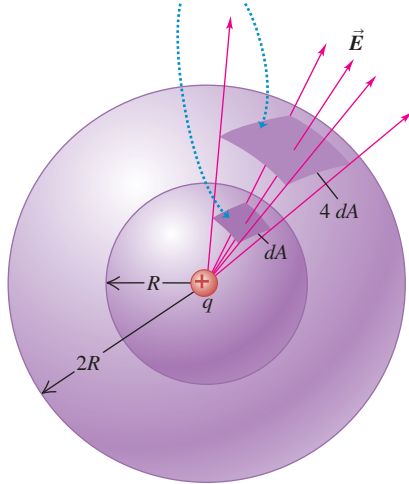
The flux is independent of the radius R of the sphere. It depends on only the charge q enclosed by the sphere.



Figure 22.10 Carl Friedrich Gauss helped develop several branches of mathematics, including differential geometry, real analysis, and number theory. The “bell curve” of statistics is one of his inventions. Gauss also made state-of-the-art investigations of the earth's magnetism and calculated the orbit of the first asteroid to be discovered.

Figure 22.11 Projection of an element of area dA of a sphere of radius R onto a concentric sphere of radius $2R$. The projection multiplies each linear dimension by 2, so the area element on the larger sphere is $4dA$.

The same number of field lines and the same flux pass through both of these area elements.



We can also interpret this result in terms of field lines. **Figure 22.11** shows two spheres with radii R and $2R$ centered on the point charge q . Every field line that passes through the smaller sphere also passes through the larger sphere, so the total flux through each sphere is the same.

What is true of the entire sphere is also true of any portion of its surface. In Fig. 22.11 an area dA is outlined on the sphere of radius R and projected onto the sphere of radius $2R$ by drawing lines from the center through points on the boundary of dA . The area projected on the larger sphere is clearly $4dA$. But since the electric field due to a point charge is inversely proportional to r^2 , the field magnitude is $\frac{1}{4}$ as great on the sphere of radius $2R$ as on the sphere of radius R . Hence the electric flux is the same for both areas and is independent of the radius of the sphere.

Point Charge Inside a Nonspherical Surface

We can extend this projection technique to nonspherical surfaces. Instead of a second sphere, let us surround the sphere of radius R by a surface of irregular shape, as in **Fig. 22.12a**. Consider a small element of area dA on the irregular surface; we note that this area is *larger* than the corresponding element on a spherical surface at the same distance from q . If a normal to dA makes an angle ϕ with a radial line from q , two sides of the area projected onto the spherical surface are foreshortened by a factor $\cos \phi$ (Fig. 22.12b). The other two sides are unchanged. Thus the electric flux through the spherical surface element is equal to the flux $E dA \cos \phi$ through the corresponding irregular surface element.

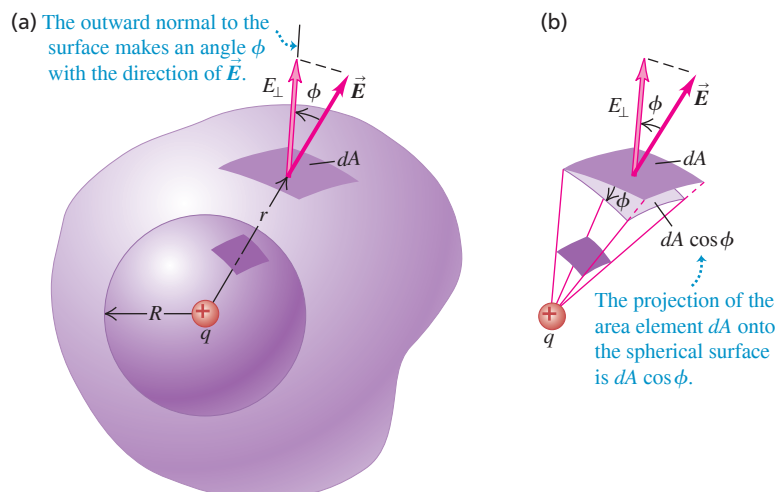
We can divide the entire irregular surface into elements dA , compute the electric flux $E dA \cos \phi$ for each, and sum the results by integrating, as in Eq. (22.5). Each of the area elements projects onto a corresponding spherical surface element. Thus the *total* electric flux through the irregular surface, given by any of the forms of Eq. (22.5), must be the same as the total flux through a sphere, which Eq. (22.6) shows is equal to q/ϵ_0 . Thus, for the irregular surface,

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0} \quad (22.7)$$

Equation (22.7) holds for a surface of *any* shape or size, provided only that it is a *closed* surface enclosing the charge q . The circle on the integral sign reminds us that the integral is always taken over a *closed* surface.

The area elements $d\vec{A}$ and the corresponding unit vectors $\hat{n}$ always point *out of* the volume enclosed by the surface. The electric flux is then positive in areas where the electric field points out of the surface and negative where it points inward. Also, $E_{\perp}$ is positive at

Figure 22.12 Calculating the electric flux through a nonspherical surface.



points where $\vec{E}$ points out of the surface and negative at points where $\vec{E}$ points into the surface.

If the point charge in Fig. 22.12 is negative, the $\vec{E}$ field is directed radially *inward*; the angle ϕ is then greater than 90° , its cosine is negative, and the integral in Eq. (22.7) is negative. But since q is also negative, Eq. (22.7) holds.

For a closed surface enclosing *no* charge,

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = 0$$

This is the mathematical statement that when a region contains no charge, any field lines caused by charges *outside* the region that enter on one side must leave again on the other side. (In Section 22.1 we came to the same conclusion by considering the special case of a rectangular box in a uniform field.) **Figure 22.13** illustrates this point. *Electric field lines can begin or end inside a region of space only when there is charge in that region.*

General Form of Gauss's Law

Now comes the final step in obtaining the general form of Gauss's law. Suppose the surface encloses not just one point charge q but several charges $q_1, q_2, q_3, \dots$. The total (resultant) electric field $\vec{E}$ at any point is the vector sum of the $\vec{E}$ fields of the individual charges. Let Q_{encl} be the *total* charge enclosed by the surface: $Q_{\text{encl}} = q_1 + q_2 + q_3 + \dots$. Also let $\vec{E}$ be the *total* field at the position of the surface area element $d\vec{A}$, and let $E_\perp$ be its component perpendicular to the plane of that element (that is, parallel to $d\vec{A}$). Then we can write an equation like Eq. (22.7) for each charge and its corresponding field and add the results. When we do, we obtain the general statement of Gauss's law:

Gauss's law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$ (22.8)

Electric flux through a closed surface of area A = surface integral of $\vec{E}$

Total charge enclosed by surface

Electric constant

The total electric flux through a closed surface is equal to the total (net) electric charge inside the surface, divided by ϵ_0 .

Using the definition of Q_{encl} and the various ways to express electric flux given in Eq. (22.5), we can express Gauss's law in the following equivalent forms:

Various forms of Gauss's law:

$$\Phi_E = \oint E \cos \phi \, dA = \oint E_\perp \, dA = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

Electric flux through a closed surface

Magnitude of electric field $\vec{E}$

Angle between $\vec{E}$ and normal to surface

Component of $\vec{E}$ perpendicular to surface

Element of surface area

Vector element of surface area

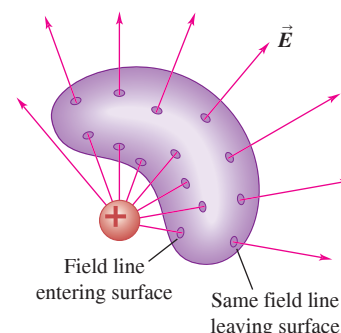
Total charge enclosed by surface

Electric constant

As in Eq. (22.5), the various forms of the integral all express the same thing, the total electric flux through the Gaussian surface, in different terms. One form is sometimes more convenient than another.

As an example, **Fig. 22.14a** (next page) shows a spherical Gaussian surface of radius r around a positive point charge $+q$. The electric field points out of the Gaussian surface, so at every point on the surface $\vec{E}$ is in the same direction as $d\vec{A}$, $\phi = 0$, and $E_\perp$ is equal to the field magnitude $E = q/4\pi\epsilon_0 r^2$. Since E is the same at all points on the surface,

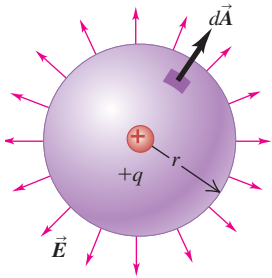
Figure 22.13 A point charge *outside* a closed surface that encloses no charge. If an electric field line from the external charge enters the surface at one point, it must leave at another.



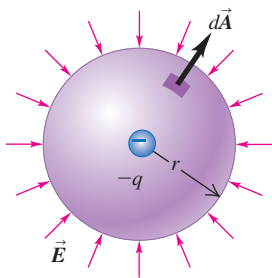
CAUTION **Gaussian surfaces are imaginary** Remember that the closed surface in Gauss's law is *imaginary*; there need not be any material object at the position of the surface. We often refer to a closed surface used in Gauss's law as a **Gaussian surface**.

Figure 22.14 Spherical Gaussian surfaces around (a) a positive point charge and (b) a negative point charge.

(a) Gaussian surface around positive charge: positive (outward) flux



(b) Gaussian surface around negative charge: negative (inward) flux



we can take it outside the integral in Eq. (22.9). Then the remaining integral is $\int dA = A = 4\pi r^2$, the area of the sphere. Hence Eq. (22.9) becomes

$$\Phi_E = \oint E_{\perp} dA = \oint \left(\frac{q}{4\pi\epsilon_0 r^2} \right) dA = \frac{q}{4\pi\epsilon_0 r^2} \oint dA = \frac{q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{q}{\epsilon_0}$$

The enclosed charge Q_{encl} is just the charge $+q$, so this agrees with Gauss's law. If the Gaussian surface encloses a *negative* point charge as in Fig. 22.14b, then $\vec{E}$ points *into* the surface at each point in the direction opposite $d\vec{A}$. Then $\phi = 180^\circ$ and $E_{\perp}$ is equal to the negative of the field magnitude: $E_{\perp} = -E = -|q|/4\pi\epsilon_0 r^2 = -q/4\pi\epsilon_0 r^2$. Equation (22.9) then becomes

$$\Phi_E = \oint E_{\perp} dA = \oint \left(\frac{-q}{4\pi\epsilon_0 r^2} \right) dA = \frac{-q}{4\pi\epsilon_0 r^2} \oint dA = \frac{-q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{-q}{\epsilon_0}$$

This again agrees with Gauss's law because the enclosed charge in Fig. 22.14b is $Q_{\text{encl}} = -q$.

In Eqs. (22.8) and (22.9), Q_{encl} is always the algebraic sum of all the positive and negative charges enclosed by the Gaussian surface, and $\vec{E}$ is the *total* field at each point on the surface. Also note that in general, this field is caused partly by charges inside the surface and partly by charges outside. But as Fig. 22.13 shows, the outside charges do *not* contribute to the total (net) flux through the surface. So Eqs. (22.8) and (22.9) are correct even when there are charges outside the surface that contribute to the electric field at the surface. When $Q_{\text{encl}} = 0$, the total flux through the Gaussian surface must be zero, even though some areas may have positive flux and others may have negative flux (see Fig. 22.3b).

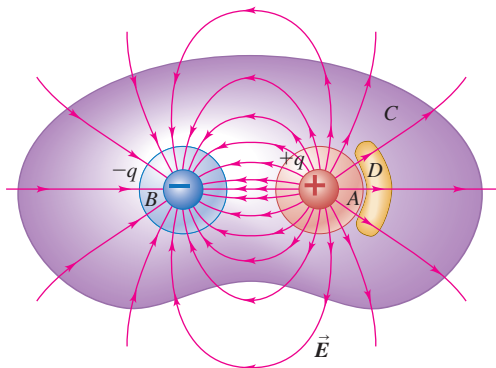
Gauss's law is the definitive answer to the question we posed at the beginning of Section 22.1: "If the electric-field pattern is known in a given region, what can we determine about the charge distribution in that region?" It provides a relationship between the electric field on a closed surface and the charge distribution within that surface. But in some cases we can use Gauss's law to answer the reverse question: "If the charge distribution is known, what can we determine about the electric field that the charge distribution produces?" Gauss's law may seem like an unappealing way to address this question, since it may look as though evaluating the integral in Eq. (22.8) is a hopeless task. Sometimes it is, but other times it is surprisingly easy. Here's an example in which *no* integration is involved at all; we'll work out several more examples in the next section.

CONCEPTUAL EXAMPLE 22.4 Electric flux and enclosed charge

WITH VARIATION PROBLEMS

Figure 22.15 shows the field produced by two point charges $+q$ and $-q$ (an electric dipole). Find the electric flux through each of the closed surfaces A , B , C , and D .

Figure 22.15 The net number of field lines leaving a closed surface is proportional to the total charge enclosed by that surface.



SOLUTION Gauss's law, Eq. (22.8), says that the total electric flux through a closed surface is equal to the total enclosed charge divided by ϵ_0 . In Fig. 22.15, surface A (shown in red) encloses the positive charge, so $Q_{\text{encl}} = +q$; surface B (in blue) encloses the negative charge, so $Q_{\text{encl}} = -q$; surface C (in purple) encloses *both* charges, so $Q_{\text{encl}} = +q + (-q) = 0$; and surface D (in yellow) encloses no charges, so $Q_{\text{encl}} = 0$. Hence, without having to do any integration, we have $\Phi_{EA} = +q/\epsilon_0$, $\Phi_{EB} = -q/\epsilon_0$, and $\Phi_{EC} = \Phi_{ED} = 0$. These results depend only on the charges enclosed within each Gaussian surface, not on the precise shapes of the surfaces.

We can draw similar conclusions by examining the electric field lines. All the field lines that cross surface A are directed out of the surface, so the flux through A must be positive. Similarly, the flux through B must be negative since all of the field lines that cross that surface point inward. For both surface C and surface D , there are as many field lines pointing into the surface as there are field lines pointing outward, so the flux through each of these surfaces is zero.

KEYCONCEPT The *net* electric flux Φ_E through a closed surface is always proportional to the *net* amount of charge q that the surface encloses. If q is zero, Φ_E is zero. These statements (Gauss's law) are true no matter what the shape of the surface.

3. If $\vec{E}$ is *perpendicular* (normal) at every point to a surface with area A , if it points *outward* from the interior of the surface, and if it has the same *magnitude* at every point on the surface, then $E_{\perp} = E = \text{constant}$, and $\int E_{\perp} dA$ over that surface is equal to EA . (If $\vec{E}$ is inward, then $E_{\perp} = -E$ and $\int E_{\perp} dA = -EA$.) This should be the case for part or all of your Gaussian surface. If $\vec{E}$ is tangent to a surface at every point, then $E_{\perp} = 0$ and the integral over that surface is zero. This may be the case for parts of a cylindrical Gaussian surface. If $\vec{E} = 0$ at every point on a surface, the integral is zero.
4. Even when there is *no* charge within a Gaussian surface, the field at any given point on the surface is not necessarily zero. In that case, however, the total electric flux through the surface is always zero.

5. The flux integral $\oint E_{\perp} dA$ can be approximated as the difference between the numbers of electric lines of force leaving and entering the Gaussian surface. In this sense the flux gives the sign of the enclosed charge, but is only proportional to it; zero flux corresponds to zero enclosed charge.
6. Once you have evaluated $\oint E_{\perp} dA$, use Eq. (22.9) to solve for your target variable.

EVALUATE your answer: If your result is a *function* that describes how the magnitude of the electric field varies with position, ensure that it makes sense.

EXAMPLE 22.5 Field of a charged conducting sphere

WITH VARIATION PROBLEMS

We place a total positive charge q on a solid conducting sphere with radius R (Fig. 22.18). Find $\vec{E}$ at any point inside or outside the sphere.

IDENTIFY and SET UP As we discussed earlier in this section, all of the charge must be on the surface of the sphere. The charge is free to move on the conductor, and there is no preferred position on the surface; the charge is therefore distributed *uniformly* over the surface, and the system is spherically symmetric. To exploit this symmetry, we take as our Gaussian surface a sphere of radius r centered on the conductor. We can calculate the field inside or outside the conductor by taking $r < R$ or $r > R$, respectively. In either case, the point at which we want to calculate $\vec{E}$ lies on the Gaussian surface.

EXECUTE The spherical symmetry means that the direction of the electric field must be radial; that's because there is no preferred direction parallel to the surface, so $\vec{E}$ can have no component parallel to the surface. There is also no preferred orientation of the sphere, so the field magnitude E can depend only on the distance r from the center and must have the same value at all points on the Gaussian surface.

For $r > R$ the entire conductor is within the Gaussian surface, so the enclosed charge is q . The area of the Gaussian surface is $4\pi r^2$, and $\vec{E}$ is uniform over the surface and perpendicular to it at each point. The flux integral $\oint E_{\perp} dA$ is then just $E(4\pi r^2)$, and Eq. (22.8) gives

$$E(4\pi r^2) = \frac{q}{\epsilon_0} \quad \text{and} \quad E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad (\text{outside a charged conducting sphere})$$

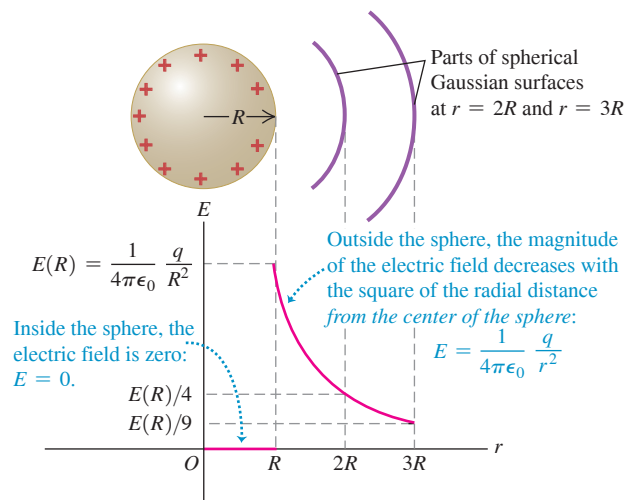
This expression is the same as that for a point charge; outside the charged sphere, its field is the same as though the entire charge were concentrated at its center. Just outside the surface of the sphere, where $r = R$,

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} \quad (\text{at the surface of a charged conducting sphere})$$

CAUTION **Flux can be positive or negative** Remember that we have chosen the charge q to be *positive*. If the charge is negative, the electric field is radially *inward* instead of radially outward, and the electric flux through the Gaussian surface is negative. The electric-field magnitudes outside and at the surface of the sphere are given by the same expressions as above, except that q denotes the *magnitude* (absolute value) of the charge. ■

For $r < R$ we again have $E(4\pi r^2) = Q_{\text{encl}}/\epsilon_0$. But now our Gaussian surface (which lies entirely within the conductor) encloses *no* charge, so $Q_{\text{encl}} = 0$. The electric field inside the conductor is therefore zero.

Figure 22.18 Calculating the electric field of a conducting sphere with positive charge q . Outside the sphere, the field is the same as if all of the charge were concentrated at the center of the sphere.



EVALUATE We already knew that $\vec{E} = 0$ inside a solid conductor (whether spherical or not) when the charges are at rest. Figure 22.18 shows E as a function of the distance r from the center of the sphere. Note that in the limit as $R \rightarrow 0$, the sphere becomes a point charge; there is then only an “outside,” and the field is everywhere given by $E = q/4\pi\epsilon_0 r^2$. Thus we have deduced Coulomb’s law from Gauss’s law. (In Section 22.3 we deduced Gauss’s law from Coulomb’s law; the two laws are equivalent.)

We can also use this method for a conducting spherical *shell* (a spherical conductor with a concentric spherical hole inside) if there is no charge inside the hole. We use a spherical Gaussian surface with radius r less than the radius of the hole. If there *were* a field inside the hole, it would have to be radial and spherically symmetric as before, so $E = Q_{\text{encl}}/4\pi\epsilon_0 r^2$. But now there is no enclosed charge, so $Q_{\text{encl}} = 0$ and $E = 0$ inside the hole.

Can you use this same technique to find the electric field in the region between a charged sphere and a concentric hollow conducting sphere that surrounds it?

KEYCONCEPT To apply Gauss’s law to a charge distribution with spherical symmetry, use a spherical Gaussian surface. At all points on such a surface, the electric field has the same magnitude and is normal to the surface.

EXAMPLE 22.6 Field of a uniform line charge

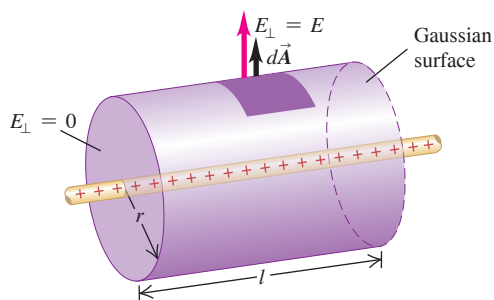
WITH VARIATION PROBLEMS

Electric charge is distributed uniformly along an infinitely long, thin wire. The charge per unit length is λ (assumed positive). Find the electric field by using Gauss's law.

IDENTIFY and SET UP We found in Example 21.10 (Section 21.5) that the field $\vec{E}$ of a uniformly charged, infinite wire is radially outward if λ is positive and radially inward if λ is negative, and that the field magnitude E depends on only the radial distance from the wire. This suggests that we use a *cylindrical* Gaussian surface, of radius r and arbitrary length l , coaxial with the wire and with its ends perpendicular to the wire (Fig. 22.19).

EXECUTE The flux through the flat ends of our Gaussian surface is zero because the radial electric field is parallel to these ends, and so $\vec{E} \cdot \hat{n} = 0$. On the cylindrical part of our surface we have $\vec{E} \cdot \hat{n} = E_{\perp} = E$ everywhere. (If λ were negative, we would have $\vec{E} \cdot \hat{n} = E_{\perp} = -E$ everywhere.) The area of the cylindrical surface is $2\pi rl$, so the flux through it—and hence the total flux Φ_E through

Figure 22.19 A coaxial cylindrical Gaussian surface is used to find the electric field outside an infinitely long, charged wire.



the Gaussian surface—is $EA = 2\pi rlE$. The total enclosed charge is $Q_{\text{encl}} = \lambda l$, and so from Gauss's law, Eq. (22.8),

$$\Phi_E = 2\pi rlE = \frac{\lambda l}{\epsilon_0} \quad \text{and}$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \quad (\text{field of an infinite line of charge})$$

We found this same result in Example 21.10 with *much* more effort.

If λ is *negative*, $\vec{E}$ is directed radially inward, and in the above expression for E we must interpret λ as the absolute value of the charge per unit length.

EVALUATE We saw in Example 21.10 that the *entire* charge on the wire contributes to the field at any point, and yet we consider only that part of the charge $Q_{\text{encl}} = \lambda l$ within the Gaussian surface when we apply Gauss's law. There's nothing inconsistent here; it takes the entire charge to give the field the properties that allow us to calculate Φ_E so easily, and Gauss's law always applies to the enclosed charge only. If the wire is short, the symmetry of the infinite wire is lost, and E is not uniform over a coaxial, cylindrical Gaussian surface. Gauss's law then *cannot* be used to find Φ_E ; we must solve the problem the hard way, as in Example 21.10.

We can use the Gaussian surface in Fig. 22.19 to show that the field outside a long, uniformly charged cylinder is the same as though all the charge were concentrated on a line along its axis (see Problem 22.39). We can also calculate the electric field in the space between a charged cylinder and a coaxial hollow conducting cylinder surrounding it (see Problem 22.37).

KEYCONCEPT To apply Gauss's law to a charge distribution with cylindrical symmetry, such as an infinite line or cylinder of charge, use a cylindrical Gaussian surface. At all points on the curved sides of the surface, the electric field has the same magnitude and is normal to the surface. On the flat ends of the cylindrical surface, the electric field is parallel to the surface, so the flux through the ends is zero.

EXAMPLE 22.7 Field of an infinite plane sheet of charge

WITH VARIATION PROBLEMS

Use Gauss's law to find the electric field caused by a thin, flat, infinite sheet with a uniform positive surface charge density σ .

IDENTIFY and SET UP In Example 21.11 (Section 21.5) we found that the field $\vec{E}$ of a uniformly charged infinite sheet is normal to the sheet, and that its magnitude is independent of the distance from the sheet. To take advantage of these symmetry properties, we use a cylindrical Gaussian surface with ends of area A and with its axis perpendicular to the sheet of charge (Fig. 22.20).

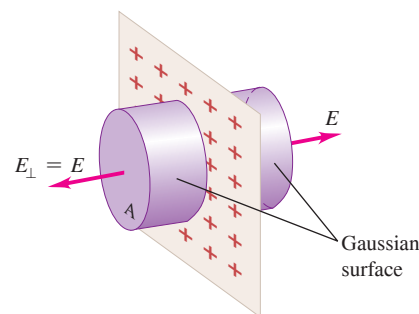
EXECUTE The flux through the cylindrical part of our Gaussian surface is zero because $\vec{E} \cdot \hat{n} = 0$ everywhere. The flux through each flat end of the surface is $+EA$ because $\vec{E} \cdot \hat{n} = E_{\perp} = E$ everywhere, so the total flux through both ends—and hence the total flux Φ_E through the Gaussian surface—is $+2EA$. The total enclosed charge is $Q_{\text{encl}} = \sigma A$, and so from Gauss's law,

$$2EA = \frac{\sigma A}{\epsilon_0} \quad \text{and}$$

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{field of an infinite sheet of charge})$$

If σ is negative, $\vec{E}$ is directed *toward* the sheet, the flux through the Gaussian surface in Fig. 22.20 is negative, and σ in the expression $E = \sigma/2\epsilon_0$ denotes the magnitude (absolute value) of the charge density.

Figure 22.20 A cylindrical Gaussian surface is used to find the field of an infinite plane sheet of charge.



EVALUATE We got the same result for the field of an infinite sheet of charge in Example 21.11 (Section 21.5). That calculation was much more complex and involved a fairly challenging integral. Thanks to the favorable symmetry, Gauss's law makes it much easier to solve this problem.

KEYCONCEPT To apply Gauss's law to a charge distribution with planar symmetry, such as an infinite sheet of charge, use a cylindrical Gaussian surface. At all points on the flat ends of the surface, the electric field has the same magnitude and is normal to the surface. On the curved sides of the cylindrical surface, the electric field is parallel to the surface, so the flux through the sides is zero.

EXAMPLE 22.8 Field between oppositely charged parallel conducting plates**WITH VARIATION PROBLEMS**

Two large plane parallel conducting plates are given charges of equal magnitude and opposite sign; the surface charge densities are $+\sigma$ and $-\sigma$. Find the electric field in the region between the plates.

IDENTIFY and SET UP Figure 22.21a shows the field. Because opposite charges attract, most of the charge accumulates at the opposing faces of the plates. A small amount of charge resides on the *outer* surfaces of the plates, and there is some spreading or “fringing” of the field at the edges. But if the plates are very large in comparison to the distance between them, the amount of charge on the outer surfaces is negligibly small, and the fringing can be ignored except near the edges. In this case we can assume that the field is uniform in the interior region between the plates, as in Fig. 22.21b, and that the charges are distributed uniformly over the opposing surfaces. To exploit this symmetry, we can use the shaded Gaussian surfaces S_1 , S_2 , S_3 , and S_4 . These surfaces are cylinders with flat ends of area A ; one end of each surface lies *within* a plate.

EXECUTE The left-hand end of surface S_1 is within the positive plate 1. Since the field is zero within the volume of any solid conductor under electrostatic conditions, there is no electric flux through this end. The electric field between the plates is perpendicular to the right-hand end, so on that end, $E_{\perp}$ is equal to E and the flux is EA ; this is positive, since $\vec{E}$ is directed out of the Gaussian surface. There is no flux through the side walls of the cylinder, since these walls are parallel to $\vec{E}$. So the total

flux integral in Gauss's law is EA . The net charge enclosed by the cylinder is σA , so Eq. (22.8) yields $EA = \sigma A/\epsilon_0$; we then have

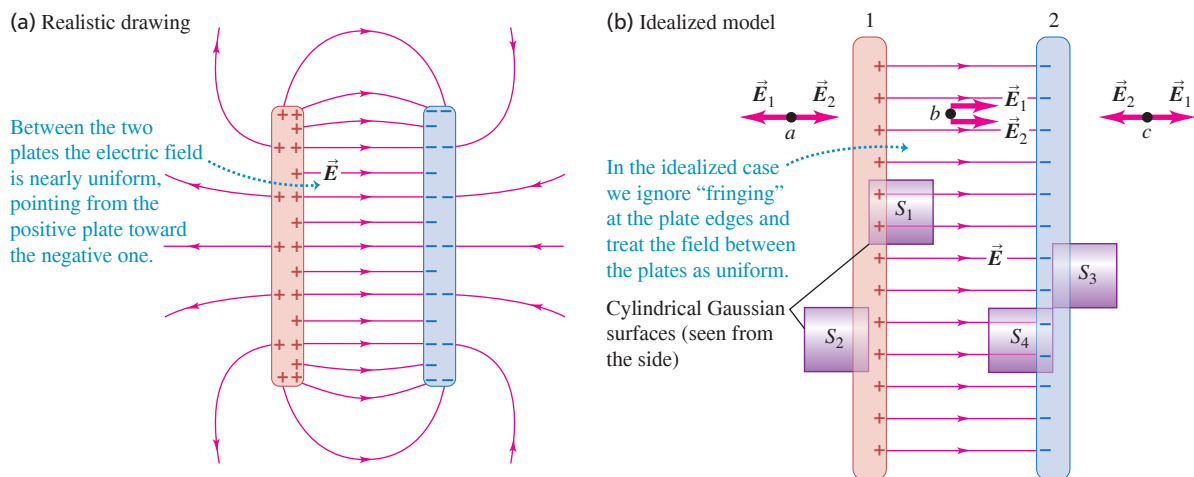
$$E = \frac{\sigma}{\epsilon_0} \quad (\text{field between oppositely charged conducting plates})$$

The field is uniform and perpendicular to the plates, and its magnitude is independent of the distance from either plate. The Gaussian surface S_4 yields the same result. Surfaces S_2 and S_3 yield $E = 0$ to the left of plate 1 and to the right of plate 2, respectively. We leave these calculations to you.

EVALUATE We obtained the same results in Example 21.12 by using the principle of superposition of electric fields. The fields due to the two sheets of charge (one on each plate) are $\vec{E}_1$ and $\vec{E}_2$; from Example 22.7, both of these have magnitude $\sigma/2\epsilon_0$. The total electric field at any point is the vector sum $\vec{E} = \vec{E}_1 + \vec{E}_2$. At points a and c in Fig. 22.21b, $\vec{E}_1$ and $\vec{E}_2$ point in opposite directions, and their sum is zero. At point b , $\vec{E}_1$ and $\vec{E}_2$ are in the same direction; their sum has magnitude $E = \sigma/\epsilon_0$, just as we found by using Gauss's law.

KEYCONCEPT The electric field between two large, parallel, oppositely charged conducting plates is uniform in the region between the plates. The field is zero in the region outside the plates as well as in the interiors of the plates. The charge on either plate lies on the surface that faces the other plate.

Figure 22.21 Electric field between oppositely charged parallel plates.

**EXAMPLE 22.9** Field of a uniformly charged sphere**WITH VARIATION PROBLEMS**

Positive electric charge Q is distributed uniformly *throughout the volume* of an *insulating* sphere with radius R . Find the magnitude of the electric field at a point P a distance r from the center of the sphere.

IDENTIFY and SET UP As in Example 22.5, the system is spherically symmetric. Hence we can use the conclusions of that example about the direction and magnitude of $\vec{E}$. To make use of the spherical symmetry, we choose as our Gaussian surface a sphere with radius r , concentric with the charge distribution.

EXECUTE From symmetry, the direction of $\vec{E}$ is radial at every point on the Gaussian surface, so $E_{\perp} = E$ and the field magnitude E is the same at every point on the surface. Hence the total electric flux through the Gaussian surface is the product of E and the total area of the surface $A = 4\pi r^2$ —that is, $\Phi_E = 4\pi r^2 E$.

The amount of charge enclosed within the Gaussian surface depends on r . To find E *inside* the sphere, we choose $r < R$. The volume

charge density ρ is the charge Q divided by the volume of the entire charged sphere of radius R :

$$\rho = \frac{Q}{4\pi R^3/3}$$

The volume V_{encl} enclosed by the Gaussian surface is $\frac{4}{3}\pi r^3$, so the total charge Q_{encl} enclosed by that surface is

$$Q_{\text{encl}} = \rho V_{\text{encl}} = \left(\frac{Q}{4\pi R^3/3} \right) \left(\frac{4}{3}\pi r^3 \right) = Q \frac{r^3}{R^3}$$

Then Gauss's law, Eq. (22.8), becomes

$$4\pi r^2 E = \frac{Q}{\epsilon_0} \frac{r^3}{R^3} \quad \text{and} \quad E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3} \quad (\text{field inside a uniformly charged sphere})$$

The field magnitude is proportional to the distance r of the field point from the center of the sphere (see the graph of E versus r in Fig. 22.22).

To find E outside the sphere, we take $r > R$. This surface encloses the entire charged sphere, so $Q_{\text{encl}} = Q$, and Gauss's law gives

$$4\pi r^2 E = \frac{Q}{\epsilon_0} \quad \text{and}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad (\text{field outside a uniformly charged sphere})$$

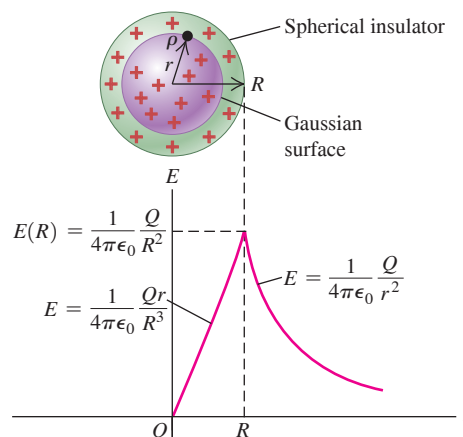
The field outside *any* spherically symmetric charged body varies as $1/r^2$, as though the entire charge were concentrated at the center. This is graphed in Fig. 22.22.

If the charge is *negative*, $\vec{E}$ is radially *inward* and in the expressions for E we interpret Q as the absolute value of the charge.

EVALUATE Notice that if we set $r = R$ in either expression for E , we get the same result $E = Q/4\pi\epsilon_0 R^2$ for the magnitude of the field at the surface of the sphere. This is because the magnitude E is a *continuous* function of r . By contrast, for the charged conducting sphere of Example 22.5 the electric-field magnitude is *discontinuous* at $r = R$ (it jumps from $E = 0$ just inside the sphere to $E = Q/4\pi\epsilon_0 R^2$ just outside the sphere). In general, the electric field $\vec{E}$ is discontinuous in magnitude, direction, or both wherever there is a *sheet* of charge, such as at the surface of a charged conducting sphere (Example 22.5), at the surface of an infinite charged sheet (Example 22.7), or at the surface of a charged conducting plate (Example 22.8).

The approach used here can be applied to *any* spherically symmetric distribution of charge, even if it is not radially uniform, as it was here.

Figure 22.22 The magnitude of the electric field of a uniformly charged insulating sphere. Compare this with the field for a conducting sphere (see Fig. 22.18).



Such charge distributions occur within many atoms and atomic nuclei, so Gauss's law is useful in atomic and nuclear physics.

KEYCONCEPT For any spherically symmetric distribution of charge, the field at a distance r from the center of the distribution is the same as if all the charge interior to r were concentrated in a point at the center. The charge at distances greater than r from the center has no effect on the field at r .

EXAMPLE 22.10 Charge on a hollow sphere

WITH VARIATION PROBLEMS

A thin-walled, hollow sphere of radius 0.250 m has an unknown charge distributed uniformly over its surface. At a distance of 0.300 m from the center of the sphere, the electric field points radially inward and has magnitude 1.80×10^2 N/C. How much charge is on the sphere?

IDENTIFY and SET UP The charge distribution is spherically symmetric. As in Examples 22.5 and 22.9, it follows that the electric field is radial everywhere and its magnitude is a function of only the radial distance r from the center of the sphere. We use a spherical Gaussian surface that is concentric with the charge distribution and has radius $r = 0.300$ m. Our target variable is $Q_{\text{encl}} = q$.

EXECUTE The charge distribution is the same as if the charge were on the surface of a 0.250-m-radius conducting sphere. Hence we can borrow the results of Example 22.5. We note that the electric field here is directed toward the sphere, so that q must be *negative*. Furthermore, the electric field is directed into the Gaussian surface, so that $E_{\perp} = -E$ and $\Phi_E = \oint E_{\perp} dA = -E(4\pi r^2)$.

By Gauss's law, the flux is equal to the charge q on the sphere (all of which is enclosed by the Gaussian surface) divided by ϵ_0 . Solving for q , we find

$$q = -E(4\pi\epsilon_0 r^2) = -(1.80 \times 10^2 \text{ N/C})(4\pi) \\ \times (8.854 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.300 \text{ m})^2 \\ = -1.80 \times 10^{-9} \text{ C} = -1.80 \text{ nC}$$

EVALUATE To determine the charge, we had to know the electric field at *all* points on the Gaussian surface so that we could calculate the flux integral. This was possible here because the charge distribution is highly symmetric. If the charge distribution is irregular or lacks symmetry, Gauss's law is not very useful for calculating the charge distribution from the field, or vice versa.

KEYCONCEPT The electric field outside a spherical distribution of charge is the same as if all of the charge in the distribution were concentrated at its center.

TEST YOUR UNDERSTANDING OF SECTION 22.4 You place a known amount of charge Q on the irregularly shaped conductor shown in Fig. 22.17. If you know the size and shape of the conductor, can you use Gauss's law to calculate the electric field at an arbitrary position outside the conductor?

ANSWER

no You might be tempted to draw a Gaussian surface that is an enlarged version of the conductor, with the same shape and placed so that it completely encloses the conductor. While you know the flux through this Gaussian surface (by Gauss's law, it's $\Phi_E = Q/\epsilon_0$), the direction of the electric field need not be perpendicular to the surface and the magnitude of the field need not be the same at all points on the surface. It's not possible to do the flux integral $\oint \vec{E} \cdot d\vec{A}$, and we can't calculate the electric field. Gauss's law is useful for calculating the electric field only when the charge distribution is highly symmetric.

BIO APPLICATION Charge**Distribution Inside a Nerve Cell**

The interior of a human nerve cell contains both positive potassium ions (K^+) and negatively charged protein molecules (Pr^-). Potassium ions can flow out of the cell through the cell membrane, but the much larger protein molecules cannot. The result is that the interior of the cell has a net negative charge. (The fluid outside the cell has a positive charge that balances this.) The fluid within the cell is a good conductor, so the Pr^- molecules distribute themselves on the outer surface of the fluid—that is, on the inner surface of the cell membrane, which is an insulator. This is true no matter what the shape of the cell.

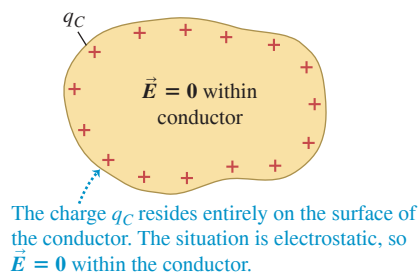
**22.5 CHARGES ON CONDUCTORS**

We have learned that in an electrostatic situation (in which there is no net motion of charge) the electric field at every point within a conductor is zero and any excess charge on a solid conductor is located entirely on its surface (**Fig. 22.23a**). But what if there is a *cavity* inside the conductor (**Fig. 22.23b**)? If there is no charge within the cavity, we can use a Gaussian surface such as *A* (which lies completely within the material of the conductor) to show that the *net* charge on the *surface of the cavity* must be zero, because $\vec{E} = \mathbf{0}$ everywhere on the Gaussian surface. In fact, we can prove in this situation that there can't be any charge *anywhere* on the cavity surface. We'll postpone detailed proof of this statement until Chapter 23.

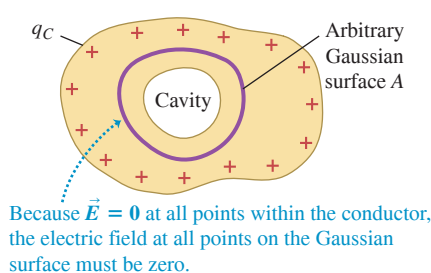
Suppose we place a small body with a charge q inside a cavity within a conductor (**Fig. 22.23c**). The conductor is uncharged and is insulated from the charge q . Again $\vec{E} = \mathbf{0}$ everywhere on surface *A*, so according to Gauss's law the *total* charge inside this surface must be zero. Therefore there must be a charge $-q$ distributed on the surface of the cavity, drawn there by the charge q inside the cavity. The *total* charge on the conductor must remain zero, so a charge $+q$ must appear either on its outer surface or inside the material. But we showed that in an electrostatic situation there can't be any excess charge within the material of a conductor. So we conclude that charge $+q$ must appear on the outer surface. By the same reasoning, if the conductor originally had a charge q_C , then the total charge on the outer surface must be $q_C + q$ after charge q is inserted into the cavity.

Figure 22.23 Finding the electric field within a charged conductor.

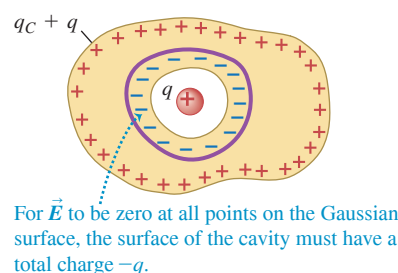
(a) Solid conductor with charge q_C



(b) The same conductor with an internal cavity



(c) An isolated charge q placed in the cavity

**CONCEPTUAL EXAMPLE 22.11 A conductor with a cavity****WITH VARIATION PROBLEMS**

A conductor with a cavity carries a total charge of $+7 \text{ nC}$. Within the cavity, insulated from the conductor, is a point charge of -5 nC . How much charge is on each surface (inner and outer) of the conductor?

SOLUTION **Figure 22.24** shows the situation. If the charge in the cavity is $q = -5 \text{ nC}$, the charge on the inner cavity surface must be $-q = -(-5 \text{ nC}) = +5 \text{ nC}$. The conductor carries a *total* charge of $+7 \text{ nC}$, none of which is in the interior of the material. If $+5 \text{ nC}$ is on the inner surface of the cavity, then there must be $(+7 \text{ nC}) - (+5 \text{ nC}) = +2 \text{ nC}$ on the outer surface of the conductor.

KEYCONCEPT If there is excess charge at rest on a conductor, all of that charge must lie on the surface of the conductor and the electric field inside the solid conductor must be zero. If there is a cavity inside the conductor, the net charge on the cavity walls equals the amount of charge enclosed by the cavity.

Figure 22.24 Our sketch for this problem. There is zero electric field inside the bulk conductor and hence zero flux through the Gaussian surface shown, so the charge on the cavity wall must be the opposite of the point charge.

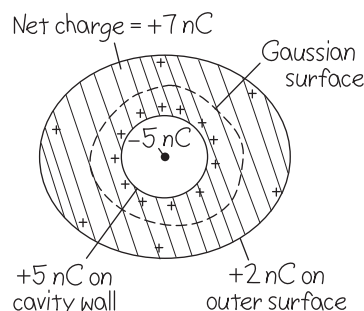
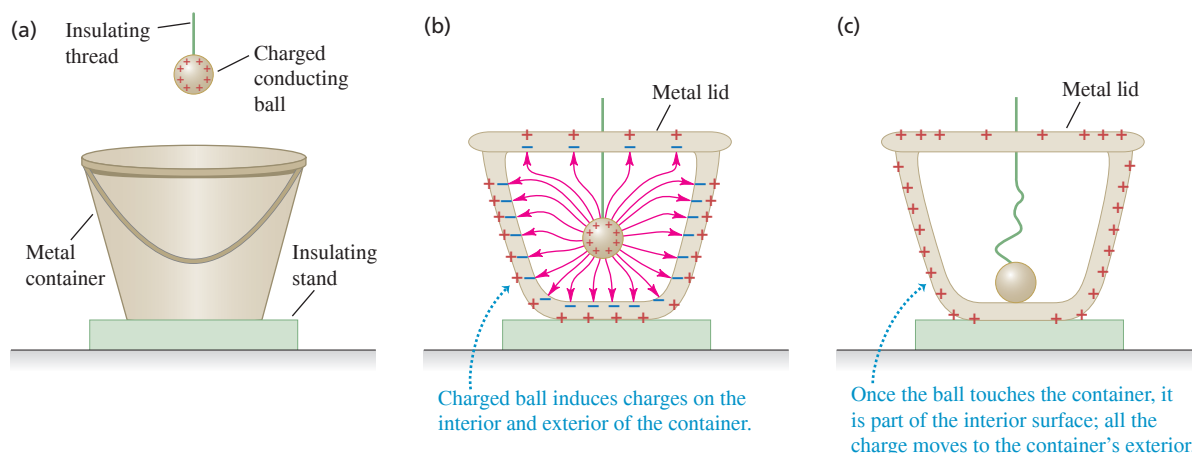


Figure 22.25 (a) A charged conducting ball suspended by an insulating thread outside a conducting container on an insulating stand. (b) The ball is lowered into the container, and the lid is put on. (c) The ball is touched to the inner surface of the container.



Testing Gauss's Law Experimentally

Figure 22.25 shows a historic experiment designed to test Gauss's law. We mount a conducting container on an insulating stand. The container is initially uncharged. Then we hang a charged metal ball from an insulating thread (Fig. 22.25a), lower it into the container, and put the lid on (Fig. 22.25b). Charges are induced on the walls of the container, as shown. But now we let the ball *touch* the inner wall (Fig. 22.25c). The surface of the ball becomes part of the cavity surface. The situation is now the same as Fig. 22.23b; if Gauss's law is correct, the net charge on the cavity surface must be zero. Thus the ball must lose all its charge. Finally, we pull the ball out; we find that it has indeed lost all its charge.

This experiment was performed in the 19th century by the English scientist Michael Faraday, using a metal icepail with a lid, and it is called **Faraday's icepail experiment**. The result confirms the validity of Gauss's law and therefore of Coulomb's law. Faraday's result was significant because Coulomb's experimental method, using a torsion balance and dividing of charges, was not very precise; it is very difficult to confirm the $1/r^2$ dependence of the electrostatic force by direct force measurements. By contrast, experiments like Faraday's test the validity of Gauss's law, and therefore of Coulomb's law, with much greater precision. Modern versions of this experiment have shown that the exponent 2 in the $1/r^2$ of Coulomb's law does not differ from precisely 2 by more than 10^{-16} . So there is no reason to believe it is anything other than exactly 2.

The same principle behind Faraday's icepail experiment is used in a *Van de Graaff electrostatic generator* (Fig. 22.26). A charged belt continuously produces a buildup of charge on the inside of a conducting shell. By Gauss's law, there can never be any charge on the inner surface of this shell, so the charge is immediately carried away to the outside surface of the shell. As a result, the charge on the shell and the electric field around it can become very large very rapidly. The Van de Graaff generator is used as an accelerator of charged particles and for physics demonstrations.

This principle also forms the basis for *electrostatic shielding*. Suppose we have a ? very sensitive electronic instrument that we want to protect from stray electric fields that might cause erroneous measurements. We surround the instrument with a conducting box, or we line the walls, floor, and ceiling of the room with a conducting material such as sheet copper. The external electric field redistributes the free electrons in the conductor, leaving a net positive charge on the outer surface in some regions and a net negative charge in others (Fig. 22.27, next page). This charge distribution causes an additional electric field such that the *total* field at every point inside the box is zero, as Gauss's law says it must be. The charge distribution on the box also alters the shapes of the field lines near the box, as the figure shows. Such a setup is often called a *Faraday cage*. The same

Figure 22.26 Cutaway view of the essential parts of a Van de Graaff electrostatic generator. The electron sink at the bottom draws electrons from the belt, giving it a positive charge; at the top the belt attracts electrons away from the conducting shell, giving the shell a positive charge.

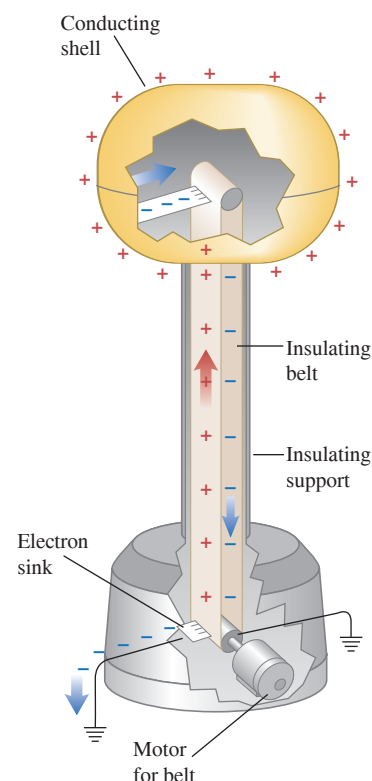


Figure 22.27 (a) A conducting box (a Faraday cage) immersed in a uniform electric field. The field of the induced charges on the box combines with the uniform field to give zero total field inside the box. (b) This person is inside a Faraday cage, and so is protected from the powerful electric discharge.

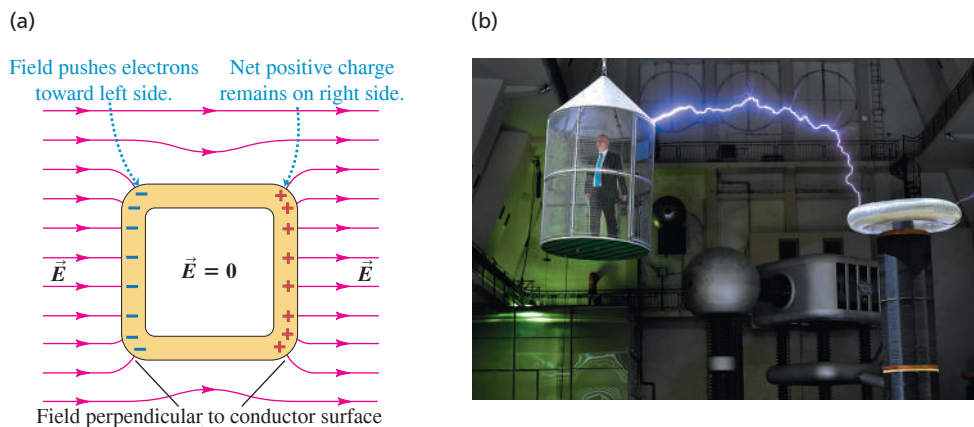
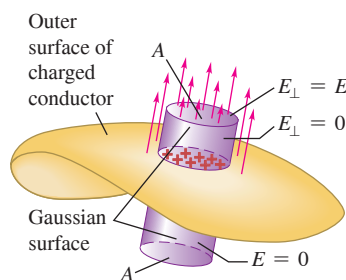


Figure 22.28 The field just outside a charged conductor is perpendicular to the surface, and its perpendicular component $E_{\perp}$ is equal to σ/ϵ_0 .



APPLICATION Why Lightning Bolts Are Vertical

Our planet is a good conductor, and its surface has a negative charge. Hence, the electric field in the atmosphere above the surface points generally downward, toward the negative charge and perpendicular to the surface (see Example 22.12). The negative charge is balanced by positive charges in the atmosphere. In a lightning storm, the vertical electric field becomes great enough to cause charges to flow vertically through the air. The air is excited and ionized by the passage of charge through it, producing a visible lightning bolt.



physics tells you that one of the safest places to be in a lightning storm is inside a car; if the car is struck by lightning, the charge tends to remain on the metal skin of the vehicle, and little or no electric field is produced inside the passenger compartment. Keep in mind this basic principle, which is a consequence of Gauss's law: *Charges outside a conductor have no effect on the interior of that conductor, even if the conductor has a cavity inside.*

Field at the Surface of a Conductor

Finally, we note that there is a direct relationship between the $\vec{E}$ field at a point just outside any conductor and the surface charge density σ at that point. In general, σ varies from point to point on the surface. We'll show in Chapter 23 that at any such point, the direction of $\vec{E}$ is always *perpendicular* to the surface. (You can see this effect in Fig. 22.27a.)

To find a relationship between σ at any point on the surface and the perpendicular component of the electric field at that point, we construct a Gaussian surface in the form of a small cylinder (Fig. 22.28). One end face, with area A , lies within the conductor and the other lies just outside. The electric field is zero at all points within the conductor. Outside the conductor the component of $\vec{E}$ perpendicular to the side walls of the cylinder is zero, and over the end face the perpendicular component is equal to $E_{\perp}$. (If σ is positive, the electric field points out of the conductor and $E_{\perp}$ is positive; if σ is negative, the field points inward and $E_{\perp}$ is negative.) Hence the total flux through the surface is $E_{\perp}A$. The charge enclosed within the Gaussian surface is σA , so from Gauss's law, $E_{\perp}A = (\sigma A)/\epsilon_0$ and

$$\text{Electric field at surface of a conductor; } \vec{E} \text{ perpendicular to surface } E_{\perp} = \frac{\sigma}{\epsilon_0} \quad (22.10)$$

Surface charge density
Electric constant

This agrees with our result for the field at the surface of a charged conducting plate (Example 22.8). We can also verify this for a charged conducting sphere. In Example 22.5 (Section 22.4) we showed that the electric field just outside the surface of a conducting sphere with radius R and total charge q is

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

The surface charge density is uniform and equal to q divided by the surface area of the sphere:

$$\sigma = \frac{q}{4\pi R^2}$$

Comparing these two expressions, we see that $E = \sigma/\epsilon_0$, which verifies Eq. (22.10).

We showed in Example 22.8 that the field magnitude between two infinite flat oppositely charged conducting plates also equals σ/ϵ_0 . In this case the field magnitude E is the same at *all* distances from the plates, but in all other cases E decreases with increasing distance from the surface.

EXAMPLE 22.12 Electric field of the earth

WITH  **ARIATION PROBLEMS**

The earth (a conductor) has a net electric charge. The resulting electric field near the surface has an average value of about 150 N/C, directed toward the center of the earth. (a) What is the corresponding surface charge density? (b) What is the *total* surface charge of the earth?

IDENTIFY and SET UP We are given the electric-field magnitude at the surface of the conducting earth. We can calculate the surface charge density σ from Eq. (22.10). The total charge Q on the earth's surface is then the product of σ and the earth's surface area.

EXECUTE (a) The direction of the field means that σ is negative (corresponding to $\vec{E}$ being directed *into* the surface, so $E_\perp$ is negative). From Eq. (22.10),

$$\begin{aligned}\sigma &= \epsilon_0 E_\perp = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(-150 \text{ N/C}) \\ &= -1.33 \times 10^{-9} \text{ C/m}^2 = -1.33 \text{ nC/m}^2\end{aligned}$$

(b) The earth's surface area is $4\pi R_E^2$, where $R_E = 6.38 \times 10^6 \text{ m}$ is the radius of the earth (see Appendix F). The total charge Q is the product $4\pi R_E^2 \sigma$, or

$$\begin{aligned}Q &= 4\pi(6.38 \times 10^6 \text{ m})^2(-1.33 \times 10^{-9} \text{ C/m}^2) \\ &= -6.8 \times 10^5 \text{ C} = -680 \text{ kC}\end{aligned}$$

EVALUATE You can check our result in part (b) by using the result of Example 22.5. Solving for Q , we find

$$\begin{aligned}Q &= 4\pi\epsilon_0 R^2 E_\perp \\ &= \frac{1}{9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2} (6.38 \times 10^6 \text{ m})^2 (-150 \text{ N/C}) \\ &= -6.8 \times 10^5 \text{ C}\end{aligned}$$

One electron has a charge of $-1.60 \times 10^{-19} \text{ C}$. Hence this much excess negative electric charge corresponds to there being $(-6.8 \times 10^5 \text{ C})/(-1.60 \times 10^{-19} \text{ C}) = 4.2 \times 10^{24}$ excess electrons on the earth, or about 7 moles of excess electrons. This is compensated by an equal *deficiency* of electrons in the earth's upper atmosphere, so the combination of the earth and its atmosphere is electrically neutral.

KEYCONCEPT The electric field $\vec{E}$ at the surface of a charged conductor is normal to the surface. The normal component of $\vec{E}$ at any point on the surface is equal to σ/ϵ_0 , where σ is the surface charge density (charge per unit area) at that point. The field points out of the surface if σ is positive and into the surface if σ is negative.

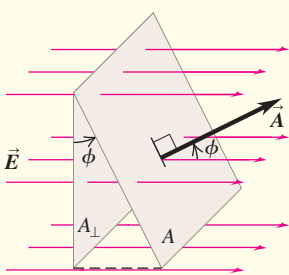
TEST YOUR UNDERSTANDING OF SECTION 22.5 A hollow conducting sphere has no net charge. There is a positive point charge q at the center of the spherical cavity within the sphere. You connect a conducting wire from the outside of the sphere to ground. Will you measure an electric field outside the sphere?

ANSWER **no** Before you connect the wire to the sphere, the presence of the point charge will induce a charge $-q$ on the inner surface of the hollow sphere and a charge q on the outer surface (the net charge on the sphere is zero). There will be an electric field outside the sphere due to the charge on the outer surface. Once you touch the conducting wire to the sphere, however, electrons will flow from ground to the outer surface of the sphere to neutralize the charge there (see Fig. 21.7c). As a result the sphere will have no charge on its outer surface and no electric field outside.

CHAPTER 22
 SUMMARY

Electric flux: Electric flux is a measure of the “flow” of electric field through a surface. It is equal to the product of an area element and the perpendicular component of $\vec{E}$, integrated over a surface. (See Examples 22.1–22.3.)

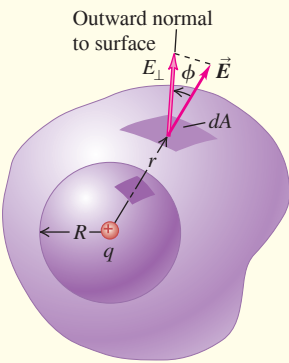
$$\begin{aligned}\Phi_E &= \int E \cos \phi \, dA \\ &= \int E_{\perp} \, dA = \int \vec{E} \cdot d\vec{A}\end{aligned}\tag{22.5}$$



Gauss’s law: Gauss’s law states that the total electric flux through a closed surface, which can be written as the surface integral of the component of $\vec{E}$ normal to the surface, equals a constant times the total charge Q_{encl} enclosed by the surface. Gauss’s law is logically equivalent to Coulomb’s law, but its use greatly simplifies problems with a high degree of symmetry. (See Examples 22.4–22.10.)

When excess charge is placed on a solid conductor and is at rest, it resides entirely on the surface, and $\vec{E} = \mathbf{0}$ everywhere in the material of the conductor. (See Examples 22.11 and 22.12.)

$$\begin{aligned}\Phi_E &= \oint E \cos \phi \, dA \\ &= \oint E_{\perp} \, dA = \oint \vec{E} \cdot d\vec{A} \\ &= \frac{Q_{\text{encl}}}{\epsilon_0}\end{aligned}\tag{22.8), (22.9)}$$



Electric field of various symmetric charge distributions: The following table lists electric fields caused by several symmetric charge distributions. In the table, q , Q , λ , and σ refer to the *magnitudes* of the quantities.

Charge Distribution	Point in Electric Field	Electric-Field Magnitude
Single point charge q	Distance r from q	$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$
Charge q on surface of conducting sphere with radius R	Outside sphere, $r > R$	$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$
	Inside sphere, $r < R$	$E = 0$
Infinite wire, charge per unit length λ	Distance r from wire	$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$
Infinite conducting cylinder with radius R , charge per unit length λ	Outside cylinder, $r > R$	$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$
	Inside cylinder, $r < R$	$E = 0$
Solid insulating sphere with radius R , charge Q distributed uniformly throughout volume	Outside sphere, $r > R$	$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$
	Inside sphere, $r < R$	$E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3}$
Infinite sheet of charge with uniform charge per unit area σ	Any point	$E = \frac{\sigma}{2\epsilon_0}$
Two oppositely charged conducting plates with surface charge densities $+\sigma$ and $-\sigma$	Any point between plates	$E = \frac{\sigma}{\epsilon_0}$
Charged conductor	Just outside the conductor	$E = \frac{\sigma}{\epsilon_0}$



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 22.2 and 22.3** (Section 22.2) and **EXAMPLE 22.4** (Section 22.3) before attempting these problems.

VP22.4.1 An imaginary cubical surface of side L has its edges parallel to the x -, y - and z -axes, one corner at the point $x = 0, y = 0, z = 0$ and the opposite corner at the point $x = L, y = L, z = L$. The cube is in a region of uniform electric field $\vec{E} = E_1\hat{i} + E_2\hat{j}$, where E_1 and E_2 are positive constants. Calculate the electric flux through (a) the cube face in the plane $x = 0$, (b) the cube face in the plane $x = L$, (c) the cube face in the plane $y = 0$, (d) the cube face in the plane $y = L$, (e) the cube face in the plane $z = 0$, (f) the cube face in the plane $z = L$, and (g) the entire cubical surface. For each face the normal points out of the cube.

VP22.4.2 A point charge q is at the point $x = 0, y = 0, z = 0$. An imaginary hemispherical surface is made by starting with a spherical surface of radius R centered on the point $x = 0, y = 0, z = 0$ and cutting off the half in the region $z < 0$. The normal to this surface points out of the hemisphere, away from its center. Calculate the electric flux through the hemisphere if (a) $q = 8.00$ nC and $R = 0.100$ m; (b) $q = -4.00$ nC and $R = 0.100$ m; (c) $q = -4.00$ nC and $R = 0.200$ m.

VP22.4.3 An imaginary cubical surface with sides of length 5.00 cm has a point charge $q = +6.00$ nC at its center. Calculate the electric flux (a) through the entire closed cubical surface and (b) through one of the six faces of the cube.

VP22.4.4 An imaginary spherical surface of radius 5.00 cm is centered on the point $x = 0, y = 0, z = 0$. Calculate the net electric flux through the surface if the following charges are present: (a) a point charge $q_1 = +3.00$ nC at $x = 1.00$ cm, $y = 1.00$ cm, $z = 0$; (b) the charge in part (a) plus a point charge $q_2 = -8.00$ nC at $x = 2.00$ cm, $y = 0, z = -4.00$ cm; (c) the charges in part (b) plus a point charge $q_3 = +2.00$ nC at $x = 4.00$ cm, $y = -2.00$ cm, $z = 3.00$ cm.

Be sure to review **EXAMPLES 22.5, 22.6, 22.7, 22.8, 22.9, and 22.10** (Section 22.4) before attempting these problems.

VP22.10.1 A sphere of radius 5.00 cm carries charge $+3.00$ nC. Calculate the electric-field magnitude at a distance 4.00 cm from the center of the sphere and at a distance 6.00 cm from the center of the sphere if the sphere is (a) a solid insulator with the charge spread uniformly throughout its volume and (b) a solid conductor.

VP22.10.2 You place a point charge $q = -4.00$ nC a distance of 9.00 cm from an infinitely long, thin wire that has linear charge density 3.00×10^{-9} C/m. What is the magnitude of the electric force that the wire exerts on the point charge? Is the force attractive or repulsive?

VP22.10.3 A point charge $q = +1.80$ nC is 8.00 cm from a thin, flat, infinite sheet that has a uniform surface charge density. If the total electric field is zero at a point halfway between the point charge and the sheet, what is the surface charge density of the sheet? Is this positive or negative?

VP22.10.4 Two large parallel conducting plates carry charges of equal magnitude and opposite charge. When you place a point charge $q = +3.60$ nC between the plates, the force on the point charge is 22.0 μ N. What is the magnitude of the surface charge density on either plate?

Be sure to review **EXAMPLES 22.11 and 22.12** (Section 22.5) before attempting these problems.

VP22.12.1 A solid block of copper, which is a good conductor, has a cavity in its interior. Within the cavity, insulated from the conductor, is a point charge of $+3.00$ nC. The block of copper itself has an excess charge of -8.00 nC. How much charge is on (a) the outer surface of the block and (b) the surface of the cavity?

VP22.12.2 A solid sphere of silver, which is a good conductor, has a spherical cavity at its center. There is a point charge at the center of the cavity. The silver sphere has a charge of $+9.00$ nC on its outer surface and a charge of -2.00 nC on the surface of the cavity. (a) What is the value of the point charge? (b) If the point charge moved to a different position within the cavity (not at the center), would this affect the total charge on the surface of the cavity or the total charge on the outer surface of the sphere?

VP22.12.3 In an electricity demonstration at the Deutsches Museum in Munich, Germany, a person sits inside a metal sphere of radius 0.90 m. Charge is applied to the sphere until the magnitude of the electric field on the outside surface of the sphere is 3.0×10^5 N/C. When the demonstration is over, the person climbs out unharmed. During the demonstration, what are the magnitudes of (a) the electric field inside the sphere, (b) the total charge on the sphere, and (c) the surface charge density on the outside of the sphere?

VP22.12.4 A spherical conductor of radius 0.330 m has a spherical cavity of radius 0.120 m at its center. The conductor carries a total charge of -6.00 nC; in addition, at the center of the spherical cavity is a point charge of $+4.00$ nC. Find (a) the total charge on the surface of the cavity, (b) the total charge on the outer surface of the conductor, (c) the magnitude of the electric field just inside the surface of the cavity, and (d) the magnitude of the electric field just outside the outer surface of the conductor.

BRIDGING PROBLEM Electric Field Inside a Hydrogen Atom

A hydrogen atom is made up of a proton of charge $+Q = 1.60 \times 10^{-19}$ C and an electron of charge $-Q = -1.60 \times 10^{-19}$ C. The proton may be regarded as a point charge at $r = 0$, the center of the atom. The motion of the electron causes its charge to be “smeared out” into a spherical distribution around the proton (**Fig. 22.29**), so that the electron is equivalent to a charge per unit volume of $\rho(r) = -(Q/\pi a_0^3)e^{-2r/a_0}$, where $a_0 = 5.29 \times 10^{-11}$ m is called the *Bohr radius*. (a) Find the total amount of the hydrogen atom’s charge that is enclosed within a sphere with radius r centered on the proton. (b) Find the electric field (magnitude and direction) caused by the charge of the hydrogen atom as a function of r .

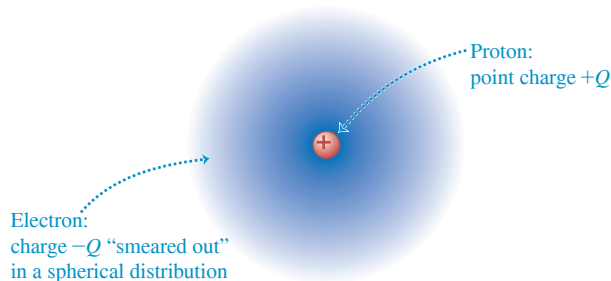
(c) Make a graph as a function of r of the ratio of the electric-field magnitude E to the magnitude of the field due to the proton alone.

SOLUTION GUIDE

IDENTIFY and SET UP

1. The charge distribution in this problem is spherically symmetric, as in Example 22.9, so you can solve it with Gauss’s law.
2. The charge within a sphere of radius r includes the proton charge $+Q$ plus the portion of the electron charge distribution

Figure 22.29 The charge distribution in a hydrogen atom.



that lies within the sphere. The difference from Example 22.9 is that the electron charge distribution is *not* uniform, so the charge enclosed within a sphere of radius r is *not* simply the charge density multiplied by the volume $4\pi r^3/3$ of the sphere. Instead, you'll have to do an integral.

3. Consider a thin spherical shell centered on the proton, with radius r' and infinitesimal thickness dr' . Since the shell is so thin, every point within the shell is at essentially the same radius from the proton. Hence the amount of electron charge within this shell is equal to the electron charge density $\rho(r')$ at this

radius multiplied by the volume dV of the shell. What is dV in terms of r' ?

4. The total electron charge within a radius r equals the integral of $\rho(r')dV$ from $r' = 0$ to $r' = r$. Set up this integral (but don't solve it yet), and use it to write an expression for the total charge (including the proton) within a sphere of radius r .

EXECUTE

5. Integrate your expression from step 4 to find the charge within radius r . (*Hint:* Integrate by substitution: Change the integration variable from r' to $x = 2r'/a_0$. You can use integration by parts to calculate the integral $\int x^2 e^{-x} dx$, or you can look it up in a table of integrals or on the Web.)
6. Use Gauss's law and your results from step 5 to find the electric field at a distance r from the proton.
7. Find the ratio referred to in part (c) and graph it versus r . (You'll actually find it simplest to graph this function versus the quantity r/a_0 .)

EVALUATE

8. How do your results for the enclosed charge and the electric-field magnitude behave in the limit $r \rightarrow 0$? In the limit $r \rightarrow \infty$? Explain your results.

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q22.1 A rubber balloon has a single point charge in its interior. Does the electric flux through the balloon depend on whether or not it is fully inflated? Explain your reasoning.

Q22.2 Suppose that in Fig. 22.15 both charges were positive. What would be the fluxes through each of the four surfaces in the example?

Q22.3 In Fig. 22.15, suppose a third point charge were placed outside the purple Gaussian surface C . Would this affect the electric flux through any of the surfaces A , B , C , or D in the figure? Why or why not?

Q22.4 A certain region of space bounded by an imaginary closed surface contains no charge. Is the electric field always zero everywhere on the surface? If not, under what circumstances is it zero on the surface?

Q22.5 A spherical Gaussian surface encloses a point charge q . If the point charge is moved from the center of the sphere to a point away from the center, does the electric field at a point on the surface change? Does the total flux through the Gaussian surface change? Explain.

Q22.6 You find a sealed box on your doorstep. You suspect that the box contains several charged metal spheres packed in insulating material. How can you determine the total net charge inside the box without opening the box? Or isn't this possible?

Q22.7 A solid copper sphere has a net positive charge. The charge is distributed uniformly over the surface of the sphere, and the electric field inside the sphere is zero. Then a negative point charge outside the sphere is brought close to the surface of the sphere. Is all the net charge on the sphere still on its surface? If so, is this charge still distributed uniformly over the surface? If it is not uniform, how is it distributed? Is the electric field inside the sphere still zero? In each case justify your answers.

Q22.8 If the electric field of a point charge were proportional to $1/r^3$ instead of $1/r^2$, would Gauss's law still be valid? Explain your reasoning. (*Hint:* Consider a spherical Gaussian surface centered on a single point charge.)

Q22.9 In a conductor, one or more electrons from each atom are free to roam throughout the volume of the conductor. Does this contradict the statement that any excess charge on a solid conductor must reside on its surface? Why or why not?

Q22.10 You charge up the Van de Graaff generator shown in Fig. 22.26, and then bring an identical but uncharged hollow conducting sphere near it, without letting the two spheres touch. Sketch the distribution of charges on the second sphere. What is the net flux through the second sphere? What is the electric field inside the second sphere?

Q22.11 A lightning rod is a rounded copper rod mounted on top of a building and welded to a heavy copper cable running down into the ground. Lightning rods are used to protect houses and barns from lightning; the lightning current runs through the copper rather than through the building. Why? Why should the end of the rod be rounded?

Q22.12 A solid conductor has a cavity in its interior. Would the presence of a point charge inside the cavity affect the electric field outside the conductor? Why or why not? Would the presence of a point charge outside the conductor affect the electric field inside the cavity? Again, why or why not?

Q22.13 Explain this statement: “In a static situation, the electric field at the surface of a conductor can have no component parallel to the surface because this would violate the condition that the charges on the surface are at rest.” Would this statement be valid for the electric field at the surface of an *insulator*? Explain your answer and the reason for any differences between the cases of a conductor and an insulator.

Q22.14 In a certain region of space, the electric field $\vec{E}$ is uniform. (a) Use Gauss’s law to prove that this region of space must be electrically neutral; that is, the volume charge density ρ must be zero. (b) Is the converse true? That is, in a region of space where there is no charge, must $\vec{E}$ be uniform? Explain.

Q22.15 (a) In a certain region of space, the volume charge density ρ has a uniform positive value. Can $\vec{E}$ be uniform in this region? Explain. (b) Suppose that in this region of uniform positive ρ there is a “bubble” within which $\rho = 0$. Can $\vec{E}$ be uniform within this bubble? Explain.

Q22.16 A negative charge $-Q$ is placed inside the cavity of a hollow metal solid. The outside of the solid is grounded by connecting a conducting wire between it and the earth. Is any excess charge induced on the inner surface of the metal? Is there any excess charge on the outside surface of the metal? Why or why not? Would someone outside the solid measure an electric field due to the charge $-Q$? Is it reasonable to say that the grounded conductor has *shielded* the region outside the conductor from the effects of the charge $-Q$? In principle, could the same thing be done for gravity? Why or why not?

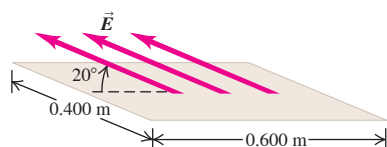
EXERCISES

Section 22.2 Calculating Electric Flux

22.1 • A flat sheet of paper of area 0.250 m^2 is oriented so that the normal to the sheet is at an angle of 60° to a uniform electric field of magnitude 14 N/C . (a) Find the magnitude of the electric flux through the sheet. (b) Does the answer to part (a) depend on the shape of the sheet? Why or why not? (c) For what angle ϕ between the normal to the sheet and the electric field is the magnitude of the flux through the sheet (i) largest and (ii) smallest? Explain your answers.

22.2 •• A flat sheet is in the shape of a rectangle with sides of lengths 0.400 m and 0.600 m . The sheet is immersed in a uniform electric field of magnitude 90.0 N/C that is directed at 20° from the plane of the sheet (Fig. E22.2). Find the magnitude of the electric flux through the sheet.

Figure E22.2



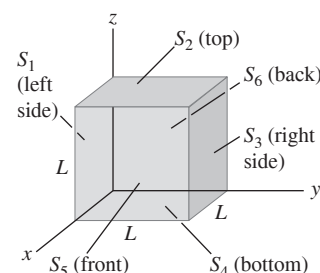
22.3 • You measure an electric field of $1.25 \times 10^6 \text{ N/C}$ at a distance of 0.150 m from a point charge. There is no other source of electric field in the region other than this point charge. (a) What is the electric flux through the surface of a sphere that has this charge at its center and that has radius 0.150 m ? (b) What is the magnitude of this charge?

22.4 • It was shown in Example 21.10 (Section 21.5) that the electric field due to an infinite line of charge is perpendicular to the line and has magnitude $E = \lambda/2\pi\epsilon_0 r$. Consider an imaginary cylinder with radius $r = 0.250 \text{ m}$ and length $l = 0.400 \text{ m}$ that has an infinite line of positive charge running along its axis. The charge per unit length on the line is $\lambda = 3.00 \mu\text{C/m}$. (a) What is the electric flux through the cylinder due to this infinite line of charge? (b) What is the flux through the cylinder if its radius is increased to $r = 0.500 \text{ m}$? (c) What is the flux through the cylinder if its length is increased to $l = 0.800 \text{ m}$?

22.5 • A uniform electric field makes an angle of 60.0° with a flat surface. The area of the surface is $6.66 \times 10^{-4} \text{ m}^2$. The resulting electric flux through the surface is $4.44 \text{ N} \cdot \text{m}^2/\text{C}$. Calculate the magnitude of the electric field.

22.6 • The cube in Fig. E22.6 has sides of length $L = 10.0 \text{ cm}$. The electric field is uniform, has magnitude $E = 4.00 \times 10^3 \text{ N/C}$, and is parallel to the xy -plane at an angle of 53.1° measured from the $+x$ -axis toward the $+y$ -axis. (a) What is the electric flux through each of the six cube faces S_1, S_2, S_3, S_4, S_5 , and S_6 ? (b) What is the total electric flux through all faces of the cube?

Figure E22.6

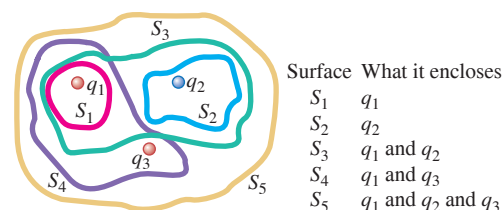


Section 22.3 Gauss’s Law

22.7 • A charge of 87.6 pC is uniformly distributed on the surface of a thin sheet of insulating material that has a total area of 29.2 cm^2 . A Gaussian surface encloses a portion of the sheet of charge. If the flux through the Gaussian surface is $5.00 \text{ N} \cdot \text{m}^2/\text{C}$, what area of the sheet is enclosed by the Gaussian surface?

22.8 • The three small spheres shown in Fig. E22.8 carry charges $q_1 = 4.00 \text{ nC}$, $q_2 = -7.80 \text{ nC}$, and $q_3 = 2.40 \text{ nC}$. Find the net electric flux through each of the following closed surfaces shown in cross section in the figure: (a) S_1 ; (b) S_2 ; (c) S_3 ; (d) S_4 ; (e) S_5 . (f) Do your answers to parts (a)–(e) depend on how the charge is distributed over each small sphere? Why or why not?

Figure E22.8



22.9 •• A charged paint is spread in a very thin uniform layer over the surface of a plastic sphere of diameter 12.0 cm , giving it a charge of $-49.0 \mu\text{C}$. Find the electric field (a) just inside the paint layer; (b) just outside the paint layer; (c) 5.00 cm outside the surface of the paint layer.

22.10 • A point charge $q_1 = 4.00 \text{ nC}$ is located on the x -axis at $x = 2.00 \text{ m}$, and a second point charge $q_2 = -6.00 \text{ nC}$ is on the y -axis at $y = 1.00 \text{ m}$. What is the total electric flux due to these two point charges through a spherical surface centered at the origin and with radius (a) 0.500 m , (b) 1.50 m , (c) 2.50 m ?

22.11 • BIO As discussed in Section 22.5, human nerve cells have a net negative charge and the material in the interior of the cell is a good conductor. If a cell has a net charge of -8.65 pC , what are the magnitude and direction (inward or outward) of the net flux through the cell boundary?

22.12 • Electric Fields in an Atom. The nuclei of large atoms, such as uranium, with 92 protons, can be modeled as spherically symmetric spheres of charge. The radius of the uranium nucleus is approximately $7.4 \times 10^{-15} \text{ m}$. (a) What is the electric field this nucleus produces just outside its surface? (b) What magnitude of electric field does it produce at the distance of the electrons, which is about $1.0 \times 10^{-10} \text{ m}$? (c) The electrons can be modeled as forming a uniform shell of negative charge. What net electric field do they produce at the location of the nucleus?

Section 22.4 Applications of Gauss's Law and

Section 22.5 Charges on Conductors

22.13 •• Two very long uniform lines of charge are parallel and are separated by 0.300 m . Each line of charge has charge per unit length $+5.20 \text{ } \mu\text{C}/\text{m}$. What magnitude of force does one line of charge exert on a 0.0500 m section of the other line of charge?

22.14 •• A solid metal sphere with radius 0.450 m carries a net charge of 0.250 nC . Find the magnitude of the electric field (a) at a point 0.100 m outside the surface of the sphere and (b) at a point inside the sphere, 0.100 m below the surface.

22.15 • (a) A conducting sphere has charge Q and radius R . If the electric field of the sphere at a distance $r = 2R$ from the center of the sphere is 1400 N/C , what is the electric field of the sphere at $r = 4R$? (b) A very long conducting cylinder of radius R has charge per unit length λ . Let r be the perpendicular distance from the axis of the cylinder. If the electric field of the cylinder at $r = 2R$ is 1400 N/C , what is the electric field at $r = 4R$? (c) A very large uniform sheet of charge has surface charge density σ . If the electric field of the sheet has a value of 1400 N/C at a perpendicular distance d from the sheet, what is the electric field of the sheet at a distance of $2d$ from the sheet?

22.16 • Some planetary scientists have suggested that the planet Mars has an electric field somewhat similar to that of the earth, producing a net electric flux of $-3.63 \times 10^{16} \text{ N} \cdot \text{m}^2/\text{C}$ at the planet's surface. Calculate: (a) the total electric charge on the planet; (b) the electric field at the planet's surface (refer to the astronomical data in Appendix F or the inside back cover); (c) the charge density on Mars, assuming all the charge is uniformly distributed over the planet's surface.

22.17 •• A very long uniform line of charge has charge per unit length $4.80 \text{ } \mu\text{C}/\text{m}$ and lies along the x -axis. A second long uniform line of charge has charge per unit length $-2.40 \text{ } \mu\text{C}/\text{m}$ and is parallel to the x -axis at $y = 0.400 \text{ m}$. What is the net electric field (magnitude and direction) at the following points on the y -axis: (a) $y = 0.200 \text{ m}$ and (b) $y = 0.600 \text{ m}$?

22.18 •• The electric field 0.400 m from a very long uniform line of charge is 840 N/C . How much charge is contained in a 2.00 cm section of the line?

22.19 •• A hollow, conducting sphere with an outer radius of 0.250 m and an inner radius of 0.200 m has a uniform surface charge density of $+6.37 \times 10^{-6} \text{ C}/\text{m}^2$. A charge of $-0.500 \text{ } \mu\text{C}$ is now introduced at the center of the cavity inside the sphere. (a) What is the new charge density on the outside of the sphere? (b) Calculate the strength of the electric field just outside the sphere. (c) What is the electric flux through a spherical surface just inside the inner surface of the sphere?

22.20 • A solid insulating sphere has total charge Q and radius R . The sphere's charge is distributed uniformly throughout its volume. Let r be the radial distance measured from the center of the sphere. If $E = 800 \text{ N/C}$ at $r = R/2$, what is E at $r = 2R$?

22.21 •• The electric field at a distance of 0.145 m from the surface of a solid insulating sphere with radius 0.355 m is 1750 N/C . (a) Assuming the sphere's charge is uniformly distributed, what is the charge density inside it? (b) Calculate the electric field inside the sphere at a distance of 0.200 m from the center.

22.22 •• A point charge of $-3.00 \text{ } \mu\text{C}$ is located in the center of a spherical cavity of radius 6.50 cm that, in turn, is at the center of an insulating charged solid sphere. The charge density in the solid is $\rho = 7.35 \times 10^{-4} \text{ C}/\text{m}^3$. Calculate the electric field inside the solid at a distance of 9.50 cm from the center of the cavity.

22.23 •• CP An electron is released from rest at a distance of 0.300 m from a large insulating sheet of charge that has uniform surface charge density $+2.90 \times 10^{-12} \text{ C}/\text{m}^2$. (a) How much work is done on the electron by the electric field of the sheet as the electron moves from its initial position to a point 0.050 m from the sheet? (b) What is the speed of the electron when it is 0.050 m from the sheet?

22.24 •• Charge Q is distributed uniformly throughout the volume of an insulating sphere of radius $R = 4.00 \text{ cm}$. At a distance of $r = 8.00 \text{ cm}$ from the center of the sphere, the electric field due to the charge distribution has magnitude $E = 940 \text{ N/C}$. What are (a) the volume charge density for the sphere and (b) the electric field at a distance of 2.00 cm from the sphere's center?

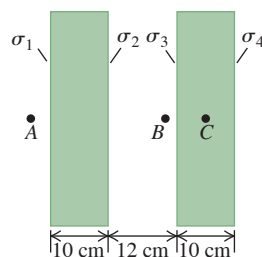
22.25 • A conductor with an inner cavity, like that shown in Fig. 22.23c, carries a total charge of $+5.00 \text{ nC}$. The charge within the cavity, insulated from the conductor, is -6.00 nC . How much charge is on (a) the inner surface of the conductor and (b) the outer surface of the conductor?

22.26 • A square insulating sheet 80.0 cm on a side is held horizontally. The sheet has 4.50 nC of charge spread uniformly over its area. (a) Estimate the electric field at a point 0.100 mm above the center of the sheet. (b) Estimate the electric field at a point 100 m above the center of the sheet. (c) Would the answers to parts (a) and (b) be different if the sheet were made of a conducting material? Why or why not?

22.27 • An infinitely long cylindrical conductor has radius R and uniform surface charge density σ . (a) In terms of σ and R , what is the charge per unit length λ for the cylinder? (b) In terms of σ , what is the magnitude of the electric field produced by the charged cylinder at a distance $r > R$ from its axis? (c) Express the result of part (b) in terms of λ and show that the electric field outside the cylinder is the same as if all the charge were on the axis. Compare your result to the result for a line of charge in Example 22.6 (Section 22.4).

22.28 • Two very large, nonconducting plastic sheets, each 10.0 cm thick, carry uniform charge densities $\sigma_1, \sigma_2, \sigma_3$, and σ_4 on their surfaces (Fig. E22.28). These surface charge densities have the values $\sigma_1 = -6.00 \text{ } \mu\text{C}/\text{m}^2$, $\sigma_2 = +5.00 \text{ } \mu\text{C}/\text{m}^2$, $\sigma_3 = +2.00 \text{ } \mu\text{C}/\text{m}^2$, and $\sigma_4 = +4.00 \text{ } \mu\text{C}/\text{m}^2$. Use Gauss's law to find the magnitude and direction of the electric field at the following points, far from the edges of these sheets: (a) point A, 5.00 cm from the left face of the left-hand sheet; (b) point B, 1.25 cm from the inner surface of the right-hand sheet; (c) point C, in the middle of the right-hand sheet.

Figure E22.28



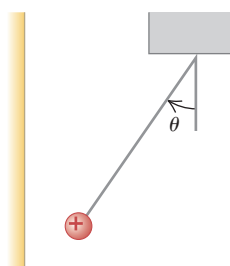
PROBLEMS

22.29 •• CP At time $t = 0$ a proton is a distance of 0.360 m from a very large insulating sheet of charge and is moving parallel to the sheet with speed 9.70×10^2 m/s. The sheet has uniform surface charge density 2.34×10^{-9} C/m². What is the speed of the proton at $t = 5.00 \times 10^{-8}$ s?

22.30 •• CP A very small object with mass 8.20×10^{-9} kg and positive charge 6.50×10^{-9} C is projected directly toward a very large insulating sheet of positive charge that has uniform surface charge density 5.90×10^{-8} C/m². The object is initially 0.400 m from the sheet. What initial speed must the object have in order for its closest distance of approach to the sheet to be 0.100 m?

22.31 •• CP A small sphere with mass 4.00×10^{-6} kg and charge 5.00×10^{-8} C hangs from a thread near a very large, charged insulating sheet (Fig. P22.31). The charge density on the surface of the sheet is uniform and equal to -2.50×10^{-9} C/m². Find the angle of the thread.

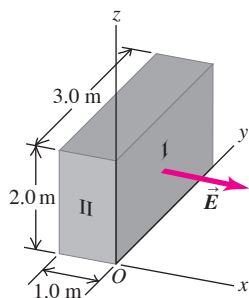
Figure P22.31



22.32 •• A cube has sides of length $L = 0.300$ m. One corner is at the origin (Fig. E22.6). The nonuniform electric field is given by $\vec{E} = (-5.00 \text{ N/C} \cdot \text{m})x\hat{i} + (3.00 \text{ N/C} \cdot \text{m})z\hat{k}$. (a) Find the electric flux through each of the six cube faces S_1, S_2, S_3, S_4, S_5 , and S_6 . (b) Find the total electric charge inside the cube.

22.33 • The electric field $\vec{E}$ in Fig. P22.33 is everywhere parallel to the x -axis, so the components E_y and E_z are zero. The x -component of the field E_x depends on x but not on y or z . At points in the yz -plane (where $x = 0$), $E_x = 125$ N/C. (a) What is the electric flux through surface I in Fig. P22.33? (b) What is the electric flux through surface II? (c) The volume shown is a small section of a very large insulating slab 1.0 m thick. If there is a total charge of -24.0 nC within the volume shown, what are the magnitude and direction of $\vec{E}$ at the face opposite surface I? (d) Is the electric field produced by charges only within the slab, or is the field also due to charges outside the slab? How can you tell?

Figure P22.33



22.34 •• CALC In a region of space there is an electric field $\vec{E}$ that is in the z -direction and that has magnitude $E = [964 \text{ N/(C} \cdot \text{m)}]x$. Find the flux for this field through a square in the xy -plane at $z = 0$ and with side length 0.350 m. One side of the square is along the $+x$ -axis and another side is along the $+y$ -axis.

22.35 • Negative charge $-Q$ is distributed uniformly over the surface of a thin spherical insulating shell with radius R . Calculate the force (magnitude and direction) that the shell exerts on a positive point charge q located a distance (a) $r > R$ from the center of the shell (outside the shell); (b) $r < R$ from the center of the shell (inside the shell).

22.36 • A long line carrying a uniform linear charge density $+50.0 \mu\text{C/m}$ runs parallel to and 10.0 cm from the surface of a large, flat plastic sheet that has a uniform surface charge density of $-100 \mu\text{C/m}^2$ on one side. Find the location of all points where an α particle would feel no force due to this arrangement of charged objects.

22.37 • The Coaxial Cable. A long coaxial cable consists of an inner cylindrical conductor with radius a and an outer coaxial cylinder with inner radius b and outer radius c . The outer cylinder is mounted on insulating supports and has no net charge. The inner cylinder has a uniform positive charge per unit length λ . Calculate the electric field (a) at any point between the cylinders a distance r from the axis and (b) at any point outside the outer cylinder. (c) Graph the magnitude of the electric field as a function of the distance r from the axis of the cable, from $r = 0$ to $r = 2c$. (d) Find the charge per unit length on the inner surface and on the outer surface of the outer cylinder.

22.38 • A very long conducting tube (hollow cylinder) has inner radius a and outer radius b . It carries charge per unit length $+\alpha$, where α is a positive constant with units of C/m. A line of charge lies along the axis of the tube. The line of charge has charge per unit length $+\alpha$. (a) Calculate the electric field in terms of α and the distance r from the axis of the tube for (i) $r < a$; (ii) $a < r < b$; (iii) $r > b$. Show your results in a graph of E as a function of r . (b) What is the charge per unit length on (i) the inner surface of the tube and (ii) the outer surface of the tube?

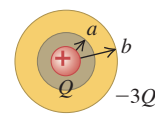
22.39 • A very long, solid cylinder with radius R has positive charge uniformly distributed throughout it, with charge per unit volume ρ . (a) Derive the expression for the electric field inside the volume at a distance r from the axis of the cylinder in terms of the charge density ρ . (b) What is the electric field at a point outside the volume in terms of the charge per unit length λ in the cylinder? (c) Compare the answers to parts (a) and (b) for $r = R$. (d) Graph the electric-field magnitude as a function of r from $r = 0$ to $r = 3R$.

22.40 • A Sphere in a Sphere. A solid conducting sphere carrying charge q has radius a . It is inside a concentric hollow conducting sphere with inner radius b and outer radius c . The hollow sphere has no net charge. (a) Derive expressions for the electric-field magnitude in terms of the distance r from the center for the regions $r < a$, $a < r < b$, $b < r < c$, and $r > c$. (b) Graph the magnitude of the electric field as a function of r from $r = 0$ to $r = 2c$. (c) What is the charge on the inner surface of the hollow sphere? (d) On the outer surface? (e) Represent the charge of the small sphere by four plus signs. Sketch the field lines of the system within a spherical volume of radius $2c$.

22.41 • A solid conducting sphere with radius R that carries positive charge Q is concentric with a very thin insulating shell of radius $2R$ that also carries charge Q . The charge Q is distributed uniformly over the insulating shell. (a) Find the electric field (magnitude and direction) in each of the regions $0 < r < R$, $R < r < 2R$, and $r > 2R$. (b) Graph the electric-field magnitude as a function of r .

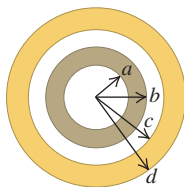
22.42 • A conducting spherical shell with inner radius a and outer radius b has a positive point charge Q located at its center. The total charge on the shell is $-3Q$, and it is insulated from its surroundings (Fig. P22.42). (a) Derive expressions for the electric-field magnitude E in terms of the distance r from the center for the regions $r < a$, $a < r < b$, and $r > b$. What is the surface charge density (b) on the inner surface of the conducting shell; (c) on the outer surface of the conducting shell? (d) Sketch the electric field lines and the location of all charges. (e) Graph E as a function of r .

Figure P22.42



22.43 • Concentric Spherical Shells. A small conducting spherical shell with inner radius a and outer radius b is concentric with a larger conducting spherical shell with inner radius c and outer radius d (Fig. P22.43). The inner shell has total charge $+2q$, and the outer shell has charge $+4q$. (a) Calculate the electric field $\vec{E}$ (magnitude and direction) in terms of q and the distance r from the common center of the two shells for (i) $r < a$; (ii) $a < r < b$; (iii) $b < r < c$; (iv) $c < r < d$; (v) $r > d$. Graph the radial component of $\vec{E}$ as a function of r . (b) What is the total charge on the (i) inner surface of the small shell; (ii) outer surface of the small shell; (iii) inner surface of the large shell; (iv) outer surface of the large shell?

Figure P22.43



22.44 • Repeat Problem 22.43, but now let the outer shell have charge $-2q$. The inner shell still has charge $+2q$.

22.45 ••• CALC An insulating hollow sphere has inner radius a and outer radius b . Within the insulating material the volume charge density is given by $\rho(r) = \alpha/r$, where α is a positive constant. (a) In terms of α and a , what is the magnitude of the electric field at a distance r from the center of the shell, where $a < r < b$? (b) A point charge q is placed at the center of the hollow space, at $r = 0$. In terms of α and a , what value must q have (sign and magnitude) in order for the electric field to be constant in the region $a < r < b$, and what then is the value of the constant field in this region?

22.46 •• CP Thomson's Model of the Atom. Early in the 20th century, a leading model of the structure of the atom was that of English physicist J. J. Thomson (the discoverer of the electron). In Thomson's model, an atom consisted of a sphere of positively charged material in which were embedded negatively charged electrons, like chocolate chips in a ball of cookie dough. Consider such an atom consisting of one electron with mass m and charge $-e$, which may be regarded as a point charge, and a uniformly charged sphere of charge $+e$ and radius R . (a) Explain why the electron's equilibrium position is at the center of the nucleus. (b) In Thomson's model, it was assumed that the positive material provided little or no resistance to the electron's motion. If the electron is displaced from equilibrium by a distance less than R , show that the resulting motion of the electron will be simple harmonic, and calculate the frequency of oscillation. (*Hint:* Review the definition of SHM in Section 14.2. If it can be shown that the net force on the electron is of this form, then it follows that the motion is simple harmonic. Conversely, if the net force on the electron does not follow this form, the motion is not simple harmonic.) (c) By Thomson's time, it was known that excited atoms emit light waves of only certain frequencies. In his model, the frequency of emitted light is the same as the oscillation frequency of the electron(s) in the atom. What radius would a Thomson-model atom need for it to produce red light of frequency 4.57×10^{14} Hz? Compare your answer to the radii of real atoms, which are of the order of 10^{-10} m (see Appendix F). (d) If the electron were displaced from equilibrium by a distance greater than R , would the electron oscillate? Would its motion be simple harmonic? Explain your reasoning. (*Historical note:* In 1910, the atomic nucleus was discovered, proving the Thomson model to be incorrect. An atom's positive charge is not spread over its volume, as Thomson supposed, but is concentrated in the tiny nucleus of radius 10^{-14} to 10^{-15} m.)

22.47 •• CALC A nonuniform, but spherically symmetric, distribution of charge has a charge density $\rho(r)$ given as follows:

$$\rho(r) = \rho_0 \left(1 - \frac{r}{R} \right) \quad \text{for } r \leq R$$

$$\rho(r) = 0 \quad \text{for } r \geq R$$

where $\rho_0 = 3Q/\pi R^3$ is a positive constant. (a) Show that the total charge contained in the charge distribution is Q . (b) Show that the electric field in the region $r \geq R$ is identical to that produced by a point charge Q at $r = 0$. (c) Obtain an expression for the electric field in the region $r \leq R$. (d) Graph the electric-field magnitude E as a function of r . (e) Find the value of r at which the electric field is maximum, and find the value of that maximum field.

22.48 •• (a) How many excess electrons must be distributed uniformly within the volume of an isolated plastic sphere 30.0 cm in diameter to produce an electric field of magnitude 1390 N/C just outside the surface of the sphere? (b) What is the electric field at a point 10.0 cm outside the surface of the sphere?

22.49 •• CALC A very long insulating cylinder has radius R and carries positive charge distributed throughout its volume. The charge distribution has cylindrical symmetry but varies with perpendicular distance from the axis of the cylinder. The volume charge density is $\rho(r) = \alpha(1 - r/R)$, where α is a constant with units C/m^3 and r is the perpendicular distance from the center line of the cylinder. Derive an expression, in terms of α and R , for $E(r)$, the electric field as a function of r . Do this for $r < R$ and also for $r > R$. Do your results agree for $r = R$?

22.50 •• CALC A solid insulating sphere has radius R and carries positive charge distributed throughout its volume. The charge distribution has spherical symmetry but varies with radial distance r from the center of the sphere. The volume charge density is $\rho(r) = \rho_0(1 - r/R)$, where ρ_0 is a constant with units of C/m^3 . (a) Derive an expression for the electric field as a function of r for $r < R$. (b) Repeat part (a) for $r > R$. (c) At what value of r , in terms of R , does the electric field have its maximum value?

22.51 • CALC A Nonuniformly Charged Slab. A slab of insulating material has thickness $2d$ and is oriented so that its faces are parallel to the yz -plane and given by the planes $x = d$ and $x = -d$. The y - and z -dimensions of the slab are very large compared to d , so treat them as infinite. The slab has charge density given by $\rho(x) = \rho_0(x/d)^2$, where ρ_0 is a positive constant. Using Gauss's law, find the electric field due to the slab (magnitude and direction) at all points in space.

22.52 • CALC A nonuniform, but spherically symmetric, distribution of charge has a charge density $\rho(r)$ given as follows:

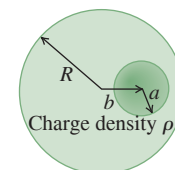
$$\rho(r) = \rho_0 \left(1 - \frac{4r}{3R} \right) \quad \text{for } r \leq R$$

$$\rho(r) = 0 \quad \text{for } r \geq R$$

where ρ_0 is a positive constant. (a) Find the total charge contained in the charge distribution. Obtain an expression for the electric field in the region (b) $r \geq R$; (c) $r \leq R$. (d) Graph the electric-field magnitude E as a function of r . (e) Find the value of r at which the electric field is maximum, and find the value of that maximum field.

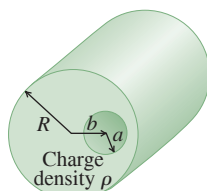
22.53 • (a) An insulating sphere with radius a has a uniform charge density ρ . The sphere is not centered at the origin but at $\vec{r} = \vec{b}$. Show that the electric field inside the sphere is given by $\vec{E} = \rho(\vec{r} - \vec{b})/3\epsilon_0$. (b) An insulating sphere of radius R has a spherical hole of radius a located within its volume and centered a distance b from the center of the sphere, where $a < b < R$ (a cross section of the sphere is shown in Fig. P22.53). The solid part of the sphere has a uniform volume charge density ρ . Find the magnitude and direction of the electric field $\vec{E}$ inside the hole, and show that $\vec{E}$ is uniform over the entire hole. [*Hint:* Use the principle of superposition and the result of part (a).]

Figure P22.53



22.54 • A very long, solid insulating cylinder has radius R ; bored along its entire length is a cylindrical hole with radius a . The axis of the hole is a distance b from the axis of the cylinder, where $a < b < R$ (Fig. P22.54). The solid material of the cylinder has a uniform volume charge density ρ . Find the magnitude and direction of the electric field $\vec{E}$ inside the hole, and show that $\vec{E}$ is uniform over the entire hole. (Hint: See Problem 22.53.)

Figure P22.54



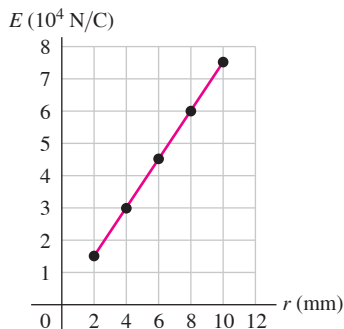
22.55 •• DATA In one experiment the electric field is measured for points at distances r from a uniform line of charge that has charge per unit length λ and length l , where $l \gg r$. In a second experiment the electric field is measured for points at distances r from the center of a uniformly charged insulating sphere that has volume charge density ρ and radius $R = 8.00$ mm, where $r > R$. The results of the two measurements are listed in the table, but you aren't told which set of data applies to which experiment:

r (cm)	E (10^5 N/C)	
	Measurement A	Measurement B
1.00	2.72	5.45
1.50	1.79	2.42
2.00	1.34	1.34
2.50	1.07	0.861
3.00	0.902	0.605
3.50	0.770	0.443
4.00	0.677	0.335

For each set of data, draw two graphs: one for Er^2 versus r and one for Er versus r . (a) Use these graphs to determine which data set, A or B, is for the uniform line of charge and which set is for the uniformly charged sphere. Explain your reasoning. (b) Use the graphs in part (a) to calculate λ for the uniform line of charge and ρ for the uniformly charged sphere.

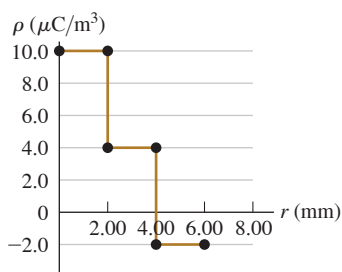
22.56 •• DATA The electric field is measured for points at distances r from the center of a uniformly charged insulating sphere that has volume charge density ρ and radius R , where $r < R$ (Fig. P22.56). Calculate ρ .

Figure P22.56



22.57 •• DATA The volume charge density ρ for a spherical charge distribution of radius $R = 6.00$ mm is not uniform. Figure P22.57 shows ρ as a function of the distance r from the center of the distribution. Calculate the electric field at these values of r : (i) 1.00 mm; (ii) 3.00 mm; (iii) 5.00 mm; (iv) 7.00 mm.

Figure P22.57



22.58 •• BIO As a honeybee flies, the passing air strips electrons from its hairs, giving the bee a net positive charge. Since flowers are negatively charged, pollen then jumps onto a bee even if the bee does not physically touch the pollen particles. (a) Estimate the diameter of the central disk of a daisy. (b) If a bee has had 75,000 electrons stripped by the air, what is its net charge? (c) If this bee lands at the edge of the daisy's central disk, determine its electric field at the far edge of the disk. Treat the bee as a thin-walled hollow sphere with its net charge distributed uniformly over its surface. (d) A pollen particle requires a force of 10 pN to dislodge from a stamen. Estimate the net charge on a pollen particle at the far end of the disk required for the particle to dislodge and jump to the bee.

22.59 •• CP CALC A very long insulating cylinder with radius R_{cylinder} has nonuniform positive charge density $\rho = (1 - r/R_{\text{cylinder}})\rho_0$, where ρ_0 is constant and r is measured radially from the axis of the cylinder. A particle with charge $-Q$ and mass M orbits the cylinder at a constant distance $R_{\text{orbit}} > R_{\text{cylinder}}$. (a) What is the linear charge density λ of the tube, in terms of R_{cylinder} and ρ_0 ? (b) Determine the period of the motion in terms of R_{orbit} . (Hint: Use Gauss's law to determine the electric field, and therefore the electric force felt by the particle, that acts centripetally.)

22.60 •• CP (a) Show that the component of the electric force normal to any flat surface with a uniform charge density σ is given by $F_{\perp} = \sigma \Phi_E$, where Φ_E is the electric flux through that surface due to an external electric field. (b) An insulating hemisphere with radius R and charge Q distributed uniformly over its flat, circular surface lies above a large plane with uniform charge density σ . The axis of the hemisphere is oriented vertically. For what mass M could the hemisphere remain stationary? (c) If the hemisphere and the plane share the same charge density of $100 \mu\text{C}/\text{m}^2$ and the hemisphere has a radius of 3.00 cm, what would be its upward acceleration if its mass were 100 g?

CHALLENGE PROBLEMS

22.61 ••• CALC A uniformly charged insulating sphere with radius r and charge $+Q$ lies at the center of a thin-walled hollow cylinder with radius $R > r$ and length $L > 2r$. The cylinder is non-conducting and carries no net charge. (a) Determine the outward electric flux through the rounded "side" of the cylinder, excluding the circular end caps. (Hint: Choose a cylindrical coordinate system with the axis of the cylinder as its z -axis and the center of the charged sphere as its origin. Note that an area element on the cylinder has magnitude $dA = 2\pi R dz$.) (b) Determine the electric flux upward through the circular cap at the top of the cylinder. (c) Determine the electric flux downward through the circular cap at the bottom of the cylinder. (d) Add the results from parts (a)–(c) to determine the outward electric flux through the closed cylinder. (e) Show that your result is consistent with Gauss's law.

22.62 ••• CP CALC A region in space contains a total positive charge Q that is distributed spherically such that the volume charge density $\rho(r)$ is given by

$$\begin{aligned} \rho(r) &= 3\alpha r/2R & \text{for } r \leq R/2 \\ \rho(r) &= \alpha[1 - (r/R)^2] & \text{for } R/2 \leq r \leq R \\ \rho(r) &= 0 & \text{for } r \geq R \end{aligned}$$

Here α is a positive constant having units of C/m^3 . (a) Determine α in terms of Q and R . (b) Using Gauss's law, derive an expression for the magnitude of the electric field as a function of r . Do this separately for all three regions. Express your answers in terms of Q . (c) What fraction of the total charge is contained within the region $R/2 \leq r \leq R$? (d) What is the magnitude of $\vec{E}$ at $r = R/2$? (e) If an electron with charge $q' = -e$ is released from rest at any point in any of the three regions, the resulting motion will be oscillatory but not simple harmonic. Why?

MCAT-STYLE PASSAGE PROBLEMS

Space Radiation Shielding. One of the hazards facing humans in space is space radiation: high-energy charged particles emitted by the sun. During a solar flare, the intensity of this radiation can reach lethal levels. One proposed method of protection for astronauts on the surface of the moon or Mars is an array of large, electrically charged spheres placed high above areas where people live and work. The spheres would produce a strong electric field $\vec{E}$ to deflect the charged particles that make up space radiation. The spheres would be similar in construction to a Mylar balloon, with a thin, electrically conducting layer on the outside surface on which a net positive or negative charge would be placed. A typical sphere might be 5 m in diameter.

22.63 Suppose that to repel electrons in the radiation from a solar flare, each sphere must produce an electric field $\vec{E}$ of magnitude 1×10^6 N/C at 25 m from the center of the sphere. What net charge on each sphere is needed? (a) -0.07 C; (b) -8 mC; (c) -80 μ C; (d) -1×10^{-20} C.

22.64 What is the magnitude of $\vec{E}$ just outside the surface of such a sphere? (a) 0; (b) 10^6 N/C; (c) 10^7 N/C; (d) 10^8 N/C.

22.65 What is the direction of $\vec{E}$ just outside the surface of such a sphere? (a) Tangent to the surface of the sphere; (b) perpendicular to the surface, pointing toward the sphere; (c) perpendicular to the surface, pointing away from the sphere; (d) there is no electric field just outside the surface.

22.66 Which statement is true about $\vec{E}$ inside a negatively charged sphere as described here? (a) It points from the center of the sphere to the surface and is largest at the center. (b) It points from the surface to the center of the sphere and is largest at the surface. (c) It is zero. (d) It is constant but not zero.

ANSWERS

Chapter Opening Question ?

(iii) The electric field inside a cavity within a conductor is zero, so there would be no electric effect on the child. (See Section 22.5.)

Key Example **✓** VARIATION Problems

VP22.4.1 (a) $-E_1 L^2$ (b) $+E_1 L^2$ (c) $-E_2 L^2$ (d) $+E_2 L^2$ (e) 0 (f) 0 (g) 0

VP22.4.2 (a) $+452$ N $\cdot$ m²/C (b) -226 N $\cdot$ m²/C (c) -226 N $\cdot$ m²/C

VP22.4.3 (a) $+678$ N $\cdot$ m²/C (b) $+113$ N $\cdot$ m²/C

VP22.4.4 (a) $+339$ N $\cdot$ m²/C (b) -565 N $\cdot$ m²/C (c) -565 N $\cdot$ m²/C

VP22.10.1 (a) 8.63×10^3 N/C at 4.00 cm, 7.49×10^3 N/C at 6.00 cm
(b) 0 at 4.00 cm, 7.49×10^3 N/C at 6.00 cm

VP22.10.2 2.42 μ N, attractive

VP22.10.3 1.79×10^{-7} C/m², positive

VP22.10.4 5.41×10^{-8} C/m²

VP22.12.1 (a) -5.00 nC (b) -3.00 nC

VP22.12.2 (a) $+2.00$ nC (b) neither would be affected

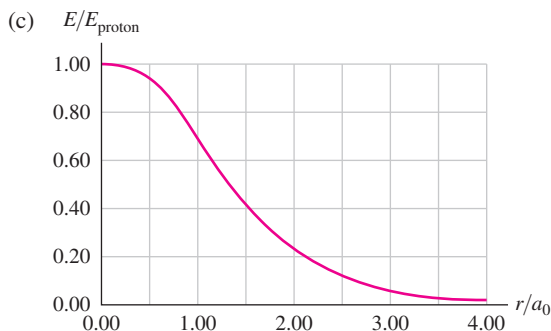
VP22.12.3 (a) 0 (b) 2.7×10^{-5} C (c) 2.7×10^{-6} C/m²

VP22.12.4 (a) $+4.00$ nC (b) -2.00 nC (c) 2.50×10^3 N/C
(d) 1.65×10^2 N/C

Bridging Problem

(a) $Q(r) = Qe^{-2r/a_0} [2(r/a_0)^2 + 2(r/a_0) + 1]$

(b) $E = \frac{kQe^{-2r/a_0}}{r^2} [2(r/a_0)^2 + 2(r/a_0) + 1]$





? In one type of welding, electric charge flows between the welding tool and the metal pieces that are to be joined. This produces a glowing arc whose high temperature fuses the pieces together. Why must the tool be held close to the metal pieces? (i) To maximize the potential difference between tool and pieces; (ii) to minimize this potential difference; (iii) to maximize the electric field between tool and pieces; (iv) to minimize this electric field; (v) more than one of these.

23 Electric Potential

This chapter is about energy associated with electrical interactions. Every time you turn on a light, use a mobile phone, or make toast in a toaster, you are using electrical energy, an indispensable ingredient of our technological society. In Chapters 6 and 7 we introduced the concepts of *work* and *energy* in the context of mechanics; now we'll combine these concepts with what we've learned about electric charge, electric forces, and electric fields. Just as we found for many problems in mechanics, using energy ideas makes it easier to solve a variety of problems in electricity.

When a charged particle moves in an electric field, the field exerts a force that can do *work* on the particle. This work can be expressed in terms of electric potential energy. Just as gravitational potential energy depends on the height of a mass above the earth's surface, electric potential energy depends on the position of the charged particle in the electric field. We'll use a new concept called *electric potential*, or simply *potential* to describe electric potential energy. In circuits, a difference in potential from one point to another is often called *voltage*. The concepts of potential and voltage are crucial to understanding how electric circuits work and have equally important applications to electron beams used in cancer radiotherapy, high-energy particle accelerators, and many other devices.

23.1 ELECTRIC POTENTIAL ENERGY

The concepts of work, potential energy, and conservation of energy proved to be extremely useful in our study of mechanics. In this section we'll show that these concepts are just as useful for understanding and analyzing electrical interactions.

Let's begin by reviewing three essential points from Chapters 6 and 7. First, when a force $\vec{F}$ acts on a particle that moves from point a to point b , the work $W_{a \rightarrow b}$ done by the force is given by a *line integral*:

$$W_{a \rightarrow b} = \int_a^b \vec{F} \cdot d\vec{l} = \int_a^b F \cos \phi \, dl \quad (\text{work done by a force}) \quad (23.1)$$

where $d\vec{l}$ is an infinitesimal displacement along the particle's path and ϕ is the angle between $\vec{F}$ and $d\vec{l}$ at each point along the path.

LEARNING OUTCOMES

In this chapter, you'll learn...

- 23.1** How to calculate the electric potential energy of a collection of charges.
- 23.2** The meaning and significance of electric potential.
- 23.3** How to calculate the electric potential that a collection of charges produces at a point in space.
- 23.4** How to use equipotential surfaces to visualize how the electric potential varies in space.
- 23.5** How to use electric potential to calculate the electric field.

You'll need to review...

- 7.1–7.4** Conservative forces and potential energy.
- 21.1–21.6** Electric force and electric fields.
- 22.4, 22.5** Applications of Gauss's law.

Second, if the force $\vec{F}$ is *conservative*, as we defined the term in Section 7.3, the work done by $\vec{F}$ can always be expressed in terms of a **potential energy** U . When the particle moves from a point where the potential energy is U_a to a point where it is U_b , the change in potential energy is $\Delta U = U_b - U_a$ and

$$W_{a \rightarrow b} = U_a - U_b = -(U_b - U_a) = -\Delta U \quad (23.2)$$

Work done by a conservative force
 Potential energy at initial position
 Potential energy at final position
 Negative of change in potential energy

Figure 23.1 The work done on a baseball moving in a uniform gravitational field.

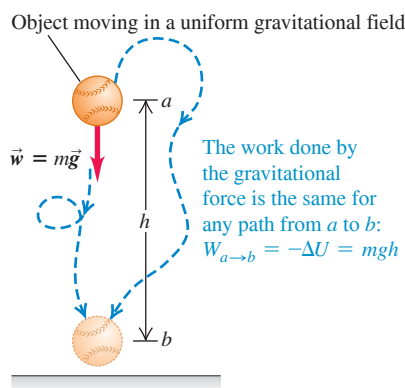
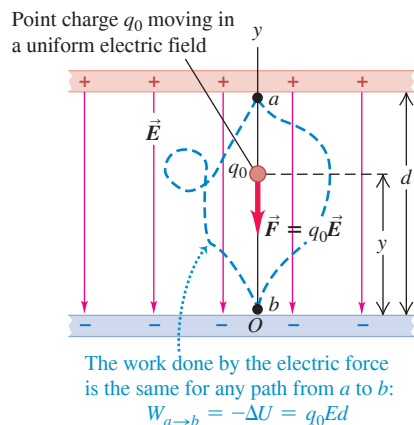


Figure 23.2 The work done on a point charge moving in a uniform electric field. Compare with Fig. 23.1.



When $W_{a \rightarrow b}$ is positive, U_a is greater than U_b , ΔU is negative, and the potential energy *decreases*. That's what happens when a baseball falls from a high point (a) to a lower point (b) under the influence of the earth's gravity; the force of gravity does positive work, and the gravitational potential energy decreases (**Fig. 23.1**). When a tossed ball is moving upward, the gravitational force does negative work during the ascent, and the potential energy increases.

Third, the work–energy theorem says that the change in kinetic energy $\Delta K = K_b - K_a$ during a displacement equals the *total* work done on the particle. If only conservative forces do work, then Eq. (23.2) gives the total work, and $K_b - K_a = -(U_b - U_a)$. We usually write this as

$$K_a + U_a = K_b + U_b \quad (23.3)$$

That is, the total mechanical energy (kinetic plus potential) is *conserved* under these circumstances.

Electric Potential Energy in a Uniform Field

Let's look at an electrical example of these concepts. In **Fig. 23.2** a pair of large, charged, parallel metal plates sets up a uniform, downward electric field with magnitude E . The field exerts a downward force with magnitude $F = q_0 E$ on a positive test charge q_0 . As the charge moves downward a distance d from point a to point b , the force on the test charge is constant and independent of its location. So the work done by the electric field is the product of the force magnitude and the component of displacement in the (downward) direction of the force:

$$W_{a \rightarrow b} = Fd = q_0 E d \quad (23.4)$$

This work is positive, since the force is in the same direction as the net displacement of the test charge.

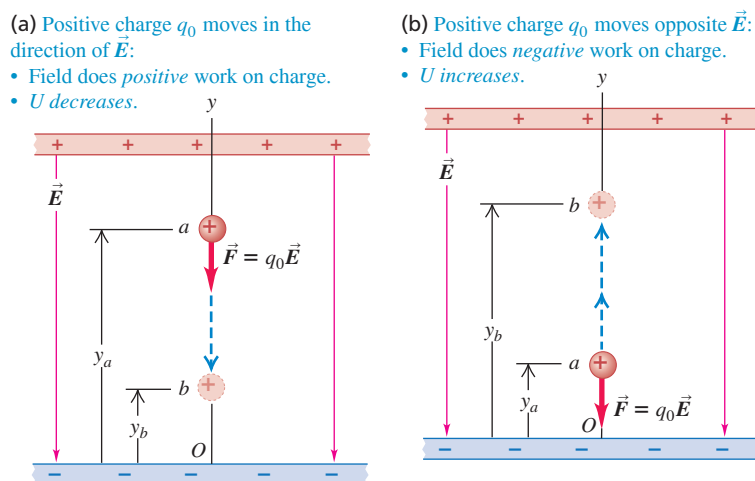
The y -component of the electric force, $F_y = -q_0 E$, is constant, and there is no x - or z -component. This is exactly analogous to the gravitational force on a mass m near the earth's surface; for this force, there is a constant y -component $F_y = -mg$ and the x - and z -components are zero. Because of this analogy, we can conclude that the force exerted on q_0 by the uniform electric field in Fig. 23.2 is *conservative*, just as is the gravitational force. This means that the work $W_{a \rightarrow b}$ done by the field is independent of the path the particle takes from a to b . We can represent this work with a *potential-energy* function U , just as we did for gravitational potential energy in Section 7.1. The potential energy for the gravitational force $F_y = -mg$ was $U = mgy$; hence the potential energy for the electric force $F_y = -q_0 E$ is

$$U = q_0 E y \quad (23.5)$$

When the test charge moves from height y_a to height y_b , the work done on the charge by the field is given by

$$W_{a \rightarrow b} = -\Delta U = -(U_b - U_a) = -(q_0 E y_b - q_0 E y_a) = q_0 E (y_a - y_b) \quad (23.6)$$

Figure 23.3 A positive charge moving (a) in the direction of the electric field $\vec{E}$ and (b) in the direction opposite $\vec{E}$.



When y_a is greater than y_b (Fig. 23.3a), the positive test charge q_0 moves downward, in the same direction as $\vec{E}$; the displacement is in the same direction as the force $\vec{F} = q_0 \vec{E}$, so the field does positive work and U decreases. [In particular, if $y_a - y_b = d$ as in Fig. 23.2, Eq. (23.6) gives $W_{a \rightarrow b} = q_0 E d$, in agreement with Eq. (23.4).] When y_a is less than y_b (Fig. 23.3b), the positive test charge q_0 moves upward, in the opposite direction to $\vec{E}$; the displacement is opposite the force, the field does negative work, and U increases.

If the test charge q_0 is *negative*, the potential energy increases when it moves with the field and decreases when it moves against the field (Fig. 23.4).

Whether the test charge is positive or negative, the following general rules apply: U *increases* if the test charge q_0 moves in the direction *opposite* the electric force $\vec{F} = q_0 \vec{E}$ (Figs. 23.3b and 23.4a); U *decreases* if q_0 moves in the *same* direction as $\vec{F} = q_0 \vec{E}$ (Figs. 23.3a and 23.4b). This is the same behavior as for gravitational potential energy, which increases if a mass m moves upward (opposite the direction of the gravitational force) and decreases if m moves downward (in the same direction as the gravitational force).

CAUTION Electric potential energy The relationship between electric potential energy change and motion in an electric field is an important one that we'll use often, but that takes some effort to understand. Studying the preceding paragraph as well as Figs. 23.3 and Fig. 23.4 carefully now will help you tremendously later!

Electric Potential Energy of Two Point Charges

The idea of electric potential energy isn't restricted to the special case of a uniform electric field. Indeed, we can apply this concept to a point charge in *any* electric field caused by a static charge distribution. Recall from Chapter 21 that we can represent any charge distribution as a collection of point charges. Therefore it's useful to calculate the work done on a test charge q_0 moving in the electric field caused by a single, stationary point charge q .

We'll consider first a displacement along the *radial* line in Fig. 23.5. The force on q_0 is given by Coulomb's law, and its radial component is

$$F_r = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \quad (23.7)$$

If q and q_0 have the same sign (+ or -), the force is repulsive and F_r is positive; if the two charges have opposite signs, the force is attractive and F_r is negative. The force is *not*

Figure 23.4 A negative charge moving (a) in the direction of the electric field $\vec{E}$ and (b) in the direction opposite $\vec{E}$. Compare with Fig. 23.3.

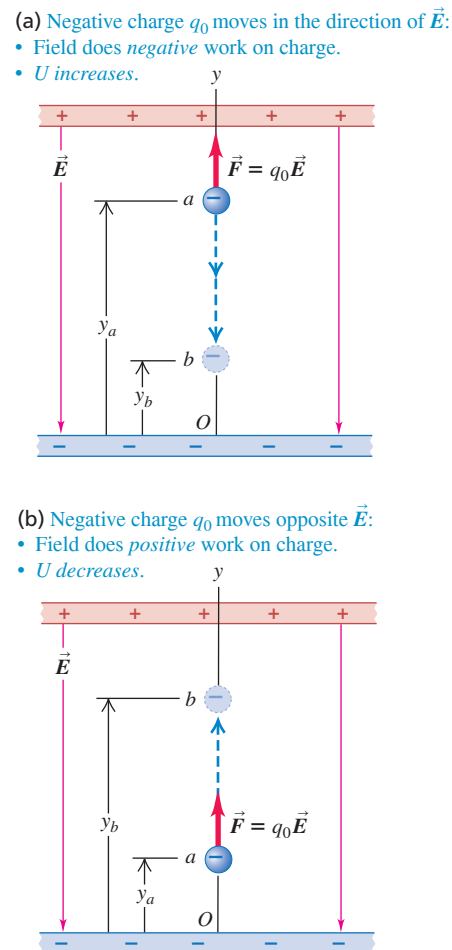


Figure 23.5 Test charge q_0 moves along a straight line extending radially from charge q . As it moves from a to b , the distance varies from r_a to r_b .

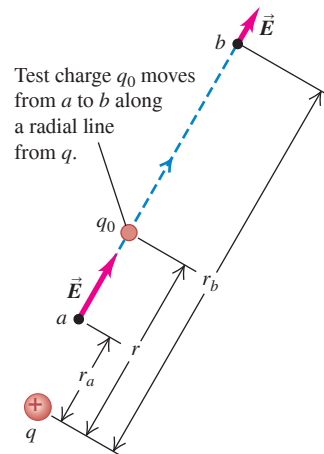
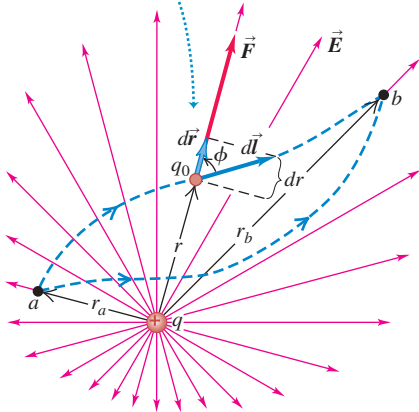


Figure 23.6 The work done on charge q_0 by the electric field of charge q does not depend on the path taken, but only on the distances r_a and r_b .

Test charge q_0 moves from a to b along an arbitrary path.



constant during the displacement, and we must integrate to calculate the work $W_{a \rightarrow b}$ done on q_0 by this force as q_0 moves from a to b :

$$W_{a \rightarrow b} = \int_{r_a}^{r_b} F_r dr = \int_{r_a}^{r_b} \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} dr = \frac{qq_0}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right) \quad (23.8)$$

The work done by the electric force for this path depends on only the endpoints.

Now let's consider a more general displacement (Fig. 23.6) in which a and b do not lie on the same radial line. From Eq. (23.1) the work done on q_0 during this displacement is given by

$$W_{a \rightarrow b} = \int_{r_a}^{r_b} F \cos \phi dl = \int_{r_a}^{r_b} \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \cos \phi dl$$

But Fig. 23.6 shows that $\cos \phi dl = dr$. That is, the work done during a small displacement $d\vec{l}$ depends only on the change dr in the distance r between the charges, which is the radial component of the displacement. Thus Eq. (23.8) is valid even for this more general displacement; the work done on q_0 by the electric field $\vec{E}$ produced by q depends only on r_a and r_b , not on the details of the path. Also, if q_0 returns to its starting point a by a different path, the total work done in the round-trip displacement is zero (the integral in Eq. (23.8) is from r_a back to r_a). These are the needed characteristics for a conservative force, as we defined it in Section 7.3. Thus the force on q_0 is a conservative force.

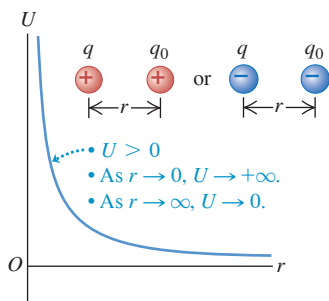
We see that Eqs. (23.2) and (23.8) are consistent if we define the potential energy to be $U_a = qq_0/4\pi\epsilon_0 r_a$ when q_0 is a distance r_a from q , and to be $U_b = qq_0/4\pi\epsilon_0 r_b$ when q_0 is a distance r_b from q . Thus

$$\text{Electric potential energy of two point charges} \quad U = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r} \quad (23.9)$$

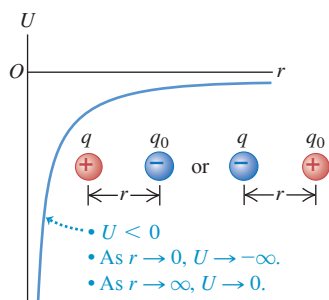
Electric constant
Values of two charges
Distance between two charges

Figure 23.7 Graphs of the potential energy U of two point charges q and q_0 versus their separation r .

(a) q and q_0 have the same sign.



(b) q and q_0 have opposite signs.



Equation (23.9) is valid no matter what the signs of the charges q and q_0 . The potential energy is positive if the charges q and q_0 have the same sign (Fig. 23.7a) and negative if they have opposite signs (Fig. 23.7b).

CAUTION Electric potential energy vs. electric force Don't confuse Eq. (23.9) for the potential energy of two point charges with Eq. (23.7) for the radial component of the electric force that one charge exerts on the other. Potential energy U is proportional to $1/r$, while the force component F_r is proportional to $1/r^2$.

Potential energy is always defined relative to some reference point where $U = 0$. In Eq. (23.9), U is zero when q and q_0 are infinitely far apart and $r = \infty$. Therefore U represents the work that would be done on the test charge q_0 by the field of q if q_0 moved from an initial distance r to infinity. If q and q_0 have the same sign, the interaction is repulsive, this work is positive, and U is positive at any finite separation (Fig. 23.7a). If the charges have opposite signs, the interaction is attractive, the work done is negative, and U is negative (Fig. 23.7b).

We emphasize that the potential energy U given by Eq. (23.9) is a shared property of the two charges. If the distance between q and q_0 is changed from r_a to r_b , the change in potential energy is the same whether q is held fixed and q_0 is moved or q_0 is held fixed and q is moved. For this reason, we never use the phrase “the electric potential energy of a point charge.” (Likewise, if a mass m is at a height h above the earth's surface, the gravitational potential energy is a shared property of the mass m and the earth. We emphasized this in Sections 7.1 and 13.3.)

Equation (23.9) also holds if the charge q_0 is outside a spherically symmetric charge distribution with total charge q ; the distance r is from q_0 to the center of the distribution. That's because Gauss's law tells us that the electric field outside such a distribution is the same as if all of its charge q were concentrated at its center (see Example 22.9 in Section 22.4).

EXAMPLE 23.1 Conservation of energy with electric forces**WITH VARIATION PROBLEMS**

A positron (the electron's antiparticle) has mass 9.11×10^{-31} kg and charge $q_0 = +e = +1.60 \times 10^{-19}$ C. Suppose a positron moves in the vicinity of an α (alpha) particle, which has charge $q = +2e = 3.20 \times 10^{-19}$ C and mass 6.64×10^{-27} kg. The α particle's mass is more than 7000 times that of the positron, so we assume that the α particle remains at rest. When the positron is 1.00×10^{-10} m from the α particle, it is moving directly away from the α particle at 3.00×10^6 m/s. (a) What is the positron's speed when the particles are 2.00×10^{-10} m apart? (b) What is the positron's speed when it is very far from the α particle? (c) Suppose the initial conditions are the same but the moving particle is an electron (with the same mass as the positron but charge $q_0 = -e$). Describe the subsequent motion.

IDENTIFY and SET UP The electric force between a positron (or an electron) and an α particle is conservative, so the total mechanical energy (kinetic plus potential) is conserved. Equation (23.9) gives the potential energy U at any separation r : The potential-energy function for parts (a) and (b) looks like that of Fig. 23.7a, and the function for part (c) looks like that of Fig. 23.7b. We are given the positron speed $v_a = 3.00 \times 10^6$ m/s when the separation between the particles is $r_a = 1.00 \times 10^{-10}$ m. In parts (a) and (b) we use Eqs. (23.3) and (23.9) to find the speed for $r = r_b = 2.00 \times 10^{-10}$ m and $r = r_c \rightarrow \infty$, respectively. In part (c) we replace the positron with an electron and reconsider the problem.

EXECUTE (a) Both particles have positive charge, so the positron speeds up as it moves away from the α particle. From the energy-conservation equation, Eq. (23.3), the final kinetic energy is

$$K_b = \frac{1}{2}mv_b^2 = K_a + U_a - U_b$$

In this expression,

$$\begin{aligned} K_a &= \frac{1}{2}mv_a^2 = \frac{1}{2}(9.11 \times 10^{-31} \text{ kg})(3.00 \times 10^6 \text{ m/s})^2 \\ &= 4.10 \times 10^{-18} \text{ J} \\ U_a &= \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r_a} = (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \\ &\quad \times \frac{(3.20 \times 10^{-19} \text{ C})(1.60 \times 10^{-19} \text{ C})}{1.00 \times 10^{-10} \text{ m}} \\ &= 4.61 \times 10^{-18} \text{ J} \\ U_b &= \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r_b} = 2.30 \times 10^{-18} \text{ J} \end{aligned}$$

Hence the positron kinetic energy and speed at $r = r_b = 2.00 \times 10^{-10}$ m are

$$\begin{aligned} K_b &= \frac{1}{2}mv_b^2 \\ &= (4.10 \times 10^{-18} \text{ J}) + (4.61 \times 10^{-18} \text{ J}) - (2.30 \times 10^{-18} \text{ J}) \\ &= 6.41 \times 10^{-18} \text{ J} \\ v_b &= \sqrt{\frac{2K_b}{m}} = \sqrt{\frac{2(6.41 \times 10^{-18} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}} \\ &= 3.8 \times 10^6 \text{ m/s} \end{aligned}$$

(b) When the positron and α particle are very far apart so that $r = r_c \rightarrow \infty$, the final potential energy U_c approaches zero. Again from energy conservation, the final kinetic energy and speed of the positron in this case are

$$\begin{aligned} K_c &= K_a + U_a - U_c = (4.10 \times 10^{-18} \text{ J}) + (4.61 \times 10^{-18} \text{ J}) - 0 \\ &= 8.71 \times 10^{-18} \text{ J} \\ v_c &= \sqrt{\frac{2K_c}{m}} = \sqrt{\frac{2(8.71 \times 10^{-18} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}} = 4.4 \times 10^6 \text{ m/s} \end{aligned}$$

(c) The electron and α particle have opposite charges, so the force is attractive and the electron slows down as it moves away. Changing the moving particle's sign from $+e$ to $-e$ means that the initial potential energy is now $U_a = -4.61 \times 10^{-18}$ J, which makes the total mechanical energy *negative*:

$$\begin{aligned} K_a + U_a &= (4.10 \times 10^{-18} \text{ J}) - (4.61 \times 10^{-18} \text{ J}) \\ &= -0.51 \times 10^{-18} \text{ J} \end{aligned}$$

The total mechanical energy would have to be positive for the electron to move infinitely far away from the α particle. Like a rock thrown upward at low speed from the earth's surface, it will reach a maximum separation $r = r_d$ from the α particle before reversing direction. At this point its speed and its kinetic energy K_d are zero, so at separation r_d we have

$$\begin{aligned} U_d &= K_a + U_a - K_d = (-0.51 \times 10^{-18} \text{ J}) - 0 \\ U_d &= \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r_d} = -0.51 \times 10^{-18} \text{ J} \\ r_d &= \frac{1}{U_d} \frac{qq_0}{4\pi\epsilon_0} \\ &= \frac{(9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)}{-0.51 \times 10^{-18} \text{ J}} (3.20 \times 10^{-19} \text{ C})(-1.60 \times 10^{-19} \text{ C}) \\ &= 9.0 \times 10^{-10} \text{ m} \end{aligned}$$

For $r_b = 2.00 \times 10^{-10}$ m we have $U_b = -2.30 \times 10^{-18}$ J, so the electron kinetic energy and speed at this point are

$$\begin{aligned} K_b &= \frac{1}{2}mv_b^2 \\ &= (4.10 \times 10^{-18} \text{ J}) + (-4.61 \times 10^{-18} \text{ J}) - (-2.30 \times 10^{-18} \text{ J}) \\ &= 1.79 \times 10^{-18} \text{ J} \\ v_b &= \sqrt{\frac{2K_b}{m}} = \sqrt{\frac{2(1.79 \times 10^{-18} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}} = 2.0 \times 10^6 \text{ m/s} \end{aligned}$$

EVALUATE Both particles behave as expected as they move away from the α particle: The positron speeds up, and the electron slows down and eventually turns around. How fast would an electron have to be moving at $r_a = 1.00 \times 10^{-10}$ m to travel infinitely far from the α particle? (*Hint*: See Example 13.5 in Section 13.3.)

KEYCONCEPT The electrostatic forces that two charged particles exert on each other are conservative. The associated electric potential energy U is proportional to the product of the two charges and inversely proportional to the distance r between them. As r approaches infinity, U approaches zero.

Figure 23.8 The potential energy associated with a charge q_0 at point a depends on the other charges q_1 , q_2 , and q_3 and on their distances r_1 , r_2 , and r_3 from point a .

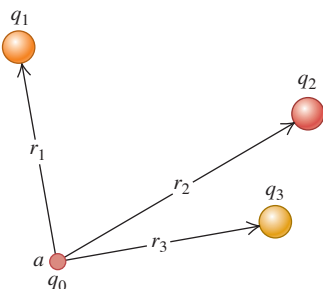
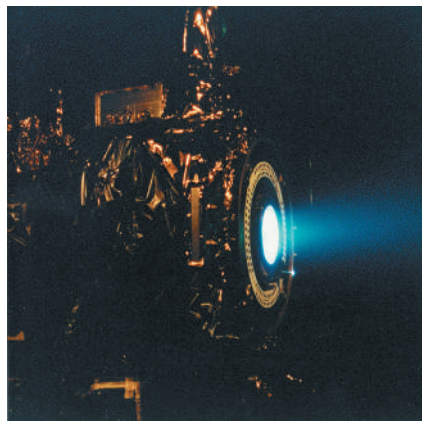


Figure 23.9 This ion engine for spacecraft uses electric forces to eject a stream of positive xenon ions (Xe^+) at speeds in excess of 30 km/s. The thrust produced is very low (about 0.09 newton) but can be maintained continuously for days, in contrast to chemical rockets, which produce a large thrust for a short time (see Fig. 8.34). Such ion engines have been used for maneuvering interplanetary spacecraft.



Electric Potential Energy with Several Point Charges

Suppose the electric field $\vec{E}$ in which charge q_0 moves is caused by *several* point charges q_1 , q_2 , q_3 , $\dots$ at distances r_1 , r_2 , r_3 , $\dots$ from q_0 , as in Fig. 23.8. For example, q_0 could be a positive ion moving in the presence of other ions (Fig. 23.9). The total electric field at each point is the *vector sum* of the fields due to the individual charges, and the total work done on q_0 during any displacement is the sum of the contributions from the individual charges. From Eq. (23.9) we conclude that the potential energy associated with the test charge q_0 at point a in Fig. 23.8 is the *algebraic sum* (not a vector sum):

$$U = \frac{q_0}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \dots \right) = \frac{q_0}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i} \quad (23.10)$$

Electric potential energy of point charge q_0 and collection of charges $q_1, q_2, q_3, \dots$
Electric constant
Distances from q_0 to $q_1, q_2, q_3, \dots$

When q_0 is at a different point b , the potential energy is given by the same expression, but r_1 , r_2 , $\dots$ are the distances from q_1 , q_2 , $\dots$ to point b . The work done on charge q_0 when it moves from a to b along any path is equal to the difference $U_a - U_b$ between the potential energies when q_0 is at a and at b .

We can represent *any* charge distribution as a collection of point charges, so Eq. (23.10) shows that we can always find a potential-energy function for *any* static electric field. It follows that **for every electric field due to a static charge distribution, the force exerted by that field is conservative.**

Equations (23.9) and (23.10) define U to be zero when distances r_1 , r_2 , $\dots$ are infinite—that is, when the test charge q_0 is very far away from all the charges that produce the field. As with any potential-energy function, the point where $U = 0$ is arbitrary; we can always add a constant to make U equal zero at any point we choose. In electrostatics problems it's usually simplest to choose this point to be at infinity. When we analyze electric circuits in Chapters 25 and 26, other choices will be more convenient.

Equation (23.10) gives the potential energy associated with the presence of the test charge q_0 in the $\vec{E}$ field produced by q_1 , q_2 , q_3 , $\dots$. But there is also potential energy involved in assembling these charges. If we start with charges q_1 , q_2 , q_3 , $\dots$ all separated from each other by infinite distances and then bring them together so that the distance between q_i and q_j is r_{ij} , the *total* potential energy U is the sum of the potential energies of interaction for each pair of charges. We can write this as

$$U = \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{r_{ij}} \quad (23.11)$$

This sum extends over all *pairs* of charges; we don't let $i = j$ (because that would be an interaction of a charge with itself), and we include only terms with $i < j$ to make sure that we count each pair only once. Thus, to account for the interaction between q_3 and q_4 , we include a term with $i = 3$ and $j = 4$ but not a term with $i = 4$ and $j = 3$.

Interpreting Electric Potential Energy

As a final comment, here are two viewpoints on electric potential energy. We have defined it in terms of the work done *by the electric field* on a charged particle moving in the field, just as in Chapter 7 we defined potential energy in terms of the work done by gravity or by a spring. When a particle moves from point a to point b , the work done on it by the electric field is $W_{a \rightarrow b} = U_a - U_b$. Thus the potential-energy difference $U_a - U_b$ equals *the work that is done by the electric force when the particle moves from a to b* . When U_a is greater than U_b , the field does positive work on the particle as it “falls” from a point of higher potential energy (a) to a point of lower potential energy (b).

An alternative but equivalent viewpoint is to consider how much work we would have to do to “raise” a particle from a point b where the potential energy is U_b to a point a where it has a greater value U_a (pushing two positive charges closer together, for example). To move the particle slowly (so as not to give it any kinetic energy), we need to exert an additional external force $\vec{F}_{\text{ext}}$ that is equal and opposite to the electric-field force and does positive work. The potential-energy difference $U_a - U_b$ is then defined as *the work that must be done by an external force to move the particle slowly from b to a against the electric force*. Because $\vec{F}_{\text{ext}}$ is the negative of the electric-field force and the displacement is in the opposite direction, this definition of the potential difference $U_a - U_b$ is equivalent to that given above. This alternative viewpoint also works if U_a is less than U_b , corresponding to “lowering” the particle; an example is moving two positive charges away from each other. In this case, $U_a - U_b$ is again equal to the work done by the external force, but now this work is negative.

We’ll use both of these viewpoints in the next section to interpret what is meant by electric *potential*, or potential energy per unit charge.

EXAMPLE 23.2 A system of point charges

WITH VARIATION PROBLEMS

Two point charges are at fixed positions on the x -axis, $q_1 = -e$ at $x = 0$ and $q_2 = +e$ at $x = a$. (a) Find the work that must be done by an external force to bring a third point charge $q_3 = +e$ from infinity to $x = 2a$. (b) Find the total potential energy of the system of three charges.

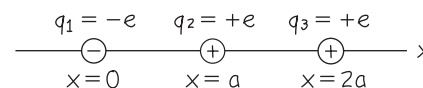
IDENTIFY and SET UP Figure 23.10 shows the final arrangement of the three charges. In part (a) we need to find the work W that must be done on q_3 by an external force $\vec{F}_{\text{ext}}$ to bring q_3 in from infinity to $x = 2a$. We do this by using Eq. (23.10) to find the potential energy associated with q_3 in the presence of q_1 and q_2 . In part (b) we use Eq. (23.11), the expression for the potential energy of a collection of point charges, to find the total potential energy of the system.

EXECUTE (a) The work W equals the difference between (i) the potential energy U associated with q_3 when it is at $x = 2a$ and (ii) the potential energy when it is infinitely far away. The second of these is zero, so the work required is equal to U . The distances between the charges are $r_{13} = 2a$ and $r_{23} = a$, so from Eq. (23.10),

$$W = U = \frac{q_3}{4\pi\epsilon_0} \left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}} \right) = \frac{+e}{4\pi\epsilon_0} \left(\frac{-e}{2a} + \frac{+e}{a} \right) = \frac{+e^2}{8\pi\epsilon_0 a}$$

This is positive, just as we should expect. If we bring q_3 in from infinity along the $+x$ -axis, it is attracted by q_1 but is repelled more strongly by q_2 . Hence we must do positive work to push q_3 to the position at $x = 2a$.

Figure 23.10 Our sketch of the situation after the third charge has been brought in from infinity.



(b) From Eq. (23.11), the total potential energy of the three-charge system is

$$\begin{aligned} U &= \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{r_{ij}} = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{(-e)(e)}{a} + \frac{(-e)(e)}{2a} + \frac{(e)(e)}{a} \right] = \frac{-e^2}{8\pi\epsilon_0 a} \end{aligned}$$

EVALUATE Our negative result in part (b) means that the system has lower potential energy than it would if the three charges were infinitely far apart. An external force would have to do *negative* work to bring the three charges from infinity to assemble this entire arrangement and would have to do *positive* work to move the three charges back to infinity.

KEYCONCEPT To find the total potential energy of a system of charges, sum the electric potential energies of all the distinct pairs of charges in the system.

TEST YOUR UNDERSTANDING OF SECTION 23.1 Consider the system of three point charges in Example 21.4 (Section 21.3) and shown in Fig. 21.14. (a) What is the sign of the total potential energy of this system? (i) Positive; (ii) negative; (iii) zero. (b) What is the sign of the total amount of work you would have to do to move these charges infinitely far from each other? (i) Positive; (ii) negative; (iii) zero.

ANSWER

(a) (i) The three charges q_1 , q_2 , and q_3 are all positive, so all three of the terms in the sum in Eq. (23.11)— $q_1 q_2 / r_{12}$, $q_1 q_3 / r_{13}$, and $q_2 q_3 / r_{23}$ —are positive. Hence the total electric potential energy U is positive. This means that it would take positive work to bring the three charges from infinity to the positions shown in Fig. 21.14, and hence *negative* work to move the three charges from these positions back to infinity.

23.2 ELECTRIC POTENTIAL

In Section 23.1 we looked at the potential energy U associated with a test charge q_0 in an electric field. Now we want to describe this potential energy on a “per unit charge” basis, just as electric field describes the force per unit charge on a charged particle in the field. This leads us to the concept of *electric potential*, often called simply *potential*. This concept is very useful in calculations involving energies of charged particles. It also facilitates many electric-field calculations because electric potential is closely related to the electric field $\vec{E}$. When we need to determine an electric field, it is often easier to determine the potential first and then find the field from it.

Potential is *potential energy per unit charge*. We define the potential V at any point in an electric field as the potential energy U *per unit charge* associated with a test charge q_0 at that point:

$$V = \frac{U}{q_0} \quad \text{or} \quad U = q_0 V \quad (23.12)$$

Potential energy and charge are both scalars, so potential is a scalar. From Eq. (23.12) its units are the units of energy divided by those of charge. The SI unit of potential, called one **volt** (1 V) in honor of the Italian electrical experimenter Alessandro Volta (1745–1827), equals 1 joule per coulomb:

$$1 \text{ V} = 1 \text{ volt} = 1 \text{ J/C} = 1 \text{ joule/coulomb}$$

Let’s put Eq. (23.2), which equates the work done by the electric force during a displacement from a to b to the quantity $-\Delta U = -(U_b - U_a)$, on a “work per unit charge” basis. We divide this equation by q_0 , obtaining

$$\frac{W_{a \rightarrow b}}{q_0} = -\frac{\Delta U}{q_0} = -\left(\frac{U_b}{q_0} - \frac{U_a}{q_0}\right) = -(V_b - V_a) = V_a - V_b \quad (23.13)$$

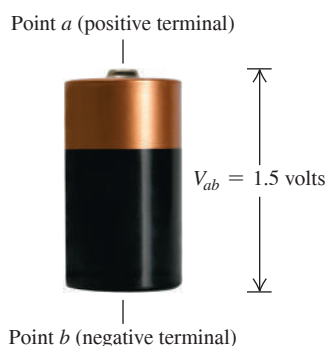
where $V_a = U_a/q_0$ is the potential energy per unit charge at point a and similarly for V_b . We call V_a and V_b the *potential at point a* and *potential at point b* , respectively. Thus the work done per unit charge by the electric force when a charged object moves from a to b is equal to the potential at a minus the potential at b .

The difference $V_a - V_b$ is called the *potential of a with respect to b* ; we sometimes abbreviate this difference as $V_{ab} = V_a - V_b$ (note the order of the subscripts). This is often called the potential difference between a and b , but that’s ambiguous unless we specify which is the reference point. In electric circuits, which we’ll analyze in later chapters, the potential difference between two points is often called **voltage** (Fig. 23.11). Equation (23.13) then states: **V_{ab} , the potential (in V) of a with respect to b , equals the work (in J) done by the electric force when a UNIT (1 C) charge moves from a to b .**

Another way to interpret the potential difference V_{ab} in Eq. (23.13) is to use the alternative viewpoint mentioned at the end of Section 23.1. In that viewpoint, $U_a - U_b$ is the amount of work that must be done by an *external* force to move a particle of charge q_0 slowly from b to a against the electric force. The work that must be done *per unit charge* by the external force is then $(U_a - U_b)/q_0 = V_a - V_b = V_{ab}$. In other words: **V_{ab} , the potential (in V) of a with respect to b , equals the work (in J) that must be done to move a UNIT (1 C) charge slowly from b to a against the electric force.**

An instrument that measures the difference of potential between two points is called a *voltmeter*. (In Chapter 26 we’ll discuss how these devices work.) Voltmeters that can measure a potential difference of 1 μV are common, and sensitivities down to 10^{-12} V can be attained.

Figure 23.11 The voltage of this battery equals the difference in potential $V_{ab} = V_a - V_b$ between its positive terminal (point a) and its negative terminal (point b).



Calculating Electric Potential

To find the potential V due to a single point charge q , we divide Eq. (23.9) by q_0 :

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (23.14)$$

Electric potential due to a point charge V is equal to the electric constant $1/(4\pi\epsilon_0)$ times the value of point charge q divided by the distance from point charge to where potential is measured r .

If q is positive, the potential that it produces is positive at all points; if q is negative, it produces a potential that is negative everywhere. In either case, V is equal to zero at $r = \infty$, an infinite distance from the point charge. Note that potential, like electric field, is independent of the test charge q_0 that we use to define it.

Similarly, we divide Eq. (23.10) by q_0 to find the potential due to a collection of point charges:

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i} \quad (23.15)$$

Electric potential due to a collection of point charges V is equal to the electric constant $1/(4\pi\epsilon_0)$ times the sum of the values of i th point charge q_i divided by the distance from i th point charge to where potential is measured r_i .

Just as the electric field due to a collection of point charges is the *vector* sum of the fields produced by each charge, the electric potential due to a collection of point charges is the *scalar* sum of the potentials due to each charge. When we have a continuous distribution of charge along a line, over a surface, or through a volume, we divide the charge into elements dq , and the sum in Eq. (23.15) becomes an integral:

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r} \quad (23.16)$$

Electric potential due to a continuous distribution of charge V is equal to the electric constant $1/(4\pi\epsilon_0)$ times the integral over charge distribution of the charge element dq divided by the distance from charge element to where potential is measured r .

We'll work out several examples of such cases. The potential defined by Eqs. (23.15) and (23.16) is zero at points that are infinitely far away from *all* the charges. Later we'll encounter cases in which the charge distribution itself extends to infinity. We'll find that in such cases we cannot set $V = 0$ at infinity, and we'll need to exercise care in using and interpreting Eqs. (23.15) and (23.16).

CAUTION What is electric potential? Before getting too involved in the details of how to calculate electric potential, remind yourself what potential is. The electric *potential* at a certain point is the potential energy per *unit* charge placed at that point. That's why potential is measured in joules per coulomb, or volts. Keep in mind, too, that there doesn't have to be a charge at a given point for a potential V to exist at that point. (In the same way, an electric field can exist at a given point even if there's no charge there to respond to it.)

Finding Electric Potential from Electric Field

When we are given a collection of point charges, Eq. (23.15) is usually the easiest way to calculate the potential V . But in some problems in which the electric field is known or can be found easily, it is easier to determine V from $\vec{E}$. The force $\vec{F}$ on a test charge q_0 can be written as $\vec{F} = q_0\vec{E}$, so from Eq. (23.1) the work done by the electric force as the test charge moves from a to b is given by

$$W_{a \rightarrow b} = \int_a^b \vec{F} \cdot d\vec{l} = \int_a^b q_0 \vec{E} \cdot d\vec{l}$$

BIO APPLICATION Electrocardiography

The electrodes used in an electrocardiogram—EKG or ECG for short—measure the potential differences (typically no greater than $1 \text{ mV} = 10^{-3} \text{ V}$) between different parts of the patient's skin. These are indicative of the potential differences between regions of the heart, and so provide a sensitive way to detect any abnormalities in the electrical activity that drives cardiac function.



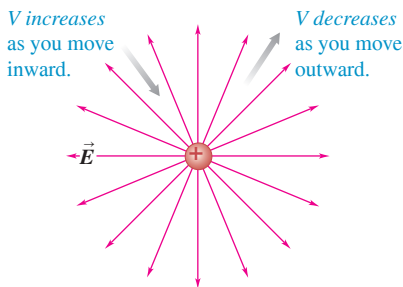
If we divide this by q_0 and compare the result with Eq. (23.13), we find

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b E \cos \phi \, dl \quad (23.17)$$

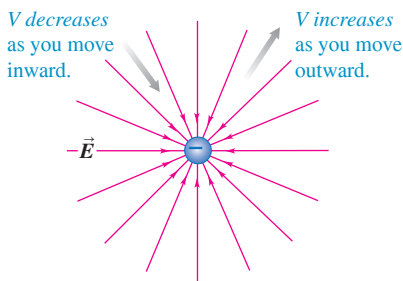
Electric potential difference $V_a - V_b$ Integral along path from a to b Displacement $d\vec{l}$ Angle between $\vec{E}$ and $d\vec{l}$ Electric-field magnitude E Scalar product of electric field and displacement vector

Figure 23.12 If you move in the direction of $\vec{E}$, electric potential V decreases; if you move in the direction opposite $\vec{E}$, V increases.

(a) A positive point charge



(b) A negative point charge



The value of $V_a - V_b$ is independent of the path taken from a to b , just as the value of $W_{a \rightarrow b}$ is independent of the path. To interpret Eq. (23.17), remember that $\vec{E}$ is the electric force per unit charge on a test charge. If the line integral $\int_a^b \vec{E} \cdot d\vec{l}$ is positive, the electric field does positive work on a positive test charge as it moves from a to b . In this case the electric potential energy decreases as the test charge moves, so the potential energy per unit charge decreases as well; hence V_b is less than V_a and $V_a - V_b$ is positive.

As an illustration, consider a positive point charge (Fig. 23.12a). The electric field is directed away from the charge, and $V = q/4\pi\epsilon_0 r$ is positive at any finite distance from the charge. If you move away from the charge, in the direction of $\vec{E}$, you move toward lower values of V ; if you move toward the charge, in the direction opposite $\vec{E}$, you move toward greater values of V . For the negative point charge in Fig. 23.12b, $\vec{E}$ is directed toward the charge and $V = q/4\pi\epsilon_0 r$ is negative at any finite distance from the charge. In this case, if you move toward the charge, you are moving in the direction of $\vec{E}$ and in the direction of decreasing (more negative) V . Moving away from the charge, in the direction opposite $\vec{E}$, moves you toward increasing (less negative) values of V . The general rule, valid for any electric field, is: Moving *with* the direction of $\vec{E}$ means moving in the direction of *decreasing* V , and moving *against* the direction of $\vec{E}$ means moving in the direction of *increasing* V .

Also, a positive test charge q_0 experiences an electric force in the direction of $\vec{E}$, toward lower values of V ; a negative test charge experiences a force opposite $\vec{E}$, toward higher values of V . Thus a positive charge tends to “fall” from a high-potential region to a lower-potential region. The opposite is true for a negative charge.

Notice that Eq. (23.17) can be rewritten as

$$V_a - V_b = - \int_b^a \vec{E} \cdot d\vec{l} \quad (23.18)$$

This has a negative sign compared to the integral in Eq. (23.17), and the limits are reversed; hence Eqs. (23.17) and (23.18) are equivalent. But Eq. (23.18) has a slightly different interpretation. To move a unit charge slowly against the electric force, we must apply an *external* force per unit charge equal to $-\vec{E}$, equal and opposite to the electric force per unit charge $\vec{E}$. Equation (23.18) says that $V_a - V_b = V_{ab}$, the potential of a with respect to b , equals the work done per unit charge by this external force to move a unit charge from b to a . This is the same alternative interpretation we discussed under Eq. (23.13).

Equations (23.17) and (23.18) show that the unit of potential difference (1 V) is equal to the unit of electric field (1 N/C) multiplied by the unit of distance (1 m). Hence the unit of electric field can be expressed as 1 volt per meter (1 V/m) as well as 1 N/C:

$$1 \text{ V/m} = 1 \text{ volt/meter} = 1 \text{ N/C} = 1 \text{ newton/coulomb}$$

In practice, the volt per meter is the usual unit of electric-field magnitude.

Electron Volts

The magnitude e of the electron charge can be used to define a unit of energy that is useful in many calculations with atomic and nuclear systems. When a particle with charge q moves from a point where the potential is V_b to a point where it is V_a , the change in the potential energy U is

$$U_a - U_b = q(V_a - V_b) = qV_{ab}$$

If charge q equals the magnitude e of the electron charge, 1.602×10^{-19} C, and the potential difference is $V_{ab} = 1$ V = 1 J/C, the change in energy is

$$U_a - U_b = (1.602 \times 10^{-19} \text{ C})(1 \text{ J/C}) = 1.602 \times 10^{-19} \text{ J}$$

This quantity of energy is defined to be 1 **electron volt** (1 eV):

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

The multiples meV, keV, MeV, GeV, and TeV are often used.

CAUTION Electron volts vs. volts Remember that the electron volt is a unit of energy, *not* a unit of potential or potential difference!

When a particle with charge e moves through a potential difference of 1 volt, the change in potential *energy* is 1 eV. If the charge is some multiple of e —say, Ne —the change in potential energy in electron volts is N times the potential difference in volts. For example, when an alpha particle, which has charge $2e$, moves between two points with a potential difference of 1000 V, the change in potential energy is $2(1000 \text{ eV}) = 2000 \text{ eV}$. To confirm this, we write

$$\begin{aligned} U_a - U_b &= qV_{ab} = (2e)(1000 \text{ V}) = (2)(1.602 \times 10^{-19} \text{ C})(1000 \text{ V}) \\ &= 3.204 \times 10^{-16} \text{ J} = 2000 \text{ eV} \end{aligned}$$

Although we defined the electron volt in terms of *potential* energy, we can use it for *any* form of energy, such as the kinetic energy of a moving particle. When we speak of a “one-million-electron-volt proton,” we mean a proton with a kinetic energy of one million electron volts (1 MeV), equal to $(10^6)(1.602 \times 10^{-19} \text{ J}) = 1.602 \times 10^{-13} \text{ J}$. The Large Hadron Collider near Geneva, Switzerland, is designed to accelerate protons to a kinetic energy of 7 TeV (7×10^{12} eV).

BIO APPLICATION Electron Volts and Cancer Radiotherapy

One way to destroy a cancerous tumor is to aim high-energy electrons directly at it. Each electron has a kinetic energy of 4 to 20 million electron volts, or MeV ($1 \text{ MeV} = 10^6 \text{ eV}$), and transfers its energy to the tumor through collisions with the tumor's atoms. Electrons in this energy range can penetrate only a few centimeters into a patient, which makes them useful for treating superficial tumors, such as those on the skin or lips.



EXAMPLE 23.3 Electric force and electric potential

WITH VARIATION PROBLEMS

A proton (charge $+e = 1.602 \times 10^{-19}$ C) moves a distance $d = 0.50$ m in a straight line between points a and b in a linear accelerator. The electric field is uniform along this line, with magnitude $E = 1.5 \times 10^7$ V/m = 1.5×10^7 N/C in the direction from a to b . Determine (a) the force on the proton; (b) the work done on it by the field; (c) the potential difference $V_a - V_b$.

IDENTIFY and SET UP This problem uses the relationship between electric field and electric force. It also uses the relationship among force, work, and potential-energy difference. We are given the electric field, so it is straightforward to find the electric force on the proton. Calculating the work is also straightforward because $\vec{E}$ is uniform, so the force on the proton is constant. Once the work is known, we find $V_a - V_b$ from Eq. (23.13).

EXECUTE (a) The force on the proton is in the same direction as the electric field, and its magnitude is

$$\begin{aligned} F &= qE = (1.602 \times 10^{-19} \text{ C})(1.5 \times 10^7 \text{ N/C}) \\ &= 2.4 \times 10^{-12} \text{ N} \end{aligned}$$

(b) The force is constant and in the same direction as the displacement, so the work done on the proton is

$$\begin{aligned} W_{a \rightarrow b} &= Fd = (2.4 \times 10^{-12} \text{ N})(0.50 \text{ m}) \\ &= 1.2 \times 10^{-12} \text{ J} \\ &= (1.2 \times 10^{-12} \text{ J}) \frac{1 \text{ eV}}{1.602 \times 10^{-19} \text{ J}} \\ &= 7.5 \times 10^6 \text{ eV} = 7.5 \text{ MeV} \end{aligned}$$

(c) From Eq. (23.13) the potential difference is the work per unit charge, which is

$$\begin{aligned} V_a - V_b &= \frac{W_{a \rightarrow b}}{q} = \frac{1.2 \times 10^{-12} \text{ J}}{1.602 \times 10^{-19} \text{ C}} \\ &= 7.5 \times 10^6 \text{ J/C} = 7.5 \times 10^6 \text{ V} = 7.5 \text{ MV} \end{aligned}$$

We can get this same result even more easily by remembering that 1 electron volt equals 1 volt multiplied by the charge e . The work done is 7.5×10^6 eV and the charge is e , so the potential difference is $(7.5 \times 10^6 \text{ eV})/e = 7.5 \times 10^6$ V.

EVALUATE We can check our result in part (c) by using Eq. (23.17) or (23.18). The angle ϕ between the constant field $\vec{E}$ and the displacement is zero, so Eq. (23.17) becomes

$$V_a - V_b = \int_a^b E \cos \phi \, dl = \int_a^b E \, dl = E \int_a^b dl$$

The integral of dl from a to b is just the distance d , so we again find

$$V_a - V_b = Ed = (1.5 \times 10^7 \text{ V/m})(0.50 \text{ m}) = 7.5 \times 10^6 \text{ V}$$

KEYCONCEPT The potential difference V_{ab} between point a and point b , equal to the difference $V_a - V_b$ of the potentials V at the two points, is the amount of work the electric force does on a unit charge as it moves from a to b . If you move in the direction of the electric field, V decreases (V_{ab} is positive); if you move opposite to the direction of the electric field, V increases (V_{ab} is negative).

EXAMPLE 23.4 Potential due to two point charges**WITH VARIATION PROBLEMS**

An electric dipole consists of point charges $q_1 = +12 \text{ nC}$ and $q_2 = -12 \text{ nC}$ placed 10.0 cm apart (**Fig. 23.13**). Compute the electric potentials at points a , b , and c .

IDENTIFY and SET UP This is the same arrangement as in Example 21.8 (Section 21.5), in which we calculated the electric *field* at each point by doing a *vector* sum. Here our target variable is the electric *potential* V at three points, which we find by doing the *algebraic* sum in Eq. (23.15).

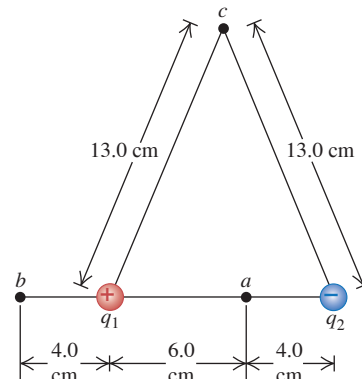
EXECUTE At point a we have $r_1 = 0.060 \text{ m}$ and $r_2 = 0.040 \text{ m}$, so Eq. (23.15) becomes

$$\begin{aligned} V_a &= \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i} = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2} \\ &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{12 \times 10^{-9} \text{ C}}{0.060 \text{ m}} \\ &\quad + (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(-12 \times 10^{-9} \text{ C})}{0.040 \text{ m}} \\ &= 1800 \text{ N} \cdot \text{m}/\text{C} + (-2700 \text{ N} \cdot \text{m}/\text{C}) \\ &= 1800 \text{ V} + (-2700 \text{ V}) = -900 \text{ V} \end{aligned}$$

In a similar way you can show that the potential at point b (where $r_1 = 0.040 \text{ m}$ and $r_2 = 0.140 \text{ m}$) is $V_b = 1930 \text{ V}$ and that the potential at point c (where $r_1 = r_2 = 0.130 \text{ m}$) is $V_c = 0$.

EVALUATE Let's confirm that these results make sense. Point a is closer to the -12 nC charge than to the $+12 \text{ nC}$ charge, so the potential at a is negative. The potential is positive at point b , which is closer to the $+12 \text{ nC}$ charge than the -12 nC charge. Finally, point c is equidistant from the $+12 \text{ nC}$ charge and the -12 nC charge, so the potential there is zero. (The potential is also equal to zero at a point infinitely far from both charges.)

Figure 23.13 What are the potentials at points a , b , and c due to this electric dipole?



Comparing this example with Example 21.8 shows that it's much easier to calculate electric potential (a scalar) than electric field (a vector). We'll take advantage of this simplification whenever possible.

KEYCONCEPT The electric potential caused at a point P by a single point charge is proportional to the charge and inversely proportional to the distance from the point charge to P . The total electric potential at P due to a system of charges is the sum of the potentials at P due to each individual charge.

EXAMPLE 23.5 Potential and potential energy**WITH VARIATION PROBLEMS**

Compute the potential energy associated with a $+4.0 \text{ nC}$ point charge if it is placed at points a , b , and c in Fig. 23.13.

IDENTIFY and SET UP The potential energy U associated with a point charge q at a location where the electric potential is V is $U = qV$. We use the values of V from Example 23.4.

EXECUTE At the three points we find

$$\begin{aligned} U_a &= qV_a = (4.0 \times 10^{-9} \text{ C})(-900 \text{ J/C}) = -3.6 \times 10^{-6} \text{ J} \\ U_b &= qV_b = (4.0 \times 10^{-9} \text{ C})(1930 \text{ J/C}) = 7.7 \times 10^{-6} \text{ J} \\ U_c &= qV_c = 0 \end{aligned}$$

All of these values correspond to U and V being zero at infinity.

EVALUATE Note that *zero* net work is done on the 4.0 nC charge if it moves from point c to infinity *by any path*. In particular, let the path be along the perpendicular bisector of the line joining the other two charges q_1 and q_2 in Fig. 23.13. As shown in Example 21.8 (Section 21.5), at points on the bisector, the direction of $\vec{E}$ is perpendicular to the bisector. Hence the force on the 4.0 nC charge is perpendicular to the path, and no work is done in any displacement along it.

KEYCONCEPT If you place a point charge q at a point P , the associated electric potential energy is equal to q multiplied by the value at P of the electric potential due to all *other* charges.

EXAMPLE 23.6 Finding potential by integration

By integrating the electric field as in Eq. (23.17), find the potential at a distance r from a point charge q .

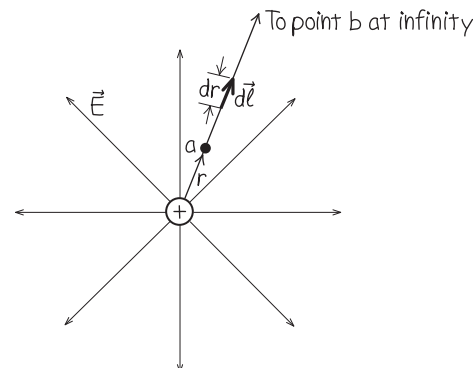
IDENTIFY and SET UP We let point a in Eq. (23.17) be at distance r and let point b be at infinity (Fig. 23.14). As usual, we choose the potential to be zero at an infinite distance from the charge q .

EXECUTE To carry out the integral, we can choose any path we like between points a and b . The most convenient path is a radial line as shown in Fig. 23.14, so that $d\vec{l}$ is in the radial direction and has magnitude dr . Writing $d\vec{l} = \hat{r}dr$, we have from Eq. (23.17)

$$\begin{aligned} V - 0 &= V = \int_r^\infty \vec{E} \cdot d\vec{l} \\ &= \int_r^\infty \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \cdot \hat{r} dr = \int_r^\infty \frac{q}{4\pi\epsilon_0 r^2} dr \\ &= -\frac{q}{4\pi\epsilon_0 r} \Big|_r^\infty = 0 - \left(-\frac{q}{4\pi\epsilon_0 r} \right) = \frac{q}{4\pi\epsilon_0 r} \end{aligned}$$

EVALUATE Our result agrees with Eq. (23.14) and is correct for positive or negative q .

Figure 23.14 Calculating the potential by integrating $\vec{E}$ for a single point charge.



KEYCONCEPT The potential difference $V_{ab} = V_a - V_b$ between point a and point b equals the integral of the electric field along a path from a to b . This integral does not depend on the path taken between the two points.

EXAMPLE 23.7 Moving through a potential difference**WITH VARIATION PROBLEMS**

In Fig. 23.15 a dust particle with mass $m = 5.0 \times 10^{-9} \text{ kg} = 5.0 \mu\text{g}$ and charge $q_0 = 2.0 \text{ nC}$ starts from rest and moves in a straight line from point a to point b . What is its speed v at point b ?

IDENTIFY and SET UP Only the conservative electric force acts on the particle, so the total mechanical energy is conserved: $K_a + U_a = K_b + U_b$. We get the potential energies U from the corresponding potentials V from Eq. (23.12): $U_a = q_0 V_a$ and $U_b = q_0 V_b$.

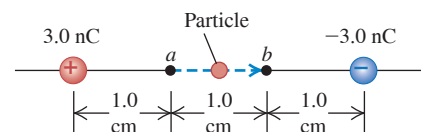
EXECUTE We have $K_a = 0$ and $K_b = \frac{1}{2}mv^2$. We substitute these and our expressions for U_a and U_b into the energy-conservation equation, then solve for v . We find

$$\begin{aligned} 0 + q_0 V_a &= \frac{1}{2}mv^2 + q_0 V_b \\ v &= \sqrt{\frac{2q_0(V_a - V_b)}{m}} \end{aligned}$$

We calculate the potentials from Eq. (23.15), $V = q/4\pi\epsilon_0 r$:

$$\begin{aligned} V_a &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \left(\frac{3.0 \times 10^{-9} \text{ C}}{0.010 \text{ m}} + \frac{(-3.0 \times 10^{-9} \text{ C})}{0.020 \text{ m}} \right) \\ &= 1350 \text{ V} \\ V_b &= (9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \left(\frac{3.0 \times 10^{-9} \text{ C}}{0.020 \text{ m}} + \frac{(-3.0 \times 10^{-9} \text{ C})}{0.010 \text{ m}} \right) \\ &= -1350 \text{ V} \\ V_a - V_b &= (1350 \text{ V}) - (-1350 \text{ V}) = 2700 \text{ V} \end{aligned}$$

Figure 23.15 The particle moves from point a to point b ; its acceleration is not constant.



Finally,

$$v = \sqrt{\frac{2(2.0 \times 10^{-9} \text{ C})(2700 \text{ V})}{5.0 \times 10^{-9} \text{ kg}}} = 46 \text{ m/s}$$

EVALUATE Our result makes sense: The positive dust particle speeds up as it moves away from the $+3.0 \text{ nC}$ charge and toward the -3.0 nC charge. To check unit consistency in the final line of the calculation, note that $1 \text{ V} = 1 \text{ J/C}$, so the numerator under the radical has units of J or $\text{kg} \cdot \text{m}^2/\text{s}^2$.

KEYCONCEPT When a particle of charge q moves from a point of higher potential to a point of lower potential, its potential energy *decreases* if q is positive; its potential energy *increases* if q is negative.

TEST YOUR UNDERSTANDING OF SECTION 23.2 If the electric *potential* at a certain point is zero, does the electric *field* at that point have to be zero? (*Hint*: Consider point c in Examples 23.4 and 21.8.)

ANSWER

no If $V = 0$ at a certain point, $\vec{E}$ does *not* have to be zero at that point. An example is point c in Figs. 21.23 and 23.13, for which there is an electric field in the $+x$ -direction (see Example 21.9 in Section 21.5) even though $V = 0$ (see Example 23.4). This isn't a surprising result because V and $\vec{E}$ are quite different quantities: V is the net amount of work required to bring a unit charge from infinity to the point in question, whereas $\vec{E}$ is the electric force that acts on a unit charge when it arrives at that point.

23.3 CALCULATING ELECTRIC POTENTIAL

When calculating the potential due to a charge distribution, we usually follow one of two routes. If we know the charge distribution, we can use Eq. (23.15) or (23.16). Or if we know how the electric field depends on position, we can use Eq. (23.17), defining the potential to be zero at some convenient place. Some problems require a combination of these approaches.

As you read through these examples, compare them with the related examples of calculating electric *field* in Section 21.5. You'll see how much easier it is to calculate scalar electric potentials than vector electric fields. The moral is clear: Whenever possible, solve problems by means of an energy approach (using electric potential and electric potential energy) rather than a dynamics approach (using electric fields and electric forces).

PROBLEM-SOLVING STRATEGY 23.1 Calculating Electric Potential

IDENTIFY *the relevant concepts*: Remember that electric potential is *potential energy per unit charge*.

SET UP *the problem* using the following steps:

1. Make a drawing showing the locations and values of the charges (which may be point charges or a continuous distribution of charge) and your choice of coordinate axes.
2. Indicate on your drawing the position of the point at which you want to calculate the electric potential V . Sometimes this position will be an arbitrary one (say, a point a distance r from the center of a charged sphere).

EXECUTE *the solution* as follows:

1. To find the potential due to a collection of point charges, use Eq. (23.15). If you are given a continuous charge distribution, devise a way to divide it into infinitesimal elements and use Eq. (23.16). Carry out the integration, using appropriate limits to include the entire charge distribution.
2. If you are given the electric field, or if you can find it from any of the methods presented in Chapters 21 or 22, it may be easier

to find the potential difference between points a and b from Eq. (23.17) or (23.18). When appropriate, make use of your freedom to define V to be zero at some convenient place, and choose this place to be point b . (For point charges, this will usually be at infinity. For other distributions of charge—especially those that themselves extend to infinity—it may be necessary to define V_b to be zero at some finite distance from the charge distribution.) Then the potential at any other point, say a , can be found from Eq. (23.17) or (23.18) with $V_b = 0$.

3. Although potential V is a *scalar* quantity, you may have to use components of the vectors $\vec{E}$ and $d\vec{l}$ when you use Eq. (23.17) or (23.18) to calculate V .

EVALUATE *your answer*: Check whether your answer agrees with your intuition. If your result gives V as a function of position, graph the function to see whether it makes sense. If you know the electric field, you can make a rough check of your result for V by verifying that V decreases if you move in the direction of $\vec{E}$.

EXAMPLE 23.8 A charged conducting sphere**WITH VARIATION PROBLEMS**

A solid conducting sphere of radius R has a total charge q . Find the electric potential everywhere, both outside and inside the sphere.

IDENTIFY and SET UP In Example 22.5 (Section 22.4) we used Gauss's law to find the electric *field* at all points for this charge distribution. We can use that result to determine the potential.

EXECUTE From Example 22.5, the field *outside* the sphere is the same as if the sphere were removed and replaced by a point charge q . We take $V = 0$ at infinity, as we did for a point charge. Then the potential at a point outside the sphere at a distance r from its center is the same as that due to a point charge q at the center:

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

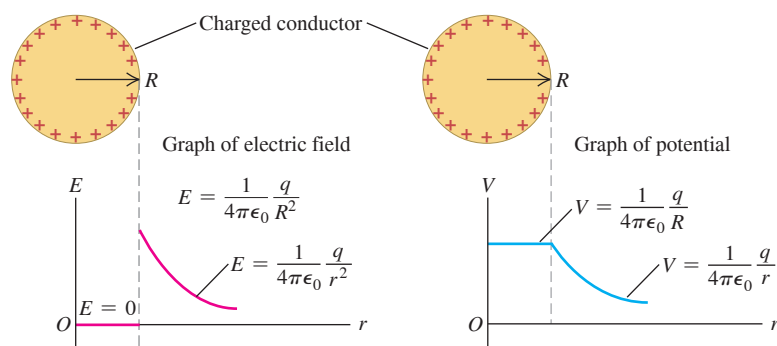
The potential at the surface of the sphere is $V_{\text{surface}} = q/4\pi\epsilon_0 R$.

Inside the sphere, $\vec{E}$ is zero everywhere. Hence no work is done on a test charge that moves from any point to any other point inside the sphere. Thus the potential is the same at every point inside the sphere and is equal to its value $q/4\pi\epsilon_0 R$ at the surface.

EVALUATE Figure 23.16 shows the field and potential for a positive charge q . In this case the electric field points radially away from the sphere. As you move away from the sphere, in the direction of $\vec{E}$, V decreases (as it should).

KEYCONCEPT If there is excess charge at rest on a solid spherical conductor, all of the charge lies on the surface of the conductor, and the electric field and electric potential outside the sphere are the same as if all the charge were concentrated at the center of the sphere. Everywhere inside the sphere the field is zero and the potential has the same value as at the surface.

Figure 23.16 Electric-field magnitude E and potential V at points inside and outside a positively charged spherical conductor.

**Ionization and Corona Discharge**

The results of Example 23.8 have numerous practical consequences. One consequence relates to the maximum potential to which a conductor in air can be raised. This potential is limited because air molecules become *ionized*, and air becomes a conductor, at an electric-field magnitude of about 3×10^6 V/m. Assume for the moment that q is positive. When we compare the expressions in Example 23.8 for the potential V_{surface} and field magnitude E_{surface} at the surface of a charged conducting sphere, we note that $V_{\text{surface}} = E_{\text{surface}} R$. Thus, if E_m represents the electric-field magnitude at which air becomes conductive (known as the *dielectric strength* of air), then the maximum potential V_m to which a spherical conductor can be raised is

$$V_m = RE_m$$

For a conducting sphere 1 cm in radius in air, $V_m = (10^{-2} \text{ m})(3 \times 10^6 \text{ V/m}) = 30,000 \text{ V}$. No amount of “charging” could raise the potential of a conducting sphere of this size in air higher than about 30,000 V; attempting to raise the potential further by adding extra charge would cause the surrounding air to become ionized and conductive, and the extra added charge would leak into the air.

To attain even higher potentials, high-voltage machines such as Van de Graaff generators use spherical terminals with very large radii (see Fig. 22.26 and the photograph that opens Chapter 22). For example, a terminal of radius $R = 2 \text{ m}$ has a maximum potential $V_m = (2 \text{ m})(3 \times 10^6 \text{ V/m}) = 6 \times 10^6 \text{ V} = 6 \text{ MV}$.

Figure 23.17 The metal mast at the top of the Empire State Building acts as a lightning rod. It is struck by lightning as many as 500 times each year.



Our result in Example 23.8 also explains what happens with a charged conductor with a very *small* radius of curvature, such as a sharp point or thin wire. Because the maximum potential is proportional to the radius, even relatively small potentials applied to sharp points in air produce sufficiently high fields just outside the point to ionize the surrounding air, making it become a conductor. The resulting current and its associated glow (visible in a dark room) are called *corona discharge*. Laser printers and photocopying machines use corona discharge from fine wires to spray charge on the imaging drum (see Fig. 21.2).

A large-radius conductor is used in situations where it's important to *prevent* corona discharge. An example is the blunt end of a metal lightning rod (Fig. 23.17). If there is an excess charge in the atmosphere, as happens during thunderstorms, a substantial charge of the opposite sign can build up on this blunt end. As a result, when the atmospheric charge is discharged through a lightning bolt, it tends to be attracted to the charged lightning rod rather than to other structures that could be damaged. (A conducting wire connecting the lightning rod to the ground then allows the acquired charge to dissipate harmlessly.) A lightning rod with a sharp end would allow less charge buildup and hence be less effective.

EXAMPLE 23.9 Oppositely charged parallel plates

WITH VARIATION PROBLEMS

Find the potential at any height y between the two oppositely charged parallel plates discussed in Section 23.1 (Fig. 23.18).

IDENTIFY and SET UP We discussed this situation in Section 23.1. From Eq. (23.5), we know the electric *potential energy* U for a test charge q_0 is $U = q_0 E y$. (We set $y = 0$ and $U = 0$ at the bottom plate.) We use Eq. (23.12), $U = q_0 V$, to find the electric *potential* V as a function of y .

EXECUTE The potential $V(y)$ at coordinate y is the potential energy per unit charge:

$$V(y) = \frac{U(y)}{q_0} = \frac{q_0 E y}{q_0} = E y$$

The potential decreases as we move in the direction of $\vec{E}$ from the upper to the lower plate. At point a , where $y = d$ and $V(y) = V_a$,

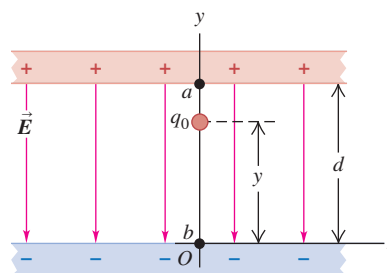
$$V_a - V_b = E d$$

and

$$E = \frac{V_a - V_b}{d} = \frac{V_{ab}}{d}$$

where V_{ab} is the potential of the positive plate with respect to the negative plate. That is, the electric field equals the potential difference between the plates divided by the distance between them. For a given potential difference V_{ab} , the smaller the distance d between the two plates, the greater the magnitude E of the electric field. (This relationship between E and V_{ab} holds *only* for the planar geometry we have described. It does *not* work for situations such as concentric cylinders or spheres in which the electric field is not uniform.)

Figure 23.18 The charged parallel plates from Fig. 23.2.



EVALUATE Our result shows that $V = 0$ at the bottom plate (at $y = 0$). This is consistent with our choice that $U = q_0 V = 0$ for a test charge placed at the bottom plate.

CAUTION “Zero potential” is arbitrary You might think that if a conducting object has zero potential, it must also have zero net charge. But that just isn't so! As an example, the plate at $y = 0$ in Fig. 23.18 has zero potential ($V = 0$) but has a nonzero charge per unit area $-\sigma$. There's nothing special about the place where potential is zero; we *define* this place to be wherever we want it to be.

KEYCONCEPT In a region of uniform electric field $\vec{E}$, such as between two oppositely charged parallel plates, the electric potential decreases linearly with position as you move in the direction of $\vec{E}$.

EXAMPLE 23.10 An infinite line charge or charged conducting cylinder**WITH VARIATION PROBLEMS**

Find the potential at a distance r from a very long line of charge with linear charge density (charge per unit length) λ .

IDENTIFY and SET UP In both Example 21.10 (Section 21.5) and Example 22.6 (Section 22.4) we found that the electric field at a radial distance r from a long straight-line charge (**Fig. 23.19a**) has only a radial component $E_r = \lambda/2\pi\epsilon_0 r$. We use this expression to find the potential by integrating $\vec{E}$ as in Eq. (23.17).

EXECUTE Since the field has only a radial component, we have $\vec{E} \cdot d\vec{l} = E_r dr$. Hence from Eq. (23.17) the potential of any point a with respect to any other point b , at radial distances r_a and r_b from the line of charge, is

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b E_r dr = \frac{\lambda}{2\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}$$

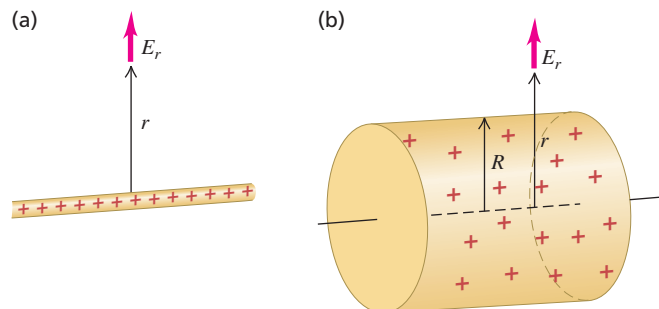
If we take point b at infinity and set $V_b = 0$, we find that V_a is *infinite* for any finite distance r_a from the line charge: $V_a = (\lambda/2\pi\epsilon_0) \ln(\infty/r_a) = \infty$. This is *not* a useful way to define V for this problem! The difficulty is that the charge distribution itself extends to infinity.

Instead, as recommended in Problem-Solving Strategy 23.1, we set $V_b = 0$ at point b at an arbitrary but *finite* radial distance r_0 . Then the potential $V = V_a$ at point a at a radial distance r is given by $V - 0 = (\lambda/2\pi\epsilon_0) \ln(r_0/r)$, or

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_0}{r}$$

EVALUATE According to our result, if λ is positive, then V decreases as r increases. This is as it should be: V decreases as we move in the direction of $\vec{E}$.

Figure 23.19 Electric field outside (a) a long, positively charged wire and (b) a long, positively charged cylinder.



From Example 22.6, the expression for E_r with which we started also applies outside a long, charged conducting cylinder with charge per unit length λ (**Fig. 23.19b**). Hence our result also gives the potential for such a cylinder, but only for values of r (the distance from the cylinder axis) equal to or greater than the radius R of the cylinder. If we choose r_0 to be the radius R , so that $V = 0$ when $r = R$, then at any point for which $r > R$,

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R}{r}$$

Inside the cylinder, $\vec{E} = 0$, and V has the same value (zero) as on the cylinder's surface.

KEYCONCEPT The electric potential outside a very long cylindrical distribution of charge or line of charge decreases logarithmically with the distance from the axis of the distribution or line.

EXAMPLE 23.11 A ring of charge**WITH VARIATION PROBLEMS**

Electric charge Q is distributed uniformly around a thin ring of radius a (**Fig. 23.20**). Find the potential at a point P on the ring axis at a distance x from the center of the ring.

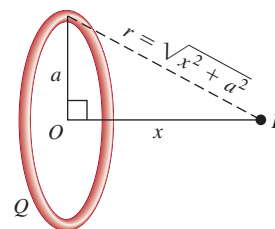
IDENTIFY and SET UP We divide the ring into infinitesimal segments and use Eq. (23.16) to find V . All parts of the ring (and therefore all elements of the charge distribution) are at the same distance from P .

EXECUTE Figure 23.20 shows that the distance from each charge element dq to P is $r = \sqrt{x^2 + a^2}$. Hence we can take the factor $1/r$ outside the integral in Eq. (23.16), and

$$\begin{aligned} V &= \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r} \\ &= \frac{1}{4\pi\epsilon_0} \frac{1}{\sqrt{x^2 + a^2}} \int dq \\ &= \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{x^2 + a^2}} \end{aligned}$$

EVALUATE When x is much larger than a , our expression for V becomes approximately $V = Q/4\pi\epsilon_0 x$, which is the potential at a distance x from a point charge Q . Very far from a charged ring, its electric potential looks like that of a point charge. We drew a similar conclusion about the electric *field* of a ring in Example 21.9 (Section 21.5).

Figure 23.20 All the charge in a ring of charge Q is the same distance r from a point P on the ring axis.



We know the electric field at all points along the x -axis from Example 21.9 (Section 21.5), so we can also find V along this axis by integrating $\vec{E} \cdot d\vec{l}$ as in Eq. (23.17).

KEYCONCEPT To find the electric potential at a point due to a continuous distribution of charge, first divide the distribution into infinitesimally small segments. Then find the potential at the point due to one such segment. Finally, integrate over all segments in the charge distribution.

EXAMPLE 23.12 Potential of a line of chargeWITH  **VARIATION PROBLEMS**

Positive electric charge Q is distributed uniformly along a line of length $2a$ lying along the y -axis between $y = -a$ and $y = +a$ (**Fig. 23.21**). Find the electric potential at a point P on the x -axis at a distance x from the origin.

IDENTIFY and SET UP This is the situation of Example 21.10 (Section 21.5), where we found an expression for the electric field $\vec{E}$ at an arbitrary point on the x -axis. We can find V at point P by using Eq. (23.16) to integrate over the charge distribution. Unlike the situation in Example 23.11, each charge element dQ is a *different* distance from point P , so the integration will take a little more effort.

EXECUTE As in Example 21.10, the element of charge dQ corresponding to an element of length dy on the rod is $dQ = (Q/2a)dy$. The distance from dQ to P is $\sqrt{x^2 + y^2}$, so the contribution dV that the charge element makes to the potential at P is

$$dV = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{dy}{\sqrt{x^2 + y^2}}$$

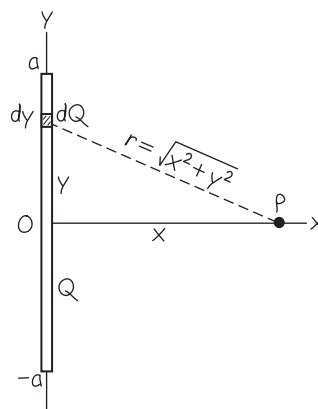
To find the potential at P due to the entire rod, we integrate dV over the length of the rod from $y = -a$ to $y = a$:

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \int_{-a}^a \frac{dy}{\sqrt{x^2 + y^2}}$$

You can look up the integral in a table. The final result is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \ln \left(\frac{\sqrt{a^2 + x^2} + a}{\sqrt{a^2 + x^2} - a} \right)$$

Figure 23.21 Our sketch for this problem.



EVALUATE We can check our result by letting x approach infinity. In this limit the point P is infinitely far from all of the charge, so we expect V to approach zero; you can verify that it does.

We know the electric field at all points along the x -axis from Example 21.10. We invite you to use this information to find V along this axis by integrating $\vec{E}$ as in Eq. (23.17).

KEYCONCEPT The electric potential due to a symmetrical distribution of charge is most easily calculated at a point of symmetry. Whenever possible, take advantage of the symmetry of the situation to check your results.

TEST YOUR UNDERSTANDING OF SECTION 23.3 If the electric *field* at a certain point is zero, does the electric *potential* at that point have to be zero? (*Hint:* Consider the center of the ring in Examples 23.11 and 21.9.)

ANSWER

no If $\vec{E} = 0$ at a certain point, V does *not* have to be zero at that point. An example is point O at the center of the charged ring in Figs. 21.23 and 23.21. From Example 21.9 (Section 21.5), the electric field is zero at O because the electric-field contributions from different parts of the ring completely cancel. From Example 23.11, however, the potential at O is *not* zero. This point corresponds to $x = 0$, so $V = (1/4\pi\epsilon_0)(Q/a)$. This value of V corresponds to the work that would have to be done to move a unit positive test charge along a path from infinity to point O ; it is nonzero because the charged ring repels the test charge, so positive work must be done to move the test charge toward the ring.

23.4 EQUIPOTENTIAL SURFACES

Field lines (see Section 21.6) help us visualize electric fields. In a similar way, the potential at various points in an electric field can be represented graphically by *equipotential surfaces*. These use the same fundamental idea as topographic maps like those used by hikers and mountain climbers (**Fig. 23.22**). On a topographic map, contour lines are drawn through points that are all at the same elevation. Any number of these could be drawn, but typically only a few contour lines are shown at equal spacings of elevation. If a mass m is moved over the terrain along such a contour line, the gravitational potential energy mgy does not change because the elevation y is constant. Thus contour lines on a topographic map are really curves of constant gravitational potential energy. Contour lines are close together where the terrain is steep and there are large changes in elevation over a small horizontal distance; the contour lines are farther apart where the terrain

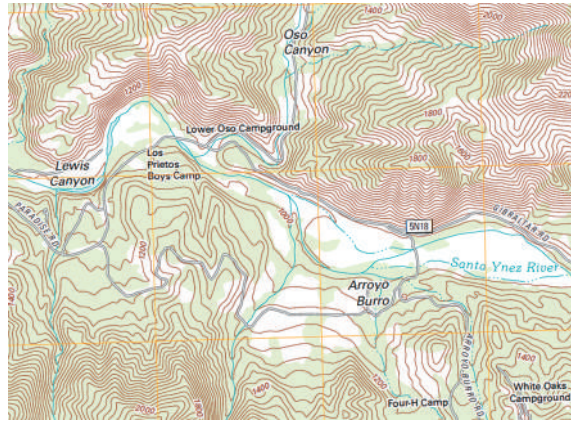


Figure 23.22 Contour lines on a topographic map are curves of constant elevation and hence of constant gravitational potential energy.

is gently sloping. A ball allowed to roll downhill will experience the greatest downhill gravitational force where contour lines are closest together.

By analogy to contour lines on a topographic map, an **equipotential surface** is a three-dimensional surface on which the *electric potential* V is the same at every point. If a test charge q_0 is moved from point to point on such a surface, the *electric potential energy* q_0V remains constant. In a region where an electric field is present, we can construct an equipotential surface through any point. In diagrams we usually show only a few representative equipotentials, often with equal potential differences between adjacent surfaces. No point can be at two different potentials, so equipotential surfaces for different potentials can never touch or intersect.

Equipotential Surfaces and Field Lines

Because potential energy does not change as a test charge moves over an equipotential surface, the electric field can do no work on such a charge. It follows that $\vec{E}$ must be perpendicular to the surface at every point so that the electric force $q_0\vec{E}$ is always perpendicular to the displacement of a charge moving on the surface. **Field lines and equipotential surfaces are always mutually perpendicular.** In general, field lines are curves, and equipotentials are curved surfaces. For the special case of a *uniform* field, in which the field lines are straight, parallel, and equally spaced, the equipotentials are parallel *planes* perpendicular to the field lines.

Figure 23.23 shows three arrangements of charges. The field lines in the plane of the charges are represented by red lines, and the intersections of the equipotential surfaces with this plane (that is, cross sections of these surfaces) are shown as blue lines. The actual equipotential surfaces are three-dimensional. At each crossing of an equipotential and a field line, the two are perpendicular.

Figure 23.23 Cross sections of equipotential surfaces (blue lines) and electric field lines (red lines) for assemblies of point charges. There are equal potential differences between adjacent surfaces. Compare these diagrams to those in Fig. 21.28, which showed only the electric field lines.

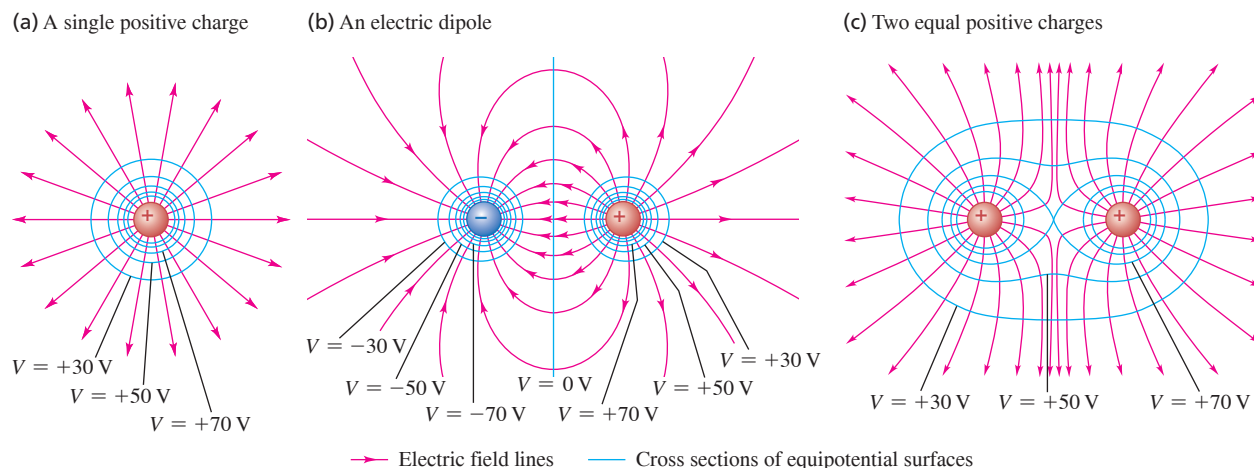


Figure 23.24 When charges are at rest, a conducting surface is always an equipotential surface. Field lines are perpendicular to a conducting surface.

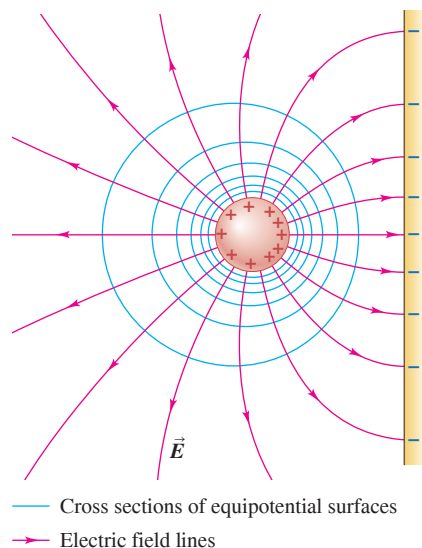


Figure 23.25 At all points on a conductor's surface, the electric field must be perpendicular to the surface. If $\vec{E}$ had a tangential component, a charge could move in a loop as shown—which is impossible because the electric force is conservative.

An impossible electric field

If the electric field just outside a conductor had a tangential component $E_{\parallel}$, a charge could move in a loop with net work done. Hence the field just outside a conductor can have only a perpendicular component.

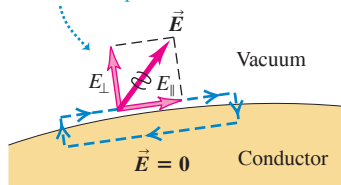
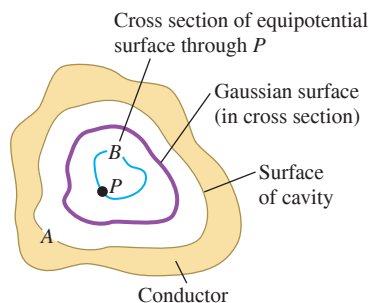


Figure 23.26 A cavity in a conductor. If the cavity contains no charge, every point in the cavity is at the same potential, the electric field is zero everywhere in the cavity, and there is no charge anywhere on the surface of the cavity.



In Fig. 23.23 we have drawn equipotentials so that there are equal potential differences between adjacent surfaces. In regions where the magnitude of $\vec{E}$ is large, the equipotential surfaces are close together because the field does a relatively large amount of work on a test charge in a relatively small displacement. This is the case near the point charge in Fig. 23.23a or between the two point charges in Fig. 23.23b; note that in these regions the field lines are also closer together. This is directly analogous to the downhill force of gravity being greatest in regions on a topographic map where contour lines are close together. Conversely, in regions where the field is weaker, the equipotential surfaces are farther apart; this happens at larger radii in Fig. 23.23a, to the left of the negative charge or the right of the positive charge in Fig. 23.23b, and at greater distances from both charges in Fig. 23.23c. (It may appear that two equipotential surfaces intersect at the center of Fig. 23.23c, in violation of the rule that this can never happen. In fact this is a single figure-8-shaped equipotential surface.)

CAUTION E need not be constant over an equipotential surface On a given equipotential surface, the potential V has the same value at every point. In general, however, the electric-field magnitude E is *not* the same at all points on an equipotential surface. For instance, on equipotential surface “ $V = -30$ V” in Fig. 23.23b, E is less to the left of the negative charge than it is between the two charges. On the figure-8-shaped equipotential surface in Fig. 23.23c, $E = 0$ at the middle point halfway between the two charges; at any other point on this surface, E is nonzero. ■

Equipotentials and Conductors

Here's an important statement about equipotential surfaces: **When all charges are at rest, the surface of a conductor is always an equipotential surface.** Since the electric field $\vec{E}$ is always perpendicular to an equipotential surface, we can prove this statement by proving that **when all charges are at rest, the electric field just outside a conductor must be perpendicular to the surface at every point** (Fig. 23.24). We know that $\vec{E} = 0$ everywhere inside the conductor; otherwise, charges would move. In particular, at any point just inside the surface the component of $\vec{E}$ tangent to the surface is zero. It follows that the tangential component of $\vec{E}$ is also zero just *outside* the surface. If it were not, a charge could move around a rectangular path partly inside and partly outside (Fig. 23.25) and return to its starting point with a net amount of work having been done on it. This would violate the conservative nature of electrostatic fields, so the tangential component of $\vec{E}$ just outside the surface must be zero at every point on the surface. Thus $\vec{E}$ is perpendicular to the surface at each point, proving our statement.

It also follows that **when all charges are at rest, the entire solid volume of a conductor is at the same potential.** Equation (23.17) states that the potential difference between two points a and b within the conductor's solid volume, $V_a - V_b$, is equal to the line integral $\int_a^b \vec{E} \cdot d\vec{l}$ of the electric field from a to b . Since $\vec{E} = 0$ everywhere inside the conductor, the integral is guaranteed to be zero for any two such points a and b . Hence the potential is the same for any two points within the solid volume of the conductor. We describe this by saying that the solid volume of the conductor is an *equipotential volume*.

We can now prove a theorem that we quoted without proof in Section 22.5. The theorem is as follows: In an electrostatic situation, if a conductor contains a cavity and if no charge is present inside the cavity, then there can be no net charge *anywhere* on the surface of the cavity. This means that if you're inside a charged conducting box, you can safely touch any point on the inside walls of the box without being shocked. To prove this theorem, we first prove that *every point in the cavity is at the same potential*. In Fig. 23.26 the conducting surface A of the cavity is an equipotential surface, as we have just proved. Suppose point P in the cavity is at a different potential; then we can construct a different equipotential surface B including point P .

Now consider a Gaussian surface, shown in Fig. 23.26, between the two equipotential surfaces. Because of the relationship between $\vec{E}$ and the equipotentials, we know that the field at every point between the equipotentials is from A toward B , or else at every point it is from B toward A , depending on which equipotential surface is at higher potential.

In either case the flux through this Gaussian surface is certainly not zero. But then Gauss's law says that the charge enclosed by the Gaussian surface cannot be zero. This contradicts our initial assumption that there is *no* charge in the cavity. So the potential at *P* *cannot* be different from that at the cavity wall.

The entire region of the cavity must therefore be at the same potential. But for this to be true, *the electric field inside the cavity must be zero everywhere*. Finally, Gauss's law shows that the electric field at any point on the surface of a conductor is proportional to the surface charge density σ at that point. We conclude that *the surface charge density on the wall of the cavity is zero at every point*. This chain of reasoning may seem tortuous, but it is worth careful study.

CAUTION Equipotential surfaces vs. Gaussian surfaces Don't confuse equipotential surfaces with the Gaussian surfaces we encountered in Chapter 22. Gaussian surfaces have relevance only when we are using Gauss's law, and we can choose *any* Gaussian surface that's convenient. We *cannot* choose equipotential surfaces; the shape is determined by the charge distribution. ■

TEST YOUR UNDERSTANDING OF SECTION 23.4 Would the shapes of the equipotential surfaces in Fig. 23.23 change if the sign of each charge were reversed?

ANSWER

If the positive charges in Fig. 23.23 were replaced by negative charges, and vice versa, the equipotential surfaces would be the same but the sign of the potential would be reversed. For example, the surfaces in Fig. 23.23b with potential $V = +30$ V and $V = -50$ V would have potential $V = -30$ V and $V = +50$ V, respectively.

23.5 POTENTIAL GRADIENT

Electric field and potential are closely related. Equation (23.17), restated here, expresses one aspect of that relationship:

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l}$$

If we know $\vec{E}$ at various points, we can use this equation to calculate potential differences. In this section we show how to turn this around; if we know the potential V at various points, we can use it to determine $\vec{E}$. Regarding V as a function of the coordinates (x, y, z) of a point in space, we'll show that the components of $\vec{E}$ are related to the *partial derivatives* of V with respect to x , y , and z .

In Eq. (23.17), $V_a - V_b$ is the potential of a with respect to b —that is, the change of potential encountered on a trip from b to a . We can write this as

$$V_a - V_b = \int_b^a dV = - \int_a^b dV$$

where dV is the infinitesimal change of potential accompanying an infinitesimal element $d\vec{l}$ of the path from b to a . Comparing to Eq. (23.17), we have

$$- \int_a^b dV = \int_a^b \vec{E} \cdot d\vec{l}$$

These two integrals must be equal for *any* pair of limits a and b , and for this to be true the *integrands* must be equal. Thus, for *any* infinitesimal displacement $d\vec{l}$,

$$-dV = \vec{E} \cdot d\vec{l}$$

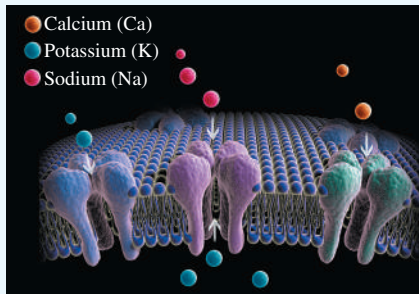
To interpret this expression, we write $\vec{E}$ and $d\vec{l}$ in terms of their components: $\vec{E} = \hat{i} E_x + \hat{j} E_y + \hat{k} E_z$ and $d\vec{l} = \hat{i} dx + \hat{j} dy + \hat{k} dz$. Then

$$-dV = E_x dx + E_y dy + E_z dz$$

Suppose the displacement is parallel to the x -axis, so $dy = dz = 0$. Then $-dV = E_x dx$ or $E_x = -(dV/dx)_{y,z \text{ constant}}$, where the subscript reminds us that only x varies in the

BIO APPLICATION Potential Gradient Across a Cell Membrane

The interior of a human cell is at a lower electric potential V than the exterior. (The potential difference when the cell is inactive is about -70 mV in neurons and about -95 mV in skeletal muscle cells.) Hence there is a potential gradient $\vec{\nabla}V$ that points from the *interior* to the *exterior* of the cell membrane, and an electric field $\vec{E} = -\vec{\nabla}V$ that points from the *exterior* to the *interior*. This field affects how ions flow into or out of the cell through special channels in the membrane.



derivative; recall that V is in general a function of x , y , and z . But this is just what is meant by the partial derivative $\partial V/\partial x$. The y - and z -components of $\vec{E}$ are related to the corresponding derivatives of V in the same way, so

Electric field components found from potential: $E_x = -\frac{\partial V}{\partial x}$ $E_y = -\frac{\partial V}{\partial y}$ $E_z = -\frac{\partial V}{\partial z}$ (23.19)

Each electric field component ... equals the negative of the corresponding partial derivative of electric potential function V .

This is consistent with the units of electric field being V/m. In terms of unit vectors we can write

Electric field vector found from potential: $\vec{E} = -\left(\hat{i}\frac{\partial V}{\partial x} + \hat{j}\frac{\partial V}{\partial y} + \hat{k}\frac{\partial V}{\partial z}\right)$ (23.20)

Partial derivatives of electric potential function V

The following operation is called the **gradient** of the function f :

$$\vec{\nabla}f = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)f \quad (23.21)$$

The operator denoted by $\vec{\nabla}$ is called “grad” or “del.” Thus in vector notation,

$$\vec{E} = -\vec{\nabla}V \quad (23.22)$$

This is read “ $\vec{E}$ is the negative of the gradient of V ” or “ $\vec{E}$ equals negative grad V .” The quantity $\vec{\nabla}V$ is called the *potential gradient*.

At each point, the potential gradient $\vec{\nabla}V$ points in the direction in which V *increases* most rapidly with a change in position. Hence at each point the direction of $\vec{E} = -\vec{\nabla}V$ is the direction in which V *decreases* most rapidly and is always perpendicular to the equipotential surface through the point. This agrees with our observation in Section 23.2 that moving in the direction of the electric field means moving in the direction of decreasing potential.

Equation (23.22) doesn’t depend on the particular choice of the zero point for V . If we were to change the zero point, the effect would be to change V at every point by the same amount; the derivatives of V would be the same.

If $\vec{E}$ has a radial component E_r with respect to a point or an axis and r is the distance from the point or axis, the relationship corresponding to Eqs. (23.19) is

$$E_r = -\frac{\partial V}{\partial r} \quad (\text{radial electric field component}) \quad (23.23)$$

Often we can compute the electric field caused by a charge distribution in either of two ways: directly, by adding the $\vec{E}$ fields of point charges, or by first calculating the potential and then taking its gradient to find the field. The second method is often easier because potential is a *scalar* quantity, requiring at worst the integration of a scalar function. Electric field is a *vector* quantity, requiring computation of components for each element of charge and a separate integration for each component. Thus, quite apart from its fundamental significance, potential offers a very useful computational technique in field calculations. In the next two examples, a knowledge of V is used to find the electric field.

We stress once more that if we know $\vec{E}$ as a function of position, we can calculate V from Eq. (23.17) or (23.18), and if we know V as a function of position, we can calculate $\vec{E}$ from Eq. (23.19), (23.20), or (23.23). Deriving V from $\vec{E}$ requires integration, and deriving $\vec{E}$ from V requires differentiation.

EXAMPLE 23.13 Potential and field of a point charge

From Eq. (23.14) the potential at a radial distance r from a point charge q is $V = q/4\pi\epsilon_0 r$. Find the vector electric field from this expression for V .

IDENTIFY and SET UP This problem uses the general relationship between the electric potential as a function of position and the electric-field vector. By symmetry, the electric field here has only a radial component E_r . We use Eq. (23.23) to find this component.

EXECUTE From Eq. (23.23),

$$E_r = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

so the vector electric field is

$$\vec{E} = \hat{r}E_r = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

EVALUATE Our result agrees with Eq. (21.7), as it must.

An alternative approach is to ignore the radial symmetry, write the radial distance as $r = \sqrt{x^2 + y^2 + z^2}$, and take the derivatives of V with respect to x , y , and z as in Eq. (23.20). We find

$$\begin{aligned} \frac{\partial V}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{x^2 + y^2 + z^2}} \right) = -\frac{1}{4\pi\epsilon_0} \frac{qx}{(x^2 + y^2 + z^2)^{3/2}} \\ &= -\frac{qx}{4\pi\epsilon_0 r^3} \end{aligned}$$

and similarly

$$\frac{\partial V}{\partial y} = -\frac{qy}{4\pi\epsilon_0 r^3} \quad \frac{\partial V}{\partial z} = -\frac{qz}{4\pi\epsilon_0 r^3}$$

Then from Eq. (23.20),

$$\begin{aligned} \vec{E} &= - \left[\hat{i} \left(-\frac{qx}{4\pi\epsilon_0 r^3} \right) + \hat{j} \left(-\frac{qy}{4\pi\epsilon_0 r^3} \right) + \hat{k} \left(-\frac{qz}{4\pi\epsilon_0 r^3} \right) \right] \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{r} \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \end{aligned}$$

This approach gives us the same answer, but with more effort. Clearly it's best to exploit the symmetry of the charge distribution whenever possible.

KEYCONCEPT If you know the electric potential as a function of position in a region of space, you can find the electric field in that region by calculating the negative gradient of the potential.

EXAMPLE 23.14 Potential and field of a ring of charge

In Example 23.11 (Section 23.3) we found that for a ring of charge with radius a and total charge Q , the potential at a point P on the ring's symmetry axis a distance x from the center (see Fig. 23.20) is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{x^2 + a^2}}$$

Find the electric field at P .

IDENTIFY and SET UP We are given V as a function of x along the x -axis, and we wish to find the electric field at a point on this axis. From the symmetry of the charge distribution, the electric field along the symmetry (x -) axis of the ring can have only an x -component. We find it by using the first of Eqs. (23.19).

EXECUTE The x -component of the electric field is

$$E_x = -\frac{\partial V}{\partial x} = \frac{1}{4\pi\epsilon_0} \frac{Qx}{(x^2 + a^2)^{3/2}}$$

EVALUATE This agrees with our result in Example 21.9.

CAUTION Don't use expressions where they don't apply In this example, V is not a function of y or z on the ring axis, so $\partial V/\partial y = \partial V/\partial z = 0$ and $E_y = E_z = 0$. But that does not mean that it's true *everywhere*; our expressions for V and E_x are valid *on the ring axis only*. If we had an expression for V valid at *all* points in space, we could use it to find the components of $\vec{E}$ at any point by using Eqs. (23.19). **|**

KEYCONCEPT If you know the electric potential as a function of position only along a line, you can find the component of the electric field parallel to that line by calculating the negative partial derivative of the potential with respect to position on that line. You cannot find the electric field at points off the line in this way.

TEST YOUR UNDERSTANDING OF SECTION 23.5 In a certain region of space the potential is given by $V = A + Bx + Cy^3 + Dxy$, where A , B , C , and D are positive constants. Which of these statements about the electric field $\vec{E}$ in this region of space is correct? (There may be more than one correct answer.) (i) Increasing the value of A will increase the value of $\vec{E}$ at all points; (ii) increasing the value of A will decrease the value of $\vec{E}$ at all points; (iii) $\vec{E}$ has no z -component; (iv) the electric field is zero at the origin ($x = 0$, $y = 0$, $z = 0$).

ANSWER

| (iii) From Eqs. (23.19), the components of the electric field are $E_x = -\partial V/\partial x = -(B + Dy)$, $E_y = -\partial V/\partial y = -(3Cy^2 + Dx)$, and $E_z = -\partial V/\partial z = 0$. The value of A has no effect, which means that we can add a constant to the electric potential at all points without changing $\vec{E}$ or the potential difference between two points. The potential does not depend on z , so the z -component of $\vec{E}$ is zero. Note that at the origin the electric field is not zero because it has a nonzero x -component: $E_x = -B$, $E_y = 0$, $E_z = 0$.

CHAPTER 23 SUMMARY

Electric potential energy: The electric force caused by any collection of charges at rest is a conservative force. The work W done by the electric force on a charged particle moving in an electric field can be represented by the change in a potential-energy function U .

The electric potential energy for two point charges q and q_0 depends on their separation r . The electric potential energy for a charge q_0 in the presence of a collection of charges q_1, q_2, q_3 depends on the distance from q_0 to each of these other charges. (See Examples 23.1 and 23.2.)

$$W_{a \rightarrow b} = U_a - U_b$$

$$U = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r}$$

(two point charges)

$$U = \frac{q_0}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \cdots \right)$$

$$= \frac{q_0}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

(q_0 in presence of other point charges)

$$(23.2) \quad U = \frac{q_0}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right)$$

(23.9)

(23.10)

Electric potential: Potential, denoted by V , is potential energy per unit charge. The potential difference between two points equals the amount of work per charge that would be required to move a positive test charge between those points. The potential V due to a quantity of charge can be calculated by summing (if the charge is a collection of point charges) or by integrating (if the charge is a distribution). (See Examples 23.3, 23.4, 23.5, 23.7, 23.11, and 23.12.)

The potential difference between two points a and b , also called the potential of a with respect to b , is given by the line integral of $\vec{E}$. The potential at a given point can be found by first finding $\vec{E}$ and then carrying out this integral. (See Examples 23.6, 23.8, 23.9, and 23.10.)

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

(due to a point charge)

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

(due to a collection of point charges)

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$

(due to a charge distribution)

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l}$$

$$= \int_a^b E \cos \phi \, dl$$

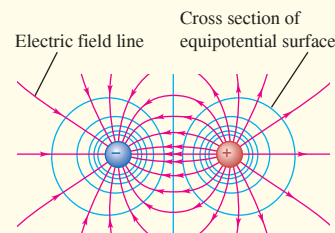
$$(23.14) \quad V = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right)$$

(23.15)

(23.16)

(23.17)

Equipotential surfaces: An equipotential surface is a surface on which the potential has the same value at every point. At a point where a field line crosses an equipotential surface, the two are perpendicular. When all charges are at rest, the surface of a conductor is always an equipotential surface and all points in the interior of a conductor are at the same potential. When a cavity within a conductor contains no charge, the entire cavity is an equipotential region and there is no surface charge anywhere on the surface of the cavity.



Finding electric field from electric potential: If the potential V is known as a function of the coordinates x , y , and z , the components of electric field $\vec{E}$ at any point are given by partial derivatives of V . (See Examples 23.13 and 23.14.)

$$E_x = -\frac{\partial V}{\partial x} \quad E_y = -\frac{\partial V}{\partial y} \quad E_z = -\frac{\partial V}{\partial z} \quad (23.19)$$

$$\vec{E} = -\left(\hat{i} \frac{\partial V}{\partial x} + \hat{j} \frac{\partial V}{\partial y} + \hat{k} \frac{\partial V}{\partial z} \right) \quad (23.20)$$

(vector form)



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 23.1 and 23.2** (Section 23.1) before attempting these problems.

VP23.2.1 An electron (charge $-e = -1.60 \times 10^{-19}$ C) is at rest at a distance 1.00×10^{-10} m from the center of a nucleus of lead (charge $+82e = +1.31 \times 10^{-17}$ C). You may treat both objects as point charges. (a) How much work would you have to do to move the electron to a distance 5.00×10^{-10} m from the center of the lead nucleus? (b) If you release the electron from the position in (a), what will be its kinetic energy when it is a distance 8.00×10^{-12} m from the center of the lead nucleus? (You can assume that the lead nucleus remains at rest, since it is much more massive than the electron.)

VP23.2.2 In a nuclear physics experiment, a proton (mass 1.67×10^{-27} kg, charge $+e = +1.60 \times 10^{-19}$ C) is fired directly at a target nucleus of unknown charge. (You can treat both objects as point charges, and assume that the nucleus remains at rest.) When it is far from its target, the proton has speed 2.50×10^6 m/s. The proton comes momentarily to rest at a distance 5.29×10^{-13} m from the center of the target nucleus, then flies back in the direction from which it came. (a) What is the electric potential energy of the proton and nucleus when they are 5.29×10^{-13} m apart? (b) What is the charge of the target nucleus?

VP23.2.3 Two point charges are at fixed positions on the y -axis: $q_1 = +e$ at $y = 0$ and $q_2 = -e$ at $y = a$. Find (a) the work you must do to bring a third charge $q_3 = -e$ from infinity to being at rest at $y = -2a$ and (b) the total potential energy of the system of three charges.

VP23.2.4 A point charge $q_1 = +5.00$ nC is at the fixed position $x = 0$, $y = 0$, $z = 0$. You find that you must do 8.10×10^{-6} J of work to bring a second point charge from infinity to the position $x = +4.00$ cm, $y = 0$, $z = 0$. (a) What is the value of the second charge? (b) Once the second charge is in place, how much additional work would you have to do to bring a third point charge $q_3 = +2.00$ nC from infinity to the position $x = 0$, $y = +3.00$ cm, $z = 0$?

Be sure to review **EXAMPLES 23.3, 23.4, 23.5, and 23.7** (Section 23.2) before attempting these problems.

VP23.7.1 In a certain region of space the electric field is uniform and given by $\vec{E} = (5.00 \times 10^2 \text{ V/m})\hat{i}$. If the electric potential at the point $x = 0$, $y = 0$, $z = 0$ is equal to V_0 , find the potential difference $V_0 - V_P$ for each of the following points P : (a) $x = +5.00$ cm, $y = 0$, $z = 0$; (b) $x = +3.00$ cm, $y = +4.00$ cm, $z = 0$; (c) $x = 0$, $y = +5.00$ cm, $z = 0$; (d) $x = -5.00$ cm, $y = 0$, $z = 0$.

VP23.7.2 You have two identical point charges of $+6.0$ nC each, one at the position $x = 3.0$ cm, $y = 0$, $z = 0$ and the other at $x = -3.0$ cm, $y = 0$, $z = 0$. Find the electric potential due to these charges at (a) $x = 0$, $y = 0$, $z = 0$; (b) $x = 0$, $y = 4.0$ cm, $z = 0$; (c) $x = 8.0$ cm, $y = 0$, $z = 0$. Take $V = 0$ at infinity. (d) At which (if any) of these points is the electric field equal to zero? If the field is zero at a certain point, is the potential necessarily zero there as well?

VP23.7.3 One point charge of $+5.90$ nC is at the position $x = 3.00$ cm, $y = 0$, $z = 0$, and one point charge of -5.90 nC is at $x = -3.00$ cm, $y = 0$, $z = 0$. Find the electric potential due to these charges at (a) $x = 3.00$ cm, $y = 8.00$ cm, $z = 0$; (b) $x = -3.00$ cm, $y = 8.00$ cm, $z = 0$. Take $V = 0$ at infinity. (c) A proton (mass 1.67×10^{-27} kg, charge $+e = +1.60 \times 10^{-19}$ C) follows a path from point (b) to point (a). If its speed at (b) is 3.00×10^5 m/s, what is its speed at (a)?

VP23.7.4 In Example 22.9 (Section 22.4) we found that the electric field *inside* a uniformly charged sphere with positive charge Q and radius R points radially outward from the center of the sphere and has magnitude $E = Qr/4\pi\epsilon_0 R^3$ at a distance r from the center. (a) By integrating the electric field, find the potential difference between the center of the sphere and its surface. (b) Which is at higher potential: the center or the surface?

Be sure to review **EXAMPLES 23.8, 23.9, 23.10, 23.11, and 23.12** (Section 23.3) before attempting these problems.

VP23.12.1 A conducting sphere of radius R carries positive charge q . Calculate the amount of work that would be required to move a small positive test charge q_0 slowly (a) from $r = 5R$ to $r = 3R$; (b) from $r = 2R$ to $r = 7R$; (c) from $r = 4R$ to $r = R/2$. In each case assume that the presence of q_0 has no effect on how the charge q is distributed over the sphere.

VP23.12.2 For the two oppositely charged parallel plates in Fig. 23.18, the potential difference between the plates is $V_{ab} = 24.0$ V, $V = 0$ at the bottom plate, and the distance between the plates is $d = 4.50$ mm. (a) What is the electric potential at a point 3.00 mm above the lower plate? (b) What is the potential energy for a dust particle of mass 5.00×10^{-9} kg and charge $+2.00$ nC at the position in part (a)? (c) If the particle in part (b) is released from rest, will it move toward the upper plate or the lower plate? What will be its speed when it hits that plate?

VP23.12.3 The uniformly charged ring shown in Fig. 23.20 has charge $+5.00$ nC and radius 2.50 cm. A point charge $q = +3.00$ nC of mass 4.00×10^{-9} kg is on the ring axis 8.00 cm from the center of the ring and is moving toward the center at 60.0 m/s. Take $V = 0$ at infinity. What is the potential energy of the point charge (a) at 8.00 cm from the ring's center and (b) at the ring's center? (c) What are the kinetic energy and speed of the point charge when it reaches the ring's center?

VP23.12.4 Positive charge Q is distributed uniformly along a rod of length L that lies along the x -axis from $x = L$ to $x = 2L$. (a) How much charge is contained within a segment of the rod of length dx ? (b) Integrate to find the total electric potential at the origin ($x = 0$) due to the rod.

BRIDGING PROBLEM A Point Charge and a Line of Charge

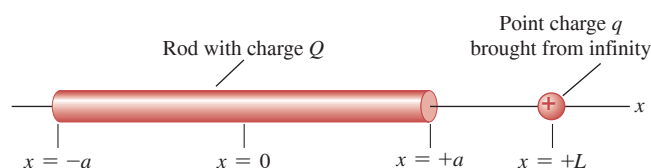
Positive electric charge Q is distributed uniformly along a thin rod of length $2a$. The rod lies along the x -axis between $x = -a$ and $x = +a$ (Fig. 23.27). Calculate how much work you must do to bring a positive point charge q from infinity to the point $x = +L$ on the x -axis, where $L > a$.

SOLUTION GUIDE

IDENTIFY and SET UP

1. In this problem you must first calculate the potential V at $x = +L$ due to the charged rod. You can then find the change in potential energy involved in bringing the point charge q from infinity (where $V = 0$) to $x = +L$.
2. To find V , divide the rod into infinitesimal segments of length dx' . How much charge is on such a segment? Consider one such

Figure 23.27 How much work must you do to bring point charge q in from infinity?



segment located at $x = x'$, where $-a \leq x' \leq a$. What is the potential dV at $x = +L$ due to this segment?

3. The total potential at $x = +L$ is the integral of dV , including contributions from all of the segments for x' from $-a$ to $+a$. Set up this integral.

EXECUTE

4. Integrate your expression from step 3 to find the potential V at $x = +L$. A simple, standard substitution will do the trick; use a table of integrals only as a last resort.
5. Use your result from step 4 to find the potential energy for a point charge q placed at $x = +L$.
6. Use your result from step 5 to find the work you must do to bring the point charge from infinity to $x = +L$.

EVALUATE

7. What does your result from step 5 become in the limit $a \rightarrow 0$? Does this make sense?
8. Suppose the point charge q were negative rather than positive. How would this affect your result in step 4? In step 5?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q23.1 A student asked, “Since electrical potential is always proportional to potential energy, why bother with the concept of potential at all?” How would you respond?

Q23.2 The potential (relative to a point at infinity) midway between two charges of equal magnitude and opposite sign is zero. Is it possible to bring a test charge from infinity to this midpoint in such a way that no work is done in any part of the displacement? If so, describe how it can be done. If it is not possible, explain why.

Q23.3 Is it possible to have an arrangement of two point charges separated by a finite distance such that the electric potential energy of the arrangement is the same as if the two charges were infinitely far apart? Why or why not? What if there are three charges? Explain.

Q23.4 Since potential can have any value you want depending on the choice of the reference level of zero potential, how does a voltmeter know what to read when you connect it between two points?

Q23.5 If $\vec{E}$ is zero everywhere along a certain path that leads from point A to point B, what is the potential difference between those two points? Does this mean that $\vec{E}$ is zero everywhere along *any* path from A to B? Explain.

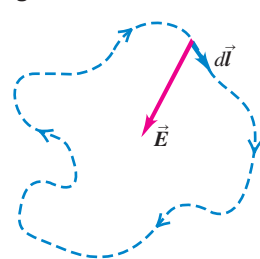
Q23.6 If $\vec{E}$ is zero throughout a certain region of space, is the potential necessarily also zero in this region? Why or why not? If not, what *can* be said about the potential?

Q23.7 Which way do electric field lines point, from high to low potential or from low to high? Explain.

Q23.8 (a) Take $V = 0$ at infinity. If the potential is zero at a point, is the electric field necessarily zero at that point? (b) If the electric field is zero at a point, is the potential necessarily zero there? Prove your answers, using simple examples.

Q23.9 If you carry out the integral of the electric field $\int \vec{E} \cdot d\vec{l}$ for a *closed* path like that shown in Fig. Q23.9, the integral will *always* be equal to zero, independent of the shape of the path and independent of where charges may be located relative to the path. Explain why.

Figure Q23.9



Q23.10 The potential difference between the two terminals of an AA battery (used in flashlights and portable stereos) is 1.5 V. If two AA batteries are placed end to end with the positive terminal of one battery touching the negative terminal of the other, what is the potential difference between the terminals at the exposed ends of the combination? What if the two positive terminals are touching each other? Explain your reasoning.

Q23.11 It is easy to produce a potential difference of several thousand volts between your body and the floor by scuffing your shoes across a nylon carpet. When you touch a metal doorknob, you get a mild shock. Yet contact with a power line of comparable voltage would probably be fatal. Why is there a difference?

Q23.12 If the electric potential at a single point is known, can $\vec{E}$ at that point be determined? If so, how? If not, why not?

Q23.13 Because electric field lines and equipotential surfaces are always perpendicular, two equipotential surfaces can never cross; if they did, the direction of $\vec{E}$ would be ambiguous at the crossing points. Yet two equipotential surfaces appear to cross at the center of Fig. 23.23c. Explain why there is no ambiguity about the direction of $\vec{E}$ in this particular case.

Q23.14 A uniform electric field is directed due east. Point B is 2.00 m west of point A , point C is 2.00 m east of point A , and point D is 2.00 m south of A . For each point, B , C , and D , is the potential at that point larger, smaller, or the same as at point A ? Give the reasoning behind your answers.

Q23.15 We often say that if point A is at a higher potential than point B , A is at positive potential and B is at negative potential. Does it necessarily follow that a point at positive potential is positively charged, or that a point at negative potential is negatively charged? Illustrate your answers with clear, simple examples.

Q23.16 A conducting sphere is to be charged by bringing in positive charge a little at a time until the total charge is Q . The total work required for this process is alleged to be proportional to Q^2 . Is this correct? Why or why not?

Q23.17 In electronics it is customary to define the potential of ground (thinking of the earth as a large conductor) as zero. Is this consistent with the fact that the earth has a net electric charge that is not zero?

Q23.18 A conducting sphere is placed between two charged parallel plates such as those shown in Fig. 23.2. Does the electric field inside the sphere depend on precisely where between the plates the sphere is placed? What about the electric potential inside the sphere? Do the answers to these questions depend on whether or not there is a net charge on the sphere? Explain your reasoning.

Q23.19 A conductor that carries a net charge Q has a hollow, empty cavity in its interior. Does the potential vary from point to point within the material of the conductor? What about within the cavity? How does the potential inside the cavity compare to the potential within the material of the conductor?

Q23.20 A high-voltage dc power line falls on a car, so the entire metal body of the car is at a potential of 10,000 V with respect to the ground. What happens to the occupants (a) when they are sitting in the car and (b) when they step out of the car? Explain your reasoning.

Q23.21 When a thunderstorm is approaching, sailors at sea sometimes observe a phenomenon called “St. Elmo’s fire,” a bluish flickering light at the tips of masts. What causes this? Why does it occur at the tips of masts? Why is the effect most pronounced when the masts are wet? (*Hint:* Seawater is a good conductor of electricity.)

Q23.22 A positive point charge is placed near a very large conducting plane. A professor of physics asserted that the field caused by this configuration is the same as would be obtained by removing the plane and placing a negative point charge of equal magnitude in the mirror-image position behind the initial position of the plane. Is this correct? Why or why not? (*Hint:* Inspect Fig. 23.23b.)

EXERCISES

Section 23.1 Electric Potential Energy

23.1 • A point charge $q_1 = +2.40 \mu\text{C}$ is held stationary at the origin. A second point charge $q_2 = -4.30 \mu\text{C}$ moves from the point $x = 0.150 \text{ m}$, $y = 0$ to the point $x = 0.250 \text{ m}$, $y = 0.250 \text{ m}$. How much work is done by the electric force on q_2 ?

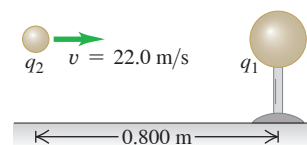
23.2 • A point charge q_1 is held stationary at the origin. A second charge q_2 is placed at point a , and the electric potential energy of the pair of charges is $+5.4 \times 10^{-8} \text{ J}$. When the second charge is moved to point b , the electric force on the charge does $-1.9 \times 10^{-8} \text{ J}$ of work. What is the electric potential energy of the pair of charges when the second charge is at point b ?

23.3 • **Energy of the Nucleus.** How much work is needed to assemble an atomic nucleus containing three protons (such as Li) if we model it as an equilateral triangle of side $2.00 \times 10^{-15} \text{ m}$ with a proton at each vertex? Assume the protons started from very far away.

23.4 •• (a) How much work would it take to push two protons very slowly from a separation of $2.00 \times 10^{-10} \text{ m}$ (a typical atomic distance) to $3.00 \times 10^{-15} \text{ m}$ (a typical nuclear distance)? (b) If the protons are both released from rest at the closer distance in part (a), how fast are they moving when they reach their original separation?

23.5 •• A small metal sphere, carrying a net charge of $q_1 = -2.80 \mu\text{C}$, is held in a stationary position by insulating supports. A second small metal sphere, with a net charge of $q_2 = -7.80 \mu\text{C}$ and mass 1.50 g, is projected toward q_1 . When the two spheres are 0.800 m apart, q_2 is moving toward q_1 with speed 22.0 m/s (**Fig. E23.5**). Assume that the two spheres can be treated as point charges. You can ignore the force of gravity. (a) What is the speed of q_2 when the spheres are 0.400 m apart? (b) How close does q_2 get to q_1 ?

Figure E23.5



23.6 •• **BIO Energy of DNA Base Pairing.** (See Exercise 21.18.)

(a) Calculate the electric potential energy of the adenine–thymine bond, using the same combinations of molecules ($\text{O}-\text{H}-\text{N}$ and $\text{N}-\text{H}-\text{N}$) as in Exercise 21.18. (b) Compare this energy with the potential energy of the proton–electron pair in the hydrogen atom.

23.7 •• Two protons, starting several meters apart, are aimed directly at each other with speeds of $2.00 \times 10^5 \text{ m/s}$, measured relative to the earth. Find the maximum electric force that these protons will exert on each other.

23.8 •• Three equal $1.20 \mu\text{C}$ point charges are placed at the corners of an equilateral triangle with sides 0.400 m long. What is the potential energy of the system? (Take as zero the potential energy of the three charges when they are infinitely far apart.)

23.9 •• Two protons are released from rest when they are 0.750 nm apart. (a) What is the maximum speed they will reach? When does this speed occur? (b) What is the maximum acceleration they will achieve? When does this acceleration occur?

23.10 •• Four electrons are located at the corners of a square 10.0 nm on a side, with an alpha particle at its midpoint. How much work is needed to move the alpha particle to the midpoint of one of the sides of the square?

Section 23.2 Electric Potential

23.11 •• **CALC** Points a and b lie in a region where the y -component of the electric field is $E_y = \alpha + \beta/y^2$. The constants in this expression have the values $\alpha = 600 \text{ N/C}$ and $\beta = 5.00 \text{ N} \cdot \text{m}^2/\text{C}$. Points a and b are on the $+y$ -axis. Point a is at $y = 2.00 \text{ cm}$ and point b is at $y = 3.00 \text{ cm}$. What is the potential difference $V_a - V_b$ between these two points and which point, a or b , is at higher potential?

23.12 • An object with charge $q = -6.00 \times 10^{-9} \text{ C}$ is placed in a region of uniform electric field and is released from rest at point A . After the charge has moved to point B , 0.500 m to the right, it has kinetic energy $3.00 \times 10^{-7} \text{ J}$. (a) If the electric potential at point A is $+30.0 \text{ V}$, what is the electric potential at point B ? (b) What are the magnitude and direction of the electric field?

23.13 • A small particle has charge $-5.00 \mu\text{C}$ and mass $2.00 \times 10^{-4} \text{ kg}$. It moves from point A , where the electric potential is $V_A = +200 \text{ V}$, to point B , where the electric potential is $V_B = +800 \text{ V}$. The electric force is the only force acting on the particle. The particle has speed 5.00 m/s at point A . What is its speed at point B ? Is it moving faster or slower at B than at A ? Explain.

23.14 • A particle with charge $+4.20 \text{ nC}$ is in a uniform electric field $\vec{E}$ directed to the left. The charge is released from rest and moves to the left; after it has moved 6.00 cm, its kinetic energy is $+2.20 \times 10^{-6} \text{ J}$. What are (a) the work done by the electric force, (b) the potential of the starting point with respect to the end point, and (c) the magnitude of $\vec{E}$?

23.15 • A charge of 28.0 nC is placed in a uniform electric field that is directed vertically upward and has a magnitude of $4.00 \times 10^4 \text{ V/m}$. What work is done by the electric force when the charge moves (a) 0.450 m to the right; (b) 0.670 m upward; (c) 2.60 m at an angle of 45.0° downward from the horizontal?

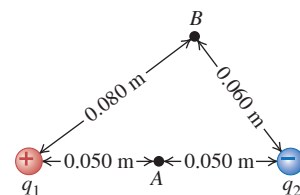
23.16 • Two stationary point charges $+3.00 \text{ nC}$ and $+2.00 \text{ nC}$ are separated by a distance of 50.0 cm . An electron is released from rest at a point midway between the two charges and moves along the line connecting the two charges. What is the speed of the electron when it is 10.0 cm from the $+3.00 \text{ nC}$ charge?

23.17 •• Point charges $q_1 = +2.00 \mu\text{C}$ and $q_2 = -2.00 \mu\text{C}$ are placed at adjacent corners of a square for which the length of each side is 3.00 cm . Point a is at the center of the square, and point b is at the empty corner closest to q_2 . Take the electric potential to be zero at a distance far from both charges. (a) What is the electric potential at point a due to q_1 and q_2 ? (b) What is the electric potential at point b ? (c) A point charge $q_3 = -5.00 \mu\text{C}$ moves from point a to point b . How much work is done on q_3 by the electric forces exerted by q_1 and q_2 ? Is this work positive or negative?

23.18 • Two point charges of equal magnitude Q are held a distance d apart. Consider only points on the line passing through both charges; take $V = 0$ at infinity. (a) If the two charges have the same sign, find the location of all points (if there are any) at which (i) the potential is zero (is the electric field zero at these points?), and (ii) the electric field is zero (is the potential zero at these points?). (b) Repeat part (a) for two point charges having opposite signs.

23.19 • Two point charges $q_1 = +2.40 \text{ nC}$ and $q_2 = -6.50 \text{ nC}$ are

Figure E23.19



0.100 m apart. Point A is midway between them; point B is 0.080 m from q_1 and 0.060 m from q_2 (Fig. E23.19). Take the electric potential to be zero at infinity. Find (a) the potential at point A ; (b) the potential at point B ; (c) the work done by the electric field on a charge

of 2.50 nC that travels from point B to point A .

23.20 •• (a) An electron is to be accelerated from $3.00 \times 10^6 \text{ m/s}$ to $8.00 \times 10^6 \text{ m/s}$. Through what potential difference must the electron pass to accomplish this? (b) Through what potential difference must the electron pass if it is to be slowed from $8.00 \times 10^6 \text{ m/s}$ to a halt?

23.21 •• Points A and B lie within a region of space where there is a uniform electric field that has no x - or z -component; only the y -component E_y is nonzero. Point A is at $y = 8.00 \text{ cm}$ and point B is at $y = 15.0 \text{ cm}$. The potential difference between B and A is $V_B - V_A = +12.0 \text{ V}$, so point B is at higher potential than point A . (a) Is E_y positive or negative? (b) What is the magnitude of the electric field? (c) Point C has coordinates $x = 5.00 \text{ cm}$, $y = 5.00 \text{ cm}$. What is the potential difference between points B and C ?

23.22 • At a certain distance from a point charge, the potential and electric-field magnitude due to that charge are 4.98 V and 16.2 V/m , respectively. (Take $V = 0$ at infinity.) (a) What is the distance to the point charge? (b) What is the magnitude of the charge? (c) Is the electric field directed toward or away from the point charge?

23.23 • A uniform electric field has magnitude E and is directed in the negative x -direction. The potential difference between point a (at $x = 0.60 \text{ m}$) and point b (at $x = 0.90 \text{ m}$) is 240 V . (a) Which point, a or b , is at the higher potential? (b) Calculate the value of E . (c) A negative point charge $q = -0.200 \mu\text{C}$ is moved from b to a . Calculate the work done on the point charge by the electric field.

23.24 •• A small sphere with charge $q = -5.00 \mu\text{C}$ is moving in a uniform electric field that has no y - or z -component. The only force on the sphere is the force exerted by the electric field. Point A is on the x -axis at $x = -0.400 \text{ m}$, and point B is at the origin. At point A the sphere has kinetic energy $K_A = 8.00 \times 10^{-4} \text{ J}$, and at point B its kinetic energy is $K_B = 3.00 \times 10^{-4} \text{ J}$. (a) What is the potential difference $V_{AB} = V_A - V_B$? Which point, A or B , is at higher potential? (b) What are the magnitude and direction of the electric field?

23.25 •• Identical point charges q_1 and q_2 each have positive charge $+6.00 \mu\text{C}$. Charge q_1 is held fixed on the x -axis at $x = +0.400 \text{ m}$, and q_2 is held fixed on the x -axis at $x = -0.400 \text{ m}$. A small sphere has charge $Q = -0.200 \mu\text{C}$ and mass 12.0 g . The sphere is initially very far from the origin. It is released from rest and moves along the y -axis toward the origin. (a) As the sphere moves from very large y to $y = 0$, how much work is done on it by the resultant force exerted by q_1 and q_2 ? (b) If the only force acting on the sphere is the force exerted by the point charges, what is its speed when it reaches the origin?

Section 23.3 Calculating Electric Potential

23.26 •• A solid conducting sphere of radius 5.00 cm carries a net charge. To find the value of the charge, you measure the potential difference $V_{AB} = V_A - V_B$ between point A , which is 8.00 cm from the center of the sphere, and point B , which is a distance r from the center of the sphere. You repeat these measurements for several values of $r > 8.00 \text{ cm}$. When you plot your data as V_{AB} versus $1/r$, the values lie close to a straight line with slope $-18.0 \text{ V} \cdot \text{m}$. What does your data give for the net charge on the sphere? Is the net charge positive or negative?

23.27 •• A thin spherical shell with radius $R_1 = 3.00 \text{ cm}$ is concentric with a larger thin spherical shell with radius $R_2 = 5.00 \text{ cm}$. Both shells are made of insulating material. The smaller shell has charge $q_1 = +6.00 \text{ nC}$ distributed uniformly over its surface, and the larger shell has charge $q_2 = -9.00 \text{ nC}$ distributed uniformly over its surface. Take the electric potential to be zero at an infinite distance from both shells. (a) What is the electric potential due to the two shells at the following distance from their common center: (i) $r = 0$; (ii) $r = 4.00 \text{ cm}$; (iii) $r = 6.00 \text{ cm}$? (b) What is the magnitude of the potential difference between the surfaces of the two shells? Which shell is at higher potential: the inner shell or the outer shell?

23.28 • A total electric charge of 3.50 nC is distributed uniformly over the surface of a metal sphere with a radius of 24.0 cm . If the potential is zero at a point at infinity, find the value of the potential at the following distances from the center of the sphere: (a) 48.0 cm ; (b) 24.0 cm ; (c) 12.0 cm .

23.29 •• A uniformly charged, thin ring has radius 15.0 cm and total charge $+24.0 \text{ nC}$. An electron is placed on the ring's axis a distance 30.0 cm from the center of the ring and is constrained to stay on the axis of the ring. The electron is then released from rest. (a) Describe the subsequent motion of the electron. (b) Find the speed of the electron when it reaches the center of the ring.

23.30 • A solid conducting sphere has net positive charge and radius $R = 0.400 \text{ m}$. At a point 1.20 m from the center of the sphere, the electric potential due to the charge on the sphere is 24.0 V . Assume that $V = 0$ at an infinite distance from the sphere. What is the electric potential at the center of the sphere?

23.31 • Charge $Q = 5.00 \mu\text{C}$ is distributed uniformly over the volume of an insulating sphere that has radius $R = 12.0 \text{ cm}$. A small sphere with charge $q = +3.00 \mu\text{C}$ and mass $6.00 \times 10^{-5} \text{ kg}$ is projected toward the center of the large sphere from an initial large distance. The large sphere is held at a fixed position and the small sphere can be treated as a point charge. What minimum speed must the small sphere have in order to come within 8.00 cm of the surface of the large sphere?

23.32 •• An infinitely long line of charge has linear charge density $5.00 \times 10^{-12} \text{ C/m}$. A proton (mass $1.67 \times 10^{-27} \text{ kg}$, charge $+1.60 \times 10^{-19} \text{ C}$) is 18.0 cm from the line and moving directly toward the line at $3.50 \times 10^3 \text{ m/s}$. (a) Calculate the proton's initial kinetic energy. (b) How close does the proton get to the line of charge?

23.33 •• A very long insulating cylindrical shell of radius 6.00 cm carries charge of linear density $8.50 \mu\text{C/m}$ spread uniformly over its outer surface. What would a voltmeter read if it were connected between (a) the surface of the cylinder and a point 4.00 cm above the surface, and (b) the surface and a point 1.00 cm from the central axis of the cylinder?

23.34 •• A very long insulating cylinder of charge of radius 2.50 cm carries a uniform linear density of 15.0 nC/m . If you put one probe of a voltmeter at the surface, how far from the surface must the other probe be placed so that the voltmeter reads 175 V ?

23.35 •• A very small sphere with positive charge $q = +8.00 \mu\text{C}$ is released from rest at a point 1.50 cm from a very long line of uniform linear charge density $\lambda = +3.00 \mu\text{C/m}$. What is the kinetic energy of the sphere when it is 4.50 cm from the line of charge if the only force on it is the force exerted by the line of charge?

23.36 • CP Two large, parallel conducting plates carrying opposite charges of equal magnitude are separated by 2.20 cm . (a) If the surface charge density for each plate has magnitude 47.0 nC/m^2 , what is the magnitude of $\vec{E}$ in the region between the plates? (b) What is the potential difference between the two plates? (c) If the separation between the plates is doubled while the surface charge density is kept constant at the value in part (a), what happens to the magnitude of the electric field and to the potential difference?

23.37 • Two large, parallel, metal plates carry opposite charges of equal magnitude. They are separated by 45.0 mm , and the potential difference between them is 360 V . (a) What is the magnitude of the electric field (assumed to be uniform) in the region between the plates? (b) What is the magnitude of the force this field exerts on a particle with charge $+2.40 \text{ nC}$? (c) Use the results of part (b) to compute the work done by the field on the particle as it moves from the higher-potential plate to the lower. (d) Compare the result of part (c) to the change of potential energy of the same charge, computed from the electric potential.

23.38 • BIO Electrical Sensitivity of Sharks. Certain sharks can detect an electric field as weak as $1.0 \mu\text{V/m}$. To grasp how weak this field is, if you wanted to produce it between two parallel metal plates by connecting an ordinary 1.5 V AA battery across these plates, how far apart would the plates have to be?

23.39 • The electric field at the surface of a charged, solid, copper sphere with radius 0.200 m is 3800 N/C , directed toward the center of the sphere. What is the potential at the center of the sphere, if we take the potential to be zero infinitely far from the sphere?

23.40 •• (a) How much excess charge must be placed on a copper sphere 25.0 cm in diameter so that the potential of its center is 3.75 kV ? Take the point where $V = 0$ to be infinitely far from the sphere. (b) What is the potential of the sphere's surface?

Section 23.4 Equipotential Surfaces and

Section 23.5 Potential Gradient

23.41 •• CALC A metal sphere with radius r_a is supported on an insulating stand at the center of a hollow, metal, spherical shell with radius r_b . There is charge $+q$ on the inner sphere and charge $-q$ on the outer spherical shell. (a) Calculate the potential $V(r)$ for (i) $r < r_a$; (ii) $r_a < r < r_b$; (iii) $r > r_b$. (Hint: The net potential is the sum of the potentials due to the individual spheres.) Take V to be zero when r is infinite. (b) Show that the potential of the inner sphere with respect to the outer is

$$V_{ab} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right)$$

(c) Use Eq. (23.23) and the result from part (a) to show that the electric field at any point between the spheres has magnitude

$$E(r) = \frac{V_{ab}}{(1/r_a - 1/r_b) r^2}$$

(d) Use Eq. (23.23) and the result from part (a) to find the electric field at a point outside the larger sphere at a distance r from the center, where $r > r_b$. (e) Suppose the charge on the outer sphere is not $-q$ but a negative charge of different magnitude, say $-Q$. Show that the answers for parts (b) and (c) are the same as before but the answer for part (d) is different.

23.42 • A very large plastic sheet carries a uniform charge density of -6.00 nC/m^2 on one face. (a) As you move away from the sheet along a line perpendicular to it, does the potential increase or decrease? How do you know, without doing any calculations? Does your answer depend on where you choose the reference point for potential? (b) Find the spacing between equipotential surfaces that differ from each other by 1.00 V . What type of surfaces are these?

23.43 • CALC In a certain region of space, the electric potential is $V(x, y, z) = Axy - Bx^2 + Cy$, where A , B , and C are positive constants. (a) Calculate the x -, y -, and z -components of the electric field. (b) At which points is the electric field equal to zero?

23.44 • CALC In a certain region of space the electric potential is given by $V = +Ax^2y - Bxy^2$, where $A = 5.00 \text{ V/m}^3$ and $B = 8.00 \text{ V/m}^3$. Calculate the magnitude and direction of the electric field at the point in the region that has coordinates $x = 2.00 \text{ m}$, $y = 0.400 \text{ m}$, and $z = 0$.

23.45 • A metal sphere with radius $r_a = 1.20 \text{ cm}$ is supported on an insulating stand at the center of a hollow, metal, spherical shell with radius $r_b = 9.60 \text{ cm}$. Charge $+q$ is put on the inner sphere and charge $-q$ on the outer spherical shell. The magnitude of q is chosen to make the potential difference between the spheres 500 V , with the inner sphere at higher potential. (a) Use the result of Exercise 23.41(b) to calculate q . (b) With the help of the result of Exercise 23.41(a), sketch the equipotential surfaces that correspond to 500 , 400 , 300 , 200 , 100 , and 0 V . (c) In your sketch, show the electric field lines. Are the electric field lines and equipotential surfaces mutually perpendicular? Are the equipotential surfaces closer together when the magnitude of $\vec{E}$ is largest?

PROBLEMS

23.46 • CP A point charge $q_1 = +5.00 \mu\text{C}$ is held fixed in space. From a horizontal distance of 6.00 cm , a small sphere with mass $4.00 \times 10^{-3} \text{ kg}$ and charge $q_2 = +2.00 \mu\text{C}$ is fired toward the fixed charge with an initial speed of 40.0 m/s . Gravity can be neglected. What is the acceleration of the sphere at the instant when its speed is 25.0 m/s ?

23.47 ••• A point charge $q_1 = 4.00 \text{ nC}$ is placed at the origin, and a second point charge $q_2 = -3.00 \text{ nC}$ is placed on the x -axis at $x = +20.0 \text{ cm}$. A third point charge $q_3 = 2.00 \text{ nC}$ is to be placed on the x -axis between q_1 and q_2 . (Take as zero the potential energy of the three charges when they are infinitely far apart.) (a) What is the potential energy of the system of the three charges if q_3 is placed at $x = +10.0 \text{ cm}$? (b) Where should q_3 be placed to make the potential energy of the system equal to zero?

23.48 •• A point charge $+8.00 \text{ nC}$ is on the $-x$ -axis at $x = -0.200 \text{ m}$, and a point charge -4.00 nC is on the $+x$ -axis at $x = 0.200 \text{ m}$. (a) In addition to $x = \pm\infty$, at what point on the x -axis is the resultant field of the two charges equal to zero? (b) Let $V = 0$ at $x = \pm\infty$. At what two other points on the x -axis is the total electric potential due to the two charges equal to zero? (c) Is $E = 0$ at either of the points in part (b) where $V = 0$? Explain.

23.49 •• A very long uniform line of charge with charge per unit length $\lambda = +5.00 \mu\text{C}/\text{m}$ lies along the x -axis, with its midpoint at the origin. A very large uniform sheet of charge is parallel to the xy -plane; the center of the sheet is at $z = +0.600 \text{ m}$. The sheet has charge per unit area $\sigma = +8.00 \mu\text{C}/\text{m}^2$, and the center of the sheet is at $x = 0$, $y = 0$. Point A is on the z -axis at $z = +0.300 \text{ m}$, and point B is on the z -axis at $z = -0.200 \text{ m}$. What is the potential difference $V_{AB} = V_A - V_B$ between points A and B ? Which point, A or B , is at higher potential?

23.50 ••• A small sphere with mass $5.00 \times 10^{-7} \text{ kg}$ and charge $+7.00 \mu\text{C}$ is released from rest a distance of 0.400 m above a large horizontal insulating sheet of charge that has uniform surface charge density $\sigma = +8.00 \text{ pC}/\text{m}^2$. Using energy methods, calculate the speed of the sphere when it is 0.100 m above the sheet.

23.51 •• A gold nucleus has a radius of $7.3 \times 10^{-15} \text{ m}$ and a charge of $+79e$. Through what voltage must an alpha particle, with charge $+2e$, be accelerated so that it has just enough energy to reach a distance of $2.0 \times 10^{-14} \text{ m}$ from the surface of a gold nucleus? (Assume that the gold nucleus remains stationary and can be treated as a point charge.)

23.52 •• CP A proton and an alpha particle are released from rest when they are 0.225 nm apart. The alpha particle (a helium nucleus) has essentially four times the mass and two times the charge of a proton. Find the maximum *speed* and maximum *acceleration* of each of these particles. When do these maxima occur: just following the release of the particles or after a very long time?

23.53 • A particle with charge $+7.60 \text{ nC}$ is in a uniform electric field directed to the left. Another force, in addition to the electric force, acts on the particle so that when it is released from rest, it moves to the right. After it has moved 8.00 cm , the additional force has done $6.50 \times 10^{-5} \text{ J}$ of work and the particle has $4.35 \times 10^{-5} \text{ J}$ of kinetic energy. (a) What work was done by the electric force? (b) What is the potential of the starting point with respect to the end point? (c) What is the magnitude of the electric field?

23.54 •• Identical charges $q = +5.00 \mu\text{C}$ are placed at opposite corners of a square that has sides of length 8.00 cm . Point A is at one of the empty corners, and point B is at the center of the square. A charge $q_0 = -3.00 \mu\text{C}$ is placed at point A and moves along the diagonal of the square to point B . (a) What is the magnitude of the net electric force on q_0 when it is at point A ? Sketch the placement of the charges and the direction of the net force. (b) What is the magnitude of the net electric force on q_0 when it is at point B ? (c) How much work does the electric force do on q_0 during its motion from A to B ? Is this work positive or negative? When it goes from A to B , does q_0 move to higher potential or to lower potential?

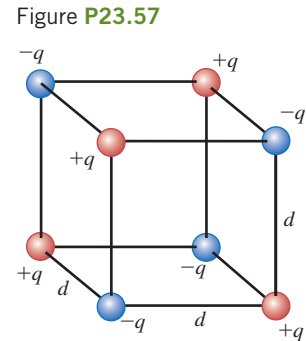
23.55 •• CALC A vacuum tube diode consists of concentric cylindrical electrodes, the negative cathode and the positive anode. Because of the accumulation of charge near the cathode, the electric potential between the electrodes is given by $V(x) = Cx^{4/3}$ where x is the distance from the cathode and C is a constant, characteristic of a particular diode and operating conditions. Assume that the distance between the cathode and anode is 13.0 mm and the potential difference between electrodes is 240 V . (a) Determine the value of C . (b) Obtain a formula for the electric field between the electrodes as a function of x . (c) Determine the force on an electron when the electron is halfway between the electrodes.

23.56 •• When you scuff your feet on a carpet, you gain electrons and become negatively charged. If you then place your finger near a metallic surface, such as a doorknob, an electric field develops between your finger and the doorknob. As your finger gets closer to the surface, the magnitude of the electric field increases. When it exceeds a threshold of $3 \text{ MV}/\text{m}$, the air breaks down, creating a small “lightning” strike, which you feel as a shock. (a) Estimate the distance between your finger and a doorknob at the point you feel the shock. (b) Using this estimate and the

threshold field strength of $3 \text{ MV}/\text{m}$, find the potential between your finger and the doorknob. (c) At this (small) distance, we can treat the tip of your finger and the end of the doorknob as infinite planar surfaces with opposite charge density. Estimate that density. (d) Estimate the effective area of your fingertip as presented to the doorknob. (e) Use these results to estimate the magnitude of charge on your finger. (f) Assuming that your net excess charge has built up on your finger, attracted there by the doorknob, estimate the number of electrons that you lost while scuffing your feet.

23.57 •• An Ionic Crystal.

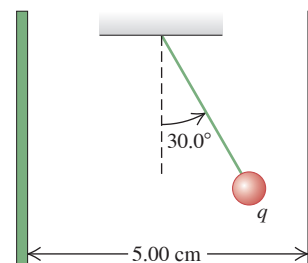
Figure P23.57 shows eight point charges arranged at the corners of a cube with sides of length d . The values of the charges are $+q$ and $-q$, as shown. This is a model of one cell of a cubic ionic crystal. In sodium chloride (NaCl), for instance, the positive ions are Na^+ and the negative ions are Cl^- . (a) Calculate the potential energy U of this arrangement. (Take as zero the potential energy of the eight charges when they are infinitely far apart.) (b) In part (a), you should have found that $U < 0$. Explain the relationship between this result and the observation that such ionic crystals exist in nature.



23.58 •• Electrical power is transmitted to our homes by overhead wires strung between poles. A typical residential utility line has a maximum potential of 22 kV relative to ground. We can treat this potential as constant and model it as generated by a net charge distributed on the wire. (a) Estimate the height of an electrical transmission line. (b) Treat the electrical wire as a long conducting cylinder with a radius of 2 cm . If the potential between the surface of the wire and a position directly beneath the wire is 22 kV , what is the linear charge density on the wire? (c) Use your estimate of the linear charge density to estimate the strength of the electric field on the ground beneath the wire.

23.59 •• CP A small sphere with mass 1.50 g hangs by a thread between two very large parallel vertical plates 5.00 cm apart (**Fig. P23.59**). The plates are insulating and have uniform surface charge densities $+\sigma$ and $-\sigma$. The charge on the sphere is $q = 8.90 \times 10^{-6} \text{ C}$. What potential difference between the plates will cause the thread to assume an angle of 30.0° with the vertical?

Figure P23.59



23.60 •• Two spherical shells have a common center. The inner shell has radius $R_1 = 5.00 \text{ cm}$ and charge $q_1 = +3.00 \times 10^{-6} \text{ C}$; the outer shell has radius $R_2 = 15.0 \text{ cm}$ and charge $q_2 = -5.00 \times 10^{-6} \text{ C}$. Both charges are spread uniformly over the shell surface. What is the electric potential due to the two shells at the following distances from their common center: (a) $r = 2.50 \text{ cm}$; (b) $r = 10.0 \text{ cm}$; (c) $r = 20.0 \text{ cm}$? Take $V = 0$ at a large distance from the shells.

23.61 • CALC Coaxial Cylinders. A long metal cylinder with radius a is supported on an insulating stand on the axis of a long, hollow, metal tube with radius b . The positive charge per unit length on the inner cylinder is λ , and there is an equal negative charge per unit length on the outer cylinder. (a) Calculate the potential $V(r)$ for (i) $r < a$; (ii) $a < r < b$; (iii) $r > b$. (Hint: The net potential is the sum of the potentials due to the individual conductors.) Take $V = 0$ at $r = b$. (b) Show that the potential of the inner cylinder with respect to the outer is

$$V_{ab} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{b}{a}$$

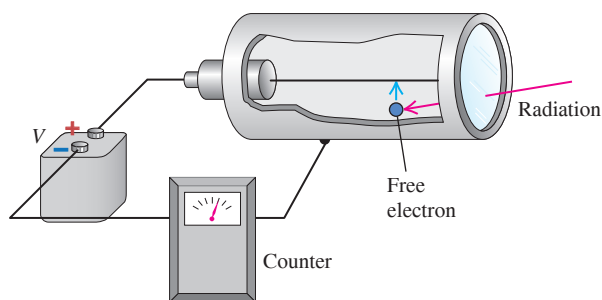
(c) Use Eq. (23.23) and the result from part (a) to show that the electric field at any point between the cylinders has magnitude

$$E(r) = \frac{V_{ab}}{\ln(b/a)} \frac{1}{r}$$

(d) What is the potential difference between the two cylinders if the outer cylinder has no net charge?

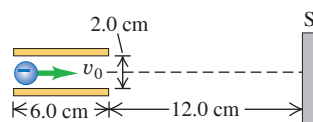
23.62 •• A Geiger counter detects radiation such as alpha particles by using the fact that the radiation ionizes the air along its path. A thin wire lies on the axis of a hollow metal cylinder and is insulated from it (Fig. P23.62). A large potential difference is established between the wire and the outer cylinder, with the wire at higher potential; this sets up a strong electric field directed radially outward. When ionizing radiation enters the device, it ionizes a few air molecules. The free electrons produced are accelerated by the electric field toward the wire and, on the way there, ionize many more air molecules. Thus a current pulse is produced that can be detected by appropriate electronic circuitry and converted to an audible “click.” Suppose the radius of the central wire is $145 \mu\text{m}$ and the radius of the hollow cylinder is 1.80 cm . What potential difference between the wire and the cylinder produces an electric field of $2.00 \times 10^4 \text{ V/m}$ at a distance of 1.20 cm from the axis of the wire? (The wire and cylinder are both very long in comparison to their radii, so the results of Problem 23.61 apply.)

Figure P23.62



23.63 • CP Deflection in a CRT. Cathode-ray tubes (CRTs) were often found in oscilloscopes and computer monitors. In Fig. P23.63 an electron with an initial speed of $6.50 \times 10^6 \text{ m/s}$ is projected along the axis midway between the deflection plates of a cathode-ray tube. The potential difference between the two plates is 22.0 V and the lower plate is the one at higher potential. (a) What is the force (magnitude and direction) on the electron when it is between the plates? (b) What is the acceleration of the electron (magnitude and direction) when acted on by the force in part (a)? (c) How far below the axis has the electron moved when it reaches the end of the plates? (d) At what angle with the axis is it moving as it leaves the plates? (e) How far below the axis will it strike the fluorescent screen S?

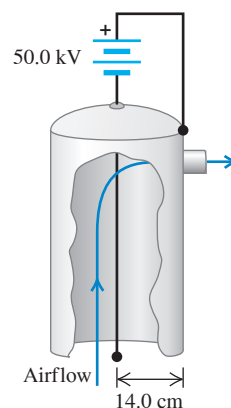
Figure P23.63



23.64 •• CP Deflecting Plates of an Oscilloscope. The vertical deflecting plates of a typical classroom oscilloscope are a pair of parallel square metal plates carrying equal but opposite charges. Typical dimensions are about 3.0 cm on a side, with a separation of about 5.0 mm . The potential difference between the plates is 25.0 V . The plates are close enough that we can ignore fringing at the ends. Under these conditions: (a) how much charge is on each plate, and (b) how strong is the electric field between the plates? (c) If an electron is ejected at rest from the negative plate, how fast is it moving when it reaches the positive plate?

23.65 •• Electrostatic precipitators use electric forces to remove pollutant particles from smoke, in particular in the smokestacks of coal-burning power plants. One form of precipitator consists of a vertical, hollow, metal cylinder with a thin wire, insulated from the cylinder, running along its axis (Fig. P23.65). A large potential difference is established between the wire and the outer cylinder, with the wire at lower potential. This sets up a strong radial electric field directed inward. The field produces a region of ionized air near the wire. Smoke enters the precipitator at the bottom, ash and dust in it pick up electrons, and the charged pollutants are accelerated toward the outer cylinder wall by the electric field. Suppose the radius of the central wire is $90.0 \mu\text{m}$, the radius of the cylinder is 14.0 cm , and a potential difference of 50.0 kV is established between the wire and the cylinder. Also assume that the wire and cylinder are both very long in comparison to the cylinder radius, so the results of Problem 23.61 apply. (a) What is the magnitude of the electric field midway between the wire and the cylinder wall? (b) What magnitude of charge must a $30.0 \mu\text{g}$ ash particle have if the electric field computed in part (a) is to exert a force ten times the weight of the particle?

Figure P23.65



23.66 •• CALC A disk with radius R has uniform surface charge density σ . (a) By regarding the disk as a series of thin concentric rings, calculate the electric potential V at a point on the disk's axis a distance x from the center of the disk. Assume that the potential is zero at infinity. (Hint: Use the result of Example 23.11 in Section 23.3.) (b) Calculate $-\partial V/\partial x$. Show that the result agrees with the expression for E_x calculated in Example 21.11 (Section 21.5).

23.67 ••• CALC Self-Energy of a Sphere of Charge. A solid sphere of radius R contains a total charge Q distributed uniformly throughout its volume. Find the energy needed to assemble this charge by bringing infinitesimal charges from far away. This energy is called the “self-energy” of the charge distribution. (Hint: After you have assembled a charge q in a sphere of radius r , how much energy would it take to add a spherical shell of thickness dr having charge dq ? Then integrate to get the total energy.)

23.68 • CALC A thin insulating rod is bent into a semicircular arc of radius a , and a total electric charge Q is distributed uniformly along the rod. Calculate the potential at the center of curvature of the arc if the potential is assumed to be zero at infinity.

23.69 •• Charge $Q = +4.00 \mu\text{C}$ is distributed uniformly over the volume of an insulating sphere that has radius $R = 5.00 \text{ cm}$. What is the potential difference between the center of the sphere and the surface of the sphere?

23.70 • (a) If a spherical raindrop of radius 0.650 mm carries a charge of -3.60 pC uniformly distributed over its volume, what is the potential at its surface? (Take the potential to be zero at an infinite distance from the raindrop.) (b) Two identical raindrops, each with radius and charge specified in part (a), collide and merge into one larger raindrop. What is the radius of this larger drop, and what is the potential at its surface, if its charge is uniformly distributed over its volume?

23.71 • CALC Electric charge is

distributed uniformly along a thin rod of length a , with total charge Q . Take the potential to be zero at infinity. Find the potential at the following points (**Fig. P23.71**):

- (a) point P , a distance x to the right of the rod, and (b) point R , a distance y above the right-hand end of the rod. (c) In parts (a) and (b), what does your result reduce to as x or y becomes much larger than a ?

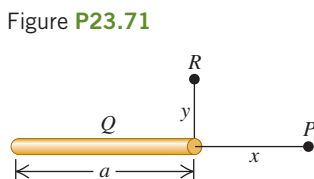
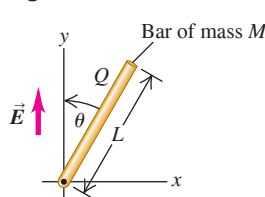


Figure P23.71

23.72 ••• CP CALC A rigid bar with

mass M , length L , and a uniformly distributed positive charge Q is free to pivot about the origin in the presence of a spatially uniform electric field $\vec{E} = E\hat{j}$, as shown in **Fig. P23.72**. Assume that the electric forces are large enough for gravity to be neglected. (a) Write the potential $V(y)$ due to the electric field as a function of y , using the convention $V(0) = V_0$, where V_0 is a constant to be determined. (b) Determine the electric potential energy U of the system as a function of θ and V_0 . (*Hint: $U = \int V dq$.*) (c) For what value of V_0 does the potential energy vanish when $\theta = 0^\circ$? (d) If the bar is released from rest at the position $\theta = 90^\circ$, what is the angular speed of the bar as it passes the position $\theta = 0^\circ$? (e) If the bar starts from rest at a small value of θ , with what frequency will the bar oscillate around the position $\theta = 0^\circ$?

Figure P23.72



23.73 ••• CP A helium nucleus, also known as an α (alpha) particle, consists of two protons and two neutrons and has a diameter of 10^{-15} m = 1 fm. The protons, with a charge of $+e$, are subject to a repulsive Coulomb force. Since the neutrons have zero charge, there must be an attractive force that counteracts the electric repulsion and keeps the protons from flying apart. This so-called strong force plays a central role in particle physics. (a) As a crude model, assume that an α particle consists of two pointlike protons attracted by a Hooke's-law spring with spring constant k , and ignore the neutrons. Assume further that in the absence of other forces, the spring has an equilibrium separation of zero. Write an expression for the potential energy when the protons are separated by distance d . (b) Minimize this potential to find the equilibrium separation d_0 in terms of e and k . (c) If $d_0 = 1.00$ fm, what is the value of k ? (d) How much energy is stored in this system, in terms of electron volts? (e) A proton has a mass of 1.67×10^{-27} kg. If the spring were to break, the α particle would disintegrate and the protons would fly off in opposite directions. What would be their ultimate speed?

23.74 • A metal sphere with radius R_1 has a charge Q_1 . Take the electric potential to be zero at an infinite distance from the sphere. (a) What are the electric field and electric potential at the surface of the sphere? This sphere is now connected by a long, thin conducting wire to another sphere of radius R_2 that is several meters from the first sphere. Before the connection is made, this second sphere is uncharged. After electrostatic equilibrium has been reached, what are (b) the total charge on each sphere; (c) the electric potential at the surface of each sphere; (d) the electric field at the surface of each sphere? Assume that the amount of charge on the wire is much less than the charge on each sphere.

23.75 • An alpha particle with kinetic energy 9.50 MeV (when far away) collides head-on with a lead nucleus at rest. What is the distance of closest approach of the two particles? (Assume that the lead nucleus remains stationary and may be treated as a point charge. The atomic number of lead is 82. The alpha particle is a helium nucleus, with atomic number 2.)

23.76 • CALC The electric potential V in a region of space is given by

$$V(x, y, z) = A(x^2 - 3y^2 + z^2)$$

where A is a constant. (a) Derive an expression for the electric field $\vec{E}$ at any point in this region. (b) The work done by the field when a $1.50 \mu\text{C}$ test charge moves from the point $(x, y, z) = (0, 0, 0.250 \text{ m})$ to the origin is measured to be $6.00 \times 10^{-5} \text{ J}$. Determine A . (c) Determine the electric field at the point $(0, 0, 0.250 \text{ m})$. (d) Show that in every plane parallel to the xz -plane the equipotential contours are circles. (e) What is the radius of the equipotential contour corresponding to $V = 1280 \text{ V}$ and $y = 2.00 \text{ m}$?

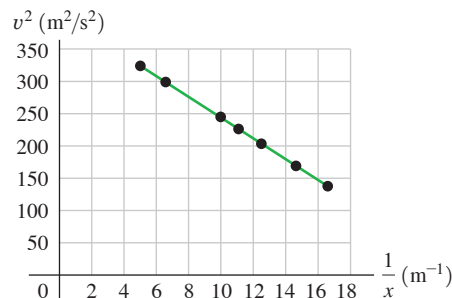
23.77 •• DATA The electric potential in a region that is within 2.00 m of the origin of a rectangular coordinate system is given by $V = Ax^l + By^m + Cz^n + D$, where A, B, C, D, l, m , and n are constants. The units of A, B, C , and D are such that if x, y , and z are in meters, then V is in volts. You measure V and each component of the electric field at four points and obtain these results:

Point	$(x, y, z) \text{ (m)}$	$V \text{ (V)}$	$E_x \text{ (V/m)}$	$E_y \text{ (V/m)}$	$E_z \text{ (V/m)}$
1	(0, 0, 0)	10.0	0	0	0
2	(1.00, 0, 0)	4.0	12.0	0	0
3	(0, 1.00, 0)	6.0	0	12.0	0
4	(0, 0, 1.00)	8.0	0	0	12.0

- (a) Use the data in the table to calculate A, B, C, D, l, m , and n . (b) What are V and the magnitude of E at the points $(0, 0, 0)$, $(0.50 \text{ m}, 0.50 \text{ m}, 0.50 \text{ m})$, and $(1.00 \text{ m}, 1.00 \text{ m}, 1.00 \text{ m})$?

23.78 •• DATA A small, stationary sphere carries a net charge Q . You perform the following experiment to measure Q : From a large distance you fire a small particle with mass $m = 4.00 \times 10^{-4} \text{ kg}$ and charge $q = 5.00 \times 10^{-8} \text{ C}$ directly at the center of the sphere. The apparatus you are using measures the particle's speed v as a function of the distance x from the sphere. The sphere's mass is much greater than the mass of the projectile particle, so you assume that the sphere remains at rest. All of the measured values of x are much larger than the radius of either object, so you treat both objects as point particles. You plot your data on a graph of v^2 versus $(1/x)$ (**Fig. P23.78**). The straight line $v^2 = 400 \text{ m}^2/\text{s}^2 - [(15.75 \text{ m}^3/\text{s}^2)/x]$ gives a good fit to the data points. (a) Explain why the graph is a straight line. (b) What is the initial speed v_0 of the particle when it is very far from the sphere? (c) What is Q ? (d) How close does the particle get to the sphere? Assume that this distance is much larger than the radii of the particle and sphere, so continue to treat them as point particles and to assume that the sphere remains at rest.

Figure P23.78



23.79 ••• DATA The Millikan Oil-Drop Experiment. The charge of an electron was first measured by the American physicist Robert Millikan during 1909–1913. In his experiment, oil was sprayed in very fine drops (about 10^{-4} mm in diameter) into the space between two parallel horizontal plates separated by a distance d . A potential difference V_{AB} was maintained between the plates, causing a downward electric field between them. Some of the oil drops acquired a negative charge because of frictional effects or because of ionization of the surrounding air by x rays or radioactivity. The drops were observed through a microscope. (a) Show that an oil drop of radius r at rest between the plates remained at rest if the magnitude of its charge was

$$q = \frac{4\pi}{3} \frac{\rho r^3 g d}{V_{AB}}$$

where ρ is oil's density. (Ignore the buoyant force of the air.) By adjusting V_{AB} to keep a given drop at rest, Millikan determined the charge on that drop, provided its radius r was known. (b) Millikan's oil drops were much too small to measure their radii directly. Instead, Millikan determined r by cutting off the electric field and measuring the *terminal speed* v_t of the drop as it fell. (We discussed terminal speed in Section 5.3.) The viscous force F on a sphere of radius r moving at speed v through a fluid with viscosity η is given by Stokes's law: $F = 6\pi\eta rv$. When a drop fell at v_t , the viscous force just balanced the drop's weight $w = mg$. Show that the magnitude of the charge on the drop was

$$q = 18\pi \frac{d}{V_{AB}} \sqrt{\frac{\eta^3 v_t^3}{2\rho g}}$$

(c) You repeat the Millikan oil-drop experiment. Four of your measured values of V_{AB} and v_t are listed in the table:

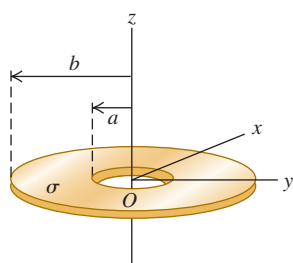
Drop	1	2	3	4
V_{AB} (V)	9.16	4.57	12.32	6.28
v_t (10^{-5} m/s)	2.54	0.767	4.39	1.52

In your apparatus, the separation d between the horizontal plates is 1.00 mm. The density of the oil you use is 824 kg/m^3 . For the viscosity η of air, use the value $1.81 \times 10^{-5} \text{ N}\cdot\text{s/m}^2$. Assume that $g = 9.80 \text{ m/s}^2$. Calculate the charge q of each drop. (d) If electric charge is *quantized* (that is, exists in multiples of the magnitude of the charge of an electron), then the charge on each drop is $-ne$, where n is the number of excess electrons on each drop. (All four drops in your table have negative charge.) Drop 2 has the smallest magnitude of charge observed in the experiment, for all 300 drops on which measurements were made, so assume that its charge is due to an excess charge of one electron. Determine the number of excess electrons n for each of the other three drops. (e) Use $q = -ne$ to calculate e from the data for each of the four drops, and average these four values to get your best experimental value of e .

CHALLENGE PROBLEMS

23.80 ••• CALC An annulus with an inner radius of a and an outer radius of b has charge density σ and lies in the xy -plane with its center at the origin, as shown in **Fig. P23.80**. (a) Using the convention that the potential vanishes at infinity, determine the potential at all points on the z -axis. (b) Determine the electric field at all points on the z -axis by differentiating the potential. (c) Show that in the limit $a \rightarrow 0$, $b \rightarrow \infty$ the

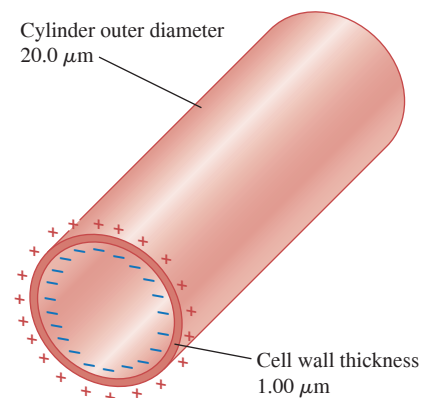
Figure P23.80



electric field reproduces the result obtained in Example 22.7 for an infinite plane sheet of charge. (d) If $a = 5.00 \text{ cm}$, $b = 10.0 \text{ cm}$ and the total charge on the annulus is $1.00 \mu\text{C}$, what is the potential at the origin? (e) If a particle with mass 1.00 g (much less than the mass of the annulus) and charge $1.00 \mu\text{C}$ is placed at the origin and given the slightest nudge, it will be projected along the z -axis. In this case, what will be its ultimate speed?

23.81 ••• BIO A heart cell can be modeled as a cylindrical shell that is $100 \mu\text{m}$ long, with an outer diameter of $20.0 \mu\text{m}$ and a cell wall thickness of $1.00 \mu\text{m}$, as shown in **Fig. P23.81**. Potassium ions move across the cell wall, depositing positive charge on the outer surface and leaving a net negative charge on the inner surface. During the so-called resting phase, the inside of the cell has a potential that is 90.0 mV lower than the potential on its outer surface. (a) If the net charge of the cell is zero, what is the magnitude of the total charge on either cell wall membrane? Ignore edge effects and treat the cell as a very long cylinder. (b) What is the magnitude of the electric field just inside the cell wall? (c) In a subsequent depolarization event, sodium ions move through channels in the cell wall, so that the inner membrane becomes positively charged. At the end of this event, the inside of the cell has a potential that is 20.0 mV higher than the potential outside the cell. If we model this event by charge moving from the outer membrane to the inner membrane, what magnitude of charge moves across the cell wall during this event? (d) If this were done entirely by the motion of sodium ions, Na^+ , how many ions have moved?

Figure P23.81



23.82 ••• CALC A hollow, thin-walled insulating cylinder of radius R and length L (like the cardboard tube in a roll of toilet paper) has charge Q uniformly distributed over its surface. (a) Calculate the electric potential at all points along the axis of the tube. Take the origin to be at the center of the tube, and take the potential to be zero at infinity. (b) Show that if $L \ll R$, the result of part (a) reduces to the potential on the axis of a ring of charge of radius R . (See Example 23.11 in Section 23.3.) (c) Use the result of part (a) to find the electric field at all points along the axis of the tube.

23.83 ••• CP In experiments in which atomic nuclei collide, head-on collisions like that described in Problem 23.75 do happen, but “near misses” are more common. Suppose the alpha particle in that problem is not “aimed” at the center of the lead nucleus but has an initial non-zero angular momentum (with respect to the stationary lead nucleus) of magnitude $L = p_0 b$, where p_0 is the magnitude of the particle's initial momentum and $b = 1.00 \times 10^{-12} \text{ m}$. What is the distance of closest approach? Repeat for $b = 1.00 \times 10^{-13} \text{ m}$ and $b = 1.00 \times 10^{-14} \text{ m}$.

MCAT-STYLE PASSAGE PROBLEMS

Materials Analysis with Ions. *Rutherford backscattering spectrometry* (RBS) is a technique used to determine the structure and composition of materials. A beam of ions (typically helium ions) is accelerated to high energy and aimed at a sample. By analyzing the distribution and energy of the ions that are scattered from (that is, deflected by collisions with) the atoms in the sample, researchers can determine the sample's composition. To accelerate the ions to high energies, a *tandem electrostatic accelerator* may be used. In this device, negative ions (He^-) start at a potential $V = 0$ and are accelerated by a high positive voltage at the midpoint of the accelerator. The high voltage produces a constant electric field in the acceleration tube through which the ions move. When accelerated ions reach the midpoint, the electrons are stripped off, turning the negative ions into doubly positively charged ions (He^{++}). These positive ions are then repelled from the midpoint by the high positive voltage there and continue to accelerate to the far end of the accelerator, where again $V = 0$.

23.84 For a particular experiment, helium ions are to be given a kinetic energy of 3.0 MeV. What should the voltage at the center of the accelerator be, assuming that the ions start essentially at rest? (a) -3.0 MV; (b) $+3.0$ MV; (c) $+1.5$ MV; (d) $+1.0$ MV.

23.85 A helium ion (He^{++}) that comes within about 10 fm of the center of the nucleus of an atom in the sample may induce a nuclear reaction instead of simply scattering. Imagine a helium ion with a kinetic energy of 3.0 MeV heading straight toward an atom at rest in the sample. Assume that the atom stays fixed. What minimum charge can the nucleus of the atom have such that the helium ion gets no closer than 10 fm from the center of the atomic nucleus? ($1 \text{ fm} = 1 \times 10^{-15} \text{ m}$, and e is the magnitude of the charge of an electron or a proton.) (a) $2e$; (b) $11e$; (c) $20e$; (d) $22e$.

23.86 The maximum voltage at the center of a typical tandem electrostatic accelerator is 6.0 MV. If the distance from one end of the acceleration tube to the midpoint is 12 m, what is the magnitude of the average electric field in the tube under these conditions? (a) 41,000 V/m; (b) 250,000 V/m; (c) 500,000 V/m; (d) 6,000,000 V/m.

ANSWERS

Chapter Opening Question ?

(iii) A large, constant potential difference V_{ab} is maintained between the welding tool (a) and the metal pieces to be welded (b). For a given potential difference between two conductors a and b , the smaller the distance d separating the conductors, the greater is the magnitude E of the field between them. Hence d must be small in order for E to be large enough to ionize the gas between the conductors (see Section 23.3) and produce an arc through this gas.

Key Example VARIATION Problems

VP23.2.1 (a) $1.51 \times 10^{-16} \text{ J}$ (b) $2.32 \times 10^{-15} \text{ J}$

VP23.2.2 (a) $5.22 \times 10^{-15} \text{ J}$ (b) $1.92 \times 10^{-18} \text{ C}$

VP23.2.3 (a) $-e^2/24\pi\epsilon_0 a$ (b) $-7e^2/24\pi\epsilon_0 a$

VP23.2.4 (a) $7.21 \times 10^{-9} \text{ C}$ (b) $5.59 \times 10^{-6} \text{ J}$

VP23.7.1 (a) $+25.0 \text{ V}$ (b) $+15.0 \text{ V}$ (c) 0 (d) -25.0 V

VP23.7.2 (a) $3.6 \times 10^3 \text{ V}$ (b) $2.2 \times 10^3 \text{ V}$ (c) $1.6 \times 10^3 \text{ V}$

(d) point (a); no

VP23.7.3 (a) $+133 \text{ V}$ (b) -133 V (c) $1.98 \times 10^5 \text{ m/s}$

VP23.7.4 (a) $Q/8\pi\epsilon_0 R$ (b) the center

VP23.12.1 (a) $qq_0/30\pi\epsilon_0 R$ (b) $-5qq_0/56\pi\epsilon_0 R$ (c) $3qq_0/16\pi\epsilon_0 R$

VP23.12.2 (a) $+16.0 \text{ V}$ (b) $3.20 \times 10^{-8} \text{ J}$ (c) lower, 3.58 m/s

VP23.12.3 (a) $1.61 \times 10^{-6} \text{ J}$ (b) $5.39 \times 10^{-6} \text{ J}$ (c) $K = 3.42 \times 10^{-6} \text{ J}$, $v = 41.3 \text{ m/s}$

VP23.12.4 (a) $dq = (Q/L)dx$ (b) $(Q/4\pi\epsilon_0 L)\ln 2$

Bridging Problem

$$\frac{qQ}{8\pi\epsilon_0 a} \ln \left(\frac{L+a}{L-a} \right)$$



? In flash photography, the energy used to make the flash is stored in a capacitor, which consists of two closely spaced conductors that carry opposite charges. If the amount of charge on the conductors is doubled, by what factor does the stored energy increase? (i) $\sqrt{2}$; (ii) 2; (iii) $2\sqrt{2}$; (iv) 4; (v) 8.

24 Capacitance and Dielectrics

When you stretch the rubber band of a slingshot or pull back the string of an archer's bow, you are storing mechanical energy as elastic potential energy. A capacitor is a device that stores *electric* potential energy and electric charge. To make a capacitor, just insulate two conductors from each other. To store energy in this device, transfer charge from one conductor to the other so that one has a negative charge and the other has an equal amount of positive charge. Work must be done to move the charges through the resulting potential difference between the conductors, and the work done is stored as electric potential energy.

Capacitors have a tremendous number of practical applications in devices such as electronic flash units for photography, mobile phones, airbag sensors for cars, and radio and television receivers. We'll encounter many of these applications in later chapters (particularly Chapter 31, in which we'll see the crucial role played by capacitors in the alternating-current circuits that pervade our technological society). In this chapter, however, our emphasis is on the fundamental properties of capacitors. For a particular capacitor, the ratio of the charge on each conductor to the potential difference between the conductors is a constant, called the *capacitance*. The capacitance depends on the sizes and shapes of the conductors and on the insulating material (if any) between them. Compared to the case in which there is only vacuum between the conductors, the capacitance increases when an insulating material (a *dielectric*) is present. This happens because a redistribution of charge, called *polarization*, takes place within the insulating material. Studying polarization will give us added insight into the electrical properties of matter.

Capacitors also give us a new way to think about electric potential energy. The energy stored in a charged capacitor is related to the electric field in the space between the conductors. We'll see that electric potential energy can be regarded as being stored *in the field itself*. The idea that the electric field is itself a storehouse of energy is at the heart of the theory of electromagnetic waves and our modern understanding of the nature of light, to be discussed in Chapter 32.

LEARNING OUTCOMES

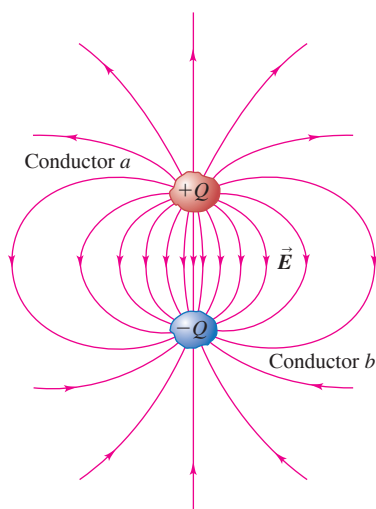
In this chapter, you'll learn...

- 24.1** The nature of capacitors, and how to calculate a quantity that measures their ability to store charge.
- 24.2** How to analyze capacitors connected in a network.
- 24.3** How to calculate the amount of energy stored in a capacitor.
- 24.4** What dielectrics are, and how they make capacitors more effective.
- 24.5** How a dielectric inside a charged capacitor becomes polarized.
- 24.6** How to use Gauss's laws when dielectrics are present.

You'll need to review...

- 21.2, 21.5, 21.7** Polarization; field of charged conductors; electric dipoles.
- 22.3–22.5** Gauss's law.
- 23.3, 23.4** Potential for charged conductors; potential due to a cylindrical charge distribution.

Figure 24.1 Any two conductors a and b insulated from each other form a capacitor.



24.1 CAPACITORS AND CAPACITANCE

Any two conductors separated by an insulator (or a vacuum) form a **capacitor** (Fig. 24.1). In most practical applications, each conductor initially has zero net charge and electrons are transferred from one conductor to the other; this is called *charging* the capacitor. Then the two conductors have charges with equal magnitude and opposite sign, and the *net* charge on the capacitor as a whole remains zero. We'll assume throughout this chapter that this is the case. When we say that a capacitor has charge Q , or that a charge Q is *stored* on the capacitor, we mean that the conductor at higher potential has charge $+Q$ and the conductor at lower potential has charge $-Q$ (assuming that Q is positive). Keep this in mind in the following discussion and examples.

In circuit diagrams a capacitor is represented by either of these symbols:



The vertical lines (straight or curved) represent the conductors, and the horizontal lines represent wires connected to either conductor. One common way to charge a capacitor is to connect these two wires to opposite terminals of a battery. Once the charges Q and $-Q$ are established on the conductors, the battery is disconnected. This gives a fixed *potential difference* V_{ab} between the conductors (that is, the potential of the positively charged conductor a with respect to the negatively charged conductor b) that is just equal to the voltage of the battery.

The electric field at any point in the region between the conductors is proportional to the magnitude Q of charge on each conductor. It follows that the potential difference V_{ab} between the conductors is also proportional to Q . If we double the magnitude of charge on each conductor, the charge density at each point doubles, the electric field at each point doubles, and the potential difference between conductors doubles; however, the *ratio* of charge to potential difference does not change. This ratio is called the **capacitance** C of the capacitor:

$$\text{Capacitance of a capacitor } C = \frac{Q}{V_{ab}} \quad (24.1)$$

$\leftarrow$ Magnitude of charge on each conductor
 $\leftarrow$ Potential difference between conductors (a has charge $+Q$, b has charge $-Q$)

The SI unit of capacitance is called one **farad** (1 F), in honor of the 19th-century English physicist Michael Faraday. From Eq. (24.1), one farad is equal to one *coulomb per volt* (1 C/V):

$$1 \text{ F} = 1 \text{ farad} = 1 \text{ C/V} = 1 \text{ coulomb/volt}$$

CAUTION Capacitance vs. coulombs Don't confuse the symbol C for capacitance (which is always in italics) with the abbreviation C for coulombs (which is never italicized).

The greater the capacitance C of a capacitor, the greater the magnitude Q of charge on either conductor for a given potential difference V_{ab} and hence the greater the amount of stored energy. (Remember that potential is potential energy per unit charge.) Thus *capacitance is a measure of the ability of a capacitor to store energy*. We'll see that the capacitance depends only on the shapes, sizes, and relative positions of the conductors and on the nature of the insulator between them. (For special types of insulating materials, the capacitance *does* depend on Q and V_{ab} . We won't discuss these materials, however.)

Calculating Capacitance: Capacitors in Vacuum

We can calculate the capacitance C of a given capacitor by finding the potential difference V_{ab} between the conductors for a given magnitude of charge Q and then using Eq. (24.1). For now we'll consider only *capacitors in vacuum*; that is, empty space separates the conductors that make up the capacitor.

The simplest form of capacitor consists of two parallel conducting plates, each with area A , separated by a distance d that is small in comparison with their dimensions (Fig. 24.2a). When the plates are charged, the electric field is almost completely localized in the region between the plates (Fig. 24.2b). As we discussed in Example 22.8 (Section 22.4), the field between such plates is essentially *uniform*, and the charges on the plates are uniformly distributed over their opposing surfaces. We call this arrangement a **parallel-plate capacitor**.

We found the electric-field magnitude E for this arrangement in Example 21.12 (Section 21.5) by using the principle of superposition of electric fields and again in Example 22.8 (Section 22.4) by using Gauss's law. It would be a good idea to review those examples. We found that $E = \sigma/\epsilon_0$, where σ is the magnitude (absolute value) of the surface charge density on each plate. This is equal to the magnitude of the total charge Q on each plate divided by the area A of the plate, or $\sigma = Q/A$, so the field magnitude E can be expressed as

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

The field is uniform and the distance between the plates is d , so the potential difference (voltage) between the two plates is

$$V_{ab} = Ed = \frac{1}{\epsilon_0} \frac{Qd}{A}$$

Thus

$$\text{Capacitance of a parallel-plate capacitor in vacuum} \quad C = \frac{Q}{V_{ab}} = \epsilon_0 \frac{A}{d} \quad (24.2)$$

Magnitude of charge on each plate
Area of each plate
Distance between plates
Electric constant

Potential difference between plates

The capacitance depends on only the geometry of the capacitor; it is directly proportional to the area A of each plate and inversely proportional to their separation d . The quantities A and d are constants for a given capacitor, and ϵ_0 is a universal constant. Thus in vacuum the capacitance C is a constant independent of the charge on the capacitor or the potential difference between the plates. If one of the capacitor plates is flexible, C changes as the plate separation d changes. This is the operating principle of a condenser microphone (Fig. 24.3).

When matter is present between the plates, its properties affect the capacitance. We'll return to this topic in Section 24.4. Meanwhile, we remark that if the space contains air at atmospheric pressure instead of vacuum, the capacitance differs from the prediction of Eq. (24.2) by less than 0.06%.

In Eq. (24.2), if A is in square meters and d in meters, then C is in farads. The units of the electric constant ϵ_0 are $\text{C}^2/\text{N} \cdot \text{m}^2$, so

$$1 \text{ F} = 1 \text{ C}^2/\text{N} \cdot \text{m} = 1 \text{ C}^2/\text{J}$$

Because $1 \text{ V} = 1 \text{ J/C}$ (energy per unit charge), this equivalence agrees with our definition $1 \text{ F} = 1 \text{ C/V}$. Finally, we can express the units of ϵ_0 as $1 \text{ C}^2/\text{N} \cdot \text{m}^2 = 1 \text{ F/m}$, so

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

This relationship is useful in capacitance calculations, and it also helps us to verify that Eq. (24.2) is dimensionally consistent.

Figure 24.2 A charged parallel-plate capacitor.

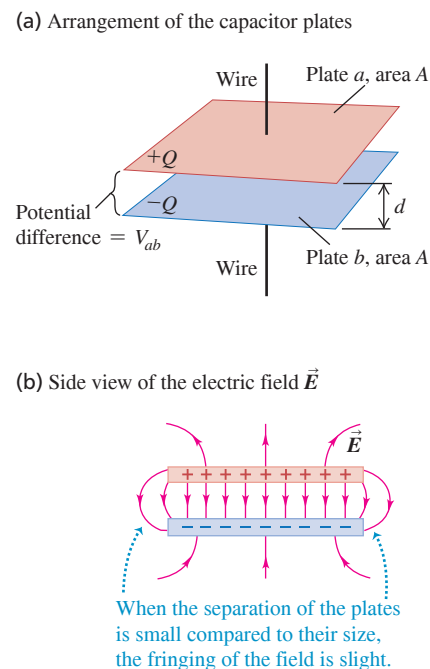
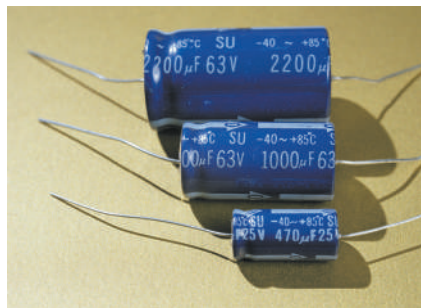


Figure 24.3 Inside a condenser microphone is a capacitor with one rigid plate and one flexible plate. The two plates are kept at a constant potential difference V_{ab} . Sound waves cause the flexible plate to move back and forth, varying the capacitance C and causing charge to flow to and from the capacitor in accordance with the relationship $C = Q/V_{ab}$. Thus a sound wave is converted to a charge flow that can be amplified and recorded digitally.



Figure 24.4 A commercial capacitor is labeled with the value of its capacitance. For these capacitors, $C = 2200\ \mu\text{F}$, $1000\ \mu\text{F}$, and $470\ \mu\text{F}$.



One farad is a very large capacitance, as the following example shows. In many applications the most convenient units of capacitance are the *microfarad* ($1\ \mu\text{F} = 10^{-6}\ \text{F}$) and the *picofarad* ($1\ \text{pF} = 10^{-12}\ \text{F}$). For example, the flash unit in a point-and-shoot camera uses a capacitor of a few hundred microfarads (Fig. 24.4), while capacitances in a radio tuning circuit are typically from 10 to 100 picofarads.

For *any* capacitor in vacuum, the capacitance C depends only on the shapes, dimensions, and separation of the conductors that make up the capacitor. If the conductor shapes are more complex than those of the parallel-plate capacitor, the expression for capacitance is more complicated than in Eq. (24.2). In the following examples we show how to calculate C for two other conductor geometries.

CAUTION **Capacitor charge and potential difference** Saying that a capacitor carries charge Q means that one conductor has positive charge $+Q$ and the other conductor has negative charge $-Q$. The *net* charge on the capacitor is zero! Furthermore, the positively charged conductor is always at higher potential than the negatively charged conductor. If the potential difference V_{ab} between the conductors is 500 V, the potential of the positive conductor is 500 V greater than that of the negative conductor. !

EXAMPLE 24.1 Size of a 1 F capacitor

The parallel plates of a 1.0 F capacitor are 1.0 mm apart. What is their area?

IDENTIFY and SET UP This problem uses the relationship among the capacitance C , plate separation d , and plate area A (our target variable) for a parallel-plate capacitor. We solve Eq. (24.2) for A .

EXECUTE From Eq. (24.2),

$$A = \frac{Cd}{\epsilon_0} = \frac{(1.0\ \text{F})(1.0 \times 10^{-3}\ \text{m})}{8.85 \times 10^{-12}\ \text{F/m}} = 1.1 \times 10^8\ \text{m}^2$$

EVALUATE This corresponds to a square about 10 km (about 6 miles) on a side! The volume of such a capacitor would be at least

$Ad = 1.1 \times 10^5\ \text{m}^3$, equivalent to that of a cube about 50 m on a side. In fact, it's possible to make 1 F capacitors a few *centimeters* on a side. The trick is to have an appropriate substance between the plates rather than a vacuum, so that (among other things) the plate separation d can be greatly reduced. We'll explore this further in Section 24.4.

KEYCONCEPT The capacitance of a capacitor is a measure of how much charge its plates will hold for a given potential difference (voltage) between the plates. Capacitance is proportional to the area of the plates and inversely proportional to the distance between them.

EXAMPLE 24.2 Properties of a parallel-plate capacitor

WITH VARIATION PROBLEMS

The plates of a parallel-plate capacitor in vacuum are 5.00 mm apart and $2.00\ \text{m}^2$ in area. A 10.0 kV potential difference is applied across the capacitor. Compute (a) the capacitance; (b) the charge on each plate; and (c) the magnitude of the electric field between the plates.

IDENTIFY and SET UP We are given the plate area A , the plate spacing d , and the potential difference $V_{ab} = 1.00 \times 10^4\ \text{V}$ for this parallel-plate capacitor. Our target variables are the capacitance C , the charge Q on each plate, and the electric-field magnitude E . We use Eq. (24.2) to calculate C and then use Eq. (24.1) and V_{ab} to find Q . We use $E = Q/\epsilon_0 A$ to find E .

EXECUTE (a) From Eq. (24.2),

$$\begin{aligned} C &= \epsilon_0 \frac{A}{d} = (8.85 \times 10^{-12}\ \text{F/m}) \frac{(2.00\ \text{m}^2)}{5.00 \times 10^{-3}\ \text{m}} \\ &= 3.54 \times 10^{-9}\ \text{F} = 0.00354\ \mu\text{F} \end{aligned}$$

(b) The charge on the capacitor is

$$\begin{aligned} Q &= CV_{ab} = (3.54 \times 10^{-9}\ \text{C/V})(1.00 \times 10^4\ \text{V}) \\ &= 3.54 \times 10^{-5}\ \text{C} = 35.4\ \mu\text{C} \end{aligned}$$

The plate at higher potential has charge $+35.4\ \mu\text{C}$, and the other plate has charge $-35.4\ \mu\text{C}$.

(c) The electric-field magnitude is

$$\begin{aligned} E &= \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A} = \frac{3.54 \times 10^{-5}\ \text{C}}{(8.85 \times 10^{-12}\ \text{C}^2/\text{N} \cdot \text{m}^2)(2.00\ \text{m}^2)} \\ &= 2.00 \times 10^6\ \text{N/C} \end{aligned}$$

EVALUATE We can also find E by recalling that the electric field is equal in magnitude to the potential gradient [Eq. (23.22)]. The field between the plates is uniform, so

$$E = \frac{V_{ab}}{d} = \frac{1.00 \times 10^4\ \text{V}}{5.00 \times 10^{-3}\ \text{m}} = 2.00 \times 10^6\ \text{V/m}$$

(Remember that $1\ \text{N/C} = 1\ \text{V/m}$.)

KEYCONCEPT The electric field between the plates of a parallel-plate capacitor is uniform. To find its magnitude E , divide the potential difference between the plates by their separation; E is also proportional to the ratio of the charge on the capacitor plates to the plate's area.

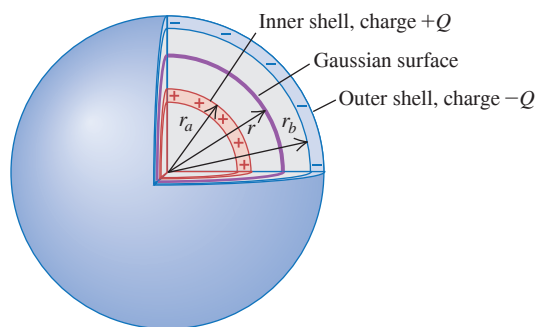
EXAMPLE 24.3 A spherical capacitor**WITH VARIATION PROBLEMS**

Two concentric spherical conducting shells are separated by vacuum (**Fig. 24.5**). The inner shell has total charge $+Q$ and outer radius r_a , and the outer shell has charge $-Q$ and inner radius r_b . Find the capacitance of this spherical capacitor.

IDENTIFY and SET UP By definition, the capacitance C is the magnitude Q of the charge on either sphere divided by the potential difference V_{ab} between the spheres. We first find V_{ab} , and then use Eq. (24.1) to find the capacitance $C = Q/V_{ab}$.

EXECUTE Using a Gaussian surface such as that shown in **Fig. 24.5**, we found in Example 22.5 (Section 22.4) that the charge on a conducting sphere produces zero field *inside* the sphere, so the outer sphere makes no contribution to the field between the spheres. Therefore the electric field *and* the electric potential between the shells are the same as those outside a charged conducting sphere with charge $+Q$. We considered that problem in Example 23.8 (Section 23.3), so the same result applies here: The potential at any point between the spheres is $V = Q/4\pi\epsilon_0 r$.

Figure 24.5 A spherical capacitor.



Hence the potential of the inner (positive) conductor at $r = r_a$ with respect to that of the outer (negative) conductor at $r = r_b$ is

$$\begin{aligned} V_{ab} &= V_a - V_b = \frac{Q}{4\pi\epsilon_0 r_a} - \frac{Q}{4\pi\epsilon_0 r_b} \\ &= \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right) = \frac{Q}{4\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b} \end{aligned}$$

The capacitance is then

$$C = \frac{Q}{V_{ab}} = 4\pi\epsilon_0 \frac{r_a r_b}{r_b - r_a}$$

As an example, if $r_a = 9.5$ cm and $r_b = 10.5$ cm,

$$\begin{aligned} C &= 4\pi(8.85 \times 10^{-12} \text{ F/m}) \frac{(0.095 \text{ m})(0.105 \text{ m})}{0.010 \text{ m}} \\ &= 1.1 \times 10^{-10} \text{ F} = 110 \text{ pF} \end{aligned}$$

EVALUATE We can relate our expression for C to that for a parallel-plate capacitor. The quantity $4\pi r_a r_b$ is intermediate between the areas $4\pi r_a^2$ and $4\pi r_b^2$ of the two spheres; in fact, it's the *geometric mean* of these two areas, which we can denote by A_{gm} . The distance between spheres is $d = r_b - r_a$, so we can write $C = 4\pi\epsilon_0 r_a r_b / (r_b - r_a) = \epsilon_0 A_{\text{gm}} / d$. This has the same form as for parallel plates: $C = \epsilon_0 A / d$. If the distance between spheres is very small in comparison to their radii, their capacitance is the same as that of parallel plates with the same area and spacing.

KEYCONCEPT To find the capacitance of any capacitor made up of two conductors, first calculate the potential difference V_{ab} between the two conductors when they carry charges $+Q$ and $-Q$. The capacitance is equal to Q divided by V_{ab} .

EXAMPLE 24.4 A cylindrical capacitor**WITH VARIATION PROBLEMS**

Two long, coaxial cylindrical conductors are separated by vacuum (**Fig. 24.6**). The inner cylinder has outer radius r_a and linear charge density $+\lambda$. The outer cylinder has inner radius r_b and linear charge density $-\lambda$. Find the capacitance per unit length for this capacitor.

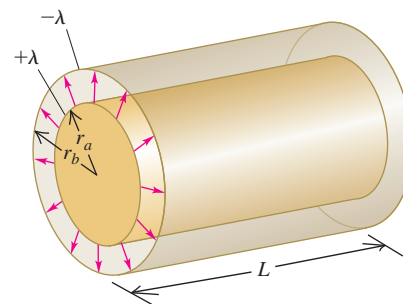
IDENTIFY and SET UP As in Example 24.3, we use the definition of capacitance, $C = Q/V_{ab}$. We use the result of Example 23.10 (Section 23.3) to find the potential difference V_{ab} between the cylinders, and find the charge Q on a length L of the cylinders from the linear charge density. We then find the corresponding capacitance C from Eq. (24.1). Our target variable is this capacitance divided by L .

EXECUTE As in Example 24.3, the potential V between the cylinders is not affected by the presence of the charged outer cylinder. Hence our result in Example 23.10 for the potential outside a charged conducting cylinder also holds in this example for potential in the space between the cylinders:

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_0}{r}$$

Here r_0 is the arbitrary, *finite* radius at which $V = 0$. We take $r_0 = r_b$, the radius of the inner surface of the outer cylinder. Then the potential at the outer surface of the inner cylinder (at which $r = r_a$) is just the potential V_{ab} of the inner (positive) cylinder a with respect to the outer (negative) cylinder b :

Figure 24.6 A long cylindrical capacitor. The linear charge density λ is assumed to be positive in this figure. The magnitude of charge in a length L of either cylinder is λL .



$$V_{ab} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}$$

If λ is positive as in **Fig. 24.6**, then V_{ab} is positive as well: The inner cylinder is at higher potential than the outer.

Continued

The total charge Q in a length L is $Q = \lambda L$, so from Eq. (24.1) the capacitance C of a length L is

$$C = \frac{Q}{V_{ab}} = \frac{\lambda L}{\frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}} = \frac{2\pi\epsilon_0 L}{\ln(r_b/r_a)}$$

The capacitance per unit length is

$$\frac{C}{L} = \frac{2\pi\epsilon_0}{\ln(r_b/r_a)}$$

Substituting $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m} = 8.85 \text{ pF/m}$, we get

$$\frac{C}{L} = \frac{55.6 \text{ pF/m}}{\ln(r_b/r_a)}$$

EVALUATE The capacitance of coaxial cylinders is determined entirely by their dimensions, just as for parallel-plate and spherical capacitors. Ordinary coaxial cables are made like this but with an insulating material instead of vacuum between the conductors. A typical cable used for connecting a television to a cable TV feed has a capacitance per unit length of 69 pF/m.

KEYCONCEPT The capacitance per unit length of two long, coaxial, conducting cylinders depends on the ratio of the inner radius of the outer conductor to the outer radius of the inner conductor.

Figure 24.7 An assortment of commercially available capacitors.



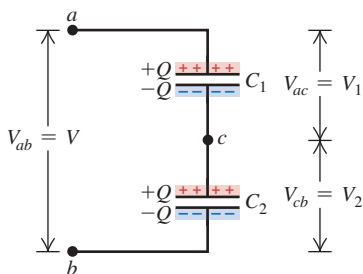
Figure 24.8 A series connection of two capacitors.

(a) Two capacitors in series

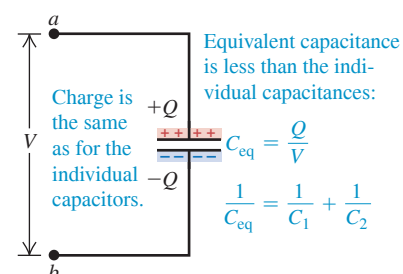
Capacitors in series:

- The capacitors have the same charge Q .
- Their potential differences add:

$$V_{ac} + V_{cb} = V_{ab}$$



(b) The equivalent single capacitor



TEST YOUR UNDERSTANDING OF SECTION 24.1 A capacitor has vacuum in the space between the conductors. If you double the amount of charge on each conductor, what happens to the capacitance? (i) It increases; (ii) it decreases; (iii) it remains the same; (iv) the answer depends on the size or shape of the conductors.

ANSWER

(iii) The capacitance does not depend on the value of the charge Q . Doubling Q causes the potential difference V_{ab} to double, so the capacitance $C = Q/V_{ab}$ remains the same. These statements are true no matter what the geometry of the capacitor.

24.2 CAPACITORS IN SERIES AND PARALLEL

Capacitors are manufactured with certain standard capacitances and working voltages (Fig. 24.7). However, these standard values may not be the ones you actually need in a particular application. You can obtain the values you need by combining capacitors; many combinations are possible, but the simplest combinations are a series connection and a parallel connection.

Capacitors in Series

Figure 24.8a is a schematic diagram of a **series connection**. Two capacitors are connected in series (one after the other) by conducting wires between points a and b . Both capacitors are initially uncharged. When a constant positive potential difference V_{ab} is applied between points a and b , the capacitors become charged; the figure shows that the charge on *all* conducting plates has the same magnitude. To see why, note first that the top plate of C_1 acquires a positive charge Q . The electric field of this positive charge pulls negative charge up to the bottom plate of C_1 until all of the field lines that begin on the top plate end on the bottom plate. This requires that the bottom plate have charge $-Q$. These negative charges had to come from the top plate of C_2 , which becomes positively charged with charge $+Q$. This positive charge then pulls negative charge $-Q$ from the connection at point b onto the bottom plate of C_2 . The total charge on the lower plate of C_1 and the upper plate of C_2 together must always be zero because these plates aren't connected to anything except each other. Thus *in a series connection the magnitude of charge on all plates is the same*.

Referring to Fig. 24.8a, we can write the potential differences between points a and c , c and b , and a and b as

$$V_{ac} = V_1 = \frac{Q}{C_1}, \quad V_{cb} = V_2 = \frac{Q}{C_2}, \quad V_{ab} = V = V_1 + V_2 = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

and so

$$\frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2} \quad (24.3)$$

Following a common convention, we use the symbols V_1 , V_2 , and V to denote the potential differences V_{ac} (across the first capacitor), V_{cb} (across the second capacitor), and V_{ab} (across the entire combination of capacitors), respectively.

The **equivalent capacitance** C_{eq} of the series combination is defined as the capacitance of a *single* capacitor for which the charge Q is the same as for the combination, when the potential difference V is the same. In other words, the combination can be replaced by an *equivalent capacitor* of capacitance C_{eq} . For such a capacitor, shown in Fig. 24.8b,

$$C_{eq} = \frac{Q}{V} \quad \text{or} \quad \frac{1}{C_{eq}} = \frac{V}{Q} \quad (24.4)$$

Combining Eqs. (24.3) and (24.4), we find

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

We can extend this analysis to any number of capacitors in series. We find the following result for the *reciprocal* of the equivalent capacitance:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \quad (24.5)$$

Capacitors in series: Equivalent capacitance of series combination Capacitances of individual capacitors

The reciprocal of the equivalent capacitance of a series combination equals the sum of the reciprocals of the individual capacitances. In a series connection the equivalent capacitance is always *less than* any individual capacitance.

CAUTION Capacitors in series The magnitude of charge is the same on all plates of all the capacitors in a series combination; however, the potential differences of the individual capacitors are *not* the same unless their individual capacitances are the same. The potential differences of the individual capacitors add to give the total potential difference across the series combination: $V_{\text{total}} = V_1 + V_2 + V_3 + \cdots$!

Capacitors in Parallel

The arrangement shown in **Fig. 24.9a** is called a **parallel connection**. Two capacitors are connected in parallel between points a and b . In this case the upper plates of the two capacitors are connected by conducting wires to form an equipotential surface, and the lower plates form another. Hence *in a parallel connection the potential difference for all individual capacitors is the same* and is equal to $V_{ab} = V$. The charges Q_1 and Q_2 are not necessarily equal, however, since charges can reach each capacitor independently from the source (such as a battery) of the voltage V_{ab} . The charges are

$$Q_1 = C_1 V \quad \text{and} \quad Q_2 = C_2 V$$

The *total* charge Q of the combination, and thus the total charge on the equivalent capacitor, is

$$Q = Q_1 + Q_2 = (C_1 + C_2)V$$

so

$$\frac{Q}{V} = C_1 + C_2 \quad (24.6)$$

The parallel combination is equivalent to a single capacitor with the same total charge $Q = Q_1 + Q_2$ and potential difference V as the combination (Fig. 24.9b). The equivalent capacitance of the combination, C_{eq} , is the same as the capacitance Q/V of this single equivalent capacitor. So from Eq. (24.6),

$$C_{eq} = C_1 + C_2$$

APPLICATION Touch Screens and Capacitance

The touch screen on a mobile phone, a tablet, or (as shown here) a medical device uses the physics of capacitors. Behind the screen are two parallel layers, one behind the other, of thin strips of a transparent conductor such as indium tin oxide. A voltage is maintained between the two layers. The strips in one layer are oriented perpendicular to those in the other layer; the points where two strips overlap act as a grid of capacitors. When you bring your finger (a conductor) up to a point on the screen, your finger and the front conducting layer act like a second capacitor in series at that point. The circuitry attached to the conducting layers detects the location of the capacitance change, and so detects where you touched the screen.

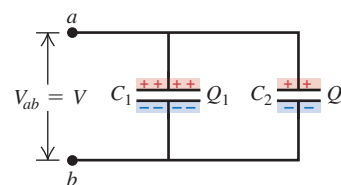


Figure 24.9 A parallel connection of two capacitors.

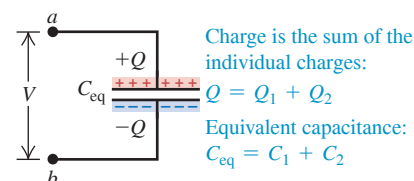
(a) Two capacitors in parallel

Capacitors in parallel:

- The capacitors have the same potential V .
- The charge on each capacitor depends on its capacitance: $Q_1 = C_1 V$, $Q_2 = C_2 V$.



(b) The equivalent single capacitor



In the same way we can show that for any number of capacitors in parallel,

$$\text{Capacitors in parallel: } C_{\text{eq}} = C_1 + C_2 + C_3 + \cdots \quad (24.7)$$

Equivalent capacitance of parallel combination Capacitances of individual capacitors

The equivalent capacitance of a parallel combination equals the *sum* of the individual capacitances. In a parallel connection the equivalent capacitance is always *greater than* any individual capacitance.

CAUTION Capacitors in parallel The potential differences are the same for all capacitors in a parallel combination; however, the charges on individual capacitors are *not* the same unless their individual capacitances are the same. The charges on the individual capacitors add to give the total charge on the parallel combination: $Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \cdots$. [Compare these statements to those in the “Caution” paragraph following Eq. (24.5).]

PROBLEM-SOLVING STRATEGY 24.1 Equivalent Capacitance

IDENTIFY *the relevant concepts:* The concept of equivalent capacitance is useful whenever two or more capacitors are connected.

SET UP *the problem* using the following steps:

1. Make a drawing of the capacitor arrangement.
2. Identify all groups of capacitors that are connected in series or in parallel.
3. Keep in mind that when we say a capacitor “has charge Q ,” we mean that the plate at higher potential has charge $+Q$ and the other plate has charge $-Q$.

EXECUTE *the solution* as follows:

1. Use Eq. (24.5) to find the equivalent capacitance of capacitors connected in series, as in Fig. 24.8. Such capacitors each have the *same charge* if they were uncharged before they were connected; that charge is the same as that on the equivalent capacitor. The potential difference across the combination is the sum of the potential differences across the individual capacitors.

2. Use Eq. (24.7) to find the equivalent capacitance of capacitors connected in parallel, as in Fig. 24.9. Such capacitors all have the *same potential difference* across them; that potential difference is the same as that across the equivalent capacitor. The total charge on the combination is the sum of the charges on the individual capacitors.
3. After replacing all the series or parallel groups you initially identified, you may find that more such groups reveal themselves. Replace those groups by using the same procedure as above until no more replacements are possible. If you then need to find the charge or potential difference for an individual original capacitor, you may have to retrace your steps.

EVALUATE *your answer:* Check whether your result makes sense. If the capacitors are connected in series, the equivalent capacitance C_{eq} must be *smaller* than any of the individual capacitances. If the capacitors are connected in parallel, C_{eq} must be *greater* than any of the individual capacitances.

EXAMPLE 24.5 Capacitors in series and in parallel

WITH VARIATION PROBLEMS

In Figs. 24.8 and 24.9, let $C_1 = 6.0 \mu\text{F}$, $C_2 = 3.0 \mu\text{F}$, and $V_{ab} = 18 \text{ V}$. Find the equivalent capacitance and the charge and potential difference for each capacitor when the capacitors are connected (a) in series (see Fig. 24.8) and (b) in parallel (see Fig. 24.9).

IDENTIFY and SET UP In both parts of this example a target variable is the equivalent capacitance C_{eq} , which is given by Eq. (24.5) for the series combination in part (a) and by Eq. (24.7) for the parallel combination in part (b). In each part we find the charge and potential difference from the definition of capacitance, Eq. (24.1), and the rules outlined in Problem-Solving Strategy 24.1.

EXECUTE (a) From Eq. (24.5) for a series combination,

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{6.0 \mu\text{F}} + \frac{1}{3.0 \mu\text{F}} \quad C_{\text{eq}} = 2.0 \mu\text{F}$$

The charge Q on each capacitor in series is the same as that on the equivalent capacitor:

$$Q = C_{\text{eq}}V = (2.0 \mu\text{F})(18 \text{ V}) = 36 \mu\text{C}$$

The potential difference across each capacitor is inversely proportional to its capacitance:

$$V_{ac} = V_1 = \frac{Q}{C_1} = \frac{36 \mu\text{C}}{6.0 \mu\text{F}} = 6.0 \text{ V}$$

$$V_{cb} = V_2 = \frac{Q}{C_2} = \frac{36 \mu\text{C}}{3.0 \mu\text{F}} = 12.0 \text{ V}$$

(b) From Eq. (24.7) for a parallel combination,

$$C_{\text{eq}} = C_1 + C_2 = 6.0 \mu\text{F} + 3.0 \mu\text{F} = 9.0 \mu\text{F}$$

The potential difference across each of the capacitors is the same as that across the equivalent capacitor, 18 V. The charge on each capacitor is directly proportional to its capacitance:

$$Q_1 = C_1V = (6.0 \mu\text{F})(18 \text{ V}) = 108 \mu\text{C}$$

$$Q_2 = C_2V = (3.0 \mu\text{F})(18 \text{ V}) = 54 \mu\text{C}$$

EVALUATE As expected, the equivalent capacitance C_{eq} for the series combination in part (a) is less than either C_1 or C_2 , while that for the parallel combination in part (b) is greater than either C_1 or C_2 . For two capacitors in series, as in part (a), the charge is the same on either capacitor and the *larger* potential difference appears across the capacitor with the *smaller* capacitance. Furthermore, the sum of the potential differences across the individual capacitors in series equals the potential difference across the equivalent capacitor: $V_{ac} + V_{cb} = V_{ab} = 18 \text{ V}$. By contrast, for two capacitors in parallel, as in part (b), each capacitor has the same potential difference and the *larger* charge appears on the capacitor with the *larger* capacitance. Can you show that the total

charge $Q_1 + Q_2$ on the parallel combination is equal to the charge $Q = C_{\text{eq}}V$ on the equivalent capacitor?

KEYCONCEPT When capacitors are combined in *series*, the charge is the same on each capacitor but the potential differences across the capacitors may not be the same. The reciprocal of the equivalent capacitance of a series combination equals the sum of the reciprocals of the individual capacitances. When capacitors are combined in *parallel*, the potential difference across each capacitor is the same but the charges on the capacitors may not be the same. The equivalent capacitance of a parallel combination equals the sum of the individual capacitances.

EXAMPLE 24.6 A capacitor network

WITH VARIATION PROBLEMS

Find the equivalent capacitance of the five-capacitor network shown in Fig. 24.10a.

IDENTIFY and SET UP These capacitors are neither all in series nor all in parallel. We can, however, identify portions of the arrangement that *are* either in series or parallel. We combine these as described in Problem-Solving Strategy 24.1 to find the net equivalent capacitance, using Eq. (24.5) for series connections and Eq. (24.7) for parallel connections.

EXECUTE The caption of Fig. 24.10 outlines our procedure. We first use Eq. (24.5) to replace the $12 \mu\text{F}$ and $6 \mu\text{F}$ series combination by its equivalent capacitance C' :

$$\frac{1}{C'} = \frac{1}{12 \mu\text{F}} + \frac{1}{6 \mu\text{F}} \quad C' = 4 \mu\text{F}$$

This gives us the equivalent combination of Fig. 24.10b. Now we see three capacitors in parallel, and we use Eq. (24.7) to replace them with their equivalent capacitance C'' :

$$C'' = 3 \mu\text{F} + 11 \mu\text{F} + 4 \mu\text{F} = 18 \mu\text{F}$$

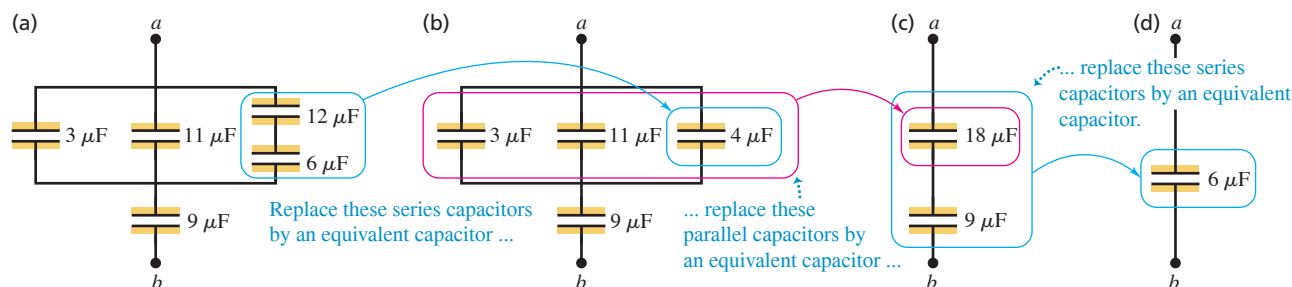
This gives us the equivalent combination of Fig. 24.10c, which has two capacitors in series. We use Eq. (24.5) to replace them with their equivalent capacitance C_{eq} , which is our target variable (Fig. 24.10d):

$$\frac{1}{C_{\text{eq}}} = \frac{1}{18 \mu\text{F}} + \frac{1}{9 \mu\text{F}} \quad C_{\text{eq}} = 6 \mu\text{F}$$

EVALUATE If the potential difference across the entire network in Fig. 24.10a is $V_{ab} = 9.0 \text{ V}$, the net charge on the network is $Q = C_{\text{eq}}V_{ab} = (6 \mu\text{F})(9.0 \text{ V}) = 54 \mu\text{C}$. Can you find the charge on, and the voltage across, each of the five individual capacitors?

KEYCONCEPT To analyze any network of capacitors, first identify which portions of the network are in series and which are in parallel. Then use the rules for series and parallel capacitors to determine the equivalent capacitance of the network.

Figure 24.10 (a) A capacitor network between points a and b . (b) The $12 \mu\text{F}$ and $6 \mu\text{F}$ capacitors in series in (a) are replaced by an equivalent $4 \mu\text{F}$ capacitor. (c) The $3 \mu\text{F}$, $11 \mu\text{F}$, and $4 \mu\text{F}$ capacitors in parallel in (b) are replaced by an equivalent $18 \mu\text{F}$ capacitor. (d) Finally, the $18 \mu\text{F}$ and $9 \mu\text{F}$ capacitors in series in (c) are replaced by an equivalent $6 \mu\text{F}$ capacitor.



TEST YOUR UNDERSTANDING OF SECTION 24.2 (a) How should you connect a $4 \mu\text{F}$ capacitor and an $8 \mu\text{F}$ capacitor so that the $4 \mu\text{F}$ capacitor has a greater potential difference across it than the $8 \mu\text{F}$ capacitor? (i) Series; (ii) parallel; (iii) either series or parallel; (iv) neither series nor parallel. (b) How should you connect them so that the $4 \mu\text{F}$ capacitor has a greater charge than the $8 \mu\text{F}$ capacitor? (i) Series; (ii) parallel; (iii) either series or parallel; (iv) neither series nor parallel.

ANSWER (i) In a series connection the two capacitors carry the same charge Q but have different potential differences $V_{ab} = Q/C$; the capacitor with the smaller capacitance C has the greater potential difference. In a parallel connection the two capacitors have the same potential difference V_{ab} but carry different charges $Q = CV_{ab}$; the capacitor with the larger capacitance C has the greater charge. Hence a $4 \mu\text{F}$ capacitor will have a greater potential difference than an $8 \mu\text{F}$ capacitor if the two are connected in series. The $4 \mu\text{F}$ capacitor cannot carry more charge than the $8 \mu\text{F}$ capacitor no matter how they are connected. In a series connection they will carry the same charge, and in a parallel connection the $8 \mu\text{F}$ capacitor will carry more charge.

24.3 ENERGY STORAGE IN CAPACITORS AND ELECTRIC-FIELD ENERGY

Many of the most important applications of capacitors depend on their ability to store energy. The electric potential energy stored in a charged capacitor is just equal to the amount of work required to charge it—that is, to separate opposite charges and place them on different conductors. When the capacitor is discharged, this stored energy is recovered as work done by electrical forces.

We can calculate the potential energy U of a charged capacitor by calculating the work W required to charge it. Suppose that when we are done charging the capacitor, the final charge is Q and the final potential difference is V . From Eq. (24.1) these quantities are related by

$$V = \frac{Q}{C}$$

Let q and v be the charge and potential difference, respectively, at an intermediate stage during the charging process; then $v = q/C$. At this stage the work dW required to transfer an additional element of charge dq is

$$dW = v dq = \frac{q dq}{C}$$

The total work W needed to increase the capacitor charge q from zero to Q is

$$W = \int_0^Q dW = \frac{1}{C} \int_0^Q q dq = \frac{Q^2}{2C} \quad (\text{work to charge a capacitor}) \quad (24.8)$$

This is also the total work done by the electric field on the charge when the capacitor discharges. Then q decreases from an initial value Q to zero as the elements of charge dq “fall” through potential differences v that vary from V down to zero.

If we define the potential energy of an *uncharged* capacitor to be zero, then W in Eq. (24.8) is equal to the potential energy U of the charged capacitor. The final stored charge is $Q = CV$, so we can express U (which is equal to W) as

$$\text{Potential energy stored in a capacitor} \quad U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV \quad (24.9)$$

Magnitude of charge on each plate
Capacitance Potential difference between plates

When Q is in coulombs, C in farads (coulombs per volt), and V in volts (joules per coulomb), U is in joules.

The last form of Eq. (24.9), $U = \frac{1}{2} QV$, shows that the total work W required to charge the capacitor is equal to the total charge Q multiplied by the *average* potential difference $\frac{1}{2} V$ during the charging process.

The expression $U = \frac{1}{2} (Q^2/C)$ in Eq. (24.9) shows that a charged capacitor is the electrical analog of a stretched spring with elastic potential energy $U = \frac{1}{2} kx^2$. The charge Q is analogous to the elongation x , and the *reciprocal* of the capacitance, $1/C$, is analogous to the force constant k . The energy supplied to a capacitor in the charging process is analogous to the work we do on a spring when we stretch it.

Equations (24.8) and (24.9) tell us that capacitance measures the ability of a capacitor to store both energy and charge. If a capacitor is charged by connecting it to a battery or other source that provides a fixed potential difference V , then increasing the value of C gives a greater charge $Q = CV$ and a greater amount of stored energy $U = \frac{1}{2} CV^2$. If instead the goal is to transfer a given quantity of charge Q from one conductor to another, Eq. (24.8) shows that the work W required is inversely proportional to C ; the greater the capacitance, the easier it is to give a capacitor a fixed amount of charge.

Applications of Capacitors: Energy Storage

Most practical applications of capacitors take advantage of their ability to store and release energy. In an electronic flash unit used in photography, the energy stored in a capacitor (see Fig. 24.4) is released when the button is pressed to take a photograph. This provides a conducting path from one capacitor plate to the other through the flash tube. Once this path is established, the stored energy is rapidly converted into a brief but intense flash of light. An extreme example of the same principle is the Z machine at Sandia National Laboratories in New Mexico, which is used in experiments in controlled nuclear fusion (**Fig. 24.11**). A bank of charged capacitors releases more than a million joules of energy in just a few billionths of a second. For that brief space of time, the power output of the Z machine is 2.9×10^{14} W, or about 80 times the power output of all the electric power plants on earth combined!

In other applications, the energy is released more slowly. Springs in the suspension of an automobile help smooth out the ride by absorbing the energy from sudden jolts and releasing that energy gradually; in an analogous way, a capacitor in an electronic circuit can smooth out unwanted variations in voltage due to power surges. We'll discuss these circuits in detail in Chapter 26.

Electric-Field Energy

We can charge a capacitor by moving electrons directly from one plate to another. This requires doing work against the electric field between the plates. Thus we can think of the energy as being stored *in the field* in the region between the plates. To see this, let's find the energy *per unit volume* in the space between the plates of a parallel-plate capacitor with plate area A and separation d . We call this the **energy density**, denoted by u . From Eq. (24.9) the total stored potential energy is $\frac{1}{2}CV^2$ and the volume between the plates is Ad ; hence

$$u = \text{Energy density} = \frac{\frac{1}{2}CV^2}{Ad} \quad (24.10)$$

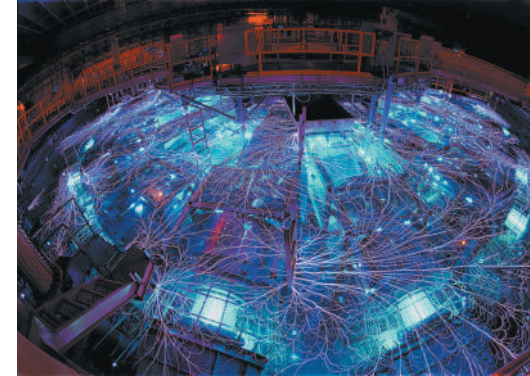
From Eq. (24.2) the capacitance C is given by $C = \epsilon_0 A/d$. The potential difference V is related to the electric-field magnitude E by $V = Ed$. If we use these expressions in Eq. (24.10), the geometric factors A and d cancel, and we find

$$\text{Electric energy density in a vacuum} \quad u = \frac{1}{2}\epsilon_0 E^2 \quad \text{Magnitude of electric field} \quad \text{Electric constant} \quad (24.11)$$

Although we have derived this relationship for a parallel-plate capacitor only, it turns out to be valid for any capacitor in vacuum and indeed *for any electric-field configuration in vacuum*. This result has an interesting implication. We think of vacuum as space with no matter in it, but vacuum can nevertheless have electric fields and therefore energy. Thus “empty” space need not be truly empty after all. We'll use this idea and Eq. (24.11) in Chapter 32 in connection with the energy transported by electromagnetic waves.

CAUTION **Electric-field energy is electric potential energy** It's a common misconception that electric-field energy is a new kind of energy, different from the electric potential energy described before. This is *not* the case; it is simply a different way of interpreting electric potential energy. We can regard the energy of a given system of charges as being a shared property of all the charges, or we can think of the energy as being a property of the electric field that the charges create. Either interpretation leads to the same value of the potential energy. ■

Figure 24.11 The Z machine uses a large number of capacitors in parallel to give a tremendous equivalent capacitance C (see Section 24.2). Hence a large amount of energy $U = \frac{1}{2}CV^2$ can be stored with even a modest potential difference V . The arcs shown here are produced when the capacitors discharge their energy into a target, which is no larger than a spool of thread. This heats the target to a temperature higher than 2×10^9 K.



EXAMPLE 24.7 Transferring charge and energy between capacitors**WITH VARIATION PROBLEMS**

We connect a capacitor $C_1 = 8.0 \mu\text{F}$ to a power supply, charge it to a potential difference $V_0 = 120 \text{ V}$, and disconnect the power supply (**Fig. 24.12**). Switch S is open. (a) What is the charge Q_0 on C_1 ? (b) What is the energy stored in C_1 ? (c) Capacitor $C_2 = 4.0 \mu\text{F}$ is initially uncharged. We close switch S . After charge no longer flows, what is the potential difference across each capacitor, and what is the charge on each capacitor? (d) What is the final energy of the system?

IDENTIFY and SET UP In parts (a) and (b) we find the charge Q_0 and stored energy U_{initial} for the single charged capacitor C_1 from Eqs. (24.1) and (24.9), respectively. After we close switch S , one wire connects the upper plates of the two capacitors and another wire connects the lower plates; the capacitors are now connected in parallel. In part (c) we use the character of the parallel connection to determine how Q_0 is shared between the two capacitors. In part (d) we again use Eq. (24.9) to find the energy stored in capacitors C_1 and C_2 ; the energy of the system is the sum of these values.

EXECUTE (a) The initial charge Q_0 on C_1 is

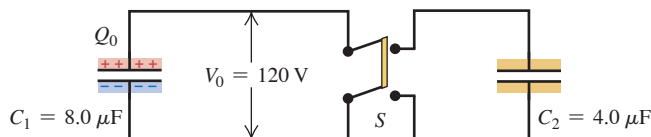
$$Q_0 = C_1 V_0 = (8.0 \mu\text{F})(120 \text{ V}) = 960 \mu\text{C}$$

(b) The energy initially stored in C_1 is

$$U_{\text{initial}} = \frac{1}{2} Q_0 V_0 = \frac{1}{2} (960 \times 10^{-6} \text{ C})(120 \text{ V}) = 0.058 \text{ J}$$

(c) When we close the switch, the positive charge Q_0 is distributed over the upper plates of both capacitors and the negative charge $-Q_0$ is distributed over the lower plates. Let Q_1 and Q_2 be the magnitudes of the final charges on the capacitors. Conservation of charge requires that $Q_1 + Q_2 = Q_0$. The potential difference V between the plates is the same for both capacitors because they are connected in parallel, so the charges are $Q_1 = C_1 V$ and $Q_2 = C_2 V$. We now have three independent equations relating the three unknowns Q_1 , Q_2 , and V . Solving these, we find

Figure 24.12 When the switch S is closed, the charged capacitor C_1 is connected to an uncharged capacitor C_2 . The center part of the switch is an insulating handle; charge can flow only between the two upper terminals and between the two lower terminals.



$$V = \frac{Q_0}{C_1 + C_2} = \frac{960 \mu\text{C}}{8.0 \mu\text{F} + 4.0 \mu\text{F}} = 80 \text{ V}$$

$$Q_1 = 640 \mu\text{C} \quad Q_2 = 320 \mu\text{C}$$

(d) The final energy of the system is

$$U_{\text{final}} = \frac{1}{2} Q_1 V + \frac{1}{2} Q_2 V = \frac{1}{2} Q_0 V \\ = \frac{1}{2} (960 \times 10^{-6} \text{ C})(80 \text{ V}) = 0.038 \text{ J}$$

EVALUATE The final energy is less than the initial energy; the difference was converted to energy of some other form. The conductors become a little warmer because of their resistance, and some energy is radiated as electromagnetic waves. We'll study the behavior of capacitors in more detail in Chapters 26 and 31.

KEYCONCEPT There are three useful ways to write the amount of electric potential energy stored in a charged capacitor: in terms of the charge Q on the plates and the potential difference V between the plates, in terms of Q and the capacitance C , and in terms of V and C .

EXAMPLE 24.8 Electric-field energy**WITH VARIATION PROBLEMS**

(a) What is the magnitude of the electric field required to store 1.00 J of electric potential energy in a volume of 1.00 m^3 in vacuum? (b) If the field magnitude is 10 times larger than that, how much energy is stored per cubic meter?

IDENTIFY and SET UP We use the relationship between the electric-field magnitude E and the energy density u . In part (a) we use the given information to find u ; then we use Eq. (24.11) to find the corresponding value of E . In part (b), Eq. (24.11) tells us how u varies with E .

EXECUTE (a) The desired energy density is $u = 1.00 \text{ J/m}^3$. Then from Eq. (24.11),

$$E = \sqrt{\frac{2u}{\epsilon_0}} = \sqrt{\frac{2(1.00 \text{ J/m}^3)}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2}} \\ = 4.75 \times 10^5 \text{ N/C} = 4.75 \times 10^5 \text{ V/m}$$

(b) Equation (24.11) shows that u is proportional to E^2 . If E increases by a factor of 10, u increases by a factor of $10^2 = 100$, so the energy density becomes $u = 100 \text{ J/m}^3$.

EVALUATE Dry air can sustain an electric field of about $3 \times 10^6 \text{ V/m}$ without experiencing *dielectric breakdown*, which we'll discuss in Section 24.4. There we'll see that field magnitudes in practical insulators can be even larger than this.

KEYCONCEPT Energy is stored in any region of space where an electric field is present. The electric energy density (energy per unit volume) at a point is proportional to the square of the electric-field magnitude at that point.

EXAMPLE 24.9 Two ways to calculate energy stored in a capacitor**WITH VARIATION PROBLEMS**

The spherical capacitor described in Example 24.3 (Section 24.1) has charges $+Q$ and $-Q$ on its inner and outer conductors. Find the electric potential energy stored in the capacitor (a) by using the capacitance C found in Example 24.3 and (b) by integrating the electric-field energy density u .

IDENTIFY and SET UP We can determine the energy U stored in a capacitor in two ways: in terms of the work done to put the charges on the two conductors, and in terms of the energy in the electric field between the conductors. The descriptions are equivalent, so they must give us the same result. In Example 24.3 we found the capacitance C and the field magnitude E in the space between the conductors. (The electric field is zero inside the inner sphere and is also zero outside the inner surface of the outer sphere, because a Gaussian surface with radius $r < r_a$ or $r > r_b$ encloses zero net charge. Hence the energy density is nonzero only in the space between the spheres, $r_a < r < r_b$.) In part (a) we use Eq. (24.9) to find U . In part (b) we use Eq. (24.11) to find u , which we integrate over the volume between the spheres to find U .

EXECUTE (a) From Example 24.3, the spherical capacitor has capacitance

$$C = 4\pi\epsilon_0 \frac{r_a r_b}{r_b - r_a}$$

where r_a and r_b are the radii of the inner and outer conducting spheres, respectively. From Eq. (24.9) the energy stored in this capacitor is

$$U = \frac{Q^2}{2C} = \frac{Q^2}{8\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b}$$

(b) The electric field in the region $r_a < r < r_b$ between the two conducting spheres has magnitude $E = Q/4\pi\epsilon_0 r^2$. The energy density in this region is

$$u = \frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\epsilon_0 \left(\frac{Q}{4\pi\epsilon_0 r^2} \right)^2 = \frac{Q^2}{32\pi^2\epsilon_0 r^4}$$

The energy density is *not* uniform; it decreases rapidly with increasing distance from the center of the capacitor. To find the total electric-field energy, we integrate u (the energy per unit volume) over the region $r_a < r < r_b$. We divide this region into spherical shells of radius r , surface area $4\pi r^2$, thickness dr , and volume $dV = 4\pi r^2 dr$. Then

$$\begin{aligned} U &= \int u dV = \int_{r_a}^{r_b} \left(\frac{Q^2}{32\pi^2\epsilon_0 r^4} \right) 4\pi r^2 dr \\ &= \frac{Q^2}{8\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = \frac{Q^2}{8\pi\epsilon_0} \left(-\frac{1}{r_b} + \frac{1}{r_a} \right) = \frac{Q^2}{8\pi\epsilon_0} \frac{r_b - r_a}{r_a r_b} \end{aligned}$$

EVALUATE Electric potential energy can be associated with either the *charges*, as in part (a), or the *field*, as in part (b); the calculated amount of stored energy is the same in either case.

KEYCONCEPT An alternative way to calculate the electric potential energy stored in a capacitor is to first find the electric field at each point in the space between the conducting plates. From this, find the electric energy density at each point. Finally, integrate the electric energy density over the entire volume where the electric field is present.

TEST YOUR UNDERSTANDING OF SECTION 24.3 You want to connect a $4\text{ }\mu\text{F}$ capacitor and an $8\text{ }\mu\text{F}$ capacitor. With which type of connection will the $4\text{ }\mu\text{F}$ capacitor have a greater amount of *stored energy* than the $8\text{ }\mu\text{F}$ capacitor? (i) Series; (ii) parallel; (iii) either series or parallel; (iv) neither series nor parallel.

ANSWER

(i) Capacitors connected in series carry the same charge Q . To compare the amount of energy stored, we use the expression $U = Q^2/2C$ from Eq. (24.9); it shows that the capacitor with the *smaller* capacitance ($C = 4\text{ }\mu\text{F}$) has more stored energy in a series combination. By contrast, capacitors in parallel have the same potential difference V , so to compare them we use $U = \frac{1}{2}CV^2$ from Eq. (24.9). It shows that in a parallel combination, the capacitor with the *larger* capacitance ($C = 8\text{ }\mu\text{F}$) has more stored energy. (If we had instead used $U = \frac{1}{2}CV^2$ to analyze the series combination, we would have to account for the different potential differences across the two capacitors. Likewise, using $U = Q^2/2C$ to study the parallel combination would require us to account for the different charges on the capacitors.)

24.4 DIELECTRICS

Most capacitors have a nonconducting material, or **dielectric**, between their conducting plates. A common type of capacitor uses long strips of metal foil for the plates, separated by strips of plastic sheet such as Mylar. A sandwich of these materials is rolled up, forming a unit that can provide a capacitance of several microfarads in a compact package (**Fig. 24.13**).

Placing a solid dielectric between the plates of a capacitor serves three functions. First, it solves the mechanical problem of maintaining two large metal sheets at a very small separation without actual contact.

Second, using a dielectric increases the maximum possible potential difference between the capacitor plates. Any insulating material, when subjected to a sufficiently large electric field, experiences a partial ionization that permits conduction through it (Section 23.3).

Figure 24.13 A common type of capacitor uses dielectric sheets to separate the conductors.

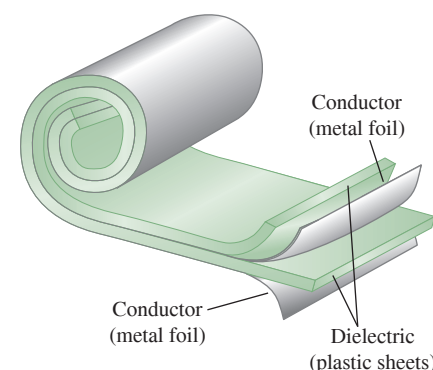
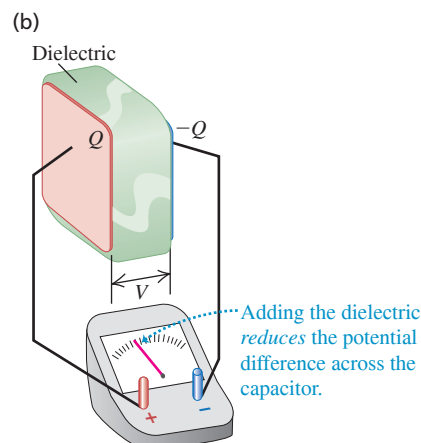
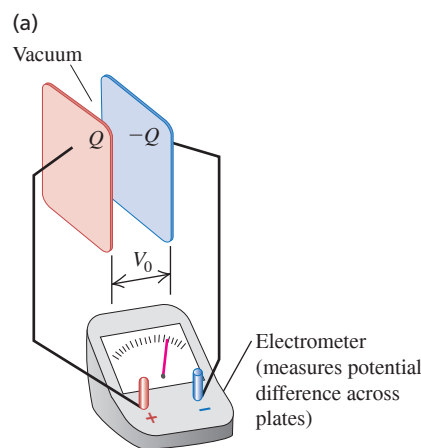


Figure 24.14 Effect of a dielectric between the plates of a parallel-plate capacitor. (a) With a given charge, the potential difference is V_0 . (b) With the same charge but with a dielectric between the plates, the potential difference V is smaller than V_0 .



This phenomenon is called **dielectric breakdown**. Many dielectric materials can tolerate stronger electric fields without breakdown than can air. Thus using a dielectric allows a capacitor to sustain a higher potential difference V and so store greater amounts of charge and energy.

Third, the capacitance of a capacitor of given dimensions is *greater* when there is a dielectric material between the plates than when there is vacuum. We can demonstrate this effect with the aid of a sensitive *electrometer*, a device that measures the potential difference between two conductors without letting any appreciable charge flow from one to the other. **Figure 24.14a** shows an electrometer connected across a charged capacitor, with magnitude of charge Q on each plate and potential difference V_0 . When we insert an uncharged sheet of dielectric, such as glass, paraffin, or polystyrene, between the plates, experiment shows that the potential difference *decreases* to a smaller value V (Fig. 24.14b). When we remove the dielectric, the potential difference returns to its original value V_0 , showing that the original charges on the plates have not changed.

The original capacitance C_0 is given by $C_0 = Q/V_0$, and the capacitance C with the dielectric present is $C = Q/V$. The charge Q is the same in both cases, and V is less than V_0 , so we conclude that the capacitance C with the dielectric present is *greater* than C_0 . When the space between plates is completely filled by the dielectric, the ratio of C to C_0 (equal to the ratio of V_0 to V) is called the **dielectric constant** of the material, K :

$$K = \frac{C}{C_0} \quad (\text{definition of dielectric constant}) \quad (24.12)$$

When the charge is constant, $Q = C_0 V_0 = CV$ and $C/C_0 = V_0/V$. In this case,

$$V = \frac{V_0}{K} \quad (\text{when } Q \text{ is constant}) \quad (24.13)$$

With the dielectric present, the potential difference for a given charge Q is *reduced* by a factor K .

The dielectric constant K is a pure number. Because C is always greater than C_0 , K is always greater than unity. **Table 24.1** gives some representative values of K . For vacuum, $K = 1$ by definition. For air at ordinary temperatures and pressures, K is about 1.0006; this is so nearly equal to 1 that for most purposes an air capacitor is equivalent to one in vacuum. Note that while water has a very large value of K , it is usually not a very practical dielectric for use in capacitors. The reason is that while pure water is a very poor conductor, it is also an excellent ionic solvent. Any ions that are dissolved in the water will cause charge to flow between the capacitor plates, so the capacitor discharges.

CAUTION Dielectric constant vs. electric constant Don't confuse the *dielectric* constant K with the *electric* constant ϵ_0 . The value of K is a pure number with no units and is different for different materials (see Table 24.1). By contrast, ϵ_0 is a universal constant with units $C^2/N \cdot m^2$ or F/m . ■

TABLE 24.1 Values of Dielectric Constant K at 20°C

Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas®	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon®	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar®	3.1	Strontium titanate	310

No real dielectric is a perfect insulator. Hence there is always some *leakage current* between the charged plates of a capacitor with a dielectric. We tacitly ignored this effect in Section 24.2 when we derived expressions for the equivalent capacitances of capacitors in series, Eq. (24.5), and in parallel, Eq. (24.7). But if a leakage current flows for a long enough time to substantially change the charges from the values we used to derive Eqs. (24.5) and (24.7), those equations may no longer be accurate.

Induced Charge and Polarization

When a dielectric material is inserted between the plates while the charge is kept constant, the potential difference between the plates decreases by a factor K . Therefore the electric field between the plates must decrease by the same factor. If E_0 is the vacuum value and E is the value with the dielectric, then

$$E = \frac{E_0}{K} \quad (\text{when } Q \text{ is constant}) \quad (24.14)$$

Since the electric-field magnitude is smaller when the dielectric is present, the surface charge density (which causes the field) must be smaller as well. The surface charge on the conducting plates does not change, but an *induced* charge of the opposite sign appears on each surface of the dielectric (Fig. 24.15). The dielectric was originally electrically neutral and is still neutral; the induced surface charges arise as a result of *redistribution* of positive and negative charges within the dielectric material, a phenomenon called **polarization**. We first encountered polarization in Section 21.2, and we suggest that you reread the discussion of Fig. 21.8. We'll assume that the induced surface charge is *directly proportional* to the electric-field magnitude E in the material; this is indeed the case for many common dielectrics. (This direct proportionality is analogous to Hooke's law for a spring.) In that case, K is a constant for any particular material. When the electric field is very strong or if the dielectric is made of certain crystalline materials, the relationship between induced charge and the electric field can be more complex; we won't consider such cases here.

We can derive a relationship between this induced surface charge and the charge on the plates. Let σ_i be the magnitude of the charge per unit area induced on the surfaces of the dielectric (the induced surface charge density). The magnitude of the surface charge density on the capacitor plates is σ , as usual. Then the *net* surface charge on each side of the capacitor has magnitude $(\sigma - \sigma_i)$; see Fig. 24.15b. As we found in Example 21.12 (Section 21.5) and in Example 22.8 (Section 22.4), the field between the plates is related to the net surface charge density by $E = \sigma_{\text{net}}/\epsilon_0$. Without and with the dielectric, respectively,

$$E_0 = \frac{\sigma}{\epsilon_0} \quad E = \frac{\sigma - \sigma_i}{\epsilon_0} \quad (24.15)$$

Using these expressions in Eq. (24.14) and rearranging the result, we find

$$\sigma_i = \sigma \left(1 - \frac{1}{K} \right) \quad (\text{induced surface charge density}) \quad (24.16)$$

This equation shows that when K is very large, σ_i is nearly as large as σ . In this case, σ_i nearly cancels σ , and the field and potential difference are much smaller than their values in vacuum.

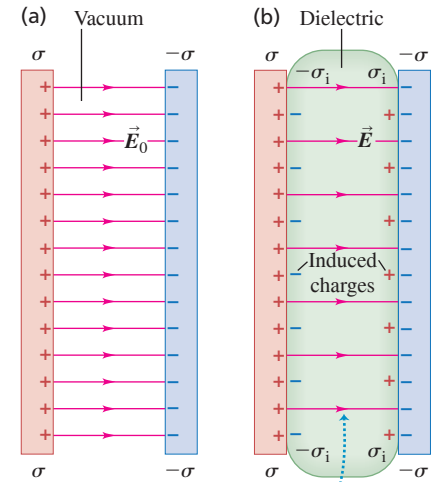
The product $K\epsilon_0$ is called the **permittivity** of the dielectric, denoted by ϵ :

$$\epsilon = K\epsilon_0 \quad (\text{definition of permittivity}) \quad (24.17)$$

In terms of ϵ we can express the electric field within the dielectric as

$$E = \frac{\sigma}{\epsilon} \quad (24.18)$$

Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.



For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

APPLICATION Capacitors in the Toolbox

Several practical devices rely on the way a capacitor responds to a change in dielectric constant. One example is an electric stud finder, used to locate metal segments hidden behind a wall's surface. It consists of a metal plate with associated circuitry. The plate acts as one half of a capacitor; the wall acts as the other half. If the stud finder moves over a metal segment, the effective dielectric constant for the capacitor changes, which changes the capacitance and triggers a signal.



Then

$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d} \quad (24.19)$$

Capacitance of a parallel-plate capacitor, dielectric between plates: C
 Dielectric constant: K
 Area of each plate: A
 Distance between plates: d
 Capacitance without dielectric: C_0
 Electric constant: ϵ_0
 Permittivity: $\epsilon = K\epsilon_0$

We can repeat the derivation of Eq. (24.11) for the energy density u in an electric field for the case in which a dielectric is present. The result is

$$u = \frac{1}{2}K\epsilon_0 E^2 = \frac{1}{2}\epsilon E^2 \quad (24.20)$$

Electric energy density in a dielectric: u
 Dielectric constant: K
 Electric constant: ϵ_0
 Permittivity: $\epsilon = K\epsilon_0$
 Magnitude of electric field: E

CAUTION Capacitance is independent of charge or voltage

Equation (24.19) shows that the capacitance C of a capacitor depends on only the capacitor geometry (the size, shape, and position of its plates) and whether or not a dielectric is present. The capacitance does *not* depend on the amount of charge placed on the plates or on the voltage between the plates!

In empty space, where $K = 1$, $\epsilon = \epsilon_0$ and Eqs. (24.19) and (24.20) reduce to Eqs. (24.2) and (24.11), respectively, for a parallel-plate capacitor in vacuum. For this reason, ϵ_0 is sometimes called the “permittivity of free space” or the “permittivity of vacuum.” Because K is a pure number, ϵ and ϵ_0 have the same units, $C^2/N \cdot m^2$ or F/m .

Equation (24.19) shows that extremely high capacitances can be obtained with plates that have a large surface area A and are separated by a small distance d by a dielectric with a large value of K . In an *electrolytic double-layer capacitor*, tiny carbon granules adhere to each plate: A is the combined surface area of the granules, which can be tremendous. The plates with granules attached are separated by a very thin dielectric sheet. Such a capacitor can have a capacitance of 5000 F yet fit in the palm of your hand (compare Example 24.1 in Section 24.1).

PROBLEM-SOLVING STRATEGY 24.2 Dielectrics

IDENTIFY the relevant concepts: The relationships in this section are useful whenever there is an electric field in a dielectric, such as a dielectric between charged capacitor plates. Typically you must relate the potential difference V_{ab} between the plates, the electric-field magnitude E in the capacitor, the charge density σ on the capacitor plates, and the induced charge density σ_i on the surfaces of the capacitor.

SET UP the problem using the following steps:

1. Make a drawing of the situation.
2. Identify the target variables, and choose which equations from this section will help you solve for those variables.

EXECUTE the solution as follows:

1. In problems such as the next example, it is easy to get lost in a blizzard of formulas. Ask yourself at each step what kind of quantity each symbol represents. For example, distinguish clearly

between charges and charge densities, and between electric fields and electric potential differences.

2. Check for consistency of units. Distances must be in meters. A microfarad is 10^{-6} farad, and so on. Don't confuse the numerical value of ϵ_0 with the value of $1/4\pi\epsilon_0$. Electric-field magnitude can be expressed in both N/C and V/m . The units of ϵ_0 are $C^2/N \cdot m^2$ or F/m .

EVALUATE your answer: With a dielectric present, (a) the capacitance is greater than without a dielectric; (b) for a given charge on the capacitor, the electric field and potential difference are less than without a dielectric; and (c) the magnitude of the induced surface charge density σ_i on the dielectric is less than that of the charge density σ on the capacitor plates.

EXAMPLE 24.10 A capacitor with and without a dielectric

WITH VARIATION PROBLEMS

Suppose the parallel plates in Fig. 24.15 each have an area of 2000 cm^2 ($2.00 \times 10^{-1} \text{ m}^2$) and are 1.00 cm ($1.00 \times 10^{-2} \text{ m}$) apart. We connect the capacitor to a power supply, charge it to a potential difference $V_0 = 3.00 \text{ kV}$, and disconnect the power supply. We then insert a sheet of insulating plastic material between the plates, completely filling the space between them. We find that the potential difference decreases to 1.00 kV while the charge on each capacitor

plate remains constant. Find (a) the original capacitance C_0 ; (b) the magnitude of charge Q on each plate; (c) the capacitance C after the dielectric is inserted; (d) the dielectric constant K of the dielectric; (e) the permittivity ϵ of the dielectric; (f) the magnitude of the induced charge Q_i on each face of the dielectric; (g) the original electric field E_0 between the plates; and (h) the electric field E after the dielectric is inserted.

IDENTIFY and SET UP This problem uses most of the relationships we have discussed for capacitors and dielectrics. (Energy relationships are treated in Example 24.11.) Most of the target variables can be obtained in several ways. The methods used below are a sample; we encourage you to think of others and compare your results.

EXECUTE (a) With vacuum between the plates, we use Eq. (24.19) with $K = 1$:

$$C_0 = \epsilon_0 \frac{A}{d} = (8.85 \times 10^{-12} \text{ F/m}) \frac{2.00 \times 10^{-1} \text{ m}^2}{1.00 \times 10^{-2} \text{ m}} \\ = 1.77 \times 10^{-10} \text{ F} = 177 \text{ pF}$$

(b) From the definition of capacitance, Eq. (24.1),

$$Q = C_0 V_0 = (1.77 \times 10^{-10} \text{ F})(3.00 \times 10^3 \text{ V}) \\ = 5.31 \times 10^{-7} \text{ C} = 0.531 \mu\text{C}$$

(c) When the dielectric is inserted, Q is unchanged but the potential difference decreases to $V = 1.00 \text{ kV}$. Hence from Eq. (24.1), the new capacitance is

$$C = \frac{Q}{V} = \frac{5.31 \times 10^{-7} \text{ C}}{1.00 \times 10^3 \text{ V}} = 5.31 \times 10^{-10} \text{ F} = 531 \text{ pF}$$

(d) From Eq. (24.12), the dielectric constant is

$$K = \frac{C}{C_0} = \frac{5.31 \times 10^{-10} \text{ F}}{1.77 \times 10^{-10} \text{ F}} = \frac{531 \text{ pF}}{177 \text{ pF}} = 3.00$$

Alternatively, from Eq. (24.13),

$$K = \frac{V_0}{V} = \frac{3000 \text{ V}}{1000 \text{ V}} = 3.00$$

(e) With K from part (d) in Eq. (24.17), the permittivity is

$$\epsilon = K\epsilon_0 = (3.00)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2) = 2.66 \times 10^{-11} \text{ C}^2/\text{N} \cdot \text{m}^2$$

(f) Multiplying both sides of Eq. (24.16) by the plate area A gives the induced charge $Q_i = \sigma_i A$ in terms of the charge $Q = \sigma A$ on each plate:

$$Q_i = Q \left(1 - \frac{1}{K} \right) = (5.31 \times 10^{-7} \text{ C}) \left(1 - \frac{1}{3.00} \right) = 3.54 \times 10^{-7} \text{ C}$$

(g) Since the electric field between the plates is uniform, its magnitude is the potential difference divided by the plate separation:

$$E_0 = \frac{V_0}{d} = \frac{3000 \text{ V}}{1.00 \times 10^{-2} \text{ m}} = 3.00 \times 10^5 \text{ V/m}$$

(h) After the dielectric is inserted,

$$E = \frac{V}{d} = \frac{1000 \text{ V}}{1.00 \times 10^{-2} \text{ m}} = 1.00 \times 10^5 \text{ V/m}$$

or, from Eq. (24.18),

$$E = \frac{\sigma}{\epsilon} = \frac{Q}{\epsilon A} = \frac{5.31 \times 10^{-7} \text{ C}}{(2.66 \times 10^{-11} \text{ C}^2/\text{N} \cdot \text{m}^2)(2.00 \times 10^{-1} \text{ m}^2)} \\ = 1.00 \times 10^5 \text{ V/m}$$

or, from Eq. (24.15),

$$E = \frac{\sigma - \sigma_i}{\epsilon_0} = \frac{Q - Q_i}{\epsilon_0 A} \\ = \frac{(5.31 - 3.54) \times 10^{-7} \text{ C}}{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(2.00 \times 10^{-1} \text{ m}^2)} \\ = 1.00 \times 10^5 \text{ V/m}$$

or, from Eq. (24.14),

$$E = \frac{E_0}{K} = \frac{3.00 \times 10^5 \text{ V/m}}{3.00} = 1.00 \times 10^5 \text{ V/m}$$

EVALUATE Inserting the dielectric increased the capacitance by a factor of $K = 3.00$ and reduced the electric field between the plates by a factor of $1/K = 1/3.00$. It did so by developing induced charges on the faces of the dielectric of magnitude $Q(1 - 1/K) = Q(1 - 1/3.00) = 0.667Q$.

KEYCONCEPT Adding a dielectric (with dielectric constant K) that fills the space between the plates of a capacitor increases the capacitance by a factor of K . For a given amount of charge, adding the dielectric also reduces both the potential difference between the plates and the electric field at each point by a factor of $1/K$.

EXAMPLE 24.11 Energy storage with and without a dielectric

WITH VARIATION PROBLEMS

Find the energy stored in the electric field of the capacitor in Example 24.10 and the energy density, both before and after the dielectric sheet is inserted.

IDENTIFY and SET UP We consider the ideas of energy stored in a capacitor and of electric-field energy density. We use Eq. (24.9) to find the stored energy and Eq. (24.20) to find the energy density.

EXECUTE From Eq. (24.9), the stored energies U_0 and U without and with the dielectric in place are

$$U_0 = \frac{1}{2} C_0 V_0^2 = \frac{1}{2} (1.77 \times 10^{-10} \text{ F})(3000 \text{ V})^2 = 7.97 \times 10^{-4} \text{ J} \\ U = \frac{1}{2} C V^2 = \frac{1}{2} (5.31 \times 10^{-10} \text{ F})(1000 \text{ V})^2 = 2.66 \times 10^{-4} \text{ J}$$

The final energy is one-third of the original energy.

Equation (24.20) gives the energy densities without and with the dielectric:

$$u_0 = \frac{1}{2} \epsilon_0 E_0^2 = \frac{1}{2} (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(3.00 \times 10^5 \text{ N/C})^2 \\ = 0.398 \text{ J/m}^3 \\ u = \frac{1}{2} \epsilon E^2 = \frac{1}{2} (2.66 \times 10^{-11} \text{ C}^2/\text{N} \cdot \text{m}^2)(1.00 \times 10^5 \text{ N/C})^2 \\ = 0.133 \text{ J/m}^3$$

The energy density with the dielectric is one-third of the original energy density.

Continued

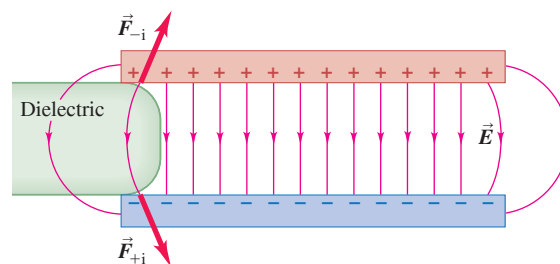
EVALUATE We can check our answer for u_0 by noting that the volume between the plates is $V_{\text{between}} = (0.200 \text{ m}^2)(0.0100 \text{ m}) = 0.00200 \text{ m}^3$. Since the electric field between the plates is uniform, u_0 is uniform as well and the energy density is just the stored energy divided by the volume:

$$u_0 = \frac{U_0}{V_{\text{between}}} = \frac{7.97 \times 10^{-4} \text{ J}}{0.00200 \text{ m}^3} = 0.398 \text{ J/m}^3$$

This agrees with our earlier answer. You can use the same approach to check our result for u .

In general, when a dielectric is inserted into a capacitor while the charge on each plate remains the same, the permittivity ϵ increases by a factor of K (the dielectric constant), and the electric field E and the energy density $u = \frac{1}{2}\epsilon E^2$ decrease by a factor of $1/K$. Where does the energy go? The answer lies in the fringing field at the edges of a real parallel-plate capacitor. As **Fig. 24.16** shows, that field tends to pull the dielectric into the space between the plates, doing work on it as it does so. We could attach a spring to the left end of the dielectric in Fig. 24.16 and use this force to stretch the spring. Because work is done by the field, the field energy density decreases.

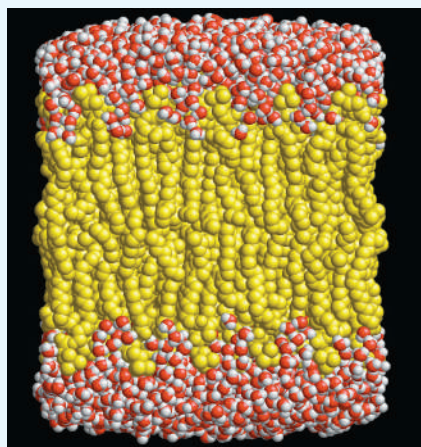
Figure 24.16 The fringing field at the edges of the capacitor exerts forces $\vec{F}_{-i}$ and $\vec{F}_{+i}$ on the negative and positive induced surface charges of a dielectric, pulling the dielectric into the capacitor.



KEYCONCEPT Adding a dielectric (with dielectric constant K) that fills the space between the plates of a capacitor reduces the electric field, the electric energy density, and the total stored energy, all by a factor of $1/K$.

BIO APPLICATION Dielectric Cell Membrane

The membrane of a living cell behaves like a dielectric between the plates of a capacitor. The membrane is made of two sheets of lipid molecules, with their water-insoluble ends in the middle and their water-soluble ends (shown in red) on the outer surfaces. Conductive fluids on either side of the membrane (water with negative ions inside the cell, water with positive ions outside) act as charged capacitor plates, and the nonconducting membrane acts as a dielectric with K of about 10. The potential difference V across the membrane is about 0.07 V and the membrane thickness d is about $7 \times 10^{-9} \text{ m}$, so the electric field $E = V/d$ in the membrane is about 10^7 V/m —close to the dielectric strength of the membrane. If the membrane were made of air, V and E would be larger by a factor of $K \approx 10$ and dielectric breakdown would occur.



Dielectric Breakdown

We mentioned earlier that when a dielectric is subjected to a sufficiently strong electric field, *dielectric breakdown* takes place and the dielectric becomes a conductor. This occurs when the electric field is so strong that electrons are ripped loose from their molecules and crash into other molecules, liberating even more electrons. This avalanche of moving charge forms a spark or arc discharge. Lightning is a dramatic example of dielectric breakdown in air.

Because of dielectric breakdown, capacitors always have maximum voltage ratings. When a capacitor is subjected to excessive voltage, an arc may form through a layer of dielectric, burning or melting a hole in it. This arc creates a conducting path (a short circuit) between the conductors. If a conducting path remains after the arc is extinguished, the device is rendered permanently useless as a capacitor.

The maximum electric-field magnitude that a material can withstand without the occurrence of breakdown is called its **dielectric strength**. This quantity is affected significantly by temperature, trace impurities, small irregularities in the metal electrodes, and other factors that are difficult to control. For this reason we can give only approximate figures for dielectric strengths. The dielectric strength of dry air is about $3 \times 10^6 \text{ V/m}$. **Table 24.2** lists the dielectric strengths of a few common insulating materials. All of the values are substantially greater than the value for air. For example, a layer of polycarbonate 0.01 mm thick (about the smallest practical thickness) has 10 times the dielectric strength of air and can withstand a maximum voltage of about $(3 \times 10^7 \text{ V/m})(1 \times 10^{-5} \text{ m}) = 300 \text{ V}$.

TABLE 24.2 Dielectric Constant and Dielectric Strength of Some Insulating Materials

Material	Dielectric Constant, K	Dielectric Strength, E_m (V/m)
Polycarbonate	2.8	3×10^7
Polyester	3.3	6×10^7
Polypropylene	2.2	7×10^7
Polystyrene	2.6	2×10^7
Pyrex® glass	4.7	1×10^7

TEST YOUR UNDERSTANDING OF SECTION 24.4 The space between the plates of an isolated parallel-plate capacitor is filled by a slab of dielectric with dielectric constant K . The two plates of the capacitor have charges Q and $-Q$. You pull out the dielectric slab. If the charges do not change, how does the energy in the capacitor change when you remove the slab? (i) It increases; (ii) it decreases; (iii) it remains the same.

ANSWER

(i) Here Q remains the same, so we use $U = Q^2/2C$ from Eq. (24.9). Removing the dielectric lowers the capacitance by a factor of $1/K$; since U is inversely proportional to C , the stored energy *increases* by a factor of K . It takes work to pull the dielectric slab out of the capacitor because the fringing field tries to pull the slab back in (Fig. 24.16). The work that you do goes into the energy stored in the capacitor.

24.5 MOLECULAR MODEL OF INDUCED CHARGE

In Section 24.4 we discussed induced surface charges on a dielectric in an electric field. Now let's look at how these surface charges can arise. If the material were a *conductor*, the answer would be simple. Conductors contain charge that is free to move, and when an electric field is present, some of the charge redistributes itself on the surface so that there is no electric field inside the conductor. But an ideal dielectric has *no* charges that are free to move, so how can a surface charge occur?

To understand this, we have to look again at rearrangement of charge at the *molecular* level. Some molecules, such as H_2O and N_2O , have equal amounts of positive and negative charges but a lopsided distribution, with excess positive charge concentrated on one side of the molecule and negative charge on the other. As we described in Section 21.7, such an arrangement is called an *electric dipole*, and the molecule is called a *polar molecule*. When no electric field is present in a gas or liquid with polar molecules, the molecules are oriented randomly (Fig. 24.17a). In an electric field, however, they tend to orient themselves as in Fig. 24.17b, as a result of the electric-field torques described in Section 21.7. Because of thermal agitation, the alignment of the molecules with $\vec{E}$ is not perfect.

Even a molecule that is *not* ordinarily polar *becomes* a dipole when it is placed in an electric field because the field pushes the positive charges in the molecules in the direction of the field and pushes the negative charges in the opposite direction. This causes a redistribution of charge within the molecule (Fig. 24.18). Such dipoles are called *induced* dipoles.

With either polar or nonpolar molecules, the redistribution of charge caused by the field leads to the formation of a layer of charge on each surface of the dielectric material (Fig. 24.19). These layers are the surface charges described in Section 24.4; their surface charge density is denoted by σ_i . The charges are *not* free to move indefinitely, as they

Figure 24.19 Polarization of a dielectric in an electric field $\vec{E}$ gives rise to thin layers of bound charges on the surfaces, creating surface charge densities σ_i and $-\sigma_i$. The sizes of the molecules are greatly exaggerated for clarity.

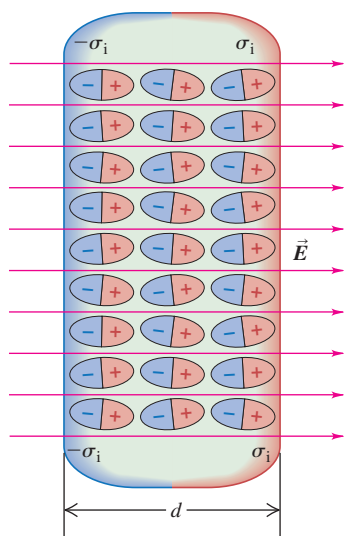


Figure 24.17 Polar molecules (a) without and (b) with an applied electric field $\vec{E}$.

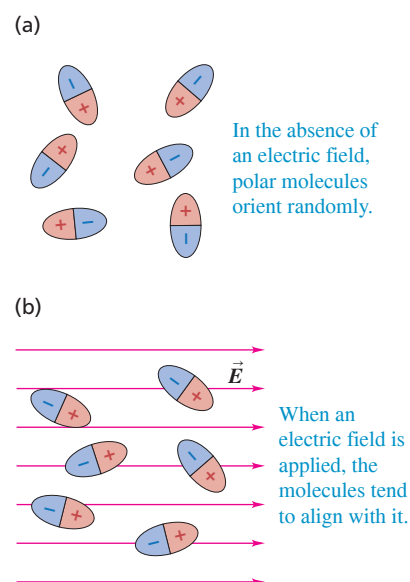
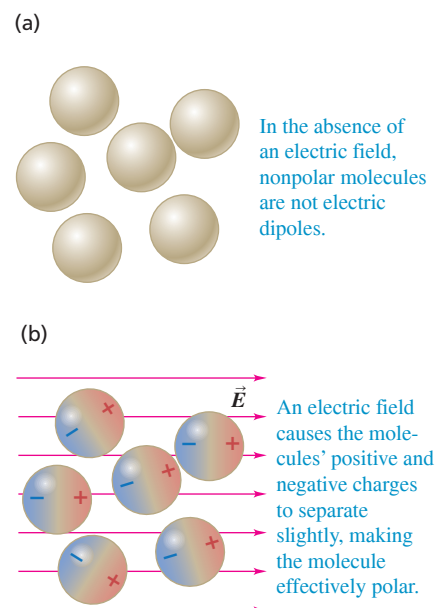


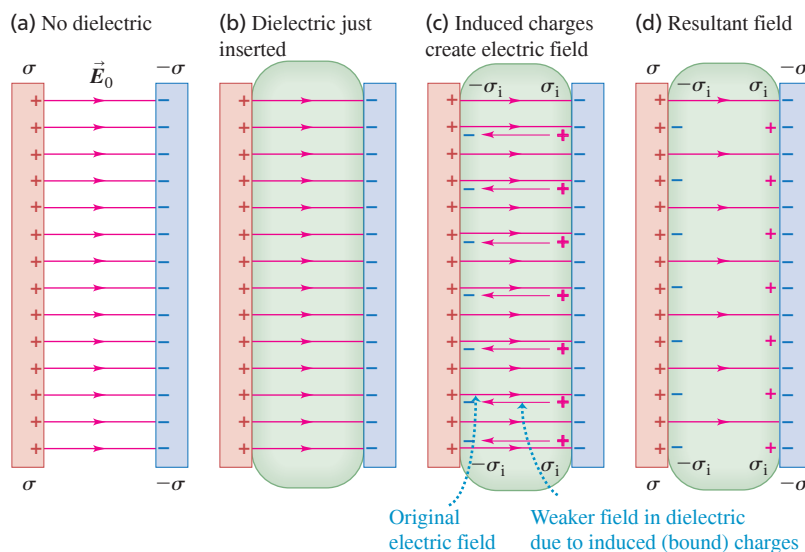
Figure 24.18 Nonpolar molecules (a) without and (b) with an applied electric field $\vec{E}$.



APPLICATION Smartphones, Capacitors, and Dielectrics A typical smartphone contains several hundred miniature capacitors, each of which uses a ceramic dielectric between its plates to boost its capacitance. These capacitors are used in every subsystem of the smartphone, including the main processor, the dynamic random access memory, the on-off and volume controls, the microphone, and the touch screen (see page 787).



Figure 24.20 (a) Electric field of magnitude E_0 between two charged plates. (b) Introduction of a dielectric of dielectric constant K . (c) The induced surface charges and their field. (d) Resultant field of magnitude E_0/K .

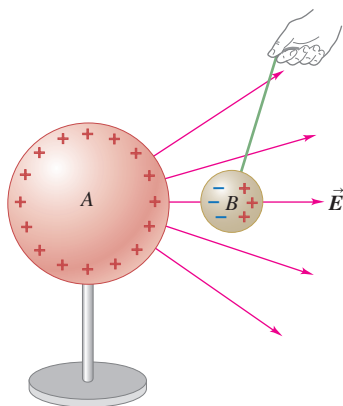


would be in a conductor, because each charge is bound to a molecule. They are in fact called **bound charges** to distinguish them from the **free charges** that are added to and removed from the conducting capacitor plates. In the interior of the material the net charge per unit volume remains zero. As we have seen, this redistribution of charge is called *polarization*, and we say that the material is *polarized*.

The four parts of Fig. 24.20 show the behavior of a slab of dielectric when it is inserted in the field between a pair of oppositely charged capacitor plates. Figure 24.20a shows the original field. Figure 24.20b is the situation after the dielectric has been inserted but before any rearrangement of charges has occurred. Figure 24.20c shows by thinner arrows the additional field set up in the dielectric by its induced surface charges. This field is *opposite* to the original field, but it is not great enough to cancel the original field completely because the charges in the dielectric are not free to move indefinitely. The resultant field in the dielectric, shown in Fig. 24.20d, is therefore decreased in magnitude. In the field-line representation, some of the field lines leaving the positive plate go through the dielectric, while others terminate on the induced charges on the faces of the dielectric.

As we discussed in Section 21.2, polarization is also the reason a charged object, such as an electrified plastic rod, can exert a force on an *uncharged* object such as a bit of paper or a pith ball. **Figure 24.21** shows an uncharged dielectric sphere B in the radial field of a positively charged object A . The induced positive charges on B experience a force toward the right, while the force on the induced negative charges is toward the left. The negative charges are closer to A , and thus are in a stronger field, than are the positive charges. The force toward the left is stronger than that toward the right, and B is attracted toward A , even though its net charge is zero. The attraction occurs whether the sign of A 's charge is positive or negative (see Fig. 21.8). Furthermore, the effect is not limited to dielectrics; an uncharged conducting object would be attracted in the same way.

Figure 24.21 A neutral sphere B in the radial electric field of a positively charged sphere A is attracted to the charge because of polarization.



TEST YOUR UNDERSTANDING OF SECTION 24.5 A parallel-plate capacitor has charges Q and $-Q$ on its two plates. A dielectric slab with $K = 3$ is then inserted into the space between the plates as shown in Fig. 24.20. Rank the following electric-field magnitudes in order from largest to smallest. (i) The field before the slab is inserted; (ii) the resultant field after the slab is inserted; (iii) the field due to the bound charges.

ANSWER

(i), (iii), (ii) Equation (24.14) says that if E_0 is the initial electric-field magnitude (before the dielectric slab is inserted), then the resultant field magnitude after the slab is inserted is $E_0/K = E_0/3$. The magnitude of the resultant field equals the difference between the initial field magnitude and the magnitude of the field due to the bound charges (see Fig. 24.20). Hence $E_0 - E_i = E_0/3$ and $E_i = 2E_0/3$.

24.6 GAUSS'S LAW IN DIELECTRICS

We can extend the analysis of Section 24.4 to reformulate Gauss's law in a form that is particularly useful for dielectrics. **Figure 24.22** is a close-up view of the left capacitor plate and left surface of the dielectric in Fig. 24.15b. Let's apply Gauss's law to the rectangular box shown in cross section by the purple line; the surface area of the left and right sides is A . The left side is embedded in the conductor that forms the left capacitor plate, and so the electric field everywhere on that surface is zero. The right side is embedded in the dielectric, where the electric field has magnitude E , and $E_{\perp} = 0$ everywhere on the other four sides. The total charge enclosed, including both the charge on the capacitor plate and the induced charge on the dielectric surface, is $Q_{\text{encl}} = (\sigma - \sigma_i)A$, so Gauss's law gives

$$EA = \frac{(\sigma - \sigma_i)A}{\epsilon_0} \quad (24.21)$$

This equation is not very illuminating as it stands because it relates two unknown quantities: E inside the dielectric and the induced surface charge density σ_i . But now we can use Eq. (24.16), developed for this same situation, to simplify this equation by eliminating σ_i . Equation (24.16) is

$$\sigma_i = \sigma \left(1 - \frac{1}{K}\right)$$

or

$$\sigma - \sigma_i = \frac{\sigma}{K}$$

Combining this with Eq. (24.21), we get

$$EA = \frac{\sigma A}{K\epsilon_0}$$

or

$$KEA = \frac{\sigma A}{\epsilon_0} \quad (24.22)$$

Equation (24.22) says that the flux of $K\vec{E}$, not $\vec{E}$, through the Gaussian surface in Fig. 24.22 is equal to the enclosed *free* charge σA divided by ϵ_0 . It turns out that for *any* Gaussian surface, whenever the induced charge is proportional to the electric field in the material, we can rewrite Gauss's law as

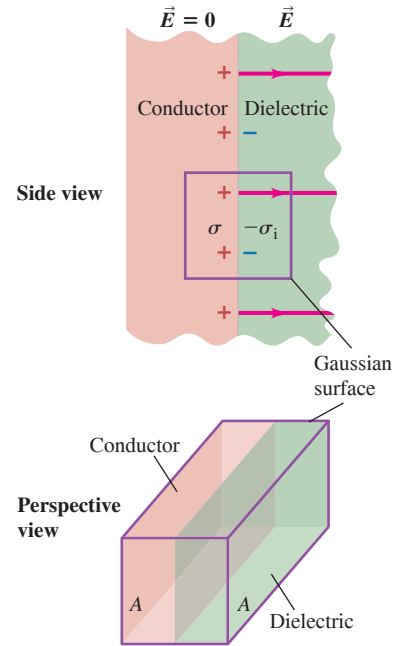
$$\oint K\vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_0} \quad (24.23)$$

Gauss's law in a dielectric:
Dielectric constant
Total free charge enclosed by surface
Electric constant

Surface integral of $K\vec{E}$ over a closed surface

where $Q_{\text{encl-free}}$ is the total *free* charge (not bound charge) enclosed by the Gaussian surface. The significance of these results is that the right sides contain only the *free* charge on the conductor, not the bound (induced) charge. In fact, although we have not proved it, Eq. (24.23) remains valid even when different parts of the Gaussian surface are embedded in dielectrics having different values of K , provided that the value of K in each dielectric is independent of the electric field (usually the case for electric fields that are not too strong) and that we use the appropriate value of K for each point on the Gaussian surface.

Figure 24.22 Gauss's law with a dielectric. This figure shows a close-up of the left-hand capacitor plate in Fig. 24.15b. The Gaussian surface is a rectangular box that lies half in the conductor and half in the dielectric.



EXAMPLE 24.12 A spherical capacitor with dielectric

Use Gauss's law to find the capacitance of the spherical capacitor of Example 24.3 (Section 24.1) if the volume between the shells is filled with an insulating oil with dielectric constant K .

IDENTIFY and SET UP The spherical symmetry of the problem is not changed by the presence of the dielectric, so as in Example 24.3, we use a concentric spherical Gaussian surface of radius r between the shells. Since a dielectric is present, we use Gauss's law in the form of Eq. (24.23).

EXECUTE From Eq. (24.23),

$$\oint \vec{K}\vec{E} \cdot d\vec{A} = \oint KE dA = KE \oint dA = (KE)(4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi K\epsilon_0 r^2} = \frac{Q}{4\pi\epsilon r^2}$$

where $\epsilon = K\epsilon_0$. Compared to the case in which there is vacuum between the shells, the electric field is reduced by a factor of $1/K$. The

potential difference V_{ab} between the shells is reduced by the same factor, and so the capacitance $C = Q/V_{ab}$ is *increased* by a factor of K , just as for a parallel-plate capacitor when a dielectric is inserted. Using the result of Example 24.3, we find that the capacitance with the dielectric is

$$C = \frac{4\pi K\epsilon_0 r_a r_b}{r_b - r_a} = \frac{4\pi\epsilon r_a r_b}{r_b - r_a}$$

EVALUATE If the dielectric fills the volume between the two conductors, the capacitance is just K times the value with no dielectric. The result is more complicated if the dielectric only partially fills this volume.

KEYCONCEPT Gauss's law can be written in a modified way if a dielectric is present. If the volume of space being studied is filled with a dielectric that has dielectric constant K , then the flux of the *product* of K and the electric field through a closed surface is proportional to the amount of *free* charge enclosed by the surface.

TEST YOUR UNDERSTANDING OF SECTION 24.6 A single point charge q is embedded in a very large block of dielectric of dielectric constant K . At a point inside the dielectric a distance r from the point charge, what is the magnitude of the electric field? (i) $q/4\pi\epsilon_0 r^2$; (ii) $Kq/4\pi\epsilon_0 r^2$; (iii) $q/4\pi K\epsilon_0 r^2$; (iv) none of these.

ANSWER

(iii) Equation (24.23) shows that this situation is the same as an isolated point charge in vacuum but with $\vec{E}$ replaced by $\vec{K}\vec{E}$. Hence KE at the point of interest is equal to $q/4\pi\epsilon_0 r^2$, and so $E = q/4\pi K\epsilon_0 r^2$. As in Example 24.12, filling the space with a dielectric reduces the electric field by a factor of $1/K$.

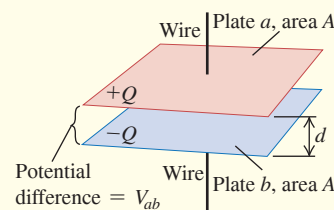
CHAPTER 24 SUMMARY

Capacitors and capacitance: A capacitor is any pair of conductors separated by an insulating material. When the capacitor is charged, there are charges of equal magnitude Q and opposite sign on the two conductors, and the potential V_{ab} of the positively charged conductor with respect to the negatively charged conductor is proportional to Q . The capacitance C is defined as the ratio of Q to V_{ab} . The SI unit of capacitance is the farad (F): $1 \text{ F} = 1 \text{ C/V}$.

A parallel-plate capacitor consists of two parallel conducting plates, each with area A , separated by a distance d . If they are separated by vacuum, the capacitance depends on only A and d . For other geometries, the capacitance can be found by using the definition $C = Q/V_{ab}$. (See Examples 24.1–24.4.)

$$C = \frac{Q}{V_{ab}} \quad (24.1)$$

$$C = \frac{Q}{V_{ab}} = \epsilon_0 \frac{A}{d} \quad (24.2)$$



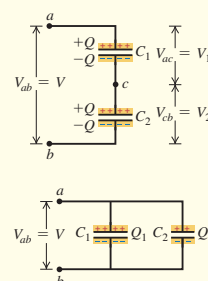
Capacitors in series and parallel: When capacitors with capacitances $C_1, C_2, C_3, \dots$ are connected in series, the reciprocal of the equivalent capacitance C_{eq} equals the sum of the reciprocals of the individual capacitances. When capacitors are connected in parallel, the equivalent capacitance C_{eq} equals the sum of the individual capacitances. (See Examples 24.5 and 24.6.)

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots \quad (24.5)$$

(capacitors in series)

$$C_{eq} = C_1 + C_2 + C_3 + \dots \quad (24.7)$$

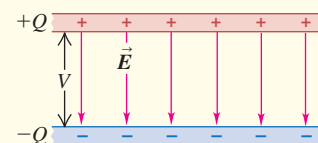
(capacitors in parallel)



Energy in a capacitor: The energy U required to charge a capacitor C to a potential difference V and a charge Q is equal to the energy stored in the capacitor. This energy can be thought of as residing in the electric field between the conductors; the energy density u (energy per unit volume) is proportional to the square of the electric-field magnitude. (See Examples 24.7–24.9.)

$$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV \quad (24.9)$$

$$u = \frac{1}{2}\epsilon_0 E^2 \quad (24.11)$$



Dielectrics: When the space between the conductors is filled with a dielectric material, the capacitance increases by a factor K , the dielectric constant of the material. The quantity $\epsilon = K\epsilon_0$ is the permittivity of the dielectric. For a fixed amount of charge on the capacitor plates, induced charges on the surface of the dielectric decrease the electric field and potential difference between the plates by the same factor K . The surface charge results from polarization, a microscopic rearrangement of charge in the dielectric. (See Example 24.10.)

Under sufficiently strong electric fields, dielectrics become conductors, a situation called dielectric breakdown. The maximum field that a material can withstand without breakdown is called its dielectric strength.

In a dielectric, the expression for the energy density is the same as in vacuum but with ϵ_0 replaced by $\epsilon = K\epsilon_0$. (See Example 24.11.)

Gauss's law in a dielectric has almost the same form as in vacuum, with two key differences: $\vec{E}$ is replaced by $K\vec{E}$ and Q_{encl} is replaced by $Q_{\text{encl-free}}$, which includes only the free charge (not bound charge) enclosed by the Gaussian surface. (See Example 24.12.)

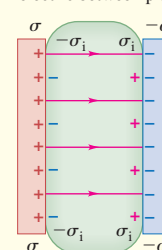
$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d} \quad (24.19)$$

(parallel-plate capacitor filled with dielectric)

$$u = \frac{1}{2}K\epsilon_0 E^2 = \frac{1}{2}\epsilon E^2 \quad (24.20)$$

$$\oint K\vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_0} \quad (24.23)$$

Dielectric between plates



Chapter 24 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 24.2, 24.3, and 24.4** (Section 24.1) before attempting these problems.

VP24.4.1 The plates of a parallel-plate capacitor in vacuum are 3.70 mm apart and 2.75 m² in area. When you apply a certain potential difference across the capacitor, the surface charge density on the positive plate is 1.40×10^{-5} C/m². Calculate (a) the capacitance of the capacitor and (b) the potential difference.

VP24.4.2 One model of a parallel-plate capacitor in vacuum is advertised as having a capacitance of 8.30 nF. When you disassemble one such capacitor, you find that its plates are 4.50 m² in area. (a) What is the spacing between this capacitor's plates? (b) When the potential difference between the plates is 3.00 kV, what are the charge on the capacitor and the magnitude of the electric field between the plates?

VP24.4.3 A spherical capacitor has vacuum between its conducting shells and a capacitance of 125 pF. The outer shell has inner radius 9.00 cm. (a) What is the outer radius of the inner shell? If the potential of the inner shell is 355 V higher than the potential of the outer shell, what are (b) the capacitor charge and (c) the surface charge densities on the inner and outer shells?

VP24.4.4 A cylindrical capacitor has a capacitance per length of 69.0 pF/m. (a) What is the ratio of the inner radius of the outer cylinder to the outer radius of the inner cylinder? (b) If the outer conductor carries charge -8.62 nC per meter of length, what is the potential difference between the two conductors? Which conductor is at higher potential?

Be sure to review **EXAMPLES 24.5 and 24.6** (Section 24.2) and **EXAMPLES 24.7, 24.8, and 24.9** (Section 24.3) before attempting these problems.

VP24.9.1 You have three capacitors that you wish to arrange in a circuit: $C_1 = 1.00$ μ F, $C_2 = 2.50$ μ F, and $C_3 = 5.00$ μ F. Find their equivalent capacitance if (a) they are connected in series; (b) they are connected in parallel; (c) C_1 and C_2 are connected in parallel, and that combination is connected to C_3 in series; (d) C_1 and C_2 are connected in series, and that combination is connected to C_3 in parallel.

VP24.9.2 You have two capacitors that you wish to use in an energy-storage device: $C_1 = 1.50$ μ F and $C_2 = 6.00$ μ F. (a) How much energy is stored in each capacitor if each has charge 4.50×10^{-4} C? Which has greater stored energy? (b) How much energy is stored in each capacitor if each has potential difference 125 V? Which has greater stored energy?

VP24.9.3 You connect two capacitors, $C_1 = 1.50$ μ F and $C_2 = 4.00$ μ F. (a) How much energy is stored in the combination if the potential difference across the combination is 145 V and the capacitors are connected in series? In parallel? In which case is the stored energy greater? (b) How much energy is stored in the combination if the charge on the combination is 4.00×10^{-4} C and the capacitors are connected in series? In parallel? In which case is the stored energy greater?

VP24.9.4 The plates of a parallel-plate capacitor in vacuum are 1.40 mm apart and 2.45 m² in area. The charge on the capacitor is 1.25×10^{-5} C. Calculate (a) the surface charge density on the positive plate, (b) the electric field magnitude between the plates, (c) the electric energy density between the plates, and (d) the total energy stored in the field between the plates.

Be sure to review **EXAMPLES 24.10 and 24.11** (Section 24.4) before attempting these problems.

VP24.11.1 The two plates of a parallel-plate capacitor each have area 0.460 m^2 , are 3.00 mm apart, and initially have vacuum between them. A power supply is attached to the capacitor, charging it to 4.00 kV , and is then disconnected. A dielectric sheet is then inserted that fills the space between the plates. The potential difference between the plates decreases to 2.50 kV , and the charge on each plate remains constant. Find (a) the original capacitance, (b) the charge on the capacitor plates, (c) the dielectric constant of the dielectric, (d) the new capacitance after you insert the dielectric, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.

VP24.11.2 The two plates of a parallel-plate capacitor each have area 1.20 m^2 , are 4.50 mm apart, and initially have vacuum between them. You connect a power supply to the capacitor, charging it to 3.50 kV . Keeping the power supply connected, you insert a sheet of dielectric with dielectric constant 2.50 that fills the space between the plates.

Find (a) the original capacitance, (b) the original charge on the capacitor plates, (c) the new capacitance once you insert the dielectric, (d) the new charge on the capacitor plates, and (e) the magnitude of the induced charge on the faces of the dielectric sheet.

VP24.11.3 With vacuum between its plates, a parallel-plate capacitor has capacitance $4.50 \text{ } \mu\text{F}$. You attach a power supply to the capacitor, charging it to 2.80 kV , and then disconnect it. You then insert a dielectric sheet that fills the space between the plates. The potential difference between the plates decreases to 1.20 kV , and the charge on each plate remains constant. Find the energy stored in the capacitor (a) before you insert the sheet and (b) after you insert the sheet.

VP24.11.4 With vacuum between its plates, a parallel-plate capacitor has capacitance $5.60 \text{ } \mu\text{F}$. You attach a power supply to the capacitor, charging it to 1.50 kV . Keeping the power supply connected, you insert a dielectric sheet with dielectric constant 2.85 that fills the space between the plates. Find the energy stored in the capacitor (a) before the sheet is inserted and (b) after the sheet is inserted.

BRIDGING PROBLEM Electric-Field Energy and Capacitance of a Conducting Sphere

A solid conducting sphere of radius R carries a charge Q . Calculate the electric-field energy density at a point a distance r from the center of the sphere for (a) $r < R$ and (b) $r > R$. (c) Calculate the total electric-field energy associated with the charged sphere. (d) How much work is required to assemble the charge Q on the sphere? (e) Use the result of part (c) to find the capacitance of the sphere. (You can think of the second conductor as a hollow conducting shell of infinite radius.)

SOLUTION GUIDE

IDENTIFY and SET UP

1. You know the electric field for this situation at all values of r from Example 22.5 (Section 22.4). You'll use this to find the electric-field energy density u and total electric-field energy U . You can then find the capacitance from the relationship $U = Q^2/2C$.
2. To find U , consider a spherical shell of radius r and thickness dr that has volume $dV = 4\pi r^2 dr$. (It will help to make a drawing of such a shell concentric with the conducting sphere.) The energy stored in this volume is $u dV$, and the total energy is the integral of $u dV$ from $r = 0$ to $r \rightarrow \infty$. Set up this integral.

EXECUTE

3. Find u for $r < R$ and for $r > R$. (*Hint:* What is the field inside a solid conductor?)
4. Substitute your results from step 3 into the expression from step 2. Then calculate the integral to find the total electric-field energy U .
5. Use your understanding of the energy stored in a charge distribution to find the work required to assemble the charge Q .
6. Find the capacitance of the sphere.

EVALUATE

7. Where is the electric-field energy density greatest? Where is it least?
8. How would the results be affected if the solid sphere were replaced by a hollow conducting sphere of the same radius R ?
9. You can find the potential difference between the sphere and infinity from $C = Q/V$. Does this agree with the result of Example 23.8 (Section 23.3)?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q24.1 Equation (24.2) shows that the capacitance of a parallel-plate capacitor becomes larger as the plate separation d decreases. However, there is a practical limit to how small d can be made, which places limits on how large C can be. Explain what sets the limit on d . (*Hint:* What happens to the magnitude of the electric field as $d \rightarrow 0$?)

Q24.2 Suppose several different parallel-plate capacitors are charged up by a constant-voltage source. Thinking of the actual movement and position of the charges on an atomic level, why does it make sense that the capacitances are proportional to the surface areas of the plates? Why does it make sense that the capacitances are *inversely* proportional to the distance between the plates?

Q24.3 Suppose the two plates of a capacitor have different areas. When the capacitor is charged by connecting it to a battery, do the charges on the two plates have equal magnitude, or may they be different? Explain your reasoning.

Q24.4 To store the maximum amount of energy in a parallel-plate capacitor with a given battery (voltage source), would it be better to have the plates far apart or close together?

Q24.5 In the parallel-plate capacitor of Fig. 24.2, suppose the plates are pulled apart so that the separation d is much larger than the size of the plates. (a) Is it still accurate to say that the electric field between the plates is uniform? Why or why not? (b) In the situation shown in Fig. 24.2, the potential difference between the plates is $V_{ab} = Qd/\epsilon_0 A$. If the plates are pulled apart as described above, is V_{ab} more or less than this formula would indicate? Explain your reasoning. (c) With the plates pulled apart as described above, is the capacitance more than, less than, or the same as that given by Eq. (24.2)? Explain your reasoning.

Q24.6 A parallel-plate capacitor is charged by being connected to a battery and is kept connected to the battery. The separation between the plates is then doubled. How does the electric field change? The charge on the plates? The total energy? Explain.

Q24.7 A parallel-plate capacitor is charged by being connected to a battery and is then disconnected from the battery. The separation between the plates is then doubled. How does the electric field change? The potential difference? The total energy? Explain.

Q24.8 Two parallel-plate capacitors, identical except that one has twice the plate separation of the other, are charged by the same voltage source. Which capacitor has a stronger electric field between the plates? Which capacitor has a greater charge? Which has greater energy density? Explain your reasoning.

Q24.9 The charged plates of a capacitor attract each other, so to pull the plates farther apart requires work by some external force. What becomes of the energy added by this work? Explain.

Q24.10 You have two capacitors and want to connect them across a voltage source (battery) to store the maximum amount of energy. Should they be connected in series or in parallel?

Q24.11 As shown in Table 24.1, water has a very large dielectric constant $K = 80.4$. Why do you think water is not commonly used as a dielectric in capacitors?

Q24.12 Is dielectric strength the same thing as dielectric constant? Explain any differences between the two quantities. Is there a simple relationship between dielectric strength and dielectric constant (see Table 24.2)?

Q24.13 A capacitor made of aluminum foil strips separated by Mylar film was subjected to excessive voltage, and the resulting dielectric breakdown melted holes in the Mylar. After this, the capacitance was found to be about the same as before, but the breakdown voltage was much less. Why?

Q24.14 Suppose you bring a slab of dielectric close to the gap between the plates of a charged capacitor, preparing to slide it between the plates. What force will you feel? What does this force tell you about the energy stored between the plates once the dielectric is in place, compared to before the dielectric is in place?

Q24.15 The freshness of fish can be measured by placing a fish between the plates of a capacitor and measuring the capacitance. How does this work? (*Hint:* As time passes, the fish dries out. See Table 24.1.)

Q24.16 Electrolytic capacitors use as their dielectric an extremely thin layer of nonconducting oxide between a metal plate and a conducting solution. Discuss the advantage of such a capacitor over one constructed using a solid dielectric between the metal plates.

Q24.17 In terms of the dielectric constant K , what happens to the electric flux through the Gaussian surface shown in Fig. 24.22 when the dielectric is inserted into the previously empty space between the plates? Explain.

Q24.18 A parallel-plate capacitor is connected to a power supply that maintains a fixed potential difference between the plates. (a) If a sheet of dielectric is then slid between the plates, what happens to (i) the electric field between the plates, (ii) the magnitude of charge on each plate, and (iii) the energy stored in the capacitor? (b) Now suppose that before the dielectric is inserted, the charged capacitor is disconnected from the power supply. In this case, what happens to (i) the electric field between the plates, (ii) the magnitude of charge on each plate, and (iii) the energy stored in the capacitor? Explain any differences between the two situations.

Q24.19 Liquid dielectrics that have polar molecules (such as water) always have dielectric constants that decrease with increasing temperature. Why?

Q24.20 A conductor is an extreme case of a dielectric, since if an electric field is applied to a conductor, charges are free to move within the conductor to set up “induced charges.” What is the dielectric constant of a perfect conductor? Is it $K = 0$, $K \rightarrow \infty$, or something in between? Explain your reasoning.

Q24.21 The two plates of a capacitor are given charges $\pm Q$. The capacitor is then disconnected from the charging device so that the charges on the plates can’t change, and the capacitor is immersed in a tank of oil. Does the electric field between the plates increase, decrease, or stay the same? Explain your reasoning. How can this field be measured?

EXERCISES

Section 24.1 Capacitors and Capacitance

24.1 • The plates of a parallel-plate capacitor are 2.50 mm apart, and each carries a charge of magnitude 80.0 nC. The plates are in vacuum. The electric field between the plates has a magnitude of 4.00×10^6 V/m. What is (a) the potential difference between the plates; (b) the area of each plate; (c) the capacitance?

24.2 • The plates of a parallel-plate capacitor are 3.28 mm apart, and each has an area of 9.82 cm^2 . Each plate carries a charge of magnitude 4.35×10^{-8} C. The plates are in vacuum. What is (a) the capacitance; (b) the potential difference between the plates; (c) the magnitude of the electric field between the plates?

24.3 • A parallel-plate air capacitor of capacitance 245 pF has a charge of magnitude $0.148 \mu\text{C}$ on each plate. The plates are 0.328 mm apart. (a) What is the potential difference between the plates? (b) What is the area of each plate? (c) What is the electric-field magnitude between the plates? (d) What is the surface charge density on each plate?

24.4 • A $5.00 \mu\text{F}$ parallel-plate capacitor is connected to a 12.0 V battery. After the capacitor is fully charged, the battery is disconnected without loss of any of the charge on the plates. (a) A voltmeter is connected across the two plates without discharging them. What does it read? (b) What would the voltmeter read if (i) the plate separation were doubled; (ii) the radius of each plate were doubled but their separation was unchanged?

24.5 • A $10.0 \mu\text{F}$ parallel-plate capacitor with circular plates is connected to a 12.0 V battery. (a) What is the charge on each plate? (b) How much charge would be on the plates if their separation were doubled while the capacitor remained connected to the battery? (c) How much charge would be on the plates if the capacitor were connected to the 12.0 V battery after the radius of each plate was doubled without changing their separation?

24.6 • A 5.00 pF , parallel-plate, air-filled capacitor with circular plates is to be used in a circuit in which it will be subjected to potentials of up to 1.00×10^2 V. The electric field between the plates is to be no greater than 1.00×10^4 N/C. As a budding electrical engineer for Live-Wire Electronics, your tasks are to (a) design the capacitor by finding what its physical dimensions and separation must be; (b) find the maximum charge these plates can hold.

24.7 • A parallel-plate air capacitor is to store charge of magnitude 240.0 pC on each plate when the potential difference between the plates is 42.0 V. (a) If the area of each plate is 6.80 cm^2 , what is the separation between the plates? (b) If the separation between the two plates is double the value calculated in part (a), what potential difference is required for the capacitor to store charge of magnitude 240.0 pC on each plate?

24.8 • A cylindrical capacitor consists of a solid inner conducting core with radius 0.250 cm, surrounded by an outer hollow conducting tube. The two conductors are separated by air, and the length of the cylinder is 12.0 cm. The capacitance is 36.7 pF. (a) Calculate the inner radius of the hollow tube. (b) When the capacitor is charged to 125 V, what is the charge per unit length λ on the capacitor?

24.9 • A capacitor is made from two hollow, coaxial, iron cylinders, one inside the other. The inner cylinder is negatively charged and the outer is positively charged; the magnitude of the charge on each is 10.0 pC. The inner cylinder has radius 0.50 mm, the outer one has radius 5.00 mm, and the length of each cylinder is 18.0 cm. (a) What is the capacitance? (b) What applied potential difference is necessary to produce these charges on the cylinders?

24.10 •• A cylindrical capacitor has an inner conductor of radius 2.2 mm and an outer conductor of radius 3.5 mm. The two conductors are separated by vacuum, and the entire capacitor is 2.8 m long. (a) What is the capacitance per unit length? (b) The potential of the inner conductor is 350 mV higher than that of the outer conductor. Find the charge (magnitude and sign) on both conductors.

24.11 •• A spherical capacitor contains a charge of 3.30 nC when connected to a potential difference of 220 V . If its plates are separated by vacuum and the inner radius of the outer shell is 4.00 cm , calculate: (a) the capacitance; (b) the radius of the inner sphere; (c) the electric field just outside the surface of the inner sphere.

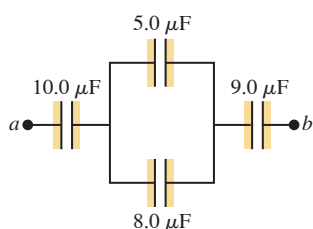
24.12 • A spherical capacitor is formed from two concentric, spherical, conducting shells separated by vacuum. The inner sphere has radius 15.0 cm and the capacitance is 116 pF . (a) What is the radius of the outer sphere? (b) If the potential difference between the two spheres is 220 V , what is the magnitude of charge on each sphere?

Section 24.2 Capacitors in Series and Parallel

24.13 •• You measure the capacitance C_1 of a capacitor by doing the following: First connect capacitors C_1 and C_2 in series to a power supply that provides a voltage V that can be varied. The capacitance of C_2 is known to be $3.00 \mu\text{F}$. Then vary the applied voltage V , and for each value of V measure the voltage V_2 across C_2 . After plotting your data as V_2 versus V , you find that the data fall close to a straight line that has slope 0.650 . What is the capacitance C_1 ?

24.14 • Figure E24.14 shows a system of four capacitors, where the potential difference across ab is 50.0 V . (a) Find the equivalent capacitance of this system between a and b . (b) How much charge is stored by this combination of capacitors? (c) How much charge is stored in each of the $10.0 \mu\text{F}$ and the $9.0 \mu\text{F}$ capacitors?

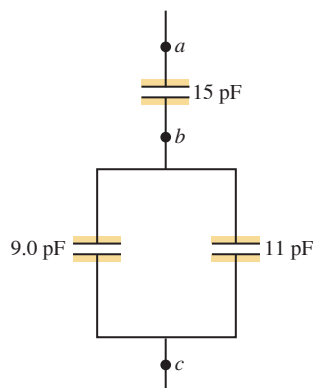
Figure E24.14



24.15 • BIO Electric Eels. Electric eels and electric fish generate large potential differences that are used to stun enemies and prey. These potentials are produced by cells that each can generate 0.10 V . We can plausibly model such cells as charged capacitors. (a) How should these cells be connected (in series or in parallel) to produce a total potential of more than 0.10 V ? (b) Using the connection in part (a), how many cells must be connected together to produce the 500 V surge of the electric eel?

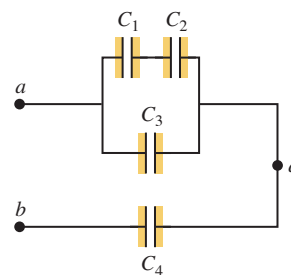
24.16 • For the system of capacitors shown in Fig. E24.16, find the equivalent capacitance (a) between b and c , and (b) between a and c .

Figure E24.16



24.17 • In Fig. E24.17, each capacitor has $C = 4.00 \mu\text{F}$ and $V_{ab} = +28.0 \text{ V}$. Calculate (a) the charge on each capacitor; (b) the potential difference across each capacitor; (c) the potential difference between points a and d .

Figure E24.17

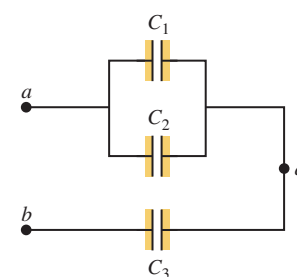


24.18 • In Fig. 24.8a, let $C_1 = 3.00 \mu\text{F}$, $C_2 = 5.00 \mu\text{F}$, and $V_{ab} = +64.0 \text{ V}$. Calculate (a) the charge on each capacitor and (b) the potential difference across each capacitor.

24.19 • In Fig. 24.9a, let $C_1 = 3.00 \mu\text{F}$, $C_2 = 5.00 \mu\text{F}$, and $V_{ab} = +52.0 \text{ V}$. Calculate (a) the charge on each capacitor and (b) the potential difference across each capacitor.

24.20 • In Fig. E24.20, $C_1 = 6.00 \mu\text{F}$, $C_2 = 3.00 \mu\text{F}$, and $C_3 = 5.00 \mu\text{F}$. The capacitor network is connected to an applied potential V_{ab} .

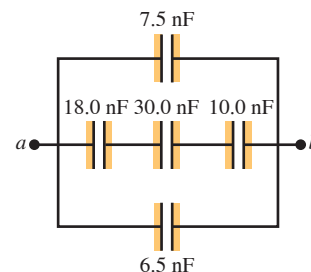
Figure E24.20



After the charges on the capacitors have reached their final values, the charge on C_2 is $30.0 \mu\text{C}$. (a) What are the charges on capacitors C_1 and C_3 ? (b) What is the applied voltage V_{ab} ?

24.21 •• For the system of capacitors shown in Fig. E24.21, a potential difference of 25 V is maintained across ab . (a) What is the equivalent capacitance of this system between a and b ? (b) How much charge is stored by this system? (c) How much charge does the 6.5 nF capacitor store? (d) What is the potential difference across the 7.5 nF capacitor?

Figure E24.21



24.22 •• A capacitor is formed from two concentric spherical conducting shells separated by vacuum. The inner sphere has radius 12.5 cm , and the outer sphere has radius 14.8 cm . A potential difference of 120 V is applied to the capacitor. (a) What is the energy density at $r = 12.6 \text{ cm}$, just outside the inner sphere? (b) What is the energy density at $r = 14.7 \text{ cm}$, just inside the outer sphere? (c) For a parallel-plate capacitor the energy density is uniform in the region between the plates, except near the edges of the plates. Is this also true for a spherical capacitor?

Section 24.3 Energy Storage in Capacitors and Electric-Field Energy

24.23 • A $5.80 \mu\text{F}$, parallel-plate, air capacitor has a plate separation of 5.00 mm and is charged to a potential difference of 400 V . Calculate the energy density in the region between the plates, in units of J/m^3 .

24.24 • A parallel-plate air capacitor has a capacitance of 920 pF . The charge on each plate is $3.90 \mu\text{C}$. (a) What is the potential difference between the plates? (b) If the charge is kept constant, what will be the potential difference if the plate separation is doubled? (c) How much work is required to double the separation?

24.25 • An air capacitor is made from two flat parallel plates 1.50 mm apart. The magnitude of charge on each plate is $0.0180 \mu\text{C}$ when the potential difference is 200 V. (a) What is the capacitance? (b) What is the area of each plate? (c) What maximum voltage can be applied without dielectric breakdown? (Dielectric breakdown for air occurs at an electric-field strength of $3.0 \times 10^6 \text{ V/m}$.) (d) When the charge is $0.0180 \mu\text{C}$, what total energy is stored?

24.26 •• A parallel-plate vacuum capacitor has 8.38 J of energy stored in it. The separation between the plates is 2.30 mm. If the separation is decreased to 1.15 mm, what is the energy stored (a) if the capacitor is disconnected from the potential source so the charge on the plates remains constant, and (b) if the capacitor remains connected to the potential source so the potential difference between the plates remains constant?

24.27 • You have two identical capacitors and an external potential source. (a) Compare the total energy stored in the capacitors when they are connected to the applied potential in series and in parallel. (b) Compare the maximum amount of charge stored in each case. (c) Energy storage in a capacitor can be limited by the maximum electric field between the plates. What is the ratio of the electric field for the series and parallel combinations?

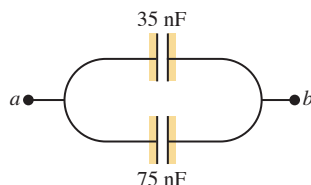
24.28 • For the capacitor network shown in Fig. E24.28, the potential difference across ab is 48 V. Find (a) the total charge stored in this network; (b) the charge on each capacitor; (c) the total energy stored in the network; (d) the energy stored in each capacitor; (e) the potential differences across each capacitor.

Figure E24.28



24.29 • For the capacitor network shown in Fig. E24.29, the potential difference across ab is 220 V. Find (a) the total charge stored in this network; (b) the charge on each capacitor; (c) the total energy stored in the network; (d) the energy stored in each capacitor; (e) the potential difference across each capacitor.

Figure E24.29



Section 24.4 Dielectrics

24.30 • Two air-filled parallel-plate capacitors with capacitances C_1 and C_2 are connected in parallel to a battery that has a voltage of 36.0 V; $C_1 = 4.00 \mu\text{F}$ and $C_2 = 6.00 \mu\text{F}$. (a) What is the total positive charge stored in the two capacitors? (b) While the capacitors remain connected to the battery, a dielectric with dielectric constant 5.00 is inserted between the plates of capacitor C_1 , completely filling the space between them. Then what is the total positive charge stored on the two capacitors? Does the insertion of the dielectric cause the total charge stored to increase or decrease?

24.31 • A $12.5 \mu\text{F}$ capacitor is connected to a power supply that keeps a constant potential difference of 24.0 V across the plates. A piece of material having a dielectric constant of 3.75 is placed between the plates, completely filling the space between them. (a) How much energy is stored in the capacitor before and after the dielectric is inserted? (b) By how much did the energy change during the insertion? Did it increase or decrease?

24.32 • A parallel-plate capacitor has capacitance $C_0 = 8.00 \text{ pF}$ when there is air between the plates. The separation between the plates is 1.50 mm. (a) What is the maximum magnitude of charge Q that can be placed on each plate if the electric field in the region between the plates is not to exceed $3.00 \times 10^4 \text{ V/m}$? (b) A dielectric with $K = 2.70$ is inserted between the plates of the capacitor, completely filling the volume between the plates. Now what is the maximum magnitude of charge on each plate if the electric field between the plates is not to exceed $3.00 \times 10^4 \text{ V/m}$?

24.33 • Two parallel plates have equal and opposite charges. When the space between the plates is evacuated, the electric field is $E = 3.20 \times 10^5 \text{ V/m}$. When the space is filled with dielectric, the electric field is $E = 2.50 \times 10^5 \text{ V/m}$. (a) What is the charge density on each surface of the dielectric? (b) What is the dielectric constant?

24.34 •• Two identical air-filled parallel-plate capacitors C_1 and C_2 , each with capacitance C , are connected in series to a battery that has voltage V . While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K > 1$ is inserted between the plates of one of the capacitors, completely filling the space between them. Let U_0 be the total energy stored in the two capacitors without the dielectric and U be the total energy stored after the dielectric is inserted. In terms of K , what is the ratio U/U_0 ? Does the total stored energy increase, decrease, or stay the same after the dielectric is inserted?

24.35 •• Two identical air-filled parallel-plate capacitors C_1 and C_2 are connected in series to a battery that has voltage V . The charge on each capacitor is Q_0 . While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K > 1$ is inserted between the plates of capacitor C_1 , completely filling the space between them. In terms of K and Q_0 , what is the charge on capacitor C_1 after the dielectric is inserted? Does the charge on C_1 increase, decrease, or stay the same?

24.36 •• Two air-filled parallel-plate capacitors with capacitances C_1 and C_2 are connected in series to a battery that has voltage V ; $C_1 = 3.00 \mu\text{F}$ and $C_2 = 6.00 \mu\text{F}$. The electric field between the plates of capacitor C_2 is E_{02} . While the two capacitors remain connected to the battery, a dielectric with dielectric constant $K = 4$ is inserted between the plates of capacitor C_1 , completely filling the space between them. After the dielectric is inserted in C_1 , the electric field between the plates of capacitor C_2 is E_2 . (a) What is the ratio E_2/E_{02} ? When the dielectric is inserted into C_1 , does the electric field in C_2 increase, decrease, or remain the same? (b) Repeat the calculation in part (a) for the two capacitors connected to the battery in parallel.

24.37 • The dielectric to be used in a parallel-plate capacitor has a dielectric constant of 3.60 and a dielectric strength of $1.60 \times 10^7 \text{ V/m}$. The capacitor is to have a capacitance of $1.25 \times 10^{-9} \text{ F}$ and must be able to withstand a maximum potential difference of 5500 V. What is the minimum area the plates of the capacitor may have?

24.38 •• BIO Potential in Human Cells. Some cell walls in the human body have a layer of negative charge on the inside surface and a layer of positive charge of equal magnitude on the outside surface. Suppose that the charge density on either surface is $\pm 0.50 \times 10^{-3} \text{ C/m}^2$, the cell wall is 5.0 nm thick, and the cell-wall material is air. (a) Find the magnitude of $\vec{E}$ in the wall between the two layers of charge. (b) Find the potential difference between the inside and the outside of the cell. Which is at the higher potential? (c) A typical cell in the human body has a volume of 10^{-16} m^3 . Estimate the total electric-field energy stored in the wall of a cell of this size. (Hint: Assume that the cell is spherical, and calculate the volume of the cell wall.) (d) In reality, the cell wall is made up, not of air, but of tissue with a dielectric constant of 5.4. Repeat parts (a) and (b) in this case.

24.39 • A constant potential difference of 12 V is maintained between the terminals of a $0.25 \mu\text{F}$, parallel-plate, air capacitor. (a) A sheet of Mylar is inserted between the plates of the capacitor, completely filling the space between the plates. When this is done, how much additional charge flows onto the positive plate of the capacitor (see Table 24.1)? (b) What is the total induced charge on either face of the Mylar sheet? (c) What effect does the Mylar sheet have on the electric field between the plates? Explain how you can reconcile this with the increase in charge on the plates, which acts to increase the electric field.

24.40 •• Polystyrene has dielectric constant 2.6 and dielectric strength 2.0×10^7 V/m. A piece of polystyrene is used as a dielectric in a parallel-plate capacitor, filling the volume between the plates. (a) When the electric field between the plates is 80% of the dielectric strength, what is the energy density of the stored energy? (b) When the capacitor is connected to a battery with voltage 500.0 V, the electric field between the plates is 80% of the dielectric strength. What is the area of each plate if the capacitor stores 0.200 mJ of energy under these conditions?

24.41 • When a 360 nF air capacitor ($1 \text{ nF} = 10^{-9} \text{ F}$) is connected to a power supply, the energy stored in the capacitor is $1.85 \times 10^{-5} \text{ J}$. While the capacitor is kept connected to the power supply, a slab of dielectric is inserted that completely fills the space between the plates. This increases the stored energy by $2.32 \times 10^{-5} \text{ J}$. (a) What is the potential difference between the capacitor plates? (b) What is the dielectric constant of the slab?

24.42 • A parallel-plate capacitor has capacitance $C = 12.5 \text{ pF}$ when the volume between the plates is filled with air. The plates are circular, with radius 3.00 cm. The capacitor is connected to a battery, and a charge of magnitude 25.0 pC goes onto each plate. With the capacitor still connected to the battery, a slab of dielectric is inserted between the plates, completely filling the space between the plates. After the dielectric has been inserted, the charge on each plate has magnitude 45.0 pC. (a) What is the dielectric constant K of the dielectric? (b) What is the potential difference between the plates before and after the dielectric has been inserted? (c) What is the electric field at a point midway between the plates before and after the dielectric has been inserted?

Section 24.6 Gauss's Law in Dielectrics

24.43 • A parallel-plate capacitor has the volume between its plates filled with plastic with dielectric constant K . The magnitude of the charge on each plate is Q . Each plate has area A , and the distance between the plates is d . (a) Use Gauss's law as stated in Eq. (24.23) to calculate the magnitude of the electric field in the dielectric. (b) Use the electric field determined in part (a) to calculate the potential difference between the two plates. (c) Use the result of part (b) to determine the capacitance of the capacitor. Compare your result to Eq. (24.12).

24.44 • A parallel-plate capacitor has plates with area 0.0225 m^2 separated by 1.00 mm of Teflon. (a) Calculate the charge on the plates when they are charged to a potential difference of 12.0 V. (b) Use Gauss's law [Eq. (24.23)] to calculate the electric field inside the Teflon. (c) Use Gauss's law to calculate the electric field if the voltage source is disconnected and the Teflon is removed.

PROBLEMS

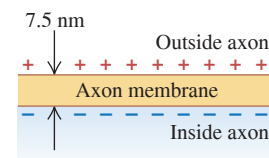
24.45 • Electronic flash units for cameras contain a capacitor for storing the energy used to produce the flash. In one such unit, the flash lasts for $\frac{1}{675} \text{ s}$ with an average light power output of $2.70 \times 10^5 \text{ W}$. (a) If the conversion of electrical energy to light is 95% efficient (the rest of the energy goes to thermal energy), how much energy must be stored in the capacitor for one flash? (b) The capacitor has a potential difference between its plates of 125 V when the stored energy equals the value calculated in part (a). What is the capacitance?

24.46 • A parallel-plate air capacitor is made by using two plates 12 cm square, spaced 3.7 mm apart. It is connected to a 12 V battery. (a) What is the capacitance? (b) What is the charge on each plate? (c) What is the electric field between the plates? (d) What is the energy stored in the capacitor? (e) If the battery is disconnected and then the plates are pulled apart to a separation of 7.4 mm, what are the answers to parts (a)–(d)?

24.47 ••• In one type of computer keyboard, each key holds a small metal plate that serves as one plate of a parallel-plate, air-filled capacitor. When the key is depressed, the plate separation decreases and the capacitance increases. Electronic circuitry detects the change in capacitance and thus detects that the key has been pressed. In one particular keyboard, the area of each metal plate is 42.0 mm^2 , and the separation between the plates is 0.700 mm before the key is depressed. (a) Calculate the capacitance before the key is depressed. (b) If the circuitry can detect a change in capacitance of 0.250 pF, how far must the key be depressed before the circuitry detects its depression?

24.48 ••• BIO Cell Membranes. Cell membranes (the walled enclosure around a cell) are typically about 7.5 nm thick. They are partially permeable to allow charged material to pass in and out, as needed. Equal but opposite charge densities build up on the inside and outside faces of such a membrane, and these charges prevent additional charges from passing through the cell wall. We can model a cell membrane as a parallel-plate capacitor, with the membrane itself containing proteins embedded in an organic material to give the membrane a dielectric constant of about 10. (See Fig. P24.48.) (a) What is the capacitance per square centimeter of such a cell wall? (b) In its normal resting state, a cell has a potential difference of 85 mV across its membrane. What is the electric field inside this membrane?

Figure P24.48

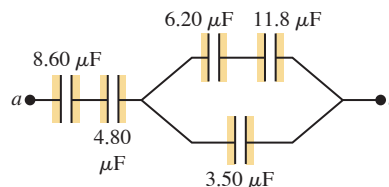


24.49 •• A $20.0 \mu\text{F}$ capacitor is charged to a potential difference of 800 V. The terminals of the charged capacitor are then connected to those of an uncharged $10.0 \mu\text{F}$ capacitor. Compute (a) the original charge of the system, (b) the final potential difference across each capacitor, (c) the final energy of the system, and (d) the decrease in energy when the capacitors are connected.

24.50 •• When lightning strikes a car, the metallic outer shell, which is insulated from the ground by its rubber tires, attains a high voltage. We can estimate how much charge is deposited by roughly modeling the car as a spherical capacitor with the outer radius taken to infinity. (a) Determine the capacitance of a sphere in terms of its radius, either by considering the potential on a sphere relative to infinity as a function of its charge, or by considering a spherical capacitor as the outer shell becomes very large. (These methods provide the same result.) (b) Estimate the radius of a sphere that corresponds to the size of a car. (c) Determine the corresponding capacitance. (d) A typical lightning strike could deliver a 100 MV potential to a car. Then what net charge would be deposited on the car?

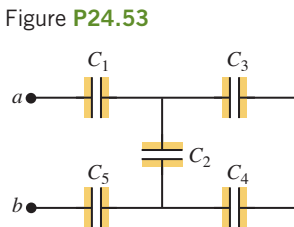
24.51 • For the capacitor network shown in Fig. P24.51, the potential difference across ab is 12.0 V. Find (a) the total energy stored in this network and (b) the energy stored in the $4.80 \mu\text{F}$ capacitor.

Figure P24.51



24.52 •• In Fig. E24.17, $C_1 = 6.00 \mu\text{F}$, $C_2 = 3.00 \mu\text{F}$, $C_3 = 4.00 \mu\text{F}$, and $C_4 = 8.00 \mu\text{F}$. The capacitor network is connected to an applied potential difference V_{ab} . After the charges on the capacitors have reached their final values, the voltage across C_3 is 40.0 V. What are (a) the voltages across C_1 and C_2 , (b) the voltage across C_4 , and (c) the voltage V_{ab} applied to the network?

24.53 • In Fig. P24.53, $C_1 = C_5 = 8.4 \mu\text{F}$ and $C_2 = C_3 = C_4 = 4.2 \mu\text{F}$. The applied potential is $V_{ab} = 220 \text{ V}$. (a) What is the equivalent capacitance of the network between points a and b ? (b) Calculate the charge on each capacitor and the potential difference across each capacitor.



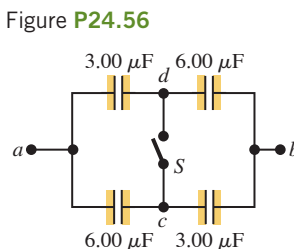
24.54 •• CP After combing your hair on a dry day, some of your hair stands up, forced away from your head by electrostatic repulsion. (a) Estimate the length L of your hair. (b) Using the average linear mass density of hair, which is $65 \mu\text{g}/\text{cm}$, estimate the mass m of one of your hairs. (c) Estimate the number N of hairs that stand after combing. (d) Assume that the comb has taken away a charge $-2Q$, and that your hair has therefore gained an amount of charge $2Q$. Assume further that half of this charge resides next to your head and the other half is distributed at the ends of the N strands that stand up. Assume the electrostatic force that lifted a hair was twice its weight. Show that this leads to

$$2mg = \frac{1}{4\pi\epsilon_0} \frac{Q^2/N}{L^2}$$

Use this equation to estimate the charge Q that resides on your head. (e) If your head were a sphere with radius R , it would have a capacitance of $4\pi\epsilon_0 R$. Estimate the radius of your head; then use your estimate to determine your head's capacitance. (f) Use your result to estimate the potential attained due to combing. (The surprising result illustrates an interesting feature of static electricity.)

24.55 •• In Fig. E24.20, $C_1 = 3.00 \mu\text{F}$ and $V_{ab} = 150 \text{ V}$. The charge on capacitor C_1 is $150 \mu\text{C}$ and the charge on C_3 is $450 \mu\text{C}$. What are the values of the capacitances of C_2 and C_3 ?

24.56 • The capacitors in Fig. P24.56 are initially uncharged and are connected, as in the diagram, with switch S open. The applied potential difference is $V_{ab} = +210 \text{ V}$. (a) What is the potential difference V_{cd} ? (b) What is the potential difference across each capacitor after switch S is closed? (c) How much charge flowed through the switch when it was closed?

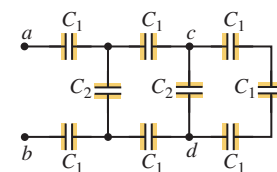


24.57 •• Three capacitors having capacitances of 8.4 , 8.4 , and $4.2 \mu\text{F}$ are connected in series across a 36 V potential difference. (a) What is the charge on the $4.2 \mu\text{F}$ capacitor? (b) What is the total energy stored in all three capacitors? (c) The capacitors are disconnected from the potential difference without allowing them to discharge. They are then reconnected in parallel with each other, with the positively charged plates connected together. What is the voltage across each capacitor in the parallel combination? (d) What is the total energy now stored in the capacitors?

24.58 • Capacitance of a Thundercloud. The charge center of a thundercloud, drifting 3.0 km above the earth's surface, contains 20 C of negative charge. Assuming the charge center has a radius of 1.0 km , and modeling the charge center and the earth's surface as parallel plates, calculate: (a) the capacitance of the system; (b) the potential difference between charge center and ground; (c) the average strength of the electric field between cloud and ground; (d) the electrical energy stored in the system.

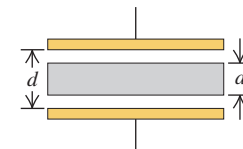
24.59 •• In Fig. P24.59, each capacitance C_1 is $6.9 \mu\text{F}$, and each capacitance C_2 is $4.6 \mu\text{F}$. (a) Compute the equivalent capacitance of the network between points a and b . (b) Compute the charge on each of the three capacitors nearest a and b when $V_{ab} = 420 \text{ V}$. (c) With 420 V across a and b , compute V_{cd} .

Figure P24.59



24.60 •• An air capacitor is made by using two flat plates, each with area A , separated by a distance d . Then a metal slab having thickness a (less than d) and the same shape and size as the plates is inserted between them, parallel to the plates and not touching either plate (Fig. P24.60).

Figure P24.60



(a) What is the capacitance of this arrangement? (b) Express the capacitance as a multiple of the capacitance C_0 when the metal slab is not present. (c) Discuss what happens to the capacitance in the limits $a \rightarrow 0$ and $a \rightarrow d$.

24.61 •• A potential difference $V_{ab} = 48.0 \text{ V}$ is applied across the capacitor network of Fig. E24.17. If $C_1 = C_2 = 4.00 \mu\text{F}$ and $C_4 = 8.00 \mu\text{F}$, what must the capacitance C_3 be if the network is to store $2.90 \times 10^{-3} \text{ J}$ of electrical energy?

24.62 • CALC The inner cylinder of a long, cylindrical capacitor has radius r_a and linear charge density $+\lambda$. It is surrounded by a coaxial cylindrical conducting shell with inner radius r_b and linear charge density $-\lambda$ (see Fig. 24.6). (a) What is the energy density in the region between the conductors at a distance r from the axis? (b) Integrate the energy density calculated in part (a) over the volume between the conductors in a length L of the capacitor to obtain the total electric-field energy per unit length. (c) Use Eq. (24.9) and the capacitance per unit length calculated in Example 24.4 (Section 24.1) to calculate U/L . Does your result agree with that obtained in part (b)?

24.63 ••• CP Dielectric elastomers are used to create voltage-dependent capacitors. These elastomers contract when subject to the stresses induced by Coulomb forces between capacitor plates. Consider a pair of parallel plates with area A separated by distance d_0 when no voltage is applied. A material with dielectric constant K and Young's modulus Y fills the space between the plates. (a) What compressive force is applied to the dielectric when the plates have opposite charges Q and $-Q$? (b) The distance between the plates is $d = d_0 - sQ^2$, where s is a squeezing coefficient. Determine s in terms of the parameters specified. (c) A capacitor of this sort has plate area 1.00 cm^2 and noncharged separation distance $d_0 = 0.400 \text{ mm}$. The plates are separated by a silicone dielectric elastomer with dielectric constant $K = 3.00$ and Young's modulus 0.0100 GPa . What is the squeezing coefficient in this case? (d) If each plate has charge of magnitude $0.700 \mu\text{C}$, what is the applied voltage? (e) What new voltage would double the amount of charge on the plates?

24.64 •• A parallel-plate capacitor is made from two plates 12.0 cm on each side and 4.50 mm apart. Half of the space between these plates contains only air, but the other half is filled with Plexiglas® of dielectric constant 3.40 (Fig. P24.64). An

Figure P24.64

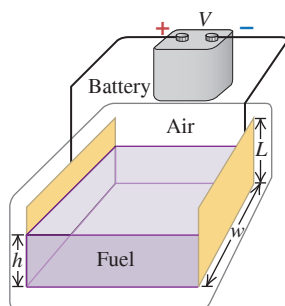


18.0 V battery is connected across the plates. (a) What is the capacitance of this combination? (Hint: Can you think of this capacitor as equivalent to two capacitors in parallel?) (b) How much energy is stored in the capacitor? (c) If we remove the Plexiglas but change nothing else, how much energy will be stored in the capacitor?

24.65 •• CP CALC A metallic circular plate with radius r is fixed to a tabletop. An identical circular plate supported from above by a cable is fixed in place a distance d above the first plate. Assume that d is much smaller than r . The two plates are attached by wires to a battery that supplies voltage V . (a) What is the tension in the cable? Neglect the weight of the plate. (b) The upper plate is slowly raised to a new height $2d$. Determine the work done by the cable by integrating $\int_d^{2d} F(z) dz$, where $F(z)$ is the cable tension when the plates are separated by a distance z . (c) Compute the energy stored in the electric field before the top plate was raised. (d) Compute the energy stored in the electric field after the top plate was raised. (e) Is the work done by the cable equal to the change in the stored electrical energy? If not, why not?

24.66 •• A fuel gauge uses a capacitor to determine the height of the fuel in a tank. The effective dielectric constant K_{eff} changes from a value of 1 when the tank is empty to a value of K , the dielectric constant of the fuel, when the tank is full. The appropriate electronic circuitry can determine the effective dielectric constant of the combined air and fuel between the capacitor plates. Each of the two rectangular plates has a width w and a length L (Fig. P24.66). The height of the fuel between the plates is h .

Figure P24.66



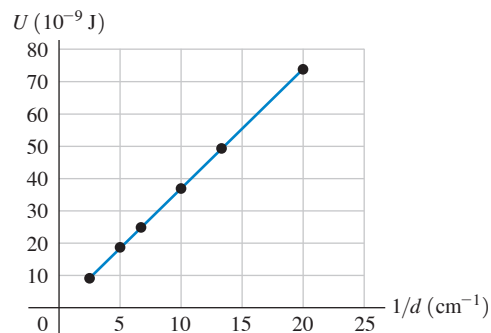
You can ignore any fringing effects. (a) Derive an expression for K_{eff} as a function of h . (b) What is the effective dielectric constant for a tank $\frac{1}{4}$ full, $\frac{1}{2}$ full, and $\frac{3}{4}$ full if the fuel is gasoline ($K = 1.95$)? (c) Repeat part (b) for methanol ($K = 33.0$). (d) For which fuel is this fuel gauge more practical?

24.67 •• DATA Your electronics company has several identical capacitors with capacitance C_1 and several others with capacitance C_2 . You must determine the values of C_1 and C_2 but don't have access to C_1 and C_2 individually. Instead, you have a network with C_1 and C_2 connected in series and a network with C_1 and C_2 connected in parallel. You have a 200.0 V battery and instrumentation that measures the total energy supplied by the battery when it is connected to the network. When the parallel combination is connected to the battery, 0.180 J of energy is stored in the network. When the series combination is connected, 0.0400 J of energy is stored. You are told that C_1 is greater than C_2 . (a) Calculate C_1 and C_2 . (b) For the series combination, does C_1 or C_2 store more charge, or are the values equal? Does C_1 or C_2 store more energy, or are the values equal? (c) Repeat part (b) for the parallel combination.

24.68 •• DATA You are designing capacitors for various applications. For one application, you want the maximum possible stored energy. For another, you want the maximum stored charge. For a third application, you want the capacitor to withstand a large applied voltage without dielectric breakdown. You start with an air-filled parallel-plate capacitor that has $C_0 = 6.00$ pF and a plate separation of 2.50 mm. You then consider the use of each of the dielectric materials listed in Table 24.2. In each application, the dielectric will fill the volume between the plates, and the electric field between the plates will be 50% of the dielectric strength given in the table. (a) For each of the five materials given in the table, calculate the energy stored in the capacitor. Which dielectric allows the maximum stored energy? (b) For each material, what is the charge Q stored on each plate of the capacitor? (c) For each material, what is the voltage applied across the capacitor? (d) Is one dielectric material in the table your best choice for all three applications?

24.69 •• DATA You are conducting experiments with an air-filled parallel-plate capacitor. You connect the capacitor to a battery with voltage 24.0 V. Initially the separation d between the plates is 0.0500 cm. In one experiment, you leave the battery connected to the capacitor, increase the separation between the plates, and measure the energy stored in the capacitor for each value of d . In a second experiment, you make the same measurements but disconnect the battery before you change the plate separation. One set of your data is given in Fig. P24.69, where you have plotted the stored energy U versus $1/d$. (a) For which experiment does this data set apply: the first (battery remains connected) or the second (battery disconnected before d is changed)? Explain. (b) Use the data plotted in Fig. P24.69 to calculate the area A of each plate. (c) For which case, the battery connected or the battery disconnected, is there more energy stored in the capacitor when $d = 0.400$ cm? Explain.

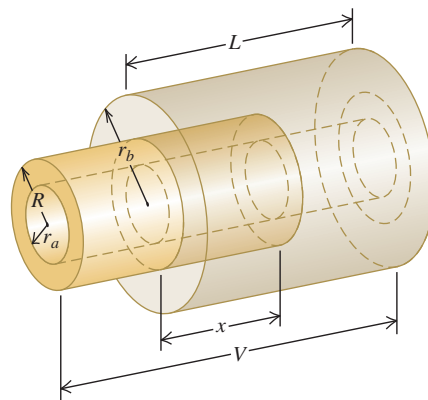
Figure P24.69



CHALLENGE PROBLEMS

24.70 •• A thin-walled, hollow, conducting cylinder with radius r_b is concentric with a solid conducting cylinder with radius $r_a < r_b$. Each has length L . The two cylinders are attached by conducting wires to the terminals of a battery that supplies potential V . A solid cylindrical shell, with inner radius r_a and outer radius $R < r_b$, made of a material with dielectric constant K , slides between the conducting cylinders, as shown in Fig. P24.70. By changing the insertion distance x , we can alter the capacitance seen by the battery and therefore alter the amount of charge stored in this device. (a) Determine the capacitance as a function of x . (b) If $L = 10.0$ cm, $r_a = 1.00$ cm, $r_b = 4.00$ cm, $R = 3.00$ cm, and $K = 3.21$, what is the capacitance when $x = 0$? (c) What is the capacitance when $x = L$? (d) What value of x results in 6.00 nC of charge on the positively charged cylinder plate when $V = 1.00$ kV?

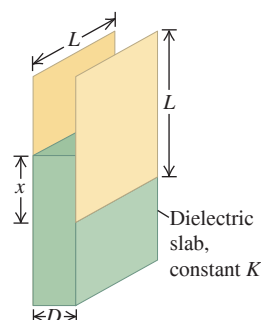
Figure P24.70



24.71 ••• **CALC** Two conducting plates with area A are separated by distance d . Between the plates is a material with a dielectric constant that varies linearly from a value of unity next to one plate to K next to the other plate. (a) What is the capacitance of this device? [Hint: We can envision this as a continuum of capacitors with differential plate separation connected in series. The reciprocal of the capacitance of a differential slice is then $dx/(K(x)\epsilon_0 A)$, where $K(x)$ is the dielectric constant specific to that locale.] (b) Show that this result matches the expected result when $K \rightarrow 1$.

24.72 ••• Two square conducting plates with sides of length L are separated by a distance D . A dielectric slab with constant K with dimensions $L \times L \times D$ is inserted a distance x into the space between the plates, as shown in **Fig. P24.72**. (a) Find the capacitance C of this system. (b) Suppose that the capacitor is connected to a battery that maintains a constant potential difference V between the plates. If the dielectric slab is inserted an additional distance dx into the space between the plates, show that the change in stored energy is

Figure P24.72



$$dU = + \frac{(K - 1)\epsilon_0 V^2 L}{2D} dx$$

(c) Suppose that before the slab is moved by dx , the plates are disconnected from the battery, so that the charges on the plates remain constant. Determine the magnitude of the charge on each plate, and then show that when the slab is moved dx farther into the space between the plates, the stored energy changes by an amount that is the *negative* of the expression for dU given in part (b). (d) If F is the force exerted on the slab by the charges on the plates, then dU should equal the work done *against* this force to move the slab a distance dx . Thus $dU = -Fdx$. Show that applying this expression to the result of part (b) suggests that the electric force on the slab pushes it *out* of the capacitor, while the result of part (c) suggests that the force pulls the slab *into* the capacitor. (e) Figure 24.16 shows that the force in fact pulls the slab into the capacitor. Explain why the result of part (b) gives an incorrect answer for the direction of this force, and calculate the magnitude of the force. (This method does not require knowledge of the nature of the fringing field.)

MCAT-STYLE PASSAGE PROBLEMS

BIO The Electric Egg. Upon fertilization, the eggs of many species undergo a rapid change in potential difference across their outer membrane. This change affects the physiological development of the eggs. The potential difference across the membrane is called the *membrane potential*, V_m , which is the potential inside the membrane minus the potential outside it. The membrane potential arises when enzymes use the energy available in ATP to expel three sodium ions (Na^+) actively and accumulate two potassium ions (K^+) inside the membrane—making the interior less positively charged than the exterior. For a sea urchin egg, V_m is about -70 mV; that is, the potential inside is 70 mV less than that outside. The egg membrane behaves as a capacitor with a capacitance of about $1 \mu\text{F}/\text{cm}^2$. The membrane of the unfertilized egg is *selectively permeable* to K^+ ; that is, K^+ can readily pass through certain channels in the membrane, but other ions cannot. When a sea urchin egg is fertilized, Na^+ channels in the membrane open, Na^+ enters the egg, and V_m rapidly increases to $+30$ mV, where it remains for several minutes. The concentration of Na^+ is about 30 mmol/L in the egg's interior but 450 mmol/L in the surrounding seawater. The K^+ concentration is about 200 mmol/L inside but 10 mmol/L outside. A useful constant that connects electrical and chemical units is the *Faraday number*, which has a value of approximately 10^5 C/mol; that is, Avogadro's number (a mole) of monovalent ions, such as Na^+ or K^+ , carries a charge of 10^5 C.

24.73 How many moles of Na^+ must move per unit area of membrane to change V_m from -70 mV to $+30$ mV, if we assume that the membrane behaves purely as a capacitor? (a) 10^{-4} mol/ cm^2 ; (b) 10^{-9} mol/ cm^2 ; (c) 10^{-12} mol/ cm^2 ; (d) 10^{-14} mol/ cm^2 .

24.74 Suppose that the egg has a diameter of $200 \mu\text{m}$. What fractional change in the internal Na^+ concentration results from the fertilization-induced change in V_m ? Assume that Na^+ ions are distributed throughout the cell volume. The concentration increases by (a) 1 part in 10^4 ; (b) 1 part in 10^5 ; (c) 1 part in 10^6 ; (d) 1 part in 10^7 .

24.75 Suppose that the change in V_m was caused by the entry of Ca^{2+} instead of Na^+ . How many Ca^{2+} ions would have to enter the cell per unit membrane to produce the change? (a) Half as many as for Na^+ ; (b) the same as for Na^+ ; (c) twice as many as for Na^+ ; (d) cannot say without knowing the inside and outside concentrations of Ca^{2+} .

24.76 What is the minimum amount of work that must be done by the cell to restore V_m to -70 mV? (a) 3 mJ; (b) 3 μJ ; (c) 3 nJ; (d) 3 pJ.

ANSWERS

Chapter Opening Question ?

(iv) Equation (24.9) shows that the energy stored in a capacitor with capacitance C and charge Q is $U = Q^2/2C$. If Q is doubled, the stored energy increases by a factor of $2^2 = 4$. Note that if the value of Q is too great, the electric-field magnitude inside the capacitor will exceed the dielectric strength of the material between the plates and dielectric breakdown will occur (see Section 24.4). This puts a practical limit on the amount of energy that can be stored.

Key Example VARIATION Problems

VP24.4.1 (a) 6.58×10^{-9} F (b) 5.85 kV

VP24.4.2 (a) 4.80 mm (b) $Q = 2.49 \times 10^{-5}$ C, $E = 6.25 \times 10^5$ V/m

VP24.4.3 (a) 8.33 cm (b) 4.44×10^{-8} C

(c) $\sigma_{\text{inner}} = +5.09 \times 10^{-7}$ C/ m^2 , $\sigma_{\text{outer}} = -4.36 \times 10^{-7}$ C/ m^2

VP24.4.4 (a) 2.24 (b) 125 V; inner conductor

VP24.9.1 (a) 0.625 μF (b) 8.50 μF (c) 2.06 μF (d) 5.71 μF

VP24.9.2 (a) $U_1 = 0.0675$ J, $U_2 = 0.0169$ J; C_1 (b) $U_1 = 0.0117$ J, $U_2 = 0.0469$ J; C_2

VP24.9.3 (a) series 0.0115 J, parallel 0.0578 J; parallel (c) series 0.0733 J, parallel 0.0145 J; series

VP24.9.4 (a) 5.10×10^{-6} C/ m^2 (b) 5.76×10^5 V/m (c) 1.47 J/ m^3 (d) 5.04×10^{-3} J

VP24.11.1 (a) 1.36 nF (b) 5.43×10^{-6} C (c) 1.60 (d) 2.17 nF (e) 2.04×10^{-6} C

VP24.11.2 (a) 2.36 nF (b) 8.26×10^{-6} C (c) 5.90 nF (d) 2.07×10^{-5} C (e) 1.24×10^{-5} C

VP24.11.3 (a) 17.6 J (b) 7.56 J

VP24.11.4 (a) 6.30 J (b) 18.0 J

Bridging Problem

(a) 0

(b) $Q^2/32\pi^2\epsilon_0 r^4$

(c) $Q^2/8\pi\epsilon_0 R$

(d) $Q^2/8\pi\epsilon_0 R$

(e) $C = 4\pi\epsilon_0 R$

? In a flashlight, how does the amount of current that flows out of the bulb compare to the amount that flows into the bulb? (i) Current out is less than current in; (ii) current out is greater than current in; (iii) current out equals current in; (iv) the answer depends on the brightness of the bulb.



25 Current, Resistance, and Electromotive Force

LEARNING OUTCOMES

In this chapter, you'll learn...

- 25.1 The meaning of electric current, and how charges move in a conductor.
- 25.2 What is meant by the resistivity and conductivity of a substance.
- 25.3 How to calculate the resistance of a conductor from its dimensions and its resistivity.
- 25.4 How an electromotive force (emf) makes it possible for current to flow in a circuit.
- 25.5 How to do calculations involving energy and power in circuits.
- 25.6 How to use a simple model to understand the flow of current in metals.

You'll need to review...

- 17.7 Thermal conductivity.
- 23.2 Voltmeters, electric field, and electric potential.
- 24.4 Dielectric breakdown in insulators.

In the past four chapters we studied the interactions of electric charges *at rest*; now we're ready to study charges *in motion*. An *electric current* consists of charges in motion from one region to another. If the charges follow a conducting path that forms a closed loop, the path is called an *electric circuit*.

Fundamentally, electric circuits are a means for conveying *energy* from one place to another. As charged particles move within a circuit, electric potential energy is transferred from a source (such as a battery or generator) to a device in which that energy is either stored or converted to another form: into sound in a stereo system or into heat and light in a toaster or light bulb. Electric circuits are useful because they allow energy to be transported without any moving parts (other than the moving charged particles themselves). They are at the heart of computers, television transmitters and receivers, and household and industrial power distribution systems. Your nervous system is a specialized electric circuit that carries vital signals from one part of your body to another.

In Chapter 26 we'll see how to analyze electric circuits and examine some practical applications of circuits. To prepare you for that, in this chapter we'll examine the basic properties of electric currents. We'll begin by describing the nature of electric conductors and considering how they are affected by temperature. We'll learn why a short, fat, cold copper wire is a better conductor than a long, skinny, hot steel wire. We'll study the properties of batteries and see how they cause current and energy transfer in a circuit. In this analysis we'll use the concepts of current, potential difference (or voltage), resistance, and electromotive force. Finally, we'll look at electric current in a material from a microscopic viewpoint.

25.1 CURRENT

A **current** is any motion of charge from one region to another. In this section we'll discuss currents in conducting materials. The vast majority of technological applications of charges in motion involve currents of this kind.

In electrostatic situations (discussed in Chapters 21 through 24) the electric field is zero everywhere within the conductor, and there is *no* current. However, this does not mean that all charges within the conductor are at rest. In an ordinary metal such as copper or aluminum, some of the electrons are free to move within the conducting material. These free electrons move randomly in all directions, somewhat like the molecules of a gas but with much greater speeds, of the order of 10^6 m/s. The electrons nonetheless do not escape from the conducting material, because they are attracted to the positive ions of the material. The motion of the electrons is random, so there is no *net* flow of charge in any direction and hence no current.

Now consider what happens if a constant, steady electric field $\vec{E}$ is established inside a conductor. (We'll see later how this can be done.) A charged particle (such as a free electron) inside the conducting material is then subjected to a steady force $\vec{F} = q\vec{E}$. If the charged particle were moving in *vacuum*, this steady force would cause a steady acceleration in the direction of $\vec{F}$, and after a time the charged particle would be moving in that direction at high speed. But a charged particle moving in a *conductor* undergoes frequent collisions with the massive, nearly stationary ions of the material. In each such collision the particle's direction of motion undergoes a random change. The net effect of the electric field $\vec{E}$ is that in addition to the random motion of the charged particles within the conductor, there is also a very slow net motion or *drift* of the moving charged particles as a group in the direction of the electric force $\vec{F} = q\vec{E}$ (Fig. 25.1). This motion is described in terms of the **drift velocity** $\vec{v}_d$ of the particles. As a result, there is a net current in the conductor.

While the random motion of the electrons has a very fast average speed of about 10^6 m/s, the drift speed is very slow, often on the order of 10^{-4} m/s. Given that the electrons move so slowly, you may wonder why the light comes on immediately when you turn on the switch of a flashlight. The reason is that the electric field is set up in the wire with a speed approaching the speed of light, and electrons start to move all along the wire at very nearly the same time. The time that it takes any individual electron to get from the switch to the light bulb isn't really relevant. A good analogy is a group of soldiers standing at attention when the sergeant orders them to start marching; the order reaches the soldiers' ears at the speed of sound, which is much faster than their marching speed, so all the soldiers start to march essentially in unison.

The Direction of Current Flow

The drift of moving charges through a conductor can be interpreted in terms of work and energy. The electric field $\vec{E}$ does work on the moving charges. The resulting kinetic energy is transferred to the material of the conductor by means of collisions with the ions, which vibrate about their equilibrium positions in the crystalline structure of the conductor. This energy transfer increases the average vibrational energy of the ions and therefore the temperature of the material. Thus much of the work done by the electric field goes into heating the conductor, *not* into making the moving charges move ever faster and faster. This heating is sometimes useful, as in an electric toaster, but in many situations is simply an unavoidable by-product of current flow.

In different current-carrying materials, the charges of the moving particles may be positive or negative. In metals the moving charges are always (negative) electrons, while in an ionized gas (plasma) or an ionic solution the moving charges may include both electrons and positively charged ions. In a semiconductor material such as germanium or silicon, conduction is partly by electrons and partly by motion of *vacancies*, also known as *holes*; these are sites of missing electrons and act like positive charges.

Figure 25.2 (next page) shows segments of two different current-carrying materials. In Fig. 25.2a the moving charges are positive, the electric force is in the same direction as

Figure 25.1 If there is no electric field inside a conductor, an electron moves randomly from point P_1 to point P_2 in a time Δt . If an electric field $\vec{E}$ is present, the electric force $\vec{F} = q\vec{E}$ imposes a small drift (greatly exaggerated here) that takes the electron to point P_2' , a distance $v_d \Delta t$ from P_2 in the direction of the force.

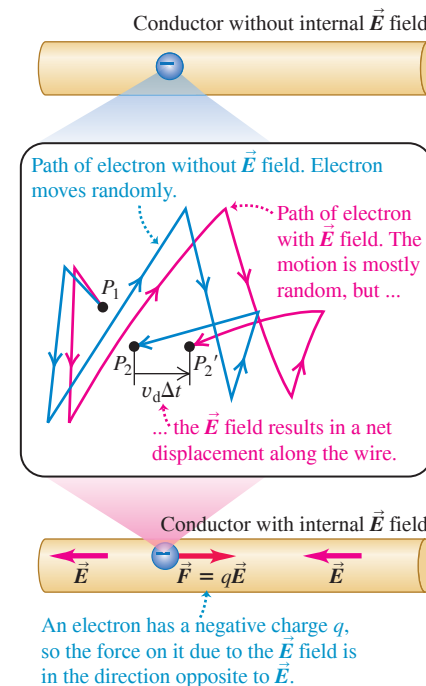
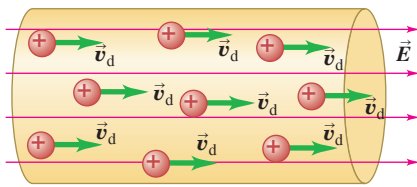


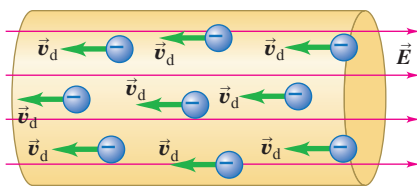
Figure 25.2 The same current is produced by (a) positive charges moving in the direction of the electric field $\vec{E}$ or (b) the same number of negative charges moving at the same speed in the direction opposite to $\vec{E}$.

(a)



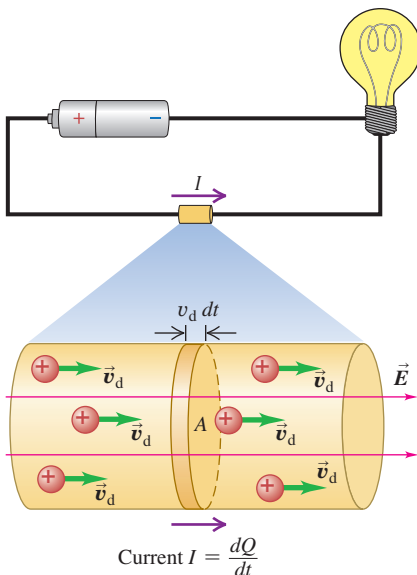
A **conventional current** is treated as a flow of positive charges, regardless of whether the free charges in the conductor are positive, negative, or both.

(b)



In a metallic conductor, the moving charges are electrons — but the **current** still points in the direction positive charges would flow.

Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).



$\vec{E}$, and the drift velocity $\vec{v}_d$ is from left to right. In Fig. 25.2b the charges are negative, the electric force is opposite to $\vec{E}$, and the drift velocity $\vec{v}_d$ is from right to left. In both cases there is a net flow of positive charge from left to right, and positive charges end up to the right of negative ones. We *define* the current, denoted by I , to be in the direction in which there is a flow of *positive* charge. Thus we describe currents as though they consisted entirely of positive charge flow, even in cases in which we know that the actual current is due to electrons. Hence the current is to the right in both Figs. 25.2a and 25.2b. This choice or convention for the direction of current flow is called **conventional current**. While the direction of the conventional current is *not* necessarily the same as the direction in which charged particles are actually moving, we'll find that the sign of the moving charges is of little importance in analyzing electric circuits.

Figure 25.3 shows a segment of a conductor in which a current is flowing. We consider the moving charges to be *positive*, so they are moving in the same direction as the current. We define the current through the cross-sectional area A to be *the net charge flowing through the area per unit time*. Thus, if a net charge dQ flows through an area in a time dt , the current I through the area is

$$I = \frac{dQ}{dt} \quad (\text{definition of current}) \quad (25.1)$$

CAUTION **Current is not a vector** Although we refer to the *direction* of a current, current as defined by Eq. (25.1) is *not* a vector quantity. In a current-carrying wire, the current is always along the length of the wire, regardless of whether the wire is straight or curved. No single vector could describe motion along a curved path. We'll usually describe the direction of current either in words (as in “the current flows clockwise around the circuit”) or by choosing a current to be positive if it flows in one direction along a conductor and negative if it flows in the other direction. **|**

The SI unit of current is the **ampere**; one ampere is defined to be *one coulomb per second* ($1 \text{ A} = 1 \text{ C/s}$). This unit is named in honor of the French scientist André Marie Ampère (1775–1836). When an ordinary flashlight (D-cell size) is turned on, the current in the flashlight is about 0.5 to 1 A; the current in the wires of a car engine's starter motor is around 200 A. Currents in radio and television circuits are usually expressed in *milliamperes* ($1 \text{ mA} = 10^{-3} \text{ A}$) or *microamperes* ($1 \mu\text{A} = 10^{-6} \text{ A}$), and currents in computer circuits are expressed in *nanoamperes* ($1 \text{ nA} = 10^{-9} \text{ A}$) or *picoamperes* ($1 \text{ pA} = 10^{-12} \text{ A}$).

Current, Drift Velocity, and Current Density

We can express current in terms of the drift velocity of the moving charges. Let's reconsider the situation of Fig. 25.3 of a conductor with cross-sectional area A and an electric field $\vec{E}$ directed from left to right. To begin with, we'll assume that the free charges in the conductor are positive; then the drift velocity is in the same direction as the field.

Suppose there are n moving charged particles per unit volume. We call n the **concentration** of particles; its SI unit is m^{-3} . Assume that all the particles move with the same drift velocity with magnitude v_d . In a time interval dt , each particle moves a distance $v_d dt$. The particles that flow out of the right end of the shaded cylinder with length $v_d dt$ during dt are the particles that were within this cylinder at the beginning of the interval dt . The volume of the cylinder is $Av_d dt$, and the number of particles within it is $nAv_d dt$. If each particle has a charge q , the charge dQ that flows out of the end of the cylinder during time dt is

$$dQ = q(nAv_d dt) = nqv_d A dt$$

and the current is

$$I = \frac{dQ}{dt} = nqv_d A$$

The current *per unit cross-sectional area* is called the **current density** J :

$$J = \frac{I}{A} = nqv_d$$

The units of current density are amperes per square meter (A/m^2).

If the moving charges are negative rather than positive, as in Fig. 25.2b, the drift velocity is opposite to $\vec{E}$. But the *current* is still in the same direction as $\vec{E}$ at each point in the conductor. Hence current I and current density J don't depend on the sign of the charge, and so we replace the charge q by its absolute value $|q|$:

$$\text{Current through an area} \quad I = \frac{dQ}{dt} = n|q|v_d A \quad (25.2)$$

Rate at which charge flows through area
 Drift speed
 Cross-sectional area
 Concentration of moving charged particles
 Charge per particle

$$J = \frac{I}{A} = n|q|v_d \quad (\text{current density}) \quad (25.3)$$

The current in a conductor is the product of the concentration of moving charged particles, the magnitude of charge of each such particle, the magnitude of the drift velocity, and the cross-sectional area of the conductor.

We can also define a *vector* current density $\vec{J}$ that includes the direction of the drift velocity:

$$\text{Vector current density} \quad \vec{J} = nq\vec{v}_d \quad (25.4)$$

Drift velocity
 Concentration of moving charged particles
 Charge per particle

There are *no* absolute value signs in Eq. (25.4). If q is positive, $\vec{v}_d$ is in the same direction as $\vec{E}$; if q is negative, $\vec{v}_d$ is opposite to $\vec{E}$. In either case, $\vec{J}$ is in the same direction as $\vec{E}$. Equation (25.3) gives the *magnitude* J of the vector current density $\vec{J}$.

CAUTION Current density vs. current Current density $\vec{J}$ is a vector, but current I is not. The difference is that the current density $\vec{J}$ describes how charges flow at a certain point, and the vector's direction tells you about the direction of the flow at that point. By contrast, the current I describes how charges flow through an extended object such as a wire. For example, I has the same value at all points in the circuit of Fig. 25.3, but $\vec{J}$ does not: $\vec{J}$ is directed downward in the left-hand side of the loop and upward in the right-hand side. The magnitude of $\vec{J}$ can also vary around a circuit. In Fig. 25.3 $J = I/A$ is less in the battery (which has a large cross-sectional area A) than in the wires (which have a small A). ■

In general, a conductor may contain several different kinds of moving charged particles having charges $q_1, q_2, \dots$, concentrations $n_1, n_2, \dots$, and drift velocities with magnitudes $v_{d1}, v_{d2}, \dots$. An example is current flow in an ionic solution (Fig. 25.4). In a sodium chloride solution, current can be carried by both positive sodium ions and negative chlorine ions; the total current I is found by adding up the currents due to each kind of charged particle, from Eq. (25.2). Likewise, the total vector current density $\vec{J}$ is found by using Eq. (25.4) for each kind of charged particle and adding the results.

We'll see in Section 25.4 that it is possible to have a current that is *steady* (constant in time) only if the conducting material forms a closed loop, called a *complete circuit*. In such a steady situation, the total charge in every segment of the conductor is constant. Hence the rate of flow of charge *out* at one end of a segment at any instant equals the rate of flow of charge *in* at the other end of the segment, and *the current is the same at all cross sections of the circuit*. We'll use this observation when we analyze electric circuits later in this chapter.

Figure 25.4 Part of the electric circuit that includes this light bulb passes through a beaker with a solution of sodium chloride. The current in the solution is carried by both positive charges (Na^+ ions) and negative charges (Cl^- ions).



In many simple circuits, such as flashlights or cordless electric drills, the direction of the current is always the same; this is called *direct current*. But home appliances such as toasters, refrigerators, and televisions use *alternating current*, in which the current continuously changes direction. In this chapter we'll consider direct current only. Alternating current has many special features worthy of detailed study, which we'll examine in Chapter 31.

EXAMPLE 25.1 Current density and drift velocity in a wire

An 18 gauge copper wire (the size usually used for lamp cords), with a diameter of 1.02 mm, carries a constant current of 1.67 A to a 200 W lamp. The free-electron density in the wire is 8.5×10^{28} per cubic meter. Find (a) the current density and (b) the drift speed.

IDENTIFY and SET UP This problem uses the relationships among current I , current density J , and drift speed v_d . We are given I and the wire diameter d , so we use Eq. (25.3) to find J . We use Eq. (25.3) again to find v_d from J and the known electron density n .

EXECUTE (a) The cross-sectional area is

$$A = \frac{\pi d^2}{4} = \frac{\pi (1.02 \times 10^{-3} \text{ m})^2}{4} = 8.17 \times 10^{-7} \text{ m}^2$$

The magnitude of the current density is then

$$J = \frac{I}{A} = \frac{1.67 \text{ A}}{8.17 \times 10^{-7} \text{ m}^2} = 2.04 \times 10^6 \text{ A/m}^2$$

(b) From Eq. (25.3) for the drift velocity magnitude v_d , we find

$$v_d = \frac{J}{n|q|} = \frac{2.04 \times 10^6 \text{ A/m}^2}{(8.5 \times 10^{28} \text{ m}^{-3})(1.60 \times 10^{-19} \text{ C})} = 1.5 \times 10^{-4} \text{ m/s} = 0.15 \text{ mm/s}$$

EVALUATE At this speed an electron would require 6700 s (almost 2 h) to travel 1 m along this wire. The speeds of random motion of the electrons are roughly 10^6 m/s, around 10^{10} times the drift speed. Picture the electrons as bouncing around frantically, with a very slow drift!

KEYCONCEPT Current is the rate at which electric charge flows through an area, and current density is the current per unit area. Current density is proportional to the concentration of moving charged particles, the charge per particle, and the drift speed of the particles.

TEST YOUR UNDERSTANDING OF SECTION 25.1 Suppose we replaced the wire in Example 25.1 with 12 gauge copper wire, which has twice the diameter of 18 gauge wire. If the current remains the same, what effect would this have on the drift speed v_d ? (i) None— v_d would be unchanged; (ii) v_d would be twice as great; (iii) v_d would be four times greater; (iv) v_d would be half as great; (v) v_d would be one-fourth as great.

ANSWER

(v) Doubling the diameter increases the cross-sectional area A by a factor of 4. Hence the current-density magnitude $J = I/A$ is reduced to $\frac{1}{4}$ of the value in Example 25.1, and the magnitude of the drift velocity $v_d = J/n|q|$ is reduced by the same factor. The new magnitude is $v_d = (0.15 \text{ mm/s})/4 = 0.038 \text{ mm/s}$. This behavior is the same as that of an incompressible fluid, which slows down when it moves from a narrow pipe to a broader one (see Section 14.4).

25.2 RESISTIVITY

The current density $\vec{J}$ in a conductor depends on the electric field $\vec{E}$ and on the properties of the material. In general, this dependence can be quite complex. But for some materials, especially metals, at a given temperature, $\vec{J}$ is nearly *directly proportional* to $\vec{E}$, and the ratio of the magnitudes of E and J is constant. This relationship, called Ohm's law, was discovered in 1826 by the German physicist Georg Simon Ohm (1787–1854). The word “law” should actually be in quotation marks, since **Ohm's law**, like the ideal-gas equation and Hooke's law, is an *idealized model* that describes the behavior of some materials quite well but is not a general description of *all* matter. In the following discussion we'll assume that Ohm's law is valid, even though there are many situations in which it is not.

We define the **resistivity** ρ of a material as

$$\text{Resistivity of a material } \rho = \frac{E}{J} \quad (25.5)$$

$\swarrow$ Magnitude of electric field in material
 $\nwarrow$ Magnitude of current density caused by electric field

TABLE 25.1 Resistivities at Room Temperature (20°C)

Substance			ρ ($\Omega \cdot \text{m}$)	Substance		ρ ($\Omega \cdot \text{m}$)
Conductors				Semiconductors		
Metals	Silver		1.47×10^{-8}	Pure carbon (graphite)		3.5×10^{-5}
	Copper		1.72×10^{-8}	Pure germanium		0.60
	Gold		2.44×10^{-8}	Pure silicon		2300
	Aluminum		2.75×10^{-8}	Insulators		
	Tungsten		5.25×10^{-8}		Amber	5×10^{14}
	Steel		20×10^{-8}		Glass	$10^{10}\text{--}10^{14}$
	Lead		22×10^{-8}		Lucite	$>10^{13}$
Alloys	Mercury		95×10^{-8}	Mica	$10^{11}\text{--}10^{15}$	
	Manganin (Cu 84%, Mn 12%, Ni 4%)		44×10^{-8}	Quartz (fused)	75×10^{16}	
	Constantan (Cu 60%, Ni 40%)		49×10^{-8}	Sulfur	10^{15}	
	Nichrome		100×10^{-8}	Teflon	$>10^{13}$	
				Wood	$10^8\text{--}10^{11}$	

The greater the resistivity, the greater the field needed to cause a given current density, or the smaller the current density caused by a given field. From Eq. (25.5) the units of ρ are $(\text{V/m})/(\text{A/m}^2) = \text{V} \cdot \text{m}/\text{A}$. As we'll discuss in Section 25.3, 1 V/A is called one *ohm* (1 Ω ; the Greek letter Ω , omega, is alliterative with “ohm”). So the SI units for ρ are $\Omega \cdot \text{m}$ (ohm-meters). **Table 25.1** lists some representative values of resistivity. A perfect conductor would have zero resistivity, and a perfect insulator would have an infinite resistivity. Metals and alloys have the smallest resistivities and are the best conductors. The resistivities of insulators are greater than those of the metals by an enormous factor, on the order of 10^{22} .

The reciprocal of resistivity is **conductivity**. Its units are $(\Omega \cdot \text{m})^{-1}$. Good conductors of electricity have larger conductivity than insulators. Conductivity is the direct electrical analog of thermal conductivity. Comparing Table 25.1 with Table 17.5 (Thermal Conductivities), we note that good electrical conductors, such as metals, are usually also good conductors of heat. Poor electrical conductors, such as ceramic and plastic materials, are also poor thermal conductors. In a metal the free electrons that carry charge in electrical conduction also provide the principal mechanism for heat conduction, so we should expect a correlation between electrical and thermal conductivity. Because of the enormous difference in conductivity between electrical conductors and insulators, it is easy to confine electric currents to well-defined paths or circuits (**Fig. 25.5**). The variation in *thermal* conductivity is much less, only a factor of 10^3 or so, and it is usually impossible to confine heat currents to that extent.

Semiconductors have resistivities intermediate between those of metals and those of insulators. These materials are important because of the way their resistivities are affected by temperature and by small amounts of impurities.

A material that obeys Ohm's law reasonably well is called an *ohmic* conductor or a *linear* conductor. For such materials, at a given temperature, ρ is a *constant* that does not depend on the value of E . Many materials show substantial departures from Ohm's-law behavior; they are *nonohmic*, or *nonlinear*. In these materials, J depends on E in a more complicated manner.

Analogies with fluid flow can be a big help in developing intuition about electric current and circuits. For example, in the making of wine or maple syrup, the product is sometimes filtered to remove sediments. A pump forces the fluid through the filter under pressure; if the flow rate (analogous to J) is proportional to the pressure difference between the upstream and downstream sides (analogous to E), the behavior is analogous to Ohm's law.

Resistivity and Temperature

The resistivity of a *metallic* conductor nearly always increases with increasing temperature, as shown in **Fig. 25.6a** (next page). As temperature increases, the ions of the conductor vibrate with greater amplitude, making it more likely that a moving electron will

Figure 25.5 The copper “wires,” or traces, on this circuit board are printed directly onto the surface of the dark-colored insulating board. Even though the traces are very close to each other (only about a millimeter apart), the board has such a high resistivity (and low conductivity) that essentially no current can flow between the traces.

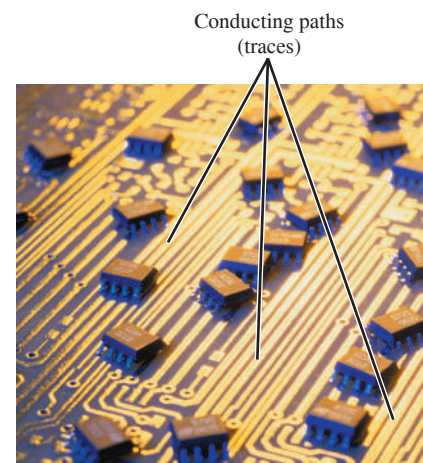
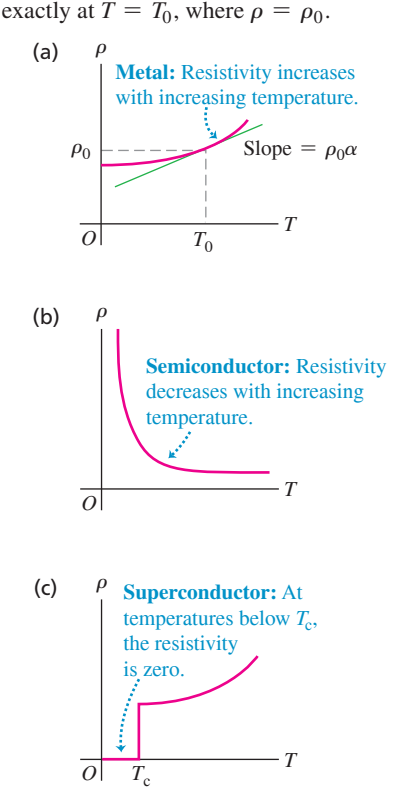


Figure 25.6 Variation of resistivity ρ with absolute temperature T for (a) a normal metal, (b) a semiconductor, and (c) a superconductor. In (a) the linear approximation to ρ as a function of T is shown as a green line; the approximation agrees exactly at $T = T_0$, where $\rho = \rho_0$.



BIO APPLICATION Resistivity and Nerve Conduction This false-color image from an electron microscope shows a cross section through a nerve fiber about $1\text{ }\mu\text{m}$ (10^{-6} m) in diameter. A layer of an insulating fatty substance called myelin is wrapped around the conductive material of the axon. The resistivity of myelin is much greater than that of the axon, so an electric signal traveling along the nerve fiber remains confined to the axon. This makes it possible for a signal to travel much more rapidly than if the myelin were absent.



collide with an ion as in Fig. 25.1; this impedes the drift of electrons through the conductor and hence reduces the current. Over a small temperature range (up to $100\text{ }^\circ\text{C}$ or so), the resistivity of a metal can be represented approximately by the equation

Temperature dependence of resistivity:

$$\rho(T) = \rho_0[1 + \alpha(T - T_0)] \quad (25.6)$$

Resistivity at temperature T
 Resistivity at reference temperature T_0
 Temperature coefficient of resistivity

The reference temperature T_0 is often taken as 0°C or 20°C ; the temperature T may be higher or lower than T_0 . The factor α is called the **temperature coefficient of resistivity**. Some representative values are given in **Table 25.2**. The resistivity of the alloy manganin is practically independent of temperature.

The resistivity of graphite (a nonmetal) *decreases* with increasing temperature, since at higher temperatures, more electrons “shake loose” from the atoms and become mobile; hence the temperature coefficient of resistivity of graphite is negative. The same behavior occurs for semiconductors (Fig. 25.6b). Measuring the resistivity of a small semiconductor crystal is therefore a sensitive measure of temperature; this is the principle of a type of thermometer called a *thermistor*.

Some materials, including several metallic alloys and oxides, show a phenomenon called *superconductivity*. As the temperature decreases, the resistivity at first decreases smoothly, like that of any metal. But then at a certain critical temperature T_c a phase transition occurs and the resistivity suddenly drops to zero, as shown in Fig. 25.6c. Once a current has been established in a superconducting ring, it continues indefinitely without the presence of any driving field.

Superconductivity was discovered in 1911 by the Dutch physicist Heike Kamerlingh Onnes (1853–1926). He discovered that at very low temperatures, below 4.2 K , the resistivity of mercury suddenly dropped to zero. For the next 75 years, the highest T_c attained was about 20 K . This meant that superconductivity occurred only when the material was cooled by using expensive liquid helium, with a boiling-point temperature of 4.2 K , or explosive liquid hydrogen, with a boiling point of 20.3 K . But in 1986 Karl Müller and Johannes Bednorz discovered an oxide of barium, lanthanum, and copper with a T_c of nearly 40 K , and the race was on to develop “high-temperature” superconducting materials.

By 1987 a complex oxide of yttrium, copper, and barium had been found that has a value of T_c well above the 77 K boiling temperature of liquid nitrogen, a refrigerant that is both inexpensive and safe. The current (2018) record for T_c is 203.5 K , and it may be possible to fabricate materials that are superconductors at room temperature. The implications of these discoveries for power-distribution systems, computer design, and transportation are enormous. Meanwhile, superconducting electromagnets cooled by liquid helium are used in particle accelerators and some experimental magnetic-levitation railroads. Superconductors have other exotic properties that require an understanding of magnetism to explore; we’ll discuss these further in Chapter 29.

TABLE 25.2 Temperature Coefficients of Resistivity (Approximate Values Near Room Temperature)

Material	$\alpha\text{ }[^\circ\text{C}^{-1}]$	Material	$\alpha\text{ }[^\circ\text{C}^{-1}]$
Aluminum	0.0039	Lead	0.0043
Brass	0.0020	Manganin	0.00000
Carbon (graphite)	−0.0005	Mercury	0.00088
Constantan	0.000008	Nichrome	0.0004
Copper	0.00393	Silver	0.0038
Iron	0.0050	Tungsten	0.0045

TEST YOUR UNDERSTANDING OF SECTION 25.2 You maintain a constant electric field inside a piece of semiconductor while lowering the semiconductor's temperature. What happens to the current density in the semiconductor? (i) It increases; (ii) it decreases; (iii) it remains the same.

ANSWER

decreases as the temperature drops and the resistivity increases. From Eq. (25.5), the magnitude of the current density is $J = E/\rho$, so the current density decreases as the resistivity increases. (iii) Figure 25.6b shows that the resistivity ρ of a semiconductor increases as the temperature decreases.

25.3 RESISTANCE

For a conductor with resistivity ρ , the current density $\vec{J}$ at a point where the electric field is $\vec{E}$ is given by Eq. (25.5), which we can write as

$$\vec{E} = \rho \vec{J} \quad (25.7)$$

When Ohm's law is obeyed, ρ is constant and independent of the magnitude of the electric field, so $\vec{E}$ is directly proportional to $\vec{J}$. Often, however, we are more interested in the total current I in a conductor than in $\vec{J}$ and more interested in the potential difference V between the ends of the conductor than in $\vec{E}$. This is so largely because I and V are much easier to measure than are $\vec{J}$ and $\vec{E}$.

Suppose our conductor is a wire with uniform cross-sectional area A and length L , as shown in **Fig. 25.7**. Let V be the potential difference between the higher-potential and lower-potential ends of the conductor, so that V is positive. (Another name for V is the voltage across the conductor.) The *direction* of the current is always from the higher-potential end to the lower-potential end. That's because current in a conductor flows in the direction of $\vec{E}$, no matter what the sign of the moving charges (Fig. 25.2), and because $\vec{E}$ points in the direction of *decreasing* electric potential (see Section 23.2). As the current flows through the potential difference, electric potential energy is lost; this energy is transferred to the ions of the conducting material during collisions.

We can also relate the *value* of the current I to the potential difference between the ends of the conductor. If the magnitudes of the current density $\vec{J}$ and the electric field $\vec{E}$ are uniform throughout the conductor, the total current I is $I = JA$, and the potential difference V between the ends is $V = EL$. We solve these equations for J and E , respectively, and substitute the results into Eq. (25.7):

$$\frac{V}{L} = \frac{\rho I}{A} \quad \text{or} \quad V = \frac{\rho L}{A} I \quad (25.8)$$

This shows that when ρ is constant, the total current I is proportional to the potential difference V .

The ratio of V to I for a particular conductor is called its **resistance** R :

$$R = \frac{V}{I} \quad (25.9)$$

Comparing this definition of R to Eq. (25.8), we see that

$$\text{Resistance of a conductor} \quad R = \frac{\rho L}{A} \quad (25.10)$$

Resistivity of conductor material
Length of conductor
Cross-sectional area of conductor

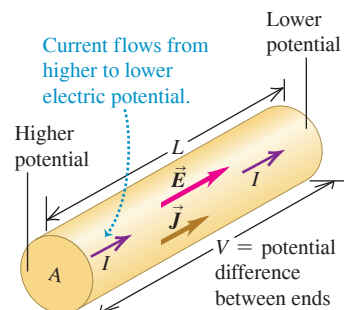
If ρ is constant, as is the case for ohmic materials, then so is R .

The following equation is often called Ohm's law:

$$\text{Relationship among voltage, current, and resistance:} \quad V = IR \quad (25.11)$$

Voltage between ends of conductor
Resistance of conductor
Current in conductor

Figure 25.7 A conductor with uniform cross section. The current density is uniform over any cross section, and the electric field is constant along the length.



However, it’s important to understand that the real content of Ohm’s law is the direct proportionality (for some materials) of V to I or of J to E . Equation (25.9) or (25.11) defines resistance R for *any* conductor, but only when R is constant can we correctly call this relationship Ohm’s law.

Figure 25.8 A long fire hose offers substantial resistance to water flow. To make water pass through the hose rapidly, the upstream end of the hose must be at much higher pressure than the end where the water emerges. In an analogous way, there must be a large potential difference between the ends of a long wire in order to cause a substantial electric current through the wire.



Figure 25.9 This resistor has a resistance of $5.7\text{ k}\Omega$ with an accuracy (tolerance) of $\pm 10\%$.

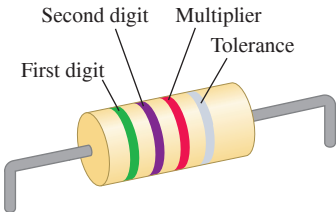


TABLE 25.3 Color Codes for Resistors

Color	Value as Digit	Value as Multiplier
Black	0	1
Brown	1	10
Red	2	10^2
Orange	3	10^3
Yellow	4	10^4
Green	5	10^5
Blue	6	10^6
Violet	7	10^7
Gray	8	10^8
White	9	10^9

Interpreting Resistance

Equation (25.10) shows that the resistance of a wire or other conductor of uniform cross section is directly proportional to its length and inversely proportional to its cross-sectional area. It is also proportional to the resistivity of the material of which the conductor is made.

The flowing-fluid analogy is again useful. In analogy to Eq. (25.10), a narrow water hose offers more resistance to flow than a fat one, and a long hose has more resistance than a short one (Fig. 25.8). We can increase the resistance to flow by stuffing the hose with cotton or sand; this corresponds to increasing the resistivity. The flow rate is approximately proportional to the pressure difference between the ends. Flow rate is analogous to current, and pressure difference is analogous to potential difference (voltage). Let’s not stretch this analogy too far, though; the water flow rate in a pipe is usually *not* proportional to its cross-sectional area (see Section 14.6).

The SI unit of resistance is the **ohm**, equal to one volt per ampere ($1\ \Omega = 1\text{ V/A}$). The *kilohm* ($1\text{ k}\Omega = 10^3\ \Omega$) and the *megohm* ($1\text{ M}\Omega = 10^6\ \Omega$) are also in common use. A 100 m length of 12 gauge copper wire, the size usually used in household wiring, has a resistance at room temperature of about $0.5\ \Omega$. A 100 W, 120 V incandescent light bulb has a resistance (at operating temperature) of $140\ \Omega$. If the same current I flows in both the copper wire and the light bulb, the potential difference $V = IR$ is much greater across the light bulb, and much more potential energy is lost per charge in the light bulb. This lost energy is converted by the light bulb filament into light and heat. You don’t want your household wiring to glow white-hot, so its resistance is kept low by using wire of low resistivity and large cross-sectional area.

Because the resistivity of a material varies with temperature, the resistance of a specific conductor also varies with temperature. For temperature ranges that are not too great, this variation is approximately linear, analogous to Eq. (25.6):

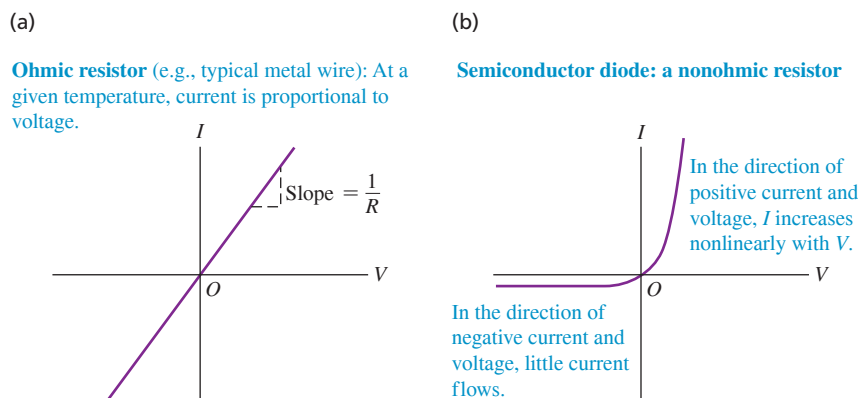
$$R(T) = R_0[1 + \alpha(T - T_0)] \tag{25.12}$$

In this equation, $R(T)$ is the resistance at temperature T and R_0 is the resistance at temperature T_0 , often taken to be 0°C or 20°C . The *temperature coefficient of resistance* α is the same constant that appears in Eq. (25.6) if the dimensions L and A in Eq. (25.10) do not change appreciably with temperature; this is indeed the case for most conducting materials. Within the limits of validity of Eq. (25.12), the *change* in resistance resulting from a temperature change $T - T_0$ is given by $R_0\alpha(T - T_0)$.

A circuit device made to have a specific value of resistance between its ends is called a **resistor**. Resistors in the range 0.01 to $10^7\ \Omega$ can be bought off the shelf. Individual resistors used in electronic circuitry are often cylindrical, a few millimeters in diameter and length, with wires coming out of the ends. The resistance may be marked with a standard code that uses three or four color bands near one end (Fig. 25.9), according to the scheme in Table 25.3. The first two bands (starting with the band nearest an end) are digits, and the third is a power-of-10 multiplier. For example, green–violet–red means $57 \times 10^2\ \Omega$, or $5.7\text{ k}\Omega$. The fourth band, if present, indicates the accuracy (tolerance) of the value; no band means $\pm 20\%$, a silver band $\pm 10\%$, and a gold band $\pm 5\%$. Another important characteristic of a resistor is the maximum *power* it can dissipate without damage. We’ll return to this point in Section 25.5.

For a resistor that obeys Ohm’s law, a graph of current as a function of potential difference (voltage) is a straight line (Fig. 25.10a). The slope of the line is $1/R$. If the sign of the potential difference changes, so does the sign of the current produced; in Fig. 25.7 this corresponds to interchanging the higher- and lower-potential ends of the conductor, so the electric field, current density, and current all reverse direction.

Figure 25.10 Current–voltage relationships for two devices. Only for a resistor that obeys Ohm’s law as in (a) is current I proportional to voltage V .



In devices that do not obey Ohm’s law, the relationship of voltage to current may not be a direct proportion, and it may be different for the two directions of current. Figure 25.10b shows the behavior of a semiconductor *diode*, a device used to convert alternating current to direct current and to perform a wide variety of logic functions in computer circuitry. For positive potentials V of the anode (one of two terminals of the diode) with respect to the cathode (the other terminal), I increases exponentially with increasing V ; for negative potentials the current is extremely small. Thus a positive V causes a current to flow in the positive direction, but a potential difference of the other sign causes little or no current. Hence a diode acts like a one-way valve in a circuit.

EXAMPLE 25.2 Electric field, potential difference, and resistance in a wire

WITH VARIATION PROBLEMS

The 18 gauge copper wire of Example 25.1 has a cross-sectional area of $8.17 \times 10^{-7} \text{ m}^2$. It carries a current of 1.67 A. Find (a) the electric-field magnitude in the wire; (b) the potential difference between two points in the wire 50.0 m apart; (c) the resistance of a 50.0 m length of this wire.

IDENTIFY and SET UP We are given the cross-sectional area A and current I . Our target variables are the electric-field magnitude E , potential difference V , and resistance R . The current density is $J = I/A$. We find E from Eq. (25.5), $E = \rho J$ (Table 25.1 gives the resistivity ρ for copper). The potential difference is then the product of E and the length of the wire. We can use either Eq. (25.10) or Eq. (25.11) to find R .

EXECUTE (a) From Table 25.1, $\rho = 1.72 \times 10^{-8} \Omega \cdot \text{m}$. Hence, from Eq. (25.5),

$$E = \rho J = \frac{\rho I}{A} = \frac{(1.72 \times 10^{-8} \Omega \cdot \text{m})(1.67 \text{ A})}{8.17 \times 10^{-7} \text{ m}^2} = 0.0352 \text{ V/m}$$

(b) The potential difference is

$$V = EL = (0.0352 \text{ V/m})(50.0 \text{ m}) = 1.76 \text{ V}$$

(c) From Eq. (25.10) the resistance of 50.0 m of this wire is

$$R = \frac{\rho L}{A} = \frac{(1.72 \times 10^{-8} \Omega \cdot \text{m})(50.0 \text{ m})}{8.17 \times 10^{-7} \text{ m}^2} = 1.05 \Omega$$

Alternatively, we can find R from Eq. (25.11):

$$R = \frac{V}{I} = \frac{1.76 \text{ V}}{1.67 \text{ A}} = 1.05 \Omega$$

EVALUATE We emphasize that the resistance of the wire is *defined* to be the ratio of voltage to current. If the wire is made of nonohmic material, then R is different for different values of V but is always given by $R = V/I$. Resistance is also always given by $R = \rho L/A$; if the material is nonohmic, ρ is not constant but depends on E (or, equivalently, on $V = EL$).

KEYCONCEPT *Resistivity* is a property of a material: The greater the resistivity, the greater the electric field that must be present in the material to produce a given current density. *Resistance* is a property of a circuit element or device: The greater the resistance, the greater the potential difference must be across the circuit element to produce a given current. Resistance depends on the dimensions of the circuit element as well as on the resistivity of the material of which it is made.

EXAMPLE 25.3 Temperature dependence of resistance

WITH VARIATION PROBLEMS

Suppose the resistance of a copper wire is 1.05Ω at 20°C . Find the resistance at 0°C and 100°C .

IDENTIFY and SET UP We are given the resistance $R_0 = 1.05 \Omega$ at a reference temperature $T_0 = 20^\circ\text{C}$. We use Eq. (25.12) to find the resistances at $T = 0^\circ\text{C}$ and $T = 100^\circ\text{C}$ (our target variables), taking the temperature coefficient of resistivity from Table 25.2.

EXECUTE From Table 25.2, $\alpha = 0.00393 (\text{C}^\circ)^{-1}$ for copper. Then from Eq. (25.12),

$$\begin{aligned} R &= R_0[1 + \alpha(T - T_0)] \\ &= (1.05 \Omega)\{1 + [0.00393 (\text{C}^\circ)^{-1}][0^\circ\text{C} - 20^\circ\text{C}]\} \\ &= 0.97 \Omega \text{ at } T = 0^\circ\text{C} \end{aligned}$$

Continued

$$R = (1.05 \, \Omega) \{1 + [0.00393 \, (\text{C}^\circ)^{-1}][100^\circ\text{C} - 20^\circ\text{C}]\}$$

$$= 1.38 \, \Omega \text{ at } T = 100^\circ\text{C}$$

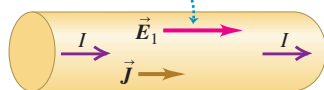
EVALUATE The resistance at 100°C is greater than that at 0°C by a factor of $(1.38 \, \Omega)/(0.97 \, \Omega) = 1.42$: Raising the temperature of copper wire from 0°C to 100°C increases its resistance by 42%. From Eq. (25.11), $V = IR$, this means that 42% more voltage is required to produce the same current at 100°C than at 0°C . Designers of electric

circuits that must operate over a wide temperature range must take this substantial effect into account.

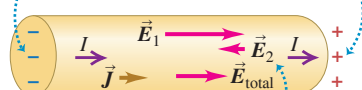
KEYCONCEPT The resistivity of a material depends on temperature, and so does the resistance of a circuit element made of that material. Over a small temperature range, the change in the resistivity of a metal is approximately proportional to the temperature change.

Figure 25.11 If an electric field is produced inside a conductor that is *not* part of a complete circuit, current flows for only a very short time.

(a) An electric field $\vec{E}_1$ produced inside an isolated conductor causes a current.



(b) The current causes charge to build up at the ends.



The charge buildup produces an opposing field $\vec{E}_2$, thus reducing the current.

(c) After a very short time $\vec{E}_2$ has the same magnitude as $\vec{E}_1$; then the total field is $\vec{E}_{\text{total}} = 0$ and the current stops completely.

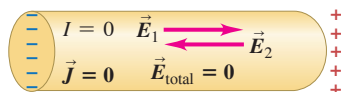


Figure 25.12 Just as a water fountain requires a pump, an electric circuit requires a source of electromotive force to sustain a steady current.



TEST YOUR UNDERSTANDING OF SECTION 25.3 Suppose you increase the voltage across the copper wire in Examples 25.2 and 25.3. The increased voltage causes more current to flow, which makes the temperature of the wire increase. (The same thing happens to the coils of an electric oven or a toaster when a voltage is applied to them. We'll explore this issue in more depth in Section 25.5.) If you double the voltage across the wire, the current in the wire increases. By what factor does it increase? (i) 2; (ii) greater than 2; (iii) less than 2.

ANSWER

(iii) Solving Eq. (25.11) for the current shows that $I = V/R$. If the resistance R of the wire remained the same, doubling the voltage V would make the current I double as well. However, we saw in Example 25.3 that the resistance is *not* constant: As the current increases and the temperature increases, R increases as well. Thus doubling the voltage produces a current that is *less* than double the original current. An ohmic conductor is one for which $R = V/I$ has the same value no matter what the voltage, so the wire is *nonohmic*. (In many practical problems the temperature change of the wire is so small that it can be ignored, so we can safely regard the wire as being ohmic. We do so in almost all examples in this book.)

25.4 ELECTROMOTIVE FORCE AND CIRCUITS

For a conductor to have a steady current, it must be part of a path that forms a closed loop or **complete circuit**. Here's why. If you establish an electric field $\vec{E}_1$ inside an isolated conductor with resistivity ρ that is *not* part of a complete circuit, a current begins to flow with current density $\vec{J} = \vec{E}_1/\rho$ (Fig. 25.11a). As a result a net positive charge quickly accumulates at one end of the conductor and a net negative charge accumulates at the other end (Fig. 25.11b). These charges themselves produce an electric field $\vec{E}_2$ in the direction opposite to $\vec{E}_1$, causing the total electric field and hence the current to decrease. Within a very small fraction of a second, enough charge builds up on the conductor ends that the total electric field $\vec{E} = \vec{E}_1 + \vec{E}_2 = 0$ inside the conductor. Then $\vec{J} = 0$ as well, and the current stops altogether (Fig. 25.11c). So there can be no steady motion of charge in such an *incomplete* circuit.

To see how to maintain a steady current in a *complete* circuit, we recall a basic fact about electric potential energy: If a charge q goes around a complete circuit and returns to its starting point, the potential energy must be the same at the end of the round trip as at the beginning. As described in Section 25.3, there is always a *decrease* in potential energy when charges move through an ordinary conducting material with resistance. So there must be some part of the circuit in which the potential energy *increases*.

The problem is analogous to an ornamental water fountain that recycles its water. The water pours out of openings at the top, cascades down over the terraces and spouts (moving in the direction of decreasing gravitational potential energy), and collects in a basin in the bottom. A pump then lifts it back to the top (increasing the potential energy) for another trip. Without the pump, the water would just fall to the bottom and stay there.

Electromotive Force

In an electric circuit there must be a device somewhere in the loop that acts like the water pump in a water fountain (Fig. 25.12). In this device a charge travels “uphill,” from lower to higher potential energy, even though the electrostatic force is trying to push it from

higher to lower potential energy. The direction of current in such a device is from lower to higher potential, just the opposite of what happens in an ordinary conductor.

The influence that makes current flow from lower to higher potential is called **electromotive force** (abbreviated **emf** and pronounced “ee-em-eff”), and a circuit device that provides emf is called a **source of emf**. Note that “electromotive force” is a poor term because emf is *not* a force but an energy-per-unit-charge quantity, like potential. The SI unit of emf is the same as that for potential, the volt ($1 \text{ V} = 1 \text{ J/C}$). A typical flashlight battery has an emf of 1.5 V; this means that the battery does 1.5 J of work on every coulomb of charge that passes through it. We’ll use the symbol $\mathcal{E}$ (a script capital E) for emf.

Every complete circuit with a steady current must include a source of emf. Batteries, electric generators, solar cells, thermocouples, and fuel cells are all examples of sources of emf. All such devices convert energy of some form (mechanical, chemical, thermal, and so on) into electric potential energy and transfer it into the circuit to which the device is connected. An *ideal* source of emf maintains a constant potential difference between its terminals, independent of the current through it. We define electromotive force quantitatively as the magnitude of this potential difference. As we’ll see, such an ideal source is a mythical beast, like the frictionless plane and the massless rope. We’ll discuss later how real-life sources of emf differ in their behavior from this idealized model.

Figure 25.13 is a schematic diagram of an ideal source of emf that maintains a potential difference between conductors a and b , called the *terminals* of the device. Terminal a , marked $+$, is maintained at *higher* potential than terminal b , marked $-$. Associated with this potential difference is an electric field $\vec{E}$ in the region around the terminals, both inside and outside the source. The electric field inside the device is directed from a to b , as shown. A charge q within the source experiences an electric force $\vec{F}_e = q\vec{E}$. But the source also provides an additional influence, which we represent as a nonelectrostatic force $\vec{F}_n$. This force, operating inside the device, pushes charge from b to a in an “uphill” direction against the electric force $\vec{F}_e$. Thus $\vec{F}_n$ maintains the potential difference between the terminals. If $\vec{F}_n$ were not present, charge would flow between the terminals until the potential difference was zero. The origin of the additional influence $\vec{F}_n$ depends on the kind of source. In a generator it results from magnetic-field forces on moving charges. In a battery or fuel cell it is associated with diffusion processes and varying electrolyte concentrations resulting from chemical reactions. In an electrostatic machine such as a Van de Graaff generator (see Fig. 22.26), an actual mechanical force is applied by a moving belt or wheel.

If a positive charge q is moved from b to a inside the source, the nonelectrostatic force $\vec{F}_n$ does a positive amount of work $W_n = q\mathcal{E}$ on the charge. This displacement is *opposite* to the electrostatic force $\vec{F}_e$, so the potential energy associated with the charge *increases* by an amount equal to qV_{ab} , where $V_{ab} = V_a - V_b$ is the (positive) potential of point a with respect to point b . For the ideal source of emf that we’ve described, $\vec{F}_e$ and $\vec{F}_n$ are equal in magnitude but opposite in direction, so the total work done on the charge q is zero; there is an increase in potential energy but *no* change in the kinetic energy of the charge. It’s like lifting a book from the floor to a high shelf at constant speed. The increase in potential energy is just equal to the nonelectrostatic work W_n , so $q\mathcal{E} = qV_{ab}$, or

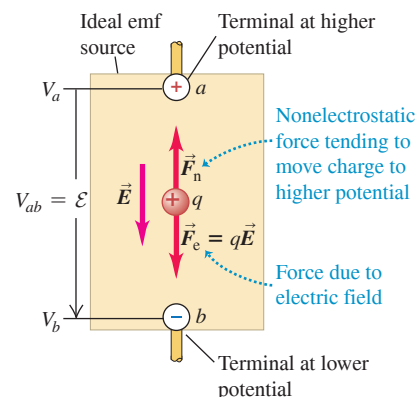
$$V_{ab} = \mathcal{E} \quad (\text{ideal source of emf}) \quad (25.13)$$

Now let’s make a complete circuit by connecting a wire with resistance R to the terminals of a source (**Fig. 25.14**). The potential difference between terminals a and b sets up an electric field within the wire; this causes current to flow around the loop from a toward b , from higher to lower potential. Where the wire bends, equal amounts of positive and negative charges persist on the “inside” and “outside” of the bend. These charges exert the forces that cause the current to follow the bends in the wire.

From Eq. (25.11) the potential difference between the ends of the wire in Fig. 25.14 is given by $V_{ab} = IR$. Combining with Eq. (25.13), we have

$$\mathcal{E} = V_{ab} = IR \quad (\text{ideal source of emf}) \quad (25.14)$$

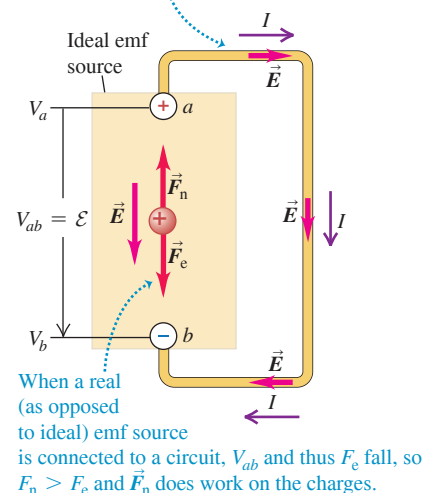
Figure 25.13 Schematic diagram of a source of emf in an “open-circuit” situation. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q .



When the emf source is not part of a closed circuit, $F_n = F_e$ and there is no net motion of charge between the terminals.

Figure 25.14 Schematic diagram of an ideal source of emf in a complete circuit. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q . The current is in the direction from a to b in the external circuit and from b to a within the source.

Potential across terminals creates electric field in circuit, causing charges to move.



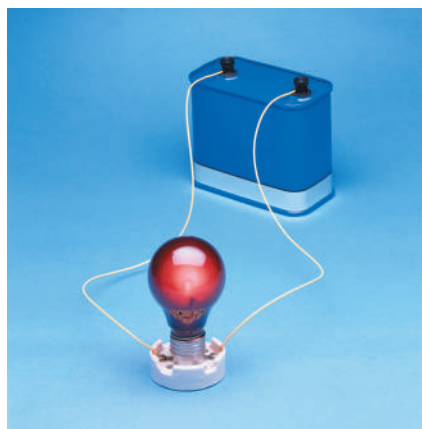
When a real (as opposed to ideal) emf source is connected to a circuit, V_{ab} and thus F_e fall, so $F_n > F_e$ and $\vec{F}_n$ does work on the charges.

BIO APPLICATION Danger: Electric Ray!

Electric rays deliver electric shocks to stun their prey and to discourage predators. (In ancient Rome, physicians practiced a primitive form of electroconvulsive therapy by placing electric rays on their patients to cure headaches and gout.) The shocks are produced by specialized flattened cells called electroplaques. Such a cell moves ions across membranes to produce an emf of about 0.05 V. Thousands of electroplaques are stacked on top of each other, so their emfs add to a total of as much as 200 V. These stacks make up more than half of an electric ray's body mass. A ray can use these to deliver an impressive current of up to 30 A for a few milliseconds.



Figure 25.15 The emf of this battery—that is, the terminal voltage when it's not connected to anything—is 12 V. But because the battery has internal resistance, the terminal voltage of the battery is less than 12 V when it is supplying current to a light bulb.



That is, when a positive charge q flows around the circuit, the potential *rise* $\mathcal{E}$ as it passes through the ideal source is numerically equal to the potential *drop* $V_{ab} = IR$ as it passes through the remainder of the circuit. Once $\mathcal{E}$ and R are known, this relationship determines the current in the circuit.

CAUTION Current is not “used up” in a circuit It's a common misconception that in a closed circuit, current squirts out of the positive terminal of a battery and is consumed or “used up” by the time it reaches the negative terminal. In fact the current is the *same* at every point in a simple loop circuit like that in Fig. 25.14, even if the wire thickness is not constant throughout the circuit. This happens because charge is conserved (it can be neither created nor destroyed) and because charge cannot accumulate in the circuit devices we have described. It's like the flow of water in an ornamental fountain; water flows out of the top at the same rate at which it reaches the bottom, no matter what the dimensions of the fountain. None of the water is “used up” along the way! **?**

Internal Resistance

Real sources of emf in a circuit don't behave in exactly the way we have described; the potential difference across a real source in a circuit is *not* equal to the emf as in Eq. (25.14). The reason is that charge moving through the material of any real source encounters *resistance*. We call this the **internal resistance** of the source, denoted by r . If this resistance behaves according to Ohm's law, r is constant and independent of the current I . As the current moves through r , it experiences an associated drop in potential equal to Ir . Thus, when a current is flowing through a source from the negative terminal b to the positive terminal a , the potential difference V_{ab} between the terminals is

$$\text{Terminal voltage, source with internal resistance} \quad V_{ab} = \mathcal{E} - Ir \quad \text{emf of source} \quad \text{Current through source} \quad \text{Internal resistance of source} \quad (25.15)$$

The potential V_{ab} , called the **terminal voltage**, is less than the emf $\mathcal{E}$ because of the term Ir representing the potential drop across the internal resistance r . Hence the increase in potential energy qV_{ab} as a charge q moves from b to a within the source is less than the work $q\mathcal{E}$ done by the nonelectrostatic force $\vec{F}_n$, since some potential energy is lost in traversing the internal resistance.

A 1.5 V battery has an emf of 1.5 V, but the terminal voltage V_{ab} of the battery is equal to 1.5 V only if no current is flowing through it so that $I = 0$ in Eq. (25.15). If the battery is part of a complete circuit through which current is flowing, the terminal voltage will be less than 1.5 V. *For a real source of emf, the terminal voltage equals the emf only if no current is flowing through the source (Fig. 25.15).* Thus we can describe the behavior of a source in terms of two properties: an emf $\mathcal{E}$, which supplies a constant potential difference independent of current, in series with an internal resistance r .

The current in the external circuit connected to the source terminals a and b is still determined by $V_{ab} = IR$. Combining this with Eq. (25.15), we find

$$\mathcal{E} - Ir = IR \quad \text{or} \quad I = \frac{\mathcal{E}}{R + r} \quad (\text{current, source with internal resistance}) \quad (25.16)$$

That is, the current equals the source emf divided by the *total* circuit resistance ($R + r$).

CAUTION A battery is not a “current source” You might have thought that a battery or other source of emf always produces the same current, no matter what circuit it's used in. Equation (25.16) shows that this isn't so! The greater the resistance R of the external circuit, the less current the source will produce. **!**

Symbols for Circuit Diagrams

An important part of analyzing any electric circuit is drawing a schematic *circuit diagram*. Table 25.4 shows the usual symbols used in circuit diagrams. We'll use these symbols extensively in this chapter and the next. We usually assume that the wires

TABLE 25.4 Symbols for Circuit Diagrams

	Conductor with negligible resistance
	Resistor
	Source of emf (longer vertical line always represents the positive terminal, usually the terminal with higher potential)
	Source of emf with internal resistance r (r can be placed on either side)
or	
	Voltmeter (measures potential difference between its terminals)
	Ammeter (measures current through it)

CAUTION Internal resistance is an intrinsic part of a source of emf. In Table 25.4 we've drawn the emf $\mathcal{E}$ and internal resistance r as distinct parts of a source of emf. In fact, both $\mathcal{E}$ and r are always present in any source. They *cannot* be separated.

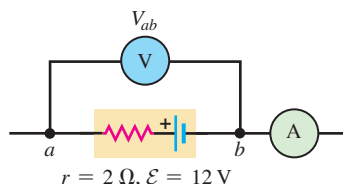
that connect the various elements of the circuit have negligible resistance; from Eq. (25.11), $V = IR$, the potential difference between the ends of such a wire is zero.

Table 25.4 includes two *meters* that are used to measure the properties of circuits. Idealized meters do not disturb the circuit in which they are connected. A **voltmeter**, introduced in Section 23.2, measures the potential difference between its terminals; an idealized voltmeter has infinitely large resistance and measures potential difference without having any current diverted through it. An ammeter measures the current passing through it; an idealized **ammeter** has zero resistance and has no potential difference between its terminals. The following examples illustrate how to analyze circuits that include meters.

CONCEPTUAL EXAMPLE 25.4 A source in an open circuit

Figure 25.16 shows a source (a battery) with emf $\mathcal{E} = 12 \text{ V}$ and internal resistance $r = 2 \Omega$. (For comparison, the internal resistance of a commercial 12 V lead storage battery is only a few thousandths of an ohm.) The wires to the left of a and to the right of the ammeter A are not connected to anything. Determine the respective readings V_{ab} and I of the idealized voltmeter V and the idealized ammeter A .

Figure 25.16 A source of emf in an open circuit.



SOLUTION There is *zero* current because there is no complete circuit. (Our idealized voltmeter has an infinitely large resistance, so no current flows through it.) Hence the ammeter reads $I = 0$. Because there is no current through the battery, there is no potential difference across its internal resistance. From Eq. (25.15) with $I = 0$, the potential difference V_{ab} across the battery terminals is equal to the emf. So the voltmeter reads $V_{ab} = \mathcal{E} = 12 \text{ V}$.

KEYCONCEPT The terminal voltage of a real, nonideal source with internal resistance equals the emf *only* if no current is flowing through the source.

EXAMPLE 25.5 A source in a complete circuit

WITH VARIATION PROBLEMS

We add a 4Ω resistor to the battery in Conceptual Example 25.4, forming a complete circuit (**Fig. 25.17**, next page). What are the voltmeter and ammeter readings V_{ab} and I now?

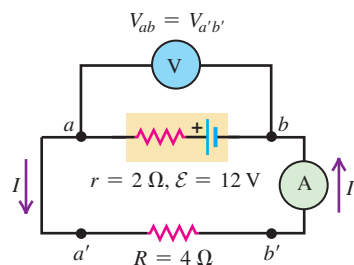
IDENTIFY and SET UP Our target variables are the current I through the circuit $aa'b'b$ and the potential difference V_{ab} . We first find I from Eq. (25.16). To find V_{ab} , we can use either Eq. (25.11) or Eq. (25.15).

EXECUTE The ideal ammeter has zero resistance, so the total resistance external to the source is $R = 4 \Omega$. From Eq. (25.16), the current through the circuit $aa'b'b$ is then

$$I = \frac{\mathcal{E}}{R + r} = \frac{12 \text{ V}}{4 \Omega + 2 \Omega} = 2 \text{ A}$$

Continued

Figure 25.17 A source of emf in a complete circuit.



Our idealized conducting wires and the idealized ammeter have zero resistance, so there is no potential difference between points a and a' or between points b and b' ; that is, $V_{ab} = V_{a'b'}$. We find V_{ab} by considering a and b as the terminals of the resistor: From Ohm's law, Eq. (25.11), we then have

$$V_{a'b'} = IR = (2 \text{ A})(4 \Omega) = 8 \text{ V}$$

Alternatively, we can consider a and b as the terminals of the source. Then, from Eq. (25.15),

$$V_{ab} = \mathcal{E} - Ir = 12 \text{ V} - (2 \text{ A})(2 \Omega) = 8 \text{ V}$$

Either way, we see that the voltmeter reading is 8 V.

EVALUATE In this example the battery is acting as a source that produces the current in the circuit. Current flows through the battery from its negative terminal toward its positive terminal, and the terminal voltage V_{ab} is less than the emf $\mathcal{E}$. But if a battery is being recharged, current flows through it in the opposite direction, from its *positive* terminal toward its *negative* terminal. In such a case I in Eq. (25.15) is negative, so the terminal voltage is actually *greater* than the emf. No matter which way current flows through the battery, the smaller the internal resistance r , the less the difference between V_{ab} and $\mathcal{E}$.

KEYCONCEPT To find the potential difference across a resistor, multiply the current through the resistor by its resistance. The potential difference across a source, such as a battery, differs from its emf by an amount determined by the source's internal resistance.

CONCEPTUAL EXAMPLE 25.6 Using voltmeters and ammeters

We move the voltmeter and ammeter in Example 25.5 to different positions in the circuit. What are the readings of the ideal voltmeter and ammeter in the situations shown in (a) Fig. 25.18a and (b) Fig. 25.18b?

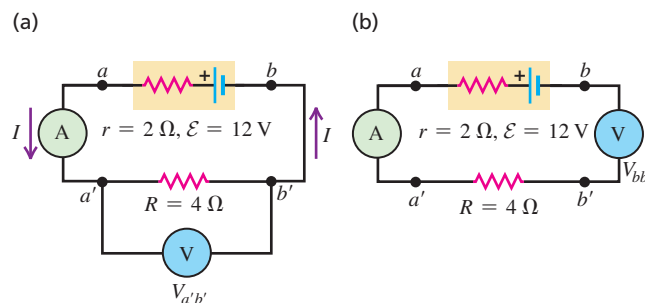
SOLUTION (a) The voltmeter now measures the potential difference between points a' and b' . As in Example 25.5, $V_{ab} = V_{a'b'}$, so the voltmeter reads the same as in Example 25.5: $V_{a'b'} = 8 \text{ V}$.

CAUTION **Current in a simple loop** As charges move through a resistor, there is a decrease in electric potential energy, but there is *no* change in the current. *The current in a simple loop is the same at every point*; it is not “used up” as it moves through a resistor. Hence the ammeter in Fig. 25.17 (“downstream” of the 4Ω resistor) and the ammeter in Fig. 25.18a (“upstream” of the resistor) both read $I = 2 \text{ A}$.

(b) There is no current through the ideal voltmeter because it has infinitely large resistance. Since the voltmeter is now part of the circuit, there is no current at all in the circuit, and the ammeter reads $I = 0$.

The voltmeter measures the potential difference $V_{bb'}$ between points b and b' . Since $I = 0$, the potential difference across the resistor is $V_{a'b'} = IR = 0$, and the potential difference between the ends a and a' of the idealized ammeter is also zero. So $V_{bb'}$ is equal to V_{ab} , the terminal voltage of the source. As in Conceptual Example 25.4, there is no current, so the terminal voltage equals the emf, and the voltmeter reading is $V_{ab} = \mathcal{E} = 12 \text{ V}$.

Figure 25.18 Different placements of a voltmeter and an ammeter in a complete circuit.



This example shows that ammeters and voltmeters are circuit elements, too. Moving the voltmeter from the position in Fig. 25.18a to that in Fig. 25.18b makes large changes in the current and potential differences in the circuit. If you want to measure the potential difference between two points in a circuit without disturbing the circuit, use a voltmeter as in Fig. 25.17 or 25.18a, *not* as in Fig. 25.18b.

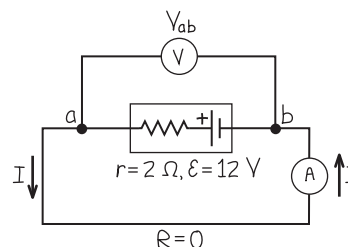
KEYCONCEPT An ideal voltmeter has infinite resistance, whereas an ideal ammeter has zero resistance. Placing these meters in inappropriate locations in a circuit can disturb the quantities you are trying to measure.

EXAMPLE 25.7 A source with a short circuit

In the circuit of Example 25.5 we replace the 4Ω resistor with a zero-resistance conductor. What are the meter readings now?

IDENTIFY and SET UP Figure 25.19 shows the new circuit. Our target variables are again I and V_{ab} . There is now a zero-resistance path between points a and b , through the lower loop, so the potential difference between these points must be zero.

Figure 25.19 Our sketch for this problem.



EXECUTE We must have $V_{ab} = IR = I(0) = 0$, no matter what the current. We can therefore find the current I from Eq. (25.15):

$$\begin{aligned} V_{ab} &= \mathcal{E} - Ir = 0 \\ I &= \frac{\mathcal{E}}{r} = \frac{12 \text{ V}}{2 \Omega} = 6 \text{ A} \end{aligned}$$

EVALUATE The current in this circuit has a different value than in Example 25.5, even though the same battery is used; the current depends on both the internal resistance r and the resistance of the external circuit.

The situation here is called a *short circuit*. The external-circuit resistance is zero, because terminals of the battery are connected directly to each other. The short-circuit current is equal to the emf $\mathcal{E}$ divided by the internal resistance r . **Warning:** Short circuits can be dangerous! An automobile battery or a household power line has very small internal resistance (much less than in these examples), and the short-circuit current can be great enough to melt a small wire or cause a storage battery to explode.

KEYCONCEPT Connecting the terminals of a source directly to each other (a short circuit) leads to a large and potentially unsafe current.

Potential Changes Around a Circuit

The net change in potential energy for a charge q making a round trip around a complete circuit must be zero. Hence the net change in *potential* around the circuit must also be zero; in other words, the algebraic sum of the potential differences and emfs around the loop is zero. We can see this by rewriting Eq. (25.16) in the form

$$\mathcal{E} - Ir - IR = 0$$

A potential gain of $\mathcal{E}$ is associated with the emf, and potential drops of Ir and IR are associated with the internal resistance of the source and the external circuit, respectively. **Figure 25.20** shows how the potential varies as we go around the complete circuit of Fig. 25.17. The horizontal axis doesn't necessarily represent actual distances, but rather various points in the loop. If we take the potential to be zero at the negative terminal of the battery, then we have a rise $\mathcal{E}$ and a drop Ir in the battery and an additional drop IR in the external resistor, and as we finish our trip around the loop, the potential is back where it started.

In this section we have considered only situations in which the resistances are ohmic. If the circuit includes a nonlinear device such as a diode (see Fig. 25.10b), Eq. (25.16) is still valid but cannot be solved algebraically because R is not a constant. In such a situation, I can be found by using numerical techniques.

Finally, we remark that Eq. (25.15) is not always an adequate representation of the behavior of a source. The emf may not be constant, and what we have described as an internal resistance may actually be a more complex voltage–current relationship that doesn't obey Ohm's law. Nevertheless, the concept of internal resistance frequently provides an adequate description of batteries, generators, and other energy converters. The principal difference between a fresh flashlight battery and an old one is not in the emf, which decreases only slightly with use, but in the internal resistance, which may increase from less than an ohm when the battery is fresh to as much as 1000Ω or more after long use. Similarly, a car battery can deliver less current to the starter motor on a cold morning than when the battery is warm, not because the emf is appreciably less but because the internal resistance increases with decreasing temperature.

TEST YOUR UNDERSTANDING OF SECTION 25.4 Rank the following circuits in order from highest to lowest current: (i) A 1.4Ω resistor connected to a 1.5 V battery that has an internal resistance of 0.10Ω ; (ii) a 1.8Ω resistor connected to a 4.0 V battery that has a terminal voltage of 3.6 V but an unknown internal resistance; (iii) an unknown resistor connected to a 12.0 V battery that has an internal resistance of 0.20Ω and a terminal voltage of 11.0 V .

ANSWER

(iii), (ii), (i) For circuit (i), we find the current from Eq. (25.16): $I = \mathcal{E}/(R + r) = (1.5 \text{ V})/(1.4 \Omega + 0.10 \Omega) = 1.0 \text{ A}$. For circuit (ii), we note that the terminal voltage $V_{ab} = 3.6 \text{ V}$ equals the voltage IR across the 1.8Ω resistor: $V_{ab} = IR$, so $I = V_{ab}/R = (3.6 \text{ V})/(1.8 \Omega) = 2.0 \text{ A}$. For circuit (iii), we use Eq. (25.15) for the terminal voltage: $V_{ab} = \mathcal{E} - Ir$, so $I = (\mathcal{E} - V_{ab})/r = (12.0 \text{ V} - 11.0 \text{ V})/(0.20 \Omega) = 5.0 \text{ A}$.

Figure 25.20 Potential rises and drops in a circuit.

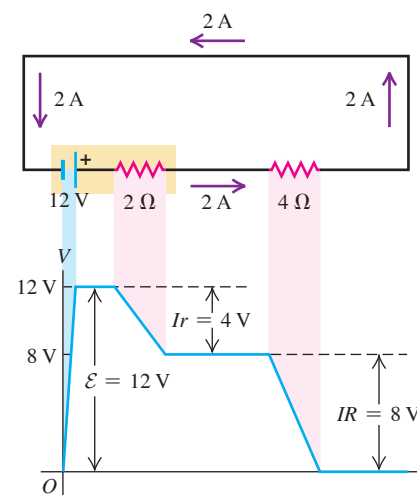
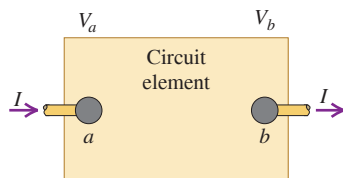


Figure 25.21 The power input to the circuit element between a and b is $P = (V_a - V_b)I = V_{ab}I$.



25.5 ENERGY AND POWER IN ELECTRIC CIRCUITS

Let's now look at some energy and power relationships in electric circuits. The box in **Fig. 25.21** represents a circuit element with potential difference $V_a - V_b = V_{ab}$ between its terminals and current I passing through it in the direction from a toward b . This element might be a resistor, a battery, or something else; the details don't matter. As charge passes through the circuit element, the electric field does work on the charge. In a source of emf, additional work is done by the force $\vec{F}_n$ that we mentioned in Section 25.4.

As an amount of charge q passes through the circuit element, there is a change in potential energy equal to qV_{ab} . For example, if $q > 0$ and $V_{ab} = V_a - V_b$ is positive, potential energy decreases as the charge “falls” from potential V_a to lower potential V_b . The moving charges don't gain *kinetic* energy, because the current (the rate of charge flow) out of the circuit element must be the same as the current into the element. Instead, the quantity qV_{ab} represents energy transferred into the circuit element. This situation occurs in the coils of a toaster or electric oven, in which electrical energy is converted to thermal energy.

If the potential at a is lower than at b , then V_{ab} is negative and there is a net transfer of energy *out* of the circuit element. The element then acts as a source, delivering electrical energy into the circuit to which it is attached. This is the usual situation for a battery, which converts chemical energy into electrical energy and delivers it to the external circuit. Thus qV_{ab} can denote a quantity of energy that is either delivered to a circuit element or extracted from that element.

In electric circuits we are most often interested in the *rate* at which energy is either delivered to or extracted from a circuit element. If the current through the element is I , then in a time interval dt an amount of charge $dQ = I dt$ passes through the element. The potential-energy change for this amount of charge is $V_{ab} dQ = V_{ab} I dt$. Dividing this expression by dt , we obtain the *rate* at which energy is transferred either into or out of the circuit element. The time rate of energy transfer is *power*, denoted by P , so we write

$$\text{Power delivered to or extracted from a circuit element} \quad P = V_{ab} I \quad (25.17)$$

Voltage across circuit element
Current in circuit element

The unit of V_{ab} is one volt, or one joule per coulomb, and the unit of I is one ampere, or one coulomb per second. Hence the unit of $P = V_{ab} I$ is one watt:

$$(1 \text{ J/C})(1 \text{ C/s}) = 1 \text{ J/s} = 1 \text{ W}$$

Let's consider a few special cases.

Power Input to a Pure Resistance

If the circuit element in Fig. 25.21 is a resistor, the potential difference is $V_{ab} = IR$. From Eq. (25.17) the power delivered to the resistor by the circuit is

$$\text{Power delivered to a resistor} \quad P = V_{ab} I = I^2 R = \frac{V_{ab}^2}{R} \quad (25.18)$$

Voltage across resistor
Current in resistor
Resistance of resistor

In this case the potential at a (where the current enters the resistor) is always higher than that at b (where the current exits). Current enters the higher-potential terminal of the device, and Eq. (25.18) represents the rate of transfer of electric potential energy *into* the circuit element.

What becomes of this energy? The moving charges collide with atoms in the resistor and transfer some of their energy to these atoms, increasing the internal energy of the material. Either the temperature of the resistor increases or there is a flow of heat out of it, or both. In any of these cases we say that energy is *dissipated* in the resistor at a rate I^2R . Every resistor has a *power rating*, the maximum power the device can dissipate without becoming overheated and damaged. Some devices, such as electric heaters, are designed to get hot and transfer heat to their surroundings. But if the power rating is exceeded, even such a device may melt or even explode.

Power Output of a Source

The upper rectangle in **Fig. 25.22a** represents a source with emf $\mathcal{E}$ and internal resistance r , connected by ideal (resistanceless) conductors to an external circuit represented by the lower box. This could describe a car battery connected to one of the car's headlights (**Fig. 25.22b**). Point a is at higher potential than point b , so $V_a > V_b$ and V_{ab} is positive. Note that the current I is *leaving* the source at the higher-potential terminal (rather than entering there). Energy is being delivered to the external circuit, at a rate given by Eq. (25.17):

$$P = V_{ab}I$$

For a source that can be described by an emf $\mathcal{E}$ and an internal resistance r , we may use Eq. (25.15):

$$V_{ab} = \mathcal{E} - Ir$$

Multiplying this equation by I , we find

$$P = V_{ab}I = \mathcal{E}I - I^2r \quad (25.19)$$

What do the terms $\mathcal{E}I$ and I^2r mean? In Section 25.4 we defined the emf $\mathcal{E}$ as the work per unit charge performed on the charges by the nonelectrostatic force as the charges are pushed “uphill” from b to a in the source. In a time dt , a charge $dQ = I dt$ flows through the source; the work done on it by this nonelectrostatic force is $\mathcal{E} dQ = \mathcal{E}I dt$. Thus $\mathcal{E}I$ is the *rate* at which work is done on the circulating charges by whatever agency causes the nonelectrostatic force in the source. This term represents the rate of conversion of nonelectrical energy to electrical energy within the source. The term I^2r is the rate at which electrical energy is *dissipated* in the internal resistance of the source. The difference $\mathcal{E}I - I^2r$ is the *net* electrical power output of the source—that is, the rate at which the source delivers electrical energy to the remainder of the circuit.

Power Input to a Source

Suppose that the lower rectangle in **Fig. 25.22a** is itself a source, with an emf *larger* than that of the upper source and opposite to that of the upper source. **Figure 25.23** shows a practical example, an automobile battery (the upper circuit element) being charged by the car's alternator (the lower element). The current I in the circuit is then *opposite* to that shown in **Fig. 25.22**; the lower source is pushing current backward through the upper source. Because of this reversal of current, instead of Eq. (25.15), we have for the upper source

$$V_{ab} = \mathcal{E} + Ir$$

and instead of Eq. (25.19), we have

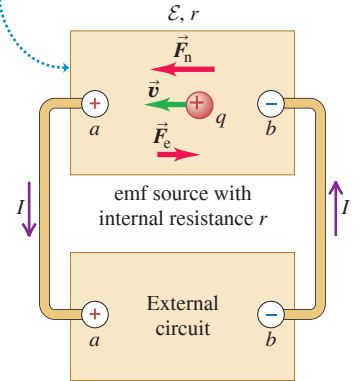
$$P = V_{ab}I = \mathcal{E}I + I^2r \quad (25.20)$$

Work is being done *on*, rather than *by*, the agent that causes the nonelectrostatic force in the upper source. There is a conversion of electrical energy into nonelectrical energy in the upper source at a rate $\mathcal{E}I$. The term I^2r in Eq. (25.20) is again the rate

Figure 25.22 Energy conversion in a simple circuit.

(a) Diagrammatic circuit

- The emf source converts nonelectrical to electrical energy at a rate $\mathcal{E}I$.
- Its internal resistance *dissipates* energy at a rate I^2r .
- The difference $\mathcal{E}I - I^2r$ is its power output.



(b) A real circuit of the type shown in (a)

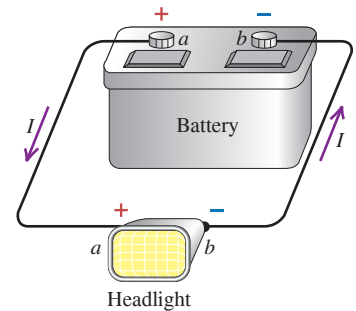
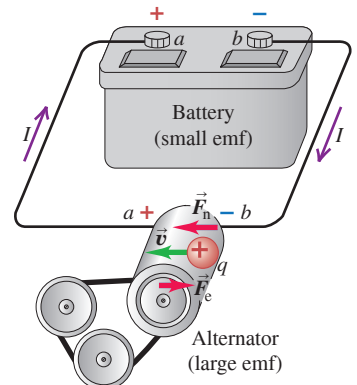


Figure 25.23 When two sources are connected in a simple loop circuit, the source with the larger emf delivers energy to the other source.



of dissipation of energy in the internal resistance of the upper source, and the sum $\mathcal{E}I + I^2r$ is the total electrical power *input* to the upper source. This is what happens when a rechargeable battery (a storage battery) is connected to a charger. The charger supplies electrical energy to the battery; part of it is converted to chemical energy, to be reconverted later, and the remainder is dissipated (wasted) in the battery's internal resistance, warming the battery and causing a heat flow out of it. If you have a power tool or laptop computer with a rechargeable battery, you may have noticed that it gets warm while it is charging.

PROBLEM-SOLVING STRATEGY 25.1 Power and Energy in Circuits

IDENTIFY *the relevant concepts:* The ideas of electrical power input and output can be applied to any electric circuit. Many problems will ask you to explicitly consider power or energy.

SET UP *the problem* using the following steps:

1. Make a drawing of the circuit.
2. Identify the circuit elements, including sources of emf and resistors. We'll introduce other circuit elements later, including capacitors (Chapter 26) and inductors (Chapter 30).
3. Identify the target variables. Typically they will be the power input or output for each circuit element, or the total amount of energy put into or taken out of a circuit element in a given time.

EXECUTE *the solution* as follows:

1. A source of emf $\mathcal{E}$ delivers power $\mathcal{E}I$ into a circuit when current I flows through the source in the direction from $-$ to $+$. (For example, energy is converted from chemical energy in a battery, or from mechanical energy in a generator.) In this case there is a *positive* power output to the circuit or, equivalently, a *negative* power input to the source.
2. A source of emf takes power $\mathcal{E}I$ from a circuit when current passes through the source from $+$ to $-$. (This occurs in charging

a storage battery, when electrical energy is converted to chemical energy.) In this case there is a *negative* power output to the circuit or, equivalently, a *positive* power input to the source.

3. There is always a *positive* power input to a resistor through which current flows, irrespective of the direction of current flow. This process removes energy from the circuit, converting it to heat at the rate $VI = I^2R = V^2/R$, where V is the potential difference across the resistor.
4. Just as in item 3, there always is a positive power input to the internal resistance r of a source through which current flows, irrespective of the direction of current flow. This process likewise removes energy from the circuit, converting it into heat at the rate I^2r .
5. If the power into or out of a circuit element is constant, the energy delivered to or extracted from that element is the product of power and elapsed time. (In Chapter 26 we'll encounter situations in which the power is not constant. In such cases, calculating the total energy requires an integral over the relevant time interval.)

EVALUATE *your answer:* Check your results; in particular, check that energy is conserved. This conservation can be expressed in either of two forms: "net power input = net power output" or "the algebraic sum of the power inputs to the circuit elements is zero."

EXAMPLE 25.8 Power input and output in a complete circuit

WITH VARIATION PROBLEMS

For the circuit that we analyzed in Example 25.5, find the rates of energy conversion (chemical to electrical) and energy dissipation in the battery, the rate of energy dissipation in the $4\ \Omega$ resistor, and the battery's net power output.

IDENTIFY and SET UP Figure 25.24 shows the circuit, gives values of quantities known from Example 25.5, and indicates how we find the target variables. We use Eq. (25.19) to find the battery's net power output, the rate of chemical-to-electrical energy conversion, and the rate of energy dissipation in the battery's internal resistance. We use Eq. (25.18) to find the power delivered to (and dissipated in) the $4\ \Omega$ resistor.

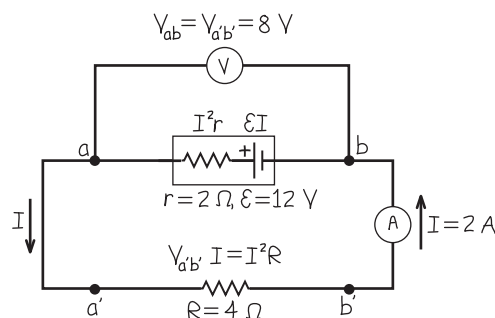
EXECUTE From the first term in Eq. (25.19), the rate of energy conversion in the battery is

$$\mathcal{E}I = (12\text{ V})(2\text{ A}) = 24\text{ W}$$

From the second term in Eq. (25.19), the rate of dissipation of energy in the battery is

$$I^2r = (2\text{ A})^2(2\ \Omega) = 8\text{ W}$$

Figure 25.24 Our sketch for this problem.



The *net* electrical power output of the battery is the difference between these: $\mathcal{E}I - I^2r = 16\text{ W}$. From Eq. (25.18), the electrical power input to, and the equal rate of dissipation of electrical energy in, the $4\ \Omega$ resistor are

$$V_{a'b'}I = (8\text{ V})(2\text{ A}) = 16\text{ W} \quad \text{and} \quad I^2R = (2\text{ A})^2(4\ \Omega) = 16\text{ W}$$

EVALUATE The rate $V_{ab}I$ at which energy is supplied to the $4\ \Omega$ resistor equals the rate I^2R at which energy is dissipated there. This is also equal to the battery's net power output: $P = V_{ab}I = (8\ \text{V})(2\ \text{A}) = 16\ \text{W}$. In summary, the rate at which the source of emf supplies energy is $\mathcal{E}I = 24\ \text{W}$, of which $I^2r = 8\ \text{W}$ is dissipated in the battery's internal resistor and $I^2R = 16\ \text{W}$ is dissipated in the external resistor.

KEYCONCEPT To find the electrical power input or output of any circuit element, multiply the current through the element by the potential difference across the element. There is a power output if the current flows through the element in the direction from low to high potential, and a power input if the current flows from high to low potential.

EXAMPLE 25.9 Increasing the resistance

WITH VARIATION PROBLEMS

Suppose we replace the external $4\ \Omega$ resistor in Fig. 25.24 with an $8\ \Omega$ resistor. How does this affect the electrical power dissipated in this resistor?

IDENTIFY and SET UP Our target variable is the power dissipated in the resistor to which the battery is connected. The situation is the same as in Example 25.8, but with a higher external resistance R .

EXECUTE According to Eq. (25.18), the power dissipated in the resistor is $P = I^2R$. You might conclude that making the resistance R twice as great as in Example 25.8 should make the power twice as great, or $2(16\ \text{W}) = 32\ \text{W}$. If instead you used the formula $P = V_{ab}^2/R$, you might conclude that the power should be one-half as great as in the preceding example, or $(16\ \text{W})/2 = 8\ \text{W}$. Which answer is correct?

In fact, *both* of these answers are *incorrect*. The first is wrong because changing the resistance R also changes the current in the circuit (remember, a source of emf does *not* generate the same current in all situations). The second answer is wrong because the potential difference V_{ab} across the resistor changes when the current changes. To get the correct answer, we first find the current just as we did in Example 25.5:

$$I = \frac{\mathcal{E}}{R + r} = \frac{12\ \text{V}}{8\ \Omega + 2\ \Omega} = 1.2\ \text{A}$$

The greater resistance causes the current to decrease. The potential difference across the resistor is

$$V_{ab} = IR = (1.2\ \text{A})(8\ \Omega) = 9.6\ \text{V}$$

which is greater than that with the $4\ \Omega$ resistor. We can then find the power dissipated in the resistor in either of two ways:

$$P = I^2R = (1.2\ \text{A})^2(8\ \Omega) = 12\ \text{W} \quad \text{or}$$

$$P = \frac{V_{ab}^2}{R} = \frac{(9.6\ \text{V})^2}{8\ \Omega} = 12\ \text{W}$$

EVALUATE Increasing the resistance R causes a *reduction* in the power input to the resistor. In the expression $P = I^2R$ the decrease in current is more important than the increase in resistance; in the expression $P = V_{ab}^2/R$ the increase in resistance is more important than the increase in V_{ab} . This same principle applies to ordinary light bulbs; a 50 W light bulb has a greater resistance than does a 100 W light bulb.

Can you show that replacing the $4\ \Omega$ resistor with an $8\ \Omega$ resistor decreases both the rate of energy conversion (chemical to electrical) in the battery and the rate of energy dissipation in the battery?

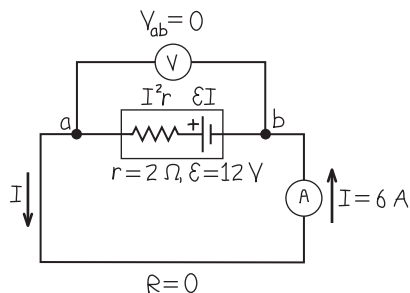
KEYCONCEPT The power input to a resistor (that is, the rate at which it dissipates energy) depends on its resistance R . For a given current through the resistor, increasing R increases the power; for a given potential difference across the resistor, increasing R decreases the power.

EXAMPLE 25.10 Power in a short circuit

For the short-circuit situation of Example 25.7, find the rates of energy conversion and energy dissipation in the battery and the net power output of the battery.

IDENTIFY and SET UP Our target variables are again the power inputs and outputs associated with the battery. **Figure 25.25** shows the circuit. This is the same situation as in Example 25.8, but now the external resistance R is zero.

Figure 25.25 Our sketch for this problem.



EXECUTE We found in Example 25.7 that the current in this situation is $I = 6\ \text{A}$. From Eq. (25.19), the rate of energy conversion (chemical to electrical) in the battery is then

$$\mathcal{E}I = (12\ \text{V})(6\ \text{A}) = 72\ \text{W}$$

and the rate of dissipation of energy in the battery is

$$I^2r = (6\ \text{A})^2(2\ \Omega) = 72\ \text{W}$$

The net power output of the source is $\mathcal{E}I - I^2r = 0$. We get this same result from the expression $P = V_{ab}I$, because the terminal voltage V_{ab} of the source is zero.

EVALUATE With ideal wires and an ideal ammeter, so that $R = 0$, *all* of the converted energy from the source is dissipated within the source. This is why a short-circuited battery is quickly ruined and may explode.

KEYCONCEPT Connecting the terminals of a source directly to each other is potentially unsafe because it can lead to a very high rate of energy dissipation in the source's internal resistance.

TEST YOUR UNDERSTANDING OF SECTION 25.5 Rank the following circuits in order from highest to lowest values of the net power output of the battery. (i) A $1.4\ \Omega$ resistor connected to a $1.5\ \text{V}$ battery that has an internal resistance of $0.10\ \Omega$; (ii) a $1.8\ \Omega$ resistor connected to a $4.0\ \text{V}$ battery that has a terminal voltage of $3.6\ \text{V}$ but an unknown internal resistance; (iii) an unknown resistor connected to a $12.0\ \text{V}$ battery that has an internal resistance of $0.20\ \Omega$ and a terminal voltage of $11.0\ \text{V}$.

ANSWER

(iii), (ii), (i) These are the same circuits that we analyzed in Test Your Understanding of Section 25.4. In each case the net power output of the battery is $P = V_{ab}I$, where V_{ab} is the battery terminal voltage. For circuit (i), we found that $I = 1.0\ \text{A}$, so $V_{ab} = \mathcal{E} - Ir = 1.5\ \text{V} - (1.0\ \text{A})(0.10\ \Omega) = 1.4\ \text{V}$, so $P = (1.4\ \text{V})(1.0\ \text{A}) = 1.4\ \text{W}$. For circuit (ii), we have $V_{ab} = 3.6\ \text{V}$ and found that $I = 2.0\ \text{A}$, so $P = (3.6\ \text{V})(2.0\ \text{A}) = 7.2\ \text{W}$. For circuit (iii), we have $V_{ab} = 11.0\ \text{V}$ and found that $I = 5.0\ \text{A}$, so $P = (11.0\ \text{V})(5.0\ \text{A}) = 55\ \text{W}$.

25.6 THEORY OF METALLIC CONDUCTION

Figure 25.26 Random motions of an electron in a metallic crystal (a) with zero electric field and (b) with an electric field that causes drift. The curvatures of the paths are greatly exaggerated.

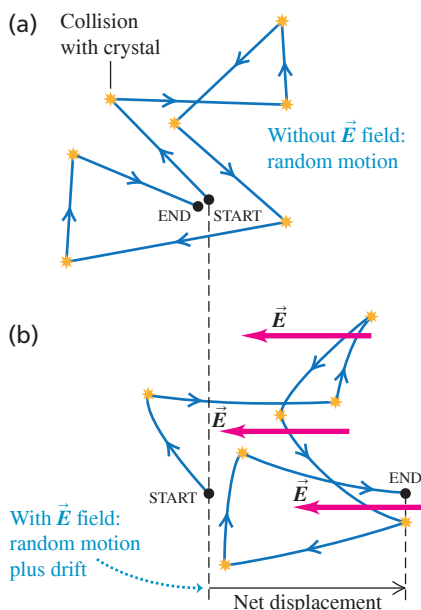
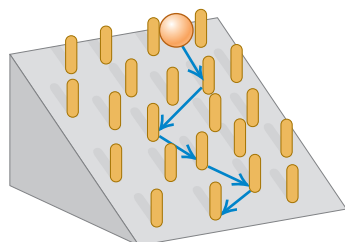


Figure 25.27 The motion of a ball rolling down an inclined plane and bouncing off pegs in its path is analogous to the motion of an electron in a metallic conductor with an electric field present.



We can gain additional insight into electrical conduction by looking at the microscopic origin of conductivity. We'll consider a very simple model that treats the electrons as classical particles and ignores their quantum-mechanical behavior in solids. Using this model, we'll derive an expression for the resistivity of a metal. Even though this model is not entirely correct, it will still help you to develop an intuitive idea of the microscopic basis of conduction.

In the simplest microscopic model of conduction in a metal, each atom in the metallic crystal gives up one or more of its outer electrons. These electrons are then free to move through the crystal, colliding at intervals with the stationary positive ions. The motion of the electrons is analogous to the motion of molecules of a gas moving through a porous bed of sand.

If there is no electric field, the electrons move in straight lines between collisions, the directions of their velocities are random, and on average they never get anywhere (Fig. 25.26a). But if an electric field is present, the paths curve slightly because of the acceleration caused by electric-field forces. Figure 25.26b shows a few paths of an electron in an electric field directed from right to left. As we mentioned in Section 25.1, the average speed of random motion is of the order of $10^6\ \text{m/s}$, while the average drift speed is *much* slower, of the order of $10^{-4}\ \text{m/s}$. The average time between collisions is called the **mean free time**, denoted by τ . Figure 25.27 shows a mechanical analog of this electron motion.

We would like to derive from this model an expression for the resistivity ρ of a material, defined by Eq. (25.5):

$$\rho = \frac{E}{J} \quad (25.21)$$

where E and J are the magnitudes of electric field and current density, respectively. The current density $\vec{J}$ is in turn given by Eq. (25.4):

$$\vec{J} = nq\vec{v}_d \quad (25.22)$$

where n is the number of free electrons per unit volume (the electron concentration), $q = -e$ is the charge of each, and $\vec{v}_d$ is their average drift velocity.

We need to relate the drift velocity $\vec{v}_d$ to the electric field $\vec{E}$. The value of $\vec{v}_d$ is determined by a steady-state condition in which, on average, the velocity *gains* of the charges due to the force of the $\vec{E}$ field are just balanced by the velocity *losses* due to collisions. To clarify this process, let's imagine turning on the two effects one at a time. Suppose that before time $t = 0$ there is no field. The electron motion is then completely random. A typical electron has velocity $\vec{v}_0$ at time $t = 0$, and the value of $\vec{v}_0$ averaged over many electrons (that is, the initial velocity of an average electron) is zero: $(\vec{v}_0)_{\text{av}} = \vec{0}$. Then at time $t = 0$ we turn on a constant electric field $\vec{E}$. The field

exerts a force $\vec{F} = q\vec{E}$ on each charge, and this causes an acceleration $\vec{a}$ in the direction of the force, given by

$$\vec{a} = \frac{\vec{F}}{m} = \frac{q\vec{E}}{m}$$

where m is the electron mass. *Every* electron has this acceleration.

After a time τ , the average time between collisions, we “turn on” the collisions. At time $t = \tau$ an electron that has velocity $\vec{v}_0$ at time $t = 0$ has a velocity

$$\vec{v} = \vec{v}_0 + \vec{a}\tau$$

The velocity $\vec{v}_{\text{av}}$ of an *average* electron at this time is the sum of the averages of the two terms on the right. As we have pointed out, the initial velocity $\vec{v}_0$ is zero for an average electron, so

$$\vec{v}_{\text{av}} = \vec{a}\tau = \frac{q\tau}{m}\vec{E} \quad (25.23)$$

After time $t = \tau$, the tendency of the collisions to decrease the velocity of an average electron (by means of randomizing collisions) just balances the tendency of the $\vec{E}$ field to increase this velocity. Thus the velocity of an average electron, given by Eq. (25.23), is maintained over time and is equal to the drift velocity $\vec{v}_d$:

$$\vec{v}_d = \frac{q\tau}{m}\vec{E}$$

Now we substitute this equation for the drift velocity $\vec{v}_d$ into Eq. (25.22):

$$\vec{J} = nq\vec{v}_d = \frac{nq^2\tau}{m}\vec{E}$$

Comparing this with Eq. (25.21), which we can rewrite as $\vec{J} = \vec{E}/\rho$, and substituting $q = -e$ for an electron, we see that

$$\rho = \frac{m}{ne^2\tau} \quad (25.24)$$

Resistivity of a metal $\rightarrow$ ρ $\leftarrow$ Electron mass
Number of free electrons per unit volume $\rightarrow$ n $\leftarrow$ Average time between collisions
 $\rightarrow$ e $\leftarrow$ Magnitude of electron charge

If n and τ are independent of $\vec{E}$, then the resistivity is independent of $\vec{E}$ and the conducting material obeys Ohm's law.

Turning the interactions on one at a time may seem artificial. But the derivation would come out the same if each electron had its own clock and the $t = 0$ times were different for different electrons. If τ is the average time between collisions, then $\vec{v}_d$ is still the average electron drift velocity, even though the motions of the various electrons aren't actually correlated in the way we postulated.

What about the temperature dependence of resistivity? In a perfect crystal with no atoms out of place, a correct quantum-mechanical analysis would let the free electrons move through the crystal with no collisions at all. But the atoms vibrate about their equilibrium positions. As the temperature increases, the amplitudes of these vibrations increase, collisions become more frequent, and the mean free time τ decreases. So this theory predicts that the resistivity of a metal increases with temperature. In a superconductor, roughly speaking, there are no inelastic collisions, τ is infinite, and the resistivity ρ is zero.

In a pure semiconductor such as silicon or germanium, the number of charge carriers per unit volume, n , is not constant but increases very rapidly with increasing temperature. This increase in n far outweighs the decrease in the mean free time, and in a semiconductor the resistivity always decreases rapidly with increasing temperature. At low

temperatures, n is very small, and the resistivity becomes so large that the material can be considered an insulator.

Electrons gain energy between collisions through the work done on them by the electric field. During collisions they transfer some of this energy to the atoms of the material of the conductor. This leads to an increase in the material's internal energy and temperature; that's why wires carrying current get warm. If the electric field in the material is large enough, an electron can gain enough energy between collisions to knock off electrons that are normally bound to atoms in the material. These can then knock off more electrons, and so on, leading to an avalanche of current. This is the basis of dielectric breakdown in insulators (see Section 24.4).

EXAMPLE 25.11 Mean free time in copper

Calculate the mean free time between collisions in copper at room temperature.

IDENTIFY and SET UP We can obtain an expression for mean free time τ in terms of n , ρ , e , and m by rearranging Eq. (25.24). From Example 25.1 and Table 25.1, for copper $n = 8.5 \times 10^{28} \text{ m}^{-3}$ and $\rho = 1.72 \times 10^{-8} \Omega \cdot \text{m}$. In addition, $e = 1.60 \times 10^{-19} \text{ C}$ and $m = 9.11 \times 10^{-31} \text{ kg}$ for electrons.

EXECUTE From Eq. (25.24), we get

$$\begin{aligned}\tau &= \frac{m}{ne^2\rho} \\ &= \frac{9.11 \times 10^{-31} \text{ kg}}{(8.5 \times 10^{28} \text{ m}^{-3})(1.60 \times 10^{-19} \text{ C})^2(1.72 \times 10^{-8} \Omega \cdot \text{m})} \\ &= 2.4 \times 10^{-14} \text{ s}\end{aligned}$$

EVALUATE The mean free time is the average time between collisions for a given electron. Taking the reciprocal, we find that each electron averages $1/\tau = 4.2 \times 10^{13}$ collisions per second!

KEYCONCEPT The resistivity of a metal is related to the average time that a current-carrying electron can travel through the metal without colliding with one of the metal's ions. The greater this mean free time, the lower the resistivity.

TEST YOUR UNDERSTANDING OF SECTION 25.6 Which of the following factors will, if increased, make it more difficult to produce a certain amount of current in a conductor? (There may be more than one correct answer.) (i) The mass of the moving charged particles in the conductor; (ii) the number of moving charged particles per cubic meter; (iii) the amount of charge on each moving particle; (iv) the average time between collisions for a typical moving charged particle.

ANSWER

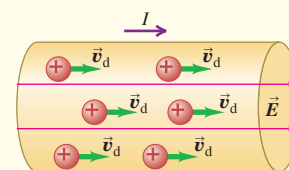
(i) The difficulty of producing a certain amount of current increases as the resistivity ρ increases. From Eq. (25.24), $\rho = m/ne^2\tau$, so increasing the mass m will increase the resistivity. That's because a more massive charged particle will respond more sluggishly to an applied electric field and hence drift more slowly. To produce the same current, a greater electric field would be needed. (Increasing n , e , or τ would decrease the resistivity and make it easier to produce a given current.)

CHAPTER 25 SUMMARY

Current and current density: Current is the amount of charge flowing through a specified area, per unit time. The SI unit of current is the ampere ($1 \text{ A} = 1 \text{ C/s}$). The current I through an area A depends on the concentration n and charge q of the charge carriers, as well as on the magnitude of their drift velocity $\vec{v}_d$. The current density is current per unit cross-sectional area. Current is usually described in terms of a flow of positive charge, even when the charges are actually negative or of both signs. (See Example 25.1.)

$$I = \frac{dQ}{dt} = n|q|v_d A \quad (25.2)$$

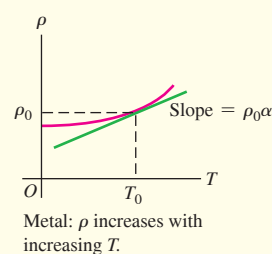
$$\vec{J} = nq\vec{v}_d \quad (25.4)$$



Resistivity: The resistivity ρ of a material is the ratio of the magnitudes of electric field and current density. Good conductors have small resistivity; good insulators have large resistivity. Ohm's law, obeyed approximately by many materials, states that ρ is a constant independent of the value of E . Resistivity usually increases with temperature; for small temperature changes this variation is represented approximately by Eq. (25.6), where α is the temperature coefficient of resistivity.

$$\rho = \frac{E}{J} \quad (25.5)$$

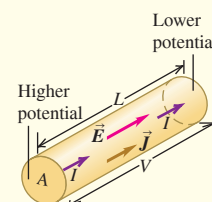
$$\rho(T) = \rho_0[1 + \alpha(T - T_0)] \quad (25.6)$$



Resistors: The potential difference V across a sample of material that obeys Ohm's law is proportional to the current I through the sample. The ratio $V/I = R$ is the resistance of the sample. The SI unit of resistance is the ohm ($1 \Omega = 1 \text{ V/A}$). The resistance of a cylindrical conductor is related to its resistivity ρ , length L , and cross-sectional area A . (See Examples 25.2 and 25.3.)

$$V = IR \quad (25.11)$$

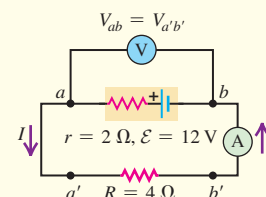
$$R = \frac{\rho L}{A} \quad (25.10)$$



Circuits and emf: A complete circuit has a continuous current-carrying path. A complete circuit carrying a steady current must contain a source of electromotive force (emf) $\mathcal{E}$. The SI unit of electromotive force is the volt (V). Every real source of emf has some internal resistance r , so its terminal potential difference V_{ab} depends on current. (See Examples 25.4–25.7.)

$$V_{ab} = \mathcal{E} - Ir \quad (25.15)$$

(source with internal resistance)



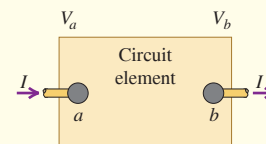
Energy and power in circuits: A circuit element puts energy into a circuit if the current direction is from lower to higher potential in the device, and it takes energy out of the circuit if the current is opposite. The power P equals the product of the potential difference $V_a - V_b = V_{ab}$ and the current I . A resistor always takes electrical energy out of a circuit. (See Examples 25.8–25.10.)

$$P = V_{ab}I \quad (25.17)$$

(general circuit element)

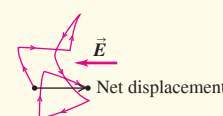
$$P = V_{ab}I = I^2R = \frac{V_{ab}^2}{R} \quad (25.18)$$

(power delivered to a resistor)



Conduction in metals: In a metal, current is due to the motion of electrons. They move freely through the metallic crystal but collide with positive ions. In a crude classical model of this motion, the resistivity of the material can be related to the electron mass, charge, speed of random motion, density, and mean free time between collisions. (See Example 25.11.)

$$\rho = \frac{m}{ne^2\tau} \quad (25.24)$$





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 25.2 and 25.3** (Section 25.3) before attempting these problems.

VP25.3.1 Some airliners use wires of aluminum (resistivity $2.75 \times 10^{-8} \Omega \cdot \text{m}$) rather than copper to reduce weight. The potential difference between the ends of an aluminum wire 35.0 m long is 1.45 V when the wire carries current 1.20 A. Find (a) the electric-field magnitude inside the wire, (b) the resistance of the wire, and (c) the cross-sectional area of the wire.

VP25.3.2 When a current of 0.218 A is present in a wire 20.0 m in length with cross-sectional area $8.19 \times 10^{-7} \text{ m}^2$, the electric field inside the wire is $6.50 \times 10^{-3} \text{ V/m}$. What are (a) the resistance of the wire and (b) the resistivity of the material of which the wire is made?

VP25.3.3 Two wires of circular cross section are made of the same metal. Wire 1 has radius r and length L ; wire 2 has radius $2r$ and length $2L$. The potential difference between the ends of the wire is the same for both wires. For each of the following quantities, find the ratio of its value for wire 2 to its value for wire 1: (a) resistance of the wire; (b) current in the wire; (c) current density in the wire; (d) electric-field magnitude in the wire.

VP25.3.4 The current in a certain wire at 20.0°C is 2.40 A when the potential difference between the ends of the wire is 1.50 V. If the wire is heated to 80.0°C and the potential difference remains the same, the current decreases to 2.00 A. (a) What is the resistance of the wire at 20.0°C ? At 80.0°C ? (b) What is the temperature coefficient of resistivity for the material of which this wire is made?

Be sure to review **EXAMPLE 25.5** (Section 25.4) before attempting these problems.

VP25.5.1 A battery with emf 9.00 V and internal resistance 1.10Ω is in a complete circuit with a resistor of resistance 15.3Ω . Find (a) the current in the circuit, (b) the potential difference across the resistor, and (c) the potential difference across the battery terminals.

VP25.5.2 When an automobile battery with an emf of 12.6 V is connected to a resistor of resistance 25.0Ω , the current in the circuit is 0.480 A. Find (a) the potential difference across the resistor and (b) the internal resistance of the battery.

BRIDGING PROBLEM Resistivity, Temperature, and Power

A toaster using a Nichrome heating element operates on 120 V. When it is switched on at 20°C , the heating element carries an initial current of 1.35 A. A few seconds later the current reaches the steady value of 1.23 A. (a) What is the final temperature of the element? The average value of the temperature coefficient of resistivity for Nichrome over the relevant temperature range is $4.5 \times 10^{-4} (\text{C}^\circ)^{-1}$. (b) What is the power dissipated in the heating element initially and when the current reaches 1.23 A?

SOLUTION GUIDE

IDENTIFY and SET UP

1. A heating element acts as a resistor that converts electrical energy into thermal energy. The resistivity ρ of Nichrome depends on temperature, and hence so does the resistance $R = \rho L/A$ of the heating element and the current $I = V/R$ that it carries.
2. We are given $V = 120 \text{ V}$ and the initial and final values of I . Select an equation that will allow you to find the initial and final values of resistance, and an equation that relates resistance to temperature [the target variable in part (a)].

VP25.5.3 An AA battery has an emf of 1.50 V. After being used for quite a while, one AA battery has a potential difference across its terminals of 1.30 V when it is connected to a light bulb with resistance 20.0Ω . Find (a) the current in this circuit and (b) the internal resistance of the battery.

VP25.5.4 The manufacturer of one brand of AAA battery states that the battery's emf is 1.50 V, its internal resistance is 0.160Ω , and the maximum allowed current for short times is 2.00 A. Find (a) the smallest resistance to which this battery can be connected for short times and (b) the potential difference across the battery terminals when it is so connected.

Be sure to review **EXAMPLES 25.8 and 25.9** (Section 25.5) before attempting these problems.

VP25.9.1 A battery with emf 24.0 V and internal resistance 1.30Ω is in a complete circuit with a resistor of resistance 18.0Ω . Find (a) the current in the circuit, (b) the rate at which energy is converted from chemical to electrical form in the battery, (c) the rate of energy dissipation in the battery, (d) the net power output of the battery, and (e) the rate at which energy is dissipated in the 18.0Ω resistor.

VP25.9.2 When you connect a battery with emf 9.00 V to a resistor of resistance 12.0Ω , energy is dissipated in the resistor at the rate of 6.54 W. Find (a) the current in the circuit, (b) the net power output of the battery, (c) the potential difference across the terminals of the battery, and (d) the internal resistance of the battery.

VP25.9.3 The current is 0.720 A in a circuit in which a battery of emf 12.0 V is connected to a resistor of resistance 16.0Ω . Find (a) the potential difference across the terminals of the battery, (b) the internal resistance of the battery, (c) the power dissipated in the resistor, (d) the rate at which chemical energy is converted to electrical energy in the battery, and (e) the rate of energy dissipation in the battery.

VP25.9.4 A battery with emf $\mathcal{E}$ and internal resistance r is first connected to a resistor of resistance $R_1 = R$, where R is greater than r . This resistor is then disconnected from the battery and replaced with a resistor of resistance $R_2 = 2R$. Find the ratio of (a) the current with the second resistor to the current with the first resistor and (b) the power dissipated in the second resistor to the power dissipated in the first resistor. (c) In which resistor is the current greater? In which is the power dissipated greater?

3. The power P dissipated in the heating element depends on I and V . Select an equation that will allow you to calculate the initial and final values of P .

EXECUTE

4. Combine your equations from step 2 to give a relationship between the initial and final values of I and the initial and final temperatures (20°C and T_{final}).
5. Solve your expression from step 4 for T_{final} .
6. Use your equation from step 3 to find the initial and final powers.

EVALUATE

7. Is the final temperature greater than or less than 20°C ? Does this make sense?
8. Is the final resistance greater than or less than the initial resistance? Again, does this make sense?
9. Is the final power greater than or less than the initial power? Does this agree with your observations in step 8?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q25.1 The definition of resistivity ($\rho = E/J$) implies that an electric field exists inside a conductor. Yet we saw in Chapter 21 that there can be no electrostatic electric field inside a conductor. Is there a contradiction here? Explain.

Q25.2 A cylindrical rod has resistance R . If we triple its length and diameter, what is its resistance, in terms of R ?

Q25.3 A cylindrical rod has resistivity ρ . If we triple its length and diameter, what is its resistivity, in terms of ρ ?

Q25.4 Two copper wires with different diameters are joined end to end. If a current flows in the wire combination, what happens to electrons when they move from the larger-diameter wire into the smaller-diameter wire? Does their drift speed increase, decrease, or stay the same? If the drift speed changes, what is the force that causes the change? Explain your reasoning.

Q25.5 When is a 1.5 V AAA battery *not* actually a 1.5 V battery? That is, when do its terminals provide a potential difference of less than 1.5 V?

Q25.6 Can the potential difference between the terminals of a battery ever be opposite in direction to the emf? If it can, give an example. If it cannot, explain why not.

Q25.7 A rule of thumb used to determine the internal resistance of a source is that it is the open-circuit voltage divided by the short-circuit current. Is this correct? Why or why not?

Q25.8 Batteries are always labeled with their emf; for instance, an AA flashlight battery is labeled “1.5 volts.” Would it also be appropriate to put a label on batteries stating how much current they provide? Why or why not?

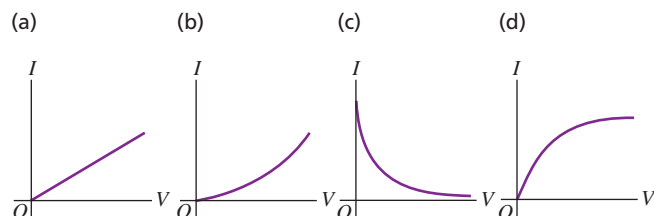
Q25.9 We have seen that a coulomb is an enormous amount of charge; it is virtually impossible to place a charge of 1 C on an object. Yet, a current of 10 A, 10 C/s, is quite reasonable. Explain this apparent discrepancy.

Q25.10 Electrons in an electric circuit pass through a resistor. The wire on either side of the resistor has the same diameter. (a) How does the drift speed of the electrons before entering the resistor compare to the speed after leaving the resistor? Explain your reasoning. (b) How does the potential energy for an electron before entering the resistor compare to the potential energy after leaving the resistor? Explain your reasoning.

Q25.11 Temperature coefficients of resistivity are given in Table 25.2. (a) If a copper heating element is connected to a source of constant voltage, does the electrical power consumed by the heating element increase or decrease as its temperature increases? Explain. (b) A resistor in the form of a carbon cylinder is connected to the voltage source. As the temperature of the cylinder increases, does the electrical power it consumes increase or decrease? Explain.

Q25.12 Which of the graphs in **Fig. Q25.12** best illustrates the current I in a real resistor as a function of the potential difference V across it? Explain.

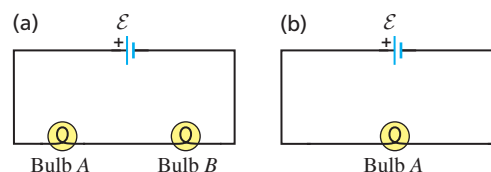
Figure Q25.12



Q25.13 Why does an electric light bulb nearly always burn out just as you turn on the light, almost never while the light is shining?

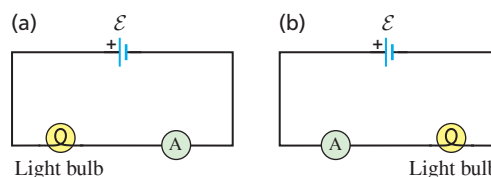
Q25.14 A light bulb glows because it has resistance. The brightness of a light bulb increases with the electrical power dissipated in the bulb. (a) In the circuit shown in **Fig. Q25.14a**, the two bulbs A and B are identical. Compared to bulb A , does bulb B glow more brightly, just as brightly, or less brightly? Explain your reasoning. (b) Bulb B is removed from the circuit and the circuit is completed as shown in **Fig. Q25.14b**. Compared to the brightness of bulb A in **Fig. Q25.14a**, does bulb A now glow more brightly, just as brightly, or less brightly? Explain your reasoning.

Figure Q25.14



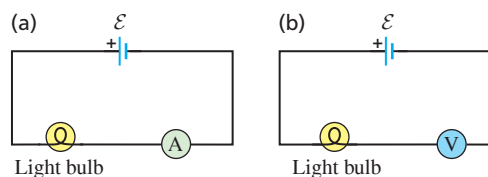
Q25.15 (See Discussion Question Q25.14.) An ideal ammeter A is placed in a circuit with a battery and a light bulb as shown in **Fig. Q25.15a**, and the ammeter reading is noted. The circuit is then reconnected as in **Fig. Q25.15b**, so that the positions of the ammeter and light bulb are reversed. (a) How does the ammeter reading in the situation shown in **Fig. Q25.15a** compare to the reading in the situation shown in **Fig. Q25.15b**? Explain your reasoning. (b) In which situation does the light bulb glow more brightly? Explain your reasoning.

Figure Q25.15



Q25.16 (See Discussion Question Q25.14.) Will a light bulb glow more brightly when it is connected to a battery as shown in **Fig. Q25.16a**, in which an ideal ammeter A is placed in the circuit, or when it is connected as shown in **Fig. Q25.16b**, in which an ideal voltmeter V is placed in the circuit? Explain your reasoning.

Figure Q25.16



Q25.17 The energy that can be extracted from a storage battery is always less than the energy that goes into it while it is being charged. Why?

Q25.18 Eight flashlight batteries in series have an emf of about 12 V, similar to that of a car battery. Could they be used to start a car with a dead battery? Why or why not?

Q25.19 Small aircraft often have 24 V electrical systems rather than the 12 V systems in automobiles, even though the electrical power requirements are roughly the same in both applications. The explanation given by aircraft designers is that a 24 V system weighs less than a 12 V system because thinner wires can be used. Explain why this is so.

Q25.20 Long-distance, electric-power, transmission lines always operate at very high voltage, sometimes as much as 750 kV. What are the advantages of such high voltages? What are the disadvantages?

Q25.21 Ordinary household electric lines in North America usually operate at 120 V. Why is this a desirable voltage, rather than a value considerably larger or smaller? On the other hand, automobiles usually have 12 V electrical systems. Why is this a desirable voltage?

Q25.22 A fuse is a device designed to break a circuit, usually by melting when the current exceeds a certain value. What characteristics should the material of the fuse have?

Q25.23 High-voltage power supplies are sometimes designed intentionally to have rather large internal resistance as a safety precaution. Why is such a power supply with a large internal resistance safer than a supply with the same voltage but lower internal resistance?

Q25.24 The text states that good thermal conductors are also good electrical conductors. If so, why don't the cords used to connect toasters, irons, and similar heat-producing appliances get hot by conduction of heat from the heating element?

EXERCISES

Section 25.1 Current

25.1 • Lightning Strikes. During lightning strikes from a cloud to the ground, currents as high as 25,000 A can occur and last for about 40 μs . How much charge is transferred from the cloud to the earth during such a strike?

25.2 • A silver wire 2.6 mm in diameter transfers a charge of 420 C in 80 min. Silver contains 5.8×10^{28} free electrons per cubic meter. (a) What is the current in the wire? (b) What is the magnitude of the drift velocity of the electrons in the wire?

25.3 • A 5.00 A current runs through a 12 gauge copper wire (diameter 2.05 mm) and through a light bulb. Copper has 8.5×10^{28} free electrons per cubic meter. (a) How many electrons pass through the light bulb each second? (b) What is the current density in the wire? (c) At what speed does a typical electron pass by any given point in the wire? (d) If you were to use wire of twice the diameter, which of the above answers would change? Would they increase or decrease?

25.4 • An 18 gauge copper wire (diameter 1.02 mm) carries a current with a current density of $3.20 \times 10^6 \text{ A/m}^2$. The density of free electrons for copper is 8.5×10^{28} electrons per cubic meter. Calculate (a) the current in the wire and (b) the magnitude of the drift velocity of electrons in the wire.

25.5 • The free-electron density in a copper wire is 8.5×10^{28} electrons/ m^3 . The electric field in the wire is 0.0600 N/C and the temperature of the wire is 20.0°C. (a) What is the drift speed v_d of the electrons in the wire? (b) What is the potential difference between two points in the wire that are separated by 20.0 cm?

25.6 •• You want to produce three 1.00-mm-diameter cylindrical wires, each with a resistance of 1.00 Ω at room temperature. One wire is gold, one is copper, and one is aluminum. Refer to Table 25.1 for the resistivity values. (a) What will be the length of each wire? (b) Gold has a density of $1.93 \times 10^4 \text{ kg/m}^3$. What will be the mass of the gold wire? If you consider the current price of gold, is this wire very expensive?

25.7 • CALC The current in a wire varies with time according to the relationship $I = 55 \text{ A} - (0.65 \text{ A/s}^2)t^2$. (a) How many coulombs of charge pass a cross section of the wire in the time interval between $t = 0$ and $t = 8.0 \text{ s}$? (b) What constant current would transport the same charge in the same time interval?

25.8 • Current passes through a solution of sodium chloride. In 1.00 s, $2.68 \times 10^{16} \text{ Na}^+$ ions arrive at the negative electrode and $3.92 \times 10^{16} \text{ Cl}^-$ ions arrive at the positive electrode. (a) What is the current passing between the electrodes? (b) What is the direction of the current?

25.9 • BIO Transmission of Nerve Impulses. Nerve cells transmit electric signals through their long tubular axons. These signals propagate due to a sudden rush of Na^+ ions, each with charge $+e$, into the axon. Measurements have revealed that typically about $5.6 \times 10^{11} \text{ Na}^+$ ions enter each meter of the axon during a time of 10 ms. What is the current during this inflow of charge in a meter of axon?

25.10 •• The mass density of silver at room temperature is $10.5 \times 10^3 \text{ kg/m}^3$ and its atomic mass is 108 g/mol. If we assume there is one free electron per silver atom, what is the free-electron density for silver, in electrons/ m^3 ? How does your answer compare to the free-electron density for copper given in Example 25.1?

Section 25.2 Resistivity and Section 25.3 Resistance

25.11 •• A metal wire has a circular cross section with radius 0.800 mm. You measure the resistivity of the wire in the following way: You connect one end of the wire to one terminal of a battery that has emf 12.0 V and negligible internal resistance. To the other terminal of the battery you connect a point along the wire so that the length of wire between the battery terminals is d . You measure the current in the wire as a function of d . The currents are small, so the temperature change of the wire is very small. You plot your results as I versus $1/d$ and find that the data lie close to a straight line that has slope 600 A $\cdot$ m. What is the resistivity of the material of which the wire is made?

25.12 • (a) At room temperature, what is the strength of the electric field in a 12 gauge copper wire (diameter 2.05 mm) that is needed to cause a 4.50 A current to flow? (b) What field would be needed if the wire were made of silver instead?

25.13 •• A 1.50 m cylindrical rod of diameter 0.500 cm is connected to a power supply that maintains a constant potential difference of 15.0 V across its ends, while an ammeter measures the current through it. You observe that at room temperature (20.0°C) the ammeter reads 18.5 A, while at 92.0°C it reads 17.2 A. You can ignore any thermal expansion of the rod. Find (a) the resistivity at 20.0°C and (b) the temperature coefficient of resistivity at 20°C for the material of the rod.

25.14 • A copper wire has a square cross 2.3 mm on a side. The wire is 4.0 m long and carries a current of 3.6 A. The density of free electrons is $8.5 \times 10^{28}/\text{m}^3$. Find the magnitudes of (a) the current density in the wire and (b) the electric field in the wire. (c) How much time is required for an electron to travel the length of the wire?

25.15 • A 14 gauge copper wire of diameter 1.628 mm carries a current of 12.5 mA. (a) What is the potential difference across a 2.00 m length of the wire? (b) What would the potential difference in part (a) be if the wire were silver instead of copper, but all else were the same?

25.16 •• A wire 6.50 m long with diameter of 2.05 mm has a resistance of 0.0290 Ω . What material is the wire most likely made of?

25.17 • A copper wire has radius 0.800 mm and carries current I at 20.0°C. A silver wire with radius 0.500 mm carries the same current and is also at 20.0°C. Let E_{Cu} be the electric field in the copper wire and E_{Ag} be the electric field in the silver wire. What is the ratio $E_{\text{Cu}}/E_{\text{Ag}}$?

25.18 •• A ductile metal wire has resistance R . What will be the resistance of this wire in terms of R if it is stretched to three times its original length, assuming that the density and resistivity of the material do not change when the wire is stretched? (*Hint:* The amount of metal does not change, so stretching out the wire will affect its cross-sectional area.)

25.19 • In household wiring, copper wire 2.05 mm in diameter is often used. Find the resistance of a 24.0 m length of this wire.

25.20 • What diameter must a copper wire have if its resistance is to be the same as that of an equal length of aluminum wire with diameter 2.14 mm?

25.21 • A current-carrying gold wire has diameter 0.84 mm. The electric field in the wire is 0.49 V/m. What are (a) the current carried by the wire; (b) the potential difference between two points in the wire 6.4 m apart; (c) the resistance of a 6.4 m length of this wire?

25.22 • You apply a potential difference of 4.50 V between the ends of a wire that is 2.50 m in length and 0.654 mm in radius. The resulting current through the wire is 17.6 A. What is the resistivity of the wire?

25.23 • (a) What is the resistance of a Nichrome wire at 0.0°C if its resistance is 100.00 Ω at 11.5°C? (b) What is the resistance of a carbon rod at 25.8°C if its resistance is 0.0160 Ω at 0.0°C?

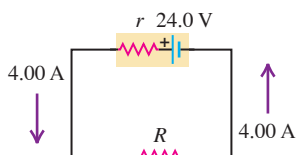
25.24 • A carbon resistor is to be used as a thermometer. On a winter day when the temperature is 4.0°C, the resistance of the carbon resistor is 217.3 Ω . What is the temperature on a spring day when the resistance is 215.8 Ω ? (Take the reference temperature T_0 to be 4.0°C.)

Section 25.4 Electromotive Force and Circuits

25.25 • A copper transmission cable 100 km long and 10.0 cm in diameter carries a current of 125 A. (a) What is the potential drop across the cable? (b) How much electrical energy is dissipated as thermal energy every hour?

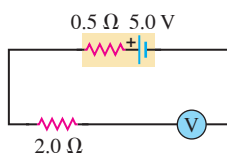
25.26 • Consider the circuit shown in **Fig. E25.26**. The terminal voltage of the 24.0 V battery is 21.2 V. What are (a) the internal resistance r of the battery and (b) the resistance R of the circuit resistor?

Figure E25.26



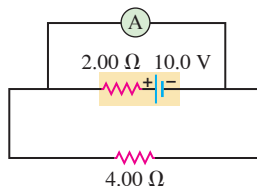
25.27 • An ideal voltmeter V is connected to a 2.0 Ω resistor and a battery with emf 5.0 V and internal resistance 0.5 Ω as shown in **Fig. E25.27**. (a) What is the current in the 2.0 Ω resistor? (b) What is the terminal voltage of the battery? (c) What is the reading on the voltmeter? Explain your answers.

Figure E25.27



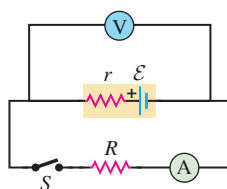
25.28 • An idealized ammeter is connected to a battery as shown in **Fig. E25.28**. Find (a) the reading of the ammeter, (b) the current through the 4.00 Ω resistor, (c) the terminal voltage of the battery.

Figure E25.28



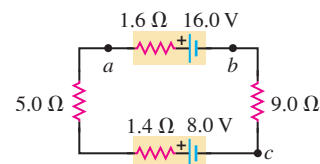
25.29 • When switch S in **Fig. E25.29** is open, the voltmeter V reads 3.08 V. When the switch is closed, the voltmeter reading drops to 2.97 V, and the ammeter A reads 1.65 A. Find the emf, the internal resistance of the battery, and the circuit resistance R . Assume that the two meters are ideal, so they don't affect the circuit.

Figure E25.29



25.30 • The circuit shown in **Fig. E25.30** contains two batteries, each with an emf and an internal resistance, and two resistors. Find (a) the current in the circuit (magnitude and direction); (b) the terminal voltage V_{ab} of the 16.0 V battery; (c) the potential difference V_{ac} of point a with respect to point c . (d) Using **Fig. 25.20** as a model, graph the potential rises and drops in this circuit.

Figure E25.30



25.31 • In the circuit shown in **Fig. E25.30**, the 16.0 V battery is removed and reinserted with the opposite polarity, so its negative terminal is now next to point a . Find (a) the current in the circuit (magnitude and direction) and (b) the terminal voltage V_{ab} of the 16.0 V battery.

25.32 • A battery has emf 30.0 V and internal resistance r . A 9.00 Ω resistor is connected to the terminals of the battery, and the voltage drop across the resistor is 27.0 V. What is the internal resistance of the battery?

Section 25.5 Energy and Power in Electric Circuits

25.33 • A battery has emf 24.0 V and internal resistance 3.00 Ω . A resistor of resistance R is connected to the battery. What are the two values of R for which 21.0 W of electrical power is consumed in the resistor?

25.34 • When a resistor with resistance R is connected to a 1.50 V flashlight battery, the resistor consumes 0.0625 W of electrical power. (Throughout, assume that each battery has negligible internal resistance.) (a) What power does the resistor consume if it is connected to a 12.6 V car battery? Assume that R remains constant when the power consumption changes. (b) The resistor is connected to a battery and consumes 5.00 W. What is the voltage of this battery?

25.35 • **Light Bulbs.** The power rating of a light bulb (such as a 100 W bulb) is the power it dissipates when connected across a 120 V potential difference. What is the resistance of (a) a 100 W bulb and (b) a 60 W bulb? (c) How much current does each bulb draw in normal use?

25.36 • If a “75 W” bulb (see Problem 25.35) is connected across a 220 V potential difference (as is used in Europe), how much power does it dissipate? Ignore the temperature dependence of the bulb's resistance.

25.37 • **European Light Bulb.** In Europe the standard voltage in homes is 220 V instead of the 120 V used in the United States. Therefore a “100 W” European bulb would be intended for use with a 220 V potential difference (see Problem 25.36). (a) If you bring a “100 W” European bulb home to the United States, what should be its U.S. power rating? (b) How much current will the 100 W European bulb draw in normal use in the United States?

25.38 • A battery-powered global positioning system (GPS) receiver operating on 9.0 V draws a current of 0.13 A. How much electrical energy does it consume during 30 minutes?

25.39 • Consider the circuit of **Fig. E25.30**. (a) What is the total rate at which electrical energy is dissipated in the 5.0 Ω and 9.0 Ω resistors? (b) What is the power output of the 16.0 V battery? (c) At what rate is electrical energy being converted to other forms in the 8.0 V battery? (d) Show that the power output of the 16.0 V battery equals the overall rate of consumption of electrical energy in the rest of the circuit.

25.40 • BIO Electric Eels. Electric eels generate electric pulses along their skin that can be used to stun an enemy when they come into contact with it. Tests have shown that these pulses can be up to 500 V and produce currents of 80 mA (or even larger). A typical pulse lasts for 10 ms. What power and how much energy are delivered to the unfortunate enemy with a single pulse, assuming a steady current?

25.41 • BIO Treatment of Heart Failure. A heart defibrillator is used to enable the heart to start beating if it has stopped. This is done by passing a large current of 12 A through the body at 25 V for a very short time, usually about 3.0 ms. (a) What power does the defibrillator deliver to the body, and (b) how much energy is transferred?

25.42 •• The battery for a certain cell phone is rated at 3.70 V. According to the manufacturer it can produce 3.15×10^4 J of electrical energy, enough for 5.25 h of operation, before needing to be recharged. Find the average current that this cell phone draws when turned on.

25.43 •• The capacity of a storage battery, such as those used in automobile electrical systems, is rated in ampere-hours (A · h). A 50 A · h battery can supply a current of 50 A for 1.0 h, or 25 A for 2.0 h, and so on. (a) What total energy can be supplied by a 12 V, 60 A · h battery if its internal resistance is negligible? (b) What volume (in liters) of gasoline has a total heat of combustion equal to the energy obtained in part (a)? (See Section 17.6; the density of gasoline is 900 kg/m^3 .) (c) If a generator with an average electrical power output of 0.45 kW is connected to the battery, how much time will be required for it to charge the battery fully?

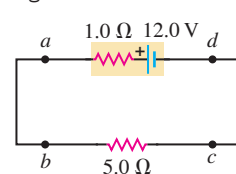
25.44 • A battery has emf $\mathcal{E}$ and internal resistance $r = 2.00 \Omega$. A 12.0Ω resistor is connected to the battery, and the resistor consumes electrical power at a rate of 96.0 J/s. What is the emf of the battery?

25.45 •• A 25.0Ω bulb is connected across the terminals of a 12.0 V battery having 3.50Ω of internal resistance. What percentage of the power of the battery is dissipated across the internal resistance and hence is not available to the bulb?

25.46 •• A typical small flashlight contains two batteries, each having an emf of 1.5 V, connected in series with a bulb having resistance 17Ω . (a) If the internal resistance of the batteries is negligible, what power is delivered to the bulb? (b) If the batteries last for 5.0 h, what is the total energy delivered to the bulb? (c) The resistance of real batteries increases as they run down. If the initial internal resistance is negligible, what is the combined internal resistance of both batteries when the power to the bulb has decreased to half its initial value? (Assume that the resistance of the bulb is constant. Actually, it will change somewhat when the current through the filament changes, because this changes the temperature of the filament and hence the resistivity of the filament wire.)

25.47 • In the circuit in Fig. E25.47, find (a) the rate of conversion of internal (chemical) energy to electrical energy within the battery; (b) the rate of dissipation of electrical energy in the battery; (c) the rate of dissipation of electrical energy in the external resistor.

Figure E25.47



25.48 • A “540 W” electric heater is designed to operate from 120 V lines. (a) What is its operating resistance? (b) What current does it draw? (c) If the line voltage drops to 110 V, what power does the heater take? (Assume that the resistance is constant. Actually, it will change because of the change in temperature.) (d) The heater coils are metallic, so that the resistance of the heater decreases with decreasing temperature. If the change of resistance with temperature is taken into account, will the electrical power consumed by the heater be larger or smaller than what you calculated in part (c)? Explain.

Section 25.6 Theory of Metallic Conduction

25.49 •• Pure silicon at room temperature contains approximately 1.0×10^{16} free electrons per cubic meter. (a) Referring to Table 25.1, calculate the mean free time τ for silicon at room temperature. (b) Your answer in part (a) is much greater than the mean free time for copper given in Example 25.11. Why, then, does pure silicon have such a high resistivity compared to copper?

PROBLEMS

25.50 •• A cell phone or computer battery has three ratings marked on it: a charge capacity listed in mAh (milliamp-hours), an energy capacity in Wh (watt-hours), and a potential rating in volts. (a) What are these three values for your cell phone? (b) Convert the charge capacity Q into coulombs. (c) Convert the energy capacity U into joules. (d) Multiply the charge rating Q by the potential rating V , and verify that this is equivalent to the energy capacity U . (e) If the charge Q were stored on a parallel-plate capacitor with air as the dielectric, at the potential V , what would be the corresponding capacitance? (f) If the energy in the battery were used to heat 1 L of water, estimate the corresponding change in the water temperature? (The heat capacity of water is $4190 \text{ J/kg} \cdot \text{K}$.)

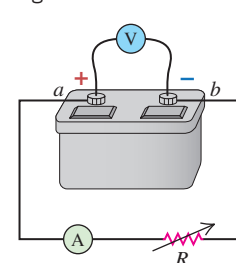
25.51 • An electrical conductor designed to carry large currents has a circular cross section 2.50 mm in diameter and is 14.0 m long. The resistance between its ends is 0.104Ω . (a) What is the resistivity of the material? (b) If the electric-field magnitude in the conductor is 1.28 V/m, what is the total current? (c) If the material has 8.5×10^{28} free electrons per cubic meter, find the average drift speed under the conditions of part (b).

25.52 •• (a) Estimate the maximum volume of water the hot-water heater in your home can hold. (b) How much heat would be required to raise the temperature of that water from 20°C to a standard household hot-water temperature of 45°C ? (Water has a density of 1.00 kg/L and a heat capacity of $4190 \text{ J/kg} \cdot \text{K}$.) (c) Suppose the water should be fully heated in 1.5 h. To what power output does this correspond? (d) If the element has a potential difference of 220 V, what current is required? (e) What should be the resistance of the element?

25.53 •• On your first day at work as an electrical technician, you are asked to determine the resistance per meter of a long piece of wire. The company you work for is poorly equipped. You find a battery, a voltmeter, and an ammeter, but no meter for directly measuring resistance (an ohmmeter). You put the leads from the voltmeter across the terminals of the battery, and the meter reads 12.6 V. You cut off a 20.0 m length of wire and connect it to the battery, with an ammeter in series with it to measure the current in the wire. The ammeter reads 7.00 A. You then cut off a 40.0 m length of wire and connect it to the battery, again with the ammeter in series to measure the current. The ammeter reads 4.20 A. Even though the equipment you have available to you is limited, your boss assures you of its high quality: The ammeter has very small resistance, and the voltmeter has very large resistance. What is the resistance of 1 meter of wire?

25.54 •• In the circuit shown in Fig. P25.54, R is a variable resistor whose value ranges from 0 to ∞ , and a and b are the terminals of a battery that has an emf $\mathcal{E} = 15.0 \text{ V}$ and an internal resistance of 4.00Ω . The ammeter and voltmeter are idealized meters. As R varies over its full range of values, what will be the largest and smallest readings of (a) the voltmeter and (b) the ammeter? (c) Sketch qualitative graphs of the readings of both meters as functions of R .

Figure P25.54



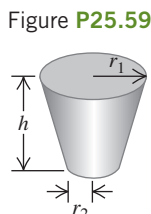
25.55 • A 3.00 m length of copper wire at 20°C has a 1.20-m-long section with diameter 1.60 mm and a 1.80-m-long section with diameter 0.80 mm. There is a current of 2.5 mA in the 1.60-mm-diameter section. (a) What is the current in the 0.80-mm-diameter section? (b) What is the magnitude of $\vec{E}$ in the 1.60-mm-diameter section? (c) What is the magnitude of $\vec{E}$ in the 0.80-mm-diameter section? (d) What is the potential difference between the ends of the 3.00 m length of wire?

25.56 •• A heating element made of tungsten wire is connected to a large battery that has negligible internal resistance. When the heating element reaches 80.0°C, it consumes electrical energy at a rate of 480 W. What is its power consumption when its temperature is 150.0°C? Assume that the temperature coefficient of resistivity has the value given in Table 25.2 and that it is constant over the temperature range in this problem. In Eq. (25.12) take T_0 to be 20.0°C.

25.57 •• CP BIO Struck by Lightning. Lightning strikes can involve currents as high as 25,000 A that last for about 40 μ s. If a person is struck by a bolt of lightning with these properties, the current will pass through his body. We shall assume that his mass is 75 kg, that he is wet (after all, he is in a rainstorm) and therefore has a resistance of 1.0 k Ω , and that his body is all water (which is reasonable for a rough, but plausible, approximation). (a) By how many degrees Celsius would this lightning bolt increase the temperature of 75 kg of water? (b) Given that the internal body temperature is about 37°C, would the person's temperature actually increase that much? Why not? What would happen first?

25.58 •• A resistor with resistance R is connected to a battery that has emf 12.0 V and internal resistance $r = 0.40 \Omega$. For what two values of R will the power dissipated in the resistor be 80.0 W?

25.59 • CALC A material of resistivity ρ is formed into a solid, truncated cone of height h and radii r_1 and r_2 at either end (**Fig. P25.59**). (a) Calculate the resistance of the cone between the two flat end faces. (*Hint: Imagine slicing the cone into very many thin disks, and calculate the resistance of one such disk.*) (b) Show that your result agrees with Eq. (25.10) when $r_1 = r_2$.



25.60 • CALC The region between two concentric conducting spheres with radii a and b is filled with a conducting material with resistivity ρ . (a) Show that the resistance between the spheres is given by

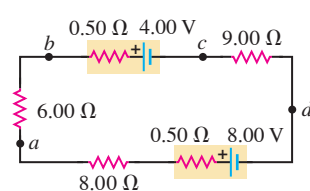
$$R = \frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$$

(b) Derive an expression for the current density as a function of radius, in terms of the potential difference V_{ab} between the spheres. (c) Show that the result in part (a) reduces to Eq. (25.10) when the separation $L = b - a$ between the spheres is small.

25.61 • The potential difference across the terminals of a battery is 8.40 V when there is a current of 1.50 A in the battery from the negative to the positive terminal. When the current is 3.50 A in the reverse direction, the potential difference becomes 10.20 V. (a) What is the internal resistance of the battery? (b) What is the emf of the battery?

25.62 • (a) What is the potential difference V_{ad} in the circuit of **Fig. P25.62**? (b) What is the terminal voltage of the 4.00 V battery? (c) A battery with emf 10.30 V and internal resistance 0.50 Ω is inserted in the circuit at d , with its negative terminal connected to the negative terminal of the 8.00 V battery. What is the difference of potential V_{bc} between the terminals of the 4.00 V battery now?

Figure P25.62



25.63 • BIO The average bulk resistivity of the human body (apart from surface resistance of the skin) is about 5.0 $\Omega \cdot \text{m}$. The conducting path between the hands can be represented approximately as a cylinder 1.6 m long and 0.10 m in diameter. The skin resistance can be made negligible by soaking the hands in salt water. (a) What is the resistance between the hands if the skin resistance is negligible? (b) What potential difference between the hands is needed for a lethal shock current of 100 mA? (Note that your result shows that small potential differences produce dangerous currents when the skin is damp.) (c) With the current in part (b), what power is dissipated in the body?

25.64 •• BIO A person with body resistance between his hands of 10 k Ω accidentally grasps the terminals of a 14 kV power supply. (a) If the internal resistance of the power supply is 2000 Ω , what is the current through the person's body? (b) What is the power dissipated in his body? (c) If the power supply is to be made safe by increasing its internal resistance, what should the internal resistance be for the maximum current in the above situation to be 1.00 mA or less?

25.65 • A typical cost for electrical power is \$0.120 per kilowatt-hour. (a) Some people leave their porch light on all the time. What is the yearly cost to keep a 75 W bulb burning day and night? (b) Suppose your refrigerator uses 400 W of power when it's running, and it runs 8 hours a day. What is the yearly cost of operating your refrigerator?

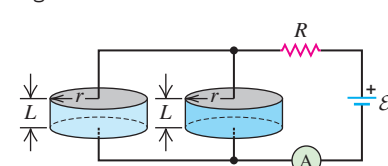
25.66 •• A cylindrical copper cable 1.50 km long is connected across a 220.0 V potential difference. (a) What should be its diameter so that it produces heat at a rate of 90.0 W? (b) What is the electric field inside the cable under these conditions?

25.67 • CALC A 1.50 m cylinder of radius 1.10 cm is made of a complicated mixture of materials. Its resistivity depends on the distance x from the left end and obeys the formula $\rho(x) = a + bx^2$, where a and b are constants. At the left end, the resistivity is $2.25 \times 10^{-8} \Omega \cdot \text{m}$, while at the right end it is $8.50 \times 10^{-8} \Omega \cdot \text{m}$. (a) What is the resistance of this rod? (b) What is the electric field at its midpoint if it carries a 1.75 A current? (c) If we cut the rod into two 75.0 cm halves, what is the resistance of each half?

25.68 •• Compact Fluorescent Bulbs. Compact fluorescent bulbs are much more efficient at producing light than are ordinary incandescent bulbs. They initially cost much more, but they last far longer and use much less electricity. According to one study of these bulbs, a compact bulb that produces as much light as a 100 W incandescent bulb uses only 23 W of power. The compact bulb lasts 10,000 hours, on the average, and costs \$11.00, whereas the incandescent bulb costs only \$0.75, but lasts just 750 hours. The study assumed that electricity costs \$0.080 per kilowatt-hour and that the bulbs are on for 4.0 h per day. (a) What is the total cost (including the price of the bulbs) to run each bulb for 3.0 years? (b) How much do you save over 3.0 years if you use a compact fluorescent bulb instead of an incandescent bulb? (c) What is the resistance of a "100 W" fluorescent bulb? (Remember, it actually uses only 23 W of power and operates across 120 V.)

25.69 ••• CP Two cylindrical cans with insulating sides and conducting end caps are filled with water, attached to the circuitry shown in **Fig. P25.69**, and used to determine salinity levels. The cans are identical, with radius $r = 5.00$ cm and length $L = 3.00$ cm. The

Figure P25.69

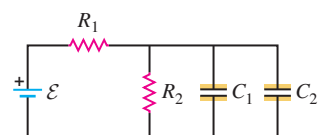


battery supplies a potential of 10.0 V, has a negligible internal resistance, and is connected in series with a resistor $R = 15.0 \Omega$. The left cylinder is filled with pure distilled water, which has infinite resistivity. The right cylinder is filled with a saltwater solution. It is known that

the resistivity of the saltwater solution is determined by the relationship $\rho = (s_0/s) \Omega \cdot \text{m}$, where s is the salinity in parts per thousand (ppt) and $s_0 = 6.30$ ppt. (a) The ammeter registers a current of 484 mA. What is the salinity of the saltwater solution? (b) The left cylinder acts as a capacitor. Use Eq. (24.19) for its capacitance. How much charge is present on its upper plate? Note that pure water has a dielectric constant of 80.4. (c) At what rate is energy dissipated by the saltwater? (d) For what salinity level would the 15.0Ω resistor dissipate half the power supplied by the battery?

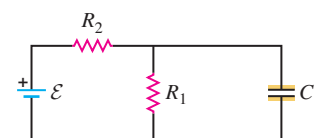
25.70 •• CP Consider the circuit Figure P25.70

shown in Fig. P25.70. The battery has emf 72.0 V and negligible internal resistance. $R_2 = 2.00 \Omega$, $C_1 = 3.00 \mu\text{F}$, and $C_2 = 6.00 \mu\text{F}$. After the capacitors have attained their final charges, the charge on C_1 is $Q_1 = 18.0 \mu\text{C}$. What is (a) the final charge on C_2 ; (b) the resistance R_1 ?



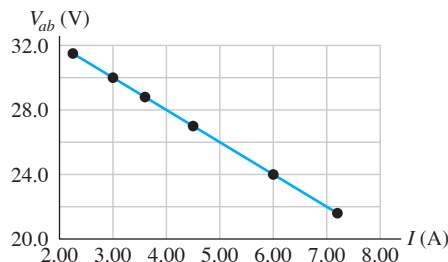
25.71 •• CP Consider the circuit shown in Fig. P25.71. The emf source has negligible internal resistance. The resistors have resistances $R_1 = 6.00 \Omega$ and $R_2 = 4.00 \Omega$. The capacitor has capacitance $C = 9.00 \mu\text{F}$. When the capacitor is fully charged, the magnitude of the charge on its plates is $Q = 36.0 \mu\text{C}$. Calculate the emf $\mathcal{E}$.

Figure P25.71



25.72 •• DATA An external resistor R is connected between the terminals of a battery. The value of R varies. For each R value, the current I in the circuit and the terminal voltage V_{ab} of the battery are measured. The results are plotted in Fig. P25.72, a graph of V_{ab} versus I that shows the best straight-line fit to the data. (a) Use the graph in Fig. P25.72 to calculate the battery's emf and internal resistance. (b) For what value of R is V_{ab} equal to 80.0% of the battery emf?

Figure P25.72



25.73 •• DATA The voltage drop V_{ab} across each of resistors A and B was measured as a function of the current I in the resistor. The results are shown in the table:

Resistor A				
I (A)	0.50	1.00	2.00	4.00
V_{ab} (V)	2.55	3.11	3.77	4.58
Resistor B				
I (A)	0.50	1.00	2.00	4.00
V_{ab} (V)	1.94	3.88	7.76	15.52

(a) For each resistor, graph V_{ab} as a function of I and graph the resistance $R = V_{ab}/I$ as a function of I . (b) Does resistor A obey Ohm's law? Explain. (c) Does resistor B obey Ohm's law? Explain. (d) What is the power dissipated in A if it is connected to a 4.00 V battery that has negligible internal resistance? (e) What is the power dissipated in B if it is connected to the battery?

25.74 •• DATA According to the U.S. National Electrical Code, copper wire used for interior wiring of houses, hotels, office buildings, and industrial plants is permitted to carry no more than a specified maximum amount of current. The table shows values of the maximum current I_{max} for several common sizes of wire with varnished cambric insulation. The "wire gauge" is a standard used to describe the diameter of wires. Note that the larger the diameter of the wire, the *smaller* the wire gauge.

Wire gauge	Diameter (cm)	I_{max} (A)
14	0.163	18
12	0.205	25
10	0.259	30
8	0.326	40
6	0.412	60
5	0.462	65
4	0.519	85

(a) What considerations determine the maximum current-carrying capacity of household wiring? (b) A total of 4200 W of power is to be supplied through the wires of a house to the household electrical appliances. If the potential difference across the group of appliances is 120 V, determine the gauge of the thinnest permissible wire that can be used. (c) Suppose the wire used in this house is of the gauge found in part (b) and has total length 42.0 m. At what rate is energy dissipated in the wires? (d) The house is built in a community where the consumer cost of electrical energy is \$0.11 per kilowatt-hour. If the house were built with wire of the next larger diameter than that found in part (b), what would be the savings in electricity costs in one year? Assume that the appliances are kept on for an average of 12 hours a day.

25.75 ••• CP CALC A material with resistivity ρ is formed into a cylinder of length L and outer radius r_{outer} . A cylindrical core with radius r_{inner} is removed from the axis of this cylinder and filled with a conducting material, which is attached to a wire. The outer surface of the cylinder is coated with a conducting material and attached to another wire. (a) If the second wire has potential V greater than the first wire, in what direction does the local electric field point inside of the cylinder? (b) The magnitude of this electric field is c/r , where c is a constant and r is the distance from the axis of the cylinder. Use the relationship $V = \int \vec{E} \cdot d\vec{l}$ to determine the constant c . (c) What is the resistance of this device? (d) A 1.00-cm-long hollow cylindrical resistor has an inner radius of 1.50 mm and an outer radius of 3.00 mm. The material is a blend of powdered carbon and ceramic whose resistivity ρ may be altered by changing the amount of carbon. If this device should have a resistance of 6.80 k Ω , what value of ρ should be selected?

25.76 ••• An incandescent light bulb uses a coiled filament of tungsten that is 580 mm long with a diameter of $46.0 \mu\text{m}$. At 20.0°C tungsten has a resistivity of $5.25 \times 10^{-8} \Omega \cdot \text{m}$. Its temperature coefficient of resistivity is $0.0045 (\text{C}^\circ)^{-1}$, and this remains accurate even at high temperatures. The temperature of the filament increases linearly with current, from 20°C when no current flows to 2520°C at 1.00 A of current. (a) What is the resistance of the light bulb at 20°C ? (b) What is the current through the light bulb when the potential difference across its terminals is 120 V? (Hint: First determine the temperature as a function of the current; then use this to determine the resistance as a function of the current. Substitute this result into the equation $V = IR$ and solve for the current I .) (c) What is the resistance when the potential is 120 V? (d) How much energy does the light bulb dissipate in 1 min when 120 V is supplied across its terminals? (e) How much energy does the light bulb dissipate in 1 min when half that voltage is supplied?

CHALLENGE PROBLEMS

25.77 ••• **CALC** The resistivity of a semiconductor can be modified by adding different amounts of impurities. A rod of semiconducting material of length L and cross-sectional area A lies along the x -axis between $x = 0$ and $x = L$. The material obeys Ohm's law, and its resistivity varies along the rod according to $\rho(x) = \rho_0 \exp(-x/L)$. The end of the rod at $x = 0$ is at a potential V_0 greater than the end at $x = L$. (a) Find the total resistance of the rod and the current in the rod. (b) Find the electric-field magnitude $E(x)$ in the rod as a function of x . (c) Find the electric potential $V(x)$ in the rod as a function of x . (d) Graph the functions $\rho(x)$, $E(x)$, and $V(x)$ for values of x between $x = 0$ and $x = L$.

25.78 ••• An external resistor with resistance R is connected to a battery that has emf $\mathcal{E}$ and internal resistance r . Let P be the electrical power output of the source. By conservation of energy, P is equal to the power consumed by R . What is the value of P in the limit that R is (a) very small; (b) very large? (c) Show that the power output of the battery is a maximum when $R = r$. What is this maximum P in terms of $\mathcal{E}$ and r ? (d) A battery has $\mathcal{E} = 64.0$ V and $r = 4.00$ Ω . What is the power output of this battery when it is connected to a resistor R , for $R = 2.00$ Ω , $R = 4.00$ Ω , and $R = 6.00$ Ω ? Are your results consistent with the general result that you derived in part (b)?

MCAT-STYLE PASSAGE PROBLEMS

10 Spiderweb Conductivity. Some types of spiders build webs that consist of threads made of dry silk coated with a solution of a variety of compounds. This coating leaves the threads, which are used to capture prey, *hygroscopic*—that is, they attract water from the atmosphere. It has been hypothesized that this aqueous coating makes the threads good electrical conductors. To test the electrical properties of coated thread, researchers placed a 5 mm length of thread between two electrical contacts. * The researchers stretched the thread in 1 mm increments to more than twice its original length, and then allowed it to return to its original length, again in 1 mm increments. Some of the resistance measurements are shown in the table:

Resistance of thread (10^9 Ω)	9	19	41	63	102	76	50	24
Length of thread (mm)	5	7	9	11	13	9	7	5

*Based on F. Vollrath and D. Edmonds, "Consequences of electrical conductivity in an orb spider's capture web," *Naturwissenschaften* (100:12, December 2013, pp. 1163–69).

25.79 What is the best explanation for the behavior exhibited in the data? (a) Longer threads can carry more current than shorter threads do and so make better electrical conductors. (b) The thread stops being a conductor when it is stretched to 13 mm, due to breaks that occur in the thin coating. (c) As the thread is stretched, the coating thins and its resistance increases; as the thread is relaxed, the coating returns nearly to its original state. (d) The resistance of the thread increases with distance from the end of the thread.

25.80 If the conductivity of the thread results from the aqueous coating only, how does the cross-sectional area A of the coating compare when the thread is 13 mm long versus the starting length of 5 mm? Assume that the resistivity of the coating remains constant and the coating is uniform along the thread. $A_{13 \text{ mm}}$ is about (a) $\frac{1}{10} A_{5 \text{ mm}}$; (b) $\frac{1}{4} A_{5 \text{ mm}}$; (c) $\frac{2}{5} A_{5 \text{ mm}}$; (d) the same as $A_{5 \text{ mm}}$.

25.81 What is the maximum current that flows in the thread during this experiment if the voltage source is a 9 V battery? (a) About 1 A; (b) about 0.1 A; (c) about 1 μ A; (d) about 1 nA.

25.82 In another experiment, a piece of the web is suspended so that it can move freely. When either a positively charged object or a negatively charged object is brought near the web, the thread is observed to move toward the charged object. What is the best interpretation of this observation? The web is (a) a negatively charged conductor; (b) a positively charged conductor; (c) either a positively or negatively charged conductor; (d) an electrically neutral conductor.

ANSWERS

Chapter Opening Question ?

(iii) The current out equals the current in. In other words, charge must enter the bulb at the same rate as it exits the bulb. It is not "used up" or consumed as it flows through the bulb.

Key Example VARIATION Problems

VP25.3.1 (a) 4.14×10^{-2} V/m (b) 1.21 Ω (c) 7.97×10^{-7} m²

VP25.3.2 (a) 0.596 Ω (b) 2.44×10^{-8} $\Omega \cdot \text{m}$

VP25.3.3 (a) $\frac{1}{2}$ (b) 2 (c) $\frac{1}{2}$ (d) $\frac{1}{2}$

VP25.3.4 (a) 0.625 Ω at 20.0°C , 0.750 Ω at 80.0°C (b) 0.00333 ($^\circ\text{C}$)⁻¹

VP25.5.1 (a) 0.549 A (b) 8.40 V (c) 8.40 V

VP25.5.2 (a) 12.0 V (b) 1.3 Ω

VP25.5.3 (a) 0.0650 A (b) 3.1 Ω

VP25.5.4 (a) 0.590 Ω (b) 1.18 V

VP25.9.1 (a) 1.24 A (b) 29.8 W (c) 2.01 W (d) 27.8 W (e) 27.8 W

VP25.9.2 (a) 0.738 A (b) 6.54 W (c) 8.86 V (d) 0.2 Ω

VP25.9.3 (a) 11.5 V (b) 0.7 Ω (c) 8.29 W (d) 8.64 W (e) 0.3 W

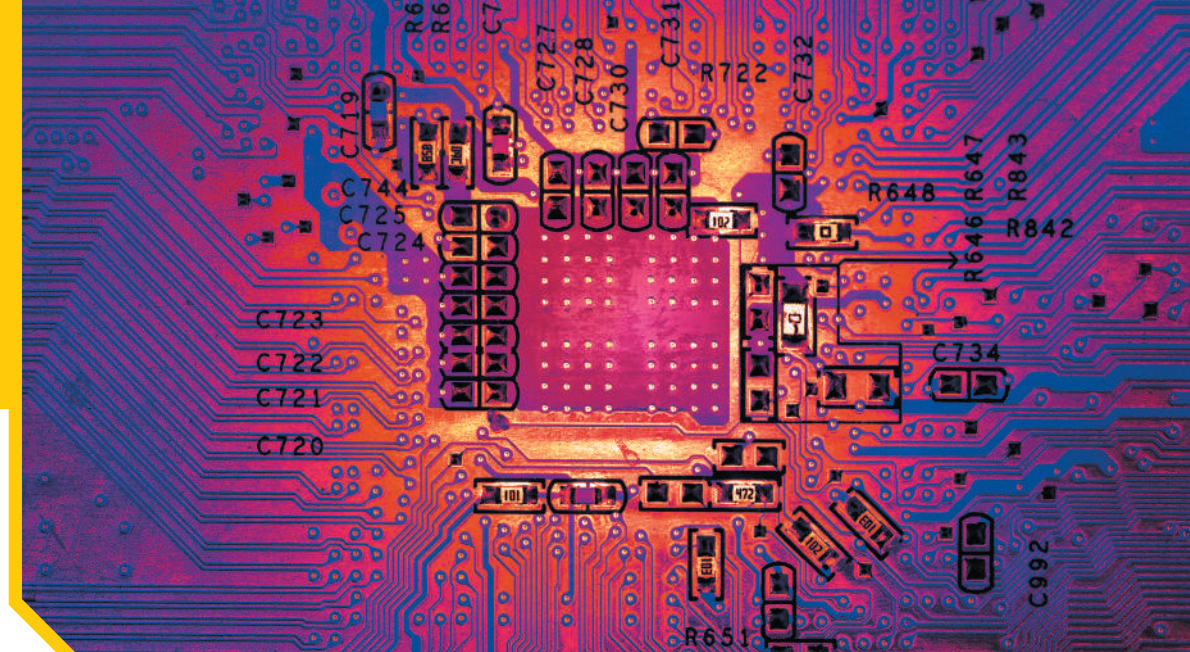
VP25.9.4 (a) $i_2/i_1 = (r + R)/(r + 2R)$
(b) $P_2/P_1 = 2(r + R)^2/(r + 2R)^2$ (c) R_1, R_1

Bridging Problem

(a) 237°C

(b) 162 W initially, 148 W at 1.23 A

? In a complex circuit like the one on this circuit board, is it possible to connect several resistors with different resistances so that all of them have the same potential difference? (i) Yes, and the current will be the same through all of the resistors; (ii) yes, but the current may be different through different resistors; (iii) no; (iv) the answer depends on the value of the potential difference.



26 Direct-Current Circuits

If you look inside a mobile phone, a computer, or under the hood of a car, you'll find circuits of much greater complexity than the simple circuits we studied in Chapter 25. Whether connected by wires or integrated in a semiconductor chip, these circuits often include several sources, resistors, and other circuit elements interconnected in a *network*.

In this chapter we study general methods for analyzing such networks, including how to find voltages and currents of circuit elements. We'll learn how to determine the equivalent resistance for several resistors connected in series or in parallel. For more general networks we need two rules called *Kirchhoff's rules*. One is based on the principle of conservation of charge applied to a junction; the other is derived from energy conservation for a charge moving around a closed loop. We'll discuss instruments for measuring various electrical quantities. We'll also look at a circuit containing resistance and capacitance, in which the current varies with time.

Our principal concern in this chapter is with **direct-current** (dc) circuits, in which the direction of the current does not change with time. Flashlights and automobile wiring systems are examples of direct-current circuits. Household electrical power is supplied in the form of **alternating current** (ac), in which the current oscillates back and forth. The same principles for analyzing networks apply to both kinds of circuits, and we conclude this chapter with a look at household wiring systems. We'll discuss alternating-current circuits in detail in Chapter 31.

26.1 RESISTORS IN SERIES AND PARALLEL

Resistors turn up in all kinds of circuits, ranging from hair dryers and space heaters to circuits that limit or divide current or reduce or divide a voltage. Such circuits often contain several resistors, so it's appropriate to consider *combinations* of resistors. A simple example is a string of light bulbs used for holiday decorations; each bulb acts as a resistor, and from a circuit-analysis perspective the string of bulbs is simply a combination of resistors.

Suppose we have three resistors with resistances R_1 , R_2 , and R_3 . **Figure 26.1** shows four different ways in which they might be connected between points *a* and *b*. When several circuit elements such as resistors, batteries, and motors are connected in sequence as

LEARNING OUTCOMES

In this chapter, you'll learn...

- 26.1** How to analyze circuits with multiple resistors in series or parallel.
- 26.2** Rules that you can apply to any circuit with more than one loop.
- 26.3** How to use an ammeter, voltmeter, ohmmeter, or potentiometer in a circuit.
- 26.4** How to analyze circuits that include both a resistor and a capacitor.
- 26.5** How electric power is distributed in the home.

You'll need to review...

- 24.2** Capacitors in series and parallel.
- 25.4** Current, ammeters, and voltmeters.
- 25.5** Power in a circuit.

in Fig. 26.1a, with only a single current path between the points, we say that they are connected in **series**. We studied *capacitors* in series in Section 24.2; we found that, because of conservation of charge, capacitors in series all have the same charge if they are initially uncharged. In circuits we're often more interested in the *current*, which is charge flow per unit time.

The resistors in Fig. 26.1b are said to be connected in **parallel** between points *a* and *b*. Each resistor provides an alternative path between the points. For circuit elements that are connected in parallel, the *potential difference* is the same across each element. We studied capacitors in parallel in Section 24.2.

In Fig. 26.1c, resistors R_2 and R_3 are in parallel, and this combination is in series with R_1 . In Fig. 26.1d, R_2 and R_3 are in series, and this combination is in parallel with R_1 .

For any combination of resistors we can always find a *single* resistor that could replace the combination and result in the same total current and potential difference. For example, a string of holiday light bulbs could be replaced by a single, appropriately chosen light bulb that would draw the same current and have the same potential difference between its terminals as the original string of bulbs. The resistance of this single resistor is called the **equivalent resistance** of the combination. If any one of the networks in Fig. 26.1 were replaced by its equivalent resistance R_{eq} , we could write

$$V_{ab} = IR_{\text{eq}} \quad \text{or} \quad R_{\text{eq}} = \frac{V_{ab}}{I}$$

where V_{ab} is the potential difference between terminals *a* and *b* of the network and I is the current at point *a* or *b*. To compute an equivalent resistance, we assume a potential difference V_{ab} across the actual network, compute the corresponding current I , and take the ratio V_{ab}/I .

Resistors in Series

We can derive general equations for the equivalent resistance of a series or parallel combination of resistors. In Fig. 26.1a, the three resistors are in *series*, so the current I is the same in all of them. (Recall from Section 25.4 that current is *not* “used up” as it passes through a circuit.) Applying $V = IR$ to each resistor, we have

$$V_{ax} = IR_1 \quad V_{xy} = IR_2 \quad V_{yb} = IR_3$$

The potential differences across each resistor need not be the same (except for the special case in which all three resistances are equal). The potential difference V_{ab} across the entire combination is the sum of these individual potential differences:

$$V_{ab} = V_{ax} + V_{xy} + V_{yb} = I(R_1 + R_2 + R_3)$$

and so

$$\frac{V_{ab}}{I} = R_1 + R_2 + R_3$$

The ratio V_{ab}/I is, by definition, the equivalent resistance R_{eq} . Therefore

$$R_{\text{eq}} = R_1 + R_2 + R_3$$

It is easy to generalize this to *any* number of resistors:

Resistors in series:

Equivalent resistance of series combination

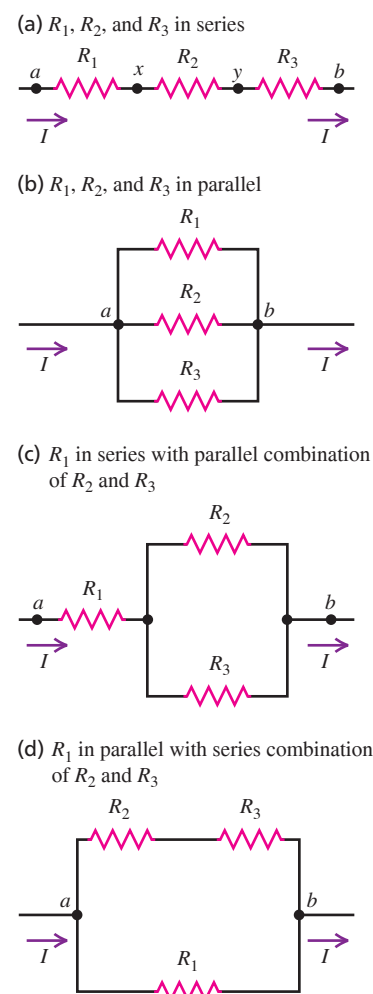
$$R_{\text{eq}} = R_1 + R_2 + R_3 + \cdots$$

Resistances of individual resistors

(26.1)

The equivalent resistance of a series combination equals the sum of the individual resistances. The equivalent resistance is *greater than* any individual resistance.

Figure 26.1 Four different ways of connecting three resistors.



CAUTION Resistors vs. capacitors in series Don't confuse *resistors* in series with *capacitors* in series. Resistors in series add *directly* [Eq. (26.1)] because the voltage across each is directly proportional to its resistance and to the common current. Capacitors in series [Eq. (24.5)] add *reciprocally*; the voltage across each is directly proportional to the common charge but *inversely* proportional to the individual capacitance. ■

Figure 26.2 A car's headlights and taillights are connected in parallel. Hence each light is exposed to the full potential difference supplied by the car's electrical system, giving maximum brightness. Another advantage is that if one headlight or taillight burns out, the other one keeps shining (see Example 26.2).



Resistors in Parallel

If the resistors are in *parallel*, as in Fig. 26.1b, the current through each resistor need **?** not be the same. But the potential difference between the terminals of each resistor must be the same and equal to V_{ab} (**Fig. 26.2**). (Remember that the potential difference between any two points does not depend on the path taken between the points.) Let's call the currents in the three resistors I_1 , I_2 , and I_3 . Then from $I = V/R$,

$$I_1 = \frac{V_{ab}}{R_1} \quad I_2 = \frac{V_{ab}}{R_2} \quad I_3 = \frac{V_{ab}}{R_3}$$

In general, the current is different through each resistor. Because charge neither accumulates at nor drains out of point a , the total current I must equal the sum of the three currents in the resistors:

$$I = I_1 + I_2 + I_3 = V_{ab} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

or

$$\frac{I}{V_{ab}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

But by the definition of the equivalent resistance R_{eq} , $I/V_{ab} = 1/R_{eq}$, so

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Again it is easy to generalize to *any* number of resistors in parallel:

$$\text{Resistors in parallel:} \quad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots \quad (26.2)$$

Equivalent resistance of parallel combination Resistances of individual resistors

The reciprocal of the equivalent resistance of a parallel combination equals the sum of the reciprocals of the individual resistances. The equivalent resistance is always *less than* any individual resistance.

CAUTION Resistors vs. capacitors in parallel Note the differences between *resistors* in parallel and *capacitors* in parallel. Resistors in parallel add *reciprocally* [Eq. (26.2)] because the current in each is proportional to the common voltage across them and *inversely* proportional to the resistance of each. Capacitors in parallel add *directly* [Eq. (24.7)] because the charge on each is proportional to the common voltage across them and *directly* proportional to the capacitance of each. ■

For the special case of *two* resistors in parallel,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1 + R_2}{R_1 R_2} \quad \text{and}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \quad (\text{two resistors in parallel}) \quad (26.3)$$

Because $V_{ab} = I_1 R_1 = I_2 R_2$, it follows that

$$\frac{I_1}{I_2} = \frac{R_2}{R_1} \quad (\text{two resistors in parallel}) \quad (26.4)$$

Thus the currents carried by two resistors in parallel are *inversely proportional* to their resistances. More current goes through the path of least resistance.

PROBLEM-SOLVING STRATEGY 26.1 Resistors in Series and Parallel

IDENTIFY the relevant concepts: As in Fig. 26.1, many resistor networks are made up of resistors in series, in parallel, or a combination thereof. Such networks can be replaced by a single equivalent resistor. The logic is similar to that of Problem-Solving Strategy 24.1 for networks of capacitors.

SET UP the problem using the following steps:

1. Make a drawing of the resistor network.
2. Identify groups of resistors connected in series or parallel.
3. Identify the target variables. They could include the equivalent resistance of the network, the potential difference across each resistor, or the current through each resistor.

EXECUTE the solution as follows:

1. Use Eq. (26.1) or (26.2), respectively, to find the equivalent resistance for series or parallel combinations.
2. If the network is more complex, try reducing it to series and parallel combinations. For example, in Fig. 26.1c we first replace the

parallel combination of R_2 and R_3 with its equivalent resistance; this then forms a series combination with R_1 . In Fig. 26.1d, the combination of R_2 and R_3 in series forms a parallel combination with R_1 .

3. Keep in mind that the total potential difference across resistors connected in series is the sum of the individual potential differences. The potential difference across resistors connected in parallel is the same for every resistor and equals the potential difference across the combination.
4. The current through resistors connected in series is the same through every resistor and equals the current through the combination. The total current through resistors connected in parallel is the sum of the currents through the individual resistors.

EVALUATE your answer: Check whether your results are consistent. The equivalent resistance of resistors connected in series should be greater than that of any individual resistor; that of resistors in parallel should be less than that of any individual resistor.

EXAMPLE 26.1 Equivalent resistance

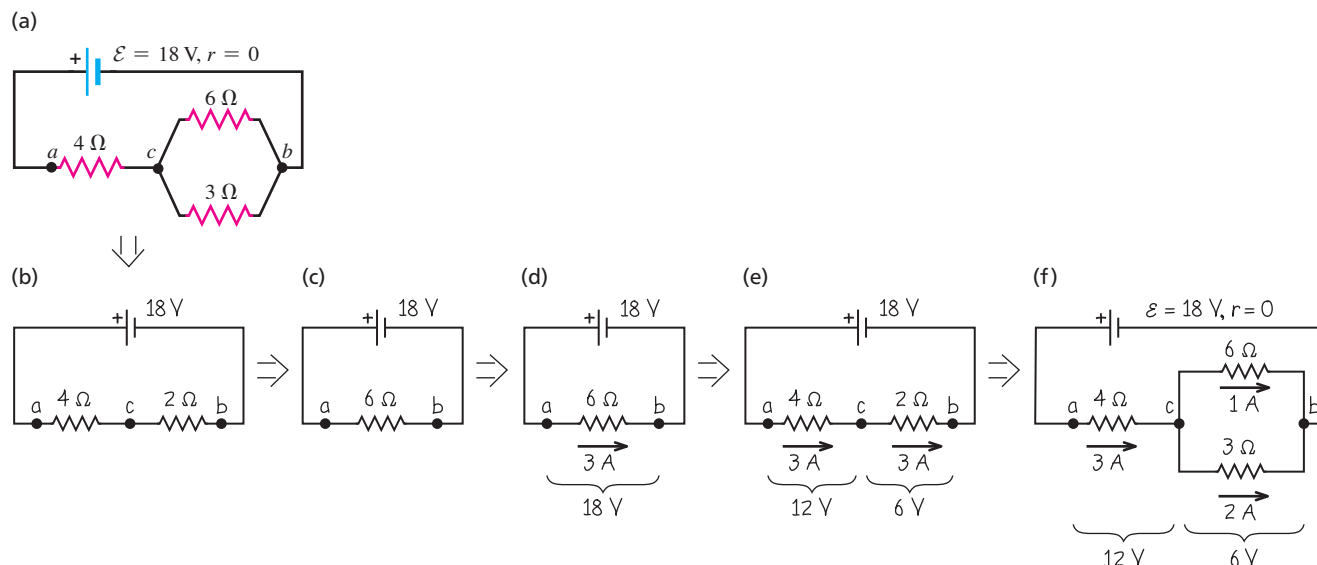
WITH VARIATION PROBLEMS

Find the equivalent resistance of the network in **Fig. 26.3a** and the current in each resistor. The source of emf has negligible internal resistance.

IDENTIFY and SET UP This network of three resistors is a *combination* of series and parallel resistances, as in Fig. 26.1c. We determine the equivalent resistance of the parallel $6\ \Omega$ and $3\ \Omega$ resistors, and then

that of their series combination with the $4\ \Omega$ resistor: This is the equivalent resistance R_{eq} of the network as a whole. We then find the current in the emf, which is the same as that in the $4\ \Omega$ resistor. The potential difference is the same across each of the parallel $6\ \Omega$ and $3\ \Omega$ resistors; we use this to determine how the current is divided between these.

Figure 26.3 Steps in reducing a combination of resistors to a single equivalent resistor and finding the current in each resistor.



Continued

EXECUTE Figures 26.3b and 26.3c show successive steps in reducing the network to a single equivalent resistance R_{eq} . From Eq. (26.2), the $6\ \Omega$ and $3\ \Omega$ resistors in parallel in Fig. 26.3a are equivalent to the single $2\ \Omega$ resistor in Fig. 26.3b:

$$\frac{1}{R_{6+3\ \Omega}} = \frac{1}{6\ \Omega} + \frac{1}{3\ \Omega} = \frac{1}{2\ \Omega}$$

[Equation (26.3) gives the same result.] From Eq. (26.1) the series combination of this $2\ \Omega$ resistor with the $4\ \Omega$ resistor is equivalent to the single $6\ \Omega$ resistor in Fig. 26.3c.

We reverse these steps to find the current in each resistor of the original network. In the circuit shown in Fig. 26.3d (identical to Fig. 26.3c), the current is $I = V_{ab}/R = (18\text{ V})/(6\ \Omega) = 3\text{ A}$. So the current in the $4\ \Omega$ and $2\ \Omega$ resistors in Fig. 26.3e (identical to Fig. 26.3b) is also 3 A . The potential difference V_{cb} across the $2\ \Omega$ resistor is therefore $V_{cb} = IR = (3\text{ A})(2\ \Omega) = 6\text{ V}$. This potential difference must also be 6 V in Fig. 26.3f (identical to Fig. 26.3a). From $I = V_{cb}/R$, the

currents in the $6\ \Omega$ and $3\ \Omega$ resistors in Fig. 26.3f are, respectively, $(6\text{ V})/(6\ \Omega) = 1\text{ A}$ and $(6\text{ V})/(3\ \Omega) = 2\text{ A}$.

EVALUATE Note that for the two resistors in parallel between points c and b in Fig. 26.3f, there is twice as much current through the $3\ \Omega$ resistor as through the $6\ \Omega$ resistor; more current goes through the path of least resistance, in accordance with Eq. (26.4). Note also that the total current through these two resistors is 3 A , the same as it is through the $4\ \Omega$ resistor between points a and c .

KEYCONCEPT When resistors are combined in *series*, the current is the same through each resistor but the potential differences across the resistors may not be the same. The equivalent resistance of a series combination equals the sum of the individual resistances. When resistors are combined in *parallel*, the potential difference across each resistor is the same but the currents through the resistors may not be the same. The reciprocal of the equivalent resistance of a parallel combination equals the sum of the reciprocals of the individual resistances.

EXAMPLE 26.2 Series versus parallel combinations

WITH VARIATION PROBLEMS

Two identical incandescent light bulbs, each with resistance $R = 2\ \Omega$, are connected to a source with $\mathcal{E} = 8\text{ V}$ and negligible internal resistance. Find the current through each bulb, the potential difference across each bulb, and the power delivered to each bulb and to the entire network if the bulbs are connected (a) in series and (b) in parallel. (c) Suppose one of the bulbs burns out; that is, its filament breaks and current can no longer flow through it. What happens to the other bulb in the series case? In the parallel case?

IDENTIFY and SET UP The light bulbs are just resistors in simple series and parallel connections (Figs. 26.4a and 26.4b). Once we find the current I through each bulb, we can find the power delivered to each bulb by using Eq. (25.18), $P = I^2R = V^2/R$.

EXECUTE (a) From Eq. (26.1) the equivalent resistance of the two bulbs between points a and c in Fig. 26.4a is $R_{\text{eq}} = 2R = 2(2\ \Omega) = 4\ \Omega$. In series, the current is the same through each bulb:

$$I = \frac{V_{ac}}{R_{\text{eq}}} = \frac{8\text{ V}}{4\ \Omega} = 2\text{ A}$$

Since the bulbs have the same resistance, the potential difference is the same across each bulb:

$$V_{ab} = V_{bc} = IR = (2\text{ A})(2\ \Omega) = 4\text{ V}$$

From Eq. (25.18), the power delivered to each bulb is

$$P = I^2R = (2\text{ A})^2(2\ \Omega) = 8\text{ W} \quad \text{or}$$

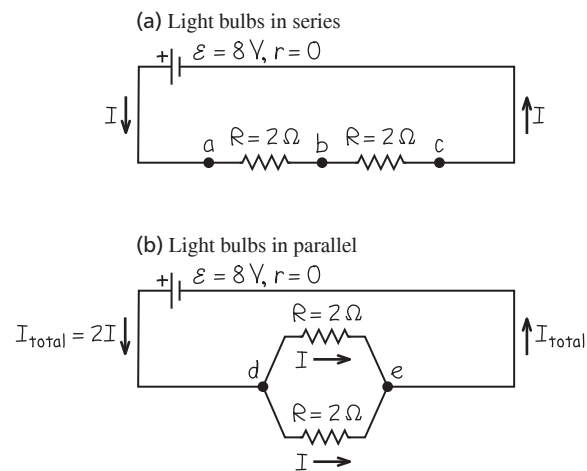
$$P = \frac{V_{ab}^2}{R} = \frac{V_{bc}^2}{R} = \frac{(4\text{ V})^2}{2\ \Omega} = 8\text{ W}$$

The total power delivered to both bulbs is $P_{\text{tot}} = 2P = 16\text{ W}$.

(b) If the bulbs are in parallel, as in Fig. 26.4b, the potential difference V_{de} across each bulb is the same and equal to 8 V , the terminal voltage of the source. Hence the current through each light bulb is

$$I = \frac{V_{de}}{R} = \frac{8\text{ V}}{2\ \Omega} = 4\text{ A}$$

Figure 26.4 Our sketches for this problem.



and the power delivered to each bulb is

$$P = I^2R = (4\text{ A})^2(2\ \Omega) = 32\text{ W} \quad \text{or}$$

$$P = \frac{V_{de}^2}{R} = \frac{(8\text{ V})^2}{2\ \Omega} = 32\text{ W}$$

Both the potential difference across each bulb and the current through each bulb are twice as great as in the series case. Hence the power delivered to each bulb is *four* times greater, and each bulb is brighter.

The total power delivered to the parallel network is $P_{\text{total}} = 2P = 64\text{ W}$, four times greater than in the series case. The increased power compared to the series case isn't obtained "for free"; energy is extracted from the source four times more rapidly in the parallel case than in the series case. If the source is a battery, it will be used up four times as fast.

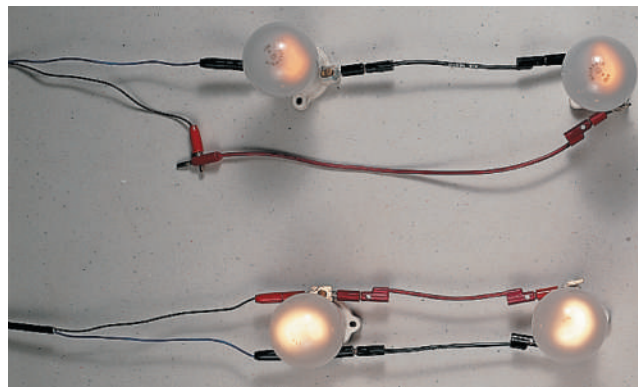
(c) In the series case the same current flows through both bulbs. If one bulb burns out, there will be no current in the circuit, and neither bulb will glow.

In the parallel case the potential difference across either bulb is unchanged if a bulb burns out. The current through the functional bulb and the power delivered to it are unchanged.

EVALUATE Our calculation isn't completely accurate, because the resistance $R = V/I$ of real light bulbs depends on the potential difference V across the bulb. That's because the filament resistance increases with increasing operating temperature and therefore with increasing V . But bulbs connected in series across a source do in fact glow less brightly than when connected in parallel across the same source (**Fig. 26.5**).

KEYCONCEPT Two or more resistors connected in parallel to a source each have more current through them and each draw more power than if they were connected in series to the same source.

Figure 26.5 When connected to the same source, two incandescent light bulbs in series (shown at top) draw less power and glow less brightly than when they are in parallel (shown at bottom).



TEST YOUR UNDERSTANDING OF SECTION 26.1 Suppose all three of the resistors shown in Fig. 26.1 have the same resistance, so $R_1 = R_2 = R_3 = R$. Rank the four arrangements shown in parts (a)–(d) of Fig. 26.1 in order of their equivalent resistance, from highest to lowest.

ANSWER

(a), (c), (d), (b) Here's why: The three resistors in Fig. 26.1a are in series, so $R_{eq} = R + R + R = 3R$. In Fig. 26.1b the three resistors are in parallel, so $1/R_{eq} = 1/R + 1/R + 1/R = 3/R$ and $R_{eq} = R/3$. In Fig. 26.1c the second and third resistors are in parallel, so their equivalent resistance R_{23} is given by $1/R_{23} = 1/R + 1/R = 2/R$; hence $R_{23} = R/2$. This combination is in series with the first resistor, so the three resistors together have equivalent resistance $R_{eq} = R + R/2 = 3R/2$. In Fig. 26.1d the second and third resistors are in series, so their equivalent resistance is $R_{23} = R + R = 2R$. This combination is in parallel with the first resistor, so the equivalent resistance of the three-resistor combination is given by $1/R_{eq} = 1/R + 1/2R = 3/2R$. Hence $R_{eq} = 2R/3$.

26.2 KIRCHHOFF'S RULES

Many practical resistor networks cannot be reduced to simple series-parallel combinations.

Figure 26.6a shows a dc power supply with emf $\mathcal{E}_1$ charging a battery with a smaller emf $\mathcal{E}_2$ and feeding current to a light bulb with resistance R . **Figure 26.6b** is a “bridge” circuit, used in many different types of measurement and control systems. (Problem 26.74 describes one important application of a “bridge” circuit.) To analyze these networks, we'll use the techniques developed by the German physicist Gustav Robert Kirchhoff (1824–1887).

First, here are two terms that we'll use often. A **junction** in a circuit is a point where three or more conductors meet. A **loop** is any closed conducting path. In Fig. 26.6a points a and b are junctions, but points c and d are not; in Fig. 26.6b points a , b , c , and d are junctions, but points e and f are not. The blue lines in Figs. 26.6a and 26.6b show some possible loops in these circuits.

Kirchhoff's rules are the following two statements:

Kirchhoff's junction rule
(valid at any junction):

The sum of the currents into any junction ...

$$\sum I = 0 \quad \dots \text{equals zero.} \quad (26.5)$$

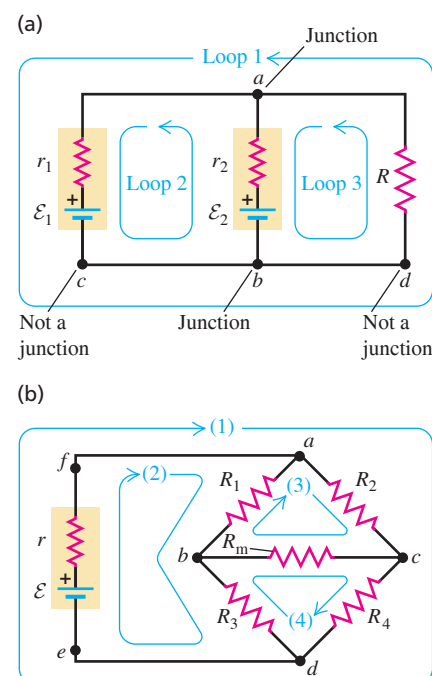
Kirchhoff's loop rule
(valid for any closed loop):

The sum of the potential differences around any loop ...

$$\sum V = 0 \quad \dots \text{equals zero.} \quad (26.6)$$

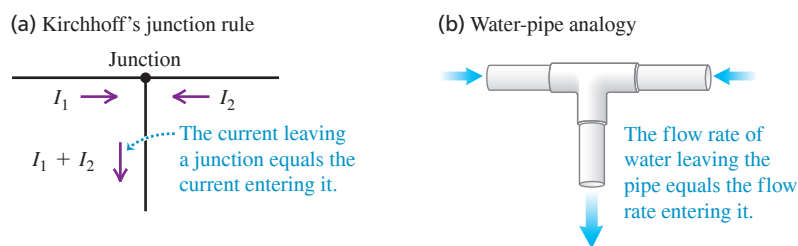
Note that the potential differences V in Eq. (26.6) include those associated with all circuit elements in the loop, including emfs and resistors.

Figure 26.6 Two networks that cannot be reduced to simple series-parallel combinations of resistors.



CAUTION **Current need not split equally at a junction** As a rule, at a junction where the current in a wire splits to follow two or more separate paths, the current does *not* split equally among the different paths. More current flows along a path that has less resistance. ■

Figure 26.7 Kirchhoff's junction rule states that as much current flows into a junction as flows out of it.



The junction rule is based on *conservation of electric charge*. No charge can accumulate at a junction, so the total charge entering the junction per unit time must equal the total charge leaving per unit time (**Fig. 26.7a**). Charge per unit time is current, so if we consider the currents entering a junction to be positive and those leaving to be negative, the algebraic sum of currents into a junction must be zero. It's like a T branch in a water pipe (**Fig. 26.7b**); if you have a total of 1 liter per minute coming in the two pipes, you can't have 3 liters per minute going out the third pipe. We used the junction rule (without saying so) in Section 26.1 in the derivation of Eq. (26.2) for resistors in parallel.

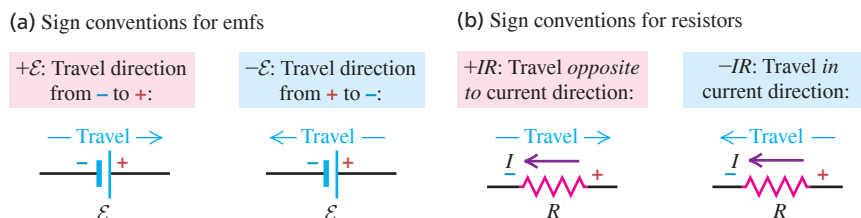
The loop rule is a statement that the electrostatic force is *conservative*. Suppose we go around a loop, measuring potential differences across successive circuit elements as we go. When we return to the starting point, we must find that the *algebraic sum* of these differences is zero; otherwise, we could not say that the potential at this point has a definite value.

Sign Conventions for the Loop Rule

In applying the loop rule, we need some sign conventions. Problem-Solving Strategy 26.2 describes in detail how to use these, but here's a quick overview. We first assume a direction for the current in each branch of the circuit and mark it on a diagram of the circuit. Then, starting at any point in the circuit, we imagine traveling around a loop, adding emfs and IR terms as we come to them. When we travel through a source in the direction from $-$ to $+$, the emf is considered to be *positive*; when we travel from $+$ to $-$, the emf is considered to be *negative* (**Fig. 26.8a**). When we travel through a resistor in the *same* direction as the assumed current, the IR term is *negative* because the current goes in the direction of decreasing potential. When we travel through a resistor in the direction *opposite* to the assumed current, the IR term is *positive* because this represents a rise of potential (**Fig. 26.8b**).

Kirchhoff's two rules are all we need to solve a wide variety of network problems. Usually, some of the emfs, currents, and resistances are known, and others are unknown. We must always obtain from Kirchhoff's rules a number of independent equations equal to the number of unknowns so that we can solve the equations simultaneously. Often the hardest part of the solution is keeping track of algebraic signs!

Figure 26.8 Use these sign conventions when you apply Kirchhoff's loop rule. In each part of the figure "Travel" is the direction that we imagine going around the loop, which is not necessarily the direction of the current.



PROBLEM-SOLVING STRATEGY 26.2 Kirchhoff's Rules

IDENTIFY the relevant concepts: Kirchhoff's rules are useful for analyzing any electric circuit.

SET UP the problem using the following steps:

1. Draw a circuit diagram, leaving room to label all quantities, known and unknown. Indicate an assumed direction for each unknown current and emf. (Kirchhoff's rules will yield the magnitudes and directions of unknown currents and emfs. If the actual direction of a quantity is opposite to your assumption, the resulting quantity will have a negative sign.)
2. As you label currents, it's helpful to use Kirchhoff's junction rule, as in Fig. 26.9, so as to express the currents in terms of as few quantities as possible.
3. Identify the target variables.

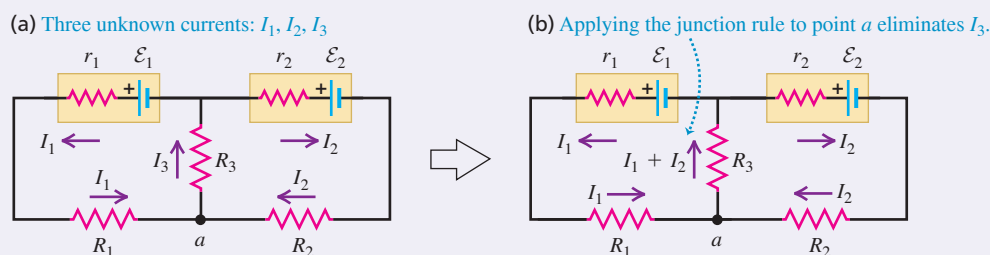
EXECUTE the solution as follows:

1. Choose any loop in the network and choose a direction (clockwise or counterclockwise) to travel around the loop as you apply Kirchhoff's loop rule. The direction need not be the same as any assumed current direction.

2. Travel around the loop in the chosen direction, adding potential differences algebraically as you cross them. Use the sign conventions of Fig. 26.8.
3. Equate the sum obtained in step 2 to zero in accordance with the loop rule.
4. If you need more independent equations, choose another loop and repeat steps 1–3; continue until you have as many independent equations as unknowns or until every circuit element has been included in at least one loop.
5. Solve the equations simultaneously to determine the unknowns.
6. You can use the loop-rule bookkeeping system to find the potential V_{ab} of any point a with respect to any other point b . Start at b and add the potential changes you encounter in going from b to a ; use the same sign rules as in step 2. The algebraic sum of these changes is $V_{ab} = V_a - V_b$.

EVALUATE your answer: Check all the steps in your algebra. Apply steps 1 and 2 to a loop you have not yet considered; if the sum of potential drops isn't zero, you've made an error somewhere.

Figure 26.9 Applying the junction rule to point a reduces the number of unknown currents from three to two.



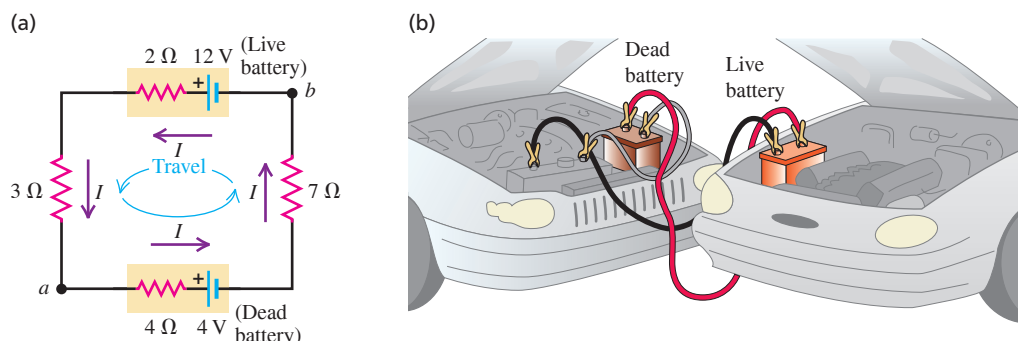
EXAMPLE 26.3 A single-loop circuit

WITH VARIATION PROBLEMS

The circuit shown in Fig. 26.10a contains two batteries, each with an emf and an internal resistance, and two resistors. Find (a) the current in the circuit, (b) the potential difference V_{ab} , and (c) the power output of the emf of each battery.

IDENTIFY and SET UP There are no junctions in this single-loop circuit, so we don't need Kirchhoff's junction rule. To apply Kirchhoff's loop rule, we first assume a direction for the current; let's assume a counterclockwise direction as shown in Fig. 26.10a.

Figure 26.10 (a) In this example we travel around the loop in the same direction as the assumed current, so all the IR terms are negative. The potential decreases as we travel from $+$ to $-$ through the bottom emf but increases as we travel from $-$ to $+$ through the top emf. (b) A real-life example of a circuit of this kind.



Continued

EXECUTE (a) Starting at a and traveling counterclockwise with the current, we add potential increases and decreases and equate the sum to zero as in Eq. (26.6):

$$-I(4\ \Omega) - 4\text{ V} - I(7\ \Omega) + 12\text{ V} - I(2\ \Omega) - I(3\ \Omega) = 0$$

Collecting like terms and solving for I , we find

$$8\text{ V} = I(16\ \Omega) \quad \text{and} \quad I = 0.5\text{ A}$$

The positive result for I shows that our assumed current direction is correct.

(b) To find V_{ab} , the potential at a with respect to b , we start at b and add potential changes as we go toward a . There are two paths from b to a ; taking the lower one, we find

$$V_{ab} = (0.5\text{ A})(7\ \Omega) + 4\text{ V} + (0.5\text{ A})(4\ \Omega) = 9.5\text{ V}$$

Point a is at 9.5 V higher potential than b . All the terms in this sum, including the IR terms, are positive because each represents an *increase* in potential as we go from b to a . For the upper path,

$$V_{ab} = 12\text{ V} - (0.5\text{ A})(2\ \Omega) - (0.5\text{ A})(3\ \Omega) = 9.5\text{ V}$$

Here the IR terms are negative because our path goes in the direction of the current, with potential decreases through the resistors. The results for V_{ab} are the same for both paths, as they must be in order for the total potential change around the loop to be zero.

(c) The power outputs of the emf of the 12 V and 4 V batteries are

$$P_{12\text{V}} = \mathcal{E}I = (12\text{ V})(0.5\text{ A}) = 6\text{ W}$$

$$P_{4\text{V}} = \mathcal{E}I = (-4\text{ V})(0.5\text{ A}) = -2\text{ W}$$

The negative sign in $\mathcal{E}$ for the 4 V battery appears because the current actually runs from the higher-potential side of the battery to the lower-potential side. The negative value of P means that we are *storing* energy in that battery; the 12 V battery is *recharging* it (if it is in fact rechargeable; otherwise, we're destroying it).

EVALUATE By applying the expression $P = I^2R$ to each of the four resistors in Fig. 26.10a, you can show that the total power dissipated in all four resistors is 4 W. Of the 6 W provided by the emf of the 12 V battery, 2 W goes into storing energy in the 4 V battery and 4 W is dissipated in the resistances.

The circuit shown in Fig. 26.10a is much like that used when a fully charged 12 V storage battery (in a car with its engine running) “jump-starts” a car with a run-down battery (Fig. 26.10b). The run-down battery is slightly recharged in the process. The 3 Ω and 7 Ω resistors in Fig. 26.10a represent the resistances of the jumper cables and of the conducting path through the car with the run-down battery. (The values of the resistances in actual automobiles and jumper cables are considerably lower, and the emf of a run-down car battery isn't much less than 12 V.)

KEYCONCEPT In any loop of a circuit, Kirchhoff's loop rule applies: The sum of the potential changes as you travel around the loop must be zero. If the circuit has only a single loop, the current has the same value everywhere in the circuit.

EXAMPLE 26.4 Charging a battery

WITH VARIATION PROBLEMS

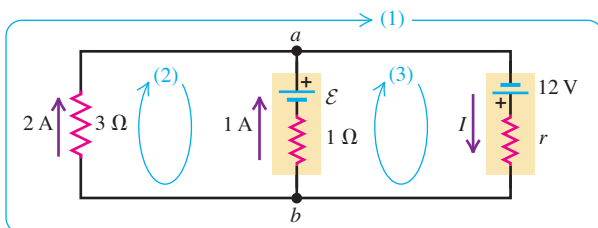
In the circuit shown in **Fig. 26.11**, a 12 V power supply with unknown internal resistance r is connected to a run-down rechargeable battery with unknown emf $\mathcal{E}$ and internal resistance 1 Ω and to an indicator light bulb of resistance 3 Ω carrying a current of 2 A. The current through the run-down battery is 1 A in the direction shown. Find r , $\mathcal{E}$, and the current I through the power supply.

IDENTIFY and SET UP This circuit has more than one loop, so we must apply both the junction and loop rules. We assume the direction of the current through the 12 V power supply, and the polarity of the run-down battery, to be as shown in Fig. 26.11. There are three target variables, so we need three equations.

EXECUTE We apply the junction rule, Eq. (26.5), to point a :

$$-I + 1\text{ A} + 2\text{ A} = 0 \quad \text{so} \quad I = 3\text{ A}$$

Figure 26.11 In this circuit a power supply charges a run-down battery and lights a bulb. An assumption has been made about the polarity of the emf $\mathcal{E}$ of the battery. Is this assumption correct?



To determine r , we apply the loop rule, Eq. (26.6), to the large, outer loop (1):

$$12\text{ V} - (3\text{ A})r - (2\text{ A})(3\ \Omega) = 0 \quad \text{so} \quad r = 2\ \Omega$$

To determine $\mathcal{E}$, we apply the loop rule to the left-hand loop (2):

$$-\mathcal{E} + (1\text{ A})(1\ \Omega) - (2\text{ A})(3\ \Omega) = 0 \quad \text{so} \quad \mathcal{E} = -5\text{ V}$$

The negative value for $\mathcal{E}$ shows that the actual polarity of this emf is opposite to that shown in Fig. 26.11. As in Example 26.3, the battery is being recharged.

EVALUATE Try applying the junction rule at point b instead of point a , and try applying the loop rule counterclockwise rather than clockwise around loop (1). You'll get the same results for I and r . We can check our result for $\mathcal{E}$ by using loop (3):

$$12\text{ V} - (3\text{ A})(2\ \Omega) - (1\text{ A})(1\ \Omega) + \mathcal{E} = 0$$

which again gives us $\mathcal{E} = -5\text{ V}$.

As an additional check, we note that $V_{ba} = V_b - V_a$ equals the voltage across the 3 Ω resistance, which is $(2\text{ A})(3\ \Omega) = 6\text{ V}$. Going from a to b by the right-hand branch, we encounter potential differences $+12\text{ V} - (3\text{ A})(2\ \Omega) = +6\text{ V}$, and going by the middle branch, we find $-(-5\text{ V}) + (1\text{ A})(1\ \Omega) = +6\text{ V}$. The three ways of getting V_{ba} give the same results.

KEYCONCEPT In any circuit that has more than one loop, Kirchhoff's junction rule applies: At each junction, the sum of the currents into the junction must be zero.

EXAMPLE 26.5 Power in a battery-charging circuit**WITH VARIATION PROBLEMS**

In the circuit of Example 26.4 (shown in Fig. 26.11), find the power delivered by the 12 V power supply and by the battery being recharged, and find the power dissipated in each resistor.

IDENTIFY and SET UP We use the results of Section 25.5, in which we found that the power delivered *from* an emf to a circuit is $\mathcal{E}I$ and the power delivered *to* a resistor from a circuit is $V_{ab}I = I^2R$. We know the values of all relevant quantities from Example 26.4.

EXECUTE The power output from the emf of the power supply is

$$P_{\text{supply}} = \mathcal{E}_{\text{supply}}I_{\text{supply}} = (12 \text{ V})(3 \text{ A}) = 36 \text{ W}$$

The power dissipated in the power supply's internal resistance r is

$$P_{r\text{-supply}} = I_{\text{supply}}^2 r_{\text{supply}} = (3 \text{ A})^2(2 \Omega) = 18 \text{ W}$$

so the power supply's *net* power output is $P_{\text{net}} = 36 \text{ W} - 18 \text{ W} = 18 \text{ W}$. Alternatively, from Example 26.4 the terminal voltage of the battery is $V_{ba} = 6 \text{ V}$, so the net power output is

$$P_{\text{net}} = V_{ba}I_{\text{supply}} = (6 \text{ V})(3 \text{ A}) = 18 \text{ W}$$

The power output of the emf $\mathcal{E}$ of the battery being charged is

$$P_{\text{emf}} = \mathcal{E}I_{\text{battery}} = (-5 \text{ V})(1 \text{ A}) = -5 \text{ W}$$

This is negative because the 1 A current runs through the battery from the higher-potential side to the lower-potential side. (As we mentioned in Example 26.4, the polarity assumed for this battery in Fig. 26.11 was wrong.) We are storing energy in the battery as we charge it. Additional power is dissipated in the battery's internal resistance; this power is

$$P_{r\text{-battery}} = I_{\text{battery}}^2 r_{\text{battery}} = (1 \text{ A})^2(1 \Omega) = 1 \text{ W}$$

The total power input to the battery is thus $1 \text{ W} + |-5 \text{ W}| = 6 \text{ W}$. Of this, 5 W represents useful energy stored in the battery; the remainder is wasted in its internal resistance. The power dissipated in the light bulb is

$$P_{\text{bulb}} = I_{\text{bulb}}^2 R_{\text{bulb}} = (2 \text{ A})^2(3 \Omega) = 12 \text{ W}$$

EVALUATE As a check, note that all of the power from the supply is accounted for. Of the 18 W of net power from the power supply, 5 W goes to recharge the battery, 1 W is dissipated in the battery's internal resistance, and 12 W is dissipated in the light bulb.

KEYCONCEPT In any circuit that contains a source (such as a battery or power supply) and resistors, the sum of the power dissipated in each resistor must equal the power supplied by the source.

EXAMPLE 26.6 A complex network**WITH VARIATION PROBLEMS**

Figure 26.12 shows a “bridge” circuit of the type described at the beginning of this section (see Fig. 26.6b). Find the current in each resistor and the equivalent resistance of the network of five resistors.

IDENTIFY and SET UP This network is neither a series combination nor a parallel combination. Hence we must use Kirchhoff's rules to find the values of the target variables. There are five unknown currents, but by applying the junction rule to junctions a and b , we can represent them in terms of three unknown currents I_1 , I_2 , and I_3 , as shown in Fig. 26.12.

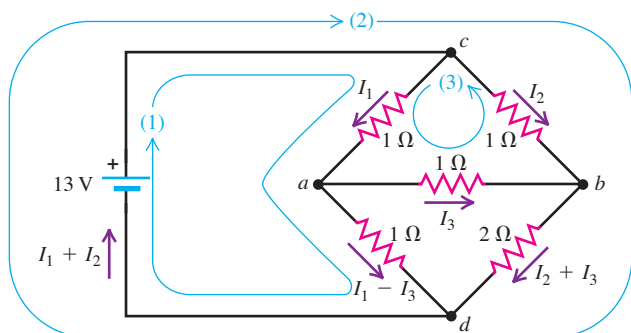
EXECUTE We apply the loop rule to the three loops shown:

$$13 \text{ V} - I_1(1 \Omega) - (I_1 - I_3)(1 \Omega) = 0 \quad (1)$$

$$-I_2(1 \Omega) - (I_2 + I_3)(2 \Omega) + 13 \text{ V} = 0 \quad (2)$$

$$-I_1(1 \Omega) - I_3(1 \Omega) + I_2(1 \Omega) = 0 \quad (3)$$

Figure 26.12 A network circuit with several resistors.



One way to solve these simultaneous equations is to solve Eq. (3) for I_2 , obtaining $I_2 = I_1 + I_3$, and then substitute this expression into Eq. (2) to eliminate I_2 . We then have

$$13 \text{ V} = I_1(2 \Omega) - I_3(1 \Omega) \quad (1')$$

$$13 \text{ V} = I_1(3 \Omega) + I_3(5 \Omega) \quad (2')$$

Now we can eliminate I_3 by multiplying Eq. (1') by 5 and adding the two equations. We obtain

$$78 \text{ V} = I_1(13 \Omega) \quad I_1 = 6 \text{ A}$$

We substitute this result into Eq. (1') to obtain $I_3 = -1 \text{ A}$, and from Eq. (3) we find $I_2 = 5 \text{ A}$. The negative value of I_3 tells us that its direction is opposite to the direction we assumed.

The total current through the network is $I_1 + I_2 = 11 \text{ A}$, and the potential drop across it is equal to the battery emf, 13 V. The equivalent resistance of the network is therefore

$$R_{\text{eq}} = \frac{13 \text{ V}}{11 \text{ A}} = 1.2 \Omega$$

EVALUATE You can check our results for I_1 , I_2 , and I_3 by substituting them back into Eqs. (1)–(3). What do you find?

KEYCONCEPT Some circuits have combinations of resistors that are neither in series nor in parallel. To analyze any such circuit, use Kirchhoff's loop rule and junction rule.

EXAMPLE 26.7 A potential difference in a complex network

WITH VARIATION PROBLEMS

In the circuit of Example 26.6 (Fig. 26.12), find the potential difference V_{ab} .

IDENTIFY and SET UP Our target variable $V_{ab} = V_a - V_b$ is the potential at point a with respect to point b . To find it, we start at point b and follow a path to point a , adding potential rises and drops as we go. We can follow any of several paths from b to a ; the result must be the same for all such paths, which gives us a way to check our result.

EXECUTE The simplest path is through the center $1\ \Omega$ resistor. In Example 26.6 we found $I_3 = -1\ \text{A}$, showing that the actual current direction through this resistor is from right to left. Thus, as we go from b to a , there is a *drop* of potential with magnitude $|I_3|R = (1\ \text{A})(1\ \Omega) = 1\ \text{V}$. Hence $V_{ab} = -1\ \text{V}$, and the potential at a is $1\ \text{V}$ less than at point b .

EVALUATE To check our result, let's try a path from b to a that goes through the lower two resistors. The currents through these are

$$I_2 + I_3 = 5\ \text{A} + (-1\ \text{A}) = 4\ \text{A} \quad \text{and}$$

$$I_1 - I_3 = 6\ \text{A} - (-1\ \text{A}) = 7\ \text{A}$$

and so

$$V_{ab} = -(4\ \text{A})(2\ \Omega) + (7\ \text{A})(1\ \Omega) = -1\ \text{V}$$

You can confirm this result by using some other paths from b to a .

KEYCONCEPT A useful way to restate Kirchhoff's loop rule is that for any path in a circuit from point a to point b , the potential difference V_{ab} must be the same.

TEST YOUR UNDERSTANDING OF SECTION 26.2 Subtract Eq. (1) from Eq. (2) in Example 26.6. To which loop in Fig. 26.12 does this equation correspond? Would this equation have simplified the solution of Example 26.6?

ANSWER

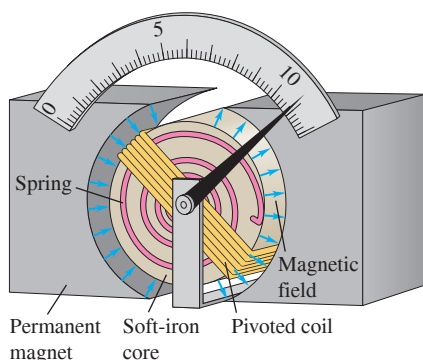
with the solution of Example 26.6.
 c to b to d to a to c in Fig. 26.12. This isn't an independent equation, so it would not have helped.
 $(1\ \Omega)(I_1) + (1\ \Omega)(I_1) = 0$. We can obtain this equation by applying the loop rule around the path from c to b to d to a to c in Fig. 26.12. This isn't an independent equation, so it would not have helped.
 $(1\ \Omega)(I_1) + (1\ \Omega)(I_1) = 0$. We can obtain this equation by applying the loop rule around the path from c to b to d to a to c in Fig. 26.12. This isn't an independent equation, so it would not have helped.

Figure 26.13 Both this voltmeter (left) and ammeter (right) are d'Arsonval galvanometers. The difference has to do with their internal connections (see Fig. 26.15).



Figure 26.14 A d'Arsonval galvanometer, showing a pivoted coil with attached pointer, a permanent magnet supplying a magnetic field that is uniform in magnitude, and a spring to provide restoring torque, which opposes magnetic-field torque.

Magnetic-field torque tends to push pointer away from zero.
 Spring torque tends to push pointer toward zero.



26.3 ELECTRICAL MEASURING INSTRUMENTS

We've been talking about potential difference, current, and resistance for two chapters, so it's about time we said something about how to *measure* these quantities. Many common devices, including car instrument panels, battery chargers, and inexpensive electrical instruments, measure potential difference (voltage), current, or resistance with a **d'Arsonval galvanometer** (Fig. 26.13). In the following discussion we'll often call it just a *meter*. A pivoted coil of fine wire is placed in the magnetic field of a permanent magnet (Fig. 26.14). Attached to the coil is a spring, similar to the hairspring on the balance wheel of a watch. In the equilibrium position, with no current in the coil, the pointer is at zero. When there is a current in the coil, the magnetic field exerts a torque on the coil that is proportional to the current. (We'll discuss this magnetic interaction in detail in Chapter 27.) As the coil turns, the spring exerts a restoring torque that is proportional to the angular displacement.

Thus the angular deflection of the coil and pointer is directly proportional to the coil current, and the device can be calibrated to measure current. The maximum deflection, typically 90° or so, is called *full-scale deflection*. The essential electrical characteristics of the meter are the current I_{fs} required for full-scale deflection (typically on the order of $10\ \mu\text{A}$ to $10\ \text{mA}$) and the resistance R_c of the coil (typically on the order of $10\ \Omega$ to $1000\ \Omega$).

The meter deflection is proportional to the *current* in the coil. If the coil obeys Ohm's law, the current is proportional to the *potential difference* between the terminals of the coil, and the deflection is also proportional to this potential difference. For example, consider a meter whose coil has a resistance $R_c = 20.0\ \Omega$ and that deflects full scale when the current in its coil is $I_{fs} = 1.00\ \text{mA}$. The corresponding potential difference for full-scale deflection is

$$V = I_{fs}R_c = (1.00 \times 10^{-3}\ \text{A})(20.0\ \Omega) = 0.0200\ \text{V}$$

Ammeters

A current-measuring instrument is usually called an **ammeter** (or milliammeter, microammeter, and so forth, depending on the range). An *ammeter always measures the current passing through it*. An *ideal* ammeter, discussed in Section 25.4, would have *zero* resistance, so including it in a branch of a circuit would not affect the current in that branch.

Real ammeters always have a finite resistance, but it is always desirable for an ammeter to have as little resistance as possible.

We can adapt any meter to measure currents that are larger than its full-scale reading by connecting a resistor in parallel with it (**Fig. 26.15a**) so that some of the current bypasses the meter coil. The parallel resistor is called a **shunt resistor** or simply a *shunt*, denoted as R_{sh} .

Suppose we want to make a meter with full-scale current I_{fs} and coil resistance R_c into an ammeter with full-scale reading I_a . To determine the shunt resistance R_{sh} needed, note that at full-scale deflection the total current through the parallel combination is I_a , the current through the coil of the meter is I_{fs} , and the current through the shunt is the difference $I_a - I_{\text{fs}}$. The potential difference V_{ab} is the same for both paths, so

$$I_{\text{fs}}R_c = (I_a - I_{\text{fs}})R_{\text{sh}} \quad (\text{for an ammeter}) \quad (26.7)$$

EXAMPLE 26.8 Designing an ammeter

What shunt resistance is required to make the 1.00 mA, 20.0 Ω meter described above into an ammeter with a range of 0 to 50.0 mA?

IDENTIFY and SET UP Since the meter is being used as an ammeter, its internal connections are as shown in Fig. 26.15a. Our target variable is the shunt resistance R_{sh} , which we'll find from Eq. (26.7). The ammeter must handle a maximum current $I_a = 50.0 \times 10^{-3}$ A. The coil resistance is $R_c = 20.0 \Omega$, and the meter shows full-scale deflection when the current through the coil is $I_{\text{fs}} = 1.00 \times 10^{-3}$ A.

EXECUTE Solving Eq. (26.7) for R_{sh} , we find

$$\begin{aligned} R_{\text{sh}} &= \frac{I_{\text{fs}}R_c}{I_a - I_{\text{fs}}} = \frac{(1.00 \times 10^{-3} \text{ A})(20.0 \Omega)}{(50.0 \times 10^{-3} \text{ A}) - (1.00 \times 10^{-3} \text{ A})} \\ &= 0.408 \Omega \end{aligned}$$

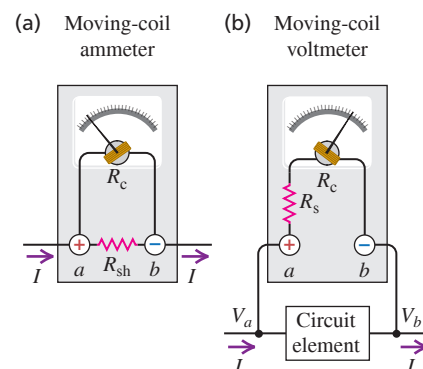
EVALUATE It's useful to consider the equivalent resistance R_{eq} of the ammeter as a whole. From Eq. (26.2),

$$\begin{aligned} R_{\text{eq}} &= \left(\frac{1}{R_c} + \frac{1}{R_{\text{sh}}} \right)^{-1} = \left(\frac{1}{20.0 \Omega} + \frac{1}{0.408 \Omega} \right)^{-1} \\ &= 0.400 \Omega \end{aligned}$$

The shunt resistance is so small in comparison to the coil resistance that the equivalent resistance is very nearly equal to the shunt resistance. The result is an ammeter with a low equivalent resistance and the desired 0–50.0 mA range. At full-scale deflection, $I = I_a = 50.0$ mA, the current through the galvanometer is 1.00 mA, the current through the shunt resistor is 49.0 mA, and $V_{ab} = 0.0200$ V. If the current I is *less* than 50.0 mA, the coil current and the deflection are proportionally less.

KEYCONCEPT An ammeter measures the current that passes through it. It should have a very small equivalent resistance in order to minimize its effect on the circuit being measured. This can be accomplished using a small shunt resistance in parallel with the meter.

Figure 26.15 Using the same meter to measure (a) current and (b) voltage.



Voltmeters

This same basic meter may also be used to measure potential difference or *voltage*. A voltage-measuring device is called a **voltmeter**. A voltmeter always measures the potential difference between two points, and its terminals must be connected to these points. (Example 25.6 in Section 25.4 described what can happen if a voltmeter is connected incorrectly.) As we discussed in Section 25.4, an ideal voltmeter would have *infinite* resistance, so connecting it between two points in a circuit would not alter any of the currents. Real voltmeters always have finite resistance, but a voltmeter should have large enough resistance that connecting it in a circuit does not change the other currents appreciably.

For the meter of Example 26.8, the voltage across the meter coil at full-scale deflection is only $I_{\text{fs}}R_c = (1.00 \times 10^{-3} \text{ A})(20.0 \Omega) = 0.0200$ V. We can extend this range by connecting a resistor R_s in *series* with the coil (Fig. 26.15b). Then only a fraction of the total potential difference appears across the coil itself, and the remainder appears across R_s . For a voltmeter with full-scale reading V_V , we need a series resistor R_s in Fig. 26.15b such that

$$V_V = I_{\text{fs}}(R_c + R_s) \quad (\text{for a voltmeter}) \quad (26.8)$$

BIO APPLICATION

Electromyography A fine needle containing two electrodes is being inserted into a muscle in this patient's hand. By using a sensitive voltmeter to measure the potential difference between these electrodes, a physician can probe the muscle's electrical activity. This is an important technique for diagnosing neurological and neuromuscular diseases.



EXAMPLE 26.9 Designing a voltmeter

What series resistance is required to make the 1.00 mA, 20.0 Ω meter described above into a voltmeter with a range of 0 to 10.0 V?

IDENTIFY and SET UP Since this meter is being used as a voltmeter, its internal connections are as shown in Fig. 26.15b. The maximum allowable voltage across the voltmeter is $V_V = 10.0$ V. We want this to occur when the current through the coil is $I_{fs} = 1.00 \times 10^{-3}$ A. Our target variable is the series resistance R_s , which we find from Eq. (26.8).

EXECUTE From Eq. (26.8),

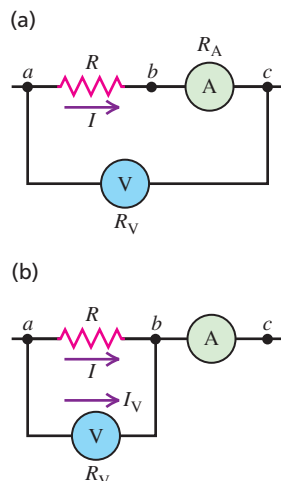
$$R_s = \frac{V_V}{I_{fs}} - R_c = \frac{10.0 \text{ V}}{0.00100 \text{ A}} - 20.0 \Omega = 9980 \Omega$$

EVALUATE At full-scale deflection, $V_{ab} = 10.0$ V, the voltage across the meter is 0.0200 V, the voltage across R_s is 9.98 V, and the current

through the voltmeter is 0.00100 A. Most of the voltage appears across the series resistor. The meter's equivalent resistance is a desirably high $R_{eq} = 20.0 \Omega + 9980 \Omega = 10,000 \Omega$. Such a meter is called a "1000 ohms-per-volt" meter, referring to the ratio of resistance to full-scale deflection. In normal operation the current through the circuit element being measured (I in Fig. 26.15b) is much greater than 0.00100 A, and the resistance between points a and b in the circuit is much less than 10,000 Ω . The voltmeter draws off only a small fraction of the current and thus disturbs the circuit being measured only slightly.

KEYCONCEPT A voltmeter measures the potential difference (voltage) between its terminals. It should have a very large equivalent resistance in order to minimize the amount of current it draws from the circuit being measured. This can be accomplished using a large shunt resistance in series with the meter.

Figure 26.16 Ammeter–voltmeter method for measuring resistance.

**Ammeters and Voltmeters in Combination**

A voltmeter and an ammeter can be used together to measure *resistance* and *power*. The resistance R of a resistor equals the potential difference V_{ab} between its terminals divided by the current I ; that is, $R = V_{ab}/I$. The power input P to any circuit element is the product of the potential difference across it and the current through it: $P = V_{ab}I$. In principle, the most straightforward way to measure R or P is to measure V_{ab} and I simultaneously.

With practical ammeters and voltmeters this isn't quite as simple as it seems. In **Fig. 26.16a**, ammeter A reads the current I in the resistor R . Voltmeter V , however, reads the *sum* of the potential difference V_{ab} across the resistor and the potential difference V_{bc} across the ammeter. If we transfer the voltmeter terminal from c to b , as in **Fig. 26.16b**, then the voltmeter reads the potential difference V_{ab} correctly, but the ammeter now reads the *sum* of the current I in the resistor and the current I_V in the voltmeter. Either way, we have to correct the reading of one instrument or the other unless the corrections are small enough to be negligible.

EXAMPLE 26.10 Measuring resistance I

The voltmeter in the circuit of **Fig. 26.16a** reads 12.0 V and the ammeter reads 0.100 A. The meter resistances are $R_V = 10,000 \Omega$ (for the voltmeter) and $R_A = 2.00 \Omega$ (for the ammeter). What are the resistance R and the power dissipated in the resistor?

IDENTIFY and SET UP The ammeter reads the current $I = 0.100$ A through the resistor, and the voltmeter reads the potential difference between a and c . If the ammeter were *ideal* (that is, if $R_A = 0$), there would be zero potential difference between b and c , the voltmeter reading $V = 12.0$ V would be equal to the potential difference V_{ab} across the resistor, and the resistance would be equal to $R = V/I = (12.0 \text{ V})/(0.100 \text{ A}) = 120 \Omega$. The ammeter is *not* ideal, however (its resistance is $R_A = 2.00 \Omega$), so the voltmeter reading V is actually the sum of the potential differences V_{bc} (across the ammeter) and V_{ab} (across the resistor). We use Ohm's law to find the voltage V_{bc} from the known current and ammeter resistance. Then we solve for V_{ab} and R . Given these, we are able to calculate the power P into the resistor.

EXECUTE From Ohm's law, $V_{bc} = IR_A = (0.100 \text{ A})(2.00 \Omega) = 0.200 \text{ V}$ and $V_{ab} = IR$. The sum of these is $V = 12.0 \text{ V}$, so the potential difference

across the resistor is $V_{ab} = V - V_{bc} = (12.0 \text{ V}) - (0.200 \text{ V}) = 11.8 \text{ V}$. Hence the resistance is

$$R = \frac{V_{ab}}{I} = \frac{11.8 \text{ V}}{0.100 \text{ A}} = 118 \Omega$$

The power dissipated in this resistor is

$$P = V_{ab}I = (11.8 \text{ V})(0.100 \text{ A}) = 1.18 \text{ W}$$

EVALUATE You can confirm this result for the power by using the alternative formula $P = I^2R$. Do you get the same answer?

KEYCONCEPT To find the resistance of a resistor and the power that the resistor dissipates requires measuring both the current through the resistor and the potential difference across the resistor. To do this, put an ammeter in series with the resistance and put a voltmeter across the resistor and ammeter.

EXAMPLE 26.11 Measuring resistance II

Suppose the meters of Example 26.10 are connected to a different resistor as shown in Fig. 26.16b, and the readings obtained on the meters are the same as in Example 26.10. What is the value of this new resistance R , and what is the power dissipated in the resistor?

IDENTIFY and SET UP In Example 26.10 the ammeter read the actual current through the resistor, but the voltmeter reading was not the same as the potential difference across the resistor. Now the situation is reversed: The voltmeter reading $V = 12.0$ V shows the actual potential difference V_{ab} across the resistor, but the ammeter reading $I_A = 0.100$ A is *not* equal to the current I through the resistor. Applying the junction rule at b in Fig. 26.16b shows that $I_A = I + I_V$, where I_V is the current through the voltmeter. We find I_V from the given values of V and the voltmeter resistance R_V , and we use this value to find the resistor current I . We then determine the resistance R from I and the voltmeter reading, and calculate the power as in Example 26.10.

EXECUTE We have $I_V = V/R_V = (12.0 \text{ V})/(10,000 \Omega) = 1.20 \text{ mA}$. The actual current I in the resistor is $I = I_A - I_V = 0.100 \text{ A} - 0.0012 \text{ A} = 0.0988 \text{ A}$, and the resistance is

$$R = \frac{V_{ab}}{I} = \frac{12.0 \text{ V}}{0.0988 \text{ A}} = 121 \Omega$$

The power dissipated in the resistor is

$$P = V_{ab}I = (12.0 \text{ V})(0.0988 \text{ A}) = 1.19 \text{ W}$$

EVALUATE Had the meters been ideal, our results would have been $R = 12.0 \text{ V}/0.100 \text{ A} = 120 \Omega$ and $P = VI = (12.0 \text{ V})(0.100 \text{ A}) = 1.2 \text{ W}$ both here and in Example 26.10. The actual (correct) results are not too different in either case. That's because the ammeter and voltmeter are nearly ideal: Compared with the resistance R under test, the ammeter resistance R_A is very small and the voltmeter resistance R_V is very large. Under these conditions, treating the meters as ideal yields pretty good results; accurate work requires calculations as in these two examples.

KEYCONCEPT An alternative way to find the resistance of a resistor and the power that the resistor dissipates is to put a voltmeter across the resistor and put an ammeter in the circuit downstream of the resistor–voltmeter combination.

Ohmmeters

An alternative method for measuring resistance is to use a d'Arsonval meter in an arrangement called an **ohmmeter**. It consists of a meter, a resistor, and a source (often a flashlight battery) connected in series (Fig. 26.17). The resistance R to be measured is connected between terminals x and y .

The series resistance R_s is variable; it is adjusted so that when terminals x and y are short-circuited (that is, when $R = 0$), the meter deflects full scale. When nothing is connected to terminals x and y , so that the circuit between x and y is *open* (that is, when $R \rightarrow \infty$), there is no current and hence no deflection. For any intermediate value of R the meter deflection depends on the value of R , and the meter scale can be calibrated to read the resistance R directly. Larger currents correspond to smaller resistances, so this scale reads backward compared to the scale showing the current.

In situations in which high precision is required, instruments containing d'Arsonval meters have been supplanted by electronic instruments with direct digital readouts. Digital voltmeters can be made with extremely high internal resistance, of the order of $100 \text{ M}\Omega$. **Figure 26.18** shows a digital *multimeter*, an instrument that can measure voltage, current, or resistance over a wide range.

Figure 26.17 Ohmmeter circuit. The resistor R_s has a variable resistance, as is indicated by the arrow through the resistor symbol. To use the ohmmeter, first connect x directly to y and adjust R_s until the meter reads zero. Then connect x and y across the resistor R and read the scale.

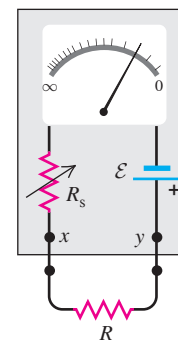


Figure 26.18 This digital multimeter can be used as a voltmeter (red arc), ammeter (yellow arc), or ohmmeter (green arc).

The Potentiometer

The *potentiometer* is an instrument that can be used to measure the emf of a source without drawing any current from the source; it also has a number of other useful applications. Essentially, it balances an unknown potential difference against an adjustable, measurable potential difference.

Figure 26.19a shows the principle of the potentiometer. A resistance wire ab of total resistance R_{ab} is permanently connected to the terminals of a source of known emf $\mathcal{E}_1$. A sliding contact c is connected through the galvanometer G to a second source whose emf $\mathcal{E}_2$ is to be measured. As contact c is moved along the resistance wire, the resistance R_{cb} between points c and b varies; if the resistance wire is uniform, R_{cb} is proportional to the length of wire between c and b . To determine the value of $\mathcal{E}_2$, contact c is moved until a position is found at which the galvanometer shows no deflection; this corresponds to zero current passing through $\mathcal{E}_2$. With $I_2 = 0$, Kirchhoff's loop rule gives

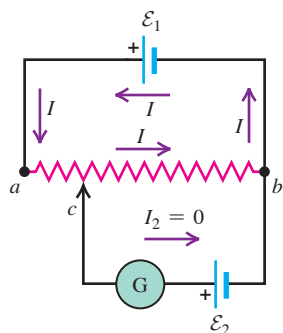
$$\mathcal{E}_2 = IR_{cb}$$

With $I_2 = 0$, the current I produced by the emf $\mathcal{E}_1$ has the same value no matter what the value of the emf $\mathcal{E}_2$. We calibrate the device by replacing $\mathcal{E}_2$ by a source of known emf; then to find any unknown emf $\mathcal{E}_2$, we measure the length of wire cb for which $I_2 = 0$. Note: For this to work, V_{ab} must be greater than $\mathcal{E}_2$.

The term *potentiometer* is also used for any variable resistor, usually having a circular resistance element and a sliding contact controlled by a rotating shaft and knob. Figure 26.19b shows the circuit symbol for a potentiometer.

Figure 26.19 A potentiometer.

(a) Potentiometer circuit



(b) Circuit symbol for potentiometer (variable resistor)



TEST YOUR UNDERSTANDING OF SECTION 26.3 You want to measure the current through and the potential difference across the $2\ \Omega$ resistor shown in Fig. 26.12 (Example 26.6 in Section 26.2). (a) How should you connect an ammeter and a voltmeter to do this? (i) Both ammeter and voltmeter in series with the $2\ \Omega$ resistor; (ii) ammeter in series with the $2\ \Omega$ resistor and voltmeter connected between points b and d ; (iii) ammeter connected between points b and d and voltmeter in series with the $2\ \Omega$ resistor; (iv) both ammeter and voltmeter connected between points b and d . (b) What resistances should these meters have? (i) Both ammeter and voltmeter resistances should be much greater than $2\ \Omega$; (ii) ammeter resistance should be much greater than $2\ \Omega$ and voltmeter resistance should be much less than $2\ \Omega$; (iii) ammeter resistance should be much less than $2\ \Omega$ and voltmeter resistance should be much greater than $2\ \Omega$; (iv) both ammeter and voltmeter resistances should be much less than $2\ \Omega$.

ANSWER

(a) (i) An ammeter must always be placed in series with the circuit element of interest, and a voltmeter must always be placed in parallel. Ideally the ammeter would have zero resistance and the voltmeter would have infinite resistance so that their presence would have no effect on either the resistor current or the voltage. Neither of these idealizations is possible, but the ammeter resistance should be much less than $2\ \Omega$ and the voltmeter resistance should be much greater than $2\ \Omega$.

26.4 R-C CIRCUITS

In the circuits we have analyzed up to this point, we have assumed that all the emfs and resistances are *constant* (time independent) so that all the potentials, currents, and powers are also independent of time. But in the simple act of charging or discharging a capacitor we find a situation in which the currents, voltages, and powers *do* change with time.

Many devices incorporate circuits in which a capacitor is alternately charged and discharged. These include flashing traffic lights, automobile turn signals, and electronic flash units. Understanding what happens in such circuits is thus of great practical importance.

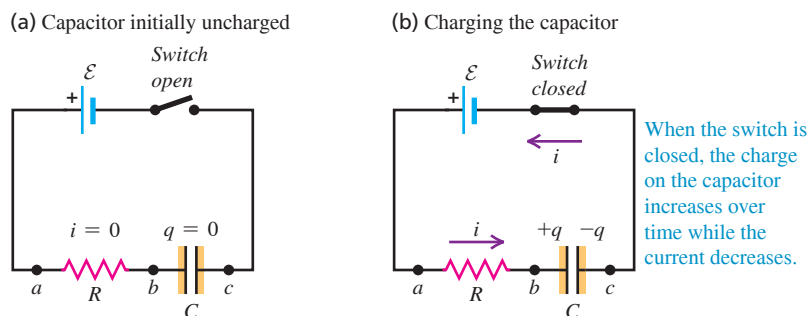


Figure 26.20 Charging a capacitor. (a) Just before the switch is closed, the charge q is zero. (b) When the switch closes (at $t = 0$), the current jumps from zero to $\mathcal{E}/R$. As time passes, q approaches Q_f and the current i approaches zero.

Charging a Capacitor

Figure 26.20 shows a simple circuit for charging a capacitor. A circuit such as this that has a resistor and a capacitor in series is called an **R-C circuit**. We idealize the battery (or power supply) to have a constant emf $\mathcal{E}$ and zero internal resistance ($r = 0$), and we ignore the resistance of all the connecting conductors.

We begin with the capacitor initially uncharged (Fig. 26.20a); then at some initial time $t = 0$ we close the switch, completing the circuit and permitting current around the loop to begin charging the capacitor (Fig. 26.20b). For all practical purposes, the current begins at the same instant in every conducting part of the circuit, and at each instant the current is the same in every part.

Because the capacitor in Fig. 26.20 is initially uncharged, the potential difference v_{bc} across it is zero at $t = 0$. At this time, from Kirchhoff's loop law, the voltage v_{ab} across the resistor R is equal to the battery emf $\mathcal{E}$. The initial ($t = 0$) current through the resistor, which we'll call I_0 , is given by Ohm's law: $I_0 = v_{ab}/R = \mathcal{E}/R$.

As the capacitor charges, its voltage v_{bc} increases and the potential difference v_{ab} across the resistor decreases, corresponding to a decrease in current. The sum of these two voltages is constant and equal to $\mathcal{E}$. After a long time the capacitor is fully charged, the current decreases to zero, and v_{ab} across the resistor becomes zero. Then the entire battery emf $\mathcal{E}$ appears across the capacitor and $v_{bc} = \mathcal{E}$.

Let q represent the charge on the capacitor and i the current in the circuit at some time t after the switch has been closed. We choose the positive direction for the current to correspond to positive charge flowing onto the left-hand capacitor plate, as in Fig. 26.20b. The instantaneous potential differences v_{ab} and v_{bc} are

$$v_{ab} = iR \quad v_{bc} = \frac{q}{C}$$

Using these in Kirchhoff's loop rule, we find

$$\mathcal{E} - iR - \frac{q}{C} = 0 \quad (26.9)$$

The potential drops by an amount iR as we travel from a to b and by q/C as we travel from b to c . Solving Eq. (26.9) for i , we find

$$i = \frac{\mathcal{E}}{R} - \frac{q}{RC} \quad (26.10)$$

At time $t = 0$, when the switch is first closed, the capacitor is uncharged, and so $q = 0$. Substituting $q = 0$ into Eq. (26.10), we find that the *initial* current I_0 is given by $I_0 = \mathcal{E}/R$, as we have already noted. If the capacitor were not in the circuit, the last term in Eq. (26.10) would not be present; then the current would be *constant* and equal to $\mathcal{E}/R$.

As the charge q increases, the term q/RC becomes larger and the capacitor charge approaches its final value, which we'll call Q_f . The current decreases and eventually becomes zero. When $i = 0$, Eq. (26.10) gives

$$\frac{\mathcal{E}}{R} = \frac{Q_f}{RC} \quad Q_f = C\mathcal{E} \quad (26.11)$$

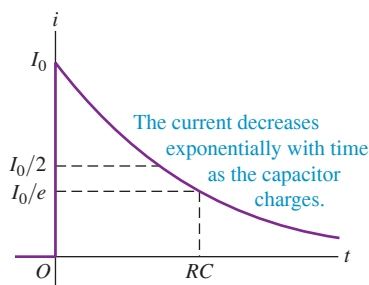
Note that the final charge Q_f does not depend on R .

CAUTION Lowercase means time-varying

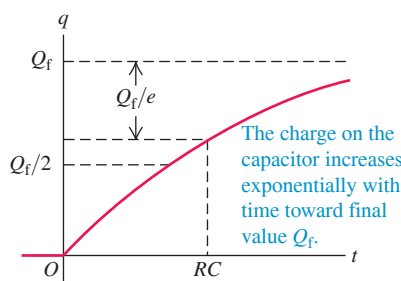
Up to this point we have been working with constant potential differences (voltages), currents, and charges, and we have used *capital* letters V , I , and Q , respectively, to denote these quantities. To distinguish between quantities that vary with time and those that are constant, we'll use *lowercase* letters v , i , and q for time-varying voltages, currents, and charges, respectively. We suggest that you follow this same convention in your own work. ■

Figure 26.21 Current i and capacitor charge q as functions of time for the circuit of Fig. 26.20. The initial current is I_0 and the initial capacitor charge is zero. The current asymptotically approaches zero, and the capacitor charge asymptotically approaches a final value of Q_f .

(a) Graph of current versus time for a charging capacitor



(b) Graph of capacitor charge versus time for a charging capacitor



BIO APPLICATION Pacemakers and Capacitors

This x-ray image shows a pacemaker implanted in a patient with a malfunctioning sinoatrial node, the part of the heart that generates the electrical signal to trigger heartbeats. The pacemaker circuit contains a battery, a capacitor, and a computer-controlled switch. To maintain regular beating, once per second the switch discharges the capacitor and sends an electrical pulse along the lead to the heart. The switch then flips to allow the capacitor to recharge for the next pulse.

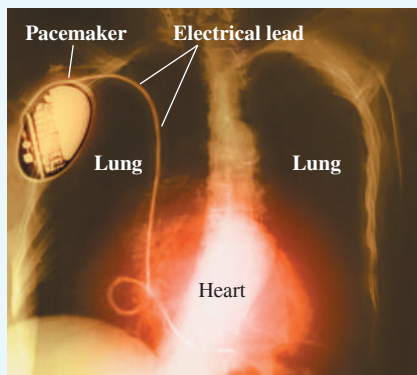


Figure 26.21 shows the current and capacitor charge as functions of time. At the instant the switch is closed ($t = 0$), the current jumps from zero to its initial value $I_0 = \mathcal{E}/R$; after that, it gradually approaches zero. The capacitor charge starts at zero and gradually approaches the final value given by Eq. (26.11), $Q_f = C\mathcal{E}$.

We can derive general expressions for charge q and current i as functions of time. With our choice of the positive direction for current (Fig. 26.20b), i equals the rate at which positive charge arrives at the left-hand (positive) plate of the capacitor, so $i = dq/dt$. Making this substitution in Eq. (26.10), we have

$$\frac{dq}{dt} = \frac{\mathcal{E}}{R} - \frac{q}{RC} = -\frac{1}{RC}(q - C\mathcal{E})$$

We can rearrange this to

$$\frac{dq}{q - C\mathcal{E}} = -\frac{dt}{RC}$$

and then integrate both sides. We change the integration variables to q' and t' so that we can use q and t for the upper limits. The lower limits are $q' = 0$ and $t' = 0$:

$$\int_0^q \frac{dq'}{q' - C\mathcal{E}} = -\int_0^t \frac{dt'}{RC}$$

When we carry out the integration, we get

$$\ln\left(\frac{q - C\mathcal{E}}{-C\mathcal{E}}\right) = -\frac{t}{RC}$$

Exponentiating both sides (that is, taking the inverse logarithm) and solving for q , we find

$$\frac{q - C\mathcal{E}}{-C\mathcal{E}} = e^{-t/RC}$$

R-C circuit, charging capacitor:

$$q = C\mathcal{E}(1 - e^{-t/RC}) = Q_f(1 - e^{-t/RC}) \quad (26.12)$$

Annotations: Capacitor charge q , Battery emf $\mathcal{E}$, Capacitance C , Time since switch closed t , Resistance R , Final capacitor charge $= C\mathcal{E}$.

The instantaneous current i is just the time derivative of Eq. (26.12):

R-C circuit, charging capacitor:

$$i = \frac{dq}{dt} = \frac{\mathcal{E}}{R} e^{-t/RC} = I_0 e^{-t/RC} = \mathcal{E}/R \quad (26.13)$$

Annotations: Current i , Battery emf $\mathcal{E}$, Time since switch closed t , Initial current $= \mathcal{E}/R$, Rate of change of capacitor charge dq/dt , Resistance R , Capacitance C .

The charge and current are both *exponential* functions of time. Figure 26.21a is a graph of Eq. (26.13) and Fig. 26.21b is a graph of Eq. (26.12).

Time Constant

After a time equal to RC , the current in the R - C circuit has decreased to $1/e$ (about 0.368) of its initial value. At this time, the capacitor charge has reached $(1 - 1/e) = 0.632$ of its final value $Q_f = C\mathcal{E}$. The product RC is therefore a measure of how quickly the capacitor charges. We call RC the **time constant**, or the **relaxation time**, of the circuit, denoted by τ :

$$\tau = RC \quad (\text{time constant for } R\text{-}C \text{ circuit}) \quad (26.14)$$

When τ is small, the capacitor charges quickly; when it is larger, the charging takes more time. If the resistance is small, it's easier for current to flow, and the capacitor charges more quickly. If R is in ohms and C in farads, τ is in seconds.

In Fig. 26.21a the horizontal axis is an *asymptote* for the curve. Strictly speaking, i never becomes exactly zero. But the longer we wait, the closer it gets. After a time equal to $10RC$, the current has decreased to 0.000045 of its initial value. Similarly, the curve in Fig. 26.21b approaches the horizontal dashed line labeled Q_f as an asymptote. The charge q never attains exactly this value, but after a time equal to $10RC$, the difference between q and Q_f is only 0.000045 of Q_f . We invite you to verify that the product RC has units of time.

Discharging a Capacitor

Now suppose that after the capacitor in Fig. 26.21b has acquired a charge Q_0 , we remove the battery from our R - C circuit and connect points a and c to an open switch (Fig. 26.22a). We then close the switch and at the same instant reset our stopwatch to $t = 0$; at that time, $q = Q_0$. The capacitor then *discharges* through the resistor, and its charge eventually decreases to zero.

Again let i and q represent the time-varying current and charge at some instant after the connection is made. In Fig. 26.22b we make the same choice of the positive direction for current as in Fig. 26.20b. Then Kirchhoff's loop rule gives Eq. (26.10) but with $\mathcal{E} = 0$; that is,

$$i = \frac{dq}{dt} = -\frac{q}{RC} \quad (26.15)$$

The current i is now negative; this is because positive charge q is leaving the left-hand capacitor plate in Fig. 26.22b, so the current is in the direction opposite to that shown. At time $t = 0$, when $q = Q_0$, the initial current is $I_0 = -Q_0/RC$.

To find q as a function of time, we rearrange Eq. (26.15), again change the variables to q' and t' , and integrate. This time the limits for q' are Q_0 to q :

$$\int_{Q_0}^q \frac{dq'}{q'} = -\frac{1}{RC} \int_0^t dt'$$

$$\ln \frac{q}{Q_0} = -\frac{t}{RC}$$

R-C circuit, discharging capacitor:

$$q = Q_0 e^{-t/RC} \quad (26.16)$$

Labels in diagram: Capacitor charge, Initial capacitor charge, Capacitance, Resistance, Time since switch closed.

The instantaneous current i is the derivative of this with respect to time:

R-C circuit, discharging capacitor:

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC} e^{-t/RC} = I_0 e^{-t/RC} \quad (26.17)$$

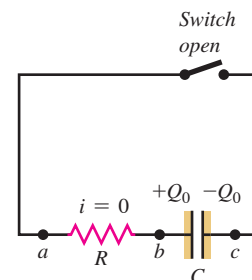
Labels in diagram: Current, Initial capacitor charge, Capacitance, Resistance, Time since switch closed, Rate of change of capacitor charge, Initial current = $-Q_0/RC$.

We graph the current and the charge in Fig. 26.23; both quantities approach zero exponentially with time. Comparing these results with Eqs. (26.12) and (26.13), we note that the expressions for the current are identical, apart from the sign of I_0 . The capacitor charge approaches zero asymptotically in Eq. (26.16), while the *difference* between q and Q approaches zero asymptotically in Eq. (26.12).

Energy considerations give us additional insight into the behavior of an R - C circuit. While the capacitor is charging, the instantaneous rate at which the battery delivers energy to the circuit is $P = \mathcal{E}i$. The instantaneous rate at which electrical energy is dissipated in

Figure 26.22 Discharging a capacitor. (a) Before the switch is closed at time $t = 0$, the capacitor charge is Q_0 and the current is zero. (b) At time t after the switch is closed, the capacitor charge is q and the current is i . The actual current direction is opposite to the direction shown; i is negative. After a long time, q and i both approach zero.

(a) Capacitor initially charged



(b) Discharging the capacitor

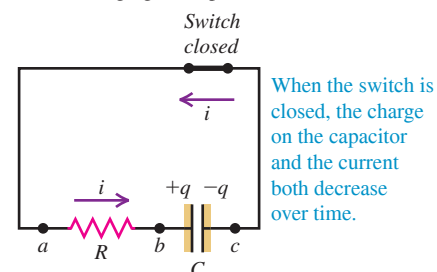
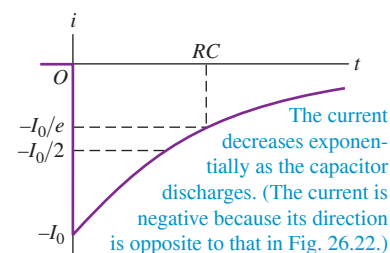
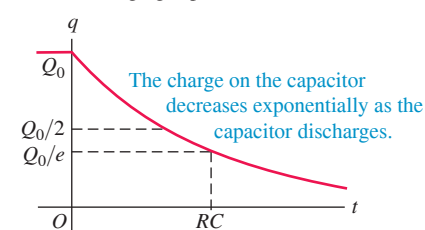


Figure 26.23 Current i and capacitor charge q as functions of time for the circuit of Fig. 26.22. The initial current is I_0 and the initial capacitor charge is Q_0 . Both i and q asymptotically approach zero.

(a) Graph of current versus time for a discharging capacitor



(b) Graph of capacitor charge versus time for a discharging capacitor



the resistor is i^2R , and the rate at which energy is stored in the capacitor is $iv_{bc} = iq/C$. Multiplying Eq. (26.9) by i , we find

$$\mathcal{E}i = i^2R + \frac{iq}{C} \quad (26.18)$$

This means that of the power $\mathcal{E}i$ supplied by the battery, part (i^2R) is dissipated in the resistor and part (iq/C) is stored in the capacitor.

The *total* energy supplied by the battery during charging of the capacitor equals the battery emf $\mathcal{E}$ multiplied by the total charge Q_f , or $\mathcal{E}Q_f$. The total energy stored in the capacitor, from Eq. (24.9), is $Q_f\mathcal{E}/2$. Thus, of the energy supplied by the battery, *exactly half* is stored in the capacitor, and the other half is dissipated in the resistor. This half-and-half division of energy doesn't depend on C , R , or $\mathcal{E}$. You can verify this result by taking the integral over time of each of the power quantities in Eq. (26.18).

EXAMPLE 26.12 Charging a capacitor

WITH VARIATION PROBLEMS

A $10\text{ M}\Omega$ resistor is connected in series with a $1.0\text{ }\mu\text{F}$ capacitor and a battery with emf 12.0 V . Before the switch is closed at time $t = 0$, the capacitor is uncharged. (a) What is the time constant? (b) What fraction of the final charge Q_f is on the capacitor at $t = 46\text{ s}$? (c) What fraction of the initial current I_0 is still flowing at $t = 46\text{ s}$?

IDENTIFY and SET UP This is the situation shown in Fig. 26.20, with $R = 10\text{ M}\Omega$, $C = 1.0\text{ }\mu\text{F}$, and $\mathcal{E} = 12.0\text{ V}$. The charge q and current i vary with time as shown in Fig. 26.21. Our target variables are (a) the time constant τ , (b) the ratio q/Q_f at $t = 46\text{ s}$, and (c) the ratio i/I_0 at $t = 46\text{ s}$. Equation (26.14) gives τ . For a capacitor being charged, Eq. (26.12) gives q and Eq. (26.13) gives i .

EXECUTE (a) From Eq. (26.14),

$$\tau = RC = (10 \times 10^6\text{ }\Omega)(1.0 \times 10^{-6}\text{ F}) = 10\text{ s}$$

(b) From Eq. (26.12),

$$\frac{q}{Q_f} = 1 - e^{-t/RC} = 1 - e^{-(46\text{ s})/(10\text{ s})} = 0.99$$

(c) From Eq. (26.13),

$$\frac{i}{I_0} = e^{-t/RC} = e^{-(46\text{ s})/(10\text{ s})} = 0.010$$

EVALUATE After 4.6 time constants the capacitor is 99% charged and the charging current has decreased to 1.0% of its initial value. The circuit would charge more rapidly if we reduced the time constant by using a smaller resistance.

KEYCONCEPT When a capacitor is charging in a circuit with a resistor and a source of emf, the capacitor charge q and the current i both vary with time: q approaches its final value asymptotically and i approaches zero asymptotically. The time constant for both quantities is the product RC of the resistance and the capacitance of the circuit elements.

EXAMPLE 26.13 Discharging a capacitor

WITH VARIATION PROBLEMS

The resistor and capacitor of Example 26.12 are reconnected as shown in Fig. 26.22. The capacitor has an initial charge of $5.0\text{ }\mu\text{C}$ and is discharged by closing the switch at $t = 0$. (a) At what time will the charge be $0.50\text{ }\mu\text{C}$? (b) What is the current at this time?

IDENTIFY and SET UP Now the capacitor is being discharged, so q and i vary with time as in Fig. 26.23, with $Q_0 = 5.0 \times 10^{-6}\text{ C}$. Again we have $RC = \tau = 10\text{ s}$. Our target variables are (a) the value of t at which $q = 0.50\text{ }\mu\text{C}$ and (b) the value of i at this time. We first solve Eq. (26.16) for t , and then solve Eq. (26.17) for i .

EXECUTE (a) Solving Eq. (26.16) for the time t gives

$$\begin{aligned} t &= -RC \ln \frac{q}{Q_0} = -(10\text{ s}) \ln \frac{0.50\text{ }\mu\text{C}}{5.0\text{ }\mu\text{C}} \\ &= 23\text{ s} = 2.3\tau \end{aligned}$$

(b) From Eq. (26.17), with $Q_0 = 5.0\text{ }\mu\text{C} = 5.0 \times 10^{-6}\text{ C}$,

$$i = -\frac{Q_0}{RC} e^{-t/RC} = -\frac{5.0 \times 10^{-6}\text{ C}}{10\text{ s}} e^{-2.3} = -5.0 \times 10^{-8}\text{ A}$$

EVALUATE The current in part (b) is negative because i has the opposite sign when the capacitor is discharging than when it is charging. Note that we could have avoided evaluating $e^{-t/RC}$ by noticing that at the time in question, $q = 0.10Q_0$; from Eq. (26.16) this means that $e^{-t/RC} = 0.10$.

KEYCONCEPT When a capacitor is discharging in a circuit with a resistor, the capacitor charge q and the current i both vary with time, and both approach zero asymptotically. The time constant for both quantities is the product RC of the resistance and the capacitance of the circuit elements.

TEST YOUR UNDERSTANDING OF SECTION 26.4 The energy stored in a capacitor is equal to $q^2/2C$. When a capacitor is discharged, what fraction of the initial energy remains after an elapsed time of one time constant? (i) $1/e$; (ii) $1/e^2$; (iii) $1 - 1/e$; (iv) $(1 - 1/e)^2$; (v) answer depends on how much energy was stored initially.

ANSWER

(ii) After one time constant, $t = RC$ and the initial charge Q_0 has decreased to $Q_0 e^{-t/RC} = Q_0 e^{-1} = Q_0/e$. Hence the stored energy has decreased from $Q_0^2/2C$ to $(Q_0/e)^2/2C = Q_0^2/2C e^2$, a fraction $1/e^2 = 0.135$ of its initial value. This result doesn't depend on the initial value of the energy.

26.5 POWER DISTRIBUTION SYSTEMS

We conclude this chapter with a brief discussion of practical household and automotive electric-power distribution systems. Automobiles use direct-current (dc) systems, while nearly all household, commercial, and industrial systems use alternating current (ac) because of the ease of stepping voltage up and down with transformers. Most of the same basic wiring concepts apply to both. We'll talk about alternating-current circuits in greater detail in Chapter 31.

The various lamps, motors, and other appliances to be operated are always connected in *parallel* to the power source (the wires from the power company for houses, or from the battery and alternator for a car). If appliances were connected in series, shutting one appliance off would shut them all off (see Example 26.2 in Section 26.1). **Figure 26.24** shows the basic idea of house wiring. One side of the “line,” as the pair of conductors is called, is called the *neutral* side; it is always connected to “ground” at the entrance panel. For houses, *ground* is an actual electrode driven into the earth (which is usually a good conductor) or sometimes connected to the household water pipes. Electricians speak of the “hot” side and the “neutral” side of the line. Most modern house wiring systems have *two* hot lines with opposite polarity with respect to the neutral. We'll return to this detail later.

Household voltage is nominally 120 V in the United States and Canada, and often 240 V in Europe. (For alternating current, which varies sinusoidally with time, these numbers represent the *root-mean-square* voltage, which is $1/\sqrt{2}$ times the peak voltage. We'll discuss this further in Section 31.1.) The amount of current I drawn by a given device is determined by its power input P , given by Eq. (25.17): $P = VI$. Hence $I = P/V$. For example, the current in a 100 W light bulb is

$$I = \frac{P}{V} = \frac{100 \text{ W}}{120 \text{ V}} = 0.83 \text{ A}$$

The power input to this bulb is actually determined by its resistance R . Using Eq. (25.18), which states that $P = VI = I^2 R = V^2/R$ for a resistor, we get the resistance of this bulb at operating temperature:

$$R = \frac{V}{I} = \frac{120 \text{ V}}{0.83 \text{ A}} = 144 \Omega \quad \text{or} \quad R = \frac{V^2}{P} = \frac{(120 \text{ V})^2}{100 \text{ W}} = 144 \Omega$$

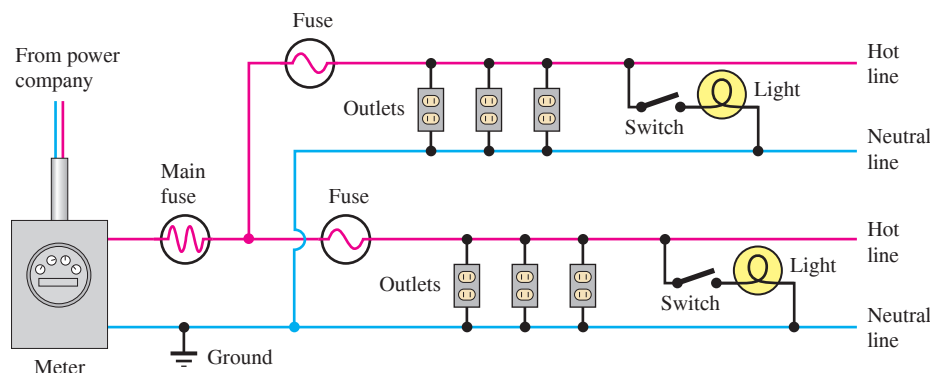


Figure 26.24 Schematic diagram of part of a house wiring system. Only two branch circuits are shown; an actual system might have four to thirty branch circuits. Lamps and appliances may be plugged into the outlets. The grounding wires, which normally carry no current, are not shown.

Similarly, a 1500 W waffle iron draws a current of $(1500 \text{ W})/(120 \text{ V}) = 12.5 \text{ A}$ and has a resistance, at operating temperature, of 9.6Ω . Because of the temperature dependence of resistivity, the resistances of these devices are considerably less when they are cold. If you measure the resistance of a 100 W light bulb with an ohmmeter (whose small current causes very little temperature rise), you'll probably get a value of about 10Ω . When a light bulb is turned on, this low resistance causes an initial surge of current until the filament heats up. That's why a light bulb that's ready to burn out nearly always does so just when you turn it on.

Circuit Overloads and Short Circuits

The maximum current available from an individual circuit is limited by the resistance of the wires. As we discussed in Section 25.5, the I^2R power loss in the wires causes them to become hot, and in extreme cases this can cause a fire or melt the wires. Ordinary lighting and outlet wiring in houses usually uses 12 gauge wire. This has a diameter of 2.05 mm and can carry a maximum current of 20 A safely (without overheating). Larger-diameter wires of the same length have lower resistance [see Eq. (25.10)]. Hence 8 gauge (3.26 mm) or 6 gauge (4.11 mm) is used for high-current appliances such as clothes dryers, and 2 gauge (6.54 mm) or larger is used for the main power lines entering a house.

Protection against overloading and overheating of circuits is provided by fuses or circuit breakers. A *fuse* contains a link of lead–tin alloy with a very low melting temperature; the link melts and breaks the circuit when its rated current is exceeded (**Fig. 26.25a**). A *circuit breaker* is an electromechanical device that performs the same function, using an electromagnet or a bimetallic strip to “trip” the breaker and interrupt the circuit when the current exceeds a specified value (**Fig. 26.25b**). Circuit breakers have the advantage that they can be reset after they are tripped, while a blown fuse must be replaced.

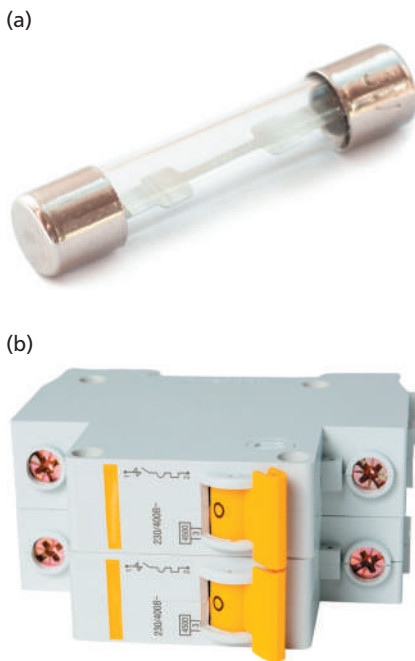
CAUTION Fuses If your system has fuses and you plug too many high-current appliances into the same outlet, the fuse blows. Do not replace the fuse with one that has a higher rating; if you do, you risk overheating the wires and starting a fire. The only safe solution is to distribute the appliances among several circuits. Modern kitchens often have three or four separate 20 A circuits. ■

Contact between the hot and neutral sides of the line causes a *short circuit*. Such a situation, which can be caused by faulty insulation or by a variety of mechanical malfunctions, provides a very low-resistance current path, permitting a very large current that would quickly melt the wires and ignite their insulation if the current were not interrupted by a fuse or circuit breaker (see Example 25.10 in Section 25.5). An equally dangerous situation is a broken wire that interrupts the current path, creating an *open circuit*. This is hazardous because of the sparking that can occur at the point of intermittent contact.

In approved wiring practice, a fuse or breaker is placed *only* in the hot side of the line, never in the neutral side. Otherwise, if a short circuit should develop because of faulty insulation or other malfunction, the ground-side fuse could blow. The hot side would still be live and would pose a shock hazard if you touched the live conductor and a grounded object such as a water pipe. For similar reasons the wall switch for a light fixture should always be in the hot side of the line, never the neutral side.

Further protection against shock hazard is provided by a third conductor called the *grounding wire*, included in all present-day wiring. This conductor corresponds to the long round or U-shaped prong of the three-prong connector plug on an appliance or power tool. It is connected to the neutral side of the line at the entrance panel. The grounding wire normally carries no current, but it connects the metal case or frame of the device to ground. If a conductor on the hot side of the line accidentally contacts the frame or case, the grounding conductor provides a current path, and the fuse blows. Without the ground wire, the frame could become “live”—that is, at a potential 120 V above ground. Then if you touched it and a water pipe (or even a damp floor) at the same time, you could get a

Figure 26.25 (a) Excess current will melt the thin wire of lead–tin alloy that runs along the length of a fuse, inside the transparent housing. (b) The switch on this circuit breaker will flip if the maximum allowable current is exceeded.



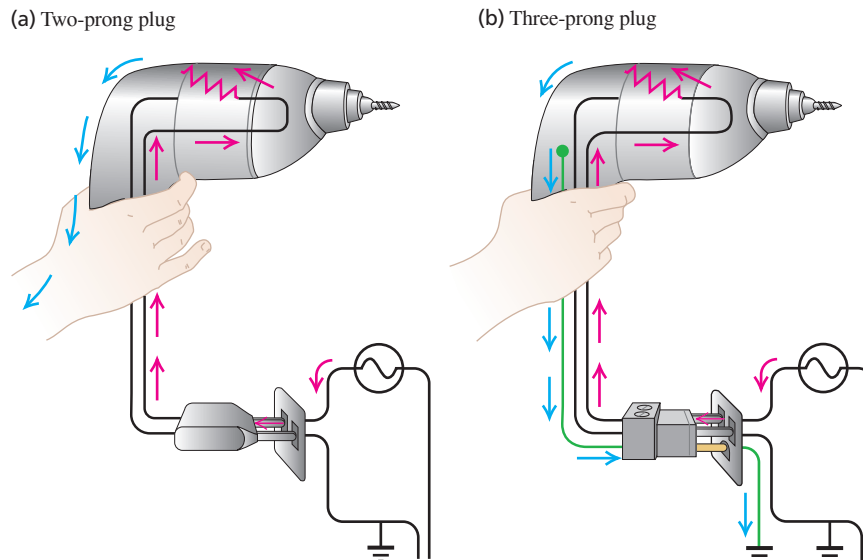


Figure 26.26 (a) If a malfunctioning electric drill is connected to a wall socket via a two-prong plug, a person may receive a shock. (b) When the drill malfunctions when connected via a three-prong plug, a person touching it receives no shock, because electric charge flows through the ground wire (shown in green) to the third prong and into the ground rather than into the person's body. If the ground current is appreciable, the fuse blows.

dangerous shock (**Fig. 26.26**). In some situations, especially outlets located outdoors or near a sink or other water pipes, a special kind of circuit breaker called a *ground-fault interrupter* (GFI or GFCI) is used. This device senses the difference in current between the hot and neutral conductors (which is normally zero) and trips when some very small value, typically 5 mA, is exceeded.

Household and Automotive Wiring

Most modern household wiring systems actually use a slight elaboration of the system described above. The power company provides *three* conductors. One is neutral; the other two are both at 120 V with respect to the neutral but with opposite polarity, giving a voltage between them of 240 V. The power company calls this a *three-wire line*, in contrast to the 120 V two-wire (plus ground wire) line described above. With a three-wire line, 120 V lamps and appliances can be connected between neutral and either hot conductor, and high-power devices requiring 240 V, such as electric ranges and clothes dryers, are connected between the two hot lines.

All of the above discussion can be applied directly to automobile wiring. The voltage is about 13 V (direct current); the power is supplied by the battery and by the alternator, which charges the battery when the engine is running. The neutral side of each circuit is connected to the body and frame of the vehicle. For this low voltage a separate grounding conductor is not required for safety. The fuse or circuit breaker arrangement is the same in principle as in household wiring. Because of the lower voltage (less energy per charge), more current (a greater number of charges per second) is required for the same power; a 100 W headlight bulb requires a current of about $(100 \text{ W})/(13 \text{ V}) = 8 \text{ A}$.

Although we spoke of *power* in the above discussion, what we buy from the power company is *energy*. Power is energy transferred per unit time, so energy is average power multiplied by time. The usual unit of energy sold by the power company is the kilowatt-hour (1 kW · h):

$$1 \text{ kW} \cdot \text{h} = (10^3 \text{ W})(3600 \text{ s}) = 3.6 \times 10^6 \text{ W} \cdot \text{s} = 3.6 \times 10^6 \text{ J}$$

In the United States, one kilowatt-hour typically costs 8 to 27 cents, depending on the location and quantity of energy purchased. To operate a 1500 W (1.5 kW) waffle iron continuously for 1 hour requires 1.5 kW · h of energy; at 10 cents per kilowatt-hour, the energy cost is 15 cents. The cost of operating any lamp or appliance for a specified time can be calculated in the same way if the power rating is known. However, many electric cooking utensils (including waffle irons) cycle on and off to maintain a constant temperature, so the average power may be less than the power rating marked on the device.

EXAMPLE 26.14 A kitchen circuit

An 1800 W toaster, a 1.3 kW electric frying pan, and a 100 W lamp are plugged into the same 20 A, 120 V circuit. (a) What current is drawn by each device, and what is the resistance of each device? (b) Will this combination trip the circuit breaker?

IDENTIFY and SET UP When plugged into the same circuit, the three devices are connected in parallel, so the voltage across each appliance is $V = 120$ V. We find the current I drawn by each device from the relationship $P = VI$, where P is the power input of the device. To find the resistance R of each device we use the relationship $P = V^2/R$.

EXECUTE (a) To simplify the calculation of current and resistance, we note that $I = P/V$ and $R = V^2/P$. Hence

$$\begin{aligned} I_{\text{toaster}} &= \frac{1800 \text{ W}}{120 \text{ V}} = 15 \text{ A} & R_{\text{toaster}} &= \frac{(120 \text{ V})^2}{1800 \text{ W}} = 8 \Omega \\ I_{\text{frying pan}} &= \frac{1300 \text{ W}}{120 \text{ V}} = 11 \text{ A} & R_{\text{frying pan}} &= \frac{(120 \text{ V})^2}{1300 \text{ W}} = 11 \Omega \\ I_{\text{lamp}} &= \frac{100 \text{ W}}{120 \text{ V}} = 0.83 \text{ A} & R_{\text{lamp}} &= \frac{(120 \text{ V})^2}{100 \text{ W}} = 144 \Omega \end{aligned}$$

For constant voltage the device with the *least* resistance (in this case the toaster) draws the most current and receives the most power.

(b) The total current through the line is the sum of the currents drawn by the three devices:

$$\begin{aligned} I &= I_{\text{toaster}} + I_{\text{frying pan}} + I_{\text{lamp}} \\ &= 15 \text{ A} + 11 \text{ A} + 0.83 \text{ A} = 27 \text{ A} \end{aligned}$$

This exceeds the 20 A rating of the line, and the circuit breaker will indeed trip.

EVALUATE We could also find the total current by using $I = P/V$ and the total power P delivered to all three devices:

$$\begin{aligned} I &= \frac{P_{\text{toaster}} + P_{\text{frying pan}} + P_{\text{lamp}}}{V} \\ &= \frac{1800 \text{ W} + 1300 \text{ W} + 100 \text{ W}}{120 \text{ V}} = 27 \text{ A} \end{aligned}$$

A third way to determine I is to use $I = V/R_{\text{eq}}$, where R_{eq} is the equivalent resistance of the three devices in parallel:

$$I = \frac{V}{R_{\text{eq}}} = (120 \text{ V}) \left(\frac{1}{8 \Omega} + \frac{1}{11 \Omega} + \frac{1}{144 \Omega} \right) = 27 \text{ A}$$

Appliances with such current demands are common, so modern kitchens have more than one 20 A circuit. To keep currents safely below 20 A, the toaster and frying pan should be plugged into different circuits.

KEYCONCEPT The voltage across any device plugged into a household wiring system has the same value. The smaller the resistance of the device, the more current it draws and the more power it receives.

TEST YOUR UNDERSTANDING OF SECTION 26.5 To prevent the circuit breaker in Example 26.14 from blowing, a home electrician replaces the circuit breaker with one rated at 40 A. Is this a reasonable thing to do?

ANSWER

no This is a very dangerous thing to do. The circuit breaker will allow currents up to 40 A, double the rated value of the wiring. The amount of power $P = I^2 R$ dissipated in a section of wire can therefore be up to four times the rated value, so the wires could get hot and start a fire. (This assumes the resistance R remains unchanged. In fact, R increases with temperature, so the dissipated power can be even greater, than we have estimated.)

CHAPTER 26 SUMMARY

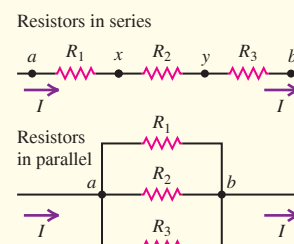
Resistors in series and parallel: When several resistors $R_1, R_2, R_3, \dots$ are connected in series, the equivalent resistance R_{eq} is the sum of the individual resistances. The same *current* flows through all the resistors in a series connection. When several resistors are connected in parallel, the reciprocal of equivalent resistance R_{eq} is the sum of the reciprocals of the individual resistances. All resistors in a parallel connection have the same *potential difference* between their terminals. (See Examples 26.1 and 26.2.)

$$R_{eq} = R_1 + R_2 + R_3 + \dots \quad (26.1)$$

(resistors in series)

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \quad (26.2)$$

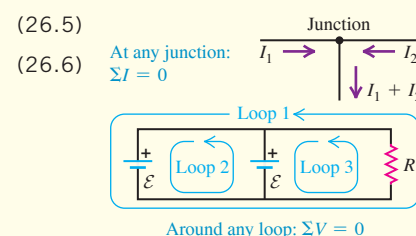
(resistors in parallel)



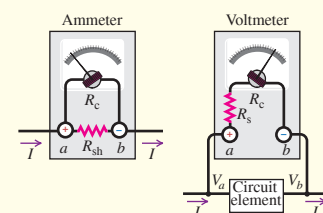
Kirchhoff's rules: Kirchhoff's junction rule is based on conservation of charge. It states that the algebraic sum of the currents into any junction must be zero. Kirchhoff's loop rule is based on conservation of energy and the conservative nature of electrostatic fields. It states that the algebraic sum of potential differences around any loop must be zero. Careful use of consistent sign rules is essential in applying Kirchhoff's rules. (See Examples 26.3–26.7.)

$$\sum I = 0 \quad (\text{junction rule})$$

$$\sum V = 0 \quad (\text{loop rule})$$



Electrical measuring instruments: In a d'Arsonval galvanometer, the deflection is proportional to the current in the coil. For a larger current range, a shunt resistor is added, so some of the current bypasses the meter coil. Such an instrument is called an ammeter. If the coil and any additional series resistance included obey Ohm's law, the meter can also be calibrated to read potential difference or voltage. The instrument is then called a voltmeter. A good ammeter has very low resistance; a good voltmeter has very high resistance. (See Examples 26.8–26.11.)



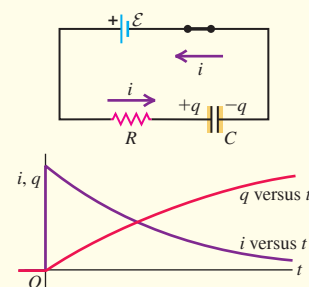
R-C circuits: When a capacitor is charged by a battery in series with a resistor, the current and capacitor charge are not constant. The charge approaches its final value asymptotically and the current approaches zero asymptotically. The charge and current in the circuit are given by Eqs. (26.12) and (26.13). After a time $\tau = RC$, the charge has approached within $1/e$ of its final value. This time is called the time constant or relaxation time of the circuit. When the capacitor discharges, the charge and current are given as functions of time by Eqs. (26.16) and (26.17). The time constant is the same for charging and discharging. (See Examples 26.12 and 26.13.)

Capacitor charging:

$$q = C\mathcal{E}(1 - e^{-t/RC}) \quad (26.12)$$

$$= Q_f(1 - e^{-t/RC})$$

$$i = \frac{dq}{dt} = \frac{\mathcal{E}}{R}e^{-t/RC} = I_0e^{-t/RC} \quad (26.13)$$

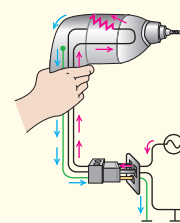


Capacitor discharging:

$$q = Q_0e^{-t/RC} \quad (26.16)$$

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC}e^{-t/RC} = I_0e^{-t/RC} \quad (26.17)$$

Household wiring: In household wiring systems, the various electrical devices are connected in parallel across the power line, which consists of a pair of conductors, one “hot” and the other “neutral.” An additional “ground” wire is included for safety. The maximum permissible current in a circuit is determined by the size of the wires and the maximum temperature they can tolerate. Protection against excessive current and the resulting fire hazard is provided by fuses or circuit breakers. (See Example 26.14.)





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 26.1 and 26.2** (Section 26.1) before attempting these problems.

VP26.2.1 You have three resistors: $R_1 = 1.00\ \Omega$, $R_2 = 2.00\ \Omega$, and $R_3 = 4.00\ \Omega$. Find the equivalent resistance for the combinations shown in (a) Fig. 26.1a, (b) Fig. 26.1b, (c) Fig. 26.1c, and (d) Fig. 26.1d.

VP26.2.2 You have three resistors, $R_1 = 4.00\ \Omega$, $R_2 = 5.00\ \Omega$, and $R_3 = 6.00\ \Omega$, connected as shown in Fig. 26.1d. The terminals of a battery with emf 24.0 V and negligible internal resistance are connected to points a and b . Find the current through (a) the battery, (b) R_1 , (c) R_2 , and (d) R_3 .

VP26.2.3 You have three resistors, $R_1 = 7.00\ \Omega$, $R_2 = 8.00\ \Omega$, and $R_3 = 9.00\ \Omega$, connected as shown in Fig. 26.1b. The terminals of a battery with emf 12.0 V and negligible internal resistance are connected to points a and b . Find (a) the power output of the battery and the power input to (b) R_1 , (c) R_2 , and (d) R_3 .

VP26.2.4 You have three resistors, $R_1 = 5.00\ \Omega$, $R_2 = 6.00\ \Omega$, and $R_3 = 7.00\ \Omega$, connected as shown in Fig. 26.1c. The terminals of a battery with emf 9.00 V and negligible internal resistance are connected to points a and b . Find (a) the power output of the battery and the power input to (b) R_1 , (c) R_2 , and (d) R_3 .

Be sure to review **EXAMPLES 26.3, 26.4, 26.5, 26.6, and 26.7** (Section 26.2) before attempting these problems.

VP26.7.1 In the circuit shown in Fig. 26.10a, you reverse the 4 V battery so that its positive terminal is on the right side instead of the left. Find (a) the current in the circuit, (b) the potential difference V_{ab} , and (c) the power output of the emf of each battery.

VP26.7.2 Figure 26.6a shows two batteries, one with emf $\mathcal{E}_1$ and one with emf $\mathcal{E}_2$, connected to a resistance R . Current I_1 flows through emf $\mathcal{E}_1$ from point c toward point a , and current I_2 flows through emf $\mathcal{E}_2$ from point b toward point a . If $\mathcal{E}_1 = 8.00\ \text{V}$, $\mathcal{E}_2 = 9.00\ \text{V}$, $R = 5.00\ \Omega$, $I_1 = 0.200\ \text{A}$, and $I_2 = 1.35\ \text{A}$, find the values of (a) the potential difference V_{ab} , (b) the internal resistance r_1 , and (c) the internal resistance r_2 .

VP26.7.3 In the circuit shown in Figure 26.12, you replace the $2\ \Omega$ resistor between points b and d with a $3\ \Omega$ resistor. Find the power dissipated in (a) the $1\ \Omega$ resistor between points c and a , (b) the $1\ \Omega$ resistor between points c and b , (c) the $1\ \Omega$ resistor between points a and b , (d) the $1\ \Omega$ resistor between points a and d , and (e) the $3\ \Omega$ resistor between points b and d . Give each answer to three significant figures.

VP26.7.4 In the circuit shown in Fig. 26.9a, let $r_1 = r_2 = 0$ (that is, the internal resistances are very small) and let $R_1 = R_2 = R_3 = R$. Find the values of (a) I_1 , (b) I_2 , and (c) I_3 .

Be sure to review **EXAMPLES 26.12 and 26.13** (Section 26.4) before attempting these problems.

VP26.13.1 You connect a $10.0\ \text{M}\Omega$ resistor in series with a $3.20\ \mu\text{F}$ capacitor and a battery with emf 9.00 V. Before you close the switch at $t = 0$ to complete the circuit, the capacitor is uncharged. Find (a) the final capacitor charge, (b) the initial current, (c) the time constant, (d) the fraction of the final charge on the capacitor at $t = 18.0\ \text{s}$, and (e) the fraction of the initial current present at $t = 18.0\ \text{s}$.

VP26.13.2 A $2.20\ \mu\text{F}$ capacitor initially has charge $4.20\ \mu\text{C}$. You connect it in series with a $4.00\ \text{M}\Omega$ resistor and an open switch. If $t = 0$ is the time when you close the switch, (a) at what time will the capacitor charge be $1.20\ \mu\text{C}$ and (b) what will be the current in the circuit at this time?

VP26.13.3 You connect an unknown resistor in series with an $8.00\ \mu\text{F}$ capacitor and an open switch. Before you close the switch at $t = 0$, the capacitor has charge $5.50\ \mu\text{C}$; at $t = 17.0\ \text{s}$, the charge has decreased to $1.10\ \mu\text{C}$. Find (a) the resistance of the resistor, (b) the current in the circuit just after you close the switch, and (c) the current in the circuit at $t = 17.0\ \text{s}$.

VP26.13.4 You connect an initially uncharged $6.40\ \mu\text{F}$ capacitor in series with a $5.00\ \text{M}\Omega$ resistor and a battery with emf 12.0 V. After letting the capacitor charge for 51.0 s, you disconnect it from this circuit and connect it in series to an open switch and a $6.00\ \text{M}\Omega$ resistor. Find the charge on the capacitor (a) when you disconnect it from the first circuit and (b) 70.0 s after you close the switch in the second circuit.

BRIDGING PROBLEM Two Capacitors and Two Resistors

A $2.40\ \mu\text{F}$ capacitor and a $3.60\ \mu\text{F}$ capacitor are connected in series. (a) A charge of $5.20\ \text{mC}$ is placed on each capacitor. What is the energy stored in the capacitors? (b) A $655\ \Omega$ resistor is connected to the terminals of the capacitor combination, and a voltmeter with resistance $4.58 \times 10^4\ \Omega$ is connected across the resistor (Fig. 26.27). What is the rate of change of the energy stored in the capacitors just after the connection is made? (c) How long after the connection is made has the energy stored in the capacitors decreased to $1/e$ of its initial value? (d) At the instant calculated in part (c), what is the rate of change of the energy stored in the capacitors?

SOLUTION GUIDE

IDENTIFY and SET UP

1. The two capacitors act as a single equivalent capacitor (see Section 24.2), and the resistor and voltmeter act as a single equivalent resistor. Select equations that will allow you to calculate the values of these equivalent circuit elements.
2. In part (a) you'll need to use Eq. (24.9), which gives the energy stored in a capacitor.
3. For parts (b), (c), and (d), you'll need to use Eq. (24.9) as well as Eqs. (26.16) and (26.17), which give the capacitor charge and

current as functions of time. (Hint: The rate at which energy is lost by the capacitors equals the rate at which energy is dissipated in the resistances.)

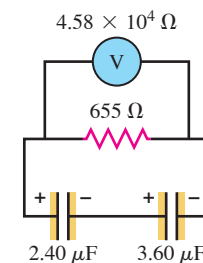
Figure 26.27 When the connection is made, the charged capacitors discharge.

EXECUTE

4. Find the stored energy at $t = 0$.
5. Find the rate of change of the stored energy at $t = 0$.
6. Find the value of t at which the stored energy has $1/e$ of the value you found in step 4.
7. Find the rate of change of the stored energy at the time you found in step 6.

EVALUATE

8. Check your results from steps 5 and 7 by calculating the rate of change in a different way. (Hint: The rate of change of the stored energy U is dU/dt .)



PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

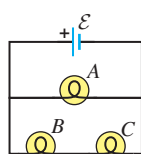
Q26.1 In which 120 V light bulb does the filament have greater resistance: a 60 W bulb or a 120 W bulb? If the two bulbs are connected to a 120 V line in series, through which bulb will there be the greater voltage drop? What if they are connected in parallel? Explain your reasoning.

Q26.2 Two 120 V light bulbs, one 25 W and one 200 W, were connected in series across a 240 V line. It seemed like a good idea at the time, but one bulb burned out almost immediately. Which one burned out, and why?

Q26.3 You connect a number of identical light bulbs to a flashlight battery. (a) What happens to the brightness of each bulb as more and more bulbs are added to the circuit if you connect them (i) in series and (ii) in parallel? (b) Will the battery last longer if the bulbs are in series or in parallel? Explain your reasoning.

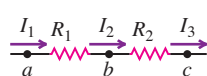
Q26.4 In the circuit shown in **Fig. Q26.4**, three identical light bulbs are connected to a flashlight battery. How do the brightnesses of the bulbs compare? Which light bulb has the greatest current passing through it? Which light bulb has the greatest potential difference between its terminals? What happens if bulb A is unscrewed? Bulb B? Bulb C? Explain your reasoning.

Figure Q26.4



Q26.5 If two resistors R_1 and R_2 ($R_2 > R_1$) are connected in series as shown in **Fig. Q26.5**, which of the following must be true? In each case justify your answer. (a) $I_1 = I_2 = I_3$. (b) The current is greater in R_1 than in R_2 .

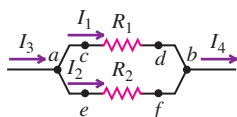
Figure Q26.5



(c) The electrical power consumption is the same for both resistors. (d) The electrical power consumption is greater in R_2 than in R_1 . (e) The potential drop is the same across both resistors. (f) The potential at point a is the same as at point c. (g) The potential at point b is lower than at point c. (h) The potential at point c is lower than at point b.

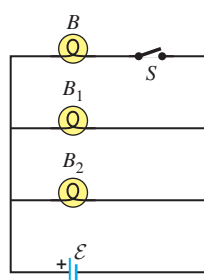
Q26.6 If two resistors R_1 and R_2 ($R_2 > R_1$) are connected in parallel as shown in **Fig. Q26.6**, which of the following must be true? In each case justify your answer. (a) $I_1 = I_2$. (b) $I_3 = I_4$. (c) The current is greater in R_1 than in R_2 . (d) The rate of electrical energy consumption is the same for both resistors. (e) The rate of electrical energy consumption is greater in R_2 than in R_1 . (f) $V_{cd} = V_{ef} = V_{ab}$. (g) Point c is at higher potential than point d. (h) Point f is at higher potential than point e. (i) Point c is at higher potential than point e.

Figure Q26.6



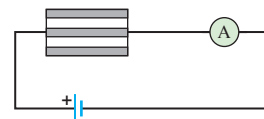
Q26.7 A battery with no internal resistance is connected across identical light bulbs as shown in **Fig. Q26.7**. When you close the switch S, will the brightness of bulbs B_1 and B_2 change? If so, how will it change? Explain.

Figure Q26.7



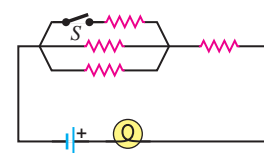
Q26.8 A resistor consists of three identical metal strips connected as shown in **Fig. Q26.8**. If one of the strips is cut out, does the ammeter reading increase, decrease, or stay the same? Why?

Figure Q26.8



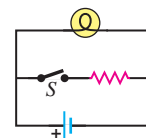
Q26.9 A light bulb is connected in the circuit shown in **Fig. Q26.9**. If we close the switch S, does the bulb's brightness increase, decrease, or remain the same? Explain why.

Figure Q26.9



Q26.10 A real battery, having nonnegligible internal resistance, is connected across a light bulb as shown in **Fig. Q26.10**. When the switch S is closed, what happens to the brightness of the bulb? Why?

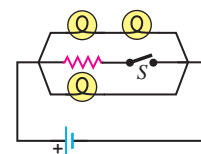
Figure Q26.10



Q26.11 If the battery in Discussion Question Q26.10 is ideal with no internal resistance, what will happen to the brightness of the bulb when S is closed? Why?

Q26.12 Consider the circuit shown in **Fig. Q26.12**. What happens to the brightnesses of the bulbs when the switch S is closed if the battery (a) has no internal resistance and (b) has nonnegligible internal resistance? Explain why.

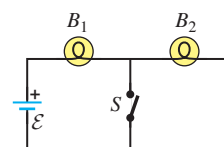
Figure Q26.12



Q26.13 Is it possible to connect resistors together in a way that cannot be reduced to some combination of series and parallel combinations? If so, give examples. If not, state why not.

Q26.14 The battery in the circuit shown in **Fig. Q26.14** has no internal resistance. After you close the switch S, will the brightness of bulb B_1 increase, decrease, or stay the same?

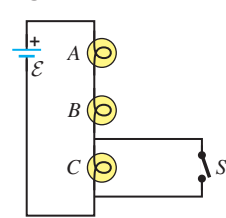
Figure Q26.14



Q26.15 In a two-cell flashlight, the batteries are usually connected in series. Why not connect them in parallel? What possible advantage could there be in connecting several identical batteries in parallel?

Q26.16 Identical light bulbs A , B , and C are connected as shown in **Fig. Q26.16**. When the switch S is closed, bulb C goes out. Explain why. What happens to the brightness of bulbs A and B ? Explain.

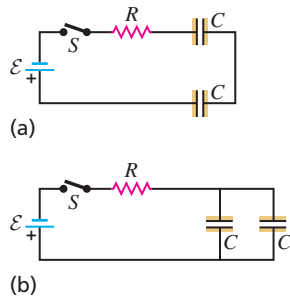
Figure Q26.16



Q26.17 The emf of a flashlight battery is roughly constant with time, but its internal resistance increases with age and use. What sort of meter should be used to test the freshness of a battery?

Q26.18 Will the capacitors in the circuits shown in **Fig. Q26.18** charge at the same rate when the switch S is closed? If not, in which circuit will the capacitors charge more rapidly? Explain.

Figure Q26.18



Q26.19 Verify that the time constant RC has units of time.

Q26.20 For very large resistances it is easy to construct R - C circuits that have time constants of several seconds or minutes. How might this fact be used to measure very large resistances, those that are too large to measure by more conventional means?

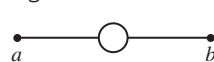
Q26.21 When a capacitor, battery, and resistor are connected in series, does the resistor affect the maximum charge stored on the capacitor? Why or why not? What purpose does the resistor serve?

EXERCISES

Section 26.1 Resistors in Series and Parallel

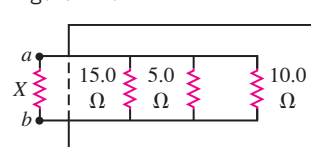
26.1 • A uniform wire of resistance R is cut into three equal lengths. One of these is formed into a circle and connected between the other two (**Fig. E26.1**). What is the resistance between the opposite ends a and b ?

Figure E26.1



26.2 • A machine part has a resistor X protruding from an opening in the side. This resistor is connected to three other resistors, as shown in **Fig. E26.2**. An ohmmeter connected across a and b reads $2.00\ \Omega$. What is the resistance of X ?

Figure E26.2

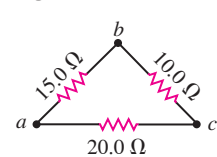


26.3 • A resistor with $R_1 = 25.0\ \Omega$ is connected to a battery that has negligible internal resistance and electrical energy is dissipated by R_1 at a rate of $36.0\ \text{W}$. If a second resistor with $R_2 = 15.0\ \Omega$ is connected in series with R_1 , what is the total rate at which electrical energy is dissipated by the two resistors?

26.4 • A $42\ \Omega$ resistor and a $20\ \Omega$ resistor are connected in parallel, and the combination is connected across a $240\ \text{V}$ dc line. (a) What is the resistance of the parallel combination? (b) What is the total current through the parallel combination? (c) What is the current through each resistor?

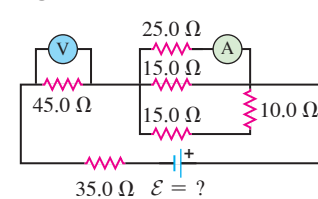
26.5 • A triangular array of resistors is shown in **Fig. E26.5**. What current will this array draw from a $35.0\ \text{V}$ battery having negligible internal resistance if we connect it across (a) ab ; (b) bc ; (c) ac ? (d) If the battery has an internal resistance of $3.00\ \Omega$, what current will the array draw if the battery is connected across bc ?

Figure E26.5



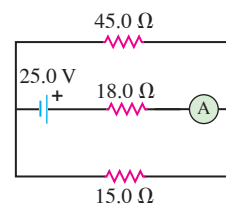
26.6 • For the circuit shown in **Fig. E26.6** both meters are idealized, the battery has no appreciable internal resistance, and the ammeter reads $1.25\ \text{A}$. (a) What does the voltmeter read? (b) What is the emf $\mathcal{E}$ of the battery?

Figure E26.6



26.7 • For the circuit shown in **Fig. E26.7** find the reading of the idealized ammeter if the battery has an internal resistance of $3.26\ \Omega$.

Figure E26.7



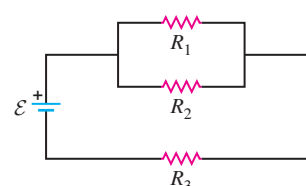
26.8 • Two resistors, R_1 and R_2 , are connected in parallel to a power supply that has voltage V and negligible internal resistance. $R_2 = 8.00\ \Omega$ and the resistance of R_1 is not known. For several values of V , you measure the current I flowing through the voltage source. You plot the data as I versus V and find that they lie close to a straight line that has slope $0.208\ \Omega^{-1}$. What is the resistance of R_1 ?

26.9 • Six identical resistors, each with resistance R , are connected to an emf $\mathcal{E}$. (a) In terms of $\mathcal{E}$ and R , what is the current I through each of the resistors if they are connected in parallel? (b) In series? (c) For which network of resistors, series or parallel, is the power consumed in each resistor greater?

26.10 • **Power Rating of a Resistor.** The *power rating* of a resistor is the maximum power the resistor can safely dissipate without too great a rise in temperature and hence damage to the resistor. (a) If the power rating of a $15\ \text{k}\Omega$ resistor is $5.0\ \text{W}$, what is the maximum allowable potential difference across the terminals of the resistor? (b) A $9.0\ \text{k}\Omega$ resistor is to be connected across a $120\ \text{V}$ potential difference. What power rating is required? (c) A $100.0\ \Omega$ and a $150.0\ \Omega$ resistor, both rated at $2.00\ \text{W}$, are connected in series across a variable potential difference. What is the greatest this potential difference can be without overheating either resistor, and what is the rate of heat generated in each resistor under these conditions?

26.11 • In **Fig. E26.11**, $R_1 = 3.00\ \Omega$, $R_2 = 6.00\ \Omega$, and $R_3 = 5.00\ \Omega$. The battery has negligible internal resistance. The current I_2 through R_2 is $4.00\ \text{A}$. (a) What are the currents I_1 and I_3 ? (b) What is the emf of the battery?

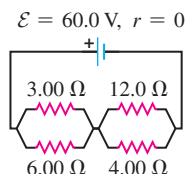
Figure E26.11



26.12 • In **Fig. E26.11** the battery has emf $35.0\ \text{V}$ and negligible internal resistance. $R_1 = 5.00\ \Omega$. The current through R_1 is $1.50\ \text{A}$, and the current through $R_3 = 4.50\ \text{A}$. What are the resistances R_2 and R_3 ?

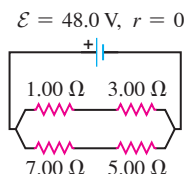
26.13 • Compute the equivalent resistance of the network in **Fig. E26.13**, and find the current in each resistor. The battery has negligible internal resistance.

Figure E26.13



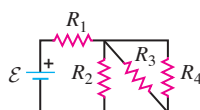
26.14 • Compute the equivalent resistance of the network in **Fig. E26.14**, and find the current in each resistor. The battery has negligible internal resistance.

Figure E26.14



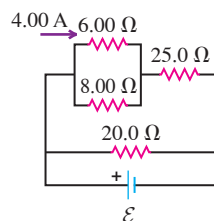
26.15 • In the circuit of **Fig. E26.15**, each resistor represents a light bulb. Let $R_1 = R_2 = R_3 = R_4 = 4.50 \, \Omega$ and $\mathcal{E} = 9.00 \, \text{V}$. (a) Find the current in each bulb. (b) Find the power dissipated in each bulb. Which bulb or bulbs glow the brightest? (c) Bulb R_4 is now removed from the circuit, leaving a break in the wire at its position. Now what is the current in each of the remaining bulbs R_1 , R_2 , and R_3 ? (d) With bulb R_4 removed, what is the power dissipated in each of the remaining bulbs? (e) Which light bulb(s) glow brighter as a result of removing R_4 ? Which bulb(s) glow less brightly? Discuss why there are different effects on different bulbs.

Figure E26.15



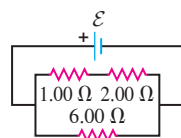
26.16 • Consider the circuit shown in **Fig. E26.16**. The current through the $6.00 \, \Omega$ resistor is $4.00 \, \text{A}$, in the direction shown. What are the currents through the $25.0 \, \Omega$ and $20.0 \, \Omega$ resistors?

Figure E26.16



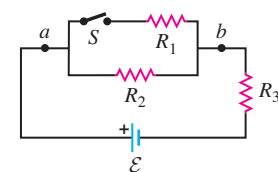
26.17 • In the circuit shown in **Fig. E26.17**, the voltage across the $2.00 \, \Omega$ resistor is $12.0 \, \text{V}$. What are the emf of the battery and the current through the $6.00 \, \Omega$ resistor?

Figure E26.17



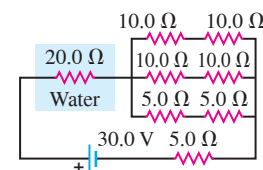
26.18 •• In the circuit shown in **Fig. E26.18**, $\mathcal{E} = 36.0 \, \text{V}$, $R_1 = 4.00 \, \Omega$, $R_2 = 6.00 \, \Omega$, and $R_3 = 3.00 \, \Omega$. (a) What is the potential difference V_{ab} between points a and b when the switch S is open and when S is closed? (b) For each resistor, calculate the current through the resistor with S open and with S closed. For each resistor, does the current increase or decrease when S is closed?

Figure E26.18



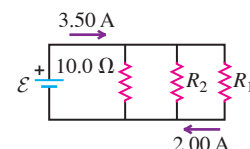
26.19 •• CP In the circuit in **Fig. E26.19**, a $20.0 \, \Omega$ resistor is inside $100 \, \text{g}$ of pure water that is surrounded by insulating Styrofoam. If the water is initially at 10.0°C , how long will it take for its temperature to rise to 58.0°C ?

Figure E26.19



26.20 • In the circuit shown in **Fig. E26.20**, the rate at which R_1 is dissipating electrical energy is $15.0 \, \text{W}$. (a) Find R_1 and R_2 . (b) What is the emf of the battery? (c) Find the current through both R_2 and the $10.0 \, \Omega$ resistor. (d) Calculate the total electrical power consumption in all the resistors and the electrical power delivered by the battery. Show that your results are consistent with conservation of energy.

Figure E26.20

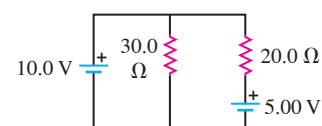


26.21 • Light Bulbs in Series and in Parallel. Two light bulbs have constant resistances of $400 \, \Omega$ and $800 \, \Omega$. If the two light bulbs are connected in series across a $120 \, \text{V}$ line, find (a) the current through each bulb; (b) the power dissipated in each bulb; (c) the total power dissipated in both bulbs. The two light bulbs are now connected in parallel across the $120 \, \text{V}$ line. Find (d) the current through each bulb; (e) the power dissipated in each bulb; (f) the total power dissipated in both bulbs. (g) In each situation, which of the two bulbs glows the brightest? (h) In which situation is there a greater total light output from both bulbs combined?

Section 26.2 Kirchhoff's Rules

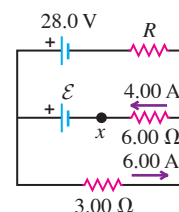
26.22 •• The batteries shown in the circuit in **Fig. E26.22** have negligibly small internal resistances. Find the current through (a) the $30.0 \, \Omega$ resistor; (b) the $20.0 \, \Omega$ resistor; (c) the $10.0 \, \text{V}$ battery.

Figure E26.22



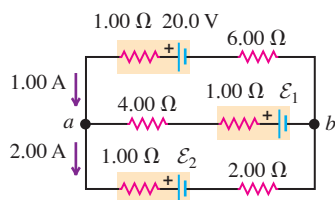
26.23 • In the circuit shown in **Fig. E26.23** find (a) the current in resistor R ; (b) the resistance R ; (c) the unknown emf $\mathcal{E}$. (d) If the circuit is broken at point x , what is the current in resistor R ?

Figure E26.23



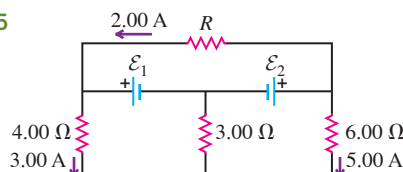
26.24 • Find the emfs $\mathcal{E}_1$ and $\mathcal{E}_2$ in the circuit of **Fig. E26.24**, and find the potential difference of point b relative to point a .

Figure E26.24



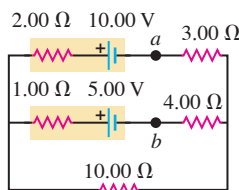
26.25 • In the circuit shown in **Fig. E26.25**, find (a) the current in the $3.00\ \Omega$ resistor; (b) the unknown emfs $\mathcal{E}_1$ and $\mathcal{E}_2$; (c) the resistance R . Note that three currents are given.

Figure E26.25



26.26 • In the circuit shown in **Fig. E26.26**, find (a) the current in each branch and (b) the potential difference V_{ab} of point a relative to point b .

Figure E26.26

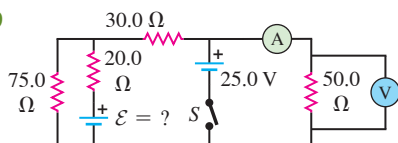


26.27 • The $10.00\ \text{V}$ battery in **Fig. E26.26** is removed from the circuit and reinserted with the opposite polarity, so that its positive terminal is now next to point a . The rest of the circuit is as shown in the figure. Find (a) the current in each branch and (b) the potential difference V_{ab} of point a relative to point b .

26.28 • The $5.00\ \text{V}$ battery in **Fig. E26.26** is removed from the circuit and replaced by a $15.00\ \text{V}$ battery, with its negative terminal next to point b . The rest of the circuit is as shown in the figure. Find (a) the current in each branch and (b) the potential difference V_{ab} of point a relative to point b .

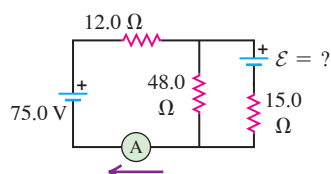
26.29 • In the circuit shown in **Fig. E26.29** the batteries have negligible internal resistance and the meters are both idealized. With the switch S open, the voltmeter reads $15.0\ \text{V}$. (a) Find the emf $\mathcal{E}$ of the battery. (b) What will the ammeter read when the switch is closed?

Figure E26.29



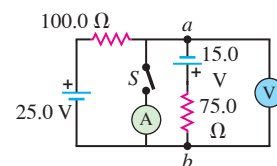
26.30 • In the circuit shown in **Fig. E26.30** both batteries have insignificant internal resistance and the idealized ammeter reads $1.50\ \text{A}$ in the direction shown. Find the emf $\mathcal{E}$ of the battery. Is the polarity shown correct?

Figure E26.30



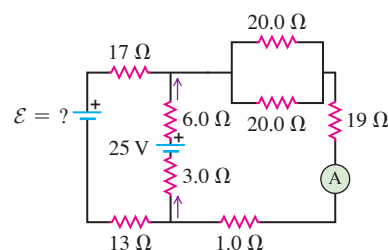
26.31 • In the circuit shown in **Fig. E26.31** all meters are idealized and the batteries have no appreciable internal resistance. (a) Find the reading of the voltmeter with the switch S open. Which point is at a higher potential: a or b ? (b) With S closed, find the reading of the voltmeter and the ammeter. Which way (up or down) does the current flow through the switch?

Figure E26.31



26.32 • In the circuit shown in **Fig. E26.32**, the $6.0\ \Omega$ resistor is consuming energy at a rate of $24\ \text{J/s}$ when the current through it flows as shown. (a) Find the current through the ammeter A . (b) What are the polarity and emf $\mathcal{E}$ of the unknown battery, assuming it has negligible internal resistance?

Figure E26.32

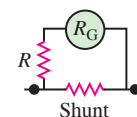


Section 26.3 Electrical Measuring Instruments

26.33 • The resistance of a galvanometer coil is $25.0\ \Omega$, and the current required for full-scale deflection is $500\ \mu\text{A}$. (a) Show in a diagram how to convert the galvanometer to an ammeter reading $20.0\ \text{mA}$ full scale, and compute the shunt resistance. (b) Show how to convert the galvanometer to a voltmeter reading $500\ \text{mV}$ full scale, and compute the series resistance.

26.34 • The resistance of the coil of a pivoted-coil galvanometer is $9.36\ \Omega$, and a current of $0.0224\ \text{A}$ causes it to deflect full scale. We want to convert this galvanometer to an ammeter reading $20.0\ \text{A}$ full scale. The only shunt available has a resistance of $0.0250\ \Omega$. What resistance R must be connected in series with the coil (**Fig. E26.34**)?

Figure E26.34



26.35 • A circuit consists of a series combination of $6.00\ \text{k}\Omega$ and $5.00\ \text{k}\Omega$ resistors connected across a $50.0\ \text{V}$ battery having negligible internal resistance. You want to measure the true potential difference (that is, the potential difference without the meter present) across the $5.00\ \text{k}\Omega$ resistor using a voltmeter having an internal resistance of $10.0\ \text{k}\Omega$. (a) What potential difference does the voltmeter measure across the $5.00\ \text{k}\Omega$ resistor? (b) What is the true potential difference across this resistor when the meter is not present? (c) By what percentage is the voltmeter reading in error from the true potential difference?

Section 26.4 R-C Circuits

26.36 • To measure the capacitance C of a capacitor, you attach the capacitor to a battery and wait until it is fully charged. You then disconnect the capacitor from the battery and let it discharge through a resistor of resistance R . You measure the time $T_{1/2}$ that it takes the voltage across the resistor to decrease to half its initial value at the instant that the connection to the capacitor is first completed. You repeat this for several different resistors. You plot the data as $T_{1/2}$ versus R and find that they lie close to a straight line that has slope $5.00\ \mu\text{F}$. What is the capacitance C of the capacitor?

26.37 • A capacitor is charged to a potential of $12.0\ \text{V}$ and is then connected to a voltmeter having a total internal resistance of $3.40\ \text{M}\Omega$. After a time of $4.00\ \text{s}$ the voltmeter reads $3.0\ \text{V}$. What are (a) the capacitance and (b) the time constant of the circuit?

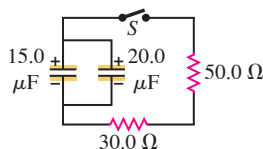
26.38 •• You connect a battery, resistor, and capacitor as in Fig. 26.20a, where $\mathcal{E} = 36.0 \text{ V}$, $C = 5.00 \mu\text{F}$, and $R = 120 \Omega$. The switch S is closed at $t = 0$. (a) When the voltage across the capacitor is 8.00 V , what is the magnitude of the current in the circuit? (b) At what time t after the switch is closed is the voltage across the capacitor 8.00 V ? (c) When the voltage across the capacitor is 8.00 V , at what rate is energy being stored in the capacitor?

26.39 • A $4.60 \mu\text{F}$ capacitor that is initially uncharged is connected in series with a $7.50 \text{ k}\Omega$ resistor and an emf source with $\mathcal{E} = 245 \text{ V}$ and negligible internal resistance. Just after the circuit is completed, what are (a) the voltage drop across the capacitor; (b) the voltage drop across the resistor; (c) the charge on the capacitor; (d) the current through the resistor? (e) A long time after the circuit is completed (after many time constants) what are the values of the quantities in parts (a)–(d)?

26.40 •• You connect a battery, resistor, and capacitor as in Fig. 26.20a, where $R = 12.0 \Omega$ and $C = 5.00 \times 10^{-6} \text{ F}$. The switch S is closed at $t = 0$. When the current in the circuit has magnitude 3.00 A , the charge on the capacitor is $40.0 \times 10^{-6} \text{ C}$. (a) What is the emf of the battery? (b) At what time t after the switch is closed is the charge on the capacitor equal to $40.0 \times 10^{-6} \text{ C}$? (c) When the current has magnitude 3.00 A , at what rate is energy being (i) stored in the capacitor, (ii) supplied by the battery?

26.41 •• CP In the circuit shown in Fig. E26.41 both capacitors are initially charged to 45.0 V . (a) How long after closing the switch S will the potential across each capacitor be reduced to 10.0 V , and (b) what will be the current at that time?

Figure E26.41



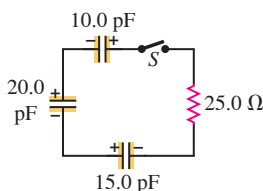
26.42 • A $12.4 \mu\text{F}$ capacitor is connected through a $0.895 \text{ M}\Omega$ resistor to a constant potential difference of 60.0 V . (a) Compute the charge on the capacitor at the following times after the connections are made: 0 , 5.0 s , 10.0 s , 20.0 s , and 100.0 s . (b) Compute the charging currents at the same instants. (c) Graph the results of parts (a) and (b) for t between 0 and 20 s .

26.43 • An emf source with $\mathcal{E} = 120 \text{ V}$, a resistor with $R = 80.0 \Omega$, and a capacitor with $C = 4.00 \mu\text{F}$ are connected in series. As the capacitor charges, when the current in the resistor is 0.900 A , what is the magnitude of the charge on each plate of the capacitor?

26.44 • A network of two identical capacitors, each with capacitance C , is charged through a resistor R using a battery with emf $\mathcal{E}$. (a) What is the time constant, in terms of R and C , for the charging circuit if the two capacitors are in series? (b) In parallel? (c) For which capacitor network, series or parallel, does the voltage across the resistor become 1% of its initial value in a shorter time?

26.45 •• CP In the circuit shown in Fig. E26.45 each capacitor initially has a charge of magnitude 3.50 nC on its plates. After the switch S is closed, what will be the current in the circuit at the instant that the capacitors have lost 80.0% of their initial stored energy?

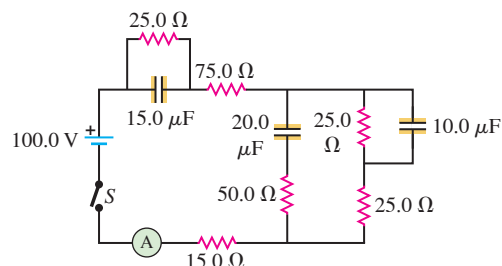
Figure E26.45



26.46 • A $1.50 \mu\text{F}$ capacitor is charging through a 12.0Ω resistor using a 10.0 V battery. What will be the current when the capacitor has acquired $\frac{1}{4}$ of its maximum charge? Will it be $\frac{1}{4}$ of the maximum current?

26.47 • In the circuit in Fig. E26.47 the capacitors are initially uncharged, the battery has no internal resistance, and the ammeter is idealized. Find the ammeter reading (a) just after the switch S is closed and (b) after S has been closed for a very long time.

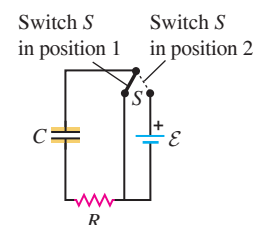
Figure E26.47



26.48 • A $12.0 \mu\text{F}$ capacitor is charged to a potential of 50.0 V and then discharged through a 225Ω resistor. How long does it take the capacitor to lose (a) half of its charge and (b) half of its stored energy?

26.49 • In the circuit shown in Fig. E26.49, $C = 5.90 \mu\text{F}$, $\mathcal{E} = 28.0 \text{ V}$, and the emf has negligible resistance. Initially the capacitor is uncharged and the switch S is in position 1. The switch is then moved to position 2, so that the capacitor begins to charge. (a) What will be the charge on the capacitor a long time after S is moved to position 2? (b) After S has been in position 2 for 3.00 ms , the charge on the capacitor is measured to be $110 \mu\text{C}$. What is the value of the resistance R ? (c) How long after S is moved to position 2 will the charge on the capacitor be equal to 99.0% of the final value found in part (a)?

Figure E26.49



26.50 • A capacitor with $C = 6.00 \mu\text{F}$ is fully charged by connecting it to a battery that has emf 50.0 V . The capacitor is disconnected from the battery. A resistor of resistance $R = 185 \Omega$ is connected across the capacitor, and the capacitor discharges through the resistor. (a) What is the charge q on the capacitor when the current in the resistor is 0.180 A ? (b) If the connection to the resistor is completed at time $t = 0$, what is the value of t when the current has the value specified in part (a)?

Section 26.5 Power Distribution Systems

26.51 • A 1500 W electric heater is plugged into the outlet of a 120 V circuit that has a 20 A circuit breaker. You plug an electric hair dryer into the same outlet. The hair dryer has power settings of 600 W , 900 W , 1200 W , and 1500 W . You start with the hair dryer on the 600 W setting and increase the power setting until the circuit breaker trips. What power setting caused the breaker to trip?

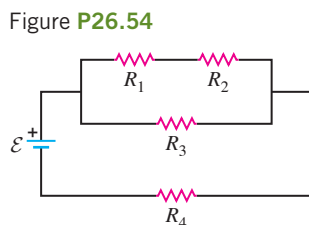
26.52 • The heating element of an electric dryer is rated at 4.1 kW when connected to a 240 V line. (a) What is the current in the heating element? Is 12 gauge wire large enough to supply this current? (b) What is the resistance of the dryer's heating element at its operating temperature? (c) At 11 cents per kWh , how much does it cost per hour to operate the dryer?

PROBLEMS

26.53 •• CALC A capacitor with capacitance C is connected in series to a resistor of resistance R and a battery with emf $\mathcal{E}$. The circuit is completed at time $t = 0$. (a) In terms of $\mathcal{E}$, R , and C , how much energy is stored in the capacitor when it is fully charged? (b) The power output of the battery is $P_{\mathcal{E}} = \mathcal{E}i$, with i given by Eq. (26.13). The electrical energy supplied in an infinitesimal time interval dt is $P_{\mathcal{E}}dt$. Integrate from $t = 0$ to $t \rightarrow \infty$ to find the total energy supplied by the battery. (c) The rate of consumption of electrical energy in the resistor is $P_R = i^2R$. In an infinitesimal time interval dt , the amount of electrical energy consumed by the resistor is P_Rdt . Integrate from $t = 0$ to $t \rightarrow \infty$ to find the total energy consumed by the resistor. (d) What fraction of the total energy supplied by the battery is stored in the capacitor? What fraction is consumed in the resistor?

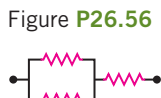
26.54 •• In **Fig. P26.54**, the battery has negligible internal resistance and $\mathcal{E} = 48.0$ V.

$R_1 = R_2 = 4.00 \Omega$ and $R_4 = 3.00 \Omega$. What must the resistance R_3 be for the resistor network to dissipate electrical energy at a rate of 295 W?



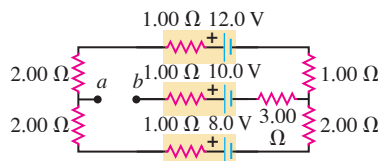
26.55 •• Consider the circuit shown in **Fig. 26.6a**. $\mathcal{E}_1 = 24.0$ V, $\mathcal{E}_2 = 36.0$ V, $r_1 = r_2 = 2.00 \Omega$, and $R = 20.0 \Omega$. (a) What is the current I_1 through $\mathcal{E}_1$? (b) What is the current I_2 through $\mathcal{E}_2$? (c) What is the current I_R through R ?

26.56 •• Each of the three resistors in **Fig. P26.56** has a resistance of 2.4Ω and can dissipate a maximum of 48 W without becoming excessively heated. What is the maximum power the circuit can dissipate?



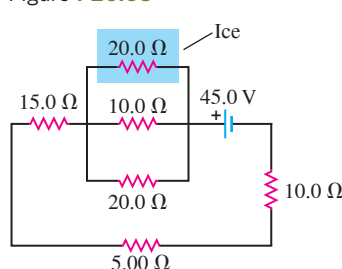
26.57 •• (a) Find the potential of point a with respect to point b in **Fig. P26.57**. (b) If points a and b are connected by a wire with negligible resistance, find the current in the 12.0 V battery.

Figure P26.57



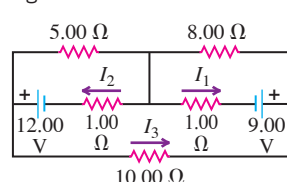
26.58 •• CP For the circuit shown in **Fig. P26.58** a 20.0Ω resistor is embedded in a large block of ice at 0.00°C , and the battery has negligible internal resistance. At what rate (in g/s) is this circuit melting the ice? (The latent heat of fusion for ice is 3.34×10^5 J/kg.)

Figure P26.58



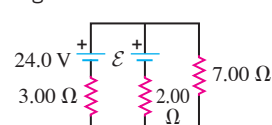
26.59 • Calculate the three currents I_1 , I_2 , and I_3 indicated in the circuit diagram shown in **Fig. P26.59**.

Figure P26.59



26.60 ••• What must the emf $\mathcal{E}$ in **Fig. P26.60** be in order for the current through the 7.00Ω resistor to be 1.80 A? Each emf source has negligible internal resistance.

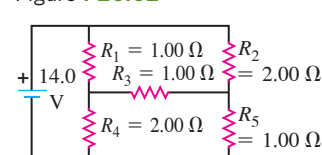
Figure P26.60



26.61 •• A standard phone charger provides 5.0 V of potential and supplies 1.0 A of current to the phone. (a) Estimate the diameter of each of the two wires inside your charge cable. (b) Assume the wire is copper, which has a resistivity of $1.72 \times 10^{-8} \Omega \cdot \text{m}$, and determine its resistance per meter. (c) If the phone draws $I = 1.0$ A, the potential supplied to the phone will be lower than the potential on the side of the charger by an amount IR , where R is the resistance of the wire. Determine the length of charging wire that would diminish the potential to 4.0 V at the phone.

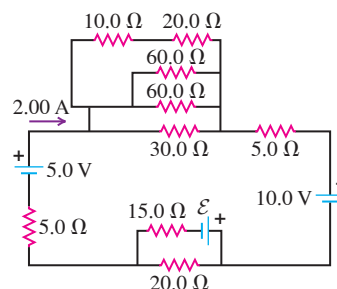
26.62 • (a) Find the current through the battery and each resistor in the circuit shown in **Fig. P26.62**. (b) What is the equivalent resistance of the resistor network?

Figure P26.62



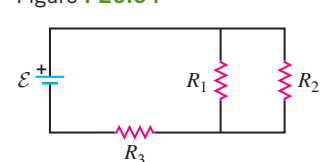
26.63 •• Consider the circuit shown in **Fig. P26.63**. (a) What must the emf $\mathcal{E}$ of the battery be in order for a current of 2.00 A to flow through the 5.00 V battery as shown? Is the polarity of the battery correct as shown? (b) How long does it take for 60.0 J of thermal energy to be produced in the 10.0Ω resistor?

Figure P26.63



26.64 •• In the circuit shown in **Fig. P26.64**, $\mathcal{E} = 24.0$ V, $R_1 = 6.00 \Omega$, $R_3 = 12.0 \Omega$, and R_2 can vary between 3.00Ω and 24.0Ω . For what value of R_2 is the power dissipated by heating element R_1 the greatest? Calculate the magnitude of the greatest power.

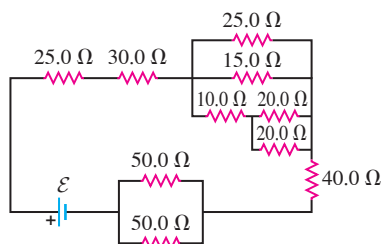
Figure P26.64



26.65 •• (a) Estimate energy usage and cost in your home by creating a table: In the first column list the household appliances that use electricity. In the second column indicate the voltage V used by each appliance. (Use 240 V for the hot-water heater, baseboard heat, and ovens; use 120 V for all other appliances.) In the third column list the current I drawn by each appliance. (This is readily obtained on the Internet.) In the fourth column indicate the number of daily hours T each appliance is used. In the fifth column compute the power of each appliance $P = IV$, and in the sixth column compute the daily energy use $U = PT$, in kilowatt-hours. At the bottom of the final column, sum the total kilowatt-hours used by these appliances. (b) Compute the monthly cost of electricity by multiplying the total daily kilowatt-hours by the cost ($\$0.12/\text{kW} \cdot \text{h}$ is typical). Then multiply the result by 30 days to obtain an estimate of your monthly cost of electricity.

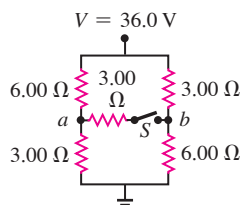
26.66 ••• In the circuit shown in **Fig. P26.66** all the resistors are rated at a maximum power of 2.00 W. What is the maximum emf $\mathcal{E}$ that the battery can have without burning up any of the resistors?

Figure P26.66



26.67 • Figure P26.67 employs a convention often used in circuit diagrams. The battery (or other power supply) is not shown explicitly. It is understood that the point at the top, labeled “36.0 V,” is connected to the positive terminal of a 36.0 V battery having negligible internal resistance, and that the ground symbol at the bottom is connected to the negative terminal of the battery. The circuit is completed through the battery, even though it is not shown. (a) What is the potential difference V_{ab} , the potential of point a relative to point b , when the switch S is open? (b) What is the current through S when it is closed? (c) What is the equivalent resistance when S is closed?

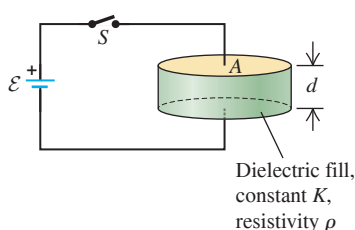
Figure P26.67



26.68 •• Three identical resistors are connected in series. When a certain potential difference is applied across the combination, the total power dissipated is 45.0 W. What power would be dissipated if the three resistors were connected in parallel across the same potential difference?

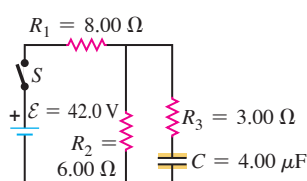
26.69 ••• CP Two capacitor plates with area A are separated by a distance d . The space between the plates is filled with dielectric material with dielectric constant K and resistivity ρ . This capacitor is attached to a battery that supplies a constant potential $\mathcal{E}$, as shown in **Fig. P26.69**, and is fully charged. At time $t = 0$ the switch S is opened. Owing to the nonzero value of ρ , this capacitor discharges by leaking. We can model this device as a capacitor and a resistor in parallel. (a) Analyze this circuit and determine the time constant, characterizing the discharge in terms of the parameters given above. Now assume the capacitor has plates with area $A = 1.00 \text{ cm}^2$ separated by $d = 30.0 \mu\text{m}$ and is filled with a ceramic of dielectric constant $K = 8.70$ and resistivity $\rho = 3.10 \times 10^{12} \Omega \cdot \text{m}$. (b) If $\mathcal{E} = 5.00 \text{ V}$, what is the charge on the capacitor at $t = 0$? (c) At what time will the capacitor have half of its original charge? (d) What is the magnitude of the leaking current at that point?

Figure P26.69



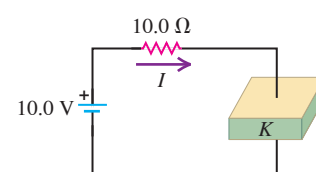
26.70 • The capacitor in **Fig. P26.70** is initially uncharged. The switch S is closed at $t = 0$. (a) Immediately after the switch is closed, what is the current through each resistor? (b) What is the final charge on the capacitor?

Figure P26.70



26.71 ••• CP CALC An R - C circuit is constructed as shown in **Fig. P26.71**. The capacitor has $2.00 \text{ cm} \times 2.00 \text{ cm}$ square plates that are parallel and separated by a distance of 3.00 mm, and it is fully charged. At time $t = 0$, a $2.00 \text{ cm} \times 2.00 \text{ cm} \times 3.00 \text{ mm}$ slab with dielectric constant $K = 12.0$ is slid between the plates as shown. (a) How much charge is on the upper plate before the slab is inserted? (b) What is the current I immediately after the slab is inserted? (c) How much energy is stored in the capacitor immediately after the slab is inserted? (d) By how much does the energy in the capacitor increase from that value after a long time has passed? (e) How much total energy is eventually supplied by the battery after the slab is inserted? (*Hint:* Determine the current as a function of time; then integrate the power delivered by the battery, $\int_0^\infty VI(t) dt$.) (f) How much energy is dissipated by the resistor?

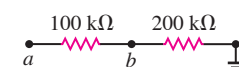
Figure P26.71



26.72 ••• A $6.00 \mu\text{F}$ capacitor that is initially uncharged is connected in series with a 5.00Ω resistor and an emf source with $\mathcal{E} = 50.0 \text{ V}$ and negligible internal resistance. At the instant when the resistor is dissipating electrical energy at a rate of 300 W, how much energy has been stored in the capacitor?

26.73 • Point a in **Fig. P26.73** is maintained at a constant potential of 400 V above ground. (See Problem 26.67.)

Figure P26.73

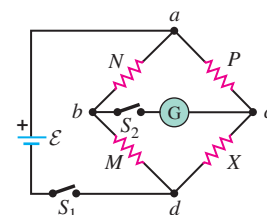


(a) What is the reading of a voltmeter with the proper range and with resistance $5.00 \times 10^4 \Omega$ when connected between point b and ground? (b) What is the reading of a voltmeter with resistance $5.00 \times 10^6 \Omega$? (c) What is the reading of a voltmeter with infinite resistance?

26.74 •• The Wheatstone Bridge.

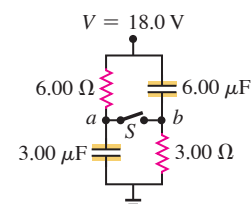
Figure P26.74

The circuit shown in **Fig. P26.74**, called a *Wheatstone bridge*, is used to determine the value of an unknown resistor X by comparison with three resistors M , N , and P whose resistances can be varied. For each setting, the resistance of each resistor is precisely known. With switches S_1 and S_2 closed, these resistors are varied until the current in the galvanometer G is zero; the bridge is then said to be *balanced*. (a) Show that under this condition the unknown resistance is given by $X = MP/N$. (This method permits very high precision in comparing resistors.) (b) If galvanometer G shows zero deflection when $M = 850.0 \Omega$, $N = 15.00 \Omega$, and $P = 33.48 \Omega$, what is the unknown resistance X ?



26.75 • (See Problem 26.67.) (a) What is the potential of point a with respect to point b in **Fig. P26.75** when the switch S is open? (b) Which point, a or b , is at the higher potential? (c) What is the final potential of point b with respect to ground when S is closed? (d) How much does the charge on each capacitor change when S is closed?

Figure P26.75

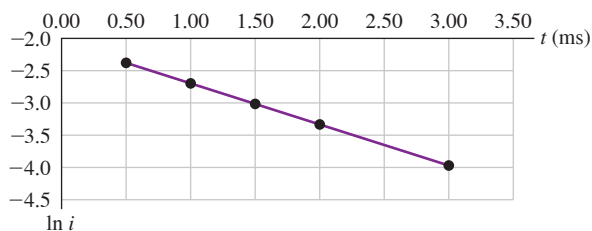


26.76 • A resistor with $R = 850 \Omega$ is connected to the plates of a charged capacitor with capacitance $C = 4.62 \mu\text{F}$. Just before the connection is made, the charge on the capacitor is 6.90 mC. (a) What is the energy initially stored in the capacitor? (b) What is the electrical power dissipated in the resistor just after the connection is made? (c) What is the electrical power dissipated in the resistor at the instant when the energy stored in the capacitor has decreased to half the value calculated in part (a)?

26.77 • A capacitor that is initially uncharged is connected in series with a resistor and an emf source with $\mathcal{E} = 110 \text{ V}$ and negligible internal resistance. Just after the circuit is completed, the current through the resistor is $6.5 \times 10^{-5} \text{ A}$. The time constant for the circuit is 5.2 s . What are the resistance of the resistor and the capacitance of the capacitor?

26.78 •• DATA You set up the circuit shown in Fig. 26.22a, where $R = 196 \Omega$. You close the switch at time $t = 0$ and measure the magnitude i of the current in the resistor R as a function of time t since the switch was closed. Your results are shown in Fig. P26.78, where you have chosen to plot $\ln i$ as a function of t . (a) Explain why your data points lie close to a straight line. (b) Use the graph in Fig. P26.78 to calculate the capacitance C and the initial charge Q_0 on the capacitor. (c) When $i = 0.0500 \text{ A}$, what is the charge on the capacitor? (d) When $q = 0.500 \times 10^{-4} \text{ C}$, what is the current in the resistor?

Figure P26.78



26.79 •• DATA You set up the circuit shown in Fig. 26.20, where $C = 5.00 \times 10^{-6} \text{ F}$. At time $t = 0$, you close the switch and then measure the charge q on the capacitor as a function of the current i in the resistor. Your results are given in the table:

$i \text{ (mA)}$	56.0	48.0	40.0	32.0	24.0
$q \text{ (}\mu\text{C)}$	10.1	19.8	30.2	40.0	49.9

(a) Graph q as a function of i . Explain why the data points, when plotted this way, fall close to a straight line. Find the slope and y-intercept of the straight line that gives the best fit to the data. (b) Use your results from part (a) to calculate the resistance R of the resistor and the emf $\mathcal{E}$ of the battery. (c) At what time t after the switch is closed is the voltage across the capacitor equal to 10.0 V ? (d) When the voltage across the capacitor is 4.00 V , what is the voltage across the resistor?

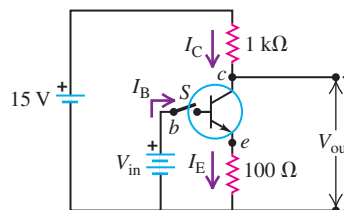
26.80 •• DATA The electronics supply company where you work has two different resistors, R_1 and R_2 , in its inventory, and you must measure the values of their resistances. Unfortunately, stock is low, and all you have are R_1 and R_2 in parallel and in series—and you can't separate these two resistor combinations. You separately connect each resistor network to a battery with emf 48.0 V and negligible internal resistance and measure the power P supplied by the battery in both cases. For the series combination, $P = 48.0 \text{ W}$; for the parallel combination, $P = 256 \text{ W}$. You are told that $R_1 > R_2$. (a) Calculate R_1 and R_2 . (b) For the series combination, which resistor consumes more power, or do they consume the same power? Explain. (c) For the parallel combination, which resistor consumes more power, or do they consume the same power?

CHALLENGE PROBLEMS

26.81 ••• The three-terminal device circled in Fig. P26.81 is an NPN transistor. It works using three simple rules: I. The net current into the device is the same as the net current out of the device, so $I_C + I_B = I_E$. II. The potential at point e is always 0.60 V less than the potential at point b . Thus, $V_{in} - V_e = 0.60 \text{ V}$. III. The current into point c is always a fixed multiple of the current into point b , so $I_C = \beta I_B$, where $\beta \gg 1$ is a parameter characteristic of the transistor. Use these rules to answer the questions. (a) What is the potential V_{out} in terms of V_{in} , in the limit $\beta \rightarrow \infty$? (Hint: Determine I_E in terms of V_{in} ,

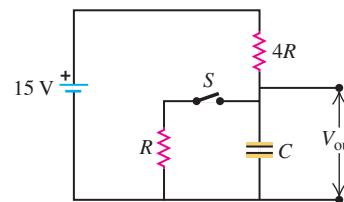
and use this to determine I_C in terms of V_{in} and β . The potential difference across the $1.0 \text{ k}\Omega$ resistor can then be used to determine V_{out} . Take the limit $\beta \rightarrow \infty$ in this result.) (b) What value of V_{in} is needed so that $V_{out} = 7.5 \text{ V}$? (c) If the input potential includes a small time-dependent "signal," so that $V_{in} = 15.0 \text{ V} + v_{in}(t)$, then the output potential is $V_{out} = 7.5 \text{ V} + Gv_{in}(t)$, where G is a "gain" factor. What is G for this circuit?

Figure P26.81



26.82 ••• In the circuit shown in Fig. P26.82, the switch S has been closed for a long time. (a) What is the output voltage V_{out} ? Now the switch is opened and the output voltage increases. When the output voltage reaches 10.0 V , the switch closes and then the capacitor begins to discharge. When the output voltage reaches 5.0 V , the switch opens again. Thus this circuit undergoes periodic cycles of charging and discharging. (b) What is the time constant during a charging phase? (c) What is the time constant during a discharging phase? (d) What length of time passes between subsequent 10.0 V output voltages? (e) If $R = 10.0 \text{ k}\Omega$ and $C = 10.0 \mu\text{F}$, what is the frequency of operation? (Note: This circuit is a simplified example of a timer circuit, a crucial element in many electronics applications.)

Figure P26.82

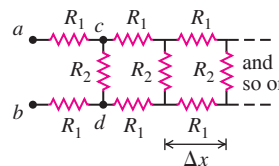


26.83 ••• An Infinite Network. As shown in Fig. P26.83, a network of resistors of resistances R_1 and R_2 extends to infinity toward the right. Prove that the total resistance R_T of the infinite network is equal to

$$R_T = R_1 + \sqrt{R_1^2 + 2R_1R_2}$$

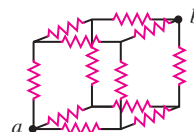
(Hint: Since the network is infinite, the resistance of the network to the right of points c and d is also equal to R_T .)

Figure P26.83



26.84 ••• Suppose a resistor R lies along each edge of a cube (12 resistors in all) with connections at the corners. Find the equivalent resistance between two diagonally opposite corners of the cube (points a and b in Fig. P26.84).

Figure P26.84



26.85 ••• BIO Attenuator Chains and Axons. The infinite network of resistors shown in Fig. P26.83 is known as an *attenuator chain*, since this chain of resistors causes the potential difference between the upper and lower wires to decrease, or attenuate, along the length of the chain. (a) Show that if the potential difference between the points a and b in Fig. 26.83 is V_{ab} , then the potential difference between points c and d is $V_{cd} = V_{ab}/(1 + \beta)$, where $\beta = 2R_1(R_T + R_2)/R_T R_2$ and R_T , the total resistance of the network, is given in Challenge Problem 26.83. (See the hint given in that problem.) (b) If the potential difference between terminals a and b at the left end of the infinite network is V_0 , show that the potential difference between the upper and lower wires n segments from the left end is $V_n = V_0/(1 + \beta)^n$. If $R_1 = R_2$, how many segments are needed to decrease the potential difference V_n to less than 1.0% of V_0 ? (c) An infinite attenuator chain provides a model of the propagation of a voltage pulse along a nerve fiber, or axon. Each segment of the network in Fig. P26.83 represents a short segment of the axon of length Δx . The resistors R_1 represent the resistance of the fluid inside and outside the membrane wall of the axon. The resistance of the membrane to current flowing through the wall is represented by R_2 . For an axon segment of length $\Delta x = 1.0 \mu\text{m}$, $R_1 = 6.4 \times 10^3 \Omega$ and $R_2 = 8.0 \times 10^8 \Omega$ (the membrane wall is a good insulator). Calculate the total resistance R_T and β for an infinitely long axon. (This is a good approximation, since the length of an axon is much greater than its width; the largest axons in the human nervous system are longer than 1 m but only about 10^{-7} m in radius.) (d) By what fraction does the potential difference between the inside and outside of the axon decrease over a distance of 2.0 mm? (e) The attenuation of the potential difference calculated in part (d) shows that the axon cannot simply be a passive, current-carrying electrical cable; the potential difference must periodically be reinforced along the axon's length. This reinforcement mechanism is slow, so a signal propagates along the axon at only about 30 m/s. In situations where faster response is required, axons are covered with a segmented sheath of fatty myelin. The segments are about 2 mm long, separated by gaps called the *nodes of Ranvier*. The myelin increases the resistance of a

$1.0\text{-}\mu\text{m}$ -long segment of the membrane to $R_2 = 3.3 \times 10^{12} \Omega$. For such a myelinated axon, by what fraction does the potential difference between the inside and outside of the axon decrease over the distance from one node of Ranvier to the next? This smaller attenuation means the propagation speed is increased.

MCAT-STYLE PASSAGE PROBLEMS

BIO Nerve Cells and R-C Circuits. The portion of a nerve cell that conducts signals is called an *axon*. Many of the electrical properties of axons are governed by ion channels, which are protein molecules that span the axon's cell membrane. When open, each ion channel has a pore that is filled with fluid of low resistivity and connects the interior of the cell electrically to the medium outside the cell. In contrast, the lipid-rich cell membrane in which ion channels reside has very high resistivity.

26.86 Assume that a typical open ion channel spanning an axon's membrane has a resistance of $1 \times 10^{11} \Omega$. We can model this ion channel, with its pore, as a 12-nm-long cylinder of radius 0.3 nm. What is the resistivity of the fluid in the pore? (a) $10 \Omega \cdot \text{m}$; (b) $6 \Omega \cdot \text{m}$; (c) $2 \Omega \cdot \text{m}$; (d) $1 \Omega \cdot \text{m}$.

26.87 In a simple model of an axon conducting a nerve signal, ions move across the cell membrane through open ion channels, which act as purely resistive elements. If a typical current density (current per unit cross-sectional area) in the cell membrane is 5 mA/cm^2 when the voltage across the membrane (the *action potential*) is 50 mV, what is the number density of open ion channels in the membrane? (a) $1/\text{cm}^2$; (b) $10/\text{cm}^2$; (c) $10/\text{mm}^2$; (d) $100/\mu\text{m}^2$.

26.88 Cell membranes across a wide variety of organisms have a capacitance per unit area of $1 \mu\text{F/cm}^2$. For the electrical signal in a nerve to propagate down the axon, the charge on the membrane "capacitor" must change. What time constant is required when the ion channels are open? (a) $1 \mu\text{s}$; (b) $10 \mu\text{s}$; (c) $100 \mu\text{s}$; (d) 1 ms.

ANSWERS

Chapter Opening Question ?

(ii) The potential difference V is the same across resistors connected in parallel. However, there is a different current I through each resistor if the resistances R are different: $I = V/R$.

Key Example VARIATION Problems

- VP26.2.1** (a) 7.00Ω (b) 0.571Ω (c) 2.33Ω (d) 0.857Ω
VP26.2.2 (a) 8.18 A (b) 6.00 A (c) 2.18 A (d) 2.18 A
VP26.2.3 (a) 54.6 W (b) 20.6 W (c) 18.0 W (d) 16.0 W
VP26.2.4 (a) 9.84 W (b) 5.98 W (c) 2.08 W (d) 1.78 W
VP26.7.1 (a) 1 A (b) 7 V (c) 12 W for 12 V battery, 4 W for 4 V battery
VP26.7.2 (a) 7.75 V (b) 1.25 V (c) 0.926 V

- VP26.7.3** (a) 33.4 W (b) 18.8 W (c) 2.09 W (d) 52.2 W (e) 25.0 W
VP26.7.4 (a) $(2\mathcal{E}_1 + \mathcal{E}_2)/3R$ (b) $-(\mathcal{E}_1 + 2\mathcal{E}_2)/3R$ (c) $(\mathcal{E}_1 - \mathcal{E}_2)/3R$
VP26.13.1 (a) $2.88 \times 10^{-5} \text{ C}$ (b) $9.00 \times 10^{-7} \text{ A}$ (c) 32.0 s (d) 0.430 (e) 0.570
VP26.13.2 (a) 11.0 s (b) $1.36 \times 10^{-7} \text{ A}$
VP26.13.3 (a) 1.32 M Ω (b) $5.21 \times 10^{-7} \text{ A}$ (c) $1.04 \times 10^{-7} \text{ A}$
VP26.13.4 (a) $6.12 \times 10^{-5} \text{ C}$ (b) $9.89 \times 10^{-6} \text{ C}$

Bridging Problem

- (a) 9.39 J (b) $2.02 \times 10^4 \text{ W}$ (c) $4.65 \times 10^{-4} \text{ s}$ (d) $7.43 \times 10^3 \text{ W}$

? The needle of a magnetic compass points north. This alignment is due to (i) a magnetic force on the needle; (ii) a magnetic torque on the needle; (iii) the magnetic field that the needle itself produces; (iv) both (i) and (ii); (v) all of (i), (ii), and (iii).



27 Magnetic Field and Magnetic Forces

LEARNING OUTCOMES

In this chapter, you'll learn...

- 27.1 The properties of magnets, and how magnets interact with each other.
- 27.2 The nature of the force that a moving charged particle experiences in a magnetic field.
- 27.3 How magnetic field lines are different from electric field lines.
- 27.4 How to analyze the motion of a charged particle in a magnetic field.
- 27.5 Some practical applications of magnetic fields in chemistry and physics.
- 27.6 How to analyze magnetic forces on current-carrying conductors.
- 27.7 How current loops behave when placed in a magnetic field.
- 27.8 How direct-current motors work.
- 27.9 How magnetic forces give rise to the Hall effect.

You'll need to review...

- 1.10 Vector product of two vectors.
- 3.4, 5.4 Uniform circular motion.
- 10.1 Torque.
- 21.6, 21.7 Electric field lines and electric dipole moment.
- 22.2, 22.3 Electric flux and Gauss's law.
- 25.1 Electric current.
- 26.3 Galvanometers.

Everybody uses magnetic forces. They are at the heart of electric motors, microwave ovens, loudspeakers, computer printers, and disk drives. The most familiar examples of magnetism are permanent magnets, which attract unmagnetized iron objects and can also attract or repel other magnets. A compass needle aligning itself with the earth's magnetism is an example of this interaction. But the *fundamental* nature of magnetism is the interaction of moving electric charges. Unlike electric forces, which act on electric charges whether they are moving or not, magnetic forces act only on *moving* charges.

We saw in Chapter 21 that the electric force arises in two stages: (1) a charge produces an electric field in the space around it, and (2) a second charge responds to this field. Magnetic forces also arise in two stages. First, a *moving* charge or a collection of moving charges (that is, an electric current) produces a *magnetic* field. Next, a second current or moving charge responds to this magnetic field, and so experiences a magnetic force.

In this chapter we study the second stage in the magnetic interaction—that is, how moving charges and currents *respond* to magnetic fields. In particular, we'll see how to calculate magnetic forces and torques, and we'll discover why magnets can pick up iron objects like paper clips. In Chapter 28 we'll complete our picture of the magnetic interaction by examining how moving charges and currents *produce* magnetic fields.

27.1 MAGNETISM

Magnetic phenomena were first observed at least 2500 years ago in fragments of magnetized iron ore found near the ancient city of Magnesia (now Manisa, in western Turkey). These fragments were what are now called **permanent magnets**; you probably have several permanent magnets on your refrigerator door at home. Permanent magnets were found to exert forces on each other as well as on pieces of iron that were not magnetized. It was discovered that when an iron rod is brought in contact with a natural magnet, the rod also becomes magnetized. When such a rod is floated on water or suspended by a

string from its center, it tends to line itself up in a north-south direction. The needle of an ordinary compass is just such a piece of magnetized iron.

Before the relationship of magnetic interactions to moving charges was understood, the interactions of permanent magnets and compass needles were described in terms of *magnetic poles*. If a bar-shaped permanent magnet, or *bar magnet*, is free to rotate, one end points north. This end is called a *north pole* or *N pole*; the other end is a *south pole* or *S pole*. Opposite poles attract each other, and like poles repel each other (Fig. 27.1). An object that contains iron but is not itself magnetized (that is, it shows no tendency to point north or south) is attracted by *either* pole of a permanent magnet (Fig. 27.2). This is the attraction that acts between a magnet and the unmagnetized steel door of a refrigerator. By analogy to electric interactions, we describe the interactions in Figs. 27.1 and 27.2 by saying that a bar magnet sets up a *magnetic field* in the space around it and a second object responds to that field. A compass needle tends to align with the magnetic field at the needle's position.

The earth itself is a magnet. Its north geographic pole is close to a magnetic *south* pole, which is why the north pole of a compass needle points north. The earth's magnetic axis is not quite parallel to its geographic axis (the axis of rotation), so a compass reading deviates somewhat from geographic north. This deviation, which varies with location, is called *magnetic declination* or *magnetic variation*. Also, the magnetic field is not horizontal at most points on the earth's surface; its angle up or down is called *magnetic inclination*. At the magnetic poles the magnetic field is vertical.

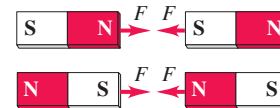
Figure 27.3 is a sketch of the earth's magnetic field. The lines, called *magnetic field lines*, show the direction that a compass would point at each location; they are discussed in detail in Section 27.3. The direction of the field at any point can be defined as the direction of the force that the field would exert on a magnetic north pole. In Section 27.2 we'll describe a more fundamental way to define the direction and magnitude of a magnetic field.

Magnetic Poles Versus Electric Charge

The concept of magnetic poles may appear similar to that of electric charge, and north and south poles may seem analogous to positive and negative charges. But the analogy can be misleading. While isolated positive and negative charges exist, there is *no* experimental evidence that one isolated magnetic pole exists; poles always appear in pairs. If a bar

Figure 27.1 (a) Two bar magnets attract when opposite poles (N and S, or S and N) are next to each other. (b) The bar magnets repel when like poles (N and N, or S and S) are next to each other.

(a) Opposite poles attract.



(b) Like poles repel.

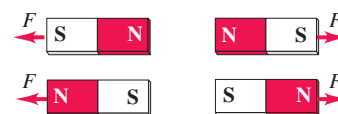
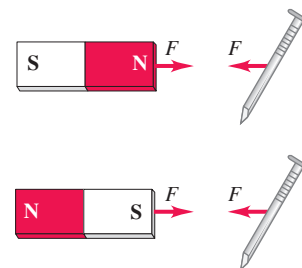


Figure 27.2 (a) Either pole of a bar magnet attracts an unmagnetized object that contains iron, such as a nail. (b) A real-life example of this effect.

(a)



(b)

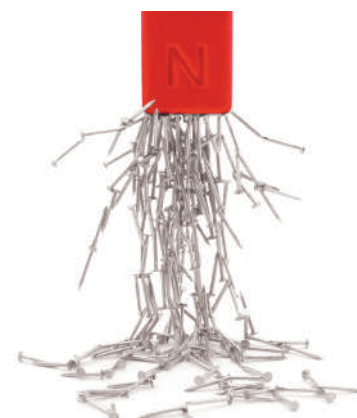


Figure 27.3 A sketch of the earth's magnetic field. The field, which is caused by currents in the earth's molten core, changes with time; geologic evidence shows that it reverses direction entirely at irregular intervals of 10^4 to 10^6 years.

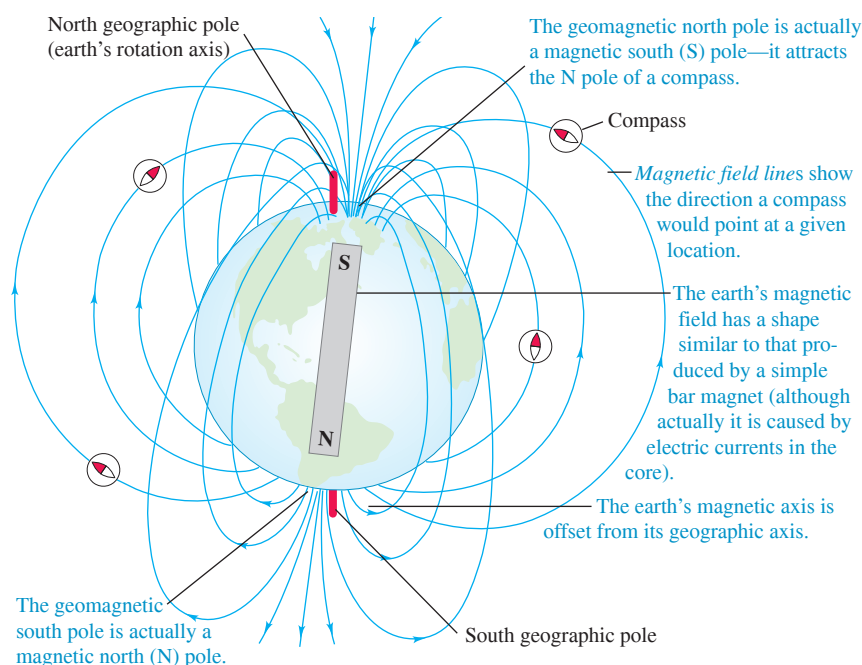


Figure 27.4 Breaking a bar magnet. Each piece has a north and south pole, even if the pieces are different sizes. (The smaller the piece, the weaker its magnetism.)

In contrast to electric charges, magnetic poles always come in pairs and can't be isolated.

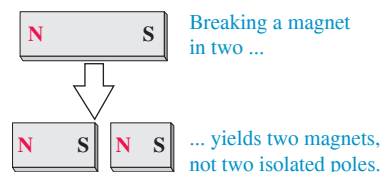
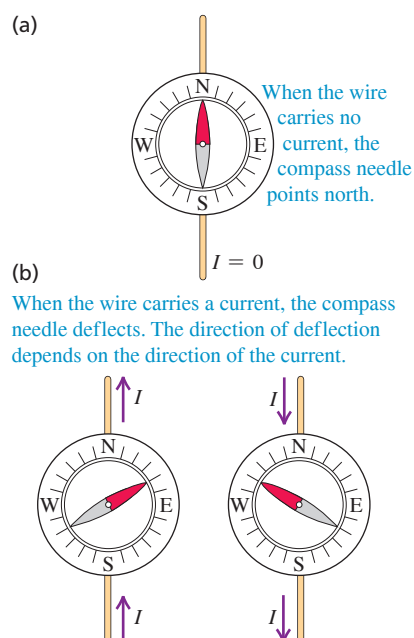


Figure 27.5 In Oersted's experiment, a compass is placed directly over a horizontal wire (here viewed from above).



BIO APPLICATION Spiny Lobsters and Magnetic Compasses Although the Caribbean spiny lobster (*Panulirus argus*) has a relatively simple nervous system, it is remarkably sensitive to magnetic fields. It has an internal magnetic “compass” that allows it to distinguish north, east, south, and west. This lobster can also sense small differences in the earth's magnetic field from one location to another and may use these differences to help it navigate.



magnet is broken in two, each broken end becomes a pole (Fig. 27.4). The existence of an isolated magnetic pole, or **magnetic monopole**, would have sweeping implications for theoretical physics. Extensive searches for magnetic monopoles have been carried out, but so far without success.

The first evidence of the relationship of magnetism to moving charges was discovered in 1820 by the Danish scientist Hans Christian Oersted. He found that a compass needle was deflected by a current-carrying wire (Fig. 27.5). Similar investigations were carried out in France by André Ampère. A few years later, Michael Faraday in England and Joseph Henry in the United States discovered that moving a magnet near a conducting loop can cause a current in the loop. We now know that the magnetic forces between two objects shown in Figs. 27.1 and 27.2 are fundamentally due to interactions between moving electrons in the atoms of the objects. (There are also *electric* interactions between the two objects, but these are far weaker than the magnetic interactions because the objects are electrically neutral.) Inside a magnetized object such as a permanent magnet, the motion of certain of the atomic electrons is *coordinated*; in an unmagnetized object these motions are not coordinated. (We'll describe these motions further in Section 27.7 and see how the interactions shown in Figs. 27.1 and 27.2 come about.)

Electric and magnetic interactions prove to be intimately connected. Over the next several chapters we'll develop the unifying principles of electromagnetism, culminating in the expression of these principles in *Maxwell's equations*. These equations represent the synthesis of electromagnetism, just as Newton's laws of motion are the synthesis of mechanics, and like Newton's laws they represent a towering achievement of the human intellect.

TEST YOUR UNDERSTANDING OF SECTION 27.1 Suppose you cut off the part of the compass needle shown in Fig. 27.5a that is painted gray. You discard this part, drill a hole in the remaining red part, and place the red part on the pivot at the center of the compass. Will the red part still swing when a current is applied as in Fig. 27.5b?

ANSWER **yes** When a magnet is cut apart, each part has a north and south pole (see Fig. 27.4). Hence the small red part behaves much like the original, full-sized compass needle.

27.2 MAGNETIC FIELD

To introduce the concept of magnetic field properly, let's review our formulation of *electric* interactions in Chapter 21, where we introduced the concept of *electric* field. We represented electric interactions in two steps:

1. A distribution of electric charge creates an electric field $\vec{E}$ in the surrounding space.
2. The electric field exerts a force $\vec{F} = q\vec{E}$ on any other charge q that is present in the field.

We can describe magnetic interactions in a similar way:

1. A moving charge or a current creates a **magnetic field** in the surrounding space (in addition to its *electric* field).
2. The magnetic field exerts a force $\vec{F}$ on any other moving charge or current that is present in the field.

In this chapter we'll concentrate on the *second* aspect of the interaction: Given the presence of a magnetic field, what force does it exert on a moving charge or a current? In Chapter 28 we'll come back to the problem of how magnetic fields are *created* by moving charges and currents.

Like electric field, magnetic field is a *vector field*—that is, a vector quantity associated with each point in space. We'll use the symbol $\vec{B}$ for magnetic field. At any position the direction of $\vec{B}$ is defined as the direction in which the north pole of a compass needle tends to point. The arrows in Fig. 27.3 suggest the direction of the earth's magnetic field; for any magnet, $\vec{B}$ points out of its north pole and into its south pole.

Magnetic Forces on Moving Charges

There are four key characteristics of the magnetic force on a moving charge. First, its magnitude is proportional to the magnitude of the charge. If a $1\text{ }\mu\text{C}$ charge and a $2\text{ }\mu\text{C}$ charge move through a given magnetic field with the same velocity, experiments show that the force on the $2\text{ }\mu\text{C}$ charge is twice as great as the force on the $1\text{ }\mu\text{C}$ charge. Second, the magnitude of the force is also proportional to the magnitude, or “strength,” of the field; if we double the magnitude of the field (for example, by using two identical bar magnets instead of one) without changing the charge or its velocity, the force doubles.

A third characteristic is that the magnetic force depends on the particle’s velocity. This is quite different from the electric-field force, which is the same whether the charge is moving or not. A charged particle at rest experiences *no* magnetic force. And fourth, we find by experiment that the magnetic force $\vec{F}$ *does not* have the same direction as the magnetic field $\vec{B}$ but instead is always *perpendicular* to both $\vec{B}$ and the velocity $\vec{v}$. The magnitude F of the force is proportional to the component of $\vec{v}$ perpendicular to the field; when that component is zero (that is, when $\vec{v}$ and $\vec{B}$ are parallel or antiparallel), the force is zero.

Figure 27.6 shows these relationships. The direction of $\vec{F}$ is always perpendicular to the plane containing $\vec{v}$ and $\vec{B}$. Its magnitude is given by

$$F = |q|v_{\perp}B = |q|vB\sin\phi \quad (27.1)$$

where $|q|$ is the magnitude of the charge and ϕ is the angle measured from the direction of $\vec{v}$ to the direction of $\vec{B}$, as shown in the figure.

This description does not specify the direction of $\vec{F}$ completely; there are always two directions, opposite to each other, that are both perpendicular to the plane of $\vec{v}$ and $\vec{B}$. To complete the description, we use the same right-hand rule that we used to define the vector product in Section 1.10. (It would be a good idea to review that section before you go on.) Draw the vectors $\vec{v}$ and $\vec{B}$ with their tails together, as in **Fig. 27.7a**. Imagine turning $\vec{v}$ until it points in the direction of $\vec{B}$ (turning through the smaller of the two possible angles). Wrap the fingers of your right hand around the line perpendicular to the plane of $\vec{v}$ and $\vec{B}$ so that they curl around with the sense of rotation from $\vec{v}$ to $\vec{B}$. Your thumb then points in the direction of the force $\vec{F}$ on a *positive* charge.

This discussion shows that the force on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is given, both in magnitude and in direction, by

$$\text{Magnetic force on a moving charged particle} \quad \vec{F} = q\vec{v} \times \vec{B} \quad (27.2)$$

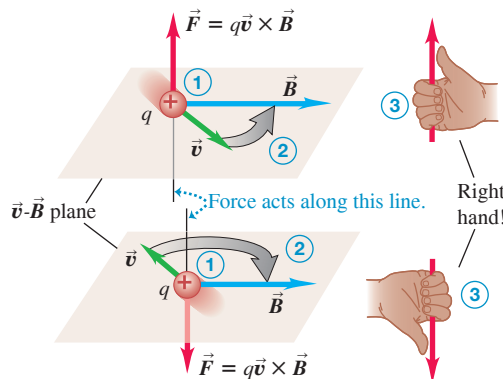
Particle's charge Particle's velocity Magnetic field

Figure 27.7 Finding the direction of the magnetic force on a moving charged particle.

(a)

Right-hand rule for the direction of magnetic force on a **positive** charge moving in a magnetic field:

- ① Place the $\vec{v}$ and $\vec{B}$ vectors tail to tail.
- ② Imagine turning $\vec{v}$ toward $\vec{B}$ in the $\vec{v}$ - $\vec{B}$ plane (through the smaller angle).
- ③ The force acts along a line perpendicular to the $\vec{v}$ - $\vec{B}$ plane. Curl the fingers of your *right hand* around this line in the same direction you rotated $\vec{v}$. Your thumb now points in the direction the force acts.



(b)

If the charge is negative, the direction of the force is **opposite** to that given by the right-hand rule.

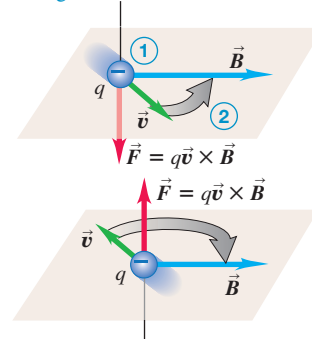
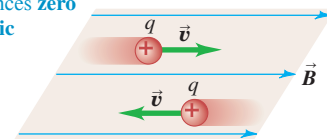


Figure 27.6 The magnetic force $\vec{F}$ acting on a positive charge q moving with velocity $\vec{v}$ is perpendicular to both $\vec{v}$ and the magnetic field $\vec{B}$. For given values of speed v and magnetic field strength B , the force is greatest when $\vec{v}$ and $\vec{B}$ are perpendicular.

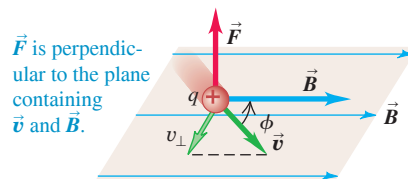
(a)

A charge moving **parallel** to a magnetic field experiences **zero magnetic force**.



(b)

A charge moving at an angle ϕ to a magnetic field experiences a magnetic force with magnitude $F = |q|v_{\perp}B = |q|vB\sin\phi$.



(c)

A charge moving **perpendicular** to a magnetic field experiences a **maximal magnetic force** with magnitude $F_{\max} = qvB$.

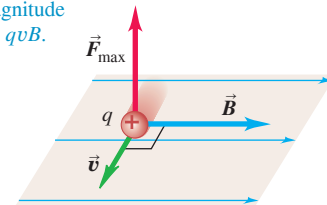
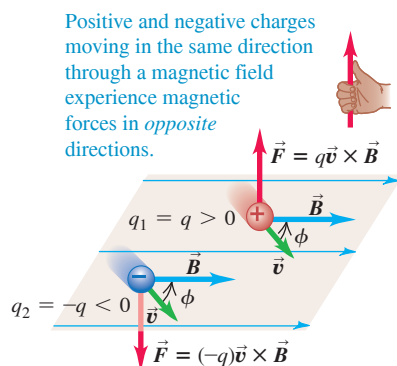


Figure 27.8 Reversing the sign of a moving charge reverses the direction of the magnetic force that acts on it.



BIO APPLICATION Magnetic Fields of the Body

All living cells are electrically active, and the feeble electric currents within your body produce weak but measurable magnetic fields. The fields produced by skeletal muscles have magnitudes less than 10^{-10} T, about one-millionth as strong as the earth's magnetic field. Your brain produces magnetic fields that are far weaker, only about 10^{-12} T.



Figure 27.9 Determining the direction of a magnetic field by using a cathode-ray tube. Because electrons have a negative charge, the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ in part (b) points opposite to the direction given by the right-hand rule (see Fig. 27.7b).

This is the first of several vector products we'll encounter in our study of magnetic-field relationships. It's important to note that Eq. (27.2) was *not* deduced theoretically; it is an observation based on *experiment*.

Equation (27.2) is valid for both positive and negative charges. When q is negative, the direction of the force $\vec{F}$ is opposite to that of $\vec{v} \times \vec{B}$ (Fig. 27.7b). If two charges with equal magnitude and opposite sign move in the same $\vec{B}$ field with the same velocity (Fig. 27.8), the forces have equal magnitude and opposite direction. Figures 27.6, 27.7, and 27.8 show several examples of the relationships of the directions of $\vec{F}$, $\vec{v}$, and $\vec{B}$ for both positive and negative charges. Be sure you understand the relationships shown in these figures.

Equation (27.1) gives the magnitude of the magnetic force $\vec{F}$ in Eq. (27.2). Since ϕ is the angle between the directions of vectors $\vec{v}$ and $\vec{B}$, we may interpret $B \sin \phi$ as the component of $\vec{B}$ perpendicular to $\vec{v}$ —that is, $B_{\perp}$. With this notation the force magnitude is

$$F = |q|vB_{\perp} \quad (27.3)$$

This form may be more convenient, especially in problems involving *currents* rather than individual particles. We'll discuss forces on currents later in this chapter.

From Eq. (27.1) the *units* of B must be the same as the units of F/qv . Therefore the SI unit of B is equivalent to $1 \text{ N} \cdot \text{s}/\text{C} \cdot \text{m}$, or, since one ampere is one coulomb per second ($1 \text{ A} = 1 \text{ C/s}$), $1 \text{ N/A} \cdot \text{m}$. This unit is called the **tesla** (abbreviated T), in honor of Nikola Tesla (1856–1943), the prominent Serbian-American scientist and inventor:

$$1 \text{ tesla} = 1 \text{ T} = 1 \text{ N/A} \cdot \text{m}$$

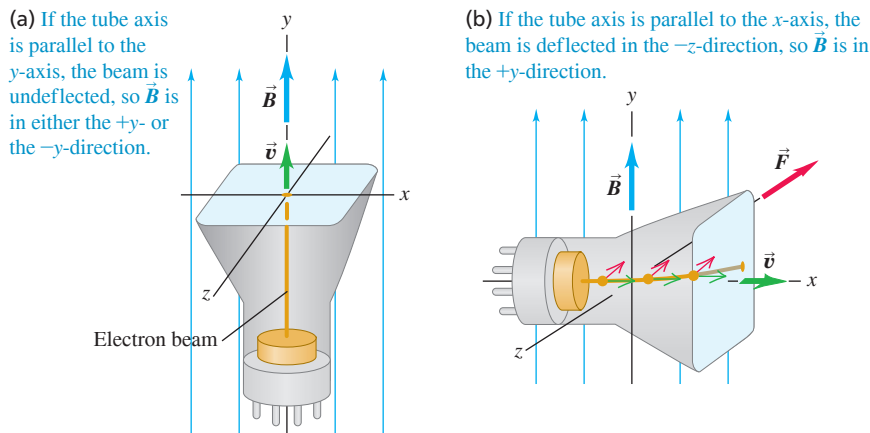
Another unit of B , the **gauss** ($1 \text{ G} = 10^{-4} \text{ T}$), is also in common use.

The magnetic field of the earth is of the order of 10^{-4} T or 1 G. Magnetic fields of the order of 10 T occur in the interior of atoms and are important in the analysis of atomic spectra. The largest steady magnetic field that can be produced at present in the laboratory is about 45 T. Some pulsed-current electromagnets can produce fields of the order of 120 T for millisecond time intervals.

Measuring Magnetic Fields with Test Charges

To explore an unknown magnetic field, we can measure the magnitude and direction of the force on a *moving* test charge and then use Eq. (27.2) to determine $\vec{B}$. The electron beam in a cathode-ray tube, such as that in an older television set (not a flat-screen TV), is a convenient device for this. The electron gun shoots out a narrow beam of electrons at a known speed. If there is no force to deflect the beam, it strikes the center of the screen.

If a magnetic field is present, in general the electron beam is deflected. But if the beam is parallel or antiparallel to the field, then $\phi = 0$ or π in Eq. (27.1) and $F = 0$; there is no force and hence no deflection. If we find that the electron beam is not deflected when its direction is parallel to a certain axis as in Fig. 27.9a, the $\vec{B}$ vector must point either up or down along that axis.



If we then turn the tube 90° (Fig. 27.9b), $\phi = \pi/2$ in Eq. (27.1) and the magnetic force is maximum; the beam is deflected in a direction perpendicular to the plane of $\vec{B}$ and $\vec{v}$. The direction and magnitude of the deflection determine the direction and magnitude of $\vec{B}$. We can perform additional experiments in which the angle between $\vec{B}$ and $\vec{v}$ is between 0° and 90° to confirm Eq. (27.1). We note that the electron has a negative charge; the force in Fig. 27.9b is opposite in direction to the force on a positive charge.

When a charged particle moves through a region of space where *both* electric and magnetic fields are present, both fields exert forces on the particle. The total force $\vec{F}$ is the vector sum of the electric and magnetic forces:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad (27.4)$$

CAUTION **No perpendicular field component, no magnetic force** When a charged particle moves in a magnetic field $\vec{B}$, the particle experiences a magnetic force *only* if there is a component of $\vec{B}$ that is perpendicular to the particle's velocity. If $\vec{B}$ and the velocity are in the same direction (as in Fig. 27.9a) or in opposite directions, there is no perpendicular component of $\vec{B}$ and hence no magnetic force. ■

PROBLEM-SOLVING STRATEGY 27.1 Magnetic Forces

IDENTIFY *the relevant concepts:* The equation $\vec{F} = q\vec{v} \times \vec{B}$ allows you to determine the magnetic force on a moving charged particle.

SET UP *the problem* using the following steps:

1. Draw the velocity $\vec{v}$ and magnetic field $\vec{B}$ with their tails together so that you can visualize the plane that contains them.
2. Determine the angle ϕ between $\vec{v}$ and $\vec{B}$.
3. Identify the target variables.

EXECUTE *the solution* as follows:

1. Use Eq. (27.2), $\vec{F} = q\vec{v} \times \vec{B}$, to express the magnetic force. Equation (27.1) gives the magnitude of the force, $F = qvB \sin \phi$.
2. Remember that $\vec{F}$ is perpendicular to the plane containing $\vec{v}$ and $\vec{B}$. The right-hand rule (see Fig. 27.7) gives the direction of $\vec{v} \times \vec{B}$. If q is negative, $\vec{F}$ is *opposite* to $\vec{v} \times \vec{B}$.

EVALUATE *your answer:* Whenever possible, solve the problem in two ways to confirm that the results agree. Do it directly from the geometric definition of the vector product. Then find the components of the vectors in some convenient coordinate system and calculate the vector product from the components. Verify that the results agree.

EXAMPLE 27.1 Magnetic force on a proton

WITH ✓ **ARIATION PROBLEMS**

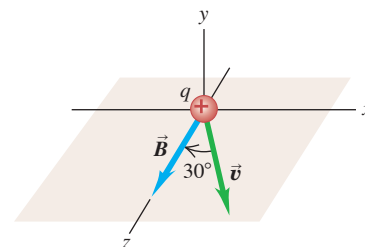
A beam of protons ($q = 1.6 \times 10^{-19}$ C) moves at 3.0×10^5 m/s through a uniform 2.0 T magnetic field directed along the positive z -axis (Fig. 27.10). The velocity of each proton lies in the xz -plane and is directed at 30° to the $+z$ -axis. Find the force on a proton.

IDENTIFY and SET UP This problem uses the expression $\vec{F} = q\vec{v} \times \vec{B}$ for the magnetic force $\vec{F}$ on a moving charged particle. The target variable is $\vec{F}$.

EXECUTE The charge is positive, so the force is in the same direction as the vector product $\vec{v} \times \vec{B}$. From the right-hand rule, this direction is along the negative y -axis. The magnitude of the force, from Eq. (27.1), is

$$\begin{aligned} F &= qvB \sin \phi \\ &= (1.6 \times 10^{-19} \text{ C})(3.0 \times 10^5 \text{ m/s})(2.0 \text{ T})(\sin 30^\circ) \\ &= 4.8 \times 10^{-14} \text{ N} \end{aligned}$$

Figure 27.10 Directions of $\vec{v}$ and $\vec{B}$ for a proton in a magnetic field.



Continued

EVALUATE To check our result, we evaluate the force by using vector language and Eq. (27.2). We have

$$\begin{aligned}\vec{v} &= (3.0 \times 10^5 \text{ m/s})(\sin 30^\circ)\hat{i} + (3.0 \times 10^5 \text{ m/s})(\cos 30^\circ)\hat{k} \\ \vec{B} &= (2.0 \text{ T})\hat{k} \\ \vec{F} &= q\vec{v} \times \vec{B} \\ &= (1.6 \times 10^{-19} \text{ C})(3.0 \times 10^5 \text{ m/s})(2.0 \text{ T}) \\ &\quad \times (\sin 30^\circ\hat{i} + \cos 30^\circ\hat{k}) \times \hat{k} \\ &= (-4.8 \times 10^{-14} \text{ N})\hat{j}\end{aligned}$$

(Recall that $\hat{i} \times \hat{k} = -\hat{j}$ and $\hat{k} \times \hat{k} = \mathbf{0}$.) We again find that the force is in the negative y -direction with magnitude $4.8 \times 10^{-14} \text{ N}$.

If the beam consists of *electrons* rather than protons, the charge is negative ($q = -1.6 \times 10^{-19} \text{ C}$) and the direction of the force is reversed. The force is now directed along the *positive* y -axis, but the magnitude is the same as before, $F = 4.8 \times 10^{-14} \text{ N}$.

KEYCONCEPT A charged particle moving with velocity $\vec{v}$ through a magnetic field $\vec{B}$ experiences a magnetic force that depends on its charge, its speed, the magnitude of $\vec{B}$, and the angle between $\vec{v}$ and $\vec{B}$. The force is zero if the directions of $\vec{v}$ and $\vec{B}$ are the same or opposite; otherwise, the force is nonzero and in a direction perpendicular to both $\vec{v}$ and $\vec{B}$.

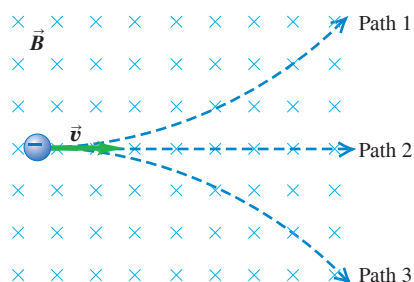
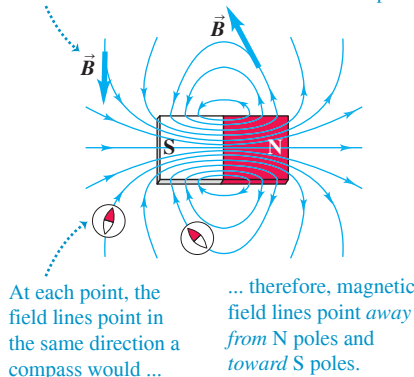


Figure 27.11 Magnetic field lines of a permanent magnet. Note that the field lines pass through the interior of the magnet.

At each point, the field line is tangent to the magnetic-field vector $\vec{B}$.

The more densely the field lines are packed, the stronger the field is at that point.



At each point, the field lines point in the same direction a compass would ...

... therefore, magnetic field lines point away from N poles and toward S poles.

TEST YOUR UNDERSTANDING OF SECTION 27.2 The accompanying figure shows a uniform magnetic field $\vec{B}$ directed into the plane of the paper (shown by the blue $\times$'s). A particle with a negative charge moves in the plane. Which path—1, 2, or 3—does the particle follow?

ANSWER

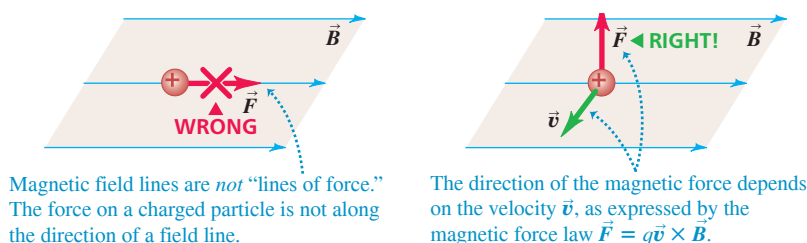
path 3 Applying the right-hand rule to the vectors $\vec{v}$ (which points to the right) and $\vec{B}$ (which points into the plane of the figure) says that the force $\vec{F} = q\vec{v} \times \vec{B}$ on a positive charge would point upward. Since the charge is negative, the force points downward and the particle follows a trajectory that curves downward.

27.3 MAGNETIC FIELD LINES AND MAGNETIC FLUX

We can represent any magnetic field by **magnetic field lines**, just as we did for the earth's magnetic field in Fig. 27.3. The idea is the same as for the electric field lines we introduced in Section 21.6. We draw the lines so that the line through any point is tangent to the magnetic-field vector $\vec{B}$ at that point (**Fig. 27.11**). Just as with electric field lines, we draw only a few representative lines; otherwise, the lines would fill up all of space. Where adjacent field lines are close together, the field magnitude is large; where these field lines are far apart, the field magnitude is small. Also, because the direction of $\vec{B}$ at each point is unique, field lines never intersect.

CAUTION **Magnetic field lines are not "lines of force"** Unlike electric field lines, magnetic field lines *do not* point in the direction of the force on a charge (**Fig. 27.12**). Equation (27.2) shows that the force on a moving charged particle is always perpendicular to the magnetic field and hence to the magnetic field line that passes through the particle's position. The direction of the force depends on the particle's velocity and the sign of its charge, so just looking at magnetic field lines cannot tell you the direction of the force on an arbitrary moving charged particle. Magnetic field lines *do* have the direction that a compass needle would point at each location; this may help you visualize them. **I**

Figure 27.12 Magnetic field lines are *not* "lines of force."

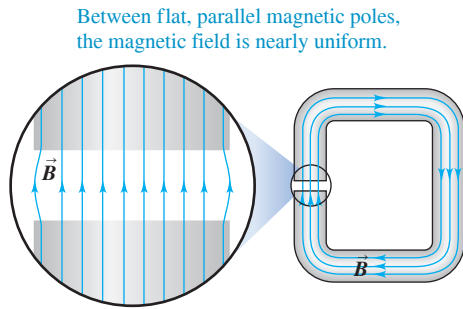


Magnetic field lines are *not* "lines of force." The force on a charged particle is not along the direction of a field line.

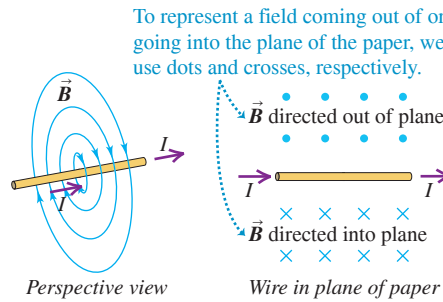
The direction of the magnetic force depends on the velocity $\vec{v}$, as expressed by the magnetic force law $\vec{F} = q\vec{v} \times \vec{B}$.

Figure 27.13 Magnetic field lines produced by some common sources of magnetic field.

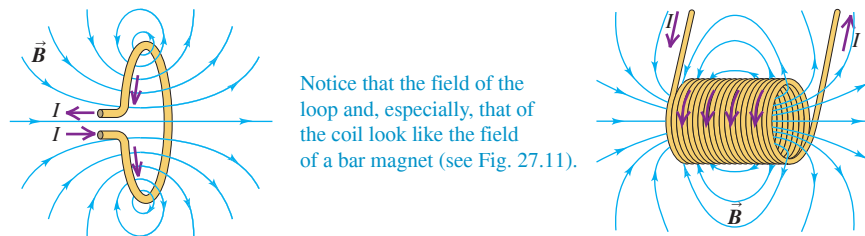
(a) Magnetic field of a C-shaped magnet



(b) Magnetic field of a straight current-carrying wire



(c) Magnetic fields of a current-carrying loop and a current-carrying coil (solenoid)



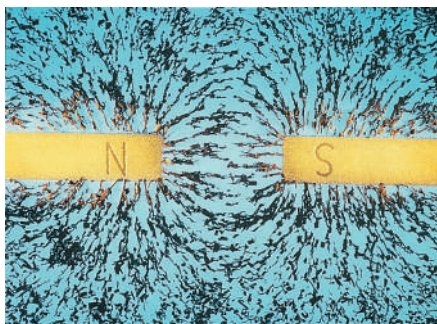
Figures 27.11 and 27.13 show magnetic field lines produced by several common sources of magnetic field. In the gap between the poles of the magnet shown in **Fig. 27.13a**, the field lines are approximately straight, parallel, and equally spaced, showing that the magnetic field in this region is approximately *uniform* (that is, constant in magnitude and direction).

Because magnetic-field patterns are three-dimensional, it's often necessary to draw magnetic field lines that point into or out of the plane of a drawing. To do this we use a dot ($\bullet$) to represent a vector directed out of the plane and a cross ($\times$) to represent a vector directed into the plane (Fig. 27.13b). To remember these, think of a dot as the head of an arrow coming directly toward you, and think of a cross as the feathers of an arrow flying directly away from you.

Iron filings, like compass needles, tend to align with magnetic field lines. Hence they provide an easy way to visualize field lines (**Fig. 27.14**).

Figure 27.14 (a) Like little compass needles, iron filings line up tangent to magnetic field lines. (b) Drawing of field lines for the situation shown in (a).

(a)



(b)

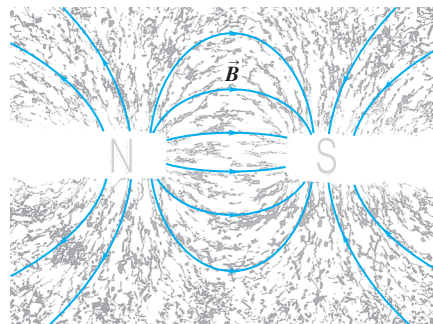
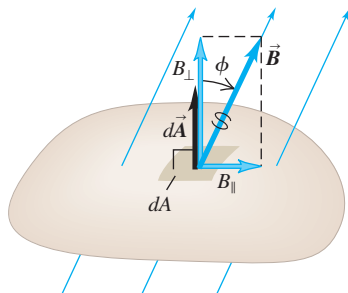


Figure 27.15 The magnetic flux through an area element dA is defined to be $d\Phi_B = B_{\perp} dA$.



Magnetic Flux and Gauss's Law for Magnetism

We define the **magnetic flux** Φ_B through a surface just as we defined electric flux in connection with Gauss's law in Section 22.2. We can divide any surface into elements of area dA (Fig. 27.15). For each element we determine $B_{\perp}$, the component of $\vec{B}$ normal to the surface at the position of that element, as shown. From the figure, $B_{\perp} = B \cos \phi$, where ϕ is the angle between the direction of $\vec{B}$ and a line perpendicular to the surface. (Be careful not to confuse ϕ with Φ_B .) In general, this component varies from point to point on the surface. We define the magnetic flux $d\Phi_B$ through this area as

$$d\Phi_B = B_{\perp} dA = B \cos \phi dA = \vec{B} \cdot d\vec{A} \quad (27.5)$$

The *total* magnetic flux through the surface is the sum of the contributions from the individual area elements:

$$\text{Magnetic flux through a surface } \Phi_B = \int B \cos \phi dA = \int B_{\perp} dA = \int \vec{B} \cdot d\vec{A} \quad (27.6)$$

Magnitude of magnetic field $\vec{B}$
Component of $\vec{B}$ perpendicular to surface

Angle between $\vec{B}$ and normal to surface
Element of surface area
Vector element of surface area

(Review the concepts of vector area and surface integral in Section 22.2.)

Magnetic flux is a *scalar* quantity. If $\vec{B}$ is uniform over a plane surface with total area A , then $B_{\perp}$ and ϕ are the same at all points on the surface, and

$$\Phi_B = B_{\perp} A = BA \cos \phi \quad (27.7)$$

If $\vec{B}$ is also perpendicular to the surface (parallel to the area vector), then $\cos \phi = 1$ and Eq. (27.7) reduces to $\Phi_B = BA$. We'll use the concept of magnetic flux extensively during our study of electromagnetic induction in Chapter 29.

The SI unit of magnetic flux is equal to the unit of magnetic field (1 T) times the unit of area (1 m^2). This unit is called the **weber** (1 Wb), in honor of the German physicist Wilhelm Weber (1804–1891):

$$1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$$

Also, $1 \text{ T} = 1 \text{ N/A} \cdot \text{m}$, so

$$1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2 = 1 \text{ N} \cdot \text{m/A}$$

In Gauss's law the total *electric* flux through a closed surface is proportional to the total electric charge enclosed by the surface. For example, if the closed surface encloses an electric dipole, the total electric flux is zero because the total charge is zero. (You may want to review Section 22.3 on Gauss's law.) By analogy, if there were such a thing as a single magnetic charge (magnetic monopole), the total *magnetic* flux through a closed surface would be proportional to the total magnetic charge enclosed. But we have mentioned that no magnetic monopole has ever been observed, despite intensive searches. This leads us to **Gauss's law for magnetism**:

CAUTION **Magnetic field lines have no ends** Unlike electric field lines, which begin and end on electric charges, magnetic field lines *never* have endpoints; such a point would indicate the presence of a monopole. You might be tempted to draw magnetic field lines that begin at the north pole of a magnet and end at a south pole. But as Fig. 27.11 shows, a magnet's field lines continue through the interior of the magnet. Like all other magnetic field lines, they form closed loops. ■

Gauss's law for magnetism:

The total magnetic flux through *any* closed surface ...

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad \dots \text{equals zero.} \quad (27.8)$$

You can verify Gauss's law for magnetism by examining Figs. 27.11 and 27.13; if you draw a closed surface anywhere in any of the field maps shown in those figures, you'll see that every field line that enters the surface also exits from it; the net flux through the surface is zero. It also follows from Eq. (27.8) that magnetic field lines always form closed loops.

For Gauss's law, which always deals with *closed* surfaces, the vector area element $d\vec{A}$ in Eq. (27.6) always points *out of* the surface. However, some applications of *magnetic* flux involve an *open* surface with a boundary line; there is then an ambiguity of sign in Eq. (27.6) because of the two possible choices of direction for $d\vec{A}$. In these cases we choose one of the two sides of the surface to be the “positive” side and use that choice consistently.

If the element of area dA in Eq. (27.5) is at right angles to the field lines, then $B_{\perp} = B$; calling the area $dA_{\perp}$, we have

$$B = \frac{d\Phi_B}{dA_{\perp}} \quad (27.9)$$

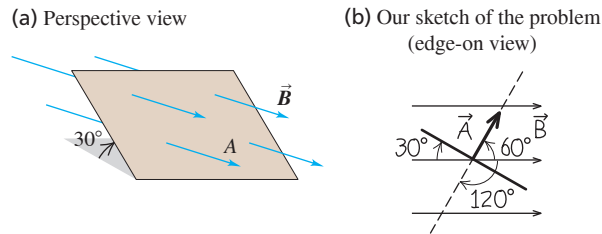
That is, the magnitude of the magnetic field is equal to *flux per unit area* across an area at right angles to the magnetic field. For this reason, magnetic field $\vec{B}$ is sometimes called **magnetic flux density**.

EXAMPLE 27.2 Magnetic flux calculations

Figure 27.16a is a perspective view of a flat surface with area 3.0 cm^2 in a uniform magnetic field $\vec{B}$. The magnetic flux through this surface is $+0.90 \text{ mWb}$. Find the magnitude of the magnetic field and the direction of the area vector $\vec{A}$.

IDENTIFY and SET UP Our target variables are the field magnitude B and the direction of the area vector. Because $\vec{B}$ is uniform, B and ϕ are the same at all points on the surface. Hence we can use Eq. (27.7), $\Phi_B = BA \cos \phi$.

Figure 27.16 (a) A flat area A in a uniform magnetic field $\vec{B}$. (b) The area vector $\vec{A}$ makes a 60° angle with $\vec{B}$. (If we had chosen $\vec{A}$ to point in the opposite direction, ϕ would have been 120° and the magnetic flux Φ_B would have been negative.)



EXECUTE The area A is $3.0 \times 10^{-4} \text{ m}^2$; the direction of $\vec{A}$ is perpendicular to the surface, so ϕ could be either 60° or 120° . But Φ_B , B , and A are all positive, so $\cos \phi$ must also be positive. This rules out 120° , so $\phi = 60^\circ$ (Fig. 27.16b). Hence we find

$$B = \frac{\Phi_B}{A \cos \phi} = \frac{0.90 \times 10^{-3} \text{ Wb}}{(3.0 \times 10^{-4} \text{ m}^2)(\cos 60^\circ)} = 6.0 \text{ T}$$

EVALUATE In many problems we are asked to calculate the flux of a given magnetic field through a given area. This example is somewhat different: It tests your understanding of the definition of magnetic flux.

KEYCONCEPT If a uniform magnetic field $\vec{B}$ is present over a surface, the magnetic flux through that surface equals the area of the surface multiplied by the component of $\vec{B}$ normal (perpendicular) to the surface. The net magnetic flux Φ_B through a closed surface is always zero.

TEST YOUR UNDERSTANDING OF SECTION 27.3 Imagine moving along the axis of the current-carrying loop in Fig. 27.13c, starting at a point well to the left of the loop and ending at a point well to the right of the loop. (a) How would the magnetic field strength vary as you moved along this path? (i) It would be the same at all points along the path; (ii) it would increase and then decrease; (iii) it would decrease and then increase. (b) Would the magnetic field direction vary as you moved along the path?

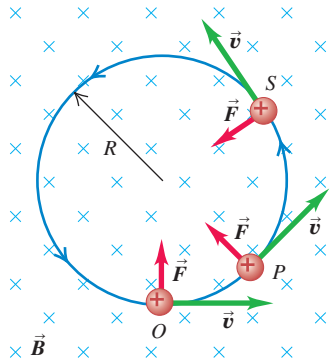
ANSWER

(a) (iii), (b) no. The magnitude of $\vec{B}$ would increase as you moved to the right, reaching a maximum as you passed through the plane of the loop. As you moved beyond the plane of the loop, the field magnitude would decrease. You can tell this from the spacing of the field lines: The closer the field lines, the stronger the field. The direction of the field would be to the right at all points along the path, since the path is along a field line and the direction of $\vec{B}$ at any point is tangent to the field line through that point.

Figure 27.17 A charged particle moves in a plane perpendicular to a uniform magnetic field $\vec{B}$.

(a) The orbit of a charged particle in a uniform magnetic field

A charge moving at right angles to a uniform $\vec{B}$ field moves in a circle at constant speed because $\vec{F}$ and $\vec{v}$ are always perpendicular to each other.



(b) An electron beam (seen as a white arc) curving in a magnetic field

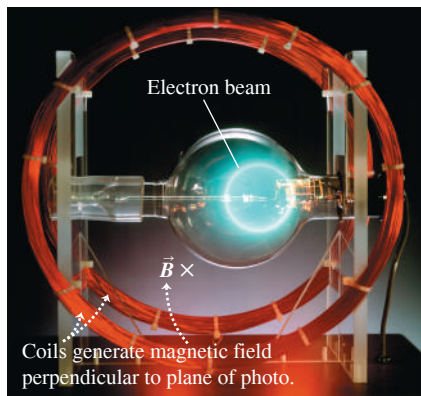
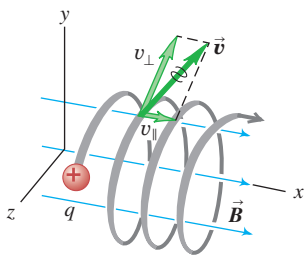


Figure 27.18 The general case of a charged particle moving in a uniform magnetic field $\vec{B}$. The magnetic field does no work on the particle, so its speed and kinetic energy remain constant.

This particle's motion has components both parallel ($v_{\parallel}$) and perpendicular ($v_{\perp}$) to the magnetic field, so it moves in a helical path.



27.4 MOTION OF CHARGED PARTICLES IN A MAGNETIC FIELD

When a charged particle moves in a magnetic field, it is acted on by the magnetic force given by Eq. (27.2), and the motion is determined by Newton's laws. **Figure 27.17a** shows a simple example. A particle with positive charge q is at point O , moving with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ directed into the plane of the figure. The vectors $\vec{v}$ and $\vec{B}$ are perpendicular, so the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ has magnitude $F = qvB$ and a direction as shown in the figure. The force is *always* perpendicular to $\vec{v}$, so it cannot change the *magnitude* of the velocity, only its direction. To put it differently, the magnetic force never has a component parallel to the particle's motion, so the magnetic force can never do *work* on the particle. This is true even if the magnetic field is not uniform.

Motion of a charged particle under the action of a magnetic field alone is always motion with constant speed.

Using this principle, we see that in Fig. 27.17a the magnitudes of both $\vec{F}$ and $\vec{v}$ are constant. As the particle of mass m moves from O to P to S , the directions of force and velocity change but their magnitudes stay the same. So the particle is acted on by a constant-magnitude force that is always at right angles to the velocity of the particle. It follows from our discussion of circular motion in Sections 3.4 and 5.4 that the particle moves in a *circle* of radius R with constant speed v . The centripetal acceleration is v^2/R and only the magnetic force acts, so from Newton's second law,

$$F = |q|vB = m \frac{v^2}{R} \quad (27.10)$$

We solve Eq. (27.10) for R :

$$\text{Radius of a circular orbit in a magnetic field} \quad R = \frac{\text{Particle's mass} \times \text{Particle's speed}}{\text{Magnetic-field magnitude} \times \text{Particle's charge}} = \frac{mv}{|q|B} \quad (27.11)$$

If the charge q is negative, the particle moves *clockwise* around the orbit in Fig. 27.17a. Figure 27.17b shows a beam of negatively charged electrons following just such an orbit.

The angular speed ω of the particle can be found from Eq. (9.13), $v = R\omega$. Combining this with Eq. (27.11), we get

$$\omega = \frac{v}{R} = v \frac{|q|B}{mv} = \frac{|q|B}{m} \quad (27.12)$$

The number of revolutions per unit time is $f = \omega/2\pi$. This frequency f is independent of the radius R of the path. It is called the **cyclotron frequency**; in a particle accelerator called a *cyclotron*, particles moving in nearly circular paths are given a boost twice each revolution, increasing their energy and their orbital radii but not their angular speed or frequency. Similarly, one type of *magnetron*, a common source of microwave radiation for microwave ovens and radar systems, emits radiation with a frequency equal to the frequency of circular motion of electrons in a vacuum chamber between the poles of a magnet.

If the direction of the initial velocity is *not* perpendicular to the field, the velocity *component* parallel to the field is constant because there is no force parallel to the field. Then the particle moves in a helix (**Fig. 27.18**). The radius of the helix is given by Eq. (27.11), where v is now the component of velocity perpendicular to the $\vec{B}$ field.

Figure 27.19 A magnetic bottle. Particles near either end of the region experience a magnetic force toward the center of the region. This is one way of containing an ionized gas that has a temperature of the order of 10^6 K, which would vaporize any material container.

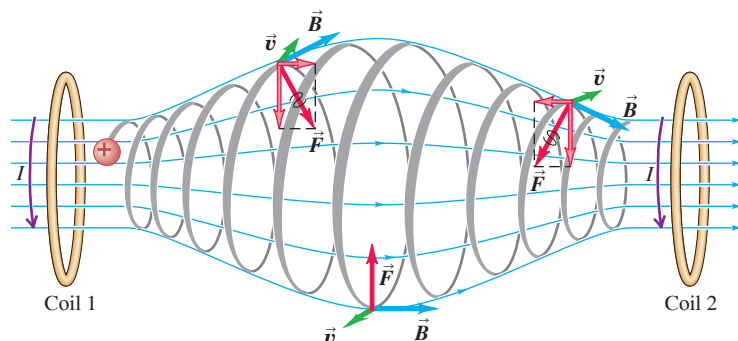
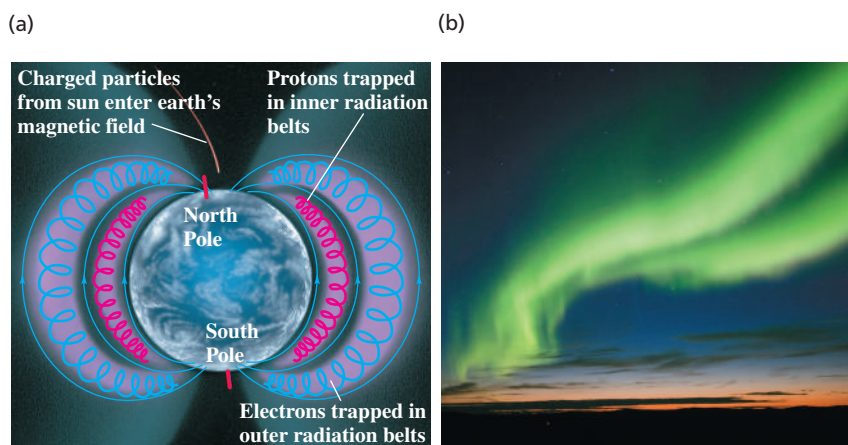


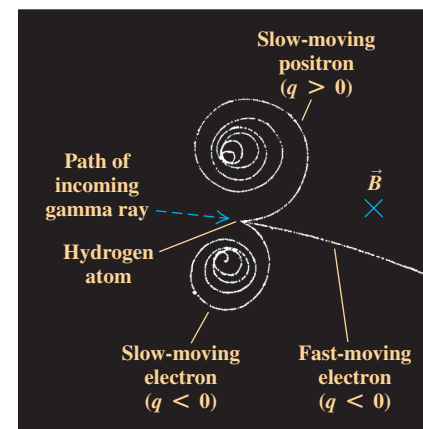
Figure 27.20 (a) The Van Allen radiation belts around the earth. Near the poles, charged particles from these belts can enter the atmosphere, producing the aurora borealis (“northern lights”) and aurora australis (“southern lights”). (b) A photograph of the aurora borealis.



Motion of a charged particle in a nonuniform magnetic field is more complex. **Figure 27.19** shows a field produced by two circular coils separated by some distance. Particles near either coil experience a magnetic force toward the center of the region; particles with appropriate speeds spiral repeatedly from one end of the region to the other and back. Because charged particles can be trapped in such a magnetic field, it is called a *magnetic bottle*. This technique is used to confine very hot plasmas with temperatures of the order of 10^6 K. In a similar way the earth's nonuniform magnetic field traps charged particles coming from the sun in doughnut-shaped regions around the earth, as shown in **Fig. 27.20**. These regions, called the *Van Allen radiation belts*, were discovered in 1958 from data obtained by instruments aboard the Explorer I satellite.

Magnetic forces on charged particles play an important role in studies of elementary particles. **Figure 27.21** shows a chamber filled with liquid hydrogen and with a magnetic field directed into the plane of the photograph. A high-energy gamma ray dislodges an electron from a hydrogen atom, sending it off at high speed and creating a visible track. The track shows the electron curving downward due to the magnetic force. The energy of the collision also produces another electron and a *positron* (a positively charged electron). Because of their opposite charges, the trajectories of the electron and the positron curve in opposite directions. As these particles plow through the liquid hydrogen, they collide with other charged particles, losing energy and speed. As a result, the radius of curvature decreases as suggested by Eq. (27.11). (The electron's speed is comparable to the speed of light, so Eq. (27.11) isn't directly applicable here.) Similar experiments allow physicists to determine the mass and charge of newly discovered particles.

Figure 27.21 This bubble chamber image shows the result of a high-energy gamma ray (which does not leave a track) that collides with an electron in a hydrogen atom. This electron flies off to the right at high speed. Some of the energy in the collision is transformed into a second electron and a positron (a positively charged electron). A magnetic field is directed into the plane of the image, which makes the positive and negative particles curve off in different directions.



PROBLEM-SOLVING STRATEGY 27.2 Motion in Magnetic Fields

IDENTIFY the relevant concepts: In analyzing the motion of a charged particle in electric and magnetic fields, you'll apply Newton's second law of motion, $\Sigma \vec{F} = m\vec{a}$, with the net force given by $\Sigma \vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. Often other forces such as gravity can be ignored. Many of the problems are similar to the trajectory and circular-motion problems in Sections 3.3, 3.4, and 5.4; it would be a good idea to review those sections.

SET UP the problem using the following steps:

1. Determine the target variable(s).
2. Often the use of components is the most efficient approach. Choose a coordinate system and then express all vector quantities in terms of their components in this system.

EXECUTE the solution as follows:

1. If the particle moves perpendicular to a uniform magnetic field, the trajectory is a circle with a radius and angular speed given by Eqs. (27.11) and (27.12), respectively.
2. If your calculation involves a more complex trajectory, use $\Sigma \vec{F} = m\vec{a}$ in component form: $\Sigma F_x = ma_x$, and so forth. This approach is particularly useful when both electric and magnetic fields are present.

EVALUATE your answer: Check whether your results are reasonable.

EXAMPLE 27.3 Electron motion in a magnetron

WITH VARIATION PROBLEMS

A magnetron in a microwave oven emits electromagnetic waves with frequency $f = 2450$ MHz. What magnetic field strength is required for electrons to move in circular paths with this frequency?

IDENTIFY and SET UP The problem refers to circular motion as shown in Fig. 27.17a. We use Eq. (27.12) to solve for the field magnitude B .

EXECUTE The angular speed that corresponds to the frequency f is $\omega = 2\pi f = (2\pi)(2450 \times 10^6 \text{ s}^{-1}) = 1.54 \times 10^{10} \text{ s}^{-1}$. Then from Eq. (27.12),

$$B = \frac{m\omega}{|q|} = \frac{(9.11 \times 10^{-31} \text{ kg})(1.54 \times 10^{10} \text{ s}^{-1})}{1.60 \times 10^{-19} \text{ C}} = 0.0877 \text{ T}$$

EVALUATE This is a moderate field strength, easily produced with a permanent magnet. Incidentally, 2450 MHz electromagnetic waves are useful for heating and cooking food because they are strongly absorbed by water molecules.

KEYCONCEPT A charged particle that travels perpendicular to a uniform magnetic field $\vec{B}$ undergoes uniform circular motion. The frequency of the motion is independent of the speed of the particle: It depends on only the magnetic field strength and the particle's charge and mass.

EXAMPLE 27.4 Helical particle motion in a magnetic field

WITH VARIATION PROBLEMS

In a situation like that shown in Fig. 27.18, the charged particle is a proton ($q = 1.60 \times 10^{-19} \text{ C}$, $m = 1.67 \times 10^{-27} \text{ kg}$) and the uniform, 0.500 T magnetic field is directed along the x -axis. At $t = 0$ the proton has velocity components $v_x = 1.50 \times 10^5 \text{ m/s}$, $v_y = 0$, and $v_z = 2.00 \times 10^5 \text{ m/s}$. Only the magnetic force acts on the proton. (a) At $t = 0$, find the force on the proton and its acceleration. (b) Find the radius of the resulting helical path, the angular speed of the proton, and the *pitch* of the helix (the distance traveled along the helix axis per revolution).

IDENTIFY and SET UP The magnetic force is $\vec{F} = q\vec{v} \times \vec{B}$; Newton's second law gives the resulting acceleration. Because $\vec{F}$ is perpendicular to $\vec{v}$, the proton's speed does not change. Hence Eq. (27.11) gives the radius of the helical path if we replace v with the velocity component perpendicular to $\vec{B}$. Equation (27.12) gives the angular speed ω , which yields the time T for one revolution (the *period*). Given the velocity component parallel to the magnetic field, we can then determine the pitch.

EXECUTE (a) With $\vec{B} = B\hat{i}$ and $\vec{v} = v_x\hat{i} + v_z\hat{k}$, Eq. (27.2) yields

$$\begin{aligned}\vec{F} &= q\vec{v} \times \vec{B} = q(v_x\hat{i} + v_z\hat{k}) \times B\hat{i} = qv_zB\hat{j} \\ &= (1.60 \times 10^{-19} \text{ C})(2.00 \times 10^5 \text{ m/s})(0.500 \text{ T})\hat{j} \\ &= (1.60 \times 10^{-14} \text{ N})\hat{j}\end{aligned}$$

(Recall: $\hat{i} \times \hat{i} = \mathbf{0}$ and $\hat{k} \times \hat{i} = \hat{j}$.) The resulting acceleration is

$$\vec{a} = \frac{\vec{F}}{m} = \frac{1.60 \times 10^{-14} \text{ N}}{1.67 \times 10^{-27} \text{ kg}}\hat{j} = (9.58 \times 10^{12} \text{ m/s}^2)\hat{j}$$

(b) Since $v_y = 0$, the component of velocity perpendicular to $\vec{B}$ is v_z ; then from Eq. (27.11),

$$\begin{aligned}R &= \frac{mv_z}{|q|B} = \frac{(1.67 \times 10^{-27} \text{ kg})(2.00 \times 10^5 \text{ m/s})}{(1.60 \times 10^{-19} \text{ C})(0.500 \text{ T})} \\ &= 4.18 \times 10^{-3} \text{ m} = 4.18 \text{ mm}\end{aligned}$$

From Eq. (27.12) the angular speed is

$$\omega = \frac{|q|B}{m} = \frac{(1.60 \times 10^{-19} \text{ C})(0.500 \text{ T})}{1.67 \times 10^{-27} \text{ kg}} = 4.79 \times 10^7 \text{ rad/s}$$

The period is $T = 2\pi/\omega = 2\pi/(4.79 \times 10^7 \text{ s}^{-1}) = 1.31 \times 10^{-7} \text{ s}$. The pitch is the distance traveled along the x -axis in this time, or

$$\begin{aligned}v_x T &= (1.50 \times 10^5 \text{ m/s})(1.31 \times 10^{-7} \text{ s}) \\ &= 0.0197 \text{ m} = 19.7 \text{ mm}\end{aligned}$$

EVALUATE Although the magnetic force has a tiny magnitude, it produces an immense acceleration because the proton mass is so small. Note that the pitch of the helix is almost five times greater than the radius R , so this helix is much more “stretched out” than that shown in Fig. 27.18.

KEYCONCEPT The most general path that a charged particle follows in a uniform magnetic field $\vec{B}$ is a helix: straight-line, constant-speed motion parallel to $\vec{B}$ combined with uniform circular motion perpendicular to $\vec{B}$.

TEST YOUR UNDERSTANDING OF SECTION 27.4 (a) If you double the speed of the charged particle in Fig. 27.17a while keeping the magnetic field the same (as well as the charge and the mass), how does this affect the radius of the trajectory? (i) The radius is unchanged; (ii) the radius is twice as large; (iii) the radius is four times as large; (iv) the radius is $\frac{1}{2}$ as large; (v) the radius is $\frac{1}{4}$ as large. (b) How does this affect the time required for one complete circular orbit? (i) The time is unchanged; (ii) the time is twice as long; (iii) the time is four times as long; (iv) the time is $\frac{1}{2}$ as long; (v) the time is $\frac{1}{4}$ as long.

ANSWER

(a) (i) The radius of the orbit as given by Eq. (27.11) is directly proportional to the speed, so doubling the particle speed causes the radius to double as well. The particle has to travel far to complete one orbit but is traveling at double the speed, so the time for one orbit is unchanged. This result also follows from Eq. (27.12), which states that the angular speed ω is independent of the linear speed v . Hence the time per orbit, $T = 2\pi/\omega$, likewise does not depend on v .

27.5 APPLICATIONS OF MOTION OF CHARGED PARTICLES

This section describes several applications of the principles introduced in this chapter. Study them carefully, watching for applications of Problem-Solving Strategy 27.2 (Section 27.4).

Velocity Selector

In a beam of charged particles produced by a heated cathode or a radioactive material, not all particles move with the same speed. Many applications, however, require a beam in which all the particle speeds are the same. Particles of a specific speed can be selected from the beam by using an arrangement of electric and magnetic fields called a *velocity selector*. In **Fig. 27.22a** a charged particle with mass m , charge q , and speed v enters a region of space where the electric and magnetic fields are perpendicular to the particle's velocity and to each other. The electric field $\vec{E}$ is to the left, and the magnetic field $\vec{B}$ is into the plane of the figure. If q is positive, the electric force is to the left, with magnitude qE , and the magnetic force is to the right, with magnitude qvB . For given field magnitudes E and B , for a particular value of v the electric and magnetic forces will be equal in magnitude; the total force is then zero, and the particle travels in a straight line with constant velocity. This will be the case if $qE = qvB$ (Fig. 27.22b), so the speed v for which there is no deflection is

$$v = \frac{E}{B} \quad (27.13)$$

Only particles with speeds equal to E/B can pass through without being deflected by the fields. By adjusting E and B appropriately, we can select particles having a particular speed for use in other experiments. Because q divides out in Eq. (27.13), a velocity selector for positively charged particles also works for electrons or other negatively charged particles.

Thomson's e/m Experiment

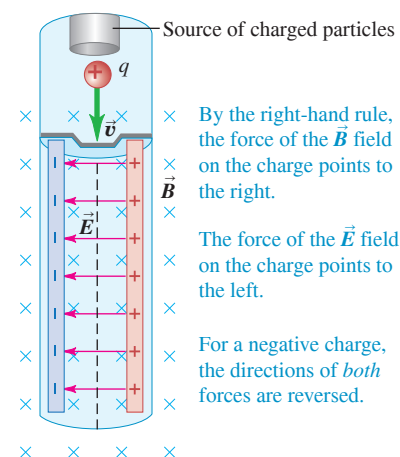
In one of the landmark experiments in physics at the end of the 19th century, J. J. Thomson (1856–1940) used the idea just described to measure the ratio of charge to mass for the electron. For this experiment, carried out in 1897 at the Cavendish Laboratory in Cambridge, England, Thomson used the apparatus shown in **Fig. 27.23** (next page). In a highly evacuated glass container, electrons from the hot cathode are accelerated and formed into a beam by a potential difference V between the two anodes A and A' . The speed v of the electrons is determined by the accelerating potential V . The gained kinetic energy $\frac{1}{2}mv^2$ equals the lost electric potential energy eV , where e is the magnitude of the electron charge:

$$\frac{1}{2}mv^2 = eV \quad \text{or} \quad v = \sqrt{\frac{2eV}{m}} \quad (27.14)$$

The electrons pass between the plates P and P' and strike the screen at the end of the tube, which is coated with a material that fluoresces (glows) at the point of impact. The

Figure 27.22 (a) A velocity selector for charged particles uses perpendicular $\vec{E}$ and $\vec{B}$ fields. Only charged particles with $v = E/B$ move through undeflected. (b) The electric and magnetic forces on a positive charge. Both forces are reversed if the charge is negative.

(a) Schematic diagram of velocity selector



(b) Free-body diagram for a positive particle

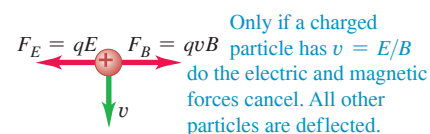
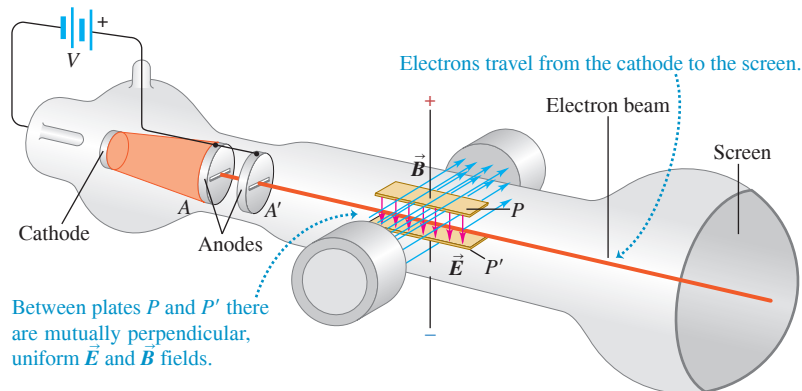


Figure 27.23 Thomson's apparatus for measuring the ratio e/m for the electron.

electrons pass straight through the plates when Eq. (27.13) is satisfied; combining this with Eq. (27.14), we get

$$\frac{E}{B} = \sqrt{\frac{2eV}{m}} \quad \text{so} \quad \frac{e}{m} = \frac{E^2}{2VB^2} \quad (27.15)$$

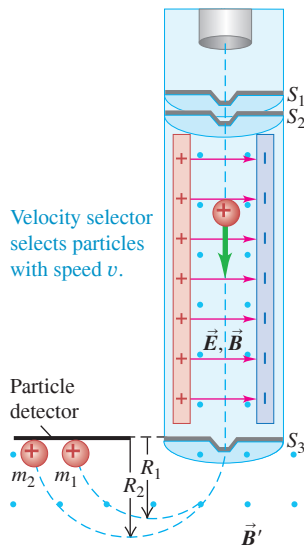
All the quantities on the right side can be measured, so the ratio e/m of charge to mass can be determined. It is *not* possible to measure e or m separately by this method, only their ratio.

The most significant aspect of Thomson's e/m measurements was that he found a *single value* for this quantity. It did not depend on the cathode material, the residual gas in the tube, or anything else about the experiment. This independence showed that the particles in the beam, which we now call electrons, are a common constituent of all matter. Thus Thomson is credited with the first discovery of a subatomic particle, the electron.

The most precise value of e/m available as of this writing is

$$e/m = 1.758820024(11) \times 10^{11} \text{ C/kg}$$

Figure 27.24 Bainbridge's mass spectrometer utilizes a velocity selector to produce particles with uniform speed v . In the region of magnetic field B' , particles with greater mass ($m_2 > m_1$) travel in paths with larger radius ($R_2 > R_1$).



In this expression, (11) indicates the likely uncertainty in the last two digits, 24.

Fifteen years after Thomson's experiments, the American physicist Robert Millikan succeeded in measuring the charge of the electron precisely (see Challenge Problem 23.81). This value, together with the value of e/m , enables us to determine the *mass* of the electron. The most precise value available at present is

$$m = 9.10938356(11) \times 10^{-31} \text{ kg}$$

Mass Spectrometers

Techniques similar to Thomson's e/m experiment can be used to measure masses of ions and thus measure atomic and molecular masses. In 1919, Francis Aston (1877–1945), a student of Thomson's, built the first of a family of instruments called **mass spectrometers**. A variation built by Bainbridge is shown in **Fig. 27.24**. Positive ions from a source pass through the slits S_1 and S_2 , forming a narrow beam. Then the ions pass through a velocity selector with crossed $\vec{E}$ and $\vec{B}$ fields, as we have described, to block all ions except those with speeds v equal to E/B . Finally, the ions pass into a region with a magnetic field $\vec{B}'$ perpendicular to the figure, where they move in circular arcs with radius R determined by Eq. (27.11): $R = mv/qB'$. Ions with different masses strike the detector at different points, and the values of R can be measured. We assume that each ion has lost one electron, so the net charge of each ion is just $+e$. With everything known in this equation except m , we can compute the mass m of the ion.

One of the earliest results from this work was the discovery that neon has two species of atoms, with atomic masses 20 and 22 g/mol. We now call these species **isotopes** of the element. Later experiments have shown that many elements have several isotopes—atoms with identical chemical behaviors but different masses due to differing numbers of neutrons in their nuclei. This is just one of the many applications of mass spectrometers in chemistry and physics.

EXAMPLE 27.5 An e/m demonstration experiment**WITH VARIATION PROBLEMS**

You set out to reproduce Thomson's e/m experiment with an accelerating potential of 150 V and a deflecting electric field of magnitude 6.0×10^6 N/C. (a) How fast do the electrons move? (b) What magnetic-field magnitude will yield zero beam deflection? (c) With this magnetic field, how will the electron beam behave if you increase the accelerating potential above 150 V?

IDENTIFY and SET UP This is the situation shown in Fig. 27.23. We use Eq. (27.14) to determine the electron speed and Eq. (27.13) to determine the required magnetic field B .

EXECUTE (a) From Eq. (27.14), the electron speed v is

$$v = \sqrt{2(e/m)V} = \sqrt{2(1.76 \times 10^{11} \text{ C/kg})(150 \text{ V})} \\ = 7.27 \times 10^6 \text{ m/s} = 0.024c$$

(b) From Eq. (27.13), the required field strength is

$$B = \frac{E}{v} = \frac{6.0 \times 10^6 \text{ N/C}}{7.27 \times 10^6 \text{ m/s}} = 0.83 \text{ T}$$

(c) Increasing the accelerating potential V increases the electron speed v . In Fig. 27.23 this doesn't change the upward electric force eE , but it increases the downward magnetic force evB . Therefore the electron beam will turn *downward* and will hit the end of the tube below the undeflected position.

EVALUATE The required magnetic field is relatively large because the electrons are moving fairly rapidly (2.4% of the speed of light). If the maximum available magnetic field is less than 0.83 T, the electric field strength E would have to be reduced to maintain the desired ratio E/B in Eq. (27.15).

KEYCONCEPT A charged particle can experience both electric and magnetic forces if both an electric field $\vec{E}$ and a magnetic field $\vec{B}$ are present. These forces can cancel if $\vec{E}$, $\vec{B}$, and the particle's velocity are all mutually perpendicular and the particle travels at one particular speed [Eq. (27.14)].

EXAMPLE 27.6 Finding leaks in a vacuum system**WITH VARIATION PROBLEMS**

There is almost no helium in ordinary air, so helium sprayed near a leak in a vacuum system will quickly show up in the output of a vacuum pump connected to such a system. You are designing a leak detector that uses a mass spectrometer to detect He^+ ions (charge $+e = +1.60 \times 10^{-19}$ C, mass 6.65×10^{-27} kg). Ions emerge from the velocity selector with a speed of 1.00×10^5 m/s. They are curved in a semicircular path by a magnetic field B' and are detected at a distance of 10.16 cm from the slit S_3 in Fig. 27.24. Calculate the magnitude of the magnetic field B' .

IDENTIFY and SET UP After it passes through the slit, the ion follows a circular path as described in Section 27.4 (see Fig. 27.17). We solve Eq. (27.11) for B' .

EXECUTE The distance given is the *diameter* of the semicircular path shown in Fig. 27.24, so the radius is $R = \frac{1}{2}(10.16 \times 10^{-2} \text{ m})$. From Eq. (27.11), $R = mv/qB'$, we get

$$B' = \frac{mv}{qR} = \frac{(6.65 \times 10^{-27} \text{ kg})(1.00 \times 10^5 \text{ m/s})}{(1.60 \times 10^{-19} \text{ C})(5.08 \times 10^{-2} \text{ m})} = 0.0818 \text{ T}$$

EVALUATE Helium leak detectors are widely used with high-vacuum systems. Our result shows that only a small magnetic field is required, so leak detectors can be relatively compact.

KEYCONCEPT The radius of the circular path that a charged particle follows in a uniform magnetic field depends on the particle's charge, mass, and speed. This dependence can be used to separate ions according to their mass.

TEST YOUR UNDERSTANDING OF SECTION 27.5 In Example 27.6 He^+ ions with charge $+e$ move at 1.00×10^5 m/s in a straight line through a velocity selector. Suppose the He^+ ions were replaced with He^{2+} ions, in which both electrons have been removed from the helium atom and the ion charge is $+2e$. At what speed must the He^{2+} ions travel through the same velocity selector in order to move in a straight line? (i) 4.00×10^5 m/s; (ii) 2.00×10^5 m/s; (iii) 1.00×10^5 m/s; (iv) 0.50×10^5 m/s; (v) 0.25×10^5 m/s.

ANSWER is required is that the particles (in this case, ions) have a nonzero charge. selector does not depend on the magnitude or sign of the charge or the mass of the particle. All that

27.6 MAGNETIC FORCE ON A CURRENT-CARRYING CONDUCTOR

What makes an electric motor work? Within the motor are conductors that carry currents (that is, whose charges are in motion), as well as magnets that exert forces on the moving charges. Hence there is a magnetic force on each current-carrying conductor, and these forces make the motor turn. The d'Arsonval galvanometer (Section 26.3) also uses magnetic forces on conductors.

We can compute the force on a current-carrying conductor starting with the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ on a single moving charge. **Figure 27.25** shows a straight segment of a conducting wire, with length l and cross-sectional area A ; the current is from bottom to

Figure 27.25 Forces on a moving positive charge in a current-carrying conductor.

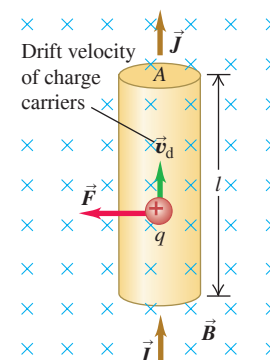


Figure 27.26 A straight wire segment of length $\vec{l}$ carries a current I in the direction of $\vec{l}$. The magnetic force on this segment is perpendicular to both $\vec{l}$ and the magnetic field $\vec{B}$.

Force $\vec{F}$ on a straight wire carrying a positive current and oriented at an angle ϕ to a magnetic field $\vec{B}$:

- Magnitude is $F = I l B_{\perp} = I l B \sin \phi$.
- Direction of $\vec{F}$ is given by the right-hand rule.

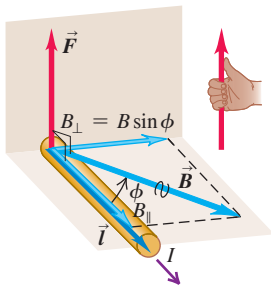
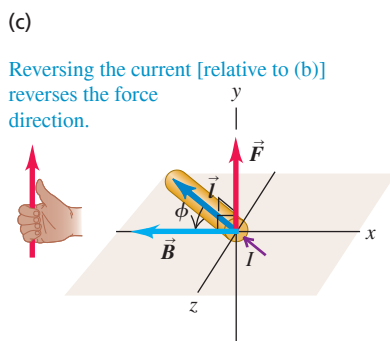
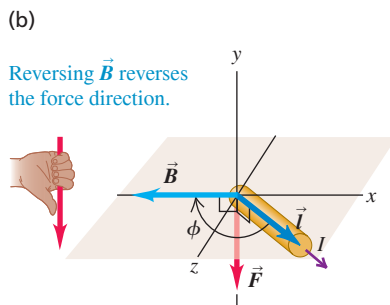
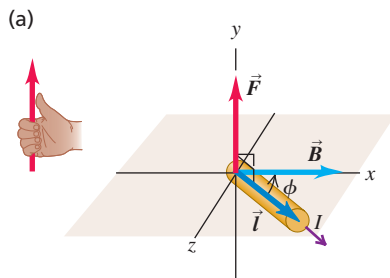


Figure 27.27 Magnetic field $\vec{B}$, length $\vec{l}$, and force $\vec{F}$ vectors for a straight wire carrying a current I .



top. The wire is in a uniform magnetic field $\vec{B}$, perpendicular to the plane of the diagram and directed *into* the plane. Let's assume first that the moving charges are positive. Later we'll see what happens when they are negative.

The drift velocity $\vec{v}_d$ is upward, perpendicular to $\vec{B}$. The average force on each charge is $\vec{F} = q\vec{v}_d \times \vec{B}$, directed to the left as shown in the figure; since $\vec{v}_d$ and $\vec{B}$ are perpendicular, the magnitude of the force is $F = qv_d B$.

We can derive an expression for the *total* force on all the moving charges in a length l of conductor with cross-sectional area A by using the same language we used in Eqs. (25.2) and (25.3) of Section 25.1. The number of charges per unit volume, or charge concentration, is n ; a segment of conductor with length l has volume Al and contains a number of charges equal to nAl . The total force $\vec{F}$ on *all* the moving charges in this segment has magnitude

$$F = (nAl)(qv_d B) = (nqv_d A)(lB) \quad (27.16)$$

From Eq. (25.3) the current density is $J = nqv_d$. The product JA is the total current I , so we can rewrite Eq. (27.16) as

$$F = IlB \quad (27.17)$$

If the $\vec{B}$ field is not perpendicular to the wire but makes an angle ϕ with it, as in Fig. 27.26, we handle the situation the same way we did in Section 27.2 for a single charge. Only the component of $\vec{B}$ perpendicular to the wire (and to the drift velocities of the charges) exerts a force; this component is $B_{\perp} = B \sin \phi$. The magnetic force on the wire segment is then

$$F = IlB_{\perp} = IlB \sin \phi \quad (27.18)$$

The force is always perpendicular to both the conductor and the field, with the direction determined by the same right-hand rule we used for a moving positive charge (Fig. 27.26). Hence this force can be expressed as a vector product, like the force on a single moving charge. We represent the segment of wire with a vector $\vec{l}$ along the wire in the direction of the current; then the force $\vec{F}$ on this segment is

$$\text{Magnetic force on a straight wire segment } \vec{F} = I \vec{l} \times \vec{B} \quad (27.19)$$

Current
Vector length of segment (points in current direction)

Figure 27.27 illustrates the directions of $\vec{B}$, $\vec{l}$, and $\vec{F}$ for several cases.

If the conductor is not straight, we can divide it into infinitesimal segments $d\vec{l}$. The force $d\vec{F}$ on each segment is

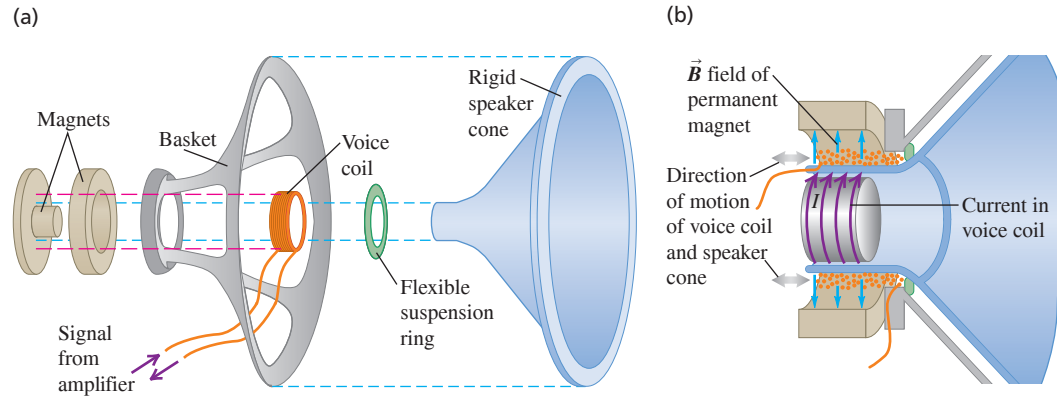
$$\text{Magnetic force on an infinitesimal wire segment } d\vec{F} = I d\vec{l} \times \vec{B} \quad (27.20)$$

Current
Vector length of segment (points in current direction)

Then we can integrate this expression along the wire to find the total force on a conductor of any shape. The integral is a *line integral*, the same mathematical operation we have used to define work (Section 6.3) and electric potential (Section 23.2).

CAUTION **Current is not a vector** Recall from Section 25.1 that the current I is not a vector. The direction of current flow is described by $d\vec{l}$, not I . If the conductor is curved, I is the same at all points along its length, but $d\vec{l}$ changes direction—it is always tangent to the conductor. ■

Figure 27.28 (a) Components of a loudspeaker. (b) The permanent magnet creates a magnetic field that exerts forces on the current in the voice coil; for a current I in the direction shown, the force is to the right. If the electric current in the voice coil oscillates, the speaker cone attached to the voice coil oscillates at the same frequency.



Finally, what happens when the moving charges are negative, such as electrons in a metal? Then in Fig. 27.25 an upward current corresponds to a downward drift velocity. But because q is now negative, the direction of the force $\vec{F}$ is the same as before. Thus Eqs. (27.17) through (27.20) are valid for *both* positive and negative charges and even when *both* signs of charge are present at once. This happens in some semiconductor materials and in ionic solutions.

A common application of the magnetic forces on a current-carrying wire is found in loudspeakers (Fig. 27.28). The radial magnetic field created by the permanent magnet exerts a force on the voice coil that is proportional to the current in the coil; the direction of the force is either to the left or to the right, depending on the direction of the current. The signal from the amplifier causes the current to oscillate in direction and magnitude. The coil and the speaker cone to which it is attached respond by oscillating with an amplitude proportional to the amplitude of the current in the coil. Turning up the volume knob on the amplifier increases the current amplitude and hence the amplitudes of the cone's oscillation and of the sound wave produced by the moving cone.

EXAMPLE 27.7 Magnetic force on a straight conductor

WITH VARIATION PROBLEMS

A straight horizontal copper rod carries a current of 50.0 A from west to east in a region between the poles of a large electromagnet. In this region there is a horizontal magnetic field toward the northeast (that is, 45° north of east) with magnitude 1.20 T. (a) Find the magnitude and direction of the force on a 1.00 m section of rod. (b) While keeping the rod horizontal, how should it be oriented to maximize the magnitude of the force? What is the force magnitude in this case?

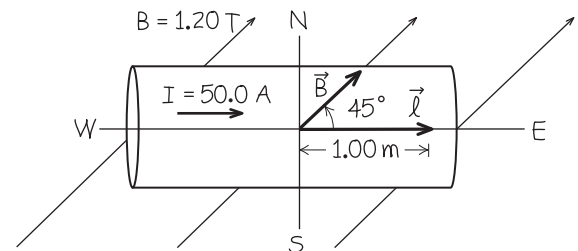
IDENTIFY and SET UP Figure 27.29 shows the situation. This is a straight wire segment in a uniform magnetic field, as in Fig. 27.26. Our target variables are the force $\vec{F}$ on the segment and the angle ϕ for which the force magnitude F is greatest. We find the magnitude of the magnetic force from Eq. (27.18) and the direction from the right-hand rule.

EXECUTE (a) The angle ϕ between the directions of current and field is 45° . From Eq. (27.18) we obtain

$$F = I l B \sin \phi = (50.0 \text{ A})(1.00 \text{ m})(1.20 \text{ T})(\sin 45^\circ) = 42.4 \text{ N}$$

The direction of the force is perpendicular to the plane of the current and the field, both of which lie in the horizontal plane. Thus the force must be vertical; the right-hand rule shows that it is vertically *upward* (out of the plane of the figure).

Figure 27.29 Our sketch of the copper rod as seen from overhead.



(b) From $F = I l B \sin \phi$, F is maximum for $\phi = 90^\circ$, so that $\vec{l}$ and $\vec{B}$ are perpendicular. To keep $\vec{F} = \vec{l} \times \vec{B}$ upward, we rotate the rod *clockwise* by 45° from its orientation in Fig. 27.29, so that the current runs toward the southeast. Then $F = I l B = (50.0 \text{ A})(1.00 \text{ m})(1.20 \text{ T}) = 60.0 \text{ N}$.

EVALUATE We check the result in part (a) by using Eq. (27.19) to calculate the force vector. If we use a coordinate system with the

Continued

x -axis pointing east, the y -axis north, and the z -axis upward, we have $\vec{l} = (1.00 \text{ m})\hat{i}$, $\vec{B} = (1.20 \text{ T})[(\cos 45^\circ)\hat{i} + (\sin 45^\circ)\hat{j}]$, and

$$\begin{aligned}\vec{F} &= \vec{l} \times \vec{B} \\ &= (50.0 \text{ A})(1.00 \text{ m})\hat{i} \times (1.20 \text{ T})[(\cos 45^\circ)\hat{i} + (\sin 45^\circ)\hat{j}] \\ &= (42.4 \text{ N})\hat{k}\end{aligned}$$

Note that the maximum upward force of 60.0 N can hold the conductor in midair against the force of gravity—that is, *magnetically levitate* the conductor—if its weight is 60.0 N and its mass is $m = w/g =$

$(60.0 \text{ N})/(9.8 \text{ m/s}^2) = 6.12 \text{ kg}$. Magnetic levitation is used in some high-speed trains to suspend the train over the tracks. Eliminating rolling friction in this way allows the train to achieve speeds of over 400 km/h.

KEYCONCEPT A straight, current-carrying conductor in a magnetic field $\vec{B}$ experiences a magnetic force that depends on its current, its length, the magnitude of $\vec{B}$, and the angle between the current direction and $\vec{B}$. The force is zero if the directions of the current and $\vec{B}$ are the same or opposite; otherwise, the force is nonzero and in a direction perpendicular to both the current and $\vec{B}$.

EXAMPLE 27.8 Magnetic force on a curved conductor

In Fig. 27.30 the magnetic field $\vec{B}$ is uniform and perpendicular to the plane of the figure, pointing out of the page. The conductor, carrying current I to the left, has three segments: (1) a straight segment with length L perpendicular to the plane of the figure, (2) a semicircle with radius R , and (3) another straight segment with length L parallel to the x -axis. Find the total magnetic force on this conductor.

IDENTIFY and SET UP The magnetic field $\vec{B} = B\hat{k}$ is uniform, so we find the forces $\vec{F}_1$ and $\vec{F}_3$ on the straight segments (1) and (3) from Eq. (27.19). We divide the curved segment (2) into infinitesimal straight segments and find the corresponding force $d\vec{F}_2$ on each straight segment from Eq. (27.20). We then integrate to find $\vec{F}_2$. The total magnetic force on the conductor is then $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$.

EXECUTE For segment (1), $\vec{L} = -L\hat{k}$. Hence from Eq. (27.19), $\vec{F}_1 = \vec{I}\vec{L} \times \vec{B} = \mathbf{0}$. For segment (3), $\vec{L} = -L\hat{i}$, so $\vec{F}_3 = \vec{I}\vec{L} \times \vec{B} = I(-L\hat{i}) \times (B\hat{k}) = ILB\hat{j}$.

For the curved segment (2), Fig. 27.30 shows a segment $d\vec{l}$ with length $dl = R d\theta$, at angle θ . The right-hand rule shows that the

direction of $d\vec{l} \times \vec{B}$ is radially outward from the center; make sure you can verify this. Because $d\vec{l}$ and $\vec{B}$ are perpendicular, the magnitude dF_2 of the force on the segment $d\vec{l}$ is $dF_2 = I dl B = I(R d\theta)B$. The components of the force on this segment are

$$dF_{2x} = IR d\theta B \cos \theta \quad dF_{2y} = IR d\theta B \sin \theta$$

To find the components of the total force, we integrate these expressions with respect to θ from $\theta = 0$ to $\theta = \pi$ to take in the whole semicircle. The results are

$$\begin{aligned}F_{2x} &= IRB \int_0^\pi \cos \theta d\theta = 0 \\ F_{2y} &= IRB \int_0^\pi \sin \theta d\theta = 2IRB\end{aligned}$$

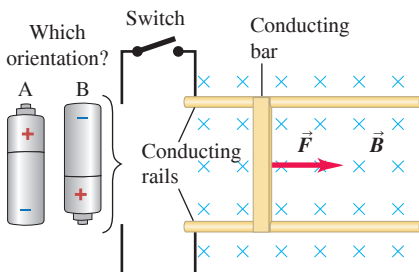
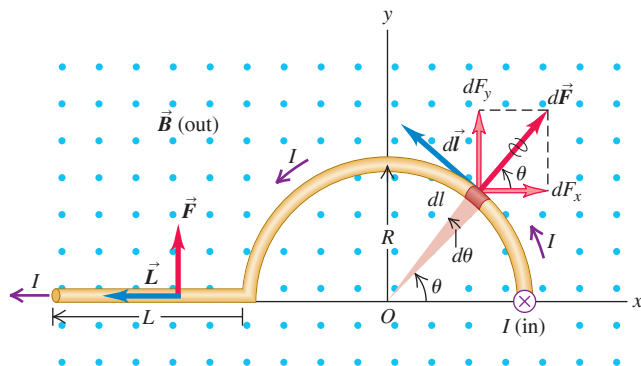
Hence $\vec{F}_2 = 2IRB\hat{j}$. Finally, adding the forces on all three segments, we find that the total force is in the positive y -direction:

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \mathbf{0} + 2IRB\hat{j} + ILB\hat{j} = IB(2R + L)\hat{j}$$

EVALUATE We could have predicted from symmetry that the x -component of $\vec{F}_2$ would be zero: On the right half of the semicircle the x -component of the force is positive (to the right) and on the left half it is negative (to the left); the positive and negative contributions to the integral cancel. The result is that $\vec{F}_2$ is the force that would be exerted if we replaced the semicircle with a *straight* segment of length $2R$ along the x -axis. Do you see why?

KEYCONCEPT To find the magnetic force on a curved, current-carrying conductor in a magnetic field, first divide the conductor into infinitesimally small straight segments. Then find the force on one such segment. Finally, integrate over all segments in the conductor to find the net force.

Figure 27.30 What is the total magnetic force on the conductor?



TEST YOUR UNDERSTANDING OF SECTION 27.6 The accompanying figure shows a top view of two conducting rails on which a conducting bar can slide. A uniform magnetic field is directed perpendicular to the plane of the figure as shown. A battery is to be connected to the two rails so that when the switch is closed, current will flow through the bar and cause a magnetic force to push the bar to the right. In which orientation, A or B, should the battery be placed in the circuit?

ANSWER

A This orientation will cause current to flow clockwise around the circuit and hence through the conducting bar in the direction from the top to the bottom of the figure. From the right-hand rule, the magnetic force $\vec{F} = I\vec{L} \times \vec{B}$ on the bar will then point to the right.

27.7 FORCE AND TORQUE ON A CURRENT LOOP

Current-carrying conductors usually form closed loops, so it is worthwhile to use the results of Section 27.6 to find the *total* magnetic force and torque on a conductor in the form of a loop. Many practical devices make use of the magnetic force or torque on a conducting loop, including loudspeakers (see Fig. 27.28) and galvanometers (see Section 26.3). Hence the results of this section are of substantial practical importance. These results will also help us understand the behavior of bar magnets described in Section 27.1.

As an example, let's look at a rectangular current loop in a uniform magnetic field. We can represent the loop as a series of straight line segments. We'll find that the *total force* on the loop is zero but that there can be a net *torque* acting on the loop, with some interesting properties.

Figure 27.31a shows a rectangular loop of wire with side lengths a and b . A line perpendicular to the plane of the loop (i.e., a *normal* to the plane) makes an angle ϕ with the direction of the magnetic field $\vec{B}$, and the loop carries a current I . The wires leading the current into and out of the loop and the source of emf are omitted to keep the diagram simple.

The force $\vec{F}$ on the right side of the loop (length a) is to the right, in the $+x$ -direction as shown. On this side, $\vec{B}$ is perpendicular to the current direction, and the force on this side has magnitude

$$F = IaB \quad (27.21)$$

A force $-\vec{F}$ with the same magnitude but opposite direction acts on the opposite side of the loop, as shown in the figure.

The sides with length b make an angle $(90^\circ - \phi)$ with the direction of $\vec{B}$. The forces on these sides are the vectors $\vec{F}'$ and $-\vec{F}'$; their magnitude F' is given by

$$F' = IbB \sin(90^\circ - \phi) = IbB \cos \phi$$

The lines of action of both forces lie along the y -axis.

The *total* force on the loop is zero because the forces on opposite sides cancel out in pairs.

The net force on a current loop in a uniform magnetic field is zero. However, the net torque is not in general equal to zero.

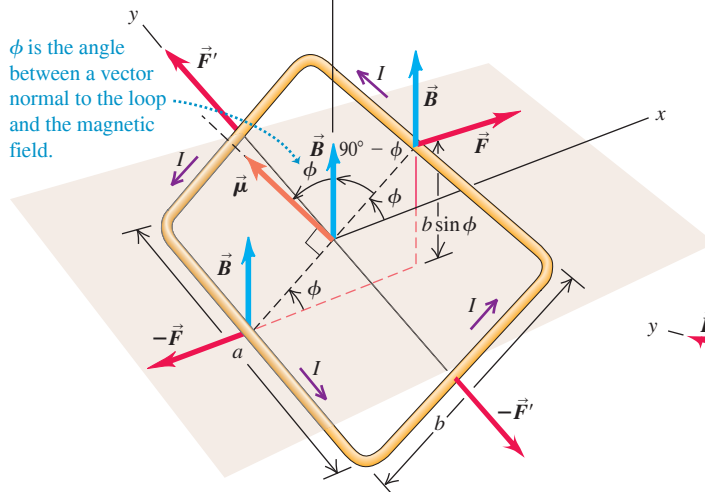
(You may find it helpful to review the discussion of torque in Section 10.1.) The two forces $\vec{F}'$ and $-\vec{F}'$ in Fig. 27.31a lie along the same line and so give rise to zero net torque with

Figure 27.31 Finding the torque on a current-carrying loop in a uniform magnetic field.

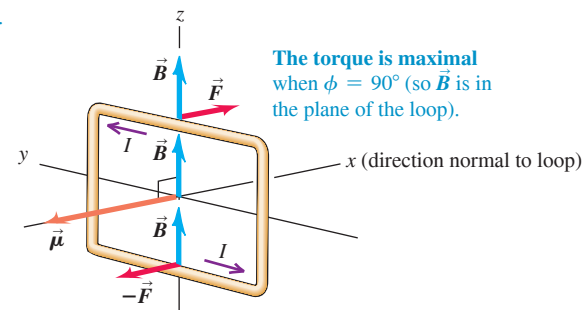
(a)

The two pairs of forces acting on the loop cancel, so no net force acts on the loop.

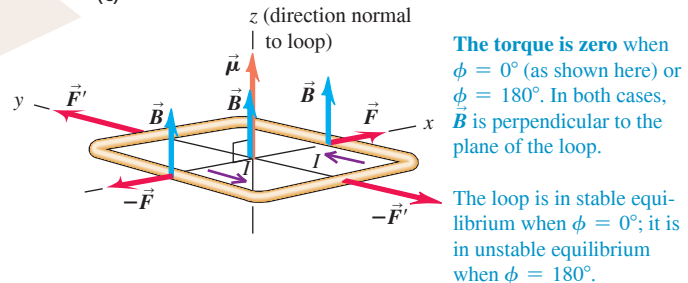
However, the forces on the a sides of the loop ($\vec{F}$ and $-\vec{F}$) produce a torque $\tau = (Ia)(b \sin \phi)$ on the loop.



(b)

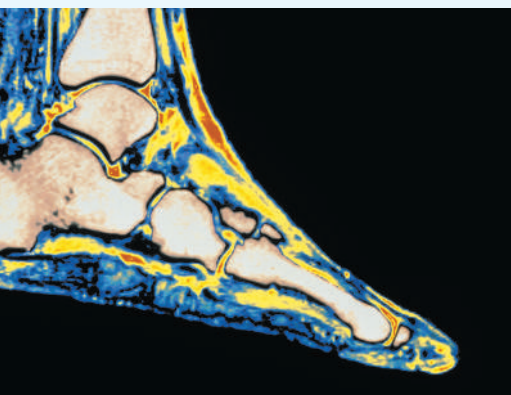


(c)



BIO APPLICATION Magnetic Resonance Imaging

In magnetic resonance imaging (MRI), a patient is placed in a strong magnetic field. Each hydrogen nucleus in the patient acts like a miniature current loop with a magnetic dipole moment that tends to align with the applied field. Radio waves of just the right frequency then flip these magnetic moments out of alignment. The extent to which the radio waves are absorbed is proportional to the amount of hydrogen present. This makes it possible to image details in hydrogen-rich soft tissue that cannot be seen in x-ray images. (X rays are superior to MRI for imaging bone, which is hydrogen deficient.)



respect to any point. The two forces $\vec{F}$ and $-\vec{F}$ lie along different lines, and each gives rise to a torque about the y-axis. According to the right-hand rule for determining the direction of torques, the vector torques due to $\vec{F}$ and $-\vec{F}$ are both in the +y-direction; hence the net vector torque $\vec{\tau}$ is in the +y-direction as well. The moment arm for each of these forces (equal to the perpendicular distance from the rotation axis to the line of action of the force) is $(b/2) \sin \phi$, so the torque due to each force has magnitude $F(b/2) \sin \phi$. If we use Eq. (27.21) for F , the magnitude of the net torque is

$$\tau = 2F(b/2) \sin \phi = (IBa)(b \sin \phi) \quad (27.22)$$

The torque is greatest when $\phi = 90^\circ$, $\vec{B}$ is in the plane of the loop, and the normal to this plane is perpendicular to $\vec{B}$ (Fig. 27.31b). The torque is zero when ϕ is 0° or 180° and the normal to the loop is parallel or antiparallel to the field (Fig. 27.31c). The value $\phi = 0^\circ$ is a stable equilibrium position because the torque is zero there, and when the loop is rotated slightly from this position, the resulting torque tends to rotate it back toward $\phi = 0^\circ$. The position $\phi = 180^\circ$ is an *unstable* equilibrium position; if displaced slightly from this position, the loop tends to move farther away from $\phi = 180^\circ$. Figure 27.31 shows rotation about the y-axis, but because the net force on the loop is zero, Eq. (27.22) for the torque is valid for *any* choice of axis. The torque always tends to rotate the loop in the direction of *decreasing* ϕ —that is, toward the stable equilibrium position $\phi = 0^\circ$.

The area A of the loop is equal to ab , so we can rewrite Eq. (27.22) as

$$\tau = IBA \sin \phi \quad (27.23)$$

Magnitude of magnetic torque on a current loop τ Current Magnetic-field magnitude Angle between normal to loop plane and field direction
Area of loop

The product IA is called the **magnetic dipole moment** or **magnetic moment** of the loop, for which we use the symbol μ (the Greek letter mu):

$$\mu = IA \quad (27.24)$$

It is analogous to the electric dipole moment introduced in Section 21.7. In terms of μ , the magnitude of the torque on a current loop is

$$\tau = \mu B \sin \phi \quad (27.25)$$

where ϕ is the angle between the normal to the loop (the direction of the vector area $\vec{A}$) and $\vec{B}$. A current loop, or any other object that experiences a magnetic torque given by Eq. (27.25), is also called a **magnetic dipole**.

Magnetic Torque: Vector Form

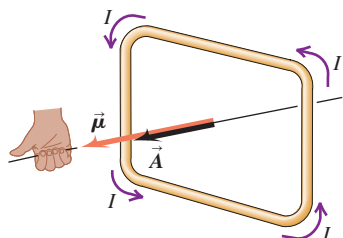
We can also define a vector magnetic moment $\vec{\mu}$ with magnitude IA : This is shown in Fig. 27.31. The direction of $\vec{\mu}$ is defined to be perpendicular to the plane of the loop, with a sense determined by a right-hand rule, as shown in **Fig. 27.32**. Wrap the fingers of your right hand around the perimeter of the loop in the direction of the current. Then extend your thumb so that it is perpendicular to the plane of the loop; its direction is the direction of $\vec{\mu}$ (and of the vector area $\vec{A}$ of the loop). The torque is greatest when $\vec{\mu}$ and $\vec{B}$ are perpendicular and is zero when they are parallel or antiparallel. In the stable equilibrium position, $\vec{\mu}$ and $\vec{B}$ are parallel.

Finally, we can express this interaction in terms of the torque vector $\vec{\tau}$, which we used for *electric*-dipole interactions in Section 21.7. From Eq. (27.25) the magnitude of $\vec{\tau}$ is equal to the magnitude of $\vec{\mu} \times \vec{B}$, and reference to Fig. 27.31 shows that the directions are also the same. So we have

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (27.26)$$

Vector magnetic torque on a current loop $\vec{\tau}$ Magnetic dipole moment Magnetic field

Figure 27.32 The right-hand rule determines the direction of the magnetic moment of a current-carrying loop. This is also the direction of the loop's area vector $\vec{A}$; $\vec{\mu} = I\vec{A}$ is a vector equation.



This result is directly analogous to the result we found in Section 21.7 for the torque exerted by an *electric* field $\vec{E}$ on an *electric* dipole with dipole moment $\vec{p}$.

Potential Energy for a Magnetic Dipole

When a magnetic dipole changes orientation in a magnetic field, the field does work on it. In an infinitesimal angular displacement $d\phi$, the work dW is given by $\tau d\phi$, and there is a corresponding change in potential energy. As the above discussion suggests, the potential energy U is least when $\vec{\mu}$ and $\vec{B}$ are parallel and greatest when they are antiparallel. To find an expression for U as a function of orientation, note that the torque on an *electric* dipole in an *electric* field is $\vec{\tau} = \vec{p} \times \vec{E}$; we found in Section 21.7 that the corresponding potential energy is $U = -\vec{p} \cdot \vec{E}$. The torque on a *magnetic* dipole in a *magnetic* field is $\vec{\tau} = \vec{\mu} \times \vec{B}$, so we can conclude immediately that

$$\text{Potential energy for a magnetic dipole in a magnetic field} \quad U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos \phi \quad (27.27)$$

Magnetic dipole moment
Angle between $\vec{\mu}$ and $\vec{B}$
Magnetic field

With this definition, U is zero when the magnetic dipole moment is perpendicular to the magnetic field ($\phi = 90^\circ$); then $U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos 90^\circ = 0$.

Magnetic Torque: Loops and Coils

Although we have derived Eqs. (27.21) through (27.27) for a rectangular current loop, all these relationships are valid for a plane loop of any shape at all. Any planar loop may be approximated as closely as we wish by a very large number of rectangular loops, as shown in **Fig. 27.33**. If these loops all carry equal currents in the same clockwise sense, then the forces and torques on the sides of two loops adjacent to each other cancel, and the only forces and torques that do not cancel are due to currents around the boundary. Thus all the above relationships are valid for a plane current loop of any shape, with the magnetic moment $\vec{\mu} = I\vec{A}$.

We can also generalize this whole formulation to a coil consisting of N planar loops close together; the effect is simply to multiply each force, the magnetic moment, the torque, and the potential energy by a factor of N .

An arrangement of particular interest is the **solenoid**, a helical winding of wire, such as a coil wound on a circular cylinder (**Fig. 27.34**). If the windings are closely spaced, the solenoid can be approximated by a number of circular loops lying in planes at right angles to its long axis. The total torque on a solenoid in a magnetic field is simply the sum of the torques on the individual turns. For a solenoid with N turns in a uniform field B , the magnetic moment is $\mu = NIA$ and

$$\tau = NIAB \sin \phi \quad (27.28)$$

where ϕ is the angle between the axis of the solenoid and the direction of the field. The magnetic-moment vector $\vec{\mu}$ is along the solenoid axis. The torque is greatest when the solenoid axis is perpendicular to the magnetic field and zero when they are parallel. The effect of this torque is to tend to rotate the solenoid into a position where its axis is parallel to the field. Solenoids are also useful as *sources* of magnetic field, as we'll discuss in Chapter 28.

The d'Arsonval galvanometer, described in Section 26.3, makes use of a magnetic torque on a coil carrying a current. As **Fig. 26.14** shows, the magnetic field is not uniform but is *radial*, so the side thrusts on the coil are always perpendicular to its plane. Thus the angle ϕ in Eq. (27.28) is always 90° , and the magnetic torque is directly proportional to the current, no matter what the orientation of the coil. A restoring torque proportional to the angular displacement of the coil is provided by two hairsprings, which also serve as current leads to the coil. When current is supplied to the coil, it rotates along with its attached pointer until the restoring spring torque just balances the magnetic torque. Thus the pointer deflection is proportional to the current.

Figure 27.33 The collection of rectangles exactly matches the irregular plane loop in the limit as the number of rectangles approaches infinity and the width of each rectangle approaches zero.

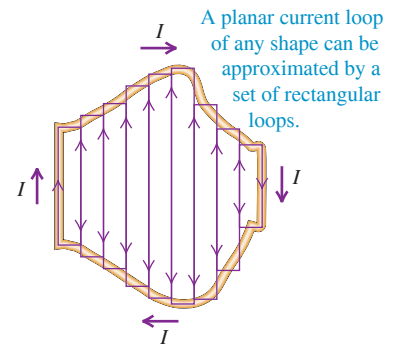
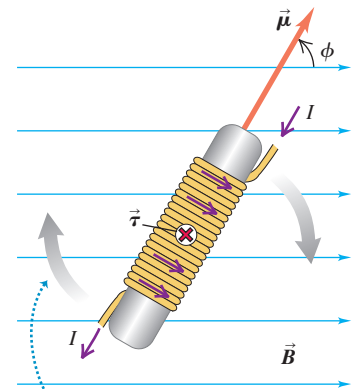


Figure 27.34 The torque $\vec{\tau} = \vec{\mu} \times \vec{B}$ on this solenoid in a uniform magnetic field is directed straight into the page. An actual solenoid has many more turns, wrapped closely together.



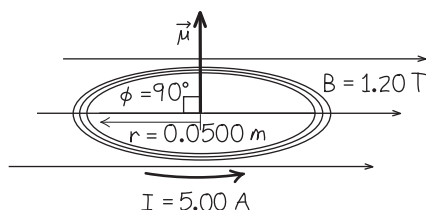
The torque tends to make the solenoid rotate clockwise in the plane of the page, aligning magnetic moment $\vec{\mu}$ with field $\vec{B}$.

EXAMPLE 27.9 Magnetic torque on a circular coil

A circular coil 0.0500 m in radius, with 30 turns of wire, lies in a horizontal plane. It carries a counterclockwise (as viewed from above) current of 5.00 A. The coil is in a uniform 1.20 T magnetic field directed toward the right. Find the magnitudes of the magnetic moment and the torque on the coil.

IDENTIFY and SET UP This problem uses the definition of magnetic moment and the expression for the torque on a magnetic dipole in a magnetic field. **Figure 27.35** shows the situation. Equation (27.24) gives the magnitude μ of the magnetic moment of a single turn of wire; for N turns, the magnetic moment is N times greater. Equation (27.25) gives the magnitude τ of the torque.

Figure 27.35 Our sketch for this problem.



EXECUTE The area of the coil is $A = \pi r^2$. From Eq. (27.24), the total magnetic moment of all 30 turns is

$$\mu_{\text{total}} = NIA = 30(5.00 \text{ A})\pi(0.0500 \text{ m})^2 = 1.18 \text{ A} \cdot \text{m}^2$$

The angle ϕ between the direction of $\vec{B}$ and the direction of $\vec{\mu}$ (which is along the normal to the plane of the coil) is 90° . From Eq. (27.25), the torque on the coil is

$$\begin{aligned}\tau &= \mu_{\text{total}} B \sin \phi = (1.18 \text{ A} \cdot \text{m}^2)(1.20 \text{ T})(\sin 90^\circ) \\ &= 1.41 \text{ N} \cdot \text{m}\end{aligned}$$

EVALUATE The torque tends to rotate the right side of the coil down and the left side up, into a position where the normal to its plane is parallel to $\vec{B}$.

KEYCONCEPT A current-carrying coil in a magnetic field $\vec{B}$ experiences a magnetic torque. The torque tends to orient the coil so its magnetic dipole moment $\vec{\mu}$ (a vector perpendicular to the plane of the coil, with magnitude equal to the product of the number of turns, the current, and the coil area) is in the same direction as $\vec{B}$.

EXAMPLE 27.10 Potential energy for a coil in a magnetic field

If the coil in Example 27.9 rotates from its initial orientation to one in which its magnetic moment $\vec{\mu}$ is parallel to $\vec{B}$, what is the change in potential energy?

IDENTIFY and SET UP Equation (27.27) gives the potential energy for each orientation. The initial position is as shown in Fig. 27.35, with $\phi_1 = 90^\circ$. In the final orientation, the coil has been rotated 90° clockwise so that $\vec{\mu}$ and $\vec{B}$ are parallel, so the angle between these vectors is $\phi_2 = 0$.

EXECUTE From Eq. (27.27), the potential energy change is

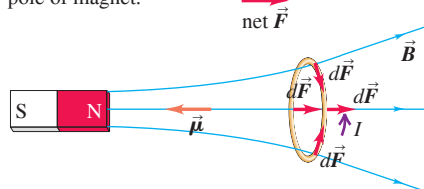
$$\begin{aligned}\Delta U &= U_2 - U_1 = -\mu B \cos \phi_2 - (-\mu B \cos \phi_1) \\ &= -\mu B (\cos \phi_2 - \cos \phi_1) \\ &= -(1.18 \text{ A} \cdot \text{m}^2)(1.20 \text{ T})(\cos 0^\circ - \cos 90^\circ) = -1.41 \text{ J}\end{aligned}$$

EVALUATE The potential energy decreases because the rotation is in the direction of the magnetic torque that we found in Example 27.9.

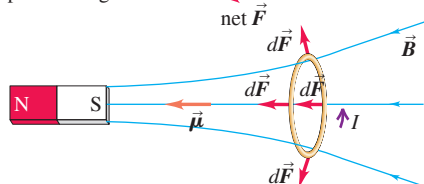
KEYCONCEPT The potential energy of a current-carrying coil in a magnetic field $\vec{B}$ depends on the magnitudes and directions of $\vec{B}$ and of the coil's magnetic dipole moment $\vec{\mu}$. The potential energy is minimum when $\vec{\mu}$ and $\vec{B}$ are in the same direction.

Figure 27.36 Forces on current loops in a nonuniform $\vec{B}$ field. In each case the axis of the bar magnet is perpendicular to the plane of the loop and passes through the center of the loop.

(a) Net force on this coil is away from north pole of magnet.



(b) Net force on same coil is toward south pole of magnet.

**Magnetic Dipole in a Nonuniform Magnetic Field**

We have seen that a current loop (that is, a magnetic dipole) experiences zero net force in a uniform magnetic field. **Figure 27.36** shows two current loops in the *nonuniform* $\vec{B}$ field of a bar magnet; in both cases the net force on the loop is *not* zero. In Fig. 27.36a the magnetic moment $\vec{\mu}$ is in the direction opposite to the field, and the force $d\vec{F} = I d\vec{l} \times \vec{B}$ on a segment of the loop has both a radial component and a component to the right. When these forces are summed to find the net force $\vec{F}$ on the loop, the radial components cancel so that the net force is to the right, away from the magnet. Note that in this case the force is toward the region where the field lines are farther apart and the field magnitude B is less. The polarity of the bar magnet is reversed in Fig. 27.36b, so $\vec{\mu}$ and $\vec{B}$ are parallel; now the net force on the loop is to the left, toward the region of greater field magnitude near the magnet. Later in this section we'll use these observations to explain why bar magnets can pick up unmagnetized iron objects.

Magnetic Dipoles and How Magnets Work

The behavior of a solenoid in a magnetic field (see Fig. 27.34) resembles that of a bar magnet or compass needle; if free to turn, both the solenoid and the magnet orient themselves with their axes parallel to the $\vec{B}$ field. In both cases this is due to the interaction of moving electric charges with a magnetic field; the difference is that in a bar magnet the motion of charge occurs on the microscopic scale of the atom.

Think of an electron as being like a spinning ball of charge. In this analogy the circulation of charge around the spin axis is like a current loop, and so the electron has a net magnetic moment. (This analogy, while helpful, is inexact; an electron isn't really a spinning sphere. A full explanation of the origin of an electron's magnetic moment involves quantum mechanics, which is beyond our scope here.) In an iron atom a substantial fraction of the electron magnetic moments align with each other, and the atom has a nonzero magnetic moment. (By contrast, the atoms of most elements have little or no net magnetic moment.) In an unmagnetized piece of iron there is no overall alignment of the magnetic moments of the atoms; their vector sum is zero, and the net magnetic moment is zero (**Fig. 27.37a**). But in an iron bar magnet the magnetic moments of many of the atoms are parallel, and there is a substantial net magnetic moment $\vec{\mu}$ (**Fig. 27.37b**). If the magnet is placed in a magnetic field $\vec{B}$, the field exerts a torque given by Eq. (27.26) that tends to align $\vec{\mu}$ with $\vec{B}$ (**Fig. 27.37c**). A bar magnet tends to align with a $\vec{B}$ field so that a line from the south pole to the north pole of the magnet is in the direction of $\vec{B}$; hence the real significance of a magnet's north and south poles is that they represent the head and tail, respectively, of the magnet's dipole moment $\vec{\mu}$.

The torque experienced by a current loop in a magnetic field also explains how an unmagnetized iron object like that in Fig. 27.37a becomes magnetized. If an unmagnetized iron paper clip is placed next to a powerful magnet, the magnetic moments of the paper clip's atoms tend to align with the $\vec{B}$ field of the magnet. When the paper clip is removed, its atomic dipoles tend to remain aligned, and the paper clip has a net magnetic moment. The paper clip can be demagnetized by being dropped on the floor or heated; the added internal energy jostles and re-randomizes the atomic dipoles.

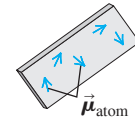
The magnetic-dipole picture of a bar magnet explains the attractive and repulsive forces between bar magnets shown in Fig. 27.1. The magnetic moment $\vec{\mu}$ of a bar magnet points from its south pole to its north pole, so the current loops in Figs. 27.36a and 27.36b are both equivalent to a magnet with its north pole on the left. Hence the situation in Fig. 27.36a is equivalent to two bar magnets with their north poles next to each other; the resultant force is repulsive, as in Fig. 27.1b. In Fig. 27.36b we again have the equivalent of two bar magnets end to end, but with the south pole of the left-hand magnet next to the north pole of the right-hand magnet. The resultant force is attractive, as in Fig. 27.1a.

Finally, we can explain how a magnet can attract an unmagnetized iron object (see Fig. 27.2). It's a two-step process. First, the atomic magnetic moments of the iron tend to align with the $\vec{B}$ field of the magnet, so the iron acquires a net magnetic dipole moment $\vec{\mu}$ parallel to the field. Second, the nonuniform field of the magnet attracts the magnetic dipole. **Figure 27.38a** shows an example. The north pole of the magnet is closer to the nail (which contains iron), and the magnetic dipole produced in the nail is equivalent to a loop with a current that circulates in a direction opposite to that shown in Fig. 27.36a. Hence the net magnetic force on the nail is opposite to the force on the loop in Fig. 27.36a, and the nail is attracted toward the magnet. Changing the polarity of the magnet, as in Fig. 27.38b, reverses the directions of both $\vec{B}$ and $\vec{\mu}$. The situation is now equivalent to that shown in Fig. 27.36b; like the loop in that figure, the nail is attracted toward the magnet. Hence a previously unmagnetized object containing iron is attracted to *either* pole of a magnet. By contrast, objects made of brass, aluminum, or wood hardly respond at all to a magnet; the atomic magnetic dipoles of these materials, if present at all, have less tendency to align with an external field.

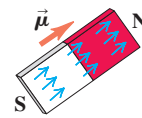
Our discussion of how magnets and pieces of iron interact has just scratched the surface of a diverse subject known as *magnetic properties of materials*. We'll discuss these properties in more depth in Section 28.8.

Figure 27.37 (a) An unmagnetized piece of iron. (Only a few representative atomic moments are shown.) (b) A magnetized piece of iron (bar magnet). The net magnetic moment of the bar magnet points from its south pole to its north pole. (c) A bar magnet in a magnetic field.

(a) Unmagnetized iron; magnetic moments are oriented randomly.



(b) In a bar magnet, the magnetic moments are aligned.



(c) A magnetic field creates a torque on the bar magnet that tends to align its dipole moment with the $\vec{B}$ field.

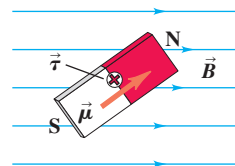
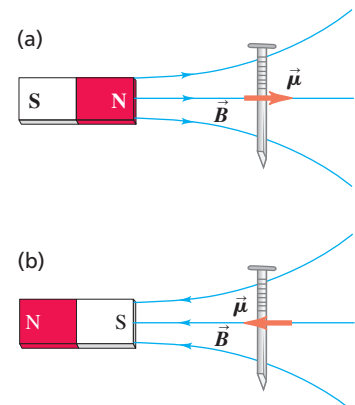


Figure 27.38 A bar magnet attracts an unmagnetized iron nail in two steps. First, the $\vec{B}$ field of the bar magnet gives rise to a net magnetic moment in the nail. Second, because the field of the bar magnet is not uniform, this magnetic dipole is attracted toward the magnet. The attraction is the same whether the nail is closer to (a) the magnet's north pole or (b) the magnet's south pole.



TEST YOUR UNDERSTANDING OF SECTION 27.7 Figure 27.13c depicts the magnetic field lines due to a circular current-carrying loop. (a) What is the direction of the magnetic moment of this loop? (b) Which side of the loop is equivalent to the north pole of a magnet, and which side is equivalent to the south pole?

ANSWER

(a) to the right; (b) North pole on the right, south pole on the left. If you wrap the fingers of your right hand around the coil in the direction of the current, your right thumb points to the right (perpendicular to the plane of the coil). This is the direction of the magnetic moment $\vec{\mu}$. The magnetic moment points from the south pole to the north pole, so the right side of the loop is equivalent to a north pole and the left side is equivalent to a south pole.

27.8 THE DIRECT-CURRENT MOTOR

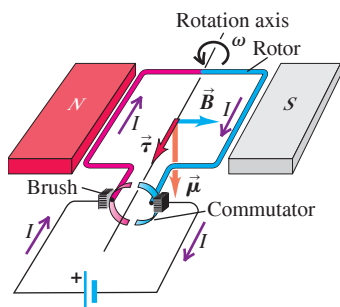
Electric motors play an important role in contemporary society. In a motor a magnetic torque acts on a current-carrying conductor, and electric energy is converted to mechanical energy. As an example, let's look at a simple type of direct-current (dc) motor, shown in **Fig. 27.39**.

The moving part of the motor is the *rotor*, a length of wire formed into an open-ended loop and free to rotate about an axis. The ends of the rotor wires are attached to circular conducting segments that form a *commutator*. In Fig. 27.39a, each of the two commutator segments makes contact with one of the terminals, or *brushes*, of an external circuit that includes a source of emf. This causes a current to flow into the rotor on one side, shown in red, and out of the rotor on the other side, shown in blue. Hence the rotor is a current loop with a magnetic moment $\vec{\mu}$. The rotor lies between opposing poles of a permanent magnet, so there is a magnetic field $\vec{B}$ that exerts a torque $\vec{\tau} = \vec{\mu} \times \vec{B}$ on the rotor. For the rotor orientation shown in Fig. 27.39a the torque causes the rotor to turn counterclockwise, in the direction that will align $\vec{\mu}$ with $\vec{B}$.

In Fig. 27.39b the rotor has rotated by 90° from its orientation in Fig. 27.39a. If the current through the rotor were constant, the rotor would now be in its equilibrium orientation; it would simply oscillate around this orientation. But here's where the commutator comes into play; each brush is now in contact with *both* segments of the commutator. There is no potential difference between the commutators, so at this instant no current flows through the rotor, and the magnetic moment is zero. The rotor continues to rotate counterclockwise because of its inertia, and current again flows through the rotor as in Fig. 27.39c. But now current enters on the *blue* side of the rotor and exits on the *red* side, just the opposite of the situation in Fig. 27.39a. While the direction of the current has reversed with respect

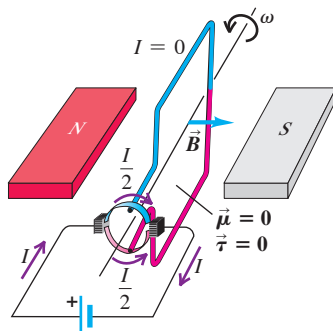
Figure 27.39 Schematic diagram of a simple dc motor. The rotor is a wire loop that is free to rotate about an axis; the rotor ends are attached to the two curved conductors that form the commutator. (The rotor halves are colored red and blue for clarity.) The commutator segments are insulated from one another.

(a) Brushes are aligned with commutator segments.



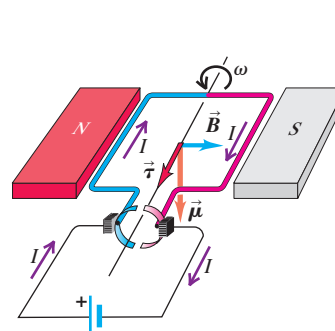
- Current flows into the red side of the rotor and out of the blue side.
- Therefore the magnetic torque causes the rotor to spin counterclockwise.

(b) Rotor has turned 90° .



- Each brush is in contact with both commutator segments, so the current bypasses the rotor altogether.
- No magnetic torque acts on the rotor.

(c) Rotor has turned 180° .



- The brushes are again aligned with commutator segments. This time the current flows into the blue side of the rotor and out of the red side.
- Therefore the magnetic torque again causes the rotor to spin counterclockwise.

to the rotor, the rotor itself has rotated 180° and the magnetic moment $\vec{\mu}$ is in the same direction with respect to the magnetic field. Hence the magnetic torque $\vec{\tau}$ is in the same direction in Fig. 27.39c as in Fig. 27.39a. Thanks to the commutator, the current reverses after every 180° of rotation, so the torque is always in the direction to rotate the rotor counterclockwise. When the motor has come “up to speed,” the average magnetic torque is just balanced by an opposing torque due to air resistance, friction in the rotor bearings, and friction between the commutator and brushes.

The simple motor shown in Fig. 27.39 has only a single turn of wire in its rotor. In practical motors, the rotor has many turns; this increases the magnetic moment and the torque so that the motor can spin larger loads. The torque can also be increased by using a stronger magnetic field, which is why many motor designs use electromagnets instead of a permanent magnet. Another drawback of the simple design in Fig. 27.39 is that the magnitude of the torque rises and falls as the rotor spins. This can be remedied by having the rotor include several independent coils of wire oriented at different angles (Fig. 27.40).

Power for Electric Motors

Because a motor converts electric energy to mechanical energy or work, it requires electric energy input. If the potential difference between its terminals is V_{ab} and the current is I , then the power input is $P = V_{ab}I$. Even if the motor coils have negligible resistance, there must be a potential difference between the terminals if P is to be different from zero. This potential difference results principally from magnetic forces exerted on the currents in the conductors of the rotor as they rotate through the magnetic field. The associated electromotive force $\mathcal{E}$ is called an *induced* emf; it is also called a *back* emf because its sense is opposite to that of the current. In Chapter 29 we'll study induced emfs resulting from motion of conductors in magnetic fields.

In a *series* motor the rotor is connected in series with the electromagnet that produces the magnetic field; in a *shunt* motor they are connected in parallel. In a series motor with internal resistance r , V_{ab} is greater than $\mathcal{E}$, and the difference is the potential drop Ir across the internal resistance. That is,

$$V_{ab} = \mathcal{E} + Ir \quad (27.29)$$

Because the magnetic force is proportional to velocity, $\mathcal{E}$ is *not* constant but is proportional to the speed of rotation of the rotor.

EXAMPLE 27.11 A series dc motor

A dc motor with its rotor and field coils connected in series has an internal resistance of $2.00\ \Omega$. When running at full load on a 120 V line, it draws a 4.00 A current. (a) What is the emf in the rotor? (b) What is the power delivered to the motor? (c) What is the rate of dissipation of energy in the internal resistance? (d) What is the mechanical power developed? (e) What is the motor's efficiency? (f) What happens if the machine being driven by the motor jams, so that the rotor suddenly stops turning?

IDENTIFY and SET UP This problem uses the ideas of power and potential drop in a series dc motor. We are given the internal resistance $r = 2.00\ \Omega$, the voltage $V_{ab} = 120\text{ V}$ across the motor, and the current $I = 4.00\text{ A}$ through the motor. We use Eq. (27.29) to determine the emf $\mathcal{E}$ from these quantities. The power delivered to the motor is $V_{ab}I$, the rate of energy dissipation is I^2r , and the power output by the motor is the difference between the power input and the power dissipated. The efficiency e is the ratio of mechanical power output to electric power input.

EXECUTE (a) From Eq. (27.29), $V_{ab} = \mathcal{E} + Ir$, we have

$$120\text{ V} = \mathcal{E} + (4.00\text{ A})(2.00\ \Omega) \quad \text{and so} \quad \mathcal{E} = 112\text{ V}$$

(b) The power delivered to the motor from the source is

$$P_{\text{input}} = V_{ab}I = (120\text{ V})(4.00\text{ A}) = 480\text{ W}$$

(c) The power dissipated in the resistance r is

$$P_{\text{dissipated}} = I^2r = (4.00\text{ A})^2(2.00\ \Omega) = 32\text{ W}$$

(d) The mechanical power output is the electric power input minus the rate of dissipation of energy in the motor's resistance (assuming that there are no other power losses):

$$P_{\text{output}} = P_{\text{input}} - P_{\text{dissipated}} = 480\text{ W} - 32\text{ W} = 448\text{ W}$$

(e) The efficiency e is the ratio of mechanical power output to electric power input:

$$e = \frac{P_{\text{output}}}{P_{\text{input}}} = \frac{448\text{ W}}{480\text{ W}} = 0.93 = 93\%$$

(f) With the rotor stalled, the back emf $\mathcal{E}$ (which is proportional to rotor speed) goes to zero. From Eq. (27.29) the current becomes

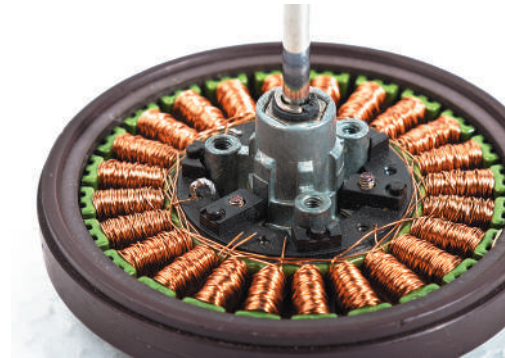
$$I = \frac{V_{ab}}{r} = \frac{120\text{ V}}{2.00\ \Omega} = 60\text{ A}$$

and the power dissipated in the resistance r becomes

$$P_{\text{dissipated}} = I^2r = (60\text{ A})^2(2.00\ \Omega) = 7200\text{ W}$$

Continued

Figure 27.40 This electric motor has multiple current-carrying coils. They interact with permanent magnets on the rotor (not shown) to make the rotor spin around the motor's shaft. (This design is the reverse of the design in Fig. 27.39, in which the permanent magnets are stationary and the coil rotates.) Because there are multiple coils, the magnetic torque is very nearly constant and the rotor spins at a constant rate. Motors of this kind are used in drones, electric cars, and many other applications.



EVALUATE If this massive overload doesn't blow a fuse or trip a circuit breaker, the coils will quickly melt. When the motor is first turned on, there's a momentary surge of current until the motor picks up speed. This surge causes greater-than-usual voltage drops ($V = IR$) in the power lines supplying the current. Similar effects are responsible for the momentary dimming of lights that can occur in a house when an air conditioner or dishwasher motor starts.

KEYCONCEPT When an electric motor is in operation, a large emf is induced in the rotor coils. To find the mechanical power developed by the motor, multiply this emf by the current in the coils.

TEST YOUR UNDERSTANDING OF SECTION 27.8 In the circuit shown in Fig. 27.39, you add a switch in series with the source of emf so that the current can be turned on and off. When you close the switch and allow current to flow, will the rotor begin to turn no matter what its original orientation?

ANSWER

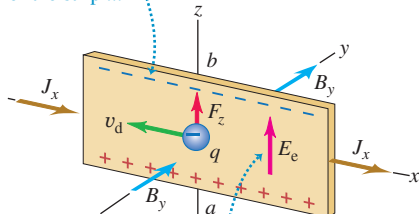
The rotor will not begin to turn when the switch is closed if the rotor is initially oriented as shown in Fig. 27.39b. In this case there is no current through the rotor and hence no magnetic torque. This situation can be remedied by using multiple rotor coils oriented at different angles around the rotation axis. With this arrangement, there is always a magnetic torque no matter what the orientation.

27.9 THE HALL EFFECT

Figure 27.41 Forces on charge carriers in a conductor in a magnetic field.

(a) Negative charge carriers (electrons)

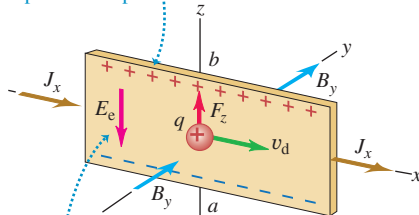
The charge carriers are pushed toward the top of the strip ...



... so point a is at a higher potential than point b.

(b) Positive charge carriers

The charge carriers are again pushed toward the top of the strip ...



... so the polarity of the potential difference is opposite to that for negative charge carriers.

The reality of the forces acting on the moving charges in a conductor in a magnetic field is strikingly demonstrated by the *Hall effect*, an effect analogous to the transverse deflection of an electron beam in a magnetic field in vacuum. (The effect was discovered by the American physicist Edwin Hall in 1879 while he was still a graduate student.) To describe this effect, let's consider a conductor in the form of a flat strip, as shown in **Fig. 27.41**. The current is in the direction of the $+x$ -axis and there is a uniform magnetic field $\vec{B}$ perpendicular to the plane of the strip, in the $+y$ -direction. The drift velocity of the moving charges (charge magnitude $|q|$) has magnitude v_d . Figure 27.41a shows the case of negative charges, such as electrons in a metal, and Fig. 27.41b shows positive charges. In both cases the magnetic force is upward, just as the magnetic force on a conductor is the same whether the moving charges are positive or negative. In either case a moving charge is driven toward the *upper* edge of the strip by the magnetic force $F_z = |q|v_d B$.

If the charge carriers are electrons, as in Fig. 27.41a, an excess negative charge accumulates at the upper edge of the strip, leaving an excess positive charge at its lower edge. This accumulation continues until the resulting transverse electrostatic field $\vec{E}_e$ becomes large enough to cause a force (magnitude $|q|E_e$) that is equal and opposite to the magnetic force (magnitude $|q|v_d B$). After that, there is no longer any net transverse force to deflect the moving charges. This electric field causes a transverse potential difference between opposite edges of the strip, called the *Hall voltage* or the *Hall emf*. The polarity depends on whether the moving charges are positive or negative. Experiments show that for metals the upper edge of the strip in Fig. 27.41a *does* become negatively charged, showing that the charge carriers in a metal are indeed negative electrons.

However, if the charge carriers are *positive*, as in Fig. 27.41b, then *positive* charge accumulates at the upper edge, and the potential difference is *opposite* to the situation with negative charges. Soon after the discovery of the Hall effect in 1879, it was observed that some materials, particularly some *semiconductors*, show a Hall emf opposite to that of the metals, as if their charge carriers were positively charged. We now know that these materials conduct by a process known as *hole conduction*. Within such a material there are locations, called *holes*, that would normally be occupied by an electron but are actually empty. A missing negative charge is equivalent to a positive charge. When an electron moves in one direction to fill a hole, it leaves another hole behind it. The hole migrates in the direction opposite to that of the electron.

In terms of the coordinate axes in Fig. 27.41b, the electrostatic field $\vec{E}_e$ for the positive- q case is in the $-z$ -direction; its z -component E_z is negative. The magnetic field is in the $+y$ -direction, and we write it as B_y . The magnetic force (in the $+z$ -direction) is $qv_d B_y$. The current density J_x is in the $+x$ -direction. In the steady state, when the forces qE_z and $qv_d B_y$ sum to zero,

$$qE_z + qv_d B_y = 0 \quad \text{or} \quad E_z = -v_d B_y$$

This confirms that when q is positive, E_z is negative. From Eq. (25.4),

$$J_x = nqv_d$$

Eliminating v_d between these equations, we find

$$\text{Hall effect: } nq = \frac{-J_x B_y}{E_z} \quad (27.30)$$

Concentration of mobile charge carriers
Current density
Magnetic field
Charge per carrier
Electrostatic field in conductor

Note that this result (as well as the entire derivation) is valid for both positive and negative q . When q is negative, E_z is positive, and conversely.

We can measure J_x , B_y , and E_z , so we can compute the product nq . In both metals and semiconductors, q is equal in magnitude to the electron charge, so the Hall effect permits a direct measurement of n , the concentration of current-carrying charges in the material. The *sign* of the charges is determined by the polarity of the Hall emf, as we have described.

The Hall effect can also be used for a direct measurement of electron drift speed v_d in metals. As we saw in Chapter 25, these speeds are very small, often of the order of 1 mm/s or less. If we move the entire conductor in the opposite direction to the current with a speed equal to the drift speed, then the electrons are at rest with respect to the magnetic field, and the Hall emf disappears. Thus the conductor speed needed to make the Hall emf vanish is equal to the drift speed.

EXAMPLE 27.12 A Hall-effect measurement

You place a strip of copper, 2.0 mm thick and 1.50 cm wide, in a uniform 0.40 T magnetic field as shown in Fig. 27.41a. When you run a 75 A current in the $+x$ -direction, you find that the potential at the bottom of the slab is $0.81 \mu\text{V}$ higher than at the top. From this measurement, determine the concentration of mobile electrons in copper.

IDENTIFY and SET UP This problem describes a Hall-effect experiment. We use Eq. (27.30) to determine the mobile electron concentration n .

EXECUTE First we find the current density J_x and the electric field E_z :

$$J_x = \frac{I}{A} = \frac{75 \text{ A}}{(2.0 \times 10^{-3} \text{ m})(1.50 \times 10^{-2} \text{ m})} = 2.5 \times 10^6 \text{ A/m}^2$$

$$E_z = \frac{V}{d} = \frac{0.81 \times 10^{-6} \text{ V}}{1.5 \times 10^{-2} \text{ m}} = 5.4 \times 10^{-5} \text{ V/m}$$

Then, from Eq. (27.30),

$$n = \frac{-J_x B_y}{qE_z} = \frac{-(2.5 \times 10^6 \text{ A/m}^2)(0.40 \text{ T})}{(-1.60 \times 10^{-19} \text{ C})(5.4 \times 10^{-5} \text{ V/m})} = 11.6 \times 10^{28} \text{ m}^{-3}$$

EVALUATE The actual value of n for copper is $8.5 \times 10^{28} \text{ m}^{-3}$. The difference shows that our simple model of the Hall effect, which ignores quantum effects and electron interactions with the ions, must be used with caution. This example also shows that with good conductors, the Hall emf is very small even with large current densities. In practice, Hall-effect devices for magnetic-field measurements use semiconductor materials, for which moderate current densities give much larger Hall emfs.

KEYCONCEPT In the Hall effect, a current in a conductor flows in a direction perpendicular to a magnetic field $\vec{B}$. The magnetic force on the current produces a charge separation in the conductor. This separation is perpendicular to both the current and $\vec{B}$, and produces a transverse electric field and potential difference in the conductor.

BIO APPLICATION Exercise Machines and the Hall Effect

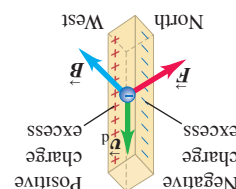
This athlete is making a flywheel spin inside a rowing machine. A gear attached to the flywheel has magnetic teeth, and every time a tooth on the spinning gear passes a current-carrying sensor, the tooth's magnetic field deflects the current and produces a Hall voltage in the sensor. The harder the athlete works, the faster the flywheel spins and the more voltage pulses there are per minute. The voltage information is passed to the rowing machine's display and tells the athlete her power output.



TEST YOUR UNDERSTANDING OF SECTION 27.9 A copper wire of square cross section is oriented vertically. The four sides of the wire face north, south, east, and west. There is a uniform magnetic field directed from east to west, and the wire carries current downward. Which side of the wire is at the highest electric potential? (i) North side; (ii) south side; (iii) east side; (iv) west side.

ANSWER

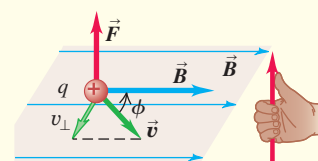
(ii) The mobile charge carriers in copper are negatively charged electrons, which move upward through the wire to give a downward current. From the right-hand rule, the force on a positively charged particle moving upward in a westward-pointing magnetic field would be to the south; hence the force on a negatively charged particle is to the north. The result is an excess of negative charge on the north side of the wire, leaving an excess of positive charge—and hence a higher electric potential—on the south side.



CHAPTER 27 SUMMARY

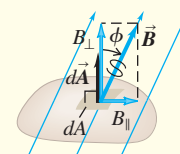
Magnetic forces: Magnetic interactions are fundamentally interactions between moving charged particles. These interactions are described by the vector magnetic field, denoted by $\vec{B}$. A particle with charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ experiences a force $\vec{F}$ that is perpendicular to both $\vec{v}$ and $\vec{B}$. The SI unit of magnetic field is the tesla ($1 \text{ T} = 1 \text{ N/A} \cdot \text{m}$). (See Example 27.1.)

$$\vec{F} = q\vec{v} \times \vec{B} \quad (27.2)$$



Magnetic field lines and flux: A magnetic field can be represented graphically by magnetic field lines. At each point a magnetic field line is tangent to the direction of $\vec{B}$ at that point. Where field lines are close together, the field magnitude is large, and vice versa. Magnetic flux Φ_B through an area is defined in an analogous way to electric flux; its SI unit is the weber ($1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$). The net magnetic flux through any closed surface is zero (Gauss's law for magnetism), so magnetic field lines always close on themselves. (See Example 27.2.)

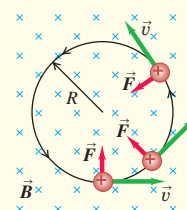
$$\begin{aligned} \Phi_B &= \int B \cos \phi \, dA = \int B_{\perp} \, dA \\ &= \int \vec{B} \cdot d\vec{A} \end{aligned} \quad (27.6)$$



$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{closed surface}) \quad (27.8)$$

Motion in a magnetic field: The magnetic force is always perpendicular to $\vec{v}$; a particle moving under the action of a magnetic field alone moves with constant speed. In a uniform field, a particle with initial velocity perpendicular to the field moves in a circle with radius R that depends on the magnetic field strength B and the particle mass m , speed v , and charge q . (See Examples 27.3 and 27.4.)

$$R = \frac{mv}{|q|B} \quad (27.11)$$

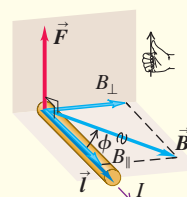


Crossed electric and magnetic fields can be used as a velocity selector. The electric and magnetic forces exactly cancel when $v = E/B$. (See Examples 27.5 and 27.6.)

Magnetic force on a conductor: A straight segment of a conductor carrying current I in a uniform magnetic field $\vec{B}$ experiences a force $\vec{F}$ that is perpendicular to both $\vec{B}$ and the vector $\vec{l}$, which points in the direction of the current and has magnitude equal to the length of the segment. A similar relationship gives the force $d\vec{F}$ on an infinitesimal current-carrying segment $d\vec{l}$. (See Examples 27.7 and 27.8.)

$$\vec{F} = I\vec{l} \times \vec{B} \quad (27.19)$$

$$d\vec{F} = I d\vec{l} \times \vec{B} \quad (27.20)$$

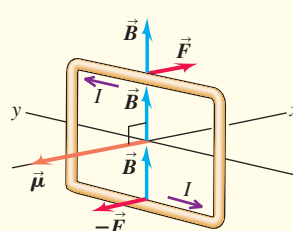


Magnetic torque: A current loop with area A and current I in a uniform magnetic field $\vec{B}$ experiences no net magnetic force, but does experience a magnetic torque of magnitude τ . The vector torque $\vec{\tau}$ can be expressed in terms of the magnetic moment $\vec{\mu} = I\vec{A}$ of the loop, as can the potential energy U of a magnetic moment in a magnetic field $\vec{B}$. The magnetic moment of a loop depends only on the current and the area; it is independent of the shape of the loop. (See Examples 27.9 and 27.10.)

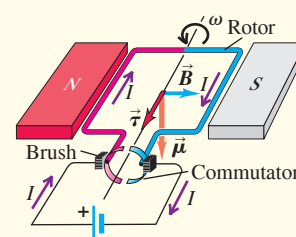
$$\tau = IBA \sin \phi \quad (27.23)$$

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (27.26)$$

$$U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos \phi \quad (27.27)$$

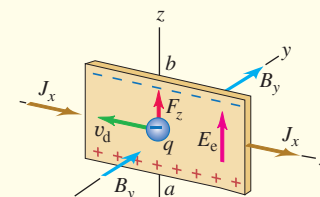


Electric motors: In a dc motor a magnetic field exerts a torque on a current in the rotor. Motion of the rotor through the magnetic field causes an induced emf called a back emf. For a series motor, in which the rotor coil is in series with coils that produce the magnetic field, the terminal voltage is the sum of the back emf and the potential drop Ir across the internal resistance. (See Example 27.11.)



The Hall effect: The Hall effect is a potential difference perpendicular to the direction of current in a conductor, when the conductor is placed in a magnetic field. The Hall potential is determined by the requirement that the associated electric field must just balance the magnetic force on a moving charge. Hall-effect measurements can be used to determine the sign of charge carriers and their concentration n . (See Example 27.12.)

$$nq = \frac{-J_x B_y}{E_z} \quad (27.30)$$



Chapter 27 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 27.1** (Section 27.2) before attempting these problems.

VP27.1.1 An electron (charge -1.60×10^{-19} C) moves at 2.20×10^5 m/s through a uniform 1.55 T magnetic field that points in the $+y$ -direction. The velocity of the electron lies in the xy -plane and is directed at 40.0° to the $+x$ -axis and 50.0° to the $+y$ -axis. Find (a) the magnitude and (b) the direction of the magnetic force on the electron.

VP27.1.2 A charged particle moving in the presence of a 0.750 T magnetic field experiences a magnetic force of magnitude 2.50×10^{-10} N. The particle is moving at 1.95×10^5 m/s at an angle of 36.0° to the direction of the magnetic field. Find the magnitude of the charge on the particle.

VP27.1.3 A particle (charge $+3.20 \times 10^{-19}$ C) with velocity $v_x = 3.75 \times 10^5$ m/s is in the presence of a $\vec{B}$ field with only a z -component. The magnetic force on the particle is 2.10×10^{-15} N in the $-y$ -direction. Find (a) the magnitude and (b) the direction of $\vec{B}$.

VP27.1.4 Highly ionized iron atoms are found in the sun's outer atmosphere, or corona. One such ion has lost 13 electrons and so has charge $+13e = +2.08 \times 10^{-18}$ C. This ion experiences a magnetic force of 2.30×10^{-15} N when it moves at 7.30×10^3 m/s in the xz -plane, 36.9° from the $+x$ -axis and 53.1° from the $+z$ -axis. The magnetic field is in the $+z$ -direction. Find (a) the magnitude of the magnetic field and (b) the direction of the force on the ion.

Be sure to review **EXAMPLES 27.3 and 27.4** (Section 27.4) and **EXAMPLES 27.5 and 27.6** (Section 27.5) before attempting these problems.

VP27.6.1 A typical magnetic field in sunspots (highly magnetized regions on the surface of the sun) is 0.300 T. For a proton (charge $+1.60 \times 10^{-19}$ C, mass 1.67×10^{-27} kg) moving at 1.25×10^4 m/s in a direction perpendicular to such a field, find (a) the radius of its circular orbit, (b) its angular speed in its orbit, and (c) the frequency of its orbital motion.

VP27.6.2 In the situation of the previous problem, suppose the 0.300 T magnetic field is in the $+y$ -direction and the proton's motion is *not* perpendicular to the field: Initially its velocity has components $v_x = 1.00 \times 10^4$ m/s, $v_y = 7.50 \times 10^3$ m/s, $v_z = 0$. Find (a) the radius of the proton's helical path, (b) how far the proton moves along the helix axis per revolution, and (c) the magnitude of the magnetic force on the proton.

VP27.6.3 A velocity selector uses an electric field $\vec{E} = (2.80 \times 10^4 \text{ N/C})\hat{i}$ and a magnetic field $\vec{B} = (0.0350 \text{ T})\hat{k}$. (a) What particle speed will yield zero deflection? (b) In what direction should a charged particle travel through these fields to have zero deflection? (c) A positively charged particle travels through these fields with the speed you found in part (a) and the direction you found in part (b), but the magnetic-field magnitude is now greater than 0.0350 T. In what direction will the particle initially be deflected?

VP27.6.4 You send a beam of oxygen ions through a mass spectrometer like the one shown in Fig. 27.24. The nuclei emerge from the velocity selector at 2.00×10^4 m/s and encounter a magnetic field of magnitude $B' = 0.0500$ T. All the ions have charge $+1.60 \times 10^{-19}$ C, but some are the isotope ^{16}O (mass 2.66×10^{-26} kg) and some are the isotope ^{18}O (mass 2.99×10^{-26} kg). Find the radius of the semicircular path followed by (a) the ^{16}O ions and (b) the ^{18}O ions.

Be sure to review **EXAMPLE 27.7** (Section 27.6) before attempting these problems.

VP27.7.1 A straight wire 0.150 m in length carries a current of 3.50 A in the $+x$ -direction. The wire is in a uniform 0.0136 T magnetic field in the xy -plane that points in a direction 20.0° from the $-x$ -axis and 70.0° from the $+y$ -axis. Find (a) the magnitude and (b) the direction of the magnetic force on the wire.

VP27.7.2 A vertical straight wire 25.0 cm in length carries a current. You do not know either the magnitude of the current or whether the current is moving upward or downward. If there is a uniform horizontal magnetic field of 0.0350 T that points due north, the wire experiences a horizontal magnetic force to the west of 0.0140 N. Find (a) the magnitude and (b) the direction of the current.

VP27.7.3 A straight wire carries a current of 1.20 A. The length vector of the wire is $\vec{l} = (0.200 \text{ m})\hat{i} + (-0.120 \text{ m})\hat{j}$, and the wire is in a uniform magnetic field $\vec{B} = (0.0175 \text{ T})\hat{k}$. Find (a) the x -component, (b) the y -component, (c) the z -component, and (d) the magnitude of the magnetic force on the wire.

VP27.7.4 A straight wire 0.280 m in length carries a current of 3.40 A. What are the *two* angles between the direction of the current and the direction of a uniform 0.0400 T magnetic field for which the magnetic force on the wire has magnitude 0.0250 N?

BRIDGING PROBLEM Magnetic Torque on a Current-Carrying Ring

A circular ring with area 4.45 cm^2 is carrying a current of 12.5 A . The ring, initially at rest, is immersed in a region of uniform magnetic field given by $\vec{B} = (1.15 \times 10^{-2} \text{ T})(12\hat{i} + 3\hat{j} - 4\hat{k})$. The ring is positioned initially such that its magnetic moment is given by $\vec{\mu}_i = \mu(-0.800\hat{i} + 0.600\hat{j})$, where μ is the (positive) magnitude of the magnetic moment. (a) Find the initial magnetic torque on the ring. (b) The ring (which is free to rotate around one diameter) is released and turns through an angle of 90.0° , at which point its magnetic moment is given by $\vec{\mu}_f = -\mu\hat{k}$. Determine the decrease in potential energy. (c) If the moment of inertia of the ring about a diameter is $8.50 \times 10^{-7} \text{ kg} \cdot \text{m}^2$, determine the angular speed of the ring as it passes through the second position.

SOLUTION GUIDE**IDENTIFY and SET UP**

1. The current-carrying ring acts as a magnetic dipole, so you can use the equations for a magnetic dipole in a uniform magnetic field.

2. There are no nonconservative forces acting on the ring as it rotates, so the sum of its rotational kinetic energy (discussed in Section 9.4) and the potential energy is conserved.

EXECUTE

3. Use the vector expression for the torque on a magnetic dipole to find the answer to part (a). (*Hint:* Review Section 1.10.)
4. Find the change in potential energy from the first orientation of the ring to the second orientation.
5. Use your result from step 4 to find the rotational kinetic energy of the ring when it is in the second orientation.
6. Use your result from step 5 to find the ring's angular speed when it is in the second orientation.

EVALUATE

7. If the ring were free to rotate around *any* diameter, in what direction would the magnetic moment point when the ring is in a state of stable equilibrium?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q27.1 Can a charged particle move through a magnetic field without experiencing any force? If so, how? If not, why not?

Q27.2 At any point in space, the electric field $\vec{E}$ is defined to be in the direction of the electric force on a positively charged particle at that point. Why don't we similarly define the magnetic field $\vec{B}$ to be in the direction of the magnetic force on a moving, positively charged particle?

Q27.3 Section 27.2 describes a procedure for finding the direction of the magnetic force using your right hand. If you use the same procedure, but with your left hand, will you get the correct direction for the force? Explain.

Q27.4 The magnetic force on a moving charged particle is always perpendicular to the magnetic field $\vec{B}$. Is the trajectory of a moving charged particle always perpendicular to the magnetic field lines? Explain your reasoning.

Q27.5 A charged particle is fired into a cubical region of space where there is a uniform magnetic field. Outside this region, there is no magnetic field. Is it possible that the particle will remain inside the cubical region? Why or why not?

Q27.6 If the magnetic force does no work on a charged particle, how can it have any effect on the particle's motion? Are there other examples of forces that do no work but have a significant effect on a particle's motion?

Q27.7 A charged particle moves through a region of space with constant velocity (magnitude and direction). If the external magnetic field is zero in this region, can you conclude that the external electric field in the region is also zero? Explain. (By "external" we mean fields other than those produced by the charged particle.) If the external electric field is zero in the region, can you conclude that the external magnetic field in the region is also zero?

Q27.8 How might a loop of wire carrying a current be used as a compass? Could such a compass distinguish between north and south? Why or why not?

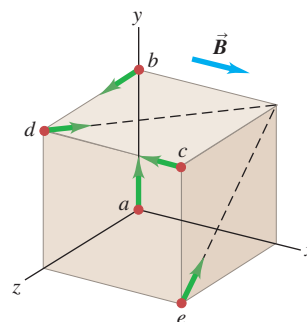
Q27.9 How could the direction of a magnetic field be determined by making only *qualitative* observations of the magnetic force on a straight wire carrying a current?

Q27.10 A loose, floppy loop of wire is carrying current I . The loop of wire is placed on a horizontal table in a uniform magnetic field $\vec{B}$ perpendicular to the plane of the table. This causes the loop of wire to expand into a circular shape while still lying on the table. In a diagram, show all possible orientations of the current I and magnetic field $\vec{B}$ that could cause this to occur. Explain your reasoning.

Q27.11 Several charges enter a uniform magnetic field directed into the page. (a) What path would a positive charge q moving with a velocity of magnitude v follow through the field? (b) What path would a positive charge q moving with a velocity of magnitude $2v$ follow through the field? (c) What path would a negative charge $-q$ moving with a velocity of magnitude v follow through the field? (d) What path would a neutral particle follow through the field?

Q27.12 Each of the lettered points at the corners of the cube in Figure Q27.12

represents a positive charge q moving with a velocity of magnitude v in the direction indicated. The region in the figure is in a uniform magnetic field $\vec{B}$, parallel to the x -axis and directed toward the right. Which charges experience a force due to $\vec{B}$? What is the direction of the force on each charge?



Q27.13 A student claims that if lightning strikes a metal flagpole, the force exerted by the earth's magnetic field on the current in the pole can be large enough to bend it. Typical lightning currents are of the order of 10^4 to 10^5 A. Is the student's opinion justified? Explain your reasoning.

Q27.14 Could an accelerator be built in which *all* the forces on the particles, for steering and for increasing speed, are magnetic forces? Why or why not?

Q27.15 The magnetic force acting on a charged particle can never do work because at every instant the force is perpendicular to the velocity. The torque exerted by a magnetic field can do work on a current loop when the loop rotates. Explain how these seemingly contradictory statements can be reconciled.

Q27.16 When the polarity of the voltage applied to a dc motor is reversed, the direction of motion does *not* reverse. Why not? How *could* the direction of motion be reversed?

Q27.17 In a Hall-effect experiment, is it possible that *no* transverse potential difference will be observed? Under what circumstances might this happen?

Q27.18 Hall-effect voltages are much greater for relatively poor conductors (such as germanium) than for good conductors (such as copper), for comparable currents, fields, and dimensions. Why?

EXERCISES

Section 27.2 Magnetic Field

27.1 • A particle with a charge of -1.24×10^{-8} C is moving with instantaneous velocity $\vec{v} = (4.19 \times 10^4 \text{ m/s})\hat{i} + (-3.85 \times 10^4 \text{ m/s})\hat{j}$. What is the force exerted on this particle by a magnetic field (a) $\vec{B} = (1.40 \text{ T})\hat{i}$ and (b) $\vec{B} = (1.40 \text{ T})\hat{k}$?

27.2 • A particle of mass 0.195 g carries a charge of -2.50×10^{-8} C. The particle is given an initial horizontal velocity that is due north and has magnitude 4.00×10^4 m/s. What are the magnitude and direction of the minimum magnetic field that will keep the particle moving in the earth's gravitational field in the same horizontal, northward direction?

27.3 • In a 1.25 T magnetic field directed vertically upward, a particle having a charge of magnitude $8.50 \mu\text{C}$ and initially moving northward at 4.75 km/s is deflected toward the east. (a) What is the sign of the charge of this particle? Make a sketch to illustrate how you found your answer. (b) Find the magnetic force on the particle.

27.4 • A particle with mass 1.81×10^{-3} kg and a charge of 1.22×10^{-8} C has, at a given instant, a velocity $\vec{v} = (3.00 \times 10^4 \text{ m/s})\hat{j}$. What are the magnitude and direction of the particle's acceleration produced by a uniform magnetic field $\vec{B} = (1.63 \text{ T})\hat{i} + (0.980 \text{ T})\hat{j}$?

27.5 • An electron is moving in the xy -plane. If at time t a magnetic field $B = 0.200$ T in the $+z$ -direction exerts a force on the electron equal to $F = 5.50 \times 10^{-18}$ N in the $-y$ -direction, what is the velocity (magnitude and direction) of the electron at this instant?

27.6 • An electron moves at 1.40×10^6 m/s through a region in which there is a magnetic field of unspecified direction and magnitude 7.40×10^{-2} T. (a) What are the largest and smallest possible magnitudes of the acceleration of the electron due to the magnetic field? (b) If the actual acceleration of the electron is one-fourth of the largest magnitude in part (a), what is the angle between the electron velocity and the magnetic field?

27.7 • A group of particles is traveling in a magnetic field of unknown magnitude and direction. You observe that a proton moving at 1.50 km/s in the $+x$ -direction experiences a force of 2.25×10^{-16} N in the $+y$ -direction, and an electron moving at 4.75 km/s in the $-z$ -direction experiences a force of 8.50×10^{-16} N in the $+y$ -direction. (a) What are the magnitude and direction of the magnetic field? (b) What are the magnitude and direction of the magnetic force on an electron moving in the $-y$ -direction at 3.20 km/s?

27.8 • CP A particle with charge -5.60 nC is moving in a uniform magnetic field $\vec{B} = -(1.25 \text{ T})\hat{k}$. The magnetic force on the particle is measured to be $\vec{F} = -(3.40 \times 10^{-7} \text{ N})\hat{i} + (7.40 \times 10^{-7} \text{ N})\hat{j}$. (a) Calculate all the components of the velocity of the particle that you can from this information. (b) Are there components of the velocity that are not determined by the measurement of the force? Explain. (c) Calculate the scalar product $\vec{v} \cdot \vec{F}$. What is the angle between $\vec{v}$ and $\vec{F}$?

Section 27.3 Magnetic Field Lines and Magnetic Flux

27.9 • A circular area with a radius of 6.50 cm lies in the xy -plane. What is the magnitude of the magnetic flux through this circle due to a uniform magnetic field $B = 0.230$ T (a) in the $+z$ -direction; (b) at an angle of 53.1° from the $+z$ -direction; (c) in the $+y$ -direction?

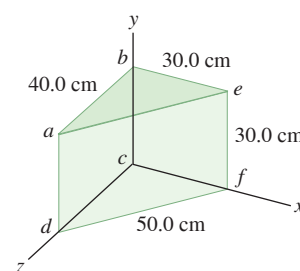
27.10 • A flat, square surface with side length 3.40 cm is in the xy -plane at $z = 0$. Calculate the magnitude of the flux through this surface produced by a magnetic field $\vec{B} = (0.200 \text{ T})\hat{i} + (0.300 \text{ T})\hat{j} - (0.500 \text{ T})\hat{k}$.

27.11 • An open plastic soda bottle with an opening diameter of 2.5 cm is placed on a table. A uniform 1.75 T magnetic field directed upward and oriented 25° from vertical encompasses the bottle. What is the total magnetic flux through the plastic of the soda bottle?

27.12 • A horizontal rectangular surface has dimensions 2.80 cm by 3.20 cm and is in a uniform magnetic field that is directed at an angle of 30.0° above the horizontal. What must the magnitude of the magnetic field be to produce a flux of 3.10×10^{-4} Wb through the surface?

27.13 • The magnetic field $\vec{B}$ in a certain region is 0.128 T, and its direction is that of the $+z$ -axis in **Fig. E27.13**. (a) What is the magnetic flux across the surface $abcd$ in the figure? (b) What is the magnetic flux across the surface $befc$? (c) What is the magnetic flux across the surface $aefd$? (d) What is the net flux through all five surfaces that enclose the shaded volume?

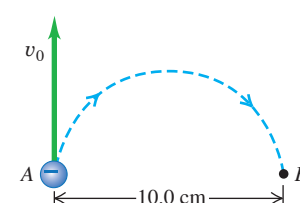
Figure E27.13



Section 27.4 Motion of Charged Particles in a Magnetic Field

27.14 • An electron at point A in **Fig. E27.14** has a speed v_0 of 1.41×10^6 m/s. Find (a) the magnitude and direction of the magnetic field that will cause the electron to follow the semicircular path from A to B, and (b) the time required for the electron to move from A to B.

Figure E27.14



27.15 • Repeat Exercise 27.14 for the case in which the particle is a proton rather than an electron.

27.16 • An alpha particle (a He nucleus, containing two protons and two neutrons and having a mass of 6.64×10^{-27} kg) traveling horizontally at 35.6 km/s enters a uniform, vertical, 1.80 T magnetic field. (a) What is the diameter of the path followed by this alpha particle? (b) What effect does the magnetic field have on the speed of the particle? (c) What are the magnitude and direction of the acceleration of the alpha particle while it is in the magnetic field? (d) Explain why the speed of the particle does not change even though an unbalanced external force acts on it.

27.17 • CP A 150 g ball containing 4.00×10^8 excess electrons is dropped into a 125 m vertical shaft. At the bottom of the shaft, the ball suddenly enters a uniform horizontal magnetic field that has magnitude 0.250 T and direction from east to west. If air resistance is negligibly small, find the magnitude and direction of the force that this magnetic field exerts on the ball just as it enters the field.

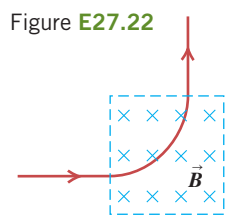
27.18 • BIO Cyclotrons are widely used in nuclear medicine for producing short-lived radioactive isotopes. These cyclotrons typically accelerate H^- (the *hydride* ion, which has one proton and two electrons) to an energy of 5 MeV to 20 MeV. This ion has a mass very close to that of a proton because the electron mass is negligible—about $\frac{1}{2000}$ of the proton's mass. A typical magnetic field in such cyclotrons is 1.9 T. (a) What is the speed of a 5.0 MeV H^- ? (b) If the H^- has energy 5.0 MeV and $B = 1.9$ T, what is the radius of this ion's circular orbit?

27.19 • In an experiment with cosmic rays, a vertical beam of particles that have charge of magnitude $3e$ and mass 12 times the proton mass enters a uniform horizontal magnetic field of 0.250 T and is bent in a semicircle of diameter 95.0 cm, as shown in **Fig. E27.19**. (a) Find the speed of the particles and the sign of their charge. (b) Is it reasonable to ignore the gravity force on the particles? (c) How does the speed of the particles as they enter the field compare to their speed as they exit the field?

27.20 • A deuteron (the nucleus of an isotope of hydrogen) has a mass of 3.34×10^{-27} kg and a charge of $+e$. The deuteron travels in a circular path with a radius of 6.96 mm in a magnetic field with magnitude 2.50 T. (a) Find the speed of the deuteron. (b) Find the time required for it to make half a revolution. (c) Through what potential difference would the deuteron have to be accelerated to acquire this speed?

27.21 • An electron in the beam of a cathode-ray tube is accelerated by a potential difference of 2.00 kV. Then it passes through a region of transverse magnetic field, where it moves in a circular arc with radius 0.180 m. What is the magnitude of the field?

27.22 •• A beam of protons traveling at 1.20 km/s enters a uniform magnetic field, traveling perpendicular to the field. The beam exits the magnetic field, leaving the field in a direction perpendicular to its original direction (**Fig. E27.22**). The beam travels a distance of 1.18 cm while in the field. What is the magnitude of the magnetic field?



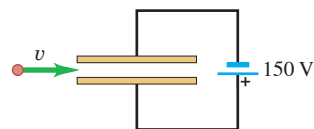
Section 27.5 Applications of Motion of Charged Particles

27.23 • Crossed $\vec{E}$ and $\vec{B}$ Fields. A particle with initial velocity $\vec{v}_0 = (5.85 \times 10^3 \text{ m/s})\hat{j}$ enters a region of uniform electric and magnetic fields. The magnetic field in the region is $\vec{B} = -(1.35 \text{ T})\hat{k}$. Calculate the magnitude and direction of the electric field in the region if the particle is to pass through undeflected, for a particle of charge (a) $+0.640$ nC and (b) -0.320 nC. You can ignore the weight of the particle.

27.24 • (a) What is the speed of a beam of electrons when the simultaneous influence of an electric field of 1.56×10^4 V/m and a magnetic field of 4.62×10^{-3} T, with both fields normal to the beam and to each other, produces no deflection of the electrons? (b) In a diagram, show the relative orientation of the vectors $\vec{v}$, $\vec{E}$, and $\vec{B}$. (c) When the electric field is removed, what is the radius of the electron orbit? What is the period of the orbit?

27.25 •• A 150 V battery is connected across two parallel metal plates of area 28.5 cm^2 and separation 8.20 mm. A beam of alpha particles (charge $+2e$, mass 6.64×10^{-27} kg) is accelerated from rest through a potential difference of 1.75 kV and enters the

Figure E27.25



region between the plates perpendicular to the electric field, as shown in **Fig. E27.25**. What magnitude and direction of magnetic field are needed so that the alpha particles emerge undeflected from between the plates?

27.26 • In the Bainbridge mass spectrometer (see **Fig. 27.22**), the magnetic-field magnitude in the velocity selector is 0.510 T, and ions having a speed of 1.82×10^6 m/s pass through undeflected. (a) What is the electric-field magnitude in the velocity selector? (b) If the separation of the plates is 5.20 mm, what is the potential difference between the plates?

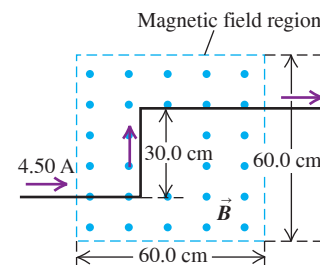
27.27 • Singly ionized (one electron removed) atoms are accelerated and then passed through a velocity selector consisting of perpendicular electric and magnetic fields. The electric field is 155 V/m and the magnetic field is 0.0315 T. The ions next enter a uniform magnetic field of magnitude 0.0175 T that is oriented perpendicular to their velocity. (a) How fast are the ions moving when they emerge from the velocity selector? (b) If the radius of the path of the ions in the second magnetic field is 17.5 cm, what is their mass?

27.28 •• BIO Ancient Meat Eating. The amount of meat in prehistoric diets can be determined by measuring the ratio of the isotopes ^{15}N to ^{14}N in bone from human remains. Carnivores concentrate ^{15}N , so this ratio tells archaeologists how much meat was consumed. For a mass spectrometer that has a path radius of 12.5 cm for ^{12}C ions (mass 1.99×10^{-26} kg), find the separation of the ^{14}N (mass 2.32×10^{-26} kg) and ^{15}N (mass 2.49×10^{-26} kg) isotopes at the detector.

Section 27.6 Magnetic Force on a Current-Carrying Conductor

27.29 •• A long wire carrying 4.50 A of current makes two 90° bends, as shown in **Fig. E27.29**. The bent part of the wire passes through a uniform 0.240 T magnetic field directed as shown in the figure and confined to a limited region of space. Find the magnitude and direction of the force that the magnetic field exerts on the wire.

Figure E27.29

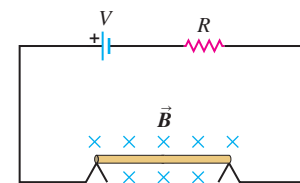


27.30 • A straight, 2.5 m wire carries a typical household current of 1.5 A (in one direction) at a location where the earth's magnetic field is 0.55 gauss from south to north. Find the magnitude and direction of the force that our planet's magnetic field exerts on this wire if it is oriented so that the current in it is running (a) from west to east,

(b) vertically upward, (c) from north to south. (d) Is the magnetic force ever large enough to cause significant effects under normal household conditions?

27.31 • A thin, 50.0-cm-long metal bar with mass 750 g rests on, but is not attached to, two metallic supports in a uniform 0.450 T magnetic field, as shown in **Fig. E27.31**. A battery and a 25.0Ω resistor in series are connected to the supports. (a) What is the highest voltage the battery can have without breaking the circuit at the supports?

Figure E27.31

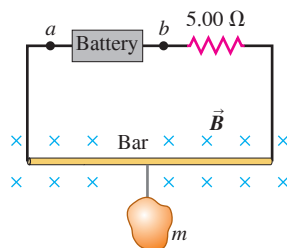


(b) The battery voltage has the maximum value calculated in part (a). If the resistor suddenly gets partially short-circuited, decreasing its resistance to 2.0Ω , find the initial acceleration of the bar.

27.32 •• An electromagnet produces a magnetic field of 0.550 T in a cylindrical region of radius 2.50 cm between its poles. A straight wire carrying a current of 10.8 A passes through the center of this region and is perpendicular to both the axis of the cylindrical region and the magnetic field. What magnitude of force does this field exert on the wire?

27.33 • Magnetic Balance. The circuit shown in **Fig. E27.33** is used to make a magnetic balance to weigh objects. The mass m to be measured is hung from the center of the bar that is in a uniform magnetic field of 1.50 T, directed into the plane of the figure. The battery voltage can be adjusted to vary the current in the circuit. The horizontal bar is 60.0 cm long and is made of extremely lightweight material. It is connected to the battery by thin vertical wires that can support no appreciable tension; all the weight of the suspended mass m is supported by the magnetic force on the bar. A resistor with $R = 5.00\ \Omega$ is in series with the bar; the resistance of the rest of the circuit is much less than this. (a) Which point, a or b , should be the positive terminal of the battery? (b) If the maximum terminal voltage of the battery is 175 V, what is the greatest mass m that this instrument can measure?

Figure E27.33



27.34 • A straight, vertical wire carries a current of 2.60 A downward in a region between the poles of a large superconducting electro-magnet, where the magnetic field has magnitude $B = 0.588\text{ T}$ and is horizontal. What are the magnitude and direction of the magnetic force on a 1.00 cm section of the wire that is in this uniform magnetic field, if the magnetic field direction is (a) east; (b) south; (c) 30.0° south of west?

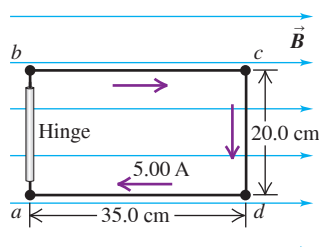
Section 27.7 Force and Torque on a Current Loop

27.35 • A flat circular coil carrying a current of 8.80 A has a magnetic dipole moment of $0.194\text{ A} \cdot \text{m}^2$ to the left. Its area vector $\vec{A}$ is 4.0 cm^2 to the left. (a) How many turns does the coil have? (b) An observer is on the coil's axis to the left of the coil and is looking toward the coil. Does the observer see a clockwise or counterclockwise current? (c) If a huge 45.0 T external magnetic field directed out of the paper is applied to the coil, what torque (magnitude and direction) results?

27.36 •• The plane of a $5.0\text{ cm} \times 8.0\text{ cm}$ rectangular loop of wire is parallel to a 0.19 T magnetic field. The loop carries a current of 6.2 A. (a) What torque acts on the loop? (b) What is the magnetic moment of the loop? (c) What is the maximum torque that can be obtained with the same total length of wire carrying the same current in this magnetic field?

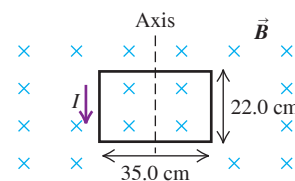
27.37 • The $20.0\text{ cm} \times 35.0\text{ cm}$ rectangular circuit shown in **Fig. E27.37** is hinged along side ab . It carries a clockwise 5.00 A current and is located in a uniform 1.20 T magnetic field oriented perpendicular to two of its sides, as shown. (a) Draw a clear diagram showing the direction of the force that the magnetic field exerts on each segment of the circuit (ab , bc , etc.). (b) Of the four forces you drew in part (a), decide which ones exert a torque about the hinge ab . Then calculate only those forces that exert this torque. (c) Use your results from part (b) to calculate the torque that the magnetic field exerts on the circuit about the hinge axis ab .

Figure E27.37



27.38 • A rectangular coil of wire, 22.0 cm by 35.0 cm and carrying a current of 1.95 A, is oriented with the plane of its loop perpendicular to a uniform 1.50 T magnetic field (**Fig. E27.38**). (a) Calculate the net force and torque that the magnetic field exerts on the coil. (b) The coil is rotated through a 30.0° angle about

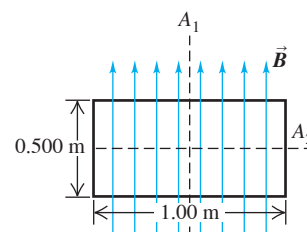
Figure E27.38



the axis shown, with the left side coming out of the plane of the figure and the right side going into the plane. Calculate the net force and torque that the magnetic field now exerts on the coil. (*Hint:* To visualize this three-dimensional problem, make a careful drawing of the coil as viewed along the rotation axis.)

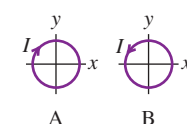
27.39 • CP A uniform rectangular coil of total mass 212 g and dimensions $0.500\text{ m} \times 1.00\text{ m}$ is oriented with its plane parallel to a uniform 3.00 T magnetic field (**Fig. E27.39**). A current of 2.00 A is suddenly started in the coil. (a) About which axis (A_1 or A_2) will the coil begin to rotate? Why? (b) Find the initial angular acceleration of the coil just after the current is started.

Figure E27.39

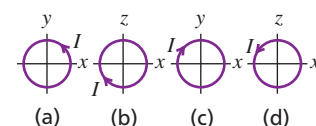


27.40 • Both circular coils A and B (**Fig. E27.40**) have area A and N turns. They are free to rotate about a diameter that coincides with the x -axis. Current I circulates in each coil in the direction shown. There is a uniform magnetic field $\vec{B}$ in the $+z$ -direction. (a) What is the direction of the magnetic moment $\vec{\mu}$ for each coil? (b) Explain why the torque on both coils due to the magnetic field is zero, so the coil is in rotational equilibrium. (c) Use Eq. (27.27) to calculate the potential energy for each coil. (d) For each coil, is the equilibrium stable or unstable? Explain.

Figure E27.40



27.41 • A circular coil with area A and N turns is free to rotate about a diameter that coincides with the x -axis. Current I is circulating in the coil. There is a uniform magnetic field $\vec{B}$ in the positive y -direction. Calculate the magnitude and direction of the torque $\vec{\tau}$ and the value of the potential energy U , as given in Eq. (27.27), when the coil is oriented as shown in parts (a) through (d) of **Fig. E27.41**.

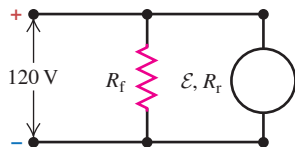


27.42 •• A coil with magnetic moment $1.45\text{ A} \cdot \text{m}^2$ is oriented initially with its magnetic moment antiparallel to a uniform 0.835 T magnetic field. What is the change in potential energy of the coil when it is rotated 180° so that its magnetic moment is parallel to the field?

Section 27.8 The Direct-Current Motor

27.43 •• In a shunt-wound dc motor with the field coils and rotor connected in parallel (**Fig. E27.43**), the resistance R_f of the field coils is $106\ \Omega$, and the resistance R_r of the rotor is $5.9\ \Omega$. When a potential difference of 120 V is applied to the brushes and the motor is running at full speed delivering mechanical power, the current supplied to it is 4.82 A . (a) What is the current in the field coils? (b) What is the current in the rotor? (c) What is the induced emf developed by the motor? (d) How much mechanical power is developed by this motor?

Figure E27.43

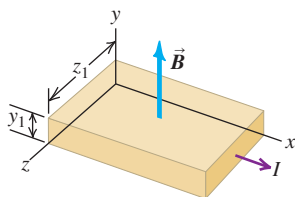


27.44 • A dc motor with its rotor and field coils connected in series has an internal resistance of $3.2\ \Omega$. When the motor is running at full load on a 120 V line, the emf in the rotor is 105 V . (a) What is the current drawn by the motor from the line? (b) What is the power delivered to the motor? (c) What is the mechanical power developed by the motor?

Section 27.9 The Hall Effect

27.45 • **Figure E27.45** shows a portion of a silver ribbon with $z_1 = 11.8\text{ mm}$ and $y_1 = 0.23\text{ mm}$, carrying a current of 120 A in the $+x$ -direction. The ribbon lies in a uniform magnetic field, in the y -direction, with magnitude 0.95 T . Apply the simplified model of the Hall effect presented in Section 27.9. If there are 5.85×10^{28} free electrons per cubic meter, find (a) the magnitude of the drift velocity of the electrons in the x -direction; (b) the magnitude and direction of the electric field in the z -direction due to the Hall effect; (c) the Hall emf.

Figure E27.45



27.46 • Let **Fig. E27.45** represent a strip of an unknown metal of the same dimensions as those of the silver ribbon in Exercise 27.45. When the magnetic field is 2.29 T and the current is 78.0 A , the Hall emf is found to be $131\ \mu\text{V}$. What does the simplified model of the Hall effect presented in Section 27.9 give for the density of free electrons in the unknown metal?

PROBLEMS

27.47 •• CP A small particle with positive charge $q = +3.75 \times 10^{-4}\text{ C}$ and mass $m = 5.00 \times 10^{-5}\text{ kg}$ is moving in a region of uniform electric and magnetic fields. The magnetic field is $B = 4.00\text{ T}$ in the $+z$ -direction. The electric field is also in the $+z$ -direction and has magnitude $E = 60.0\text{ N/C}$. At time $t = 0$ the particle is on the y -axis at $y = +1.00\text{ m}$ and has velocity $v = 30.0\text{ m/s}$ in the $+x$ -direction. Neglect gravity. (a) What are the x -, y -, and z -coordinates of the particle at $t = 0.0200\text{ s}$? (b) What is the speed of the particle at $t = 0.0200\text{ s}$?

27.48 • A particle with charge $7.26 \times 10^{-8}\text{ C}$ is moving in a region where there is a uniform 0.650 T magnetic field in the $+x$ -direction. At a particular instant, the velocity of the particle has components $v_x = -1.68 \times 10^4\text{ m/s}$, $v_y = -3.11 \times 10^4\text{ m/s}$, and $v_z = 5.85 \times 10^4\text{ m/s}$. What are the components of the force on the particle at this time?

27.49 •• CP Fusion Reactor. If two deuterium nuclei (charge $+e$, mass $3.34 \times 10^{-27}\text{ kg}$) get close enough together, the attraction of the strong nuclear force will fuse them to make an isotope of helium, releasing vast amounts of energy. The range of this force is about 10^{-15} m . This is the principle behind the fusion reactor. The deuterium nuclei are moving much too fast to be contained by physical walls, so they are confined magnetically. (a) How fast would two nuclei have to move so that in a head-on collision they would get close enough to fuse? (Assume their speeds are equal. Treat the nuclei as point charges, and assume that a separation of $1.0 \times 10^{-15}\text{ m}$ is required for fusion.) (b) What strength magnetic field is needed to make deuterium nuclei with this speed travel in a circle of diameter 2.50 m ?

27.50 •• Magnetic Moment of the Hydrogen Atom. In the Bohr model of the hydrogen atom (see Section 39.3), in the lowest energy state the electron orbits the proton at a speed of $2.2 \times 10^6\text{ m/s}$ in a circular orbit of radius $5.3 \times 10^{-11}\text{ m}$. (a) What is the orbital period of the electron? (b) If the orbiting electron is considered to be a current loop, what is the current I ? (c) What is the magnetic moment of the atom due to the motion of the electron?

27.51 •• You measure the charge-to-mass ratio q/m for a particle with positive charge in the following way: The particle starts from rest, is accelerated through a potential difference ΔV , and attains a velocity with magnitude v . It then enters a region of uniform magnetic field $B = 0.200\text{ T}$ that is directed perpendicular to the velocity; the particle moves in a path that is an arc of a circle of radius R . You measure R as a function of ΔV . You plot your data as R^2 (in units of m^2) versus ΔV (in V) and find that the values lie close to a straight line that has slope $1.04 \times 10^{-6}\text{ m}^2/\text{V}$. What is the value of q/m for this particle?

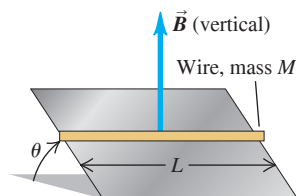
27.52 • The magnetic poles of a small cyclotron produce a magnetic field with magnitude 0.85 T . The poles have a radius of 0.40 m , which is the maximum radius of the orbits of the accelerated particles. (a) What is the maximum energy to which protons ($q = 1.60 \times 10^{-19}\text{ C}$, $m = 1.67 \times 10^{-27}\text{ kg}$) can be accelerated by this cyclotron? Give your answer in electron volts and in joules. (b) What is the time for one revolution of a proton orbiting at this maximum radius? (c) What would the magnetic-field magnitude have to be for the maximum energy to which a proton can be accelerated to be twice that calculated in part (a)? (d) For $B = 0.85\text{ T}$, what is the maximum energy to which alpha particles ($q = 3.20 \times 10^{-19}\text{ C}$, $m = 6.64 \times 10^{-27}\text{ kg}$) can be accelerated by this cyclotron? How does this compare to the maximum energy for protons?

27.53 •• Suppose the electric field between the plates in **Fig. 27.22** is $1.88 \times 10^4\text{ V/m}$ and the magnetic field in both regions is 0.682 T . If the source contains the three isotopes of krypton, ^{82}Kr , ^{84}Kr , and ^{86}Kr , and the ions are singly charged, find the distance between the lines formed by the three isotopes on the particle detector. Assume the atomic masses of the isotopes (in atomic mass units) are equal to their mass numbers, 82, 84, and 86. (One atomic mass unit $= 1\text{ u} = 1.66 \times 10^{-27}\text{ kg}$.)

27.54 •• Mass Spectrograph. A mass spectrograph is used to measure the masses of ions, or to separate ions of different masses (see Section 27.5). In one design for such an instrument, ions with mass m and charge q are accelerated through a potential difference V . They then enter a uniform magnetic field that is perpendicular to their velocity, and they are deflected in a semicircular path of radius R . A detector measures where the ions complete the semicircle and from this it is easy to calculate R . (a) Derive the equation for calculating the mass of the ion from measurements of B , V , R , and q . (b) What potential difference V is needed so that singly ionized ^{12}C atoms will have $R = 50.0\text{ cm}$ in a 0.150 T magnetic field? (c) Suppose the beam consists of a mixture of ^{12}C and ^{14}C ions. If v and B have the same values as in part (b), calculate the separation of these two isotopes at the detector. Do you think that this beam separation is sufficient for the two ions to be distinguished? (Make the assumption described in Problem 27.53 for the masses of the ions.)

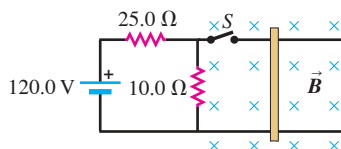
27.55 •• A straight piece of conducting wire with mass M and length L is placed on a frictionless incline tilted at an angle θ from the horizontal (**Fig. P27.55**). There is a uniform, vertical magnetic field $\vec{B}$ at all points (produced by an arrangement of magnets not shown in the figure). To keep the wire from sliding down the incline, a voltage source is attached to the ends of the wire. When just the right amount of current flows through the wire, the wire remains at rest. Determine the magnitude and direction of the current in the wire that will cause the wire to remain at rest. Copy the figure and draw the direction of the current on your copy. In addition, show in a free-body diagram all the forces that act on the wire.

Figure P27.55



27.56 •• CP A 2.60 N metal bar, 0.850 m long and having a resistance of $10.0 \, \Omega$, rests horizontally on conducting wires connecting it to the circuit shown in **Fig. P27.56**. The bar is in a uniform, horizontal, $1.60 \, \text{T}$ magnetic field and is not attached to the wires in the circuit. What is the acceleration of the bar just after the switch S is closed?

Figure P27.56

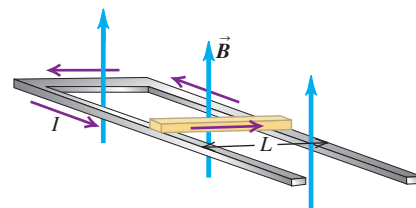


27.57 •• BIO Determining Diet. One method for determining the amount of corn in early Native American diets is the *stable isotope ratio analysis* (SIRA) technique. As corn photosynthesizes, it concentrates the isotope carbon-13, whereas most other plants concentrate carbon-12. Overreliance on corn consumption can then be correlated with certain diseases, because corn lacks the essential amino acid lysine. Archaeologists use a mass spectrometer to separate the ^{12}C and ^{13}C isotopes in samples of human remains. Suppose you use a velocity selector to obtain singly ionized (missing one electron) atoms of speed $8.50 \, \text{km/s}$, and you want to bend them within a uniform magnetic field in a semicircle of diameter $25.0 \, \text{cm}$ for the ^{12}C . The measured masses of these isotopes are $1.99 \times 10^{-26} \, \text{kg}$ (^{12}C) and $2.16 \times 10^{-26} \, \text{kg}$ (^{13}C). (a) What strength of magnetic field is required? (b) What is the diameter of the ^{13}C semicircle? (c) What is the separation of the ^{12}C and ^{13}C ions at the detector at the end of the semicircle? Is this distance large enough to be easily observed?

27.58 •• CP A plastic circular loop has radius R , and a positive charge q is distributed uniformly around the circumference of the loop. The loop is then rotated around its central axis, perpendicular to the plane of the loop, with angular speed ω . If the loop is in a region where there is a uniform magnetic field $\vec{B}$ directed parallel to the plane of the loop, calculate the magnitude of the magnetic torque on the loop.

27.59 •• CP An Electromagnetic Rail Gun. A conducting bar with mass m and length L slides over horizontal rails that are connected to a voltage source. The voltage source maintains a constant current I in the rails and bar, and a constant, uniform, vertical magnetic field $\vec{B}$ fills the region between the rails (**Fig. P27.59**). (a) Find the magnitude and direction of the net force on the conducting bar. Ignore friction, air resistance, and electrical resistance. (b) If the bar has mass m , find the distance d that the bar must move along the rails from rest to attain speed v . (c) It has been suggested that rail guns based on this principle could accelerate payloads into earth orbit or beyond. Find the distance the bar must travel along the rails if it is to reach the escape speed for the earth ($11.2 \, \text{km/s}$). Let $B = 0.80 \, \text{T}$, $I = 2.0 \times 10^3 \, \text{A}$, $m = 25 \, \text{kg}$, and $L = 50 \, \text{cm}$. For simplicity assume the net force on the object is equal to the magnetic force, as in parts (a) and (b), even though gravity plays an important role in an actual launch in space.

Figure P27.59



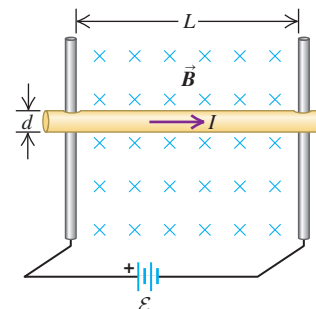
27.60 • A wire $25.0 \, \text{cm}$ long lies along the z -axis and carries a current of $7.40 \, \text{A}$ in the $+z$ -direction. The magnetic field is uniform and has components $B_x = -0.242 \, \text{T}$, $B_y = -0.985 \, \text{T}$, and $B_z = -0.336 \, \text{T}$. (a) Find the components of the magnetic force on the wire. (b) What is the magnitude of the net magnetic force on the wire?

27.61 •• CALC Point a is on the $+y$ -axis at $y = +0.200 \, \text{m}$ and point b is on the $+x$ -axis at $x = +0.200 \, \text{m}$. A wire in the shape of a circular arc of radius $0.200 \, \text{m}$ and centered on the origin goes from a to b and carries current $I = 5.00 \, \text{A}$ in the direction from a to b . (a) If the wire is in a uniform magnetic field $B = 0.800 \, \text{T}$ in the $+z$ -direction, what are the magnitude and direction of the net force that the magnetic field exerts on the wire segment? (b) What are the magnitude and direction of the net force on the wire if the field is $B = 0.800 \, \text{T}$ in the $+x$ -direction?

27.62 •• Ceiling fans use electric motors that involve stationary permanent magnets (called stators) attached to the central hub and with a typical strength of $1.0 \, \text{T}$, that supply torque to current-carrying coils (called rotors) fixed to the fan blades. Think of a ceiling fan at your home or workplace. (a) Estimate the diameter of the central hub at the inner edge of the blades. (b) If 12 rotors are fixed around the central hub, and if the diameters of the rotors make up one-third of the circumference of the hub, what is the diameter of a single rotor? (c) A typical ceiling fan supplies $1.4 \, \text{N} \cdot \text{m}$ of torque. If this torque is supplied equally by the 12 rotors in the presence of $1.0 \, \text{T}$ stator fields parallel to the axis of each rotor, estimate the magnitude of the magnetic moment of each rotor. (d) The rotors are connected in parallel, so each receives $\frac{1}{12}$ of the fan current, which is typically $0.75 \, \text{A}$. Estimate the number of windings in each rotor.

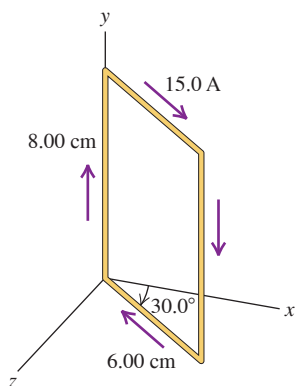
27.63 •• Earth's magnetic field near the ground is typically $0.50 \, \text{G}$ ($0.50 \, \text{gauss}$), where $1 \, \text{G} = 10^{-4} \, \text{tesla}$. In temperate northern latitudes, this field is inclined downward at an angle of approximately 45° . Consider the feasibility of using the earth's magnetic field to enable the levitation device shown in **Fig. P27.63** (next page). (a) Estimate the current needed to lift a copper bar with diameter $d = 5.0 \, \text{mm}$ using the horizontal component of the earth's magnetic field. The density of copper is $8900 \, \text{kg/m}^3$. (b) A copper bar with diameter $5.0 \, \text{mm}$ will melt if it carries a current greater than the fusing current of $900 \, \text{A}$. Is it feasible for our device? (c) Estimate the minimum strength of the magnetic field needed to levitate our copper bar. (d) Suppose we use easily obtainable $1.0 \, \text{T}$ permanent magnets to supply the horizontal magnetic field and suppose our bar has length $10 \, \text{cm}$. Estimate how much extra weight we could levitate with our device using a current of $10 \, \text{A}$.

Figure P27.63



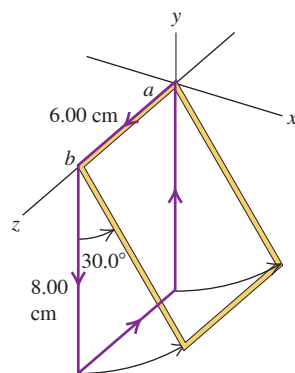
27.64 •• The rectangular loop shown in **Fig. P27.64** is pivoted about the y -axis and carries a current of 15.0 A in the direction indicated. (a) If the loop is in a uniform magnetic field with magnitude 0.48 T in the $+x$ -direction, find the magnitude and direction of the torque required to hold the loop in the position shown. (b) Repeat part (a) for the case in which the field is in the $-z$ -direction. (c) For each of the above magnetic fields, what torque would be required if the loop were pivoted about an axis through its center, parallel to the y -axis?

Figure P27.64



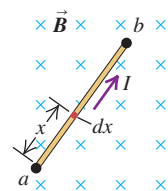
27.65 •• CP The rectangular loop of wire shown in **Fig. P27.65** has a mass of 0.15 g per centimeter of length and is pivoted about side ab on a frictionless axis. The current in the wire is 8.2 A in the direction shown. Find the magnitude and direction of the magnetic field parallel to the y -axis that will cause the loop to swing up until its plane makes an angle of 30.0° with the yz -plane.

Figure P27.65



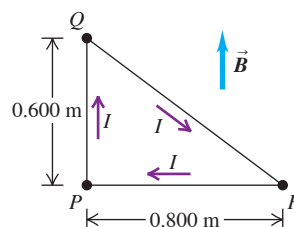
27.66 •• CALC A uniform bar of length L carries a current I in the direction from point a to point b (**Fig. P27.66**). The bar is in a uniform magnetic field that is directed into the page. Consider the torque about an axis perpendicular to the bar at point a that is due to the force that the magnetic field exerts on the bar. (a) Suppose that an infinitesimal section of the bar has length dx and is located a distance x from point a . Calculate the torque $d\tau$ about point a due to the magnetic force on this infinitesimal section. (b) Use $\tau = \int_a^b d\tau$ to calculate the total torque τ on the bar. (c) Show that τ is the same as though all of the magnetic force acted at the midpoint of the bar.

Figure P27.66



27.67 •• The loop of wire shown in **Fig. P27.67** forms a right triangle and carries a current $I = 5.00$ A in the direction shown. The loop is in a uniform magnetic field that has magnitude $B = 3.00$ T and the same direction as the current in side PQ of the loop. (a) Find the force exerted by the magnetic field on each side of the triangle. If the force is not zero, specify its direction. (b) What is the net force on the loop? (c) The loop is pivoted about an axis that lies along side PR . Use the forces calculated in part (a) to calculate the torque on each side of the loop (see Problem 27.66). (d) What is the magnitude of the net torque on the loop? Calculate the net torque from the torques calculated in part (c) and also from Eq. (27.28). Do these two results agree? (e) Is the net torque directed to rotate point Q into the plane of the figure or out of the plane of the figure?

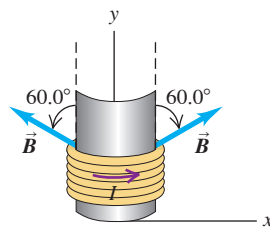
Figure P27.67



27.68 •• Repeat Problem 27.67 but with the magnetic field $B = 3.00$ T directed into the page in **Fig. P27.67**.

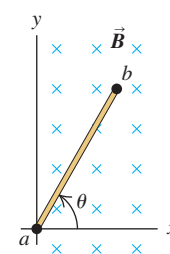
27.69 •• CALC A Voice Coil. It was shown in Section 27.7 that the net force on a current loop in a *uniform* magnetic field is zero. The magnetic force on the voice coil of a loudspeaker (see **Fig. 27.28**) is nonzero because the magnetic field at the coil is not uniform. A voice coil in a loudspeaker has 50 turns of wire and a diameter of 1.56 cm, and the current in the coil is 0.950 A. Assume that the magnetic field at each point of the coil has a constant magnitude of 0.220 T and is directed at an angle of 60.0° outward from the normal to the plane of the coil (**Fig. P27.69**). Let the axis of the coil be in the y -direction. The current in the coil is in the direction shown (counterclockwise as viewed from a point above the coil on the y -axis). Calculate the magnitude and direction of the net magnetic force on the coil.

Figure P27.69



27.70 •• CP A uniform bar has mass 0.0120 kg and is 30.0 cm long. It pivots without friction about an axis perpendicular to the bar at point a (**Fig. P27.70**). The gravitational force on the bar acts in the $-y$ -direction. The bar is in a uniform magnetic field that is directed into the page and has magnitude $B = 0.150$ T. (a) What must be the current I in the bar for the bar to be in rotational equilibrium when it is at an angle $\theta = 30.0^\circ$ above the horizontal? Use your result from Problem 27.66. (b) For the bar to be in rotational equilibrium, should I be in the direction from a to b or b to a ?

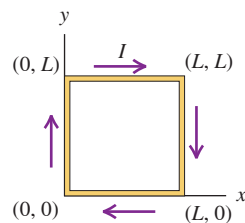
Figure P27.70



27.71 •• CALC Force on a Current Loop in a Nonuniform Magnetic Field.

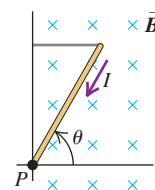
It was shown in Section 27.7 that the net force on a current loop in a *uniform* magnetic field is zero. But what if $\vec{B}$ is *not* uniform? **Figure P27.71** shows a square loop of wire that lies in the xy -plane. The loop has corners at $(0, 0)$, $(0, L)$, $(L, 0)$, and (L, L) and carries a constant current I in the clockwise direction. The magnetic field has no x -component but has both y - and z -components: $\vec{B} = (B_0 z/L)\hat{j} + (B_0 y/L)\hat{k}$, where B_0 is a positive constant. (a) Sketch the magnetic field lines in the yz -plane. (b) Find the magnitude and direction of the magnetic force exerted on each of the sides of the loop by integrating Eq. (27.20). (c) Find the magnitude and direction of the net magnetic force on the loop.

Figure P27.71



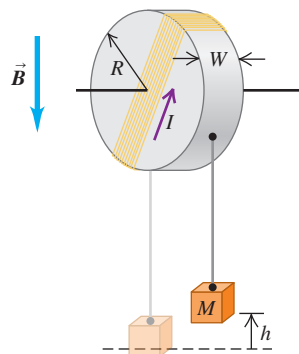
27.72 •• CP The lower end of the thin uniform rod in **Fig. P27.72** is attached to the floor by a frictionless hinge at point P . The rod has mass 0.0840 kg and length 18.0 cm and is in a uniform magnetic field $B = 0.120$ T that is directed into the page. The rod is held at an angle $\theta = 53.0^\circ$ above the horizontal by a horizontal string that connects the top of the rod to the wall. The rod carries a current $I = 12.0$ A in the direction toward P . Calculate the tension in the string. Use your result from Problem 27.66 to calculate the torque due to the magnetic-field force.

Figure P27.72



27.73 ••• CP A magnetic lift uses a cylinder with radius R and width W wrapped by conducting wire N times on a diameter, as shown in **Fig. P27.73**. A uniform external magnetic field $\vec{B}$ points downward while current I flows in the sense indicated. A mass M is hung from the cylinder by a cable attached to its rim on the axis of the coil. The height above the lowest possible position of the mass is h . (a) What is the minimum current I for which the mass can be suspended with $h > 0$? (b) What is the height of the mass $h = h_{\text{top}}$ when the cable attachment rotates one-half turn to the top of the cylinder, in terms of R ? Define a dimensionless parameter $\sigma = 2NIWB/Mg$ and express subsequent results in terms of σ and other given quantities. (c) What is the net torque on the cylinder if $0 \leq h \leq R$? (d) What is the net torque on the cylinder if $R \leq h \leq h_{\text{top}}$? (e) Define potential energy $U(h)$ that is zero when $h = 0$. What is the potential energy for $0 \leq h \leq R$? (f) What is the potential energy for $R \leq h \leq h_{\text{top}}$? (g) If $\sigma > 1$, for what value of h will the mass remain suspended motionless? (h) For what values of σ will the cylinder rotate more than 180° if the mass is released from rest at $h = h_{\text{top}}$? (Hint: This corresponds to $U(h_{\text{top}}) > 0$.)

Figure P27.73



27.74 ••• CP CALC A particle with charge q and mass m is dropped at time $t = 0$ from rest at its origin in a region of constant magnetic field $\vec{B}$ that points horizontally. What happens? To answer, construct a Cartesian coordinate system with the y -axis pointing downward and the z -axis pointing in the direction of the magnetic field. At time $t \geq 0$ the particle has velocity $\vec{v} = v_x\hat{i} + v_y\hat{j}$. The net force $\vec{F} = F_x\hat{i} + F_y\hat{j}$ on the particle is the vector sum of its weight and the magnetic force. (a) Using Newton's second law, write equations for a_x and a_y , where $\vec{a} = a_x\hat{i} + a_y\hat{j}$ is the acceleration of the particle. (b) Differentiate the second of these

equations with respect to time. Then substitute your expression for $a_x = dv_x/dt$ to determine an equation for dv_y^2/dt^2 in terms of v_y . (c) This result shows that v_y is a simple harmonic oscillator. Use the initial conditions to determine $v_y(t)$. Write your answer in terms of the angular frequency $\omega = qB/m$. Note that $(dv_y/dt)_0 = g$. (d) Substitute your result for $v_y(t)$ into your equation for dv_x/dt . Integrate using the initial conditions to determine $v_x(t)$. (e) Integrate your expressions for $v_x(t)$ and $v_y(t)$ to determine $x(t)$ and $y(t)$. (f) If $m = 1.00$ mg, $q = 19.6$ μC , and $B = 10.0$ T, what maximum vertical distance does the particle drop before returning upward?

27.75 •• A circular loop of wire with area A lies in the xy -plane. As viewed along the z -axis looking in the $-z$ -direction toward the origin, a current I is circulating clockwise around the loop. The torque produced by an external magnetic field $\vec{B}$ is given by $\vec{\tau} = D(4\hat{i} - 3\hat{j})$, where D is a positive constant, and for this orientation of the loop the magnetic potential energy $U = -\vec{\mu} \cdot \vec{B}$ is negative. The magnitude of the magnetic field is $B_0 = 13D/IA$. (a) Determine the vector magnetic moment of the current loop. (b) Determine the components B_x , B_y , and B_z of $\vec{B}$.

27.76 •• DATA You are using a type of mass spectrometer to measure charge-to-mass ratios of atomic ions. In the device, atoms are ionized with a beam of electrons to produce positive ions, which are then accelerated through a potential difference V . (The final speed of the ions is great enough that you can ignore their initial speed.) The ions then enter a region in which a uniform magnetic field $\vec{B}$ is perpendicular to the velocity of the ions and has magnitude $B = 0.250$ T. In this $\vec{B}$ region, the ions move in a semicircular path of radius R . You measure R as a function of the accelerating voltage V for one particular atomic ion:

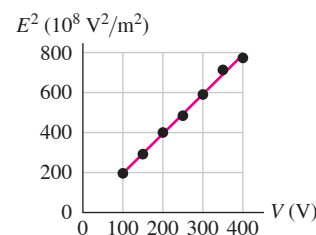
V (kV)	10.0	12.0	14.0	16.0	18.0
R (cm)	19.9	21.8	23.6	25.2	26.8

(a) How can you plot the data points so that they will fall close to a straight line? Explain. (b) Construct the graph described in part (a). Use the slope of the best-fit straight line to calculate the charge-to-mass ratio (q/m) for the ion. (c) For $V = 20.0$ kV, what is the speed of the ions as they enter the $\vec{B}$ region? (d) If ions that have $R = 21.2$ cm for $V = 12.0$ kV are singly ionized, what is R when $V = 12.0$ kV for ions that are doubly ionized?

27.77 •• DATA You are a research scientist working on a high-energy particle accelerator. Using a modern version of the Thomson e/m apparatus, you want to measure the mass of a muon (a fundamental particle that has the same charge as an electron but greater mass). The magnetic field between the two charged plates is 0.340 T. You measure the electric field for zero particle deflection as a function of the accelerating potential V . This potential is large enough that you can assume the initial speed of the muons to be zero.

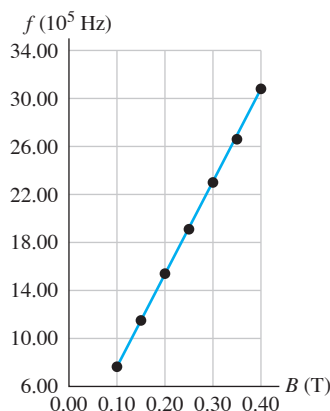
Figure P27.77 is an E^2 -versus- V graph of your data. (a) Explain why the data points fall close to a straight line. (b) Use the graph in **Fig. P27.77** to calculate the mass m of a muon. (c) If the two charged plates are separated by 6.00 mm, what must be the voltage between the plates in order for the electric field between the plates to be 2.00×10^5 V/m? Assume that the dimensions of the plates are much larger than the separation between them. (d) When $V = 400$ V, what is the speed of the muons as they enter the region between the plates?

Figure P27.77



27.78 •• DATA You are a technician testing the operation of a cyclotron. An alpha particle in the device moves in a circular path in a magnetic field $\vec{B}$ that is directed perpendicular to the path of the alpha particle. You measure the number of revolutions per second (the frequency f) of the alpha particle as a function of the magnetic field strength B . **Figure P27.78** shows your results and the best straight-line fit to your data. (a) Use the graph in Fig. P27.78 to calculate the charge-to-mass ratio of the alpha particle, which has charge $+2e$. On the basis of your data, what is the mass of an alpha particle? (b) With $B = 0.300$ T, what are the cyclotron frequencies f of a proton and of an electron? How do these f values compare to the frequency of an alpha particle? (c) With $B = 0.300$ T, what speed and kinetic energy does an alpha particle have if the radius of its path is 12.0 cm?

Figure P27.78



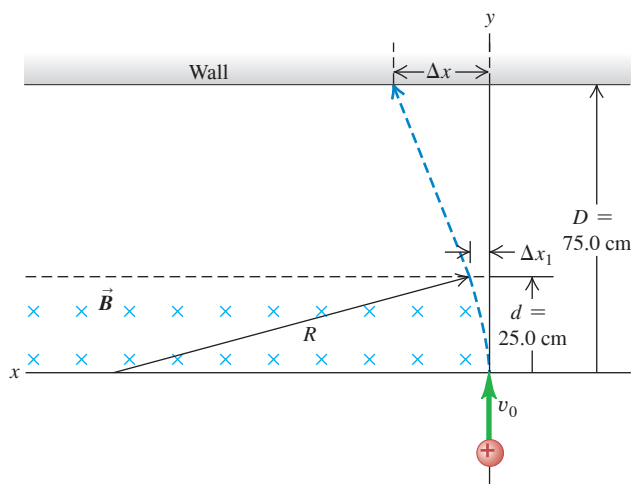
CHALLENGE PROBLEMS

27.79 ••• CALC Determine the magnetic moment $\vec{\mu}$ of a spherical shell with radius R and uniform charge Q rotating with angular speed $\vec{\Omega} = \omega \hat{k}$. Use the following steps: (a) Consider a coordinate system with the origin at the center of the sphere. Parameterize each latitude on the sphere with the angle θ measured from the positive z -axis. There is a circular current loop at each value of θ for $0 \leq \theta \leq \pi$. What is the radius of the loop at latitude θ ? (b) The differential current carried by that loop is $dI = \sigma v dW$, where σ is the charge density of the sphere, v is the tangential speed of the loop, and $dW = R d\theta$ is its differential width. Express dI in terms of σ , R , ω , θ , and $d\theta$. (c) The differential magnetic moment of the loop is $d\mu = A dI$, where A is the area enclosed by the loop. Express $d\mu$ in terms of R , ω , θ , and $d\theta$. (d) Integrate over the sphere to determine the magnetic moment. Express your result as a vector; use the total charge Q rather than the charge density σ . (e) If the sphere is in a uniform magnetic field $\vec{B} = (\sin \alpha \hat{i} + \cos \alpha \hat{j})B$, what is the torque on the sphere?

27.80 ••• An electron traveling from the sun as part of the solar wind strikes the earth's magnetosphere at latitude 80.0° N in a region where the magnetic field has a strength of $15.0 \mu\text{T}$ and is directed toward the earth's center. The electron has a speed of 400 km/s and is directed toward the earth's axis parallel to the equator. Magnetic forces send the electron on a helical trajectory. (a) What is the radius of this helix? (b) With what speed does the electron approach the surface of the earth? (c) If you are looking downward toward the earth from space, is the electron's motion clockwise or counterclockwise? (d) What is the frequency of the motion? (e) This electron strikes the ionosphere, where it is further accelerated by an electric field with strength 20.0 mV/m directed northward parallel to the earth's surface. What is the electron's new speed after it has been deflected 100 km southward by this field? (f) By what factor has its kinetic energy been increased by the electric field?

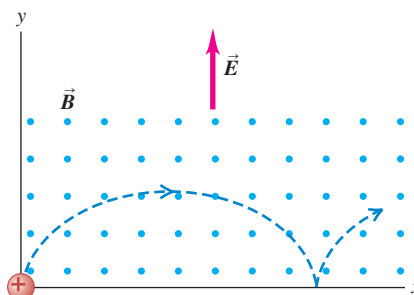
27.81 ••• A particle with charge $2.15 \mu\text{C}$ and mass 3.20×10^{-11} kg is initially traveling in the $+y$ -direction with a speed $v_0 = 1.45 \times 10^5$ m/s. It then enters a region containing a uniform magnetic field that is directed into, and perpendicular to, the page in **Fig. P27.81**. The magnitude of the field is 0.420 T. The region extends a distance of 25.0 cm along the initial direction of travel; 75.0 cm from the point of entry into the magnetic field region is a wall. The length of the field-free region is thus 50.0 cm. When the charged particle enters the magnetic field, it follows a curved path whose radius of curvature is R . It then leaves the magnetic field after a time t_1 , having been deflected a distance Δx_1 . The particle then travels in the field-free region and strikes the wall after undergoing a total deflection Δx . (a) Determine the radius R of the curved part of the path. (b) Determine t_1 , the time the particle spends in the magnetic field. (c) Determine Δx_1 , the horizontal deflection at the point of exit from the field. (d) Determine Δx , the total horizontal deflection.

Figure P27.81



27.82 ••• CP A Cycloidal Path. A particle with mass m and positive charge q starts from rest at the origin shown in **Fig. P27.82**. There is a uniform electric field $\vec{E}$ in the $+y$ -direction and a uniform magnetic field $\vec{B}$ directed out of the page. It is shown in more advanced books that the path is a *cycloid* whose radius of curvature at the top points is twice the y -coordinate at that level. (a) Explain why the path has this general shape and why it is repetitive. (b) Prove that the speed at any point is equal to $\sqrt{2qEy/m}$. (Hint: Use energy conservation.) (c) Applying Newton's second law at the top point and taking as given that the radius of curvature here equals $2y$, prove that the speed at this point is $2E/B$.

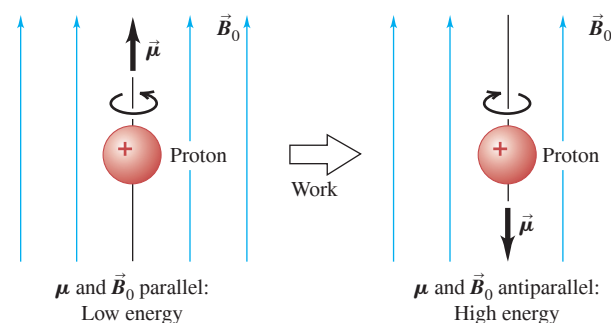
Figure P27.82



MCAT-STYLE PASSAGE PROBLEMS

BIO Magnetic Fields and MRI. *Magnetic resonance imaging* (MRI) is a powerful imaging method that, unlike x-ray imaging, allows sharp images of soft tissue to be made without exposing the patient to potentially damaging radiation. A rudimentary understanding of this method can be achieved by the relatively simple application of the classical (that is, non-quantum) physics of magnetism. The starting point for MRI is *nuclear magnetic resonance* (NMR), a technique that depends on the fact that protons in the atomic nucleus have a magnetic field $\vec{B}$. The origin of the proton's magnetic field is the spin of the proton. Being charged, the spinning proton constitutes an electric current analogous to a wire loop through which current flows. Like the wire loop, the proton has a magnetic moment $\vec{\mu}$; thus it will experience a torque when it is subjected to an external magnetic field $\vec{B}_0$. The magnitude of $\vec{\mu}$ is about 1.4×10^{-26} J/T. The proton can be thought of as being in one of two states, with $\vec{\mu}$ oriented parallel or antiparallel to the applied magnetic field, and work must be done to flip the proton from the low-energy state to the high-energy state, as the accompanying figure below shows.

An important consideration is that the net magnetic field of any nucleus, except for that of hydrogen (which has a proton only), consists of contributions from both protons and neutrons. If a nucleus has an even number of protons and neutrons, they will pair in such a way that half of the protons have spins in one orientation and half have spins in the other orientation. Thus the net magnetic moment of the nucleus is zero. Only nuclei with a net magnetic moment are candidates for MRI. Hydrogen is the atom that is most commonly imaged.



27.83 If a proton is exposed to an external magnetic field of 2 T that has a direction perpendicular to the axis of the proton's spin, what will be the torque on the proton? (a) 0; (b) 1.4×10^{-26} N·m; (c) 2.8×10^{-26} N·m; (d) 0.7×10^{-26} N·m.

27.84 Which of following elements is a candidate for MRI? (a) $^{12}\text{C}_6$; (b) $^{16}\text{O}_8$; (c) $^{40}\text{Ca}_{20}$; (d) $^{31}\text{P}_{15}$.

27.85 The large magnetic fields used in MRI can produce forces on electric currents within the human body. This effect has been proposed as a possible method for imaging “biocurrents” flowing in the body, such as the current that flows in individual nerves. For a magnetic field strength of 2 T, estimate the magnitude of the maximum force on a 1-mm-long segment of a single cylindrical nerve that has a diameter of 1.5 mm. Assume that the entire nerve carries a current due to an applied voltage of 100 mV (that of a typical action potential). The resistivity of the nerve is $0.6 \Omega \cdot \text{m}$. (a) 6×10^{-7} N; (b) 1×10^{-6} N; (c) 3×10^{-4} N; (d) 0.3 N.

ANSWERS

Chapter Opening Question?

(ii) A magnetized compass needle has a magnetic dipole moment along its length, and the earth's magnetic field (which points generally northward) exerts a torque that tends to align that dipole moment with the field. See Section 27.7 for details.

Key Example **VP27.1** VARIATION Problems

VP27.1.1 (a) 4.18×10^{-14} N (b) $-z$ -direction

VP27.1.2 2.91×10^{-15} C

VP27.1.3 (a) 0.0175 T (b) $+z$ -direction

VP27.1.4 (a) 0.189 T (b) $-y$ -direction

VP27.6.1 (a) 0.435 mm (b) 2.87×10^7 rad/s (c) 4.57×10^6 Hz

VP27.6.2 (a) 0.348 mm (b) 1.64 mm (c) 4.80×10^{-16} N

VP27.6.3 (a) 8.00×10^5 m/s (b) $-y$ -direction (c) $-x$ -direction

VP27.6.4 (a) 6.64 cm (b) 7.47 cm

VP27.7.1 (a) 2.44×10^{-3} N (b) $+z$ -direction

VP27.7.2 (a) 1.60 A (b) upward

VP27.7.3 (a) -2.52×10^{-3} N (b) -4.20×10^{-3} N (c) 0 N

(d) 4.90×10^{-3} N

VP27.7.4 41.0° and 139.0°

Bridging Problem

(a) $\tau_x = -1.54 \times 10^{-4}$ N·m,

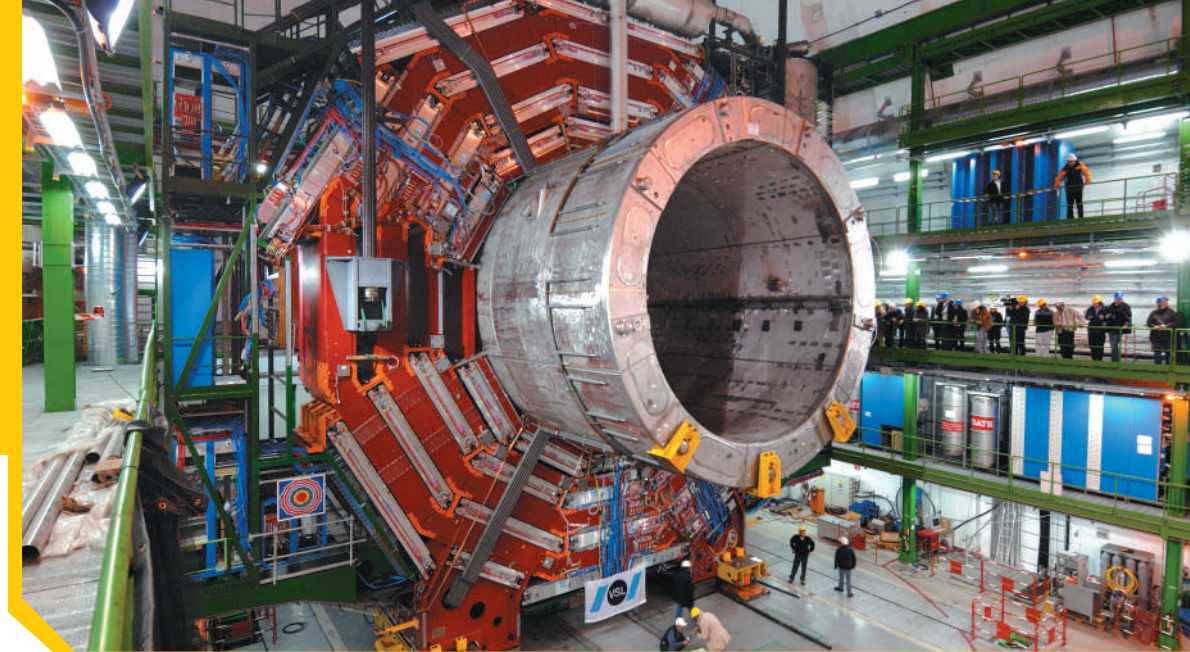
$\tau_y = -2.05 \times 10^{-4}$ N·m,

$\tau_z = -6.14 \times 10^{-4}$ N·m

(b) -7.55×10^{-4} J

(c) 42.1 rad/s

? The immense cylinder in this photograph is a current-carrying coil, or solenoid, that generates a uniform magnetic field in its interior as part of an experiment at CERN, the European Organization for Nuclear Research. If two such solenoids were joined end to end, the magnetic field along their common axis would (i) become four times stronger; (ii) double in strength; (iii) become $\sqrt{2}$ times stronger; (iv) not change; (v) weaken.



28 Sources of Magnetic Field

LEARNING OUTCOMES

In this chapter, you'll learn...

- 28.1 The nature of the magnetic field produced by a single moving charged particle.
- 28.2 How to describe the magnetic field produced by an element of a current-carrying conductor.
- 28.3 How to calculate the magnetic field produced by a long, straight, current-carrying wire.
- 28.4 Why wires carrying current in the same direction attract, while wires carrying opposing currents repel.
- 28.5 How to calculate the magnetic field produced by a current-carrying wire bent into a circle.
- 28.6 What Ampere's law is, and what it tells us about magnetic fields.
- 28.7 How to use Ampere's law to calculate the magnetic field of symmetric current distributions.
- 28.8 How microscopic currents within materials give them their magnetic properties.

You'll need to review...

- 10.5 Angular momentum of a particle.
- 21.3–21.5 Coulomb's law and electric-field calculations.
- 22.4 Solving problems with Gauss's law.
- 27.2–27.9 Magnetic field and magnetic force.

In Chapter 27 we studied the forces exerted on moving charges and on current-carrying conductors in a magnetic field. We didn't worry about how the magnetic field got there; we simply took its existence as a given fact. But how are magnetic fields *created*? We know that both permanent magnets and electric currents in electromagnets create magnetic fields. In this chapter we'll study these sources of magnetic field in detail.

We've learned that a charge creates an electric field and that an electric field exerts a force on a charge. But a *magnetic* field exerts a force on only a *moving* charge. Similarly, we'll see that only *moving* charges *create* magnetic fields. We'll begin our analysis with the magnetic field created by a single moving point charge. We can use this analysis to determine the field created by a small segment of a current-carrying conductor. Once we can do that, we can in principle find the magnetic field produced by *any* shape of conductor.

Then we'll introduce Ampere's law, which plays a role in magnetism analogous to the role of Gauss's law in electrostatics. Ampere's law lets us exploit symmetry properties in relating magnetic fields to their sources.

Moving charged particles within atoms respond to magnetic fields and can also act as sources of magnetic field. We'll use these ideas to understand how certain magnetic materials can be used to intensify magnetic fields as well as why some materials such as iron act as permanent magnets.

28.1 MAGNETIC FIELD OF A MOVING CHARGE

Let's start with the basics, the magnetic field of a single point charge q moving with a constant velocity $\vec{v}$. In practical applications, such as the solenoid shown in the photo that opens this chapter, magnetic fields are produced by tremendous numbers of charged particles moving together in a current. But once we understand how to calculate the magnetic field due to a single point charge, it's a small leap to calculate the field due to a current-carrying wire or collection of wires.

As we did for electric fields, we call the location of the moving charge at a given instant the **source point** and the point P where we want to find the field the **field point**. In Section 21.4 we found that at a field point a distance r from a point charge q , the magnitude of the

electric field $\vec{E}$ caused by the charge is proportional to the charge magnitude $|q|$ and to $1/r^2$, and the direction of $\vec{E}$ (for positive q) is along the line from source point to field point. The corresponding relationship for the *magnetic* field $\vec{B}$ of a point charge q moving with constant velocity has some similarities and some interesting differences.

Experiments show that the magnitude of $\vec{B}$ is also proportional to $|q|$ and to $1/r^2$. But the *direction* of $\vec{B}$ is *not* along the line from source point to field point. Instead, $\vec{B}$ is perpendicular to the plane containing this line and the particle's velocity vector $\vec{v}$, as shown in **Fig. 28.1**. Furthermore, the field *magnitude* B is also proportional to the particle's speed v and to the sine of the angle ϕ . Thus the magnetic-field magnitude at point P is

$$B = \frac{\mu_0}{4\pi} \frac{|q|v \sin \phi}{r^2} \quad (28.1)$$

The quantity μ_0 (read as “mu-nought” or “mu-sub-zero”) is called the **magnetic constant**. The reason for including the factor of 4π will emerge shortly. We did something similar with Coulomb's law in Section 21.3.

Moving Charge: Vector Magnetic Field

We can incorporate both the magnitude and direction of $\vec{B}$ into a single vector equation by using the vector product. To avoid having to say “the direction from the source q to the field point P ” over and over, we introduce a *unit* vector $\hat{r}$ (“r-hat”) that points from the source point to the field point. (We used $\hat{r}$ for the same purpose in Section 21.4.) This unit vector is equal to the vector $\vec{r}$ from the source to the field point divided by its magnitude: $\hat{r} = \vec{r}/r$. Then

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2} \quad (28.2)$$

Magnetic field due to a point charge with constant velocity $\vec{B}$ = $\frac{\mu_0}{4\pi}$ $\frac{q\vec{v} \times \hat{r}}{r^2}$

Magnetic constant μ_0 Charge q Velocity $\vec{v}$ Unit vector from point charge toward where field is measured $\hat{r}$ Distance from point charge to where field is measured r

Figure 28.1 shows the relationship of $\hat{r}$ to P and shows the magnetic field $\vec{B}$ at several points in the vicinity of the charge. At all points along a line through the charge parallel to the velocity $\vec{v}$, the field is zero because $\sin \phi = 0$ at all such points. At any distance r from q , $\vec{B}$ has its greatest magnitude at points lying in the plane perpendicular to $\vec{v}$, because there $\phi = 90^\circ$ and $\sin \phi = 1$. If q is negative, the directions of $\vec{B}$ are opposite to those shown in Fig. 28.1.

Moving Charge: Magnetic Field Lines

A point charge in motion also produces an *electric* field, with field lines that radiate outward from a positive charge. The *magnetic* field lines are completely different. For a point charge moving with velocity $\vec{v}$, the magnetic field lines are *circles* centered on the line of $\vec{v}$ and lying in planes perpendicular to this line. The field-line directions for a positive charge are given by the following *right-hand rule*, one of several that we'll encounter in this chapter: Grasp the velocity vector $\vec{v}$ with your right hand so that your right thumb points in the direction of $\vec{v}$; your fingers then curl around the line of $\vec{v}$ in the same sense as the magnetic field lines, assuming q is positive. Figure 28.1a shows parts of a few field lines; Fig. 28.1b shows some field lines in a plane through q , perpendicular to $\vec{v}$. If the point charge is negative, the directions of the field and field lines are the opposite of those shown in Fig. 28.1.

Equations (28.1) and (28.2) describe the $\vec{B}$ field of a point charge moving with *constant* velocity. If the charge *accelerates*, the field can be much more complicated. We won't need these more complicated results for our purposes. (The moving charged particles that make up a current in a wire accelerate at points where the wire bends and the direction of $\vec{v}$ changes. But because the magnitude v_d of the drift velocity in a conductor is typically very small, the centripetal acceleration v_d^2/r is so small that we can ignore its effects.)

As we discussed in Section 27.2, the unit of B is one tesla (1 T):

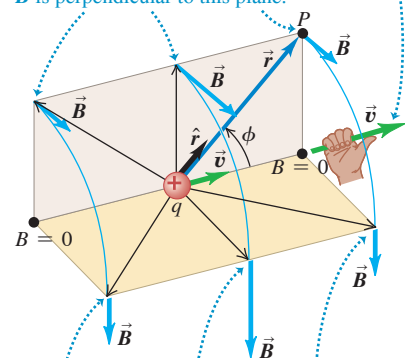
$$1 \text{ T} = 1 \text{ N} \cdot \text{s} / \text{C} \cdot \text{m} = 1 \text{ N} / \text{A} \cdot \text{m}$$

Figure 28.1 (a) Magnetic-field vectors due to a moving positive point charge q . At each point, $\vec{B}$ is perpendicular to the plane of $\vec{r}$ and $\vec{v}$, and its magnitude is proportional to the sine of the angle between them. (b) Magnetic field lines in a plane containing a moving positive charge.

(a) Perspective view

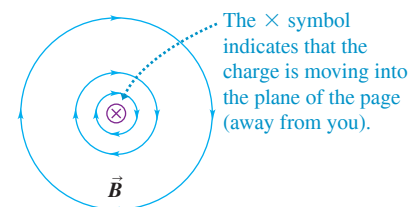
Right-hand rule for the magnetic field due to a positive charge moving at constant velocity: Point the thumb of your right hand in the direction of the velocity. Your fingers now curl around the charge in the direction of the magnetic field lines. (If the charge is negative, the field lines are in the opposite direction.)

For these field points, $\vec{r}$ and $\vec{v}$ both lie in the beige plane, and $\vec{B}$ is perpendicular to this plane.



For these field points, $\vec{r}$ and $\vec{v}$ both lie in the gold plane, and $\vec{B}$ is perpendicular to this plane.

(b) View from behind the charge



Using this with Eq. (28.1) or (28.2), we find that the units of the constant μ_0 are

$$1 \text{ N} \cdot \text{s}^2/\text{C}^2 = 1 \text{ N}/\text{A}^2 = 1 \text{ Wb}/\text{A} \cdot \text{m} = 1 \text{ T} \cdot \text{m}/\text{A}$$

In SI units the numerical value of μ_0 is, to nine significant figures,

$$\begin{aligned}\mu_0 &= 1.25663706 \times 10^{-6} \text{ N} \cdot \text{s}^2/\text{C}^2 = 1.25663706 \times 10^{-6} \text{ Wb}/\text{A} \cdot \text{m} \\ &\cong 4\pi \times 10^{-7} \text{ T} \cdot \text{m}/\text{A}\end{aligned}\quad (28.3)$$

It may seem incredible that μ_0 is equal to $4\pi \times 10^{-7} \text{ T} \cdot \text{m}/\text{A}$ to nine significant figures! In fact this is a consequence of how the ampere and coulomb were previously defined. We'll explore this further in Section 28.4.

Here's another remarkable aspect of the value of the magnetic constant μ_0 . Recall from Section 21.3 that the *electric* constant ϵ_0 that appears in Coulomb's law and Gauss's law has the value $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$. If we multiply ϵ_0 by $\mu_0 = 1.257 \times 10^{-6} \text{ N} \cdot \text{s}^2/\text{C}^2$, take the square root of this product, then take the reciprocal of the result, we get

$$\frac{1}{\sqrt{\epsilon_0 \mu_0}} = 2.998 \times 10^8 \text{ m/s}$$

But $2.998 \times 10^8 \text{ m/s}$ is just the value of the speed of light in vacuum! We conclude that

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \quad (28.4)$$

This suggests that electric and magnetic fields are intimately related to the nature of light. We'll explore this relationship when we study electromagnetic waves in Chapter 32.

EXAMPLE 28.1 Forces between two moving protons

WITH VARIATION PROBLEMS

Two protons move parallel to the x -axis in opposite directions (**Fig. 28.2**) at the same speed v (small compared to the speed of light c). At the instant shown, find the electric and magnetic forces on the upper proton and compare their magnitudes.

IDENTIFY and SET UP Coulomb's law [Eq. (21.2)] gives the electric force F_E on the upper proton. The magnetic force law [Eq. (27.2)] gives the magnetic force on the upper proton; to use it, we must first use Eq. (28.2) to find the magnetic field that the lower proton produces at the position of the upper proton. The unit vector from the lower proton (the source) to the position of the upper proton is $\hat{r} = \hat{j}$.

EXECUTE From Coulomb's law, the magnitude of the electric force on the upper proton is

$$F_E = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2}$$

The forces are repulsive, and the force on the upper proton is vertically upward (in the $+y$ -direction).

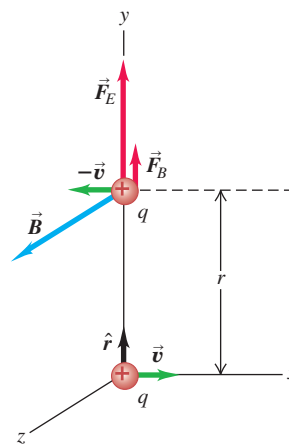
The velocity of the lower proton is $\vec{v} = v\hat{i}$. From the right-hand rule for the cross product $\vec{v} \times \hat{r}$ in Eq. (28.2), the $\vec{B}$ field due to the lower proton at the position of the upper proton is in the $+z$ -direction (see Fig. 28.2). From Eq. (28.2), the field is

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q(v\hat{i}) \times \hat{j}}{r^2} = \frac{\mu_0}{4\pi} \frac{qv}{r^2} \hat{k}$$

The velocity of the upper proton is $-\vec{v} = -v\hat{i}$, so the magnetic force on it is

$$\vec{F}_B = q(-\vec{v}) \times \vec{B} = q(-v\hat{i}) \times \frac{\mu_0}{4\pi} \frac{qv}{r^2} \hat{k} = \frac{\mu_0}{4\pi} \frac{q^2 v^2}{r^2} \hat{j}$$

Figure 28.2 Electric and magnetic forces between two moving protons.



The magnetic interaction in this situation is also repulsive.

The ratio of the force magnitudes is

$$\frac{F_B}{F_E} = \frac{\mu_0 q^2 v^2 / 4\pi r^2}{q^2 / 4\pi \epsilon_0 r^2} = \frac{\mu_0 v^2}{1/\epsilon_0} = \epsilon_0 \mu_0 v^2$$

With the relationship $\epsilon_0 \mu_0 = 1/c^2$, Eq. (28.4), this becomes

$$\frac{F_B}{F_E} = \frac{v^2}{c^2}$$

When v is small in comparison to the speed of light, the magnetic force is much smaller than the electric force.

EVALUATE We have described the velocities, fields, and forces as they are measured by an observer who is stationary in the coordinate system of Fig. 28.2. In a coordinate system that moves with one of the charges, one of the velocities would be zero, so there would be *no* magnetic force. The explanation of this apparent paradox provided one of the paths that led to the special theory of relativity.

KEYCONCEPT A moving charged particle produces a magnetic field $\vec{B}$. The magnitude of $\vec{B}$ at a given point is proportional to the charge magnitude and to the particle's speed, and inversely proportional to the square of the distance from the charge; it also depends on the direction from the particle to that point compared to the direction of the particle's velocity. The direction of $\vec{B}$ is given by a right-hand rule.

TEST YOUR UNDERSTANDING OF SECTION 28.1 (a) If two protons are traveling parallel to each other in the *same* direction and at the same speed, is the magnetic force between them (i) attractive or (ii) repulsive? (b) Is the net force between them (i) attractive, (ii) repulsive, or (iii) zero? (Assume that the protons' speed is much slower than the speed of light.)

ANSWER (a) (i), (b) (ii) The situation is the same as shown in Fig. 28.2 except that the upper proton has velocity $\vec{v}$ rather than $-\vec{v}$. The magnetic field due to the lower proton is the same as shown in Fig. 28.2, but the direction of the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ on the upper proton is reversed. Hence the magnetic force is attractive. Since the speed v is small compared to c , the magnetic force is much smaller in magnitude than the repulsive electric force and the net force is still repulsive.

28.2 MAGNETIC FIELD OF A CURRENT ELEMENT

As for electric fields, there is a *principle of superposition of magnetic fields*:

PRINCIPLE OF SUPERPOSITION OF MAGNETIC FIELDS The total magnetic field caused by several moving charges is the vector sum of the fields caused by the individual charges.

We can use this principle with the results of Section 28.1 to find the magnetic field produced by a current in a conductor.

We begin by calculating the magnetic field caused by a short segment $d\vec{l}$ of a current-carrying conductor, as shown in Fig. 28.3a. The volume of the segment is $A dl$, where A is the cross-sectional area of the conductor. If there are n moving charged particles per unit volume, each of charge q , the total moving charge dQ in the segment is

$$dQ = nqA dl$$

The moving charges in this segment are equivalent to a single charge dQ , traveling with a velocity equal to the *drift* velocity $\vec{v}_d$. (Magnetic fields due to the *random* motions of the charges will, on average, cancel out at every point.) From Eq. (28.1) the magnitude of the resulting field $d\vec{B}$ at any field point P is

$$dB = \frac{\mu_0}{4\pi} \frac{|dQ|v_d \sin \phi}{r^2} = \frac{\mu_0}{4\pi} \frac{n|q|v_d A dl \sin \phi}{r^2}$$

But from Eq. (25.2), $n|q|v_d A$ equals the current I in the element. So

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \phi}{r^2} \quad (28.5)$$

Current Element: Vector Magnetic Field

In vector form, using the unit vector $\hat{r}$ as in Section 28.1, we have

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2} \quad (28.6)$$

Magnetic field due to an infinitesimal current element

Magnetic constant

Current

Vector length of element (points in current direction)

Unit vector from element toward where field is measured

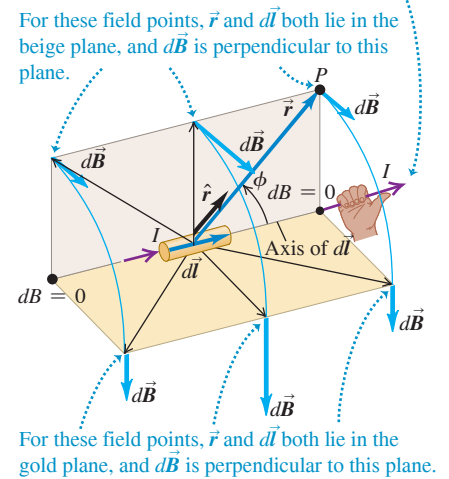
Distance from element to where field is measured

where $d\vec{l}$ is a vector with length dl , in the same direction as the current.

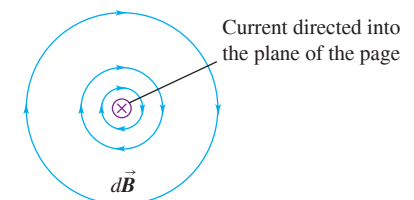
Figure 28.3 (a) Magnetic-field vectors due to a current element $d\vec{l}$. (b) Magnetic field lines in a plane containing the current element $d\vec{l}$. Compare this figure to Fig. 28.1 for the field of a moving point charge.

(a) Perspective view

Right-hand rule for the magnetic field due to a current element: Point the thumb of your right hand in the direction of the current. Your fingers now curl around the current element in the direction of the magnetic field lines.



(b) View along the axis of the current element



APPLICATION Currents and Planetary Magnetism

The earth's magnetic field is caused by currents circulating within its molten, conducting interior. These currents are stirred by our planet's relatively rapid spin (one rotation per 24 hours). The planet Jupiter's internal currents are much stronger: Jupiter is much larger than the earth, has an interior that is mostly liquid hydrogen (which is a very good conductor under high pressures), and spins very rapidly (one rotation per 10 hours). Hence at the same distance from the center of each planet, Jupiter's magnetic field is about 2×10^4 times greater than the earth's field.



Equations (28.5) and (28.6) are called the **law of Biot and Savart** (pronounced “Bee-oh” and “Suh-var”). We can use this law to find the total magnetic field $\vec{B}$ at any point in space due to the current in a complete circuit. To do this, we integrate Eq. (28.6) over all segments $d\vec{l}$ that carry current; symbolically,

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l} \times \hat{r}}{r^2} \quad (28.7)$$

In the following sections we'll carry out this vector integration for several examples.

Current Element: Magnetic Field Lines

As Fig. 28.3 shows, the field vectors $d\vec{B}$ and the magnetic field lines of a current element are exactly like those set up by a positive charge dQ moving in the direction of the drift velocity $\vec{v}_d$. The field lines are circles in planes perpendicular to $d\vec{l}$ and centered on the line of $d\vec{l}$. Their directions are given by the same right-hand rule that we introduced for point charges in Section 28.1.

We can't verify Eq. (28.5) or (28.6) directly because we can never experiment with an isolated segment of a current-carrying circuit. What we measure experimentally is the *total* $\vec{B}$ for a complete circuit. But we can still verify these equations indirectly by calculating $\vec{B}$ for various current configurations with Eq. (28.7) and comparing the results with experimental measurements.

If matter is present in the space around a current-carrying conductor, the field at a field point P in its vicinity will have an additional contribution resulting from the *magnetization* of the material. We'll return to this point in Section 28.8. However, unless the material is iron or some other ferromagnetic material, the additional field is small and is usually negligible. Additional complications arise if time-varying electric or magnetic fields are present or if the material is a superconductor; we'll return to these topics later.

PROBLEM-SOLVING STRATEGY 28.1 Magnetic-Field Calculations

IDENTIFY *the relevant concepts:* The law of Biot and Savart [Eqs. (28.5) and (28.6)] allows you to calculate the magnetic field at a field point P due to a current-carrying wire of any shape. The idea is to calculate the field element $d\vec{B}$ at P due to a representative current element in the wire and integrate all such field elements to find the field $\vec{B}$ at P .

SET UP *the problem* using the following steps:

1. Make a diagram showing a representative current element and the field point P .
2. Draw the current element $d\vec{l}$, being careful that it points in the direction of the current.
3. Draw the unit vector $\hat{r}$ directed *from* the current element (the source point) to P .
4. Identify the target variable (usually $\vec{B}$).

EXECUTE *the solution* as follows:

1. Use Eq. (28.5) or (28.6) to express the magnetic field $d\vec{B}$ at P from the representative current element.
2. Add up all the $d\vec{B}$'s using the rules of *vector* addition to find the total field at point P . In some situations the $d\vec{B}$'s at point P have

the same direction for all the current elements; then the magnitude of the total $\vec{B}$ field is the sum of the magnitudes of the $d\vec{B}$'s. But often the $d\vec{B}$'s have different directions for different current elements. Then you have to set up a coordinate system and represent each $d\vec{B}$ in terms of its components. The integral for the total $\vec{B}$ is then expressed in terms of an integral for each component.

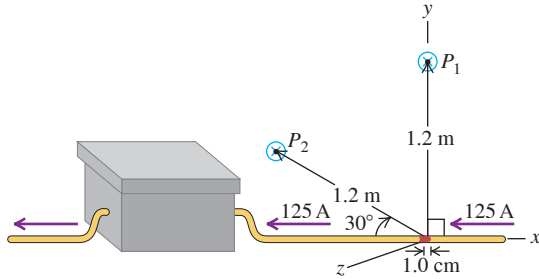
3. Sometimes you can use the symmetry of the situation to prove that one component of $\vec{B}$ must vanish. Always be alert for ways to use symmetry to simplify the problem.
4. Look for ways to use the principle of superposition of magnetic fields. Later in this chapter we'll determine the fields produced by certain simple conductor shapes; if you encounter a conductor of a complex shape that can be represented as a combination of these simple shapes, you can use superposition to find the field of the complex shape. Examples include a rectangular loop and a semi-circle with straight line segments on both sides.

EVALUATE *your answer:* Often your answer will be a mathematical expression for $\vec{B}$ as a function of the position of the field point. Check the answer by examining its behavior in as many limits as you can.

EXAMPLE 28.2 Magnetic field of a current segment**WITH VARIATION PROBLEMS**

A copper wire carries a steady 125 A current to an electroplating tank (Fig. 28.4). Find the magnetic field due to a 1.0 cm segment of this wire at a point 1.2 m away from it, if the point is (a) point P_1 , straight out to the side of the segment, and (b) point P_2 , in the xy -plane and on a line at 30° to the segment.

Figure 28.4 Finding the magnetic field at two points due to a 1.0 cm segment of current-carrying wire (not shown to scale).



IDENTIFY and SET UP Although Eqs. (28.5) and (28.6) apply only to infinitesimal current elements, we may use either of them here because the segment length is much less than the distance to the field point. The current element is shown in red in Fig. 28.4 and points in the $-x$ -direction (the direction of the current), so $d\vec{l} = dl(-\hat{i})$. The unit vector $\hat{r}$ for each field point is directed from the current element toward that point: $\hat{r}$ is in the $+y$ -direction for point P_1 and at an angle of 30° above the $-x$ -direction for point P_2 .

EXECUTE (a) At point P_1 , $\hat{r} = \hat{j}$, so

$$\begin{aligned}\vec{B} &= \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \frac{I dl(-\hat{i}) \times \hat{j}}{r^2} = -\frac{\mu_0}{4\pi} \frac{I dl}{r^2} \hat{k} \\ &= -(10^{-7} \text{ T} \cdot \text{m/A}) \frac{(125 \text{ A})(1.0 \times 10^{-2} \text{ m})}{(1.2 \text{ m})^2} \hat{k} \\ &= -(8.7 \times 10^{-8} \text{ T}) \hat{k}\end{aligned}$$

The direction of $\vec{B}$ at P_1 is into the xy -plane of Fig. 28.4.

(b) At P_2 , the unit vector is $\hat{r} = (-\cos 30^\circ)\hat{i} + (\sin 30^\circ)\hat{j}$. From Eq. (28.6),

$$\begin{aligned}\vec{B} &= \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \frac{I dl(-\hat{i}) \times (-\cos 30^\circ \hat{i} + \sin 30^\circ \hat{j})}{r^2} \\ &= -\frac{\mu_0 I dl \sin 30^\circ}{4\pi r^2} \hat{k} \\ &= -(10^{-7} \text{ T} \cdot \text{m/A}) \frac{(125 \text{ A})(1.0 \times 10^{-2} \text{ m})(\sin 30^\circ)}{(1.2 \text{ m})^2} \hat{k} \\ &= -(4.3 \times 10^{-8} \text{ T}) \hat{k}\end{aligned}$$

The direction of $\vec{B}$ at P_2 is also into the xy -plane of Fig. 28.4.

EVALUATE We can check our results for the direction of $\vec{B}$ by comparing them with Fig. 28.3. The xy -plane in Fig. 28.4 corresponds to the beige plane in Fig. 28.3, but here the direction of the current and hence of $d\vec{l}$ is the reverse of that shown in Fig. 28.3. Consequently the direction of the magnetic field is reversed as well. It follows that the field at points in the xy -plane in Fig. 28.4 must point *into*, not *out of*, that plane, just as we concluded above.

KEYCONCEPT The magnetic field $\vec{B}$ due to a short current segment of length dl is like that of a moving charged particle. The difference is that the product of the moving particle's charge and velocity is replaced by the product of the current and the vector $d\vec{l}$ in the direction of the current.

TEST YOUR UNDERSTANDING OF SECTION 28.2 An infinitesimal current element located at the origin ($x = y = z = 0$) carries current I in the positive y -direction. Rank the following locations in order of the strength of the magnetic field that the current element produces at that location, from largest to smallest value. (i) $x = L$, $y = 0$, $z = 0$; (ii) $x = 0$, $y = L$, $z = 0$; (iii) $x = 0$, $y = 0$, $z = L$; (iv) $x = L/\sqrt{2}$, $y = L/\sqrt{2}$, $z = 0$.

ANSWER

(i) 0, (ii) $1/\sqrt{2}$, (iii) 1 , and (iv) $1/\sqrt{2}$.
 $\phi = 45^\circ$, so the values of $\sin \phi$ are (i) 1 ,
 points the angle is (i) 0° , (ii) 90° , (iii) 0° , and (iv)
 so the value of dB is proportional to the value of $\sin \phi$. For the four
 All four points are the same distance $r = L$ from the current element.

of the current and a vector from the current element to the field point.
 the element to the field point, and ϕ is the angle between the direction
 $dB = (\mu_0/4\pi)(I dl \sin \phi/r^2)$. In this expression r is the distance from
 field dB due to a current element of length dl carrying current I is
 (i) and (iii) (ii), (iv), (iii) From Eq. (28.5), the magnitude of the

28.3 MAGNETIC FIELD OF A STRAIGHT CURRENT-CARRYING CONDUCTOR

Let's use the law of Biot and Savart to find the magnetic field produced by a straight current-carrying conductor. This result is useful because straight conducting wires are found in essentially all electric and electronic devices. **Figure 28.5** (next page) shows such a conductor with length $2a$ carrying a current I . We'll find $\vec{B}$ at a point a distance x from the conductor on its perpendicular bisector.

We first use the law of Biot and Savart, Eq. (28.5), to find the field $d\vec{B}$ caused by the element of conductor of length $dl = dy$ shown in Fig. 28.5. From the figure, $r = \sqrt{x^2 + y^2}$ and $\sin \phi = \sin(\pi - \phi) = x/\sqrt{x^2 + y^2}$. The right-hand rule for the vector product $d\vec{l} \times \hat{r}$

Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.

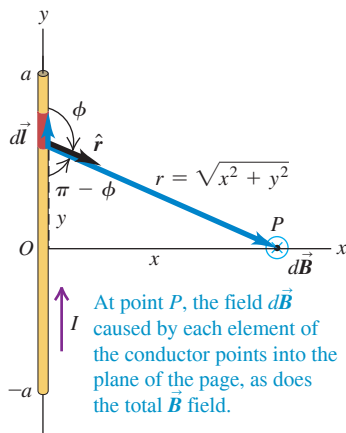
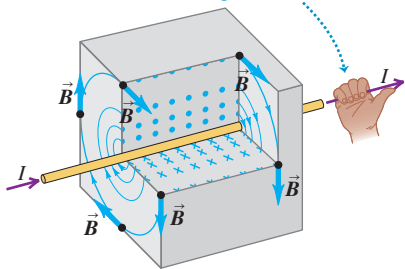


Figure 28.6 Magnetic field around a long, straight, current-carrying conductor. The field lines are circles, with directions determined by the right-hand rule.

Right-hand rule for the magnetic field around a current-carrying wire: Point the thumb of your right hand in the direction of the current. Your fingers now curl around the wire in the direction of the magnetic field lines.



shows that the *direction* of $d\vec{B}$ is into the plane of the figure, perpendicular to the plane; furthermore, the directions of the $d\vec{B}$'s from *all* elements of the conductor are the same. Thus in integrating Eq. (28.7), we can just add the *magnitudes* of the $d\vec{B}$'s, a significant simplification.

Putting the pieces together, we find that the magnitude of the total $\vec{B}$ field is

$$B = \frac{\mu_0 I}{4\pi} \int_{-a}^a \frac{x dy}{(x^2 + y^2)^{3/2}}$$

We can integrate this by trigonometric substitution or by using an integral table:

$$B = \frac{\mu_0 I}{4\pi} \frac{2a}{x\sqrt{x^2 + a^2}} \quad (28.8)$$

When the length $2a$ of the conductor is much greater than its distance x from point P , we can consider it to be infinitely long. When a is much larger than x , $\sqrt{x^2 + a^2}$ is approximately equal to a ; hence in the limit $a \rightarrow \infty$, Eq. (28.8) becomes

$$B = \frac{\mu_0 I}{2\pi x}$$

The physical situation has axial symmetry about the y -axis. Hence $\vec{B}$ must have the same *magnitude* at all points on a circle centered on the conductor and lying in a plane perpendicular to it, and the *direction* of $\vec{B}$ must be everywhere tangent to such a circle (**Fig. 28.6**). Thus, at all points on a circle of radius r around the conductor, the magnitude B is

$$\text{Magnetic field near a long, straight, current-carrying conductor} \quad B = \frac{\mu_0 I}{2\pi r} \quad (28.9)$$

Magnetic constant μ_0 Current I Distance from conductor r

The geometry in this case is similar to that of Example 21.10 (Section 21.5), in which we solved the problem of the *electric* field caused by an infinite line of charge. The same integral appears in both problems, and the field magnitudes in both problems are proportional to $1/r$. But the lines of $\vec{B}$ in the magnetic problem have completely different shapes than the lines of $\vec{E}$ in the analogous electrical problem. Electric field lines radiate outward from a positive line charge distribution (inward for negative charges). By contrast, magnetic field lines *encircle* the current that acts as their source. Electric field lines due to charges begin and end at those charges, but magnetic field lines always form closed loops and *never* have endpoints, irrespective of the shape of the current-carrying conductor that sets up the field. As we discussed in Section 27.3, this is a consequence of Gauss's law for magnetism, which states that the total magnetic flux through *any* closed surface is always zero:

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{magnetic flux through any closed surface}) \quad (28.10)$$

Any magnetic field line that enters a closed surface must emerge from that surface.

EXAMPLE 28.3 Magnetic field of a single wire

WITH VARIATION PROBLEMS

A long, straight conductor carries a 1.0 A current. At what distance from the axis of the conductor does the resulting magnetic field have magnitude $B = 0.5 \times 10^{-4}$ T (about that of the earth's magnetic field in Pittsburgh)?

IDENTIFY and SET UP The length of a “long” conductor is much greater than the distance from the conductor to the field point. Hence we can use the ideas of this section. The geometry is the same as that of Fig. 28.6, so we use Eq. (28.9). All of the quantities in this equation are known except the target variable, the distance r .

EXECUTE We solve Eq. (28.9) for r :

$$r = \frac{\mu_0 I}{2\pi B} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(1.0 \text{ A})}{(2\pi)(0.5 \times 10^{-4} \text{ T})} = 4 \times 10^{-3} \text{ m} = 4 \text{ mm}$$

EVALUATE As we saw in Example 26.14, currents of an ampere or more are typical of those found in the wiring of home appliances. This example shows that the magnetic fields produced by these appliances are very weak even very close to the wire; the fields are proportional to $1/r$, so they become even weaker at greater distances.

KEYCONCEPT The magnetic field $\vec{B}$ of a long, straight, current-carrying wire is proportional to the current and inversely proportional to the distance from the wire. The field lines are concentric circles around the wire, with the field direction given by a right-hand rule.

EXAMPLE 28.4 Magnetic field of two wires

WITH  **ARIATION PROBLEMS**

Figure 28.7a is an end-on view of two long, straight, parallel wires perpendicular to the xy -plane, each carrying a current I but in opposite directions. (a) Find $\vec{B}$ at points P_1 , P_2 , and P_3 . (b) Find an expression for $\vec{B}$ at any point on the x -axis to the right of wire 2.

IDENTIFY and SET UP We can find the magnetic fields $\vec{B}_1$ and $\vec{B}_2$ due to wires 1 and 2 at each point by using the ideas of this section. By the superposition principle, the magnetic field at each point is then $\vec{B} = \vec{B}_1 + \vec{B}_2$. We use Eq. (28.9) to find the magnitudes B_1 and B_2 of these fields and the right-hand rule to find the corresponding directions. Figure 28.7a shows $\vec{B}_1$, $\vec{B}_2$, and $\vec{B} = \vec{B}_{\text{total}}$ at each point; you should confirm that the directions and relative magnitudes shown are correct. Figure 28.7b shows some of the magnetic field lines due to this two-wire system.

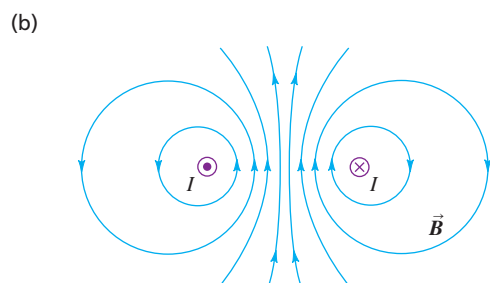
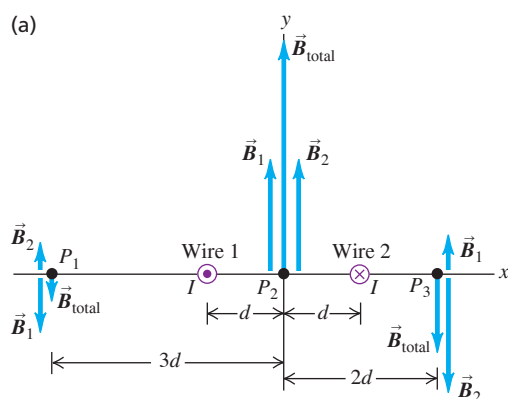
EXECUTE (a) Since point P_1 is a distance $2d$ from wire 1 and a distance $4d$ from wire 2, $B_1 = \mu_0 I / 2\pi(2d) = \mu_0 I / 4\pi d$ and $B_2 = \mu_0 I / 2\pi(4d) = \mu_0 I / 8\pi d$. The right-hand rule shows that $\vec{B}_1$ is in the negative y -direction and $\vec{B}_2$ is in the positive y -direction, so

$$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 = -\frac{\mu_0 I}{4\pi d} \hat{j} + \frac{\mu_0 I}{8\pi d} \hat{j} = -\frac{\mu_0 I}{8\pi d} \hat{j} \quad (\text{point } P_1)$$

At point P_2 , a distance d from both wires, $\vec{B}_1$ and $\vec{B}_2$ are both in the positive y -direction, and both have the same magnitude $B_1 = B_2 = \mu_0 I / 2\pi d$. Hence

$$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{2\pi d} \hat{j} + \frac{\mu_0 I}{2\pi d} \hat{j} = \frac{\mu_0 I}{\pi d} \hat{j} \quad (\text{point } P_2)$$

Figure 28.7 (a) Two long, straight conductors carrying equal currents in opposite directions. The conductors are seen end-on. (b) Map of the magnetic field produced by the two conductors. The field lines are closest together between the conductors, where the field is strongest.



Finally, at point P_3 the right-hand rule shows that $\vec{B}_1$ is in the positive y -direction and $\vec{B}_2$ is in the negative y -direction. This point is a distance $3d$ from wire 1 and a distance d from wire 2, so $B_1 = \mu_0 I / 2\pi(3d) = \mu_0 I / 6\pi d$ and $B_2 = \mu_0 I / 2\pi d$. The total field at P_3 is

$$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{6\pi d} \hat{j} - \frac{\mu_0 I}{2\pi d} \hat{j} = -\frac{\mu_0 I}{3\pi d} \hat{j} \quad (\text{point } P_3)$$

The same technique can be used to find $\vec{B}_{\text{total}}$ at any point; for points off the x -axis, caution must be taken in vector addition, since $\vec{B}_1$ and $\vec{B}_2$ need no longer be simply parallel or antiparallel.

(b) At any point on the x -axis to the right of wire 2 (that is, for $x > d$), $\vec{B}_1$ and $\vec{B}_2$ are in the same directions as at P_3 . Such a point is a distance $x + d$ from wire 1 and a distance $x - d$ from wire 2, so the total field is

$$\begin{aligned} \vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 &= \frac{\mu_0 I}{2\pi(x + d)} \hat{j} - \frac{\mu_0 I}{2\pi(x - d)} \hat{j} \\ &= -\frac{\mu_0 I d}{\pi(x^2 - d^2)} \hat{j} \end{aligned}$$

where we used a common denominator to combine the two terms.

EVALUATE Consider our result from part (b) at a point very far from the wires, so that x is much larger than d . Then the d^2 term in the denominator can be ignored, and the magnitude of the total field is approximately $B_{\text{total}} = \mu_0 I d / \pi x^2$. For one wire, Eq. (28.9) shows that the magnetic field decreases with distance in proportion to $1/x$; for two wires carrying opposite currents, $\vec{B}_1$ and $\vec{B}_2$ partially cancel each other, and so B_{total} decreases more rapidly, in proportion to $1/x^2$. This effect is used in communication systems such as telephone or computer networks. The wiring is arranged so that a conductor carrying a signal in one direction and the conductor carrying the return signal are side by side, as in Fig. 28.7a, or twisted around each other (Fig. 28.8). As a result, the magnetic field due to these signals *outside* the conductors is very small, making it less likely to exert unwanted forces on other information-carrying currents.

Figure 28.8 Computer cables, or cables for audio-video equipment, create little or no magnetic field. This is because within each cable, closely spaced wires carry current in both directions along the length of the cable. The magnetic fields from these opposing currents cancel each other.



KEYCONCEPT Magnetic fields obey the superposition principle: The net magnetic field due to two or more sources is the vector sum of the fields due to the individual sources.

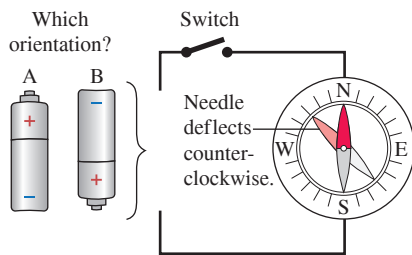
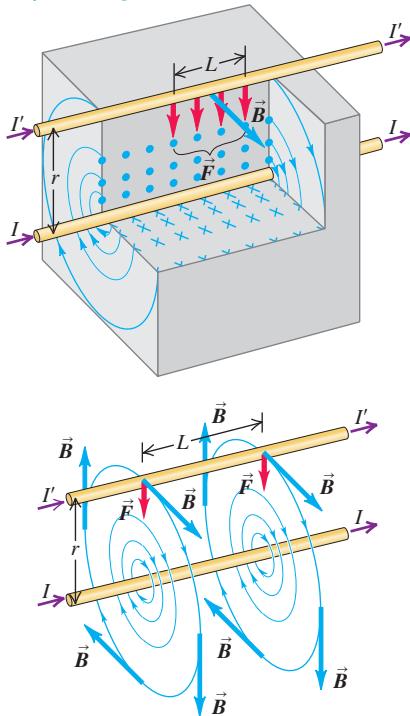


Figure 28.9 Parallel conductors carrying currents in the same direction attract each other. The diagrams show how the magnetic field $\vec{B}$ caused by the current in the lower conductor exerts a force $\vec{F}$ on the upper conductor.

The magnetic field of the lower wire exerts an attractive force on the upper wire. By the same token, the upper wire attracts the lower one.

If the wires had currents in *opposite* directions, they would *repel* each other.



TEST YOUR UNDERSTANDING OF SECTION 28.3 The accompanying figure shows a circuit that lies on a horizontal table. A compass is placed on top of the circuit as shown. A battery is to be connected to the circuit so that when the switch is closed, the compass needle deflects counter-clockwise. In which orientation, A or B, should the battery be placed in the circuit?

ANSWER

A This orientation will cause current to flow clockwise around the circuit. Hence current will flow south through the wire under the compass. From the right-hand rule for the magnetic field produced by a long, straight, current-carrying conductor, this will produce a magnetic field that points to the left at the position of the compass (which lies atop the wire). The combination of the northward magnetic field of the earth and the westward field produced by the current gives a net magnetic field to the north-west, so the compass needle will swing counterclockwise to align with this field.

28.4 FORCE BETWEEN PARALLEL CONDUCTORS

Now that we know how to calculate the magnetic field produced by a long, current-carrying conductor, we can find the *magnetic force* that one such conductor exerts on another. This force plays a role in many practical situations in which current-carrying wires are close to each other. **Figure 28.9** shows segments of two long, straight, parallel conductors separated by a distance r and carrying currents I and I' in the same direction. Each conductor lies in the magnetic field set up by the other, so each experiences a force. The figure shows some of the field lines set up by the current in the lower conductor.

From Eq. (28.9) the lower conductor produces a $\vec{B}$ field that, at the position of the upper conductor, has magnitude

$$B = \frac{\mu_0 I}{2\pi r}$$

From Eq. (27.19) the force that this field exerts on a length L of the upper conductor is $\vec{F} = I'\vec{L} \times \vec{B}$, where the vector $\vec{L}$ is in the direction of the current I' and has magnitude L . Since $\vec{B}$ is perpendicular to the length of the conductor and hence to $\vec{L}$, the magnitude of this force is

$$F = I'LB = \frac{\mu_0 II' L}{2\pi r}$$

and the force *per unit length* F/L is

$$\text{Magnetic force per unit length between two long, straight, parallel, current-carrying conductors} \quad \frac{F}{L} = \frac{\mu_0 II'}{2\pi r} \quad (28.11)$$

μ_0 Magnetic constant
 I Current in first conductor
 I' Current in second conductor
 r Distance between conductors

Applying the right-hand rule to $\vec{F} = I'\vec{L} \times \vec{B}$ shows that the force on the upper conductor is directed *downward*.

The current in the *upper* conductor also sets up a $\vec{B}$ field at the position of the *lower* conductor. Two successive applications of the right-hand rule for vector products (one to find the direction of the $\vec{B}$ field due to the upper conductor, as in Section 28.2, and one to find the direction of the force that this field exerts on the lower conductor, as in Section 27.6) show that the force on the lower conductor is *upward*. Thus *two parallel conductors carrying current in the same direction attract each other*. If the direction of either current is reversed, the forces also reverse. *Parallel conductors carrying currents in opposite directions repel each other*.

Magnetic Forces and the Value of μ_0

We saw in Section 28.1 that the magnetic constant μ_0 is very nearly equal to $4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$. If μ_0 had *exactly* this value, Eq. (28.11) shows that the force per unit length on each of two infinitely long parallel conductors one meter apart, each carrying a current of one ampere, would be

$$\frac{F}{L} = \frac{\mu_0 II'}{2\pi r} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(1 \text{ A})(1 \text{ A})}{2\pi(1 \text{ m})} = 2 \times 10^{-7} \text{ T} \cdot \text{A} = 2 \times 10^{-7} \text{ N/m}$$

Figure 28.12 Magnetic field on the axis of a circular loop. The current in the segment $d\vec{l}$ causes the field $d\vec{B}$, which lies in the xy -plane. The currents in other $d\vec{l}$'s cause $d\vec{B}$'s with different components perpendicular to the x -axis; these components add to zero. The x -components of the $d\vec{B}$'s combine to give the total $\vec{B}$ field at point P .

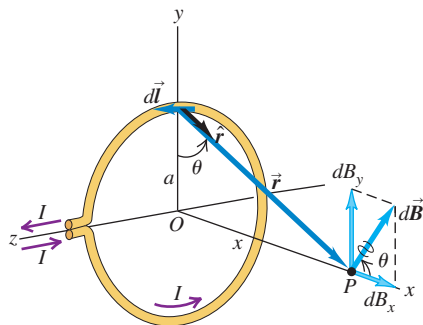


Figure 28.13 The right-hand rule for the direction of the magnetic field produced on the axis of a current-carrying coil.

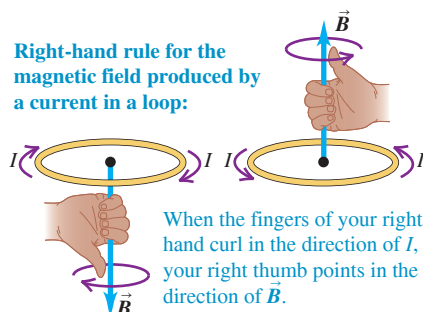
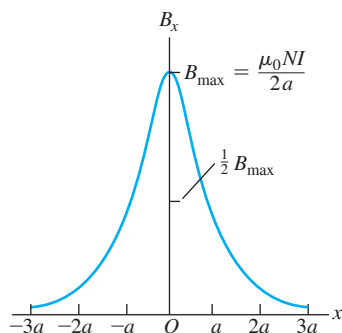


Figure 28.14 Graph of the magnetic field along the axis of a circular coil with N turns. When x is much larger than a , the field magnitude decreases approximately as $1/x^3$.



We can use the law of Biot and Savart, Eq. (28.5) or (28.6), to find the magnetic field at a point P on the axis of the loop, at a distance x from the center. As the figure shows, $d\vec{l}$ and $\vec{r}$ are perpendicular, and the direction of the field $d\vec{B}$ caused by this particular element $d\vec{l}$ lies in the xy -plane. Since $r^2 = x^2 + a^2$, the magnitude dB of the field due to element $d\vec{l}$ is

$$dB = \frac{\mu_0 I}{4\pi} \frac{dl}{(x^2 + a^2)} \quad (28.12)$$

The components of the vector $d\vec{B}$ are

$$dB_x = dB \cos \theta = \frac{\mu_0 I}{4\pi} \frac{dl}{(x^2 + a^2)} \frac{a}{(x^2 + a^2)^{1/2}} \quad (28.13)$$

$$dB_y = dB \sin \theta = \frac{\mu_0 I}{4\pi} \frac{dl}{(x^2 + a^2)} \frac{x}{(x^2 + a^2)^{1/2}} \quad (28.14)$$

The total field $\vec{B}$ at P has only an x -component (it is perpendicular to the plane of the loop). Here's why: For every element $d\vec{l}$ there is a corresponding element on the opposite side of the loop, with opposite direction. These two elements give equal contributions to the x -component of $d\vec{B}$, given by Eq. (28.13), but *opposite* components perpendicular to the x -axis. Thus all the perpendicular components cancel and only the x -components survive.

To obtain the x -component of the total field $\vec{B}$, we integrate Eq. (28.13), including all the $d\vec{l}$'s around the loop. Everything in this expression except dl is constant and can be taken outside the integral, and we have

$$B_x = \int \frac{\mu_0 I}{4\pi} \frac{a dl}{(x^2 + a^2)^{3/2}} = \frac{\mu_0 I a}{4\pi(x^2 + a^2)^{3/2}} \int dl$$

The integral of dl is just the circumference of the circle, $\int dl = 2\pi a$, and so

$$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}} \quad (28.15)$$

Magnetic field on axis of a circular current-carrying loop $\rightarrow B_x$
 Magnetic constant μ_0
 Current I
 Radius of loop a
 Distance along axis from center of loop to field point x

The direction of this magnetic field is given by a right-hand rule. If you curl the fingers of your right hand around the loop in the direction of the current, your right thumb points in the direction of the field (Fig. 28.13).

Magnetic Field on the Axis of a Coil

Now suppose that instead of the single loop in Fig. 28.12 we have a coil consisting of N loops, all with the same radius. The loops are closely spaced so that the plane of each loop is essentially the same distance x from the field point P . Then the total field is N times the field of a single loop:

$$B_x = \frac{\mu_0 N I a^2}{2(x^2 + a^2)^{3/2}} \quad (\text{on the axis of } N \text{ circular loops}) \quad (28.16)$$

The factor N in Eq. (28.16) is the reason coils of wire, not single loops, are used to produce strong magnetic fields; for a desired field strength, using a single loop might require a current I so great as to exceed the rating of the loop's wire.

Figure 28.14 shows a graph of B_x as a function of x . The maximum value of the field is at $x = 0$, the center of the loop or coil:

$$B_x = \frac{\mu_0 N I}{2a} \quad (28.17)$$

Magnetic field at center of N circular current-carrying loops $\rightarrow B_x$
 Magnetic constant μ_0
 Number of loops N
 Current I
 Radius of loop a

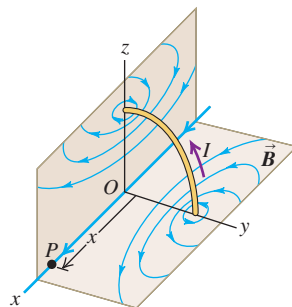
In Section 27.7 we defined the *magnetic dipole moment* μ (or *magnetic moment*) of a current-carrying loop to be equal to IA , where A is the cross-sectional area of the loop. If there are N loops, the total magnetic moment is NIA . The circular loop in Fig. 28.12 has area $A = \pi a^2$, so the magnetic moment of a single loop is $\mu = I\pi a^2$; for N loops, $\mu = NI\pi a^2$. Substituting these results into Eqs. (28.15) and (28.16), we find

$$B_x = \frac{\mu_0 \mu}{2\pi(x^2 + a^2)^{3/2}} \quad (\text{on the axis of any number of circular loops}) \quad (28.18)$$

We described a magnetic dipole in Section 27.7 in terms of its response to a magnetic field produced by currents outside the dipole. But a magnetic dipole is also a *source* of magnetic field; Eq. (28.18) describes the magnetic field *produced* by a magnetic dipole for points along the dipole axis. This field is directly proportional to the magnetic dipole moment μ . Note that the field at all points on the x -axis is in the same direction as the vector magnetic moment $\vec{\mu}$.

Figure 28.15 shows some of the magnetic field lines surrounding a circular current loop (magnetic dipole) in planes through the axis. The directions of the field lines are given by the same right-hand rule as for a long, straight conductor. Grab the conductor with your right hand, with your thumb in the direction of the current; your fingers curl around in the same direction as the field lines. The field lines for the circular current loop are closed curves that encircle the conductor; they are *not* circles, however.

Figure 28.15 Magnetic field lines produced by the current in a circular loop. At points on the axis the $\vec{B}$ field has the same direction as the magnetic moment of the loop.



CAUTION **Magnetic field of a coil** Equations (28.15), (28.16), and (28.18) are valid only on the *axis* of a loop or coil. Don't attempt to apply these equations at other points!

BIO APPLICATION Magnetic Fields for MRI

Magnetic resonance imaging (see Section 27.7) requires a magnetic field of about 1.5 T. In a typical MRI scan, the patient lies inside a coil that produces the intense field. The currents required are very high, so the coils are bathed in liquid helium at a temperature of 4.2 K to keep them from overheating.



EXAMPLE 28.6 Magnetic field of a coil

A coil consisting of 100 circular loops with radius 0.60 m carries a 5.0 A current. (a) Find the magnetic field at a point along the axis of the coil, 0.80 m from the center. (b) Along the axis, at what distance from the center of the coil is the field magnitude $\frac{1}{8}$ as great as it is at the center?

IDENTIFY and SET UP This problem concerns the magnetic-field magnitude B along the axis of a current-carrying coil, so we can use Eq. (28.16). We are given $N = 100$, $I = 5.0$ A, and $a = 0.60$ m. In part (a) our target variable is B_x at a given value of x . In part (b) the target variable is the value of x at which the field has $\frac{1}{8}$ of the magnitude that it has at the origin.

EXECUTE (a) Using $x = 0.80$ m, from Eq. (28.16) we have

$$\begin{aligned} B_x &= \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(100)(5.0 \text{ A})(0.60 \text{ m})^2}{2[(0.80 \text{ m})^2 + (0.60 \text{ m})^2]^{3/2}} \\ &= 1.1 \times 10^{-4} \text{ T} \end{aligned}$$

(b) Considering Eq. (28.16), we seek a value of x such that

$$\frac{1}{(x^2 + a^2)^{3/2}} = \frac{1}{8} \frac{1}{(0^2 + a^2)^{3/2}}$$

To solve this for x , we take the reciprocal of the whole thing and then take the $\frac{2}{3}$ power of both sides; the result is

$$x = \pm \sqrt{3}a = \pm 1.04 \text{ m}$$

EVALUATE We check our answer in part (a) by finding the coil's magnetic moment and substituting the result into Eq. (28.18):

$$\begin{aligned} \mu &= NI\pi a^2 = (100)(5.0 \text{ A})\pi(0.60 \text{ m})^2 = 5.7 \times 10^2 \text{ A} \cdot \text{m}^2 \\ B_x &= \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(5.7 \times 10^2 \text{ A} \cdot \text{m}^2)}{2\pi[(0.80 \text{ m})^2 + (0.60 \text{ m})^2]^{3/2}} = 1.1 \times 10^{-4} \text{ T} \end{aligned}$$

The magnetic moment μ is relatively large, yet it produces a rather small field, comparable to that of the earth. This illustrates how difficult it is to produce strong fields of 1 T or more.

KEYCONCEPT The magnetic field $\vec{B}$ on the axis of a current-carrying coil points along the axis, with a direction given by a right-hand rule. The magnitude of $\vec{B}$ is proportional to the magnetic dipole moment of the coil and is greatest at the center of the coil.

TEST YOUR UNDERSTANDING OF SECTION 28.5 Figure 28.12 shows the magnetic field $d\vec{B}$ produced at point P by a segment $d\vec{l}$ that lies on the positive y -axis (at the top of the loop). This field has components $dB_x > 0$, $dB_y > 0$, $dB_z = 0$. (a) What are the signs of the components of the field $d\vec{B}$ produced at P by a segment $d\vec{l}$ on the negative y -axis (at the bottom of the loop)? (i) $dB_x > 0$, $dB_y > 0$, $dB_z = 0$; (ii) $dB_x > 0$, $dB_y < 0$, $dB_z = 0$; (iii) $dB_x < 0$, $dB_y > 0$, $dB_z = 0$; (iv) $dB_x < 0$, $dB_y < 0$, $dB_z = 0$; (v) none of these. (b) What are the signs of the components of the field $d\vec{B}$ produced at P by a segment $d\vec{l}$ on the negative z -axis (at the right-hand side of the loop)? (i) $dB_x > 0$, $dB_y > 0$, $dB_z = 0$; (ii) $dB_x > 0$, $dB_y < 0$, $dB_z = 0$; (iii) $dB_x < 0$, $dB_y > 0$, $dB_z = 0$; (iv) $dB_x < 0$, $dB_y < 0$, $dB_z = 0$; (v) none of these.

ANSWER

(a) (i) The vector $d\vec{B}$ is in the direction of $d\vec{l} \times \vec{r}$. For a segment on the negative y -axis, $d\vec{l}$ points in the negative y -direction and $\vec{r}$ has a positive x -component, a negative y -component, and a negative z -component. Hence $d\vec{l} \times \vec{r} = 1(a)(-b)\hat{i} - 1(-b)(-c)\hat{j} + 1(a)(-c)\hat{k} = -ab\hat{i} - bc\hat{j} - ac\hat{k}$. Hence $d\vec{B}$ has components $dB_x < 0$, $dB_y < 0$, and $dB_z < 0$. (ii) For a segment on the negative z -axis, $d\vec{l}$ points in the negative z -direction and $\vec{r}$ has a positive x -component, a negative y -component, and zero z -component. Hence $d\vec{l} \times \vec{r} = 1(-a)(-b)\hat{i} - 1(-a)(-c)\hat{j} + 1(-b)(-c)\hat{k} = ab\hat{i} - ac\hat{j} + bc\hat{k}$. Hence $d\vec{B}$ has components $dB_x > 0$, $dB_y < 0$, and $dB_z > 0$. (iii) For a segment on the positive y -axis, $d\vec{l}$ points in the positive y -direction and $\vec{r}$ has a positive x -component, a positive y -component, and zero z -component. Hence $d\vec{l} \times \vec{r} = 1(a)(-b)\hat{i} - 1(a)(-c)\hat{j} + 1(b)(-c)\hat{k} = -ab\hat{i} + ac\hat{j} - bc\hat{k}$. Hence $d\vec{B}$ has components $dB_x < 0$, $dB_y > 0$, and $dB_z < 0$. (iv) For a segment on the positive z -axis, $d\vec{l}$ points in the positive z -direction and $\vec{r}$ has a positive x -component, a positive y -component, and a positive z -component. Hence $d\vec{l} \times \vec{r} = 1(a)(-b)\hat{i} - 1(a)(-c)\hat{j} + 1(b)(-c)\hat{k} = -ab\hat{i} + ac\hat{j} - bc\hat{k}$. Hence $d\vec{B}$ has components $dB_x < 0$, $dB_y > 0$, and $dB_z < 0$. (v) none of these.

28.6 AMPERE'S LAW

So far our calculations of the magnetic field due to a current have involved finding the infinitesimal field $d\vec{B}$ due to a current element and then summing all the $d\vec{B}$'s to find the total field. This approach is directly analogous to our *electric*-field calculations in Chapter 21.

For the electric-field problem we found that in situations with a highly symmetric charge distribution, it was often easier to use Gauss's law to find $\vec{E}$. There is likewise a law that allows us to more easily find the *magnetic* fields caused by highly symmetric *current* distributions. But the law that allows us to do this, called *Ampere's law*, is rather different in character from Gauss's law.

Gauss's law for electric fields (Chapter 22) involves the flux of $\vec{E}$ through a closed surface; it states that this flux is equal to the total charge enclosed within the surface, divided by the constant ϵ_0 . Thus this law relates electric fields and charge distributions. By contrast, Gauss's law for *magnetic* fields, Eq. (28.10), is *not* a relationship between magnetic fields and current distributions; it states that the flux of $\vec{B}$ through *any* closed surface is always zero, whether or not there are currents within the surface. So Gauss's law for $\vec{B}$ can't be used to determine the magnetic field produced by a particular current distribution.

Ampere's law is formulated not in terms of magnetic flux, but rather in terms of the *line integral* of $\vec{B}$ around a closed path, denoted by

$$\oint \vec{B} \cdot d\vec{l}$$

We used line integrals to define work in Chapter 6 and to calculate electric potential in Chapter 23. To evaluate this integral, we divide the path into infinitesimal segments $d\vec{l}$, calculate the scalar product of $\vec{B} \cdot d\vec{l}$ for each segment, and sum these products. In general, $\vec{B}$ varies from point to point, and we must use the value of $\vec{B}$ at the location of each $d\vec{l}$. An alternative notation is $\oint B_{\parallel} dl$, where $B_{\parallel}$ is the component of $\vec{B}$ parallel to $d\vec{l}$ at each point. The circle on the integral sign indicates that this integral is always computed for a *closed* path, one whose beginning and end points are the same.

Ampere's Law for a Long, Straight Conductor

To introduce the basic idea of Ampere's law, let's consider again the magnetic field caused by a long, straight conductor carrying a current I . We found in Section 28.3 that the field at a distance r from the conductor has magnitude

$$B = \frac{\mu_0 I}{2\pi r}$$

The magnetic field lines are circles centered on the conductor. Let's take the line integral of $\vec{B}$ around a circle with radius r , as in **Fig. 28.16a**. At every point on the circle, $\vec{B}$ and $d\vec{l}$ are parallel, and so $\vec{B} \cdot d\vec{l} = B dl$; since r is constant around the circle, B is constant as well. Alternatively, we can say that $B_{\parallel}$ is constant and equal to B at every point on the circle. Hence we can take B outside of the integral. The remaining integral is just the circumference of the circle, so

$$\oint \vec{B} \cdot d\vec{l} = \oint B_{\parallel} dl = B \oint dl = \frac{\mu_0 I}{2\pi r} (2\pi r) = \mu_0 I$$

The line integral is thus independent of the radius of the circle and is equal to μ_0 multiplied by the current passing through the area bounded by the circle.

In **Fig. 28.16b** the situation is the same, but the integration path now goes around the circle in the opposite direction. Now $\vec{B}$ and $d\vec{l}$ are antiparallel, so $\vec{B} \cdot d\vec{l} = -B dl$ and the line integral equals $-\mu_0 I$. We get the same result if the integration path is the same as in **Fig. 28.16a**, but the direction of the current is reversed. Thus $\oint \vec{B} \cdot d\vec{l}$ equals μ_0 multiplied by the current passing through the area bounded by the integration path, with a positive or negative sign depending on the direction of the current relative to the direction of integration.

There's a simple rule for the sign of the current; you won't be surprised to learn that it uses your right hand. Curl the fingers of your right hand around the integration path so that they curl in the direction of integration (that is, the direction that you use to evaluate $\oint \vec{B} \cdot d\vec{l}$). Then your right thumb indicates the positive current direction. Currents that pass through the integration path in this direction are positive; those in the opposite direction are negative. Using this rule, convince yourself that the current is positive in **Fig. 28.16a** and negative in **Fig. 28.16b**. Here's another way to say the same thing: Looking at the surface bounded by the integration path, integrate counterclockwise around the path as in **Fig. 28.16a**. Currents moving toward you through the surface are positive, and those going away from you are negative.

An integration path that does *not* enclose the conductor is used in **Fig. 28.16c**. Along the circular arc ab of radius r_1 , $\vec{B}$ and $d\vec{l}$ are parallel and $B_{\parallel} = B_1 = \mu_0 I / 2\pi r_1$; along the circular arc cd of radius r_2 , $\vec{B}$ and $d\vec{l}$ are antiparallel and $B_{\parallel} = -B_2 = -\mu_0 I / 2\pi r_2$. The $\vec{B}$ field is perpendicular to $d\vec{l}$ at each point on the straight sections bc and da , so $B_{\parallel} = 0$ and these sections contribute zero to the line integral. The total line integral is then

$$\begin{aligned} \oint \vec{B} \cdot d\vec{l} &= \oint B_{\parallel} dl = B_1 \int_a^b dl + (0) \int_b^c dl + (-B_2) \int_c^d dl + (0) \int_d^a dl \\ &= \frac{\mu_0 I}{2\pi r_1} (r_1 \theta) + 0 - \frac{\mu_0 I}{2\pi r_2} (r_2 \theta) + 0 = 0 \end{aligned}$$

The magnitude of $\vec{B}$ is greater on arc cd than on arc ab , but the arc length is less, so the contributions from the two arcs exactly cancel. Even though there is a magnetic field everywhere along the integration path, the line integral $\oint \vec{B} \cdot d\vec{l}$ is zero if there is no current passing through the area bounded by the path.

We can also derive these results for more general integration paths, such as the one in **Fig. 28.17a**. At the position of the line element $d\vec{l}$, the angle between $d\vec{l}$ and $\vec{B}$ is ϕ , and

$$\vec{B} \cdot d\vec{l} = B dl \cos \phi$$

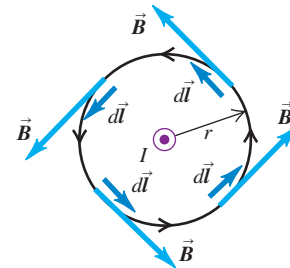
From the figure, $dl \cos \phi = r d\theta$, where $d\theta$ is the angle subtended by $d\vec{l}$ at the position of the conductor and r is the distance of $d\vec{l}$ from the conductor. Thus

$$\oint \vec{B} \cdot d\vec{l} = \oint \frac{\mu_0 I}{2\pi r} (r d\theta) = \frac{\mu_0 I}{2\pi} \oint d\theta$$

Figure 28.16 Three integration paths for the line integral of $\vec{B}$ in the vicinity of a long, straight conductor carrying current I out of the plane of the page (as indicated by the circle with a dot). The conductor is seen end-on.

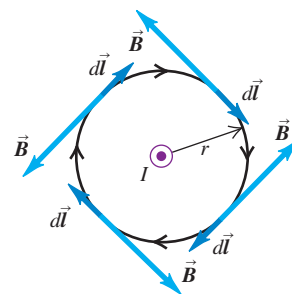
(a) Integration path is a circle centered on the conductor; integration goes around the circle counterclockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$



(b) Same integration path as in (a), but integration goes around the circle clockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = -\mu_0 I$



(c) An integration path that does not enclose the conductor

Result: $\oint \vec{B} \cdot d\vec{l} = 0$

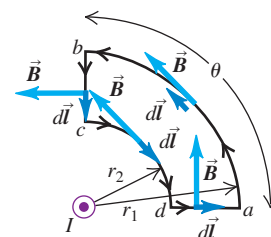


Figure 28.17 (a) A more general integration path for the line integral of $\vec{B}$ around a long, straight conductor carrying current I out of the plane of the page. The conductor is seen end-on. (b) A more general integration path that does not enclose the conductor.

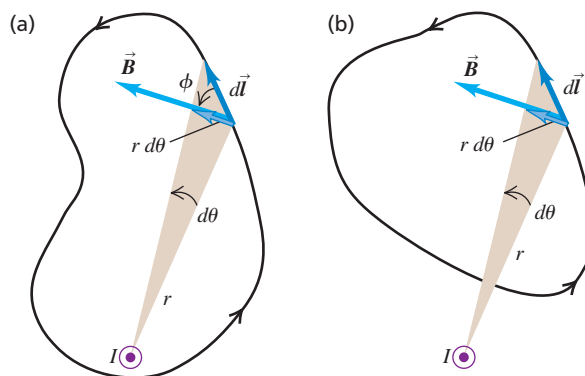
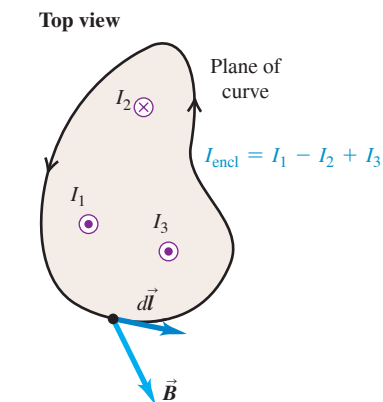
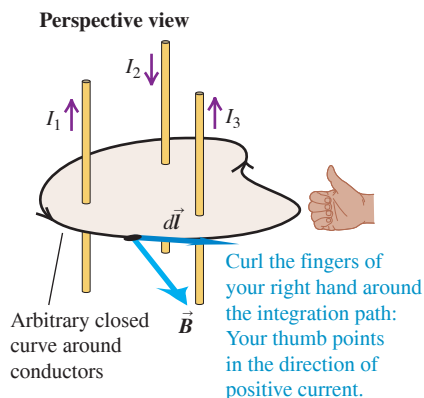
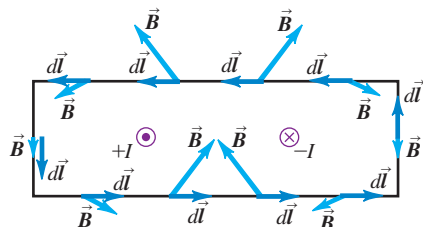


Figure 28.18 Ampere's law.



Ampere's law: If we calculate the line integral of the magnetic field around a closed curve, the result equals μ_0 times the total enclosed current: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$.

Figure 28.19 An end-on view of two long, straight conductors carrying equal currents in opposite directions. The line integral $\oint \vec{B} \cdot d\vec{l}$ gets zero contribution from the upper and lower segments, a positive contribution from the left segment, and a negative contribution from the right segment; the net integral is zero.



But $\oint d\theta$ is just equal to 2π , the total angle swept out by the radial line from the conductor to $d\vec{l}$ during a complete trip around the path. So we get

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I \quad (28.19)$$

This result doesn't depend on the shape of the path or on the position of the wire inside it. If the current in the wire is opposite to that shown, the integral has the opposite sign. But if the path doesn't enclose the wire (Fig. 28.17b), then the net change in θ during the trip around the integration path is zero; $\oint d\theta$ is zero instead of 2π and the line integral is zero.

Ampere's Law: General Statement

We can generalize Ampere's law even further. Suppose *several* long, straight conductors pass through the surface bounded by the integration path. The total magnetic field $\vec{B}$ at any point on the path is the vector sum of the fields produced by the individual conductors. Thus the line integral of the total $\vec{B}$ equals μ_0 times the *algebraic sum* of the currents. In calculating this sum, we use the sign rule for currents described above. If the integration path does not enclose a particular wire, the line integral of the $\vec{B}$ field of that wire is zero, because the angle θ for that wire sweeps through a net change of zero rather than 2π during the integration. Any conductors present that are not enclosed by a particular path may still contribute to the value of $\vec{B}$ at every point, but the *line integrals* of their fields around the path are zero.

Thus we can replace I in Eq. (28.19) with I_{encl} , the algebraic sum of the currents *enclosed* or *linked* by the integration path, with the sum evaluated by using the sign rule just described (Fig. 28.18). Then **Ampere's law** says

Line integral around a closed path

Magnetic constant

Net current enclosed by path

Scalar product of magnetic field and vector segment of path

Ampere's law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$ (28.20)

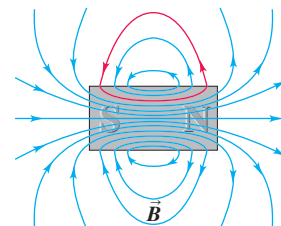
While we have derived Ampere's law only for the special case of the field of several long, straight, parallel conductors, Eq. (28.20) is in fact valid for conductors and paths of *any* shape. The general derivation is no different in principle from what we have presented, but the geometry is more complicated.

If $\oint \vec{B} \cdot d\vec{l} = 0$, it *does not* necessarily mean that $\vec{B} = \mathbf{0}$ everywhere along the path, only that the total current through an area bounded by the path is zero. In Figs. 28.16c and 28.17b, the integration paths enclose no current at all; in **Fig. 28.19** there are positive and negative currents of equal magnitude through the area enclosed by the path. In both cases, $I_{\text{encl}} = 0$ and the line integral is zero.

Equation (28.20) turns out to be valid *only* if the currents are steady and if no time-varying electric fields are present. In Chapter 29 we'll see how to generalize Ampere's law for time-varying fields.

CAUTION **Line integrals of electric and magnetic fields** In Chapter 23 we saw that the line integral of the electrostatic field $\vec{E}$ around any closed path is equal to zero; this is a statement that the electrostatic force $\vec{F} = q\vec{E}$ on a point charge q is conservative, so this force does zero work on a charge that moves around a closed path that returns to the starting point. The value of the line integral $\oint \vec{B} \cdot d\vec{l}$ is not similarly related to the question of whether the *magnetic* force is conservative. Remember that the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ on a moving charged particle is always *perpendicular* to $\vec{B}$, so $\oint \vec{B} \cdot d\vec{l}$ is *not* related to the work done by the magnetic force; as stated in Ampere's law, this integral is related only to the total current through a surface bounded by the integration path. In fact, the magnetic force on a moving charged particle is *not* conservative. A conservative force depends on only the position of the object on which the force is exerted, but the magnetic force on a moving charged particle also depends on the *velocity* of the particle. ■

TEST YOUR UNDERSTANDING OF SECTION 28.6 The accompanying figure shows magnetic field lines through the center of a permanent magnet. The magnet is not connected to a source of emf. One of the field lines is colored red. What can you conclude about the currents inside the permanent magnet within the region enclosed by this field line? (i) There are no currents inside the magnet; (ii) there are currents directed out of the plane of the page; (iii) there are currents directed into the plane of the page; (iv) not enough information is given to decide.



ANSWER

(ii) Imagine carrying out the integral $\oint \vec{B} \cdot d\vec{l}$ along an integration path that goes counterclockwise around the red magnetic field line. At each point along the path the magnetic field $\vec{B}$ and the infinitesimal segment $d\vec{l}$ are both tangent to the path, so $\vec{B} \cdot d\vec{l}$ is positive at each point and the integral $\oint \vec{B} \cdot d\vec{l}$ is likewise positive. It follows from Ampere's law $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$ and the right-hand rule that the integration path encloses a current directed out of the plane of the page. There are no currents in the empty space outside the magnet, so there must be currents inside the magnet (see Section 28.8). ■

28.7 APPLICATIONS OF AMPERE'S LAW

Below are several examples of using Ampere's law to calculate the magnetic field due to a current distribution. Problem-Solving Strategy 28.2 is directly analogous to Problem-Solving Strategy 22.1 (Section 22.4) for using Gauss's law to calculate the electric field due to a charge distribution; we suggest you review that strategy now.

CAUTION **Ampere's law is powerful but limited** Like Gauss's law for $\vec{E}$, Ampere's law for $\vec{B}$ can be used to calculate the field *only* in highly symmetric situations. As the next examples show, in such situations you can exploit the symmetry to calculate the line integral of $\vec{B}$ around a simple path. ■

PROBLEM-SOLVING STRATEGY 28.2 Ampere's Law

IDENTIFY the relevant concepts: Ampere's law, $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$, can yield the magnitude of $\vec{B}$ as a function of position if the geometry of the field-generating electric current is highly symmetric.

SET UP the problem using the following steps:

1. Determine the target variable(s). Usually one will be the magnitude of the $\vec{B}$ field as a function of position.
2. Select the integration path to use with Ampere's law. To determine the magnetic field at a certain point, the path must pass through that point. The integration path doesn't have to be any actual physical boundary; it may be in empty space, embedded in a solid object, or some of each. The integration path has to have enough *symmetry* to make evaluation of the integral possible. Ideally the path will be tangent to $\vec{B}$ in regions of interest; elsewhere the path should be perpendicular to $\vec{B}$ or should run through regions where $\vec{B} = 0$.

EXECUTE the solution as follows:

1. Carry out the integral $\oint \vec{B} \cdot d\vec{l}$ along the chosen path. If $\vec{B}$ is tangent to all or some portion of the path and has the same magnitude B at every point, then its line integral is the product of B and the length of that portion of the path. If $\vec{B}$ is perpendicular

to some portion of the path, or if $\vec{B} = 0$, that portion makes no contribution to the integral.

2. In the integral $\oint \vec{B} \cdot d\vec{l}$, $\vec{B}$ is the *total* magnetic field at each point on the path; it can be caused by currents enclosed *or not enclosed* by the path. If *no* net current is enclosed by the path, the field at points on the path need not be zero, but the integral $\oint \vec{B} \cdot d\vec{l}$ is always zero.
3. Determine the current I_{enc} enclosed by the integration path. A right-hand rule gives the sign of this current: If you curl the fingers of your right hand so that they follow the path in the direction of integration, then your right thumb points in the direction of positive current. If $\vec{B}$ is tangent to the path everywhere and I_{enc} is positive, the direction of $\vec{B}$ is the same as the direction of integration. If instead I_{enc} is negative, $\vec{B}$ is in the direction opposite to that of the integration.
4. Use Ampere's law $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$ to solve for the target variable.

EVALUATE your answer: If your result is an expression for the field magnitude as a function of position, check it by examining how the expression behaves in different limits.

EXAMPLE 28.7 Field of a long, straight, current-carrying conductor

In Section 28.6 we derived Ampere's law from Eq. (28.9) for the field $\vec{B}$ of a long, straight, current-carrying conductor. Reverse this process, and use Ampere's law to find $\vec{B}$ for this situation.

IDENTIFY and SET UP The situation has cylindrical symmetry, so in Ampere's law we take our integration path to be a circle with radius r centered on the conductor and lying in a plane perpendicular to it, as in Fig. 28.16a. The field $\vec{B}$ is everywhere tangent to this circle and has the same magnitude B everywhere on the circle.

EXECUTE With our choice of the integration path, Ampere's law [Eq. (28.20)] becomes

$$\oint \vec{B} \cdot d\vec{l} = \oint B_{\parallel} dl = B(2\pi r) = \mu_0 I$$

Equation (28.9), $B = \mu_0 I / 2\pi r$, follows immediately.

Ampere's law determines the direction of $\vec{B}$ as well as its magnitude. Since we chose to go counterclockwise around the integration path, the positive direction for current is out of the plane of Fig. 28.16a; this is the same as the actual current direction in the figure, so I is positive and the integral $\oint \vec{B} \cdot d\vec{l}$ is also positive. Since the $d\vec{l}$'s run counterclockwise, the direction of $\vec{B}$ must be counterclockwise as well, as shown in Fig. 28.16a.

EVALUATE Our results are consistent with those in Section 28.6.

KEYCONCEPT Ampere's law is easiest to use if the magnetic field $\vec{B}$ is everywhere tangent to an integration path and has the same magnitude at all points along that path.

EXAMPLE 28.8 Field of a long cylindrical conductor**WITH VARIATION PROBLEMS**

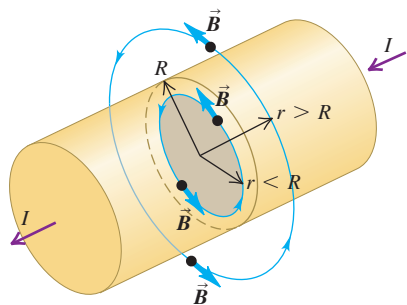
A cylindrical conductor with radius R carries a current I (Fig. 28.20). The current is uniformly distributed over the cross-sectional area of the conductor. Find the magnetic field as a function of the distance r from the conductor axis for points both inside ($r < R$) and outside ($r > R$) the conductor.

IDENTIFY and SET UP As in Example 28.7, the current distribution has cylindrical symmetry, and the magnetic field lines must be circles concentric with the conductor axis. To find the magnetic field inside and outside the conductor, we choose circular integration paths with radii $r < R$ and $r > R$, respectively (see Fig. 28.20).

EXECUTE In either case the field $\vec{B}$ has the same magnitude at every point on the circular integration path and is tangent to the path. Thus the magnitude of the line integral is simply $B(2\pi r)$. To find the current I_{encl} enclosed by a circular integration path inside the conductor ($r < R$), note that the current density (current per unit area) is $J = I/\pi R^2$ so $I_{\text{encl}} = J(\pi r^2) = Ir^2/R^2$. Hence Ampere's law gives $B(2\pi r) = \mu_0 Ir^2/R^2$, or

$$B = \frac{\mu_0 I}{2\pi} \frac{r}{R^2} \quad (\text{inside the conductor, } r < R) \quad (28.21)$$

Figure 28.20 To find the magnetic field at radius $r < R$, we apply Ampere's law to the circle enclosing the gray area. The current through the gray area is $(r^2/R^2)I$. To find the magnetic field at radius $r > R$, we apply Ampere's law to the circle enclosing the entire conductor.



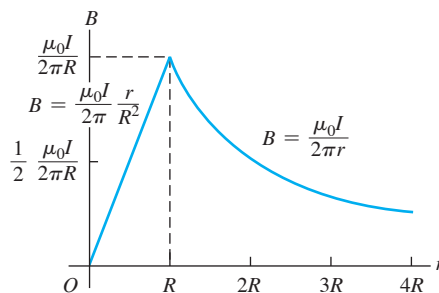
A circular integration path outside the conductor encloses the total current in the conductor, so $I_{\text{encl}} = I$. Applying Ampere's law gives the same equation as in Example 28.7, with the same result for B :

$$B = \frac{\mu_0 I}{2\pi r} \quad (\text{outside the conductor, } r > R) \quad (28.22)$$

Outside the conductor, the magnetic field is the same as that of a long, straight conductor carrying current I , independent of the radius R over which the current is distributed. Indeed, the magnetic field outside *any* cylindrically symmetric current distribution is the same as if the entire current were concentrated along the axis of the distribution. This is analogous to the results of Examples 22.5 and 22.9 (Section 22.4), in which we found that the *electric* field outside a spherically symmetric *charged* object is the same as though the entire charge were concentrated at the center.

EVALUATE At the surface of the conductor ($r = R$), Eqs. (28.21) and (28.22) agree, as they must. **Figure 28.21** shows a graph of B as a function of r .

Figure 28.21 Magnitude of the magnetic field inside and outside a long, straight cylindrical conductor with radius R carrying a current I .



KEYCONCEPT The magnetic field lines produced by a solid, cylindrical, current-carrying conductor are concentric circles around the conductor, with the field direction given by a right-hand rule. Inside the conductor the field magnitude increases with increasing distance from the center; outside the conductor the magnitude decreases with distance.

EXAMPLE 28.9 Field of a solenoid

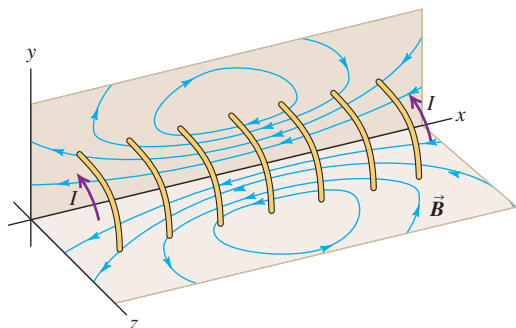
WITH VARIATION PROBLEMS

A **solenoid** consists of a helical winding of wire on a cylinder, usually circular in cross section. There can be thousands of closely spaced turns (often in several layers), each of which can be regarded as a circular loop. For simplicity, **Fig. 28.22** shows a solenoid with only a few turns. All turns carry the same current I , and the total $\vec{B}$ field at every point is the vector sum of the fields caused by the individual turns. The figure shows field lines in the xy - and xz -planes. We draw field lines that are uniformly spaced at the center of the solenoid. Exact calculations show that for a long, closely wound solenoid, half of these field lines emerge from the ends and half “leak out” through the windings between the center and the end, as the figure suggests.

If the solenoid is long in comparison with its cross-sectional diameter and the coils are tightly wound, the field inside the solenoid near its midpoint is very nearly uniform over the cross section and parallel to the axis; the *external* field near the midpoint is very small.

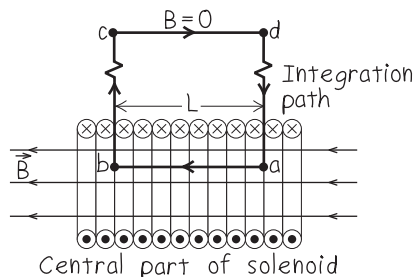
Use Ampere's law to find the field at or near the center of such a solenoid if it has n turns per unit length and carries current I .

Figure 28.22 Magnetic field lines produced by the current in a solenoid. For clarity, only a few turns are shown.



IDENTIFY and SET UP We assume that $\vec{B}$ is uniform inside the solenoid and zero outside. **Figure 28.23** shows the situation and our chosen integration path, rectangle $abcd$. Side ab , with length L , is parallel to the axis of the solenoid. Sides bc and da are taken to be very long so that side cd is far from the solenoid; then the field at side cd is negligibly small.

Figure 28.23 Our sketch for this problem.



EXECUTE Along side ab , $\vec{B}$ is parallel to the path and is constant. Our Ampere's-law integration takes us along side ab in the same direction as $\vec{B}$, so here $B_{\parallel} = +B$ and

$$\int_a^b \vec{B} \cdot d\vec{l} = BL$$

Along sides bc and da , $\vec{B}$ is perpendicular to the path, and so $B_{\parallel} = 0$; along side cd , $\vec{B} = 0$ and so $B_{\parallel} = 0$. Around the entire closed path, then, we have $\oint \vec{B} \cdot d\vec{l} = BL$.

In a length L there are nL turns, each of which passes once through $abcd$ carrying current I . Hence the total current enclosed by the rectangle is $I_{\text{encl}} = nLI$. The integral $\oint \vec{B} \cdot d\vec{l}$ is positive, so from Ampere's law I_{encl} must be positive as well. This means that the current passing through the surface bounded by the integration path must be in the direction shown in Fig. 28.23. Ampere's law then gives $BL = \mu_0 nLI$, or

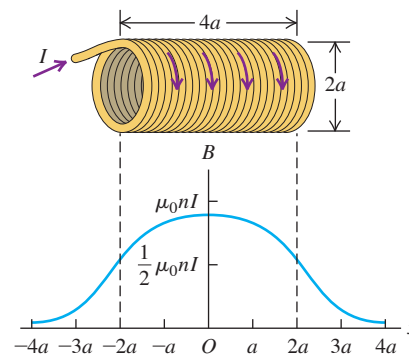
$$B = \mu_0 nI \quad (\text{solenoid}) \quad (28.23)$$

Side ab need not lie on the axis of the solenoid, so this result demonstrates that the field is uniform over the entire cross section at the center of the solenoid's length.

EVALUATE Note that the *direction* of $\vec{B}$ inside the solenoid is in the same direction as the solenoid's vector magnetic moment $\vec{\mu}$, as we found in Section 28.5 for a single current-carrying loop.

For points along the axis, the field is strongest at the center of the solenoid and drops off near the ends. For a solenoid very long in comparison to its diameter, the field magnitude at each end is exactly half that at the center. This is approximately the case even for a relatively short solenoid, as **Fig. 28.24** shows.

Figure 28.24 Magnitude of the magnetic field at points along the axis of a solenoid with length $4a$, equal to four times its radius a . The field magnitude at each end is about half its value at the center. (Compare with Fig. 28.14 for the field of N circular loops.)



KEYCONCEPT The magnetic field $\vec{B}$ of an ideal solenoid is uniform inside the solenoid and zero outside. The direction of $\vec{B}$ inside the solenoid is given by the same right-hand rule as for the field along the axis of a current-carrying coil (see Example 28.6).

EXAMPLE 28.10 Field of a toroidal solenoid

WITH VARIATION PROBLEMS

Figure 28.25a shows a doughnut-shaped **toroidal solenoid**, tightly wound with N turns of wire carrying a current I . (In a practical solenoid the turns would be much more closely spaced than they are in the figure.) Find the magnetic field at all points.

IDENTIFY and SET UP Ignoring the slight pitch of the helical windings, we can consider each turn of a tightly wound toroidal solenoid as a loop lying in a plane perpendicular to the large, circular axis of the toroid. The symmetry of the situation then tells us that the magnetic field lines must be circles concentric with the toroid axis. Therefore we choose circular integration paths for use with Ampere's law, so that the field $\vec{B}$ (if any) is tangent to each path at all points along the path. Figure 28.25b shows three such paths.

EXECUTE Along each path, $\oint \vec{B} \cdot d\vec{l}$ equals the product of B and the path circumference $l = 2\pi r$. The total current enclosed by path 1 is zero, so from Ampere's law the field $\vec{B} = \vec{0}$ everywhere on this path.

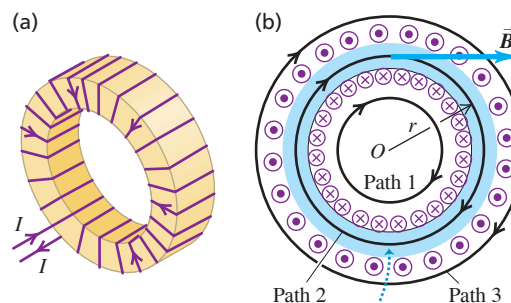
Each turn of the winding passes *twice* through the area bounded by path 3, carrying equal currents in opposite directions. The *net* current enclosed is therefore zero, and hence $\vec{B} = \vec{0}$ at all points on this path as well. We conclude that *the field of an ideal toroidal solenoid is confined to the space enclosed by the windings*. We can think of such a solenoid as a tightly wound, straight solenoid that has been bent into a circle.

For path 2, we have $\oint \vec{B} \cdot d\vec{l} = 2\pi rB$. Each turn of the winding passes *once* through the area bounded by this path, so $I_{\text{enc}} = NI$. We note that I_{enc} is positive for the clockwise direction of integration in Fig. 28.25b, so $\vec{B}$ is in the direction shown. Ampere's law then says that $2\pi rB = \mu_0 NI$, so

$$B = \frac{\mu_0 NI}{2\pi r} \quad (\text{toroidal solenoid}) \quad (28.24)$$

EVALUATE Equation (28.24) indicates that B is *not* uniform over the interior of the core, because different points in the interior are different distances r from the toroid axis. However, if the radial extent of the core is small in comparison to r , the variation is slight. In that case, considering that $2\pi r$ is the circumferential length of the toroid and that $N/2\pi r$

Figure 28.25 (a) A toroidal solenoid. For clarity, only a few turns of the winding are shown. (b) Integration paths (black circles) used to compute the magnetic field $\vec{B}$ set up by the current (shown as dots and crosses).



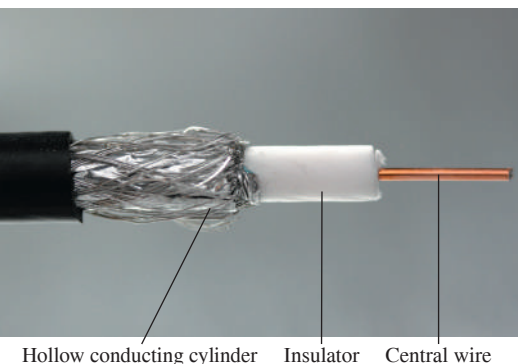
The magnetic field is confined almost entirely to the space enclosed by the windings (in blue).

is the number of turns per unit length n , the field may be written as $B = \mu_0 nI$, just as it is at the center of a long, *straight* solenoid.

In a real toroidal solenoid the turns are not precisely circular loops but rather segments of a bent helix. As a result, the external field is not exactly zero. To estimate its magnitude, we imagine Fig. 28.25a as being *very* roughly equivalent, for points outside the torus, to a *single-turn* circular loop with radius r . At the center of such a loop, Eq. (28.17) gives $B = \mu_0 I/2r$; this is smaller than the field inside the solenoid by the factor N/π .

The equations we have derived for the field in a closely wound straight or toroidal solenoid are strictly correct only for windings in *vacuum*. For most practical purposes, however, they can be used for windings in air or on a core of any nonmagnetic, nonsuperconducting material. In the next section we'll show how these equations are modified if the core is a magnetic material.

KEYCONCEPT Inside the space enclosed by the windings of an ideal toroidal solenoid, the magnetic field lines are circles that follow the curvature of the toroid. Outside the windings, the magnetic field $\vec{B}$ is zero.



TEST YOUR UNDERSTANDING OF SECTION 28.7 In a *coaxial cable* (see photo), a conducting wire runs along the central axis of a hollow conducting cylinder. (The cable that connects a television set to a local cable provider is an example of a coaxial cable.) In such a cable a current I runs in one direction along the hollow conducting cylinder and is spread uniformly over the cylinder's cross-sectional area. An equal current runs in the opposite direction along the central wire. How does the magnitude B of the magnetic field outside such a cable depend on the distance r from the central axis of the cable? (i) B is proportional to $1/r$; (ii) B is proportional to $1/r^2$; (iii) B is zero at all points outside the cable.

ANSWER

(iii) By symmetry, any $\vec{B}$ field outside the cable must circulate around the cable, with circular field lines like those surrounding the solid cylindrical conductor in Fig. 28.20. Choose an integration path like the one shown in Fig. 28.20 with radius $r > R$, so that the path completely encloses the cable. As in Example 28.8, the integral $\oint \vec{B} \cdot d\vec{l}$ for this path has magnitude $B(2\pi r)$. From Ampere's law this is equal to $\mu_0 I_{\text{enc}}$. The net enclosed current I_{enc} is zero because it includes two currents of equal magnitude but opposite direction: one in the central wire and one in the hollow cylinder. Hence $B(2\pi r) = 0$, and so $B = 0$ for any value of r outside the cable. (The field is non-zero *inside* the cable; see Exercise 28.39.)

28.8 MAGNETIC MATERIALS

In discussing how currents cause magnetic fields, we have assumed that the conductors are surrounded by vacuum. But the coils in transformers, motors, generators, and electromagnets nearly always have iron cores to increase the magnetic field and confine it to desired regions. Permanent magnets, magnetic recording tapes, and computer disks depend directly on the magnetic properties of materials; when you store information on a computer disk, you are actually setting up an array of microscopic permanent magnets on the disk. So it is worthwhile to examine some aspects of the magnetic properties of materials. After describing the atomic origins of magnetic properties, we'll discuss three broad classes of magnetic behavior that occur in materials; these are called *paramagnetism*, *diamagnetism*, and *ferromagnetism*.

The Bohr Magneton

As we discussed briefly in Section 27.7, the atoms that make up all matter contain moving electrons, and these electrons form microscopic current loops that produce magnetic fields of their own. In many materials these currents are randomly oriented and cause no net magnetic field. But in some materials an external field (a field produced by currents outside the material) can cause these loops to become oriented preferentially with the field, so their magnetic fields *add* to the external field. We then say that the material is *magnetized*.

Let's look at how these microscopic currents come about. **Figure 28.26** shows a primitive model of an electron in an atom. We picture the electron (mass m , charge $-e$) as moving in a circular orbit with radius r and speed v . This moving charge is equivalent to a current loop. In Section 27.7 we found that a current loop with area A and current I has a magnetic dipole moment μ given by $\mu = IA$; for the orbiting electron the area of the loop is $A = \pi r^2$. To find the current associated with the electron, we note that the orbital period T (the time for the electron to make one complete orbit) is the orbit circumference divided by the electron speed: $T = 2\pi r/v$. The equivalent current I is the total charge passing any point on the orbit per unit time, which is just the magnitude e of the electron charge divided by the orbital period T :

$$I = \frac{e}{T} = \frac{ev}{2\pi r}$$

The magnetic moment $\mu = IA$ is then

$$\mu = \frac{ev}{2\pi r}(\pi r^2) = \frac{evr}{2} \quad (28.25)$$

It is useful to express μ in terms of the *angular momentum* L of the electron. For a particle moving in a circular path, the magnitude of angular momentum equals the magnitude of momentum mv multiplied by the radius r —that is, $L = mvr$ (see Section 10.5). Comparing this with Eq. (28.25), we can write

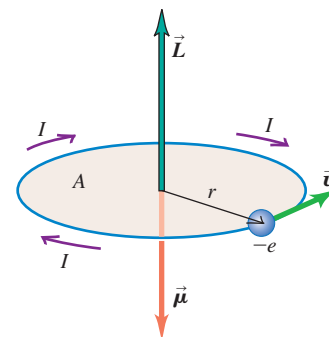
$$\mu = \frac{e}{2m}L \quad (28.26)$$

Equation (28.26) is useful in this discussion because atomic angular momentum is *quantized*; its component in a particular direction is always an integer multiple of $h/2\pi$, where h is a fundamental physical constant called *Planck's constant*. The numerical value of h is

$$h = 6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$$

The quantity $h/2\pi$ thus represents a fundamental unit of angular momentum in atomic systems, just as e is a fundamental unit of charge. Associated with the quantization of $\vec{L}$ is a fundamental uncertainty in the *direction* of $\vec{L}$ and therefore of $\vec{\mu}$. In the following discussion, when we speak of the magnitude of a magnetic moment, a more precise statement would be “maximum component in a given direction.” Thus, to say that a magnetic moment $\vec{\mu}$ is aligned with a magnetic field $\vec{B}$ really means that $\vec{\mu}$ has its maximum possible component in the direction of $\vec{B}$; such components are always quantized.

Figure 28.26 An electron moving with speed v in a circular orbit of radius r has an angular momentum $\vec{L}$ and an oppositely directed orbital magnetic dipole moment $\vec{\mu}$. It also has a spin angular momentum and an oppositely directed spin magnetic dipole moment.



CAUTION Planck's constant and SI units

We first introduced Planck's constant h in Section 1.3. Since h has units of $\text{J} \cdot \text{s} = \text{kg} \cdot \text{m}^2/\text{s}$, its *defined* value, combined with the definitions of the meter and the second, serves as the basis for the definition of the kilogram. Physicists define the kilogram this way because the value of h is relatively easy to measure in any physics laboratory, so it provides a useful standard for determining masses in kilograms. **|**

Equation (28.26) shows that associated with the fundamental unit of angular momentum is a corresponding fundamental unit of magnetic moment. If $L = h/2\pi$, then

$$\mu = \frac{e}{2m} \left(\frac{h}{2\pi} \right) = \frac{eh}{4\pi m} \quad (28.27)$$

This quantity is called the **Bohr magneton**, denoted by μ_B . Its numerical value is

$$\mu_B = 9.274 \times 10^{-24} \text{ A} \cdot \text{m}^2 = 9.274 \times 10^{-24} \text{ J/T}$$

You should verify that these two sets of units are consistent. The second set is useful when we compute the potential energy $U = -\vec{\mu} \cdot \vec{B}$ for a magnetic moment in a magnetic field.

Electrons also have an intrinsic angular momentum, called *spin*, that is not related to orbital motion but that can be pictured in a classical model as spinning on an axis. This angular momentum also has an associated magnetic moment, and its magnitude turns out to be almost exactly one Bohr magneton. (Effects having to do with quantization of the electromagnetic field cause the spin magnetic moment to be about $1.001 \mu_B$.)

Paramagnetism

In an atom, most of the various orbital and spin magnetic moments of the electrons add up to zero. However, in some cases the atom has a net magnetic moment that is of the order of μ_B . When such a material is placed in a magnetic field, the field exerts a torque on each magnetic moment, as given by Eq. (27.26): $\vec{\tau} = \vec{\mu} \times \vec{B}$. These torques tend to align the magnetic moments with the field, as we discussed in Section 27.7. In this position, the directions of the current loops are such as to *add* to the externally applied magnetic field.

We saw in Section 28.5 that the $\vec{B}$ field produced by a current loop is proportional to the loop's magnetic dipole moment. In the same way, the additional $\vec{B}$ field produced by microscopic electron current loops is proportional to the total magnetic moment $\vec{\mu}_{\text{total}}$ per unit volume V in the material. We call this vector quantity the **magnetization** of the material, denoted by $\vec{M}$:

$$\vec{M} = \frac{\vec{\mu}_{\text{total}}}{V} \quad (28.28)$$

The additional magnetic field due to magnetization of the material turns out to be equal simply to $\mu_0 \vec{M}$, where μ_0 is the same constant that appears in the law of Biot and Savart and Ampere's law. When such a material completely surrounds a current-carrying conductor, the total magnetic field $\vec{B}$ in the material is

$$\vec{B} = \vec{B}_0 + \mu_0 \vec{M} \quad (28.29)$$

where $\vec{B}_0$ is the field caused by the current in the conductor.

To check that the units in Eq. (28.29) are consistent, note that magnetization $\vec{M}$ is the magnetic moment per unit volume. The units of magnetic moment are current times area ($\text{A} \cdot \text{m}^2$), so the units of magnetization are $(\text{A} \cdot \text{m}^2)/\text{m}^3 = \text{A/m}$. From Section 28.1, the units of the constant μ_0 are $\text{T} \cdot \text{m/A}$. So the units of $\mu_0 \vec{M}$ are the same as the units of $\vec{B}$: $(\text{T} \cdot \text{m/A})(\text{A/m}) = \text{T}$.

A material showing the behavior just described is said to be **paramagnetic**. The result is that the magnetic field at any point in such a material is greater by a dimensionless factor K_m , called the **relative permeability** of the material, than it would be if the material were replaced by vacuum. The value of K_m is different for different materials; for common paramagnetic solids and liquids at room temperature, K_m typically ranges from 1.00001 to 1.003.

All of the equations in this chapter that relate magnetic fields to their sources can be adapted to the situation in which the current-carrying conductor is embedded in a paramagnetic material. All that need be done is to replace μ_0 by $K_m \mu_0$. This product is usually denoted as μ and is called the **permeability** of the material:

$$\mu = K_m \mu_0 \quad (28.30)$$

CAUTION Two meanings of the symbol μ

Equation (28.30) involves dangerous notation because we use μ for magnetic dipole moment as well as for permeability, as is customary. But beware: From now on, every time you see a μ , make sure you know whether it is permeability or magnetic moment. You can usually tell from the context. ■

The amount by which the relative permeability differs from unity is called the **magnetic susceptibility**, denoted by χ_m :

$$\chi_m = K_m - 1 \quad (28.31)$$

Both K_m and χ_m are dimensionless quantities. **Table 28.1** lists values of magnetic susceptibility for several materials. For example, for aluminum, $\chi_m = 2.2 \times 10^{-5}$ and $K_m = 1.000022$. The first group in the table consists of paramagnetic materials; we'll soon discuss the second group, which contains *diamagnetic* materials.

The tendency of atomic magnetic moments to align themselves parallel to the magnetic field (where the potential energy is minimum) is opposed by random thermal motion, which tends to randomize their orientations. For this reason, paramagnetic susceptibility always decreases with increasing temperature. In many cases it is inversely proportional to the absolute temperature T , and the magnetization M can be expressed as

$$M = C \frac{B}{T} \quad (28.32)$$

This relationship is called *Curie's law*, after its discoverer, Pierre Curie (1859–1906). The quantity C is a constant, different for different materials, called the *Curie constant*.

As we described in Section 27.7, an object with atomic magnetic dipoles is attracted to the poles of a magnet. In most paramagnetic substances this attraction is very weak due to thermal randomization of the atomic magnetic moments. But at very low temperatures the thermal effects are reduced, the magnetization increases in accordance with Curie's law, and the attractive forces are greater.

TABLE 28.1 Magnetic Susceptibilities of Paramagnetic and Diamagnetic Materials at $T = 20^\circ\text{C}$

Material	$\chi_m = K_m - 1 (\times 10^{-5})$
Paramagnetic	
Iron ammonium alum	66
Uranium	40
Platinum	26
Aluminum	2.2
Sodium	0.72
Oxygen gas	0.19
Diamagnetic	
Bismuth	−16.6
Mercury	−2.9
Silver	−2.6
Carbon (diamond)	−2.1
Lead	−1.8
Sodium chloride	−1.4
Copper	−1.0

EXAMPLE 28.11 Magnetic dipoles in a paramagnetic material

Nitric oxide (NO) is a paramagnetic compound. The magnetic moment of each NO molecule has a maximum component in any direction of about one Bohr magneton. Compare the interaction energy of such magnetic moments in a 1.5 T magnetic field with the average translational kinetic energy of molecules at 300 K.

IDENTIFY and SET UP This problem involves the energy of a magnetic moment in a magnetic field and the average thermal kinetic energy. We have Eq. (27.27), $U = -\vec{\mu} \cdot \vec{B}$, for the interaction energy of a magnetic moment $\vec{\mu}$ with a $\vec{B}$ field, and Eq. (18.16), $K = \frac{3}{2}kT$, for the average translational kinetic energy of a molecule at temperature T .

EXECUTE We can write $U = -\mu_{\parallel}B$, where $\mu_{\parallel}$ is the component of the magnetic moment $\vec{\mu}$ in the direction of the $\vec{B}$ field. Here the maximum value of $\mu_{\parallel}$ is about μ_B , so

$$\begin{aligned} |U|_{\max} &\approx \mu_B B = (9.27 \times 10^{-24} \text{ J/T})(1.5 \text{ T}) \\ &= 1.4 \times 10^{-23} \text{ J} = 8.7 \times 10^{-5} \text{ eV} \end{aligned}$$

The average translational kinetic energy K is

$$\begin{aligned} K &= \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23} \text{ J/K})(300 \text{ K}) \\ &= 6.2 \times 10^{-21} \text{ J} = 0.039 \text{ eV} \end{aligned}$$

EVALUATE At 300 K the magnetic interaction energy is only about 0.2% of the thermal kinetic energy, so we expect only a slight degree of alignment. This is why paramagnetic susceptibilities at ordinary temperature are usually very small.

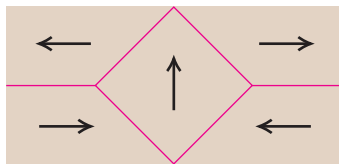
KEYCONCEPT In a paramagnetic material the magnetic dipole moments of its atoms tend to align with an externally applied magnetic field. Random thermal motions tend to cancel out this alignment, however. Paramagnetic materials are weakly attracted to the poles of a magnet.

Diamagnetism

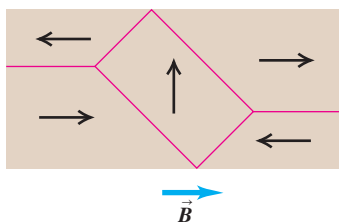
In some materials the total magnetic moment of all the atomic current loops is zero when no magnetic field is present. But even these materials have magnetic effects because an external field alters electron motions within the atoms, causing additional current loops and induced magnetic dipoles comparable to the induced *electric* dipoles we studied in Section 28.5. In this case the additional field caused by these current loops is always *opposite* in direction to that of the external field. (This behavior is explained by Faraday's law of induction, which we'll study in Chapter 29. An induced current always tends to cancel the field change that caused it.)

Figure 28.27 In this drawing adapted from a magnified photo, the arrows show the directions of magnetization in the domains of a single crystal of nickel. Domains that are magnetized in the direction of an applied magnetic field grow larger.

(a) No field



(b) Weak field



(c) Stronger field

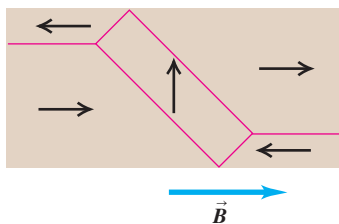
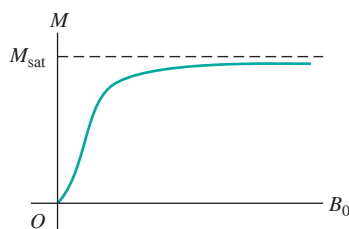


Figure 28.28 A magnetization curve for a ferromagnetic material. The magnetization M approaches its saturation value M_{sat} as the magnetic field B_0 (caused by external currents) becomes large.



Such materials are said to be **diamagnetic**. They always have negative susceptibility, as shown in Table 28.1, and relative permeability K_m slightly *less* than unity, typically of the order of 0.99990 to 0.99999 for solids and liquids. Diamagnetic susceptibilities are very nearly temperature independent.

Ferromagnetism

There is a third class of materials, called **ferromagnetic** materials, that includes iron, nickel, cobalt, and many alloys containing these elements. In these materials, strong interactions between atomic magnetic moments cause them to line up parallel to each other in regions called **magnetic domains**, even when no external field is present. Figure 28.27 shows an example of magnetic domain structure. Within each domain, nearly all of the atomic magnetic moments are parallel.

When there is no externally applied field, the domain magnetizations are randomly oriented. But when a field $\vec{B}_0$ (caused by external currents) is present, the domains tend to orient themselves parallel to the field. The domain boundaries also shift; the domains that are magnetized in the field direction grow, and those that are magnetized in other directions shrink. Because the total magnetic moment of a domain may be many thousands of Bohr magnetons, the torques that tend to align the domains with an external field are much stronger than occur with paramagnetic materials. The relative permeability K_m is *much* larger than unity, typically of the order of 1000 to 100,000. As a result, an object made of a ferromagnetic material such as iron is strongly magnetized by the field from a permanent magnet and is attracted to the magnet (see Fig. 27.38). A paramagnetic material such as aluminum is also attracted to a permanent magnet, but K_m for paramagnetic materials is so much smaller for such a material than for ferromagnetic materials that the attraction is very weak. Thus a magnet can pick up iron nails, but not aluminum cans.

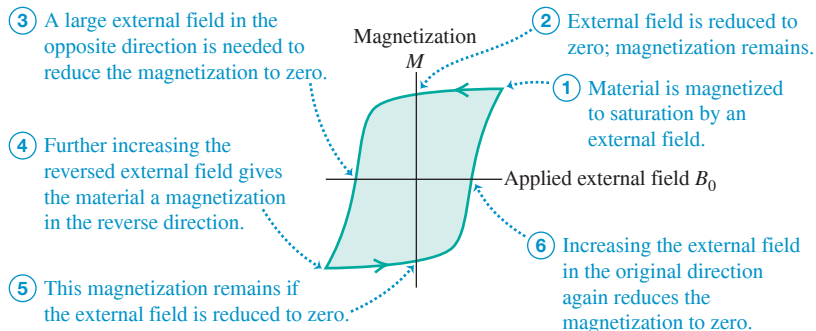
As the external field is increased, a point is eventually reached at which nearly *all* the magnetic moments in the ferromagnetic material are aligned parallel to the external field. This condition is called **saturation magnetization**; after it is reached, further increase in the external field causes no increase in magnetization or in the additional field caused by the magnetization.

Figure 28.28 shows a “magnetization curve,” a graph of magnetization M as a function of external magnetic field B_0 , for soft iron. An alternative description of this behavior is that K_m is not constant but decreases as B_0 increases. (Paramagnetic materials also show saturation at sufficiently strong fields. But the magnetic fields required are so large that departures from a linear relationship between M and B_0 in these materials can be observed only at very low temperatures, 1 K or so.)

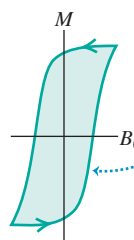
For many ferromagnetic materials the relationship of magnetization to external magnetic field is different when the external field is increasing from when it is decreasing. Figure 28.29a shows this relationship for such a material. When the material is magnetized to saturation and then the external field is reduced to zero, some magnetization

Figure 28.29 Hysteresis loops. The materials of both (a) and (b) remain strongly magnetized when B_0 is reduced to zero. Since (a) is also hard to demagnetize, it would be good for permanent magnets. Since (b) magnetizes and demagnetizes more easily, it could be used as a computer memory material. The material of (c) would be useful for transformers and other alternating-current devices where zero hysteresis would be optimal.

(a)



(b)



(c)

These materials can be magnetized to saturation and demagnetized by smaller external fields than in (a).

remains. This behavior is characteristic of permanent magnets, which retain most of their saturation magnetization when the magnetizing field is removed. To reduce the magnetization to zero requires a magnetic field in the reverse direction.

This behavior is called **hysteresis**, and the curves in Fig. 28.29 are called *hysteresis loops*. Magnetizing and demagnetizing a material that has hysteresis involve the dissipation of energy, and the temperature of the material increases during such a process.

Ferromagnetic materials are widely used in electromagnets, transformer cores, and motors and generators, in which it is desirable to have as large a magnetic field as possible for a given current. Because hysteresis dissipates energy, materials that are used in these applications should usually have as narrow a hysteresis loop as possible. Soft iron is often used; it has high permeability without appreciable hysteresis. For permanent magnets a broad hysteresis loop is usually desirable, with large zero-field magnetization and large reverse field needed to demagnetize. Many kinds of steel and many alloys, such as Alnico, are used for permanent magnets. The remaining magnetic field in such a material, after it has been magnetized to near saturation, is typically of the order of 1 T, corresponding to a remaining magnetization $M = B/\mu_0$ of about 800,000 A/m.

EXAMPLE 28.12 A ferromagnetic material

A cube-shaped permanent magnet is made of a ferromagnetic material with a magnetization M of about 8×10^5 A/m. The side length is 2 cm. (a) Find the magnetic dipole moment of the magnet. (b) Estimate the magnetic field due to the magnet at a point 10 cm from the magnet along its axis.

IDENTIFY and SET UP This problem uses the relationship between magnetization M and magnetic dipole moment μ_{total} and the idea that a magnetic dipole produces a magnetic field. We find μ_{total} from Eq. (28.28). To estimate the field, we approximate the magnet as a current loop with this same magnetic moment and use Eq. (28.18).

EXECUTE (a) From Eq. (28.28),

$$\mu_{\text{total}} = MV = (8 \times 10^5 \text{ A/m})(2 \times 10^{-2} \text{ m})^3 = 6 \text{ A} \cdot \text{m}^2$$

(b) From Eq. (28.18), the magnetic field on the axis of a current loop with magnetic moment μ_{total} is

$$B = \frac{\mu_0 \mu_{\text{total}}}{2\pi(x^2 + a^2)^{3/2}}$$

where x is the distance from the loop and a is its radius. We can use this expression here if we take a to refer to the size of the permanent magnet. Strictly speaking, there are complications because our magnet does not have the same geometry as a circular current loop. But because $x = 10$ cm is fairly large in comparison to the 2 cm size of the magnet, the term a^2 is negligible in comparison to x^2 and can be ignored. So

$$B \approx \frac{\mu_0 \mu_{\text{total}}}{2\pi x^3} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(6 \text{ A} \cdot \text{m}^2)}{2\pi(0.1 \text{ m})^3} = 1 \times 10^{-3} \text{ T} = 10 \text{ G}$$

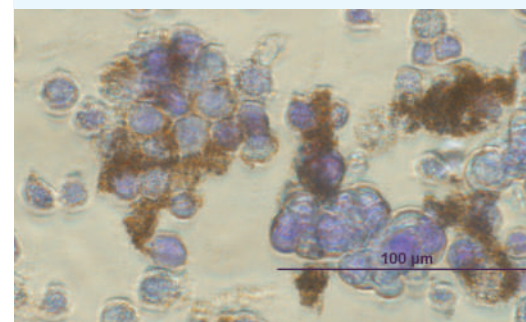
which is about ten times stronger than the earth's magnetic field.

EVALUATE We calculated B at a point *outside* the magnetic material and therefore used μ_0 , not the permeability μ of the magnetic material, in our calculation. You would substitute permeability μ for μ_0 if you were calculating B *inside* a material with relative permeability K_m , for which $\mu = K_m \mu_0$.

KEYCONCEPT In a ferromagnetic material there are strong interactions between the magnetic dipole moments of neighboring atoms, and the material can maintain a large overall magnetic dipole even if there is no externally applied magnetic field.

BIO APPLICATION Ferromagnetic Nanoparticles for Cancer Therapy

The violet blobs in this microscope image are cancer cells that have broken away from a tumor and threaten to spread throughout a patient's body. An experimental technique for fighting these cells uses particles of a ferromagnetic material (shown in brown) injected into the body. These particles are coated with a chemical that preferentially attaches to cancer cells. A magnet outside the patient then "steers" the particles out of the body, taking the cancer cells with them. (Photo courtesy of cancer researcher Dr. Kenneth Scarberry.)



TEST YOUR UNDERSTANDING OF SECTION 28.8 Which of the following materials are attracted to a magnet? (i) Sodium; (ii) bismuth; (iii) lead; (iv) uranium.

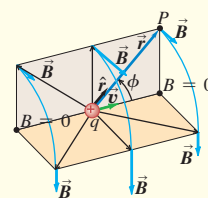
ANSWER

(i) and (iv) Sodium and uranium are paramagnetic materials and hence are attracted to a magnet, while bismuth and lead are diamagnetic materials that are repelled by a magnet. (See Table 28.1.)

CHAPTER 28 SUMMARY

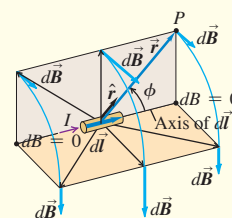
Magnetic field of a moving charge: The magnetic field $\vec{B}$ created by a charge q moving with velocity $\vec{v}$ depends on the distance r from the source point (the location of q) to the field point (where $\vec{B}$ is measured). The $\vec{B}$ field is perpendicular to $\vec{v}$ and to $\hat{r}$, the unit vector directed from the source point to the field point. The principle of superposition of magnetic fields states that the total $\vec{B}$ field produced by several moving charges is the vector sum of the fields produced by the individual charges. (See Example 28.1.)

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2} \quad (28.2)$$



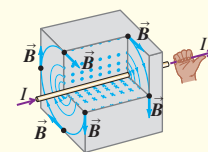
Magnetic field of a current-carrying conductor: The law of Biot and Savart gives the magnetic field $d\vec{B}$ created by an element $d\vec{l}$ of a conductor carrying current I . The field $d\vec{B}$ is perpendicular to both $d\vec{l}$ and $\hat{r}$, the unit vector from the element to the field point. The $\vec{B}$ field created by a finite current-carrying conductor is the integral of $d\vec{B}$ over the length of the conductor. (See Example 28.2.)

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2} \quad (28.6)$$



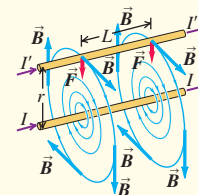
Magnetic field of a long, straight, current-carrying conductor: The magnetic field $\vec{B}$ at a distance r from a long, straight conductor carrying a current I has a magnitude that is inversely proportional to r . The magnetic field lines are circles coaxial with the wire, with directions given by the right-hand rule. (See Examples 28.3 and 28.4.)

$$B = \frac{\mu_0 I}{2\pi r} \quad (28.9)$$



Magnetic force between current-carrying conductors: Two long, parallel, current-carrying conductors attract if the currents are in the same direction and repel if the currents are in opposite directions. The magnetic force per unit length between the conductors depends on their currents I and I' and separation r . The definition of the ampere is based on this relationship. (See Example 28.5.)

$$\frac{F}{L} = \frac{\mu_0 II'}{2\pi r} \quad (28.11)$$



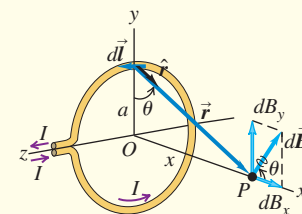
Magnetic field of a current loop: The law of Biot and Savart allows us to calculate the magnetic field produced along the axis of a circular conducting loop of radius a carrying current I . The field depends on the distance x along the axis from the center of the loop to the field point. If there are N loops, the field is multiplied by N . At the center of the loop, $x = 0$. (See Example 28.6.)

$$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}} \quad (28.15)$$

(circular loop)

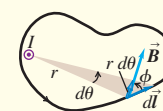
$$B_x = \frac{\mu_0 N I}{2a} \quad (28.17)$$

(center of N circular loops)



Ampere's law: Ampere's law states that the line integral of $\vec{B}$ around any closed path equals μ_0 times the net current through the area enclosed by the path. The positive sense of current is determined by a right-hand rule. (See Examples 28.7–28.10.)

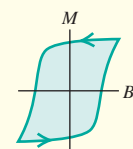
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}} \quad (28.20)$$



Magnetic fields due to current distributions: The table lists magnetic fields caused by several current distributions. In each case the conductor is carrying current I .

Current Distribution	Point in Magnetic Field	Magnetic-Field Magnitude
Long, straight conductor	Distance r from conductor	$B = \frac{\mu_0 I}{2\pi r}$
Circular loop of radius a (for N loops, multiply these expressions for magnetic-field amplitude by N)	On axis of loop	$B = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$
	At center of loop	$B = \frac{\mu_0 I}{2a}$
Long cylindrical conductor of radius R	Inside conductor, $r < R$	$B = \frac{\mu_0 I}{2\pi} \frac{r}{R^2}$
	Outside conductor, $r > R$	$B = \frac{\mu_0 I}{2\pi r}$
Long, closely wound solenoid with n turns per unit length, near its midpoint	Inside solenoid, near center	$B = \mu_0 n I$
	Outside solenoid	$B \approx 0$
Tightly wound toroidal solenoid (toroid) with N turns	Within the space enclosed by the windings, distance r from symmetry axis	$B = \frac{\mu_0 N I}{2\pi r}$
	Outside the space enclosed by the windings	$B \approx 0$

Magnetic materials: When magnetic materials are present, the magnetization of the material causes an additional contribution to $\vec{B}$. For paramagnetic and diamagnetic materials, μ_0 is replaced in magnetic-field expressions by $\mu = K_m \mu_0$, where μ is the permeability of the material and K_m is its relative permeability. The magnetic susceptibility χ_m is defined as $\chi_m = K_m - 1$. Magnetic susceptibilities for paramagnetic materials are small positive quantities; those for diamagnetic materials are small negative quantities. For ferromagnetic materials, K_m is much larger than unity and is not constant. Some ferromagnetic materials are permanent magnets, retaining their magnetization even after the external magnetic field is removed. (See Examples 28.11 and 28.12.)



Chapter 28 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 28.1** (Section 28.1) and **EXAMPLE 28.2** (Section 28.2) before attempting these problems.

VP28.2.1 A proton with charge $+1.60 \times 10^{-19}$ C is at the origin ($x = y = z = 0$) and is moving at 2.00×10^5 m/s in the $+x$ -direction. Find the magnetic field $\vec{B}$ that this proton produces at (a) $x = 0$, $y = 1.00$ mm, $z = 0$; (b) $x = 0$, $y = 0$, $z = 2.00$ mm; (c) $x = 1.00$ mm, $y = 0$, $z = 0$; (d) $x = 1.00$ mm, $y = 1.00$ mm, $z = 0$.

VP28.2.2 A proton (charge $+1.60 \times 10^{-19}$ C) and an electron (charge -1.60×10^{-19} C) are both moving in the xy -plane with the same speed, 4.20×10^5 m/s. The proton is moving in the $+y$ -direction along the line $x = 0$, and the electron is moving in the $-y$ -direction along the line $x = +2.00$ mm. At the instant when the proton and electron are at their closest approach, what are the magnitude and direction of the magnetic force that (a) the proton exerts on the electron and (b) the electron exerts on the proton?

VP28.2.3 A segment of wire centered at the origin ($x = y = z = 0$) is 2.00 mm in length and carries a current of 6.00 A in the $+y$ -direction. You measure the magnetic field due to this segment at the point $x = 3.00$ m, $y = 4.00$ m, $z = 0$. Find (a) the unit vector from the wire segment to this point and (b) the magnetic field at this point.

VP28.2.4 A segment of wire centered at the origin ($x = y = z = 0$) is 8.00 mm in length and carries a current of 4.00 A in the $+z$ -direction. Find the magnetic force that this segment exerts on an electron (charge -1.60×10^{-19} C) at the point $x = 1.25$ m, $y = 0$, $z = 0$ that is moving at 3.00×10^5 m/s in the $-x$ -direction.

Be sure to review **EXAMPLES 28.3 and 28.4** (Section 28.3) and **EXAMPLE 28.5** (Section 28.4) before attempting these problems.

VP28.5.1 How much current must a long, straight wire carry in order for the magnetic field that it produces 1.40 cm from the central axis of the wire to have magnitude 3.20×10^{-5} T?

VP28.5.2 Two long, straight, conducting wires are both perpendicular to the xy -plane. Wire 1 passes through the point $x = 0$, $y = 1.00$ cm, $z = 0$ and carries current 1.00 A in the $+z$ -direction. Wire 2 passes through the point $x = 0$, $y = -1.00$ cm, $z = 0$ and carries current 4.00 A also in the $+z$ -direction. Find the net magnetic field due to both wires at (a) $x = 0$, $y = 0$, $z = 0$; (b) $x = 0$, $y = 2.00$ cm, $z = 0$; (c) $x = 0$, $y = -2.00$ cm, $z = 0$.

VP28.5.3 A long, straight, conducting wire carries a current of 2.00 A. A second, identical wire is parallel to the first and separated from it by 3.00 cm. The magnetic field at a point halfway between the two wires has magnitude 4.00×10^{-5} T. What is the current in the second wire if its direction is (a) the same as in the first wire? (b) Opposite to that in the first wire?